SUPPLEMENTARY MATERIAL TO "EXPLICIT HAMILTONIAN BALL PACKINGS IN EVERY DIMENSION"

LIANG SONG

Abstract. This supplement contains the complete construction and every proof that the main text states without proof or proves only in outline: the finite comparator and the explicit planar rearrangements, the handle construction on closed and bordered bases, the strip and layer maps, the four model families and their transfers to the target, the common isotopy and the global Hamiltonian in every dimension, the input-name programs, the finite compilation, and the termination of the algorithm. The source records the names of the corresponding Lean declarations; the appendix describes the Lean explicitness check and the formal statements that differ in form from the text.

Authorship and the role of AI. See the corresponding section at the end of the main text. Lean declaration names in §1 and in the source tags (indexed in `verification/PAPER-INDEX.md`) locate the formal proofs.

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1. A checking route

This section lists, in the order of the construction, the statements that have to be checked for Theorem A, and with it Theorem C(i) for $n = 2$ and Corollary F; then the steps that replace the four-dimensional ones for $n \geq 3$, which complete Theorem C. Each step gives its input and output, where it is proved, and what it adds to the existence proofs it replaces. Numbers with the prefix M refer to the main text, as do the lettered results A–G; numbers without prefix refer to this supplement. The conventions used at each interface are in Table 3, and Table 1 at the end of this section lists the Lean declarations of each step; the full list of source tags is `verification/PAPER-INDEX.md`.

Organisation and reading budget. After the conventions of §2, the sections fall into four groups: the comparison data and the planar maps (§§3–7); the four-dimensional model, the handle construction, the strip and layer maps and the return to the ball (§§8–13), where §§10, 11 and 12 also contain the bordered handle used for $n \geq 3$; the models in dimension $2n \geq 6$ (§§14–19); and the global construction, the worked instance, the finite form and the algorithm (§§20–26).

For Theorem A (equal balls, the linear datum) a reader needs §2 (pp. 7–18), §3.7 (pp. 26–27), §§4 and 5 (pp. 27–54; in §4 only the equal-ball family), §§8 and 9 (pp. 73–85), §§10.9–10.10 (pp. 89–97), §§11.1–11.3 (pp. 97–102), §13 (pp. 108–117), and §§20–21 (pp. 149–159). The worked instance of §22 follows the same data numerically. The other four-dimensional inputs of Theorem C add the scalar comparator of §6 (pp. 54–62).

For Theorem C with $n \geq 3$ a reader needs, besides §§2, 20 and 21: the comparator, §§3.1–3.6 (pp. 18–26); the planar maps, §4 for the slice family (pp. 27–43), Lemma 6.4 with the wells (22) (p. 57) and §7 (pp. 62–73); the bordered handle, §§10.1–10.8 and 10.10 (pp. 85–89, pp. 96–97), §§11 and 12 (pp. 97–108); the ball chart, Lemma 13.20 (p. 114), and Lemma 23.2 (p. 162), which supplies the rational auxiliary data; and the models, §§14–15 for capacities below one half (pp. 117–127), §16.1 and §17 for one large ball and $n \geq 4$ (pp. 128–129, pp. 131–138), and §§16.2, 18 and 19 for one large ball and $n = 3$ (pp. 129–131, pp. 138–149); the opening of §16 (p. 127) serves both. Part (iii) of Theorem C adds §§23–26 (pp. 162–173).

1.1. **The four-dimensional chain.** The data are $n = 2$ and admissible $(2, k, C, c)$, i.e. $\sum_i c_i^2 < C^2$ and $3c_i < C$. The data of Theorem A, and those of Corollary F with $k \geq 10$ and $\varepsilon < \frac{7}{8}$, are equal capacities π with $\frac{3}{4} < \rho\sqrt{k} < 1$, $\rho = \pi/C$; for them steps F2–F4 take the linear form, and for all other data the scalar form. The scalar route (Proposition 6.8) also covers equal capacities; the linear route is kept for its closed form, in which every parameter is an algebraic expression in $\sqrt{k}$ and ρ (Remark M3.9, Remark 13.25).

F1 *Normalisation and case selection.* In: $(2, k, C, c)$ admissible. Out: $\rho_i = c_i/C < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$, and the choice of the linear or the scalar datum. Proved: Definition M6.1, Construction 20.1.

F2 *Auxiliary capacities and profile.* In: ρ. Out: $a_i \in (\rho_i, \frac{1}{3})$, height $h < \frac{1}{3}$, τ_Δ, and the concave function $P_{\mathrm{b}} = H_{\mathrm{b}} - \sum_i (a_i - p)_+$ on $[0, h]$ with $\int_0^h P_{\mathrm{b}} > 0$. Proved: Lemma 8.5; equal case Remark 13.25 ($a = t_0/\sqrt{k}$, $h = t_0/3$). New: closed formulas, rational for rational ρ in the general case.

F3 *Comparison function.* In: P_{b}. Out: a convex function K and a gap $g > 0$ for the rearrangement inequality. Proved: Proposition M3.8 (linear, $K(p) = (3\sqrt{k} - 9)p$); Proposition 6.2 (scalar). New: a finite iterate with an explicit gap in place of the fixed point of the rearrangement contraction of [Ope26]; in the linear case no iteration.

F4 *Planar realisation.* In: K, $F = K - P_{\mathrm{b}}$, g. Out: valid cubic data (Definition 10.13): a Hamiltonian isotopy q_t of $(\mathbb{C}, \omega_{\mathbb{C}})$, with generator Q_t supported in $\{\pi|z|^2 \leq P_{\mathrm{supp}}\}$, $P_{\mathrm{supp}} < h$, satisfying (63). Proved: Theorems 5.34 and 4.21 (linear); Propositions 6.7 and 6.8 (scalar). New: the area-preserving map is built from Morse charts of an explicit convex potential, and its Hamiltonian is a radial source primitive.

F5 *Model and handle data.* In: $h < d_n < \frac{1}{3}$. Out: the cubic cover model $(\mathbb{R}^2 \times \mathbb{C}, \Omega_0)$ with deck maps, marked fibres, annulus and dual one-form. Proved: Definition 8.1, Propositions 8.3 and 9.16. New: an explicit theta lift Z of the Fermat torus, so that Biran's disc bundle [Bir01] is given by formulas on the cover.

F6 *Handle.* In: valid cubic data and the model. Out: maps f_i^N on $B^\circ(\hat\rho_i)$ into $\{p < h\}$ of the cover with $(f_i^N)^*\Omega_0 = \omega_0$, injective, with disjoint images and no two image points deck equivalent. Proved: §10.9 (Propositions 10.21, 10.25, 10.28 and 10.29) with the strip map, Proposition 10.32; outline in Remark M4.7. New: the three Moser paths have explicit primitives that vanish on the marked fibres and are invariant under the deck maps.

F7 *Transfer.* In: compact subsets of $\{p < h\}$ off the marked fibres. Out: $\mathcal{T}$ into $B^\circ(1)$ with $\mathcal{T}^*\omega_0 = \Omega_0$, identifying exactly deck-equivalent points, with a compact margin. Proved: Theorem 13.21 (§13). New: injection, transport and Moser return are explicit flows; a cube-root symmetry keeps the image off the line removed by the ball chart.

F8 *Embeddings.* In: steps F6 and F7. Out: normalised embeddings $f_i^\natural$ on $B^\circ(\rho_i^{\mathrm{dom}})$, injective on $B(\rho_i^+)$, with disjoint images in $B(1-m)$. Proved: Theorem 13.23; statement Theorem M5.1.

F9 *Common isotopy.* In: the rescaled embedding datum (Proposition M6.3) and centres with disjoint closed balls. Out: $F_{i,t}$, symplectic, with disjoint images at every time, from $u \mapsto s_i + u$ to f_i. Proved: Proposition M6.6, Proposition 21.14. New: an explicit route, an explicit path in $\mathrm{Sp}(2n)$ and a common Alexander tail, the same in every dimension.

F10 *Hamiltonian.* In: $F_{i,t}$. Out: $H_{\mathfrak{E}}$, smooth, supported in $(0,1)\times\mathbb{B}(R_F)$, with $X_H = \dot F_{i,t}$ on the moving balls, satisfying Pack. Proved: Definition M6.10, Lemma M6.12, Proposition M6.15, Theorem M6.14. New: the Hamiltonian is the pushforward of a cut-off radial primitive computed in the source balls.

F11 *Statements.* $H[2,k,C,c,s] = H_{\mathfrak{E}}$ (Definition M6.16); Theorem C(i) for $n = 2$, Theorem A and Corollary F follow after checking admissibility (§M6.4).

F12 *Finite form and algorithm* (only for Theorem C(iii)). In: any H satisfying Pack. Out: (B, P) with $\beta_B P$ satisfying Pack, and a machine that finds such a pair from input names. Proved: Theorem M7.2 (Theorem 24.15), Theorem M7.4 (Theorems 26.4 and 25.5). New: a sound rational Euler certificate, and termination proved from the explicit H rather than from an existence theorem.

1.2. **The higher-dimensional branches.** For $n \geq 3$ step F1 normalises in the same way, with the conditions $\sum_i \rho_i^n < 1$ and $\rho_i + \rho_j < 1$, and the case distinction of Construction 20.1 selects one of three branches. Steps F9–F12 are unchanged, and steps F2–F8 are replaced as follows.

H1 *Comparison datum on a polytope.* In: a rational downward polytope Δ and a concave minorant P. Out: K and a Hamiltonian isotopy of $\mathbb{C}^m$ preserving V_Δ, Poisson-commuting with the face moments on a collar, with gap γ. Proved: Theorems M3.5 and M3.4; the lift is Theorem 7.13 (§§7 and 4). New: a finite polyhedral iterate with log-sum-exp smoothing and an explicit realisation of each coordinate rearrangement.

H2 *Packing in a model.* In: a model input with handle data and collar bounds, a comparison datum, and $\rho_i < a_i$ with the profile minorant. Out: embeddings ψ_i of $B(\rho_i)$ into $\operatorname{int} V_\Delta(E)$. Proved: Theorem M2.6 in §M4.5 and Theorem 10.12 (Propositions 10.8–10.11, 11.15 and 11.17); the composition with the transfer is Proposition M2.7. New: Moser maps given by explicit fields on a bordered base with a finite atlas.

H3 *All* $\rho_i < \frac{1}{2}$. In: ρ with $\sum_i \rho_i^n < 1$. Out: embeddings into $B^\circ(1)$ with disjoint images. The diagonal affine quadric curve, §§14 and 15. Proved: transfer Theorem 15.12; embeddings Theorem 14.7, statement Theorem M5.2. New: a fixed explicit curve in place of a generic complete intersection, and a transfer by explicit flows.

H4 *One* $\rho_j \geq \frac{1}{2}$, $n \geq 4$. In: $A = \rho_j$ and valid auxiliary data. Out: the identity on the large ball and embeddings of the others into the shell S_a. The mixed model and the slit-shell return,

§§16.1 and 17. Proved: Lemma 17.7, Theorem 17.9; statement Theorem M5.3. New: the return to the shell about the reserved ball is a closed formula.

H5 *One* $\rho_j \geq \frac{1}{2}$, $n = 3$. In: $A = \rho_j$ and valid rational auxiliary data. Out: embeddings into $B^\circ(1)$ with disjoint images. The polynomial cubic model and the bubble, §§16.2, 18 and 19. Proved: transfer Proposition 19.3; Theorem 19.7. New: an area profile with a quantitative tail (Proposition 18.6) and a polynomial bubble that inserts the large ball.

Steps F9–F12 then give Theorem C(i) and Corollaries D and E.

1.3. **What the formal proof covers.** The scope of the Lean verification, the trusted base and the unverified metaprograms are described in §M8; the explicitness check and the formal statements whose form differs from the text are in Appendix A.

Table 1: Lean declarations for the steps above. Names are relative to `SymplecticPacking`, except the theorems of `verification/`, which have no namespace.

step	Lean declarations
F1	`General.selectEmbeddings`, `Branch2.EqualCase`
F2	`Branch2.aux`, `Branch2.height`, `Branch2.Hyp.integral_Pb`; `Params.t0`, `Params.a`, `Params.h`
F3	`Params.K_sub_F_ge`; `Branch2.Cmp.Input.Kcmp`, `Branch2.Cmp.Input.Hyp.Kcmp_gap`
F4	`Qhat.Cubic.qhatP_spec`, `Qhat.qhat_interface`, `Qhat.K_sub_F_q_ge`; `Branch2.Hyp.cubicDataOf_valid`
F5	`Bundle.Omega0loc`, `Embed.deckU`, `Embed.deckV`, `Bundle.annulus`
F6	`Handle.Cubic.handleMoser`, `Strip.Cubic.fN_symplectic`, `Strip.Cubic.fN_disjoint`, `Strip.Cubic.fN_avoid_fibres`
F7	`Return.compositeT_pullback`, `Return.compositeT_inj_mod_gauge`, `Return.avoid_L_compact`, `Return.ballChart_symplectic_rep`
F8	`Branch2.embeddings2_spec`, `Branch2.embeddingsEq_spec`, `Branch2.embeddingsGen_spec`
F9	`General.Ham.Isotopy.IsoInput.commonIsotopy`, `…isoF_sympl`, `…isoF_disj`
F10	`HamEnd.hamiltonianOfEmbeddings`, `General.Ham.hamVF_ofIsotopy_image`, `HamEnd.hamiltonian_end`
F11	`main_theorem`, `four_dim_theorem`, `epoch_packing`, `epoch_theorem` in `verification/Statement.lean`
F12	`General.Compile.main_finite`, `original_name_algorithm`, `epoch_name_algorithm`
H1	`HighDim.CmpInput.comparatorDatum`, `HighDim.ComparisonDatum`, `HighDim.PlanarGen.Input`
H2	the tags of Theorem 10.12 in `PAPER-INDEX.md`; composition `HighDim.assembly_spec`
H3	`BranchSmall.vectorTransfer_spec`, `BranchSmall.embeddings_spec`
H4	`BranchLargeHigh.shell_pullback`, `BranchLargeHigh.embeddings_spec`
H5	`BranchLarge3.cubicTransfer_spec`, `BranchLarge3.embeddings_spec`; corollaries `main_theorem`, `neighbourhood_criterion`, `equal_balls`

2. Conventions and elementary tools

This section fixes the conventions of both documents and proves the elementary lemmas that the other sections use without comment: the flat step and the flat concatenation of paths and isotopies, cutoffs, the top power and the local form of toric symplectic forms, and the Moser lemma with the sign conventions of the paper. Lemma 2.8 contains Lemma M4.2 as a special case. Tables 2 and 3 (§2.6) collect the coordinate, sign and normalisation conventions of all parts of the construction and say where one is converted into another; Table 4 extends Table M3.

2.1. **Conventions.** This subsection expands the conventions of §M1.8 with the identities used in the proofs.

Manifolds are finite dimensional, Hausdorff and second countable; boundaries are stated explicitly.

Phase space. We write $\mathbb{R}^{2n} = \mathbb{C}^n$ with $z_j = x_j + iy_j$ and $\omega_0 = \sum_{j=1}^n dx_j \wedge dy_j$. Let J_0 be the matrix of ω_0, i.e. $\omega_0(v,w) = \langle v, J_0 w\rangle$; in the coordinates (x,y), $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, which is multiplication by $-i$ on $\mathbb{C}^n$. Where $\mathbb{R}^{2n}$ is regarded as a complex vector space with i acting by J_0 (§M6), this complex structure is the conjugate of that of $\mathbb{C}^n$. For a function $H(t,z)$, smooth in (t,z), the Hamiltonian vector field $X_H = X_{H_t}$ is defined by $\iota_{X_H}\omega_0 = dH_t$, where d is the spatial differential. Equivalently

$$X_H = J_0 \nabla H, \qquad \dot{x}_j = \frac{\partial H}{\partial y_j}, \quad \dot{y}_j = -\frac{\partial H}{\partial x_j}, \qquad \dot{z}_j = -2i\,\frac{\partial H}{\partial \bar{z}_j},$$

since $\omega_0(X,w) = \langle X, J_0 w\rangle = -\langle J_0 X, w\rangle$ and $J_0^2 = -I$. We write φ_H^t for the flow of X_H from time 0 to time t. For example $H = \pi|z_1|^2$ generates $z_1 \mapsto e^{-2\pi i t} z_1$, a rotation of period one. The letter p is reserved for moment maps.

Capacities and balls. For $c > 0$ and $s \in \mathbb{C}^n$,

$$B(c) = \{z \in \mathbb{C}^n : \pi|z|^2 \le c\}, \qquad B^\circ(c) = \{\pi|z|^2 < c\}, \qquad B(s,c) = s + B(c),$$

the closed and open balls of capacity c; the Euclidean radius of $B(c)$ is $\sqrt{c/\pi}$. We write $\mathbb{B}(r)$ and $\mathbb{B}^\circ(r)$ for the closed and open Euclidean balls of radius r about the origin. Volumes are taken with respect to $\omega^n/n!$; for ω_0 this is Lebesgue measure, and $\operatorname{vol} B(c) = \pi^n (c/\pi)^n/n! = c^n/n!$.

Lemma 2.1 (Dilations)**.** *Let $\lambda > 0$, $\delta_\lambda(z) = \lambda z$, and let $H(t,z)$ be smooth. Then $\delta_\lambda^* \omega_0 = \lambda^2 \omega_0$, $\delta_\lambda(B(s,c)) = B(\lambda s, \lambda^2 c)$, and the function $\tilde{H}(t,z) = \lambda^2 H(t, z/\lambda)$ satisfies $X_{\tilde{H}}(t,z) = \lambda X_H(t, z/\lambda)$; hence an integral curve γ of X_H gives the integral curve $\lambda\gamma$ of $X_{\tilde{H}}$, and $\varphi_{\tilde{H}}^t = \delta_\lambda \circ \varphi_H^t \circ \delta_\lambda^{-1}$ wherever either side is defined.*

Proof. The first two statements are immediate. Put $X(t,z) = \lambda X_H(t, z/\lambda)$. For $w \in \mathbb{C}^n$, $\omega_0(X(t,z), w) = \lambda\, dH_{(t,z/\lambda)}(w)$, while $d\tilde{H}_{(t,z)}(w) = \lambda^2\, dH_{(t,z/\lambda)}(w/\lambda)$ is the same number; so $X = X_{\tilde{H}}$. If $\dot{\gamma} = X_H(t,\gamma)$ then $\frac{d}{dt}(\lambda\gamma) = \lambda X_H(t,\gamma) = X_{\tilde{H}}(t, \lambda\gamma)$. □

With $\lambda = 1/\sqrt{C}$ this is the normalisation of the target capacity to 1 used throughout (§M6).

Moments and angles. On a fibre $\mathbb{C}^m$ with coordinates $w_j = \xi_j + i\eta_j$ we use

$$p_j = \pi|w_j|^2, \qquad \theta_j = \frac{\arg w_j}{2\pi} \in \mathbb{R}/\mathbb{Z}, \qquad |p| = \sum_{j=1}^m p_j, \qquad \omega_V = \sum_{j=1}^m dp_j \wedge d\theta_j = \sum_{j=1}^m d\xi_j \wedge d\eta_j.$$

The identity $dp_j \wedge d\theta_j = d\xi_j \wedge d\eta_j$ holds where $w_j \neq 0$: in polar coordinates $w_j = re^{i\phi}$ both sides equal $r\, dr \wedge d\phi$. The one-form $p_j\, d\theta_j = \frac{1}{2}(\xi_j\, d\eta_j - \eta_j\, d\xi_j)$ is smooth on all of $\mathbb{C}$, with differential $d\xi_j \wedge d\eta_j$. A function $f(p)$ with f smooth on an open subset of $\mathbb{R}^m$ is a smooth function of w, because each p_j is a polynomial in (ξ_j, η_j); such functions are called *toric*.

Fibre Hamiltonians. On fibres, and on the models built from them, Hamiltonian fields of fibre Hamiltonians Q follow the convention of [Ope26],

$$\iota_{X_Q}\omega_V = -dQ,$$

which is opposite to the convention for H. With this convention $X_{p_j} = \partial_{\theta_j}$ generates $w_j \mapsto e^{2\pi i t} w_j$, the rotation of period one in the positive direction, because $\iota_{\partial_{\theta_j}}(dp_j \wedge d\theta_j) = -dp_j$. The field X_Q of the fibre convention is the field X_{-Q} of the convention of ω_0. Two functions *Poisson-commute* if $\omega_V(X_F, X_G) = 0$; this condition does not depend on the convention, and it

holds if and only if F is constant along the flow of X_G. The convention $\iota_{X_Q}\omega_V = -dQ$ is used for fibre Hamiltonians, for Hamiltonians on the toric models and for Moser fields; the convention $\iota_{X_H}\omega_0 = dH$ is used for the global Hamiltonian on $\mathbb{R}^{2n}$ (§M6, §M7, §21, §24).

Complex conventions and the Fubini–Study form. We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial,$$

so that $d^c u$ is a real one-form for real u. Since ∂ and $\bar\partial$ are derivations, $d^c F(u) = F'(u)\, d^c u$ for smooth F and smooth real u. For a function $u = F(|z|^2)$ of one complex variable, $\bar\partial u - \partial u = F'(|z|^2)(z\,d\bar z - \bar z\,dz) = -2i\,F'(|z|^2)|z|^2\,d\phi$ with $\phi = \arg z$, hence

$$d^c F(|z|^2) = F'(|z|^2)\,|z|^2\,d\theta, \qquad \text{in particular} \qquad d^c \log|z|^2 = d\theta, \tag{1}$$

with $\theta = \arg z/2\pi$. The one-form $|z|^2 d\theta = (x\,dy - y\,dx)/2\pi$ is smooth, so (1) holds on all of $\mathbb{C}$ when F is smooth, and on $\mathbb{C}\setminus 0$ for $F = \log$. Since $\int_{|z|=\varepsilon} d\theta = 1$, the current $dd^c \log|z|^2$ has unit mass at the origin. On $\mathbb{CP}^n$ we put $\omega_{\mathrm{FS}} = dd^c \log\|Z\|^2$.

Lemma 2.2 (The affine Fubini–Study chart)**.** *On $\mathbb{C}^N$, put $D = 1 + |z|^2$ and*

$$\lambda_{\mathrm{FS}} = \frac{1}{2\pi D}\sum_{j=1}^{N}(x_j\,dy_j - y_j\,dx_j), \qquad \omega_{\mathrm{FS}} = d\lambda_{\mathrm{FS}}.$$

Then:

- *(a) $\lambda_{\mathrm{FS}} = d^c \log D$ is smooth. Where every coordinate is nonzero it equals $\sum_j p_j^{\mathrm{FS}}\,d\theta_j$, with $p_j^{\mathrm{FS}} = |z_j|^2/D$ and $\theta_j = \arg z_j/(2\pi)$; its differential is $\sum_j dp_j^{\mathrm{FS}} \wedge d\theta_j$ there.*
- *(b) The map $\mathrm{ch}(z) = z/\sqrt{\pi D}$ is a smooth bijection of $\mathbb{C}^N$ onto $B^\circ(1)$ with $\mathrm{ch}^*\omega_0 = \omega_{\mathrm{FS}}$, with smooth inverse $w \mapsto \sqrt{\pi}\,w/\sqrt{1-\pi|w|^2}$.*
- *(c) $\mathrm{ch}^*\lambda_{\mathrm{std}} = \lambda_{\mathrm{FS}}$, where $\lambda_{\mathrm{std}} = \frac{1}{2}\sum_j(x_j\,dy_j - y_j\,dx_j)$, and $\pi|\mathrm{ch}(z)|^2 = 1 - D^{-1} < 1$.*

Proof. The identity $d^c|z|^2 = \sum_j(x_j\,dy_j - y_j\,dx_j)/(2\pi)$ and the chain rule give (a). The inverse in (b) follows by solving $\pi|w|^2 = |z|^2/(1+|z|^2)$ for the radial factor. Both maps are smooth on their stated domains. Under a variable real radial rescaling $z \mapsto r(z)z$, λ_{std} pulls back to $r(z)^2\lambda_{\mathrm{std}}$: the derivative of r contributes $\omega_0(z,z) = 0$. Taking $r = (\pi D)^{-1/2}$ gives (c), and applying d gives $\mathrm{ch}^*\omega_0 = \omega_{\mathrm{FS}}$. □

The model forms of Table M2 are written with d^c and ω_{FS} in this normalisation.

Bidegree and orientation. Let U be an open subset of a surface Σ with oriented coordinates (x, y) over which a bundle $E = L_1 \oplus \cdots \oplus L_m$ has a unitary frame, so that $E|_U = U \times \mathbb{C}^m$. Every form on an open subset of $U \times \mathbb{C}^m$ splits uniquely into components of bidegree (r, s), where r counts dx, dy and s counts $d\xi_j, d\eta_j$; the coefficients may depend on all variables. A form of bidegree $(r, 0)$ is called *basic*, and a component vanishes if $r > 2$ or $s > 2m$. A top-degree form is *positive* if it is a positive multiple of $\omega_V^m \wedge dx \wedge dy$ for an oriented coordinate system (x, y) of Σ; when κ is a positive area form on Σ, this is the orientation of $\omega_V^m \wedge \kappa$. A one-form on E is *horizontal* if it annihilates the vertical tangent vectors.

2.2. **The flat step.** We use the following flat step for clocks and cutoffs, composing it with the explicitly defined smooth functions and taking products.

Definition 2.3 (Flat step)**.** Let $a_\flat(x) = 0$ for $x \le 0$ and $a_\flat(x) = e^{-1/x}$ for $x > 0$, and

$$S(x) = \frac{a_\flat(x)}{a_\flat(x) + a_\flat(1-x)} \qquad (x \in \mathbb{R}).$$

The denominator is positive because $x > 0$ or $1 - x > 0$. On $(0,1)$, $S(x) = e^{-1/x}/(e^{-1/x} + e^{-1/(1-x)})$; $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$.

Lemma 2.4 (Properties of S)**.** *S is C^∞ on $\mathbb{R}$, takes values in $[0,1]$, is nondecreasing on $\mathbb{R}$ and strictly increasing on $[0,1]$, and $S(1-x) = 1 - S(x)$. For every $j \geq 1$, $S^{(j)} = 0$ on $(-\infty, 0] \cup [1, \infty)$; in particular $S^{(j)}(0) = S^{(j)}(1) = 0$.*

Proof. On $x > 0$ every derivative of $a_\flat$ has the form $Q_j(1/x)e^{-1/x}$ with Q_j a polynomial, and $Q_j(1/x)e^{-1/x} \to 0$ as $x \downarrow 0$; by induction all derivatives of $a_\flat$ exist at 0 and vanish there, so $a_\flat$ is C^∞. Since the denominator of S is smooth and positive, S is C^∞. The identity $S(1-x) = a_\flat(1-x)/(a_\flat(1-x)+a_\flat(x)) = 1-S(x)$ is immediate. On $(0,1)$, $S(x) = 1/(1+\exp(1/x-1/(1-x)))$ and the exponent is strictly decreasing, so S is strictly increasing on $[0,1]$; with the constant values outside $(0,1)$ this gives monotonicity and $0 \leq S \leq 1$. Finally S is constant on $(-\infty, 0]$ and on $[1, \infty)$, so $S^{(j)}$, $j \geq 1$, vanishes on the interiors of these sets, and by continuity of $S^{(j)}$ also at 0 and 1. □

The vanishing of all derivatives at 0 and 1 is what allows paths to be concatenated in time without corners (Lemma 2.5).

Lemma 2.5 (Flat concatenation)**.** *Let $0 = t_0 < t_1 < \cdots < t_N = 1$ and let $\sigma_j(t) = S\big((t - t_{j-1})/(t_j - t_{j-1})\big)$, $1 \leq j \leq N$, be the clocks.*

(a) Let $\mathcal{P}$ and M be smooth manifolds and $\gamma_1, \dots, \gamma_N \colon [0,1] \times \mathcal{P} \to M$ smooth maps (smooth up to the boundary, i.e. restrictions of smooth maps on $(-\delta, 1+\delta) \times \mathcal{P}$ for some $\delta > 0$) such that $\gamma_j(1, y) = \gamma_{j+1}(0, y)$ for all y. Define $\bar\gamma \colon \mathbb{R} \times \mathcal{P} \to M$ by $\bar\gamma(t, y) = \gamma_j(\sigma_j(t), y)$ for $t \in [t_{j-1}, t_j]$, $\bar\gamma(t, y) = \gamma_1(0, y)$ for $t \leq 0$ and $\bar\gamma(t, y) = \gamma_N(1, y)$ for $t \geq 1$. Then $\bar\gamma$ is well defined and C^∞, and at every point (t_j, y), $0 \leq j \leq N$, every partial derivative of $\bar\gamma$ (in charts) that contains at least one derivative in t vanishes.

(b) Let $\psi^{(1)}, \dots, \psi^{(N)} \colon [0,1] \times M \to M$ be smooth isotopies of M (smooth up to the boundary as in (a)) with $\psi_0^{(j)} = \mathrm{id}$, generated by smooth time-dependent vector fields $Y_s^{(j)}$, so that $\partial_s \psi_s^{(j)} = Y_s^{(j)} \circ \psi_s^{(j)}$. Put $\Phi_0 = \mathrm{id}$, $\Phi_j = \psi_1^{(j)} \circ \Phi_{j-1}$, and

$$\varphi_t = \psi^{(j)}_{\sigma_j(t)} \circ \Phi_{j-1} \qquad (t \in [t_{j-1}, t_j]),$$

$\varphi_t = \mathrm{id}$ for $t \leq 0$ and $\varphi_t = \Phi_N$ for $t \geq 1$. Then φ is a smooth isotopy with $\varphi_0 = \mathrm{id}$ and $\varphi_1 = \Phi_N = \psi_1^{(N)} \circ \cdots \circ \psi_1^{(1)}$, generated by the field $Y_t = \sigma_j'(t)\, Y^{(j)}_{\sigma_j(t)}$ on $[t_{j-1}, t_j]$ and $Y_t = 0$ for $t \notin (0,1)$. The field Y is smooth on $\mathbb{R} \times M$, and Y and all its partial derivatives vanish at $t = t_j$. If M is symplectic and $Y_s^{(j)}$ is the Hamiltonian field of $Q_s^{(j)}$ (in either sign convention), with $Q^{(j)}$ smooth on $[0,1] \times M$ (up to the boundary), then Y_t is the Hamiltonian field of $Q_t = \sigma_j'(t)\, Q^{(j)}_{\sigma_j(t)}$ on $[t_{j-1}, t_j]$, $Q_t = 0$ for $t \notin (0,1)$, and Q is smooth on $\mathbb{R} \times M$.

Proof. (a) At $t = t_j$ the two definitions give $\gamma_j(1, y) = \gamma_{j+1}(0, y)$, and at $t = 0, 1$ the constant extensions agree with $\gamma_1(0, y)$ and $\gamma_N(1, y)$; so $\bar\gamma$ is well defined and continuous. Fix j and a point (t_j, y_0), and work in charts around $\bar\gamma(t_j, y_0)$ and y_0 (by continuity $\bar\gamma$ maps a neighbourhood into the chart). On $[t_{j-1}, t_j] \times \mathcal{P}$ the map $\bar\gamma$ is the smooth map $\gamma_j \circ (\sigma_j \times \mathrm{id})$. By the chain rule and Faà di Bruno's formula, a partial derivative of it containing at least one t-derivative is a sum of terms each of which contains a factor $\sigma_j^{(i)}(t)$ with $i \geq 1$; at $t = t_j$ (and at $t = t_{j-1}$) these factors are $(t_j - t_{j-1})^{-i} S^{(i)}(1) = 0$ (resp. $S^{(i)}(0) = 0$) by Lemma 2.4. The derivatives containing only y-derivatives are the y-derivatives of $\gamma_j(1, \cdot) = \gamma_{j+1}(0, \cdot)$ from both sides. Hence the one-sided jets at $t = t_j$ from the left and from the right agree (the same holds at $t = 0$ and

$t = 1$ against the constant extensions). A continuous map which is C^∞ up to the boundary on two closed half-spaces and whose one-sided jets agree along the common boundary is C^∞; so $\bar\gamma$ is C^∞.

(b) The maps $\gamma_j(s,x) = \psi_s^{(j)}(\Phi_{j-1}(x))$ satisfy the hypotheses of (a), since $\gamma_j(1,\cdot) = \Phi_j = \psi_0^{(j+1)} \circ \Phi_j = \gamma_{j+1}(0,\cdot)$; so φ is smooth, and each φ_t is a diffeomorphism as a composition of diffeomorphisms. On $[t_{j-1}, t_j]$,

$$\partial_t \varphi_t = \sigma_j'(t)\,(\partial_s \psi^{(j)})_{\sigma_j(t)} \circ \Phi_{j-1} = \sigma_j'(t)\, Y^{(j)}_{\sigma_j(t)} \circ \psi^{(j)}_{\sigma_j(t)} \circ \Phi_{j-1} = \sigma_j'(t)\, Y^{(j)}_{\sigma_j(t)} \circ \varphi_t.$$

The field $(t,x) \mapsto \sigma_j'(t) Y^{(j)}(\sigma_j(t), x)$ is smooth on $[t_{j-1}, t_j] \times M$, and every partial derivative of it (in t, in x, or mixed) at $t = t_{j-1}$ or $t = t_j$ is a sum of terms containing a factor $\sigma_j^{(i)}$ with $i \geq 1$, hence vanishes there; on $t \notin (0,1)$ the field is 0. So the one-sided jets at the seams are all zero and Y is smooth. The statement on Hamiltonians follows because $X_{cQ} = cX_Q$ for constants c in either convention, and Q is smooth by the same argument applied to the smooth function $Q^{(j)}$ on $[0,1] \times M$. □

2.3. **Cutoffs.**

Definition 2.6 (Radial and interval cutoffs)**.** For $0 \leq r < r'$ and u in a Euclidean space $\mathbb{R}^d$ put

$$\chi_{r,r'}(u) = 1 - S\Big(\frac{|u|^2 - r^2}{r'^2 - r^2}\Big).$$

For $\ell_0 < \ell_1$ and $0 < w < (\ell_1 - \ell_0)/2$ put $\chi^{\mathrm{I}}_{\ell_0,\ell_1,w}(x) = S\big((x-\ell_0)/w\big)\, S\big((\ell_1 - x)/w\big)$ for $x \in \mathbb{R}$.

The argument of S in $\chi_{r,r'}$ is a polynomial in u, so every derivative of a cutoff is an explicit expression in the derivatives of S. We write

$$s_1 = \max_{\mathbb{R}} S',$$

a finite constant determined by S, since S' is continuous and vanishes outside $[0,1]$.

Lemma 2.7 (Cutoffs)**.** *(a) $\chi_{r,r'}$ is C^∞, $O(d)$-invariant and nonincreasing in $|u|$, takes values in $[0,1]$, equals 1 on $\mathbb{B}(r)$ and 0 outside $\mathbb{B}^\circ(r')$, and all its derivatives of order ≥ 1 vanish on $\mathbb{B}(r) \cup (\mathbb{R}^d \setminus \mathbb{B}^\circ(r'))$, in particular on the spheres $|u| = r$ and $|u| = r'$. Its gradient is*

$$\nabla \chi_{r,r'}(u) = -\frac{2u}{r'^2 - r^2}\, S'\Big(\frac{|u|^2 - r^2}{r'^2 - r^2}\Big), \qquad |\nabla \chi_{r,r'}| \leq \frac{2r' s_1}{r'^2 - r^2}.$$

(b) $\chi^{\mathrm{I}}_{\ell_0,\ell_1,w}$ is C^∞ with values in $[0,1]$, equals 1 on $[\ell_0 + w, \ell_1 - w]$ and 0 outside (ℓ_0, ℓ_1), and all its derivatives of order ≥ 1 vanish outside $(\ell_0, \ell_0 + w) \cup (\ell_1 - w, \ell_1)$.

Proof. (a) Write $\chi_{r,r'} = 1 - S \circ \ell$ with $\ell(u) = (|u|^2 - r^2)/(r'^2 - r^2)$, a polynomial. Smoothness, invariance, monotonicity and the range follow from Lemma 2.4. If $|u| \leq r$ then $\ell(u) \leq 0$ and $\chi = 1$; if $|u| \geq r'$ then $\ell(u) \geq 1$ and $\chi = 0$. By Faà di Bruno's formula every derivative of order ≥ 1 of $S \circ \ell$ at u is a sum of terms each containing a factor $S^{(i)}(\ell(u))$ with $i \geq 1$, which vanishes when $\ell(u) \leq 0$ or $\ell(u) \geq 1$ (Lemma 2.4). The gradient is the chain rule, and $|u| \leq r'$ wherever $S'(\ell(u)) \neq 0$. (b) If $x \leq \ell_0$ or $x \geq \ell_1$ one factor vanishes. If $\ell_0 + w \leq x \leq \ell_1 - w$ both arguments are ≥ 1. On $(\ell_0, \ell_0 + w)$ we have $x < \ell_1 - w$ because $2w < \ell_1 - \ell_0$, so the second factor is 1 and only the first varies; symmetrically on $(\ell_1 - w, \ell_1)$. The statement on derivatives follows as in (a). □

A cutoff around a coordinate disc always means the squared-distance cutoff $\chi_{r,r'}$ in that coordinate.

2.4. **Toric forms.**

Lemma 2.8 (Top power)**.** *Let $m \geq 1$, $n = m+1$, and let U be an oriented surface chart with coordinates (x, y). On an open subset of $U \times \mathbb{C}^m$, write $w_j = \xi_j + i\eta_j$, $\omega_V = \sum_j d\xi_j \wedge d\eta_j$, and $\kappa' = \varkappa\, dx \wedge dy$. The mixed term ϑ is of one of the two types used in the construction:*

(a) $\vartheta = \sum_j dp_j \wedge a'_j$, where $p_j = \pi|w_j|^2$ and the a'_j are basic one-forms; their coefficients may depend on the base and fibre variables.

(b) $\vartheta = \zeta \wedge \beta'$, where ζ is vertical and β' is basic.

In both cases $\omega_V^{m-1} \wedge \vartheta^2 = 0$, and

$$(\omega_V + \kappa' + \vartheta)^n = n\,\omega_V^m \wedge \kappa' = n!\,\varkappa\, d\xi_1 \wedge d\eta_1 \wedge \cdots \wedge d\xi_m \wedge d\eta_m \wedge dx \wedge dy.$$

Thus the two-form is nondegenerate exactly where $\varkappa \neq 0$, and positive where $\varkappa > 0$. In particular, on the handle annulus take $\beta = b(v)\,dv$ closed, $\nu = c_\nu\, ds \wedge dv$, $e > 0$, $c_\nu > 0$, $b \geq 0$, and a smooth $G(s, w)$ independent of v. Then

$$\big(\omega_V + e\nu + d(G\beta)\big)^n = n\big(ec_\nu + b(v)\partial_s G\big)\omega_V^m \wedge ds \wedge dv.$$

It is symplectic and positive where $\partial_s G \geq 0$. Lemma M4.2 uses precisely these two cases.

Proof. Forms of even degree commute, so the multinomial formula gives

$$(\omega_V + \kappa' + \vartheta)^n = \sum_{i_1+i_2+i_3=n} \frac{n!}{i_1!\,i_2!\,i_3!}\,\omega_V^{i_1} \wedge \kappa'^{i_2} \wedge \vartheta^{i_3},$$

and the summand of index (i_1, i_2, i_3) has bidegree $(2i_2+i_3,\ 2i_1+i_3)$. It vanishes unless $2i_2+i_3 \leq 2$ and $2i_1 + i_3 \leq 2m$. The first condition leaves $(i_2, i_3) \in \{(0,0), (0,1), (0,2), (1,0)\}$, and then $i_1 = n - i_2 - i_3$. For $(i_2, i_3) = (0,0)$ and $(0,1)$ the fibre degree is $2n$ and $2m+1$, both larger than $2m$. For $(i_2, i_3) = (1,0)$, $i_1 = m$, and the summand is $n\,\omega_V^m \wedge \kappa'$. For $(i_2, i_3) = (0,2)$, $i_1 = m-1$, and the summand is $\frac{n!}{2\,(m-1)!}\,\omega_V^{m-1} \wedge \vartheta^2$, which vanishes by the identity proved below in cases (a) and (b). Since $\omega_V^m = m!\,d\xi_1 \wedge d\eta_1 \wedge \cdots \wedge d\xi_m \wedge d\eta_m$, this proves the formula. A two-form on a vector space of dimension $2n$ is nondegenerate if and only if its n-th power is nonzero.

(a) $\vartheta \wedge \vartheta = -\sum_{j,l} dp_j \wedge dp_l \wedge a'_j \wedge a'_l$, and $\omega_V^{m-1} = (m-1)! \sum_{i=1}^m \bigwedge_{r\neq i} d\xi_r \wedge d\eta_r$. The one-form $dp_j = 2\pi(\xi_j d\xi_j + \eta_j d\eta_j)$ lies in the span of $d\xi_j, d\eta_j$, so $d\xi_j \wedge d\eta_j \wedge dp_j = 0$. For $j = l$, $dp_j \wedge dp_j = 0$. For $j \neq l$ and any i, one of j, l differs from i, say j; then the summand $\bigwedge_{r\neq i} d\xi_r \wedge d\eta_r$ contains $d\xi_j \wedge d\eta_j$, and its product with dp_j vanishes. Hence the mixed-square identity. (For $m = 1$ only $j = l = 1$ occurs.)

(b) $(\zeta \wedge \beta') \wedge (\zeta \wedge \beta') = -\zeta \wedge \zeta \wedge \beta' \wedge \beta' = 0$ because $\beta' \wedge \beta' = 0$. For the displayed case: $d\beta = 0$, $\beta = b(v)\,dv$ and $\partial_v G = 0$, so $d(G\beta) = dG \wedge \beta = \partial_s G\, b(v)\, ds \wedge dv + d_w G \wedge \beta$, where $d_w G = \sum_j (\partial_{\xi_j} G\, d\xi_j + \partial_{\eta_j} G\, d\eta_j)$ has bidegree $(0,1)$. With $\nu = c_\nu\, ds \wedge dv$ the form is $\omega_V + \kappa' + \vartheta$ with $\kappa' = (ec_\nu + b\,\partial_s G)\, ds \wedge dv$ and $\vartheta = d_w G \wedge \beta$, of the type just treated. Since $e > 0$, $c_\nu > 0$ and $b \geq 0$, the coefficient is positive where $\partial_s G \geq 0$. □

The non-toric term $d_w G \wedge \beta$ of (b) is harmless because it is divisible by the single basic one-form β, so it squares to zero (closedness of β is used only for $d(G\beta) = dG \wedge \beta$). A sum of a toric mixed term and a term $\zeta \wedge \beta$ need not satisfy the mixed-square identity: for $m = 1$, $\omega_V^0 \wedge (dp \wedge a' + \zeta \wedge \beta)^2 = 2\, dp \wedge a' \wedge \zeta \wedge \beta$ is in general nonzero. The handle construction therefore uses (a) off the annulus $\mathcal{A}$ and (b) on it, never a mixture.

The model calculations below are local coordinate calculations. On an oriented base chart U, let a_j be smooth real one-forms, $F_j = -da_j$, and let σ be a smooth two-form. On $U \times \mathbb{C}^m$ put $p_j = \pi|w_j|^2$ and

$$\lambda_j = \tfrac{1}{2}(\xi_j\, d\eta_j - \eta_j\, d\xi_j), \qquad p_j\alpha_j = \lambda_j + p_j a_j.$$

Here $\alpha_j = d\theta_j + a_j$ where $w_j \neq 0$; the product $p_j\alpha_j$ is defined everywhere by the displayed smooth expression. These formulas require no choice of a global frame or identification with a projective normal bundle. In a cover presentation their descent is checked by the stated deck transformations.

Lemma 2.9 (Local form of the model form)**.** *On the above product chart put*

$$\Omega_0 = \sigma + \sum_{j=1}^{m} d(p_j\alpha_j).$$

(a) The one-forms $p_j\alpha_j$ are smooth, and Ω_0 is closed and invariant under fibre rotations.
(b)

$$\Omega_0 = \omega_V + \kappa(p) + \sum_j dp_j \wedge a_j, \qquad \kappa(p) = \sigma - \sum_j p_j F_j.$$

(c) For the circle generator $X_{p_j} = \partial_{\theta_j}$, $\iota_{X_{p_j}}\Omega_0 = -dp_j$, including the fibre axis.
(d) With $n = m + 1$, $\Omega_0^n = n\,\omega_V^m \wedge \kappa(p)$. Thus it is nondegenerate where $\kappa(p) \neq 0$ and positive where $\kappa(p) > 0$. If $F_j = d_j\sigma$ for constants d_j, its horizontal coefficient is $\kappa(p) = (1 - \sum_j d_j p_j)\sigma$.
(e) On a fixed compact base patch, integration at fixed moment gives $\int \kappa(p) = \int \sigma - \sum_j p_j \int F_j$.

Proof. The defining expression $\lambda_j + p_j a_j$ is smooth, and

$$d(\lambda_j + p_j a_j) = d\xi_j \wedge d\eta_j + dp_j \wedge a_j - p_j F_j.$$

This proves (a),(b), since two-forms on the two-dimensional base are closed. Contraction with a fibre rotation gives (c); both sides vanish on its axis. Part (d) is Lemma 2.8 for toric mixed terms. Linearity of integration gives (e). □

For the periodic cubic input, $m = 1$, $F_1 = 3\sigma$ and $\kappa = (1 - 3p)\sigma$; for the quadric curves, $F_j = 2\sigma$ and $\kappa = (1 - 2|p|)\sigma$ (§8, §14).

2.5. **The Moser lemma.**

Lemma 2.10 (Moser lemma)**.** *Let M be a smooth manifold, $U \subset M$ open and $I \subset \mathbb{R}$ an open interval containing $[0, 1]$. Let Ω_t and $\mathfrak{v}_t$, $t \in I$, be two-forms and one-forms on U depending smoothly on $(t, x) \in I \times U$, such that each Ω_t is closed and nondegenerate at every point of U and*

$$\partial_t\Omega_t = d\mathfrak{v}_t.$$

Let Y_t be the vector field on U defined by $\iota_{Y_t}\Omega_t = -\mathfrak{v}_t$, and let Ψ_t be its flow from time 0 ($\Psi_0 = \mathrm{id}$), where defined.

(a) Y is smooth on $I \times U$, and $Y_t(x) = 0$ whenever $\mathfrak{v}_t(x) = 0$ as an element of T_x^*M.
(b) If $\Psi_t(x)$ is defined for $t \in [0, 1]$, then Ψ_t is defined and smooth on a neighbourhood of x and $(\Psi_t^*\Omega_t)_x = (\Omega_0)_x$ for $t \in [0, 1]$.
(c) If $Y_t(x) = 0$ for all $t \in [0, 1]$, then $\Psi_t(x) = x$ for $t \in [0, 1]$. In particular Ψ_t fixes pointwise every set at whose points $\mathfrak{v}_t$ is the zero covector for all $t \in [0, 1]$.
(d) (Tangency.) Let $U' \subset U$ be open, $\mu \in C^\infty(U')$ and V a vector field on U' with $\iota_V\Omega_t = -d\mu$ and $\mathfrak{v}_t(V) = 0$ on U' for $t \in [0, 1]$. Then $d\mu(Y_t) = 0$ on U', and μ is constant along every integral curve of Y on a subinterval of $[0, 1]$ as long as the curve stays in U'.
(e) (Completeness.) Let $\mathcal{K} \subset U$ be compact, and suppose that every integral curve $\gamma\colon J \to U$ of Y, $J \subset [0, 1]$ an interval, which meets $\mathcal{K}$ satisfies $\gamma(J) \subset \mathcal{K}$. Then Ψ_t is defined on $\mathcal{K}$ for every $t \in [0, 1]$, $\Psi_t(\mathcal{K}) = \mathcal{K}$, $\Psi_t|_{\mathcal{K}}$ is the restriction of a diffeomorphism between open neighbourhoods of $\mathcal{K}$, and $\Psi_t^*\Omega_t = \Omega_0$ at every point of $\mathcal{K}$.

(f) *(Reparametrisation and concatenation.) If* $\varsigma\colon I \to I$ *is smooth with* $\varsigma(0) = 0$*, the pair* $(\Omega_{\varsigma(t)}, \varsigma'(t)\mathfrak{v}_{\varsigma(t)})$ *satisfies the hypotheses, its field is* $\varsigma'(t)Y_{\varsigma(t)}$*, and its flow is* $\Psi_{\varsigma(t)}$*. Consequently, let* $(\Omega_s^{(j)}, \mathfrak{v}_s^{(j)})$*,* $1 \le j \le N$*, be pairs on* U*, smooth on* $[0,1] \times U$ *(up to the boundary), with every* $\Omega_s^{(j)}$ *closed and nondegenerate on* U*,* $\partial_s \Omega_s^{(j)} = d\mathfrak{v}_s^{(j)}$*, and* $\Omega_1^{(j)} = \Omega_0^{(j+1)}$*. With the clocks* σ_j *of Lemma 2.5 put* $(\Omega_t, \mathfrak{v}_t) = (\Omega^{(j)}_{\sigma_j(t)}, \sigma_j'(t)\mathfrak{v}^{(j)}_{\sigma_j(t)})$ *on* $[t_{j-1}, t_j]$*,* $(\Omega_t, \mathfrak{v}_t) = (\Omega_0^{(1)}, 0)$ *for* $t \le 0$ *and* $(\Omega_t, \mathfrak{v}_t) = (\Omega_1^{(N)}, 0)$ *for* $t \ge 1$*. Then* $(\Omega_t, \mathfrak{v}_t)_{t\in\mathbb{R}}$ *satisfies the hypotheses with* $I = \mathbb{R}$*, and its time-one map is* $\Psi_1 = \Psi_1^{(N)} \circ \cdots \circ \Psi_1^{(1)}$ *wherever the right side is defined.*

(g) *(Equivariance.) If a diffeomorphism* g *of* U *satisfies* $g^*\Omega_t = \Omega_t$ *and* $g^*\mathfrak{v}_t = \mathfrak{v}_t$ *for all* t*, then* $g_* Y_t = Y_t$ *and* $g \circ \Psi_t = \Psi_t \circ g$ *wherever defined.*

Proof. (a) In local coordinates Ω_t is a smooth family of invertible skew matrices and $Y_t = -\Omega_t^{-1}\mathfrak{v}_t$; the inverse is smooth by Cramer's rule, and $Y_t(x) = 0$ if $\mathfrak{v}_t(x) = 0$.

(b) The domain of the flow is open, so if $\Psi_t(x)$ is defined for $t \in [0,1]$, then Ψ_t is defined and smooth on a neighbourhood of x for t in a neighbourhood of $[0,1]$. For time-dependent flows, $\frac{d}{dt}\Psi_t^*\Omega_t = \Psi_t^*\big(\partial_t\Omega_t + \mathcal{L}_{Y_t}\Omega_t\big)$. By Cartan's formula and $d\Omega_t = 0$, $\mathcal{L}_{Y_t}\Omega_t = d\iota_{Y_t}\Omega_t = -d\mathfrak{v}_t$, so the derivative vanishes, and $\Psi_0^*\Omega_0 = \Omega_0$.

(c) The constant curve at x is an integral curve; by uniqueness it is the integral curve through x. The second statement follows by (a).

(d) $d\mu(Y_t) = -(\iota_V\Omega_t)(Y_t) = -\Omega_t(V, Y_t) = \Omega_t(Y_t, V) = (\iota_{Y_t}\Omega_t)(V) = -\mathfrak{v}_t(V) = 0$, and $\frac{d}{dt}\mu(\gamma(t)) = d\mu(Y_t) = 0$ along integral curves in U'.

(e) Choose a smooth compactly supported cutoff equal to 1 near $[0,1] \times \mathcal{K}$, with support in $I \times U$, and extend the cut-off field by zero. Its flow Φ_t and inverse are defined on all of M for $t \in [0,1]$ and are jointly smooth near that interval. For any trajectory of this field, the times at which it lies in $\mathcal{K}$ form a relatively closed subset of $[0,1]$. This subset is also relatively open: near a point of $[0,1] \times \mathcal{K}$ the cutoff is 1, so the trajectory solves the original equation, and the trapping hypothesis applies on a time interval about that point. Connectedness of $[0,1]$ shows that a trajectory meeting $\mathcal{K}$ stays there. This applies both to a trajectory starting in $\mathcal{K}$ and to the trajectory through a point of $\mathcal{K}$ at time t, proving $\Phi_t(\mathcal{K}) = \mathcal{K}$ in both directions. The tube lemma gives an open neighbourhood of $\mathcal{K}$ whose trajectories remain where the cutoff is 1 throughout $[0,1]$. There Φ is the original flow Ψ, with its smooth inverse; part (b) gives the pullback identity.

(f) $\iota_{\varsigma'(t)Y_{\varsigma(t)}}\Omega_{\varsigma(t)} = -\varsigma'(t)\mathfrak{v}_{\varsigma(t)}$ and $\partial_t\Omega_{\varsigma(t)} = \varsigma'(t)\, d\mathfrak{v}_{\varsigma(t)}$. The curve $t \mapsto \Psi_{\varsigma(t)}(x)$ starts at x and has derivative $\varsigma'(t)Y_{\varsigma(t)}(\Psi_{\varsigma(t)}(x))$, so by uniqueness it is the flow of the reparametrised field. For the concatenation, each Ω_t is one of the closed nondegenerate forms $\Omega_s^{(j)}$; Ω is smooth on $\mathbb{R} \times U$ by Lemma 2.5(a), applied to the maps $(s,x) \mapsto \Omega_s^{(j)}(x)$ into the bundle of two-forms; and $\mathfrak{v}$ is smooth on $\mathbb{R} \times U$ by the argument of Lemma 2.5(b): on each $[t_{j-1}, t_j]$ every partial derivative of $\sigma_j'(t)\mathfrak{v}^{(j)}_{\sigma_j(t)}$, including the zeroth, contains a factor $\sigma_j^{(i)}$ with $i \ge 1$, so all one-sided t-jets vanish at every t_j, as they do for the constant extension 0. On $[t_{j-1}, t_j]$, $\partial_t\Omega_t = \sigma_j'(t)\,(\partial_s\Omega^{(j)})_{\sigma_j(t)} = d\mathfrak{v}_t$, and off $(0,1)$ both sides vanish. On $[t_{j-1}, t_j]$ the curve $t \mapsto \Psi^{(j)}_{\sigma_j(t)}(x)$ starts at x for $t = t_{j-1}$ and has derivative $\sigma_j'(t)Y^{(j)}_{\sigma_j(t)} = Y_t$ at its points, as in the first part; so by uniqueness the flow of Y from time t_{j-1} to time t is $\Psi^{(j)}_{\sigma_j(t)}$, and composing the stages gives Ψ_1.

(g) Since $g^*\mathfrak{v}_t = \mathfrak{v}_t$, also $(g^{-1})^*\mathfrak{v}_t = \mathfrak{v}_t$, and $\iota_{g_*Y_t}\Omega_t = (g^{-1})^*\big(\iota_{Y_t}g^*\Omega_t\big) = (g^{-1})^*\big(\iota_{Y_t}\Omega_t\big) = -(g^{-1})^*\mathfrak{v}_t = -\mathfrak{v}_t$; by uniqueness $g_*Y_t = Y_t$. Flows of g-related fields are conjugated by g. □

Signs. The Moser field is defined by $\iota_{Y_t}\Omega_t = -\mathfrak{v}_t$, the same sign as the fibre convention: if $\Omega_t = \omega$ is constant and $\mathfrak{v}_t = dQ_t$, then $Y_t = X_{Q_t}$ with $\iota_{X_Q}\omega = -dQ$. The field depends on the chosen primitive $\mathfrak{v}_t$, not only on $\partial_t\Omega_t$; properties (c) and (d) are properties of the primitive. Part (c) needs $\mathfrak{v}_t$ to vanish as a covector on M; vanishing of its pullback to a submanifold only makes Y_t Ω_t-orthogonal to that submanifold.

2.6. **Conventions across the construction.** Table 2 lists the conventions by setting. Two of them differ in a way that matters for checking: the global Hamiltonian uses $\iota_{X_H}\omega_0 = dH$, while fibre Hamiltonians and Moser fields use the opposite sign. Table 3 lists, for each interface between parts of the construction, the convention on each side and the place where the conversion is made.

Table 2: Conventions by setting.

setting	convention	where
phase space $\mathbb{R}^{2n}$	$z_j = x_j + iy_j$; real coordinates $\zeta = (x_1, \dots, x_n, y_1, \dots, y_n)$, the order in which $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$; $\omega_0 = \sum_j dx_j \wedge dy_j$, $\omega_0(v,w) = \langle v, J_0 w\rangle$; multiplication by i is $J_{\mathbb{C}} = -J_0$	§M1, §M6, §2.1, §21
global Hamiltonian	$\iota_{X_H}\omega_0 = dH$, $X_H = J_0\nabla H$, $\dot{x}_j = \partial_{y_j}H$, $\dot{y}_j = -\partial_{x_j}H$; $H = \pi\lvert z_1\rvert^2$ rotates z_1 by $e^{-2\pi i t}$. Used for $H[n,k,C,c,s]$, for $H_{\mathrm{fin}} = \beta_B P$ and for the certificate fields $J_0\nabla P_e$	§M1, §24, §25
capacities and radii	$B(c) = \{\pi\lvert z\rvert^2 \le c\}$ has Euclidean radius $\sqrt{c/\pi}$ and volume $c^n/n!$; $B(s,c) = \mathbb{B}(s,\sqrt{c/\pi})$. The models, transfers and comparison data work with capacities; the common isotopy works with the radii $r_i = \sqrt{c_i/\pi}$, $r_{\ell,i}$, r_i', $\bar{r}_i$. In the Epoch problem the unit ball is $B(s_i,\pi)$ and $B^\circ(\pi R^2) \subset \mathbb{B}(R)$	§M1, Def. M6.4, Cor. F
normalisation	target capacity 1: $\rho_i = c_i/C$; $\delta(z) = z/\sqrt{C}$ maps $B^\circ(C)$ to $B^\circ(1)$, admissibility is homogeneous, $\tilde{H}(t,z) = \lambda^2 H(t, z/\lambda)$ under $z \mapsto \lambda z$	Def. M6.1, Lemma 2.1
toric fibre $\mathbb{C}^m$	$w_j = \xi_j + i\eta_j$ (written $x_j + iy_j$ in §M2.1); $p_j = \pi\lvert w_j\rvert^2$, $\theta_j = \arg w_j/2\pi \in \mathbb{R}/\mathbb{Z}$, $\lvert p\rvert = \sum_j p_j$; $\omega_V = \sum_j d\xi_j \wedge d\eta_j = \sum_j dp_j \wedge d\theta_j$	§M1.8, §2.1
fibre Hamiltonian	$\iota_{X_Q}\omega_V = -dQ$; $X_{p_j} = \partial_{\theta_j}$ rotates w_j by $e^{+2\pi i t}$; the field X_Q of this convention is X_{-Q} of the global one. The same sign is used in the plane, $\iota_X\omega_{\mathbb{C}} = -dQ$, for the comparison data and the cubic data	§M3, Def. M2.4, §4, Def. 10.13
Moser fields	$\iota_{Y_t}\Omega_t = -\mathfrak{v}_t$ for $\partial_t\Omega_t = d\mathfrak{v}_t$; for constant Ω_t and $\mathfrak{v}_t = dQ_t$ this is the fibre convention. Used for the handle paths and for the returns of the transfers	Lemma 2.10
planar machine	disc $D(P) = \{\pi\lvert z\rvert^2 < P\}$ of area P; $\omega_{\mathbb{C}} = dp \wedge d\vartheta$ with $p = \pi\lvert z\rvert^2$, $\vartheta = \arg z/2\pi$; Morse plane (u,v) with $du \wedge dv$; rectangle (x,θ) with $dx \wedge d\theta$ and $\theta = \vartheta - \frac{1}{2}$, chart $(v,\theta) \mapsto -\sqrt{v/\pi}\, e^{2\pi i\theta}$; symplectic means Jacobian 1	§4, §5.1, §M3.3
model balls	$\mathbb{C}^n = \mathbb{C}\times\mathbb{C}^m \ni (z_{\mathrm{b}}, z_{\mathrm{f}})$, $\varpi = \pi\lvert z_{\mathrm{b}}\rvert^2$, $B(c) = \{\varpi + \lvert p\rvert \le c\}$, $\omega_0 = \omega_{\mathbb{C}}(z_{\mathrm{b}}) + \omega_V(z_{\mathrm{f}})$; slice height $(c - \lvert p\rvert)_+$	Def. M2.1, §M4.5
model forms	$\Omega_0 = \sigma + \sum_j d(p_j\alpha_j) = \omega_V + \kappa(p) + \sum_j dp_j \wedge a_j$, $\iota_{X_{p_j}}\Omega_0 = -dp_j$; cubic cover $\mathbb{R}^2\times\mathbb{C} \ni (u,v,w)$: $\alpha = d\arg w/2\pi - 3\eta$, $\Omega_0 = \omega_V + (1-3p)\sigma - 3\,dp\wedge\eta$, $\int_\Gamma \sigma = 3$; quadric: $\Omega_0 = \omega_V + d[(1-2\lvert p\rvert)\lambda]$; large ball, $n \ge 4$: $u = p_1$, $w = \sum_{j\ge 2} p_j$, $\Omega_{\mathrm{mix}} = \omega_V + d[(1-u-2w)\lambda]$	Lemma 2.9, Def. 8.1, Table M2, §16.1
Kähler forms	$d^c = \frac{i}{4\pi}(\bar\partial - \partial)$, so $dd^c \log\lvert z\rvert^2$ has unit mass and a line in $\mathbb{CP}^n$ has area 1; $\omega_{\mathrm{FS}} = dd^c \log\lVert Z\rVert^2$, in the affine chart $dd^c\log(1+\lvert z\rvert^2)$; $\mathrm{ch}(z) = z/\sqrt{\pi(1+\lvert z\rvert^2)}$ maps $\mathbb{C}^N$ onto $B^\circ(1)$ with $\mathrm{ch}^*\omega_0 = \omega_{\mathrm{FS}}$; cubic representatives $(Z,W) \sim (e^{i\chi}Z, e^{3i\chi}W)$	Lemma 2.2, §13.1

Table 3: Conventions at the interfaces and where they are converted.

interface	conventions and conversion	where
comparison → planar realisation	The comparison is a statement on the moment polytope, in the moments p; the rearrangement R_j acts on p_j. The planar map lives on a disc of area 1 in the rectangle chart, with the fibre sign on both sides. The conversion is the rescaling of the inner slice to area 1: $Q_t^{(j)}(w) = \chi(y)S'(t)\ell_{\mathrm{in}}(y)\,Q^{\mathrm{pl}}\big(y, \chi(y)S(t), w_j/\sqrt{\ell_{\mathrm{in}}(y)}\big)$, the dilation rule with $\lambda^2 = \ell_{\mathrm{in}}$. For $n = 2$ no rescaling occurs: the planar path acts on the fibre $\mathbb{C}$ itself	§M3.3, Thm. 7.13, Prop. 6.7
fibre Hamiltonian → bundle	The isotopy q_t, generated with the fibre sign, is suspended over the annulus by $\mathcal{D}_\lambda(s, v, w) = (s, v, q_{\lambda f(s)}(w))$ with generator $Q_s^\lambda = \lambda f'(s) Q_{\lambda f(s)}$; then $\mathcal{D}_\lambda^*\omega_V = \omega_V + d[(Q_s^\lambda \circ q_{\lambda f(s)})\,ds]$. The time of q becomes the annulus coordinate s; the sign is unchanged, and the Moser fields of the three paths use the same sign	Lemma M4.4, §10.9
model → transfer to $B^\circ(1)$	The model embeddings satisfy $\psi_i^*\Omega = \omega_0$ for the splitting $\mathbb{C}\times\mathbb{C}^m$ of the source; the transfer satisfies $\mathcal{T}^*\omega_0 = \Omega$ (resp. Ω_0) and lands in $B^\circ(1)$, the target normalised to capacity 1. The passage from ω_{FS} to ω_0 is the chart ch (quadric) or b_{ball}, with $b_{\mathrm{ball}}^*\omega_0 = \omega_{\mathrm{FS}}$ on unit representatives (cubic)	Prop. M2.7, Thms. 13.21, 15.12, §13.5
embeddings → common isotopy	The models give normalised maps $f_i^\natural$ for $(n, k, 1, \rho)$ with capacities $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$. The conversion to the physical target is $f_i(u) = \sqrt{C}\, f_i^\natural(u/\sqrt{C})$, $c_i' = C\rho_i^+$, $\bar{c}_i = C\rho_i^{\mathrm{dom}}$; from here on balls are described by Euclidean radii and linear maps by matrices in (x, y) blocks	Def. M6.1, Prop. M6.3, Def. M6.4
Hamiltonian → finite form	Same coordinates and sign. $P \in \mathbb{Q}[t, z]$ is a polynomial in t and the real coordinates $\zeta_1, \dots, \zeta_{2n}$ (the variables $x_0 = t, x_1, \dots, x_{2n}$ of a polynomial program), and $\beta_B(z) = \prod_l \beta_{B,1}(\zeta_l)$. The Epoch interface reads $(Q, P) = (x, y)$	§M7, Def. 25.1, Rem. M7.7

2.7. **Notation.** Table 4 lists the symbols used in more than one section. Local symbols are introduced where they are used; a statement that reuses a letter of the table redefines it there.

Table 4: Notation used in more than one section. The letters s, y, b and J have more than one row or a local meaning (for instance in Table M2 and in §16); the section fixes which is meant.

symbol	meaning	defined in
$n \geq 2$, $m = n - 1$	complex dimension of phase space; fibre rank of the toric models	§M1
$z_j = x_j + iy_j$	coordinates of $\mathbb{C}^n = \mathbb{R}^{2n}$	§M1
ω_0, X_H	$\sum dx_j \wedge dy_j$; Hamiltonian field, $\iota_{X_H}\omega_0 = dH$	§M1
J_0, φ_H^t	matrix of ω_0 (multiplication by $-i$); flow of X_H	§M6, §2
$B(c)$, $B^\circ(c)$, $B(s, c)$	closed / open ball $\{\pi\lvert z\rvert^2 \leq c\}$ of capacity c; $s + B(c)$	§M1
$\mathbb{B}(r)$, $\mathbb{B}^\circ(r)$	closed / open Euclidean ball of radius r	§M6
k, c_i, s_i, C	number of balls, capacities, centres, target capacity	§M1
$\rho_i = c_i/C$	normalised capacities	§M1
a_i; $\bar{\rho}_i$, $\hat{\rho}_i$	auxiliary capacities; closed enlarged source capacity and open domain capacity of the layer construction, $\rho_i < \bar{\rho}_i < \hat{\rho}_i < a_i$; with $\delta_i = \frac{1}{2}(a_i - \rho_i)$, $(\bar{\rho}_i, \hat{\rho}_i) = (a_i - \delta_i, a_i - \frac{3}{4}\delta_i)$ for $n = 2$ and $(a_i - \frac{3}{4}\delta_i, a_i - \frac{1}{2}\delta_i)$ for $n \geq 3$	§M4, Def. 11.9

Continued on the next page

Table 4 (continued)

symbol	meaning	defined in
$\eta_\star = 1-\rho\sqrt{k}$	volume defect of the equal-capacity branch in dimension four; $t_0 = 1-\eta_\star/2$	§3.7, Rem. 13.25
ρ_i^+, ρ_i^{dom}	enlarged closed source capacity and capacity of the open embedding domain, chosen separately in each branch	§M6
A, a, $s = 1-a$	large given capacity, reserved capacity, shell parameter	§16
$p = (p_j)$, $\lvert p\rvert$, θ_j	moments $\pi\lvert w_j\rvert^2$, their sum, angles $\arg w_j/2\pi$	§M1, §2
ω_V, X_{p_j}	fibre form $\sum dp_j \wedge d\theta_j$; rotation generator, $\iota_{X_{p_j}}\omega_V = -dp_j$	§M2, §2
$w = (w_j)$; ϖ	fibre coordinates; moment of the base disc in $\mathbb{C}\times\mathbb{C}^m$	§M2
Δ, $\langle b_\nu, p\rangle \le \ell_\nu$, τ_Δ	downward moment polytope, its outer faces, core fraction	§§M2, M3
h; h_1, h_2	fibre height(s) of Δ	§M2
V_Δ, $V_\Delta(E)$	toric domain $\{p(w)\in\Delta\}\subset\mathbb{C}^m$; its fibrewise version in E	§M2
Γ, Z	base of a model and its marked set; the model section fixes the affine or cover presentation	§§8–16
Σ	base surface (Γ, or a compact bordered subsurface)	§M2
d^c, ω_{FS}, σ	$\frac{i}{4\pi}(\bar\partial-\partial)$; normalised affine Fubini–Study form; base area form	§2
E, α_j, Ω_0	model space in its affine or cover presentation; angular forms; $\sigma+\sum d(p_j\alpha_j)$	§§8–16
a_j, $a_j(p)$, a_j'; F_j	connection one-forms, toric mixed terms (not the capacities a_i); curvature (Chern) forms	§M2, §2
$\kappa(p)$	horizontal area form at moment p	§M2
$H_{\mathrm{b}}(p) = \int_\Sigma \kappa(p)$	base area profile	§M2
$P_{\mathrm{b}}(p)$	unused area $H_{\mathrm{b}}(p) - \sum_i (a_i-\lvert p\rvert)_+$	§M2
P	minorant of P_{b} in a comparison datum	§§M2, M3
K, $F = K-P$, γ	comparison function, comparison defect, gap	§§M2, M3
R_j, $\mathcal{R}$	coordinatewise increasing rearrangement, $\mathcal{R} = R_m\circ\cdots\circ R_1$	§M3
q_t, $q = q_1$, Q_t	fibre isotopy, its endpoint, generator ($\iota_{X_{Q_t}}\omega_V = -dQ_t$)	§§M2, M3
$\mathcal{A}\subset\mathcal{A}^+$, $\mathcal{B}$, (s,v), β	handle annulus, dual band, handle coordinates, closed one-form of period 1	§M4
ν, c_ν; Λ	periodic unit-area reference form and its local density; finite affine reference form of the handle on a bordered base	§M4
$e_\Lambda\Lambda$, H^*, J	positive background form, handle mass $H_{\mathrm{b}} - \epsilon_{\mathrm{sm}}$, collar coefficient $K-H^*$	§M4
$h_i(p)$, R^*	smooth caps, $0\le h_i\le(a_i-\lvert p\rvert)_+$, with $h_i\ge(a_i+\rho_i^\sharp)/2-\lvert p\rvert$; residual	§M4
$\mathcal{D}$, Φ; Λ_i, E_i	twist, Moser map; layers, layer maps ($\mathcal{E}_i$ in the main text)	§M4
$\mathcal{T}$, $\mathcal{C}_\mathcal{K}$	transfer from a model to the target, compact set containing the image of $\mathcal{K}$	§M2, §§8–16
S, s_1, σ_j; $\chi_{r,r'}$, χ^{I}	flat step, $\max S'$, clocks; radial and interval cutoffs	§2
$\mathfrak{v}_t$	Moser primitive, $\iota_{Y_t}\Omega_t = -\mathfrak{v}_t$	§2
the(h)	definite description	§M1
$r_{1,i}<r_{2,i}<r_{3,i}<r_{4,i}$	enlarged Euclidean source radii for the common isotopy and its cutoff	§M6
$y_i = f_i(0)$, $Y_i = y_i/\tau$, L, δ_c, τ	isotopy centres, far centres, Lipschitz bound, separation, common tail scale	§M6
$F_{i,t}$	common isotopy	§M6
$\mathfrak{a}_{i,t}$, $\mathfrak{h}_{i,t}$	source one-form, source primitive	§M6
$H = H[n,k,C,c,s]$	the global Hamiltonian	§M6
β_B, P, H_{fin}	box cutoff, rational polynomial (this meaning of P only in §M7), $H_{\mathrm{fin}} = \beta_B\cdot P$	§M7

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Table 4 (continued)

symbol	meaning	defined in
b_R, L_X, η	radial budget, Lipschitz bound of X_H, field error (other uses of η, such as the connection form of the cubic model, are local)	§M7
L1, L2, L3	levels of explicitness	§M1

3. The finite comparator

This section proves the scalar part of the comparison theorem: the iterated coordinate rearrangement of a rational convex polyhedral function is again rational convex polyhedral, and after an explicit number of iterations it gives, once smoothed, a comparison function with an explicit gap. It proves Lemma M3.2, gives the detailed form of Proposition M3.3 and the composition lemma used in the proof of Theorem M3.5, and records the scalar identities behind Proposition M3.8.

3.1. **Setting.** Throughout, $m \geq 1$ and

$$\Delta = \{p \in \mathbb{R}^m_{\geq 0} : \langle b_\nu, p\rangle \leq \ell_\nu,\ \nu = 1, \dots, N_\Delta\}, \qquad b_\nu \in \mathbb{Q}^m_{\geq 0}, \quad \ell_\nu \in \mathbb{Q}_{>0},$$

is a compact full-dimensional polytope. It is *downward closed*: if $p \in \Delta$ and $0 \leq p' \leq p$ coordinatewise, then $p' \in \Delta$, because all b_ν are nonnegative. In particular $0 \in \Delta$ and $\tau\Delta \subset \Delta$ for $0 \leq \tau \leq 1$. We use the moments p, $|p| = \sum_j p_j$ and the form ω_V of §2, put $V_\Delta = p^{-1}(\Delta) \subset \mathbb{C}^m$, write $\mathbf{e}_1, \dots, \mathbf{e}_m$ for the standard basis of $\mathbb{R}^m$, and write $\|\cdot\|$ for the uniform norm on Δ. As in §M3, the letter P denotes an arbitrary concave function on Δ, to be applied with $P = P_\mathrm{b}$ or with a concave minorant of P_b; it is unrelated to the compiled polynomial. The letters y, L, λ, η_Δ, M_P, β_P, and in §3.7 the letters $s, a, c_s, g, A, B, \Lambda$, are local to this section and to §M3, where they have the same meaning.

Fix a coordinate j. We write $p = (y, x)$ with $x = p_j$ and $y = (p_i)_{i \neq j} \in \mathbb{R}^{m-1}$ (for $m = 1$ the variable y is absent). If $b_{\nu j} = 0$ for every ν, then $t\mathbf{e}_j \in \Delta$ for all $t \geq 0$, which contradicts compactness; so the set $\{\nu : b_{\nu j} > 0\}$ is nonempty and

$$L(y) = \min_{\nu:\, b_{\nu j} > 0} L_\nu(y), \qquad L_\nu(y) = \frac{\ell_\nu - \sum_{i \neq j} b_{\nu i} y_i}{b_{\nu j}},$$

is a minimum of finitely many rational affine functions, hence concave, continuous and piecewise affine on $\mathbb{R}^{m-1}$. Let $\Delta_j = \{y : (y, 0) \in \Delta\}$ be the projection of Δ along $\mathbf{e}_j$; it is a compact convex polytope (a point if $m = 1$). For $y \in \Delta_j$ we have $L(y) \geq 0$, and

$$\Delta = \{(y, x) : y \in \Delta_j,\ 0 \leq x \leq L(y)\}. \tag{2}$$

Indeed, for $y \in \Delta_j$ the constraints with $b_{\nu j} = 0$ do not involve x and hold because $(y, 0) \in \Delta$, while those with $b_{\nu j} > 0$ say $x \leq L_\nu(y)$; conversely $(y, x) \in \Delta$ implies $(y, 0) \in \Delta$ by downward closure. The *slice* through $y \in \Delta_j$ is $\{y\} \times [0, L(y)]$; it may have length zero.

A function on Δ is *rational convex polyhedral* if it is the restriction to Δ of $\max_{\iota \in I} \phi_\iota$ for a finite nonempty family of affine functions ϕ_ι with rational coefficients, and *rational concave piecewise affine* if it is the restriction of $\min_{u \in U} \psi_u$ for such a family. Sums, maxima and nonnegative rational multiples of rational convex polyhedral functions are rational convex polyhedral (for sums, $\max_\iota \phi_\iota + \max_{\iota'} \phi'_{\iota'} = \max_{\iota,\iota'}(\phi_\iota + \phi'_{\iota'})$), and $-P$ is rational convex polyhedral when P is rational concave piecewise affine. A sum of rational concave piecewise affine functions is rational concave piecewise affine, and for rational a so is $-(a - |p|)_+ = \min(0, |p| - a)$. Hence each profile $P_\mathrm{b} = H_\mathrm{b} - \sum_i (a_i - |p|)_+$ with affine H_b is concave piecewise affine, and it is rational when H_b and the auxiliary capacities a_i are rational. This applies to the cubic model (Lemma 8.5), the quadric model (§14) and the large-ball model in dimension at least eight (§16). In dimension

six the profile of the bordered model need not be concave, and the comparison is applied to the concave affine minorant of Proposition 19.1.

Rationality is used only in Lemma 3.3 and Proposition 3.4(a),(d), to conclude that the iterates have rational coefficients and that μ, λ and N are rational. Every other statement of this section, and every inequality in Proposition 3.4, holds verbatim for real b_ν, ℓ_ν and real affine pieces; for real auxiliary data this real-coefficient version is the one used. The four-dimensional equal case, where $a = t_0/\sqrt{k}$ is irrational for non-square k, uses the closed-form datum of Proposition M3.8 (§3.7).

3.2. **Coordinate rearrangement.** For a convex function on an interval, the increasing rearrangement has the min–max form of Lemma 3.1. We use this form throughout because it turns rearrangement into projection: the epigraph of R_jF is the image of a rational polyhedron (Lemma 3.3). Rearranging a rational convex polyhedral function therefore takes one Fourier–Motzkin elimination, and the fixed-point iteration of §3.3 stays inside a class of finite rational objects. We first treat one variable. For $L \geq 0$ and a continuous function f on $[0, L]$ put

$$\mathcal{L}_f(b) = \big|\{v \in [0, L] : f(v) \leq b\}\big| \qquad (b \in \mathbb{R}),$$

where $|\cdot|$ is Lebesgue measure; $\mathcal{L}_f$ is the distribution function of f.

Lemma 3.1 (Increasing rearrangement on an interval)**.** *Let $L \geq 0$, let $f\colon [0, L] \to \mathbb{R}$ be continuous and convex, and for $0 \leq x \leq L$ put*

$$f^\sharp(x) = \min_{0 \leq a \leq L - x} \max\big(f(a), f(a + x)\big).$$

(a) For $b \in \mathbb{R}$ and $0 \leq x \leq L$: $f^\sharp(x) \leq b$ if and only if $\{f \leq b\} \neq \emptyset$ and $\mathcal{L}_f(b) \geq x$. Hence $f^\sharp(x) = \min\{b \geq \min f : \mathcal{L}_f(b) \geq x\}$, the increasing rearrangement of f.
(b) $\mathcal{L}_{f^\sharp} = \mathcal{L}_f$; in particular $\int_0^L f^\sharp = \int_0^L f$.
(c) $f^\sharp$ is nondecreasing, $f^\sharp(0) = \min f$ and $f^\sharp(L) = \max f$.

Proof. The minimum exists because the admissible a form the compact interval $[0, L - x]$ and the function minimised is continuous. Since f is convex, every sublevel set $\{f \leq b\}$ is a closed interval, possibly empty, and on any interval $[a, a + x]$ we have $f \leq \max(f(a), f(a + x))$.

(a) If $f^\sharp(x) \leq b$, choose a with $\max(f(a), f(a + x)) \leq b$; then $[a, a + x] \subset \{f \leq b\}$, so this set is nonempty and has length at least x. Conversely, if $\{f \leq b\} = [u, v]$ with $v - u \geq x$, then $a = u$ is admissible and both endpoints of $[u, u + x] \subset [u, v]$ lie in $\{f \leq b\}$, so $f^\sharp(x) \leq b$. Thus $\{b : f^\sharp(x) \leq b\} = \{b : \{f \leq b\} \neq \emptyset,\ \mathcal{L}_f(b) \geq x\}$, and the minimum of the right-hand set is $f^\sharp(x)$.

(b) If $b < \min f$, then $\{f^\sharp \leq b\} = \emptyset$ by (a), and $\mathcal{L}_f(b) = 0 = \mathcal{L}_{f^\sharp}(b)$. If $b \geq \min f$, then by (a) $\{x \in [0, L] : f^\sharp(x) \leq b\} = [0, \mathcal{L}_f(b)]$ (note $\mathcal{L}_f(b) \leq L$), whose length is $\mathcal{L}_f(b)$. Two bounded measurable functions on $[0, L]$ with the same distribution function have the same integral, since for c_0 below both, $\int_0^L \varphi = c_0 L + \int_{c_0}^\infty \big(L - \mathcal{L}_\varphi(b)\big)\, db$.

(c) Let $x \leq x'$ and let $[a, a + x']$ be a minimising interval for x'. Then a is admissible for x, and both a and $a + x$ lie in $[a, a + x']$, so $\max(f(a), f(a + x)) \leq \max(f(a), f(a + x')) = f^\sharp(x')$. For $x = 0$ the admissible a range over $[0, L]$ and $f^\sharp(0) = \min f$. For $x = L$ only $a = 0$ is admissible, and $f^\sharp(L) = \max(f(0), f(L)) = \max f$ by convexity. □

Let R_j be the coordinate rearrangement of Definition M3.1. For F continuous on Δ put

$$E = \{(y, x, a) : (y, x) \in \Delta,\ a \geq 0,\ a + x \leq L(y)\},$$
$$\Phi_F(y, x, a) = \max\big(F(y, a), F(y, a + x)\big) \quad \text{on } E,$$

so that $(R_jF)(y, x) = \min_a \Phi_F(y, x, a)$, the minimum over the a with $(y, x, a) \in E$. By (2) the points (y, a) and $(y, a + x)$ lie in Δ for $(y, x, a) \in E$; the set E is compact and Φ_F is continuous.

Lemma 3.2 (Properties of R_j)**.** *Let F, G be continuous convex functions on Δ and $j \in \{1, \dots, m\}$.*

(i) For every $y \in \Delta_j$, $x \mapsto (R_jF)(y,x)$ is the function $f^\sharp$ of Lemma 3.1 for $f = F(y,\cdot)$ on $[0, L(y)]$: the increasing rearrangement of F on the slice through y. It is nondecreasing, $(R_jF)(y,0) = \min_{[0,L(y)]} F(y,\cdot)$ and $(R_jF)(y,L(y)) = \max_{[0,L(y)]} F(y,\cdot)$.

(ii) R_jF is continuous and convex on Δ.

(iii) $\int_\Delta R_jF = \int_\Delta F$.

(iv) If $F \le G$ then $R_jF \le R_jG$. For $c \in \mathbb{R}$ and $\alpha \ge 0$, $R_j(F+c) = R_jF + c$ and $R_j(\alpha F) = \alpha R_jF$; in particular $R_j 0 = 0$. Moreover $\|R_jF - R_jG\| \le \|F - G\|$, $\max_\Delta R_jF = \max_\Delta F$ and $\min_\Delta R_jF = \min_\Delta F$.

(v) If $i \ne j$ and F is nondecreasing in p_i (that is, $F(p) \le F(p')$ whenever $p, p' \in \Delta$ differ only in the i-th coordinate and $p_i \le p_i'$), then so is R_jF. Consequently $\mathcal{R}F = R_m \circ \cdots \circ R_1 F$ is nondecreasing in every coordinate, and $\mathcal{R}F(0) = \min_\Delta \mathcal{R}F$.

Proof. (i) By (2), for $(y,x) \in \Delta$ the admissible a are exactly $0 \le a \le L(y) - x$, and the points (y,a), $(y, a+x)$ lie on the slice through y. So the definition of R_j is the formula of Lemma 3.1 applied to $F(y,\cdot)$ on $[0, L(y)]$, and the remaining statements are Lemma 3.1(a),(c).

(ii) *Continuity.* Let $(y_n, x_n) \to (y,x)$ in Δ. Upper semicontinuity: let a be a minimiser at (y,x) and put $a_n = \min\big(a, L(y_n) - x_n\big)$. Then $a_n \ge 0$, $(y_n, x_n, a_n) \in E$, and $a_n \to a$ because $L(y_n) - x_n \to L(y) - x \ge a$; hence $\limsup_n (R_jF)(y_n,x_n) \le \lim_n \Phi_F(y_n,x_n,a_n) = \Phi_F(y,x,a) = (R_jF)(y,x)$. Lower semicontinuity: pass to a subsequence along which $(R_jF)(y_n,x_n)$ converges to the $\liminf$, choose minimisers a_n, and pass to a further subsequence with $a_n \to a^*$. Since E is closed, $(y,x,a^*) \in E$, so $(R_jF)(y,x) \le \Phi_F(y,x,a^*) = \lim \Phi_F(y_n,x_n,a_n) = \liminf (R_jF)(y_n,x_n)$. This covers slices of length zero, where $a^* = 0$.

Convexity. Let $p_0 = (y_0, x_0)$ and $p_1 = (y_1, x_1)$ be points of Δ with minimisers a_0, a_1, let $0 \le \vartheta \le 1$, and put $y_\vartheta = (1-\vartheta)y_0 + \vartheta y_1$, $x_\vartheta = (1-\vartheta)x_0 + \vartheta x_1$, $a_\vartheta = (1-\vartheta)a_0 + \vartheta a_1$. Then $a_\vartheta \ge 0$ and, by concavity of L, $a_\vartheta + x_\vartheta \le (1-\vartheta)L(y_0) + \vartheta L(y_1) \le L(y_\vartheta)$, so a_ϑ is admissible at $(y_\vartheta, x_\vartheta)$. Convexity of F gives $F(y_\vartheta, a_\vartheta) \le (1-\vartheta)F(y_0,a_0) + \vartheta F(y_1,a_1) \le (1-\vartheta)(R_jF)(p_0) + \vartheta(R_jF)(p_1)$, and the same bound for $F(y_\vartheta, a_\vartheta + x_\vartheta)$. Hence $(R_jF)(y_\vartheta, x_\vartheta) \le (1-\vartheta)(R_jF)(p_0) + \vartheta(R_jF)(p_1)$.

(iii) By (2) and Fubini, $\int_\Delta G = \int_{\Delta_j} \int_0^{L(y)} G(y,x)\,dx\,dy$ for continuous G (for $m = 1$ there is no outer integral). By (i) and Lemma 3.1(b) the inner integrals of F and R_jF agree for every y.

(iv) If $F \le G$, then $\Phi_F \le \Phi_G$ pointwise on E and the minima satisfy the same inequality. Since $\max(u+c, v+c) = \max(u,v) + c$ and $\max(\alpha u, \alpha v) = \alpha \max(u,v)$ for $\alpha \ge 0$, we have $\Phi_{F+c} = \Phi_F + c$ and $\Phi_{\alpha F} = \alpha \Phi_F$. Applying monotonicity and translation to $G - \|F-G\| \le F \le G + \|F-G\|$ gives $\|R_jF - R_jG\| \le \|F - G\|$. By (i), $\max_\Delta R_jF = \max_y \max_{[0,L(y)]} F(y,\cdot) = \max_\Delta F$, and likewise for the minimum.

(v) Let (y,x) and (y',x) be points of Δ with $y' = y + t\mathbf{e}_i$, $t \ge 0$. Since $b_{\nu i} \ge 0$, $L(y) \ge L(y')$. A minimiser a' at (y',x) satisfies $0 \le a'$ and $a' + x \le L(y') \le L(y)$, so it is admissible at (y,x), and $F(y,a') \le F(y',a')$, $F(y,a'+x) \le F(y',a'+x)$ because F is nondecreasing in p_i. Hence $(R_jF)(y,x) \le (R_jF)(y',x)$. For the consequence, R_1F is nondecreasing in p_1 by (i); each further R_j makes the output nondecreasing in p_j and, by what was just shown, keeps it nondecreasing in $p_1, \dots, p_{j-1}$. For $p \in \Delta$ the path from 0 to p that raises one coordinate at a time stays in Δ by downward closure, so $\mathcal{R}F(0) \le \mathcal{R}F(p)$. □

Lemma 3.3 (Polyhedral form)**.** *Let $F = \max_{\iota \in I} \phi_\iota$ be rational convex polyhedral on Δ. Let $\Pi \subset \mathbb{R}^{m-1} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ be the rational polyhedron of points (y,x,t,a) satisfying*

$$(y,x) \in \Delta, \qquad a \ge 0, \qquad b_{\nu j}(a+x) + \textstyle\sum_{i \ne j} b_{\nu i} y_i \le \ell_\nu \quad (b_{\nu j} > 0),$$
$$t \ge \phi_\iota(y,a), \qquad t \ge \phi_\iota(y, a+x) \qquad (\iota \in I).$$

Then the epigraph $\{(p,t) : p \in \Delta,\ t \geq (R_jF)(p)\}$ *of* R_jF *over* Δ *is the image of* Π *under* $(y,x,t,a) \mapsto (y,x,t)$. *Eliminating* a *from the defining inequalities of* Π *by one step of Fourier–Motzkin elimination gives finitely many rational inequalities* $\langle \alpha_l, p\rangle + \zeta_l t \leq \delta_l$ $(l \in \mathcal{S})$ *that define this epigraph, every* ζ_l *is* ≤ 0, *the set* $\mathcal{S}_- = \{l : \zeta_l < 0\}$ *is nonempty, and*

$$R_jF = \max_{l \in \mathcal{S}_-} \frac{\langle \alpha_l, p\rangle - \delta_l}{-\zeta_l} \qquad \text{on } \Delta.$$

In particular R_jF *is rational convex polyhedral, and so is* $\mathcal{R}F$.

Proof. Let $(y,x) \in \Delta$ and $a \geq 0$ with $a + x \leq L(y)$. Then (y,a) and $(y,a+x)$ lie in Δ by (2), and $t \geq \max(F(y,a), F(y,a+x))$ is equivalent to the $2|I|$ inequalities in the second line. Hence $(y,x,t,a) \in \Pi$ if and only if $(y,x,a) \in E$ and $t \geq \Phi_F(y,x,a)$. Since the minimum defining R_jF is attained, $t \geq (R_jF)(y,x)$ holds if and only if some a with $(y,x,t,a) \in \Pi$ exists. This proves the first assertion.

Write each inequality of Π as $\kappa a \leq \Psi(y,x,t)$ with $\kappa \in \mathbb{Q}$ and Ψ rational affine. Those with $\kappa > 0$ give upper bounds $a \leq \Psi/\kappa$, those with $\kappa < 0$ lower bounds $a \geq \Psi/\kappa$, and those with $\kappa = 0$ do not involve a. A real a satisfying all of them exists if and only if every lower bound is at most every upper bound and the inequalities without a hold; this is Fourier–Motzkin elimination of one variable [Sch86, §12.2], and it produces finitely many rational affine inequalities in (y,x,t) that define the projection.

Fix $p \in \Delta$. The set $\{t : (p,t) \in \text{epigraph}\}$ is $[(R_jF)(p), \infty)$. An inequality with $\zeta_l > 0$ would fail for large t; so $\zeta_l \leq 0$ for every l. The inequalities with $\zeta_l = 0$ hold at every $p \in \Delta$, because Δ is the projection of the epigraph. If $\mathcal{S}_-$ were empty, every t would be admissible at p, contradicting the lower bound $(R_jF)(p)$; and for $p \in \Delta$ the admissible t are exactly those with $t \geq (\langle \alpha_l, p\rangle - \delta_l)/(-\zeta_l)$ for all $l \in \mathcal{S}_-$. Taking the least such t gives the formula. The statement for $\mathcal{R}F$ follows by applying this m times. □

Proof of Lemma M3.2. Item (i) there is items (ii) and (iii) of Lemma 3.2; item (ii) there is item (iv); item (iii) there is item (i) together with the last assertion of item (iv); item (iv) there is item (v); and item (v) there is Lemma 3.3. □

3.3. **The finite fixed point.** For $0 < \lambda < 1$ and a concave continuous P on Δ, let $\mathcal{C}$ be the set of continuous convex functions on Δ and $T_\lambda(F) = \mathcal{R}(\lambda F - P)$. The fixed point K_λ of T_λ is used as in [Ope26, §2.2]. Here λ and the number N of iterations are given by rational formulas, and the N-th iterate K_N, a rational convex polyhedral function obtained by finitely many Fourier–Motzkin eliminations, already satisfies the comparison inequality with gap $\frac{7}{8}d$. The following proposition is the detailed form of Proposition M3.3.

Proposition 3.4 (Finite comparator, detailed form)**.** *Let* $\tau_\Delta \in (0,1)$ *be rational and let* P *be rational concave piecewise affine on* Δ *with mean* $\mu = \mathrm{vol}(\Delta)^{-1}\int_\Delta P > 0$ *and* $P \geq \beta_P$ *on* $\Delta \setminus \tau_\Delta\Delta$ *for some* $\beta_P > 0$. *Let* $d \in \mathbb{Q}$ *with* $0 < d < \min(\mu, \beta_P)$, *and put*

$$\eta_\Delta = (1-\tau_\Delta)(\mu - d), \qquad M_P = \max\bigl(1,\ d - \min_\Delta P + 1\bigr), \qquad \lambda = 1 - \frac{\eta_\Delta}{2(M_P + \eta_\Delta)}.$$

- *(a)* μ, $\min_\Delta P$, $\max_\Delta P$ *and* $\|P\|$ *are rational and are computed from* (Δ, P) *by finitely many rational operations; hence so are* η_Δ, M_P *and* λ, *and* $\frac{1}{2} < \lambda < 1$, $\lambda\eta_\Delta/(1-\lambda) = 2M_P + \eta_\Delta > d - \min_\Delta P$.
- *(b)* T_λ *maps* $\mathcal{C}$ *to itself and is a contraction with constant* λ *for the uniform norm. Its unique fixed point* K_λ *is convex, nondecreasing in every coordinate, has mean* $-\mu/(1-\lambda)$, *and* $\|K_\lambda\| \leq \|P\|/(1-\lambda)$.
- *(c)* $(1-\lambda)\max_\Delta K_\lambda < -d$, *and* $\mathcal{R}(K_\lambda - P) \leq K_\lambda - d$ *on* Δ.

(d) *The iterates* $K_0 = 0$, $K_{r+1} = T_\lambda(K_r)$ *are rational convex polyhedral and satisfy* $\|K_r - K_\lambda\| \le \lambda^r \|P\|/(1-\lambda)$. *Each* K_{r+1} *is obtained from* K_r *by* m *Fourier–Motzkin eliminations (Lemma 3.3).*

(e) *Let* N_0 *be the least integer* $N_0 \ge 0$ *with* $\lambda^{N_0}\|P\|/(1-\lambda) < d/16$. *For every* $N \ge N_0$,

$$N_0 \le 1 + \frac{1}{1-\lambda}\log^+ \frac{16\|P\|}{d(1-\lambda)} \qquad \text{and} \qquad \mathcal{R}(K_N - P) \le K_N - \tfrac{7}{8}d \quad \text{on } \Delta.$$

Proof. (a) Write $G = -P = \max_{u\in U}(-\psi_u)$ with rational affine ψ_u.

Extrema. For a rational convex polyhedral H, $\mathcal{R}H$ is nondecreasing in every coordinate and has the same minimum as H (Lemma 3.2(iv),(v)). Since $0 \in \Delta$ and $\Delta \subset \mathbb{R}^m_{\ge 0}$,

$$\operatorname{Min}(H) := (\mathcal{R}H)(0) = \min_{\Delta} H,$$

and $(\mathcal{R}H)(0)$ is the maximum of the finitely many rational affine functions produced by Lemma 3.3, evaluated at 0. For an affine ϕ, $\max_\Delta \phi = -\operatorname{Min}(-\phi)$, and $\max_\Delta \max_u \phi_u = \max_u \max_\Delta \phi_u$. Hence $\min_\Delta P = -\max_\Delta G$ and $\max_\Delta P = -\operatorname{Min}(G)$ are rational and are obtained by finitely many eliminations and comparisons, and $\|P\| = \max(|\min_\Delta P|, |\max_\Delta P|)$.

Integrals. We integrate a rational polynomial f over a region $D \subset \mathbb{R}^r$ given by finitely many rational closed halfspaces and contained in a cube $[-M, M]^r$; we add the $2r$ cube inequalities to the list. For Δ one may take $M = 1 + \sum_j \sum_{\nu:\, b_{\nu j}>0} \ell_\nu / b_{\nu j}$, since $0 \le p_j \le \ell_\nu / b_{\nu j}$ on Δ whenever $b_{\nu j} > 0$, and such ν exist for every j (§3.1). If $r = 0$, the ambient space is a point of volume 1: return f if the (constant) inequalities hold and 0 otherwise. If $r > 0$, write $p = (x, t)$ with $x \in \mathbb{R}^{r-1}$. Each inequality with positive (negative) t-coefficient, divided by that coefficient, gives an upper bound $t \le u_\alpha(x)$ (a lower bound $t \ge l_\beta(x)$) with u_α, l_β rational affine; inequalities with zero t-coefficient are constraints on x. The cube inequalities provide at least one bound of each kind. Delete repeated affine functions from each of the two lists. For each pair (α, β) let $D_{\alpha\beta} \subset \mathbb{R}^{r-1}$ be the region defined by the constraints on x, the projected cube, and

$$l_{\beta'}(x) \le l_\beta(x) \text{ for all } \beta', \qquad u_\alpha(x) \le u_{\alpha'}(x) \text{ for all } \alpha', \qquad l_\beta(x) \le u_\alpha(x).$$

For $x \in D_{\alpha\beta}$ the fibre $\{t : (x,t) \in D\}$ is $[l_\beta(x), u_\alpha(x)]$, and the sets $D_{\alpha\beta}$ cover the projection of D (choose α, β attaining the minimum and maximum). If $(\alpha,\beta) \ne (\alpha',\beta')$, then $D_{\alpha\beta} \cap D_{\alpha'\beta'}$ lies in the zero set of a nonzero difference $u_\alpha - u_{\alpha'}$ or $l_\beta - l_{\beta'}$ of distinct affine functions, which is empty or an affine hyperplane, hence a null set. Expanding $f(x,t) = \sum_{k=0}^n f_k(x) t^k$ and integrating in t,

$$\int_D f = \sum_{\alpha,\beta} \int_{D_{\alpha\beta}} g_{\alpha\beta}, \qquad g_{\alpha\beta}(x) = \sum_{k=0}^{n} f_k(x)\, \frac{u_\alpha(x)^{k+1} - l_\beta(x)^{k+1}}{k+1},$$

by Fubini's theorem on the compact set D; fibres of length zero contribute zero. Each $g_{\alpha\beta}$ is a rational polynomial and each $D_{\alpha\beta}$ is again a rational halfspace region in a cube, of dimension $r-1$. By induction on r the recursion terminates after finitely many rational operations and returns $\int_D f$ exactly, also when D is empty or lower-dimensional.

The mean. Choose a rational R with $R > |G|$ on $[-M, M]^m$, for instance $R = 1 + \max_u\big(|\psi_u(0)| + M\sum_j |\partial_j \psi_u|\big)$. The slab

$$E = \{(p,t) : p \in \Delta,\ -\psi_u(p) \le t \ (u \in U),\ t \le R\}$$

is a rational halfspace region in $[-\max(M,R), \max(M,R)]^{m+1}$ whose fibre over $p \in \Delta$ is $[G(p), R]$. By Fubini,

$$\operatorname{vol}(E) = R\operatorname{vol}(\Delta) - \int_\Delta G, \qquad \text{so} \qquad \int_\Delta P = \operatorname{vol}(E) - R\operatorname{vol}(\Delta).$$

Both volumes are computed by the recursion with $f = 1$, and $\mathrm{vol}(\Delta) > 0$ because Δ is full-dimensional. Hence μ is rational and computed by finitely many rational operations.

Since $d < \mu$, $\eta_\Delta > 0$; since $M_P \geq 1$, $1 - \lambda = \eta_\Delta/(2(M_P + \eta_\Delta)) \in (0, \frac{1}{2})$. Then $\lambda\eta_\Delta/(1-\lambda) = \lambda \cdot 2(M_P + \eta_\Delta) = 2(M_P + \eta_\Delta) - \eta_\Delta = 2M_P + \eta_\Delta$, and $2M_P + \eta_\Delta > M_P \geq d - \min_\Delta P + 1$.

(b) $\mathcal{C}$ is a closed subset of the Banach space $C(\Delta)$, hence complete. If $F \in \mathcal{C}$, then $\lambda F - P$ is continuous and convex, and $\mathcal{R}(\lambda F - P) \in \mathcal{C}$ by Lemma 3.2(ii). By Lemma 3.2(iv), $\|T_\lambda F - T_\lambda G\| \leq \|\lambda(F - G)\| = \lambda\|F - G\|$. The contraction principle gives a unique fixed point $K_\lambda \in \mathcal{C}$. Since $K_\lambda = \mathcal{R}(\cdots)$, it is nondecreasing in every coordinate by Lemma 3.2(v). Integrating $K_\lambda = \mathcal{R}(\lambda K_\lambda - P)$ with Lemma 3.2(iii) gives $\int K_\lambda = \lambda \int K_\lambda - \int P$, so the mean of K_λ is $-\mu/(1-\lambda)$. Finally $\mathcal{R}0 = 0$, so $\|K_\lambda\| = \|\mathcal{R}(\lambda K_\lambda - P) - \mathcal{R}0\| \leq \lambda\|K_\lambda\| + \|P\|$.

(c) Let $M_\lambda = \max_\Delta K_\lambda$ and suppose $M_\lambda \geq -d/(1-\lambda)$, that is, $\lambda M_\lambda \leq M_\lambda + d$. We show $\lambda K_\lambda - P < M_\lambda$ on Δ.

If $p \in \Delta \setminus \tau_\Delta\Delta$, then $\lambda K_\lambda(p) - P(p) \leq \lambda M_\lambda - \beta_P \leq M_\lambda + d - \beta_P < M_\lambda$, since $d < \beta_P$.

If $p \in \tau_\Delta\Delta$, write $p = \tau_\Delta v$ with $v \in \Delta$. By (b), $K_\lambda(0) = \min K_\lambda$, and the minimum is at most the mean, so $K_\lambda(0) \leq -\mu/(1-\lambda)$. Convexity on the segment from 0 to v gives

$$K_\lambda(p) \leq \tau_\Delta K_\lambda(v) + (1 - \tau_\Delta)K_\lambda(0) \leq M_\lambda - (1 - \tau_\Delta)\big(M_\lambda - K_\lambda(0)\big),$$

and $M_\lambda - K_\lambda(0) \geq -d/(1-\lambda) + \mu/(1-\lambda) = (\mu - d)/(1-\lambda)$. Hence $K_\lambda(p) \leq M_\lambda - \eta_\Delta/(1-\lambda)$, and by (a)

$$\lambda K_\lambda(p) - P(p) \leq \lambda M_\lambda - \frac{\lambda\eta_\Delta}{1-\lambda} - \min_\Delta P \leq M_\lambda + d - \min_\Delta P - (2M_P + \eta_\Delta) < M_\lambda.$$

Since Δ is compact, $\max_\Delta(\lambda K_\lambda - P) < M_\lambda$. But $\max \mathcal{R}G = \max G$ by Lemma 3.2(iv) applied m times, so $M_\lambda = \max \mathcal{R}(\lambda K_\lambda - P) = \max(\lambda K_\lambda - P)$, a contradiction. Therefore $(1-\lambda)M_\lambda < -d$, so $(1-\lambda)K_\lambda < -d$ on Δ and $K_\lambda - P \leq (\lambda K_\lambda - P) - d$. By Lemma 3.2(iv), $\mathcal{R}(K_\lambda - P) \leq \mathcal{R}(\lambda K_\lambda - P) - d = K_\lambda - d$.

(d) $K_0 = 0$ is rational convex polyhedral. If K_r is, then $\lambda K_r - P$ is (§3.1; λ is rational), and $K_{r+1} = R_m \cdots R_1(\lambda K_r - P)$ is, by m applications of Lemma 3.3. The contraction gives $\|K_r - K_\lambda\| \leq \lambda^r\|K_0 - K_\lambda\| = \lambda^r\|K_\lambda\| \leq \lambda^r\|P\|/(1-\lambda)$ by (b).

(e) N_0 is found by comparing rationals. Put $n_0 = (1-\lambda)^{-1}\log^+\big(16\|P\|/(d(1-\lambda))\big)$. If $16\|P\| < d(1-\lambda)$, then $N_0 = 0$. Otherwise every integer $n > n_0$ satisfies $\lambda^n \leq e^{-(1-\lambda)n} < e^{-(1-\lambda)n_0} = d(1-\lambda)/(16\|P\|)$; so $N_0 \leq \lfloor n_0 \rfloor + 1$ in both cases. For $N \geq N_0$ the same tail test holds since $0 < \lambda < 1$. Put $\delta_N = \|K_N - K_\lambda\| < d/16$. By Lemma 3.2(iv) and (c),

$$\mathcal{R}(K_N - P) \leq \mathcal{R}(K_\lambda - P) + \delta_N \leq K_\lambda - d + \delta_N \leq K_N - d + 2\delta_N < K_N - \tfrac{7}{8}d. \qquad \square$$

The construction uses the arithmetic count

$$N = \left\lceil \frac{16\|P\|}{d(1-\lambda)(\lambda^{-1} - 1)} \right\rceil + 1.$$

Bernoulli's inequality gives $\lambda^{-N} \geq 1 + N(\lambda^{-1} - 1) > 16\|P\|/[d(1-\lambda)]$, so this count satisfies the tail test and is at least N_0.

Only an upper bound for $\|P\|$ enters the argument: if $\bar{P} \geq \|P\|$ is rational, the least N with $\lambda^N\bar{P}/(1-\lambda) < d/16$ has the same property. The number $\#I$ of affine pieces of $K_N = \max_{\iota \in I} \phi_\iota$ is finite and is read off from the eliminations; it enters later only through the smoothing parameter c_K of Lemma 3.5.

3.4. **Smoothing.**

Lemma 3.5 (Log-sum-exp smoothing)**.** *Let ϕ_ι ($\iota \in I$, $1 \le \#I < \infty$) be affine functions on $\mathbb{R}^m$, $c_K > 0$, and*

$$K = \frac{1}{c_K} \log \sum_{\iota \in I} e^{c_K \phi_\iota}.$$

(a) $\max_\iota \phi_\iota \le K \le \max_\iota \phi_\iota + \log(\#I)/c_K$ *on $\mathbb{R}^m$; if $\#I = 1$, K is the single ϕ_ι.*
(b) K is real-analytic and convex on $\mathbb{R}^m$, and $K \circ p$ is real-analytic on $\mathbb{C}^m$.
(c) $\nabla K = \sum_\iota \varpi_\iota \nabla \phi_\iota$ *with weights* $\varpi_\iota = e^{c_K\phi_\iota} / \sum_{\iota'} e^{c_K \phi_{\iota'}} \in (0,1]$, $\sum_\iota \varpi_\iota = 1$*; in particular* $|\partial_i K| \le \max_\iota |\partial_i \phi_\iota|$*, and K is nondecreasing in p_i if every ϕ_ι is.*

If $d > 0$ and c_K is an integer with $c_K > 32\,\#I/d$, then $\log(\#I)/c_K < d/32$.

Proof. (a) With $\phi_{\max} = \max_\iota \phi_\iota$, $e^{c_K\phi_{\max}} \le \sum_\iota e^{c_K\phi_\iota} \le \#I\, e^{c_K\phi_{\max}}$; take $\frac{1}{c_K}\log$.

(c) Differentiating, $\nabla K = \sum_\iota \varpi_\iota \nabla\phi_\iota$, where each $\nabla\phi_\iota$ is a constant vector; the bound and the monotonicity follow because the ϖ_ι are convex weights.

(b) The sum of exponentials of affine functions is real-analytic and positive, and log is real-analytic on $(0,\infty)$. Differentiating (c) once more,

$$\partial_a\partial_b K = c_K\Big(\sum_\iota \varpi_\iota\, \partial_a\phi_\iota\, \partial_b\phi_\iota - \Big(\sum_\iota \varpi_\iota\, \partial_a\phi_\iota\Big)\Big(\sum_\iota \varpi_\iota\, \partial_b\phi_\iota\Big)\Big),$$

which is c_K times the covariance matrix of the vectors $\nabla\phi_\iota$ under the probability weights ϖ_ι, hence positive semidefinite. So K is convex. Since each $p_j = \pi|w_j|^2$ is a polynomial in the real coordinates of w, $K \circ p$ is real-analytic.

Finally $\log x < x$ for $x \ge 1$, so $\log(\#I)/c_K < \#I/c_K < d/32$. □

3.5. **Composition of coordinate realisations.** The next lemma concatenates one isotopy per coordinate and adds up their errors. The clock is the flat step S of Definition 2.3, all of whose derivatives vanish at 0 and 1.

Lemma 3.6 (Composition of coordinate realisations)**.** *Let F_0 be continuous and convex on Δ, $F_j = R_j F_{j-1}$ for $1 \le j \le m$, and $e_1, \dots, e_m \ge 0$. For each j let $Q^{(j)} \colon [0,1] \times \mathbb{C}^m \to \mathbb{R}$ be smooth (with one-sided derivatives in s at $s = 0, 1$), and let $(q_s^{(j)})_{s\in[0,1]}$ be a smooth isotopy of $\mathbb{C}^m$ (smooth in (s,w) in the same sense) generated by $Q_s^{(j)}$, that is, $\partial_s q_s^{(j)} = X_s^{(j)} \circ q_s^{(j)}$ with $\iota_{X_s^{(j)}}\omega_V = -dQ_s^{(j)}$, with $q_0^{(j)} = \mathrm{id}$ and $q_s^{(j)}(V_\Delta) = V_\Delta$. Assume:*

(1) there is $\varsigma_j > 0$ such that $\{Q_s^{(j)}, \langle b_\nu, p\rangle\} = 0$ on $p^{-1}\{\langle b_\nu, p\rangle > \ell_\nu - \varsigma_j\}$ for every ν and s;
(2) $F_{j-1}\big(p(q_1^{(j)}(w))\big) \le F_j\big(p(w)\big) + e_j$ for every $w \in V_\Delta$.

For $t \in [\frac{r-1}{m}, \frac{r}{m}]$, $r = 1, \dots, m$, and $j = m + 1 - r$, put

$$q_t = q^{(j)}_{S(mt-r+1)} \circ q_1^{(j+1)} \circ \cdots \circ q_1^{(m)}.$$

Then:

(a) q_t is well defined, $q_0 = \mathrm{id}$, $q_1 = q_1^{(1)} \circ \cdots \circ q_1^{(m)}$, $q_t(V_\Delta) = V_\Delta$, and (q_t) is a smooth isotopy of $\mathbb{C}^m$ generated by the smooth function $Q_t = mS'(mt - r + 1)\, Q^{(j)}_{S(mt-r+1)}$ on $[\frac{r-1}{m}, \frac{r}{m}]$; $Q_t = 0$ at $t = r/m$;
(b) with $\varsigma = \min_j \varsigma_j$, $\{Q_t, \langle b_\nu, p\rangle\} = 0$ on $p^{-1}\{\langle b_\nu, p\rangle > \ell_\nu - \varsigma\}$ for every ν and t;
(c) $F_0\big(p(q_1(w))\big) \le (\mathcal{R}F_0)\big(p(w)\big) + \sum_j e_j$ for every $w \in V_\Delta$.

Proof. (a) Apply Lemma 2.5(b) with $X = \mathbb{C}^m$, $N = m$, $t_r = r/m$ and $\psi^{(r)} = q^{(m+1-r)}$, the isotopy generated by $Y_s^{(r)} = X_s^{(m+1-r)}$, the Hamiltonian field of $Q_s^{(m+1-r)}$. The clocks there are $\sigma_r(t) = S(mt - r + 1)$, and on $[\frac{r-1}{m}, \frac{r}{m}]$ the concatenation is $\psi^{(r)}_{\sigma_r(t)} \circ \psi_1^{(r-1)} \circ \cdots \circ \psi_1^{(1)} =$

q_t. That lemma gives that q_t is well defined and a smooth isotopy of $\mathbb{C}^m$ with $q_0 = \mathrm{id}$ and $q_1 = \psi_1^{(m)} \circ \cdots \circ \psi_1^{(1)} = q_1^{(1)} \circ \cdots \circ q_1^{(m)}$, generated on $[\frac{r-1}{m}, \frac{r}{m}]$ by the Hamiltonian $\sigma_r'(t) Q^{(j)}_{\sigma_r(t)} = mS'(mt - r + 1) Q^{(j)}_{S(mt-r+1)}$, which is smooth in (t, w) and vanishes at the seams. Each factor maps V_Δ onto itself, hence so does q_t.

(b) On the r-th interval Q_t is a multiple of $Q_s^{(j)}$ with $s = S(mt - r + 1)$, and $\{\langle b_\nu, p\rangle > \ell_\nu - \varsigma\} \subset \{\langle b_\nu, p\rangle > \ell_\nu - \varsigma_j\}$.

(c) Fix $w \in V_\Delta$ and put $u_m = w$, $u_{j-1} = q_1^{(j)}(u_j)$ for $j = m, \dots, 1$. All u_j lie in V_Δ, and $u_0 = q_1(w)$. By (2), $F_{j-1}(p(u_{j-1})) \le F_j(p(u_j)) + e_j$. Chaining these inequalities from $j = 1$ to $j = m$ gives $F_0(p(q_1(w))) \le F_m(p(w)) + \sum_j e_j$, and $F_m = \mathcal{R} F_0$. □

In the proof of Theorem M3.5, the isotopies $q^{(j)}$ of Theorem M3.4 for F_{j-1}, the coordinate j and $e_j = d/(8(m+1))$ satisfy the hypotheses of Lemma 3.6, with $F_0 = K_N - P$: they are smooth Hamiltonian isotopies of $\mathbb{C}^m$ with smooth compactly supported generators; hypothesis (2) is Theorem M3.4(c); and (1) holds with the collar ς_j of Theorem M3.4(b), because a function vanishing on an open set has zero Poisson bracket there, while for $b_{\nu j} = 0$ the bracket vanishes everywhere. Lemma 3.6(a),(b) is then the smoothness of (q_t) and (Q_t) across the seams and condition (1) of Definition M2.4, and Lemma 3.6(c) with Proposition 3.4(e) gives the estimate $(K_N - P)(p(q_1(w))) \le K_N(p(w)) - \frac{3}{4} d$ used there.

3.6. **One fibre coordinate.** For $m = 1$ we have $\Delta = [0, h]$ for some rational $h > 0$ (a single outer face $p = h$), $L \equiv h$, and R_1 is the increasing rearrangement of Lemma 3.1 on $[0, h]$. This is the case of the four-dimensional models. Two further facts are used there: the rearrangement does not increase Lipschitz constants along slices, which the planar realisation of §7 uses to convert a level error into a value error, and for $m = 1$ the elimination of Lemma 3.3 can be replaced by the inversion of a distribution function.

Lemma 3.7 (Lipschitz bounds)**.** *Let $L \ge 0$, let f be continuous and convex on $[0, L]$ with $|f(v) - f(v')| \le \Lambda_f |v - v'|$, and let $f^\sharp$ be as in Lemma 3.1. Then $0 \le f^\sharp(x + b) - f^\sharp(x) \le \Lambda_f b$ whenever $0 \le x \le x + b \le L$. Consequently, for every m, if F is continuous and convex on Δ and $F(y, \cdot)$ is Λ_f-Lipschitz on every slice, then so is $(R_j F)(y, \cdot)$; if $F = \max_\iota \phi_\iota$, one may take $\Lambda_f = \max_\iota |\partial_j \phi_\iota|$.*

Proof. The lower bound is Lemma 3.1(c). Let $[a, a + x]$ be a minimising interval for x and put $a' = \min(a, L - x - b)$. Then $0 \le a' \le a$ and $a' + x + b \le L$, so a' is admissible for $x + b$. If $a' = a$, the new endpoints are a and $a + x + b$; if $a' = L - x - b < a$, then $a - a' = a + x + b - L \le b$ and $a' + x + b = L \ge a + x$. In both cases $|a' - a| \le b$ and $0 \le (a' + x + b) - (a + x) \le b$, so $f^\sharp(x + b) \le \max(f(a'), f(a' + x + b)) \le \max(f(a), f(a + x)) + \Lambda_f b = f^\sharp(x) + \Lambda_f b$. The statement for R_j follows from Lemma 3.2(i). For $F = \max_\iota \phi_\iota$, $F(y, \cdot)$ is a maximum of affine functions of slopes $\partial_j \phi_\iota$. □

Lemma 3.8 (Rearrangement of a piecewise affine function)**.** *Let f be continuous, convex and piecewise affine on $[0, h]$, with knots $0 = \xi_0 < \xi_1 < \cdots < \xi_n = h$ and rational knots and values. Put $f_{\min} = \min f$. Then $\mathcal{L}_f$ is continuous, piecewise affine and strictly increasing on $[f_{\min}, \max f]$, with $\mathcal{L}_f(\max f) = h$ and knots among the values $f(\xi_i)$, and*

$$R_1 f(x) = f_{\min} \quad (0 \le x \le \mathcal{L}_f(f_{\min})), \qquad R_1 f(x) = \mathcal{L}_f^{-1}(x) \quad (\mathcal{L}_f(f_{\min}) \le x \le h).$$

$R_1 f$ is piecewise affine with rational values and with knots among 0, h and the rational numbers $\mathcal{L}_f(f(\xi_i))$, and it is convex by Lemma 3.2(ii).

Proof. Put $w_i = \xi_{i+1} - \xi_i > 0$, $v_i = f(\xi_i)$, $s_i = (v_{i+1} - v_i)/w_i$ and $c_i = v_i - s_i\xi_i$. The length contributed by the i-th affine segment is

$$d_i(b) = \begin{cases} \max\left(0, \min\left(\xi_{i+1}, \dfrac{b-c_i}{s_i}\right) - \xi_i\right), & s_i > 0, \\ \max\left(0, \xi_{i+1} - \max\left(\xi_i, \dfrac{b-c_i}{s_i}\right)\right), & s_i < 0, \\ w_i\, \mathbf{1}_{\{b \geq c_i\}}, & s_i = 0. \end{cases}$$

Indeed, solve the affine inequality on $[\xi_i, \xi_{i+1}]$, using the left endpoint for positive slope and the right endpoint for negative slope. Consecutive closed segments overlap only at endpoints, which have measure zero; hence $\mathcal{L}_f(b) = \sum_i d_i(b)$. All constants in this formula are rational. By convexity, a segment with $s_i = 0$ lies at the level $f_{\min}$ (its zero slope lies between the slopes on either side, which are therefore ≤ 0 to the left and ≥ 0 to the right), so on $[f_{\min}, \max f]$ its contribution is the constant w_i. Every other contribution is continuous and affine between the values v_i and v_{i+1}. Hence $\mathcal{L}_f$ is continuous and piecewise affine on $[f_{\min}, \max f]$, with knots among the v_i, and the values $\mathcal{L}_f(v_i)$ are rational.

Let $f_{\min} \leq b < b' \leq \max f$. By the intermediate value theorem some $v_0 \in [0, h]$ has $f(v_0) = \frac{1}{2}(b + b')$, so $v_0 \in \{f \leq b'\} \setminus \{f \leq b\}$. Both sets are closed intervals, the first containing the second (which is nonempty), so the convex hull of $\{f \leq b\} \cup \{v_0\}$ lies in $\{f \leq b'\}$ and is strictly longer than $\{f \leq b\}$. Thus $\mathcal{L}_f$ is strictly increasing on $[f_{\min}, \max f]$, and $\mathcal{L}_f(\max f) = h$.

Lemma 3.1(a) now gives $R_1 f = f_{\min}$ on $[0, \mathcal{L}_f(f_{\min})]$ and $R_1 f = \mathcal{L}_f^{-1}$ on $[\mathcal{L}_f(f_{\min}), h]$. On each interval between consecutive values v_i, $\mathcal{L}_f(b) = C + \Sigma b$ with rational C, Σ and $\Sigma > 0$, whose inverse is $x \mapsto (x - C)/\Sigma$. So $R_1 f$ is piecewise affine with rational coefficients and knots among 0, h and the $\mathcal{L}_f(v_i)$. Convexity is Lemma 3.2(ii). □

3.7. **Equal balls in dimension four.** When $k \geq 10$ balls of equal capacity are packed, the scalar problem has the closed-form solution of Proposition M3.8. We record the identities and margins used there and in the explicit inner map of §5. Let $k \geq 10$ be an integer, $\eta_\star \in (0, \frac{1}{4})$ and $t_0 = 1 - \eta_\star/2$, and let $s = \sqrt{k}$, $a = t_0/s$, $h = t_0/3$, $c_s = 3s - 9$, $g = 3(1 - t_0) = \frac{3}{2}\eta_\star$, $A = h/a$, $B = h/(h - a)$ and the tent Λ be as in §M3.4. In the equal-capacity branch $\eta_\star = 1 - \rho\sqrt{k}$ is the volume defect of the normalised capacity ρ (Remark 13.25). Put $K_{\mathrm{lin}}(p) = c_s p$, $P_{\mathrm{b}}(p) = 3(1 - 3p) - k(a - p)_+$ and $F = K_{\mathrm{lin}} - P_{\mathrm{b}}$ on $\mathbb{R}$, and let $\bar{D}(h) = \{z \in \mathbb{C} : \pi|z|^2 \leq h\}$. By §M3.4, $0 < a < h$, $c_s > 0$, $A, B > 0$, $1/A + 1/B = 1$ (M8), and $F = -g + c_s\Lambda$ on $\mathbb{R}$ (M9).

Lemma 3.9 (The slopes of the tent)**.** *$A + B = AB$, $h/A = a$, $h/B = h - a$, and*

$$c_s A = k - 3s, \qquad c_s B = 3s, \qquad 3sa = 3 - g.$$

Proof. Multiplying $1/A + 1/B = 1$ by AB gives $A + B = AB$; $h/A = a$ and $h/B = h - a$ are the definitions. Using $h - a = t_0(s - 3)/(3s)$,

$$c_s A = (3s - 9)\frac{s}{3} = s^2 - 3s = k - 3s, \qquad c_s B = (3s - 9)\frac{s}{s - 3} = 3s, \qquad 3sa = 3t_0 = 3 - g. \ \square$$

The tent takes the values $\Lambda(0) = Aa = h$, $\Lambda(a) = 0$ and $\Lambda(h) = B(h - a) = h$, and by Corollary M3.7 its increasing rearrangement on $[0, h]$ is $x \mapsto x$. No area-preserving homeomorphism q of $\bar{D}(h)$ satisfies $\Lambda(p(q(z))) \leq \pi|z|^2$ exactly. Indeed, for $0 < b < h = Aa$ this would give $q(\bar{D}(b)) \subset \{z : \Lambda(\pi|z|^2) \leq b\}$, which by Lemma M3.6 is the closed annulus $\{a - b/A \leq \pi|z|^2 \leq a + b/B\}$ of area b with $a - b/A > 0$. The compact set $q(\bar{D}(b))$ has area b, so its relatively open complement in the annulus has measure zero; as every nonempty relatively open subset of this annulus has positive measure, $q(\bar{D}(b))$ is the whole annulus. A closed disc is not homeomorphic to an annulus. The inner map of §5 achieves the inequality up to an error

$4e_q$, where e_q is chosen as follows. The bound $e_q \le g/(128c_s)$ makes the cost $c_s \cdot 4e_q$ of the inner error at most $g/32$; the bound $e_q \le h/100$ keeps the error terms of the inner map small compared with the fibre height.

Definition 3.10 (Inner smoothing parameter)**.** Put

$$e_q = \min\Big(\frac{h}{100},\ \frac{g}{128\,c_s}\Big).$$

Then $0 < e_q \le h/100$.

Lemma 3.11 (Margin)**.** *$g - 4c_s e_q \ge \frac{31}{32}g$. Consequently, for all real u and x,*

$$\Lambda(x) \le u + 4e_q \quad\Longrightarrow\quad K_{\mathrm{lin}}(u) - F(x) \ge \tfrac{31}{32}g.$$

In particular, if q satisfies $\Lambda(p(q(z))) \le \pi|z|^2 + 4e_q$ on $\bar D(h)$, then $(K_{\mathrm{lin}} - P_{\mathrm{b}})(p(q(z))) \le K_{\mathrm{lin}}(\pi|z|^2) - \frac{31}{32}g$ there.

Proof. $4c_s e_q \le 4c_s \cdot g/(128c_s) = g/32$. By (M9) and $c_s > 0$, $K_{\mathrm{lin}}(u) - F(x) = c_s u + g - c_s\Lambda(x) \ge g - 4c_s e_q \ge \frac{31}{32}g$. The last statement is the case $u = \pi|z|^2$, $x = p(q(z))$. □

4. Area-preserving disc maps from Morse charts

The fibre map of the handle construction (§M4) must be the endpoint of a compactly supported Hamiltonian isotopy of a disc which is the identity near the boundary and satisfies a comparison inequality. In [Ope26] such an isotopy is obtained from a rearrangement lemma by nested discs, a construction it credits to [Sch99, Lemma 3.2], and Moser's method. This section constructs it explicitly. The construction is used twice: for the explicit inner map for equal balls (§5) and for the realisation of coordinate rearrangements of convex functions (§7). The input is a pair of potentials, U in a moment variable p and V in an angle θ, each with a single nondegenerate minimum near which it is exactly quadratic; in the slice family both depend smoothly on a parameter y (the other moments). The output is a compactly supported Hamiltonian isotopy of the plane which preserves a disc of prescribed area, is the identity on an outer collar of it, and whose endpoint on a prescribed core disc is an explicit area-preserving map carrying round circles onto level sets of $U + V$. The two families of potentials are defined later, in §5 and in Definition 7.3; this section uses only the properties listed in Definition 4.1, and a reader of the equal-ball case can ignore every statement about the slice family.

The construction has two halves. The first (§§4.2–4.4) produces the endpoint. Signed square roots of the potentials give a Morse chart. A rotation normalisation on each circle and a radial area normalisation make it area preserving, and the polar chart places it in the disc (Theorem 4.7). The second half (§§4.5–4.10) joins the identity to this endpoint without changing it on the core. Interpolating the Morse chart linearly with its linearisation, and rescaling the polar chart, gives a path of symplectic embeddings. A five-segment path of embeddings, all with image in a fixed smaller disc, connects it to the identity. The ambient flow of this path, cut off in the source variable, is made area preserving by one explicit normalisation of the whole path. The Hamiltonian is the radial primitive of the resulting flow, normalised at a fixed point of the collar (Theorems 4.21 and 4.23).

Conventions. For $P > 0$, $D(P) = \{z \in \mathbb{C} : \pi|z|^2 < P\}$ is the open disc of *area* P and $\bar D(P)$ its closure; $D(\infty) = \mathbb{C}$. Every plane carries its standard area form: $\omega_{\mathbb{C}}$ on the source and target discs $\mathbb{C}$, $du \wedge dv$ on the Morse plane $\mathbb{R}^2 \ni w = (u, v) = u + iv$, and $dx \wedge d\theta$ on the rectangle coordinates (x, θ). A map between open subsets of planes is *symplectic* if its Jacobian determinant is identically 1. In polar area coordinates $p = \pi|z|^2$, $\vartheta = \arg(z)/2\pi \in \mathbb{R}/\mathbb{Z}$ (the fibre angle θ_j of the main text) one has $\omega_{\mathbb{C}} = dp \wedge d\vartheta$ on $\mathbb{C} \setminus \{0\}$; under the polar chart C_y of

§4.4 the rectangle angle is $\theta = \vartheta - \frac{1}{2}$. Rot_α is the rotation by the angle α (radians). S is the flat step of Definition 2.3; by Lemma 2.4 it is C^∞ and nondecreasing, $S = 0$ on $(-\infty, 0]$, $S = 1$ on $[1, \infty)$, and all derivatives of S vanish at 0 and 1.

In this section E always denotes an *energy level* (a value of a potential), J a positive density on a plane, and $h > 0$ the area of the disc $D(h)$ preserved by the isotopy. A function on an arbitrary subset $\Omega \subset \mathbb{R}^N$ is called *smooth* if every point of Ω has an open neighbourhood in $\mathbb{R}^N$ on which it is the restriction of a C^∞ function; on open sets this is the usual notion, and at the endpoints of a parameter interval it means smoothness with one-sided derivatives. Fibre Hamiltonians follow the convention $\iota_X \omega_{\mathbb{C}} = -dQ$ of §M3.

4.1. **Admissible potential pairs.**

Definition 4.1 (Admissible potential pair)**.** An *admissible potential pair* is one of the following two families: the modified equal-ball wells of §5 ($k \geq 10$, $\frac{7}{8} < t_0 < 1$; Lemma 5.15), or the modified slice wells of Definition 7.3 over the open parameter set $\mathcal{O}$ of the planar data (Lemma 7.4). For either family let $\mathcal{Y} \subset \mathbb{R}^d$ be its open parameter domain ($d = 0$ meaning a single point), $h > 0$ its disc area, and $U_y(p)$, $V_y(\theta)$ its potentials on open intervals I_U, I_V. The following properties are proved for both families in the lemmas just cited:

- (P1) The potentials are jointly smooth. There is a unique $p^*(y) \in I_U \cap (0, h)$ with $U_y'(p)(p - p^*(y)) > 0$ away from the minimum, $U_y''(p^*(y)) > 0$, and $V_y(0) = 0$, $V_y'(\theta)\theta > 0$ away from zero, $V_y''(0) > 0$.
- (P2) Both potentials tend to $+\infty$ at their interval endpoints. Their centred wells are exactly quadratic on neighbourhoods of zero; the widths are positive uniformly on the parameter domain used in a construction. The quadratic coefficients are smooth positive functions of the parameters.

We write $U_{\min}(y) = U_y(p^*(y))$. The pair is *exactly quadratic near the minimum* if there are $\varepsilon_U, \varepsilon_V > 0$ with $U_y(p) - U_{\min}(y) = \frac{1}{2}U_y''(p^*(y))(p - p^*(y))^2$ for $|p - p^*(y)| \leq \varepsilon_U$ and $V_y(\theta) = \frac{1}{2}V_y''(0)\theta^2$ for $|\theta| \leq \varepsilon_V$.

The modified wells need not be convex; only the signs of their derivatives and their quadratic germs are used.

4.2. **Morse inverse charts.**

Lemma 4.2 (Morse inverse charts)**.** *Let (U, V) be an admissible potential pair over $\mathcal{Y}$.*

(i) The minimum point $y \mapsto p^*(y)$ and $y \mapsto U_{\min}(y)$ are C^∞ on $\mathcal{Y}$.

(ii) On the open set $\mathcal{D}_U = \{(y, x) : y \in \mathcal{Y},\ p^*(y) + x \in I_U\}$ put

$$\Sigma_U(y, x) = \operatorname{sign}(x)\sqrt{U_y(p^*(y) + x) - U_{\min}(y)}.$$

This is the signed square root, with its removable singularity at $x = 0$; and Σ_U is C^∞ with $\partial_x \Sigma_U > 0$. For each y, $\Sigma_U(y, \cdot)$ is an increasing diffeomorphism of $I_U - p^*(y)$ onto $\mathbb{R}$; its inverse X_y is C^∞ jointly in $(y, u) \in \mathcal{Y} \times \mathbb{R}$, and

$$U_y\big(p^*(y) + X_y(u)\big) - U_{\min}(y) = u^2,$$
$$\operatorname{sign} X_y(u) = \operatorname{sign} u, \qquad X_y'(0) = a_0(y) := \sqrt{2/U_y''(p^*(y))}.$$

(iii) In the same way $\Sigma_V(y, \theta) = \operatorname{sign}(\theta)\sqrt{V_y(\theta)}$ is an increasing diffeomorphism of I_V onto $\mathbb{R}$ for each y, and its inverse Y_y is C^∞ in (y, v), with $V_y(Y_y(v)) = v^2$, $\operatorname{sign} Y_y(v) = \operatorname{sign} v$ and $Y_y'(0) = b_0(y) := \sqrt{2/V_y''(0)}$.

(iv) The Morse chart $F_y(u,v) = (X_y(u), Y_y(v))$ is a C^∞ diffeomorphism of $\mathbb{R}^2$ onto the Morse rectangle $\mathcal{R}_y = (I_U - p^*(y)) \times I_V$, jointly smooth in (y,w), with

$$F_y^*(dx \wedge d\theta) = J_y\, du \wedge dv, \qquad J_y(u,v) = X_y'(u)\, Y_y'(v) > 0, \qquad J_y(0) = j_0(y) := a_0(y) b_0(y),$$

and $U_y(p^*(y)+x) - U_{\min}(y) + V_y(\theta) = u^2 + v^2$ for $(x,\theta) = F_y(u,v)$. Consequently, for $E \geq 0$,

$$\int_{|w|^2 \leq E} J_y(w)\, dw = \operatorname{area}\big\{(p,\theta) \in I_U \times I_V :\ U_y(p) - U_{\min}(y) + V_y(\theta) \leq E\big\},$$

and these areas increase to $|I_U|\,|I_V| \in (0,\infty]$ as $E \to \infty$.

(v) If the pair is exactly quadratic near the minimum, then $X_y(u) = a_0(y)u$ for $|u| \leq \varepsilon_U/a_0(y)$ and $Y_y(v) = b_0(y)v$ for $|v| \leq \varepsilon_V/b_0(y)$; hence $J_y = j_0(y)$ on the disc $|w| \leq \rho_T(y)$, $\rho_T(y) = \min(\varepsilon_U/a_0(y), \varepsilon_V/b_0(y))$.

Proof. (i) $\partial_p U_y'(p) = U_y''(p) > 0$ at $p = p^*(y)$, so the implicit function theorem gives, near each y_0, a smooth solution of $U_y'(p) = 0$ through $p^*(y_0)$; by (P1) the zero of U_y' is unique, so this solution is $p^*(y)$. Then $U_{\min}(y) = U_y(p^*(y))$ is smooth.

(ii) $\mathcal{D}_U$ is open since p^* is continuous. The quadratic germ gives $\Sigma_U(y,x) = \sqrt{U_y''(p^*)/2}\, x$ on a common neighbourhood of $x = 0$, which is jointly smooth there. Away from $x = 0$, positivity of $U_y(p^*+x) - U_{\min}$ gives smoothness directly from the signed square-root formula. For $x \neq 0$, differentiating $\Sigma_U^2 = U_y(p^*+x) - U_{\min}$ gives $2\Sigma_U \partial_x \Sigma_U = U_y'(p^*+x)$; both Σ_U and $U_y'(p^*+x)$ have the sign of x, so $\partial_x \Sigma_U > 0$; at $x = 0$, $\partial_x \Sigma_U = \sqrt{U_y''(p^*)/2} > 0$. By (P2), $\Sigma_U(y,\cdot)$ tends to $\mp\infty$ at the two endpoints of $I_U - p^*(y)$, so it is an increasing bijection onto $\mathbb{R}$. The map $(y,x) \mapsto (y, \Sigma_U(y,x))$ is a smooth bijection of $\mathcal{D}_U$ onto $\mathcal{Y} \times \mathbb{R}$ with invertible (triangular) derivative; by the inverse function theorem its local smooth inverses exist everywhere, and by injectivity they agree with the global inverse $(y,u) \mapsto (y, X_y(u))$, which is therefore smooth. The identities follow from $\Sigma_U(y, X_y(u)) = u$, and $X_y'(0) = 1/\partial_x \Sigma_U(y,0) = \sqrt{2/U_y''(p^*)}$.

(iii) is (ii) for V, whose minimum point is 0 and minimum value 0.

(iv) F_y is a product of diffeomorphisms; its Jacobian is $X_y' Y_y' > 0$, and $J_y(0) = a_0 b_0$. The level identity is the sum of the identities in (ii) and (iii). The change of variables formula applied to F_y on $\{|w|^2 \leq E\}$ gives the area formula, after the translation $p = p^*(y) + x$. As $E \to \infty$ the discs exhaust $\mathbb{R}^2$, so their images exhaust $\mathcal{R}_y$ and the areas increase to $\operatorname{area} \mathcal{R}_y = |I_U||I_V|$.

(v) The exact quadratic formula on $|x| \leq \varepsilon_U$ gives $\Sigma_U(y,x) = x/a_0$; hence $X_y(u) = a_0 u$ for $|u| \leq \varepsilon_U/a_0$. The same holds for V. For $|w| \leq \rho_T$ both $|u|$ and $|v|$ are at most ρ_T, so $J_y(w) = a_0 b_0$. □

4.3. **Rotation and area normalisation.** The two normalisations are defined for the positive densities used in the construction. Throughout this subsection we fix the following setting, which we call *setting (N)*: $\mathcal{P} \subset \mathbb{R}^N$ is an arbitrary subset (the parameters), $\mathcal{U} \subset \mathbb{R}^N \times \mathbb{R}^2$ is an open set containing $\mathcal{P} \times \mathbb{R}^2$, and $J\colon \mathcal{U} \to \mathbb{R}$, $(\sigma, w) \mapsto J_\sigma(w)$, is C^∞ and positive on $\mathcal{P} \times \mathbb{R}^2$, and J has a constant germ at $w = 0$: locally in the parameter there are $\rho_0 > 0$ and a smooth positive function $j(\sigma)$ with $J_\sigma(w) = j(\sigma)$ for $|w| \leq \rho_0$. The densities used are the Morse densities of the two potential families and their interpolations (§4.5), the Morse density of §6, and the whole-path densities of §4.8. Points of $\mathbb{R}^2$ are identified with complex numbers, so that $w\zeta$ is the complex product.

Definition 4.3 (Rotation and area normalisation)**.** In setting (N), for $\sigma \in \mathcal{P}$, $w \in \mathbb{R}^2$ and $E \geq 0$ put

$$\bar{J}_\sigma(|w|^2) = \frac{1}{2\pi}\int_0^{2\pi} J_\sigma(\mathrm{Rot}_\alpha w)\,d\alpha, \qquad \mathcal{M}_\sigma(w) = \frac{1}{2\pi}\int_0^{2\pi} (\alpha-\pi)\,J_\sigma(\mathrm{Rot}_\alpha w)\,d\alpha,$$
$$T_\sigma(w) = \mathrm{Rot}_{\tau_\sigma(w)}\,w, \qquad \tau_\sigma(w) = \mathcal{M}_\sigma(w)/\bar{J}_\sigma(|w|^2),$$
$$\mathcal{A}_\sigma(E) = \int_{|w|^2\leq E} J_\sigma(w)\,dw, \qquad \mathcal{A}_\sigma(\infty) = \lim_{E\to\infty}\mathcal{A}_\sigma(E) \in (0,\infty],$$
$$\mathcal{G}_\sigma(w) = \frac{1}{\pi}\int_{|\zeta|\leq 1} J_\sigma(w\zeta)\,d\zeta.$$

Let $E_\sigma\colon [0,\mathcal{A}_\sigma(\infty)) \to [0,\infty)$ be the inverse of $\mathcal{A}_\sigma$ (Lemma 4.5) and, for $z \in D(\mathcal{A}_\sigma(\infty))$ with $I = \pi|z|^2$,

$$D_\sigma(z) = \sqrt{\pi E_\sigma(I)/I}\;z \quad (z\neq 0), \qquad D_\sigma(0) = 0.$$

T_σ is the *rotation normalisation* and D_σ the *area normalisation* of J_σ. When a single density J is meant we write $\bar{J}$, $\mathcal{M}_J$, T_J, $\mathcal{A}_J$, E_J, $\mathcal{G}_J$, D_J.

The normalisations of J_σ at a point w with $|w| \leq R$, or at a point z with $E_\sigma(\pi|z|^2) \leq R^2$, depend only on J_σ on the disc $|w| \leq R$: the circle and disc integrals defining $\bar{J}_\sigma$, $\mathcal{M}_\sigma$ and $\mathcal{A}_\sigma$ on $[0,R^2]$ stay in that disc. Hence on such points they coincide with the normalisations of

$$J^{\mathrm{ext}}_\sigma(w) = \vartheta_R(w)J_\sigma(w) + (1-\vartheta_R(w))j(\sigma), \qquad \vartheta_R(w) = 1 - S\left(\frac{|w|^2-R^2}{3R^2}\right),$$

which is smooth, positive, has the same germ, equals J_σ for $|w| \leq R$, and has $\mathcal{A}_{J^{\mathrm{ext}}_\sigma}(\infty) = \infty$. For the slice family we use this with $R^2 = \bar{E}$ (Definition 7.3), so that the source disc $D(p^\dagger)$ is covered by (D5); it is a device of proof only, and all maps are defined with the original densities.

On the circle of radius ρ the density defines a measure of total mass $2\pi\rho\bar{J}_\sigma(\rho^2)$. On each such circle T_σ is determined up to a rotation by the condition that it carries this measure to the uniform measure of the same mass, i.e. by the derivative of its angle (Lemma 4.4(ii)). The rotation is fixed by the weighted mean $\mathcal{M}_\sigma$; this choice needs no base point on the circle and is a Cartesian function of w. Likewise $\mathcal{G}_\sigma$ writes the mean of J_σ over the disc of radius $|w|$ as an integral over a fixed disc, which is what makes the area normalisation smooth at the origin.

Lemma 4.4 (Rotation normalisation)**.** *In setting (N), let $\sigma \in \mathcal{P}$.*

- *(i) $\partial_{\arg}\mathcal{M}_\sigma = J_\sigma - \bar{J}_\sigma(|w|^2)$, where $\partial_{\arg}$ is the derivative along ψ in $w = \rho e^{i\psi}$.*
- *(ii) T_σ maps each circle $\{|w| = \rho\}$ onto itself; for $\rho > 0$, writing $T_\sigma(\rho e^{i\psi}) = \rho e^{i\psi^\sharp(\psi)}$, one has $(\psi^\sharp)' = J_\sigma/\bar{J}_\sigma > 0$ and $\psi^\sharp(\psi+2\pi) = \psi^\sharp(\psi)+2\pi$. T_σ is a bijection of $\mathbb{R}^2$ and $T_\sigma(0) = 0$.*
- *(iii) The maps $(\sigma,w)\mapsto T_\sigma(w)$ and $(\sigma,w)\mapsto T_\sigma^{-1}(w)$ are smooth on $\mathcal{P}\times\mathbb{R}^2$, each T_σ is a diffeomorphism of $\mathbb{R}^2$ with $DT_\sigma(0) = \mathrm{Id}$, and $T_\sigma^*\big(\bar{J}_\sigma(|w|^2)\,du\wedge dv\big) = J_\sigma\,du\wedge dv$ on all of $\mathbb{R}^2$.*
- *(iv) If $J_\sigma(w) = j$ for $|w| \leq \rho_0$, then $\mathcal{M}_\sigma = 0$ and $T_\sigma = T_\sigma^{-1} = \mathrm{id}$ on $\{|w|\leq\rho_0\}$.*

Proof. Local formulas. Let $\mathcal{U}' = \mathcal{U}\cap\{J>0\}$, an open set containing $\mathcal{P}\times\mathbb{R}^2$. Given $\sigma_0 \in \mathcal{P}$ and $\rho_1 > 0$, the compact set $\{\sigma_0\}\times\bar{B}(0,\rho_1)$ lies in $\mathcal{U}'$, so there is an open ball $N_0 \ni \sigma_0$ with $N_0\times B(0,\rho_1) \subset \mathcal{U}'$. On $N_0\times B(0,\rho_1)$, which is invariant under the rotations of the second factor, the integrands defining $\bar{J}_\sigma(|w|^2)$ and $\mathcal{M}_\sigma(w)$ are C^∞ in $(\alpha,\sigma,w) \in [0,2\pi]\times N_0\times B(0,\rho_1)$ and positive, respectively smooth, so differentiation under the integral sign shows that both formulas define C^∞ functions there, with $\bar{J} > 0$. Hence the formula for T_σ defines a C^∞ map $\tilde{T}$ on $N_0\times B(0,\rho_1)$, which is a local smooth extension of $(\sigma,w)\mapsto T_\sigma(w)$.

(i) Fix ρ and put $j(\psi) = J_\sigma(\rho e^{i\psi})$, a 2π-periodic function. Then $\mathcal{M}_\sigma(\rho e^{i\psi}) = \frac{1}{2\pi}\int_0^{2\pi}(\alpha - \pi)j(\psi+\alpha)\,d\alpha$, and integration by parts gives

$$\partial_\psi \mathcal{M}_\sigma = \frac{1}{2\pi}\Big[(\alpha-\pi)j(\psi+\alpha)\Big]_0^{2\pi} - \frac{1}{2\pi}\int_0^{2\pi} j(\psi+\alpha)\,d\alpha = j(\psi) - \bar{J}_\sigma(\rho^2).$$

(ii) T_σ preserves $|w|$. With $\psi^\sharp = \psi + \mathcal{M}_\sigma/\bar{J}_\sigma$, (i) gives $(\psi^\sharp)' = 1 + (J_\sigma - \bar{J}_\sigma)/\bar{J}_\sigma = J_\sigma/\bar{J}_\sigma > 0$, and $\psi^\sharp(\psi+2\pi) = \psi^\sharp(\psi) + 2\pi$ because $\mathcal{M}_\sigma$ is a single-valued function of w. So T_σ restricts to an orientation-preserving diffeomorphism of each circle of positive radius, and $T_\sigma(0) = 0$; hence it is a bijection of $\mathbb{R}^2$.

(iii) Smoothness of $(\sigma, w) \mapsto T_\sigma(w)$ was shown above. At $w = 0$, $\mathcal{M}_\sigma(0) = J_\sigma(0)\frac{1}{2\pi}\int_0^{2\pi}(\alpha - \pi)d\alpha = 0$, so $\tau_\sigma(0) = 0$, and differentiating $T_\sigma(w) = \mathrm{Rot}_{\tau_\sigma(w)}w$ at $w = 0$ gives $DT_\sigma(0) = \mathrm{Rot}_0 = \mathrm{Id}$. On $w \neq 0$, in polar coordinates $du \wedge dv = \rho\, d\rho \wedge d\psi$, and $\bar{J}_\sigma(|T_\sigma w|^2) = \bar{J}_\sigma(\rho^2)$, so

$$T_\sigma^*\big(\bar{J}_\sigma\,\rho\,d\rho\wedge d\psi\big) = \bar{J}_\sigma\,\rho\,d\rho\wedge((\psi^\sharp)'d\psi + \partial_\rho\psi^\sharp\,d\rho) = J_\sigma\,\rho\,d\rho\wedge d\psi.$$

Thus $\det DT_\sigma = J_\sigma/\bar{J}_\sigma(|w|^2)$ on $w \neq 0$; both sides are continuous, so this holds at 0 as well (where both equal 1), and the pull-back identity holds on $\mathbb{R}^2$. In particular DT_σ is everywhere invertible, and T_σ is a diffeomorphism by the inverse function theorem. For the inverse, let $(\sigma_0, x_0) \in \mathcal{P}\times\mathbb{R}^2$, $w_0 = T_{\sigma_0}^{-1}(x_0)$, and choose $\rho_1 > |w_0|$ and N_0 as above. The map $(\sigma, w) \mapsto (\sigma, \tilde{T}(\sigma, w))$ is C^∞ on $N_0 \times B(0,\rho_1)$ with derivative of determinant $\det D_w\tilde{T}(\sigma_0, w_0) > 0$ at (σ_0, w_0), so it has a C^∞ inverse $\mathcal{L}$ from an open neighbourhood N of (σ_0, x_0) onto an open neighbourhood of (σ_0, w_0). If $(\sigma, x) \in N$ with $\sigma \in \mathcal{P}$, then $\mathcal{L}(\sigma, x) = (\sigma, w')$ with $T_\sigma(w') = x$, so $w' = T_\sigma^{-1}(x)$ by injectivity of T_σ. Hence $(\sigma, x) \mapsto T_\sigma^{-1}(x)$ is smooth.

(iv) For $|w| \le \rho_0$ every point $\mathrm{Rot}_\alpha w$ lies in the disc, so $\bar{J}_\sigma(|w|^2) = j$ and $\mathcal{M}_\sigma(w) = j\frac{1}{2\pi}\int_0^{2\pi}(\alpha - \pi)d\alpha = 0$; hence $T_\sigma(w) = w$. Since T_σ is a bijection preserving every circle and fixing the closed disc pointwise, T_σ^{-1} fixes it too. □

Lemma 4.5 (Area normalisation)**.** *In setting (N), let $\sigma \in \mathcal{P}$ and put $\Omega = \{(\sigma, z) : \sigma \in \mathcal{P},\ \pi|z|^2 < \mathcal{A}_\sigma(\infty)\}$.*

- *(i) $\mathcal{A}_\sigma(E) = \pi\int_0^E \bar{J}_\sigma(t)\,dt$; $\mathcal{A}_\sigma$ is a continuous strictly increasing bijection of $[0,\infty)$ onto $[0, \mathcal{A}_\sigma(\infty))$, so E_σ is well defined.*
- *(ii) $(\sigma, w) \mapsto \mathcal{G}_\sigma(w)$ is smooth and positive on $\mathcal{P}\times\mathbb{R}^2$, $\mathcal{G}_\sigma(0) = J_\sigma(0)$, and $\mathcal{A}_\sigma(|w|^2) = \pi|w|^2\mathcal{G}_\sigma(w)$.*
- *(iii) $\Psi_\sigma(w) = \sqrt{\mathcal{G}_\sigma(w)}\,w$ is a diffeomorphism of $\mathbb{R}^2$ onto $D(\mathcal{A}_\sigma(\infty))$ with $\pi|\Psi_\sigma(w)|^2 = \mathcal{A}_\sigma(|w|^2)$ and $\Psi_\sigma^*\omega_{\mathbb{C}} = \bar{J}_\sigma(|w|^2)\,du\wedge dv$; its inverse is D_σ. The map $(\sigma, z) \mapsto D_\sigma(z)$ is smooth on Ω, including $z = 0$; $|D_\sigma(z)|^2 = E_\sigma(\pi|z|^2)$, $D_\sigma^*(\bar{J}_\sigma(|w|^2)\,du\wedge dv) = \omega_{\mathbb{C}}$, $DD_\sigma(0) = J_\sigma(0)^{-1/2}\,\mathrm{Id}$, and the quotient in the definition of D_σ is*

$$\frac{\pi E_\sigma(\pi|z|^2)}{\pi|z|^2} = \frac{1}{\mathcal{G}_\sigma(D_\sigma(z))} \qquad (z \neq 0),$$

  a smooth positive function of $(\sigma, z) \in \Omega$ with value $1/J_\sigma(0)$ at $z = 0$.
- *(iv) (Composite.) Suppose $(\sigma, w) \mapsto F_\sigma(w)$ is C^∞ on an open neighbourhood of $\mathcal{P}\times\mathbb{R}^2$ and each F_σ is a diffeomorphism of $\mathbb{R}^2$ onto an open set $\mathcal{V}_\sigma$ with $F_\sigma^*(dx\wedge d\theta) = J_\sigma\,du\wedge dv$. Then $f_\sigma = F_\sigma \circ T_\sigma^{-1} \circ D_\sigma$ is a symplectic diffeomorphism of $D(\mathcal{A}_\sigma(\infty))$ onto $\mathcal{V}_\sigma$, $(\sigma, z) \mapsto f_\sigma(z)$ is smooth on Ω, f_σ maps the circle $\{\pi|z|^2 = I\}$ onto $F_\sigma(\{|w|^2 = E_\sigma(I)\})$, $f_\sigma(0) = F_\sigma(0)$ and $Df_\sigma(0) = J_\sigma(0)^{-1/2}DF_\sigma(0)$.*
- *(v) (Constant near the origin.) If $J_\sigma(w) = j$ for $|w| \le \rho_0$, then $\mathcal{G}_\sigma = j$ there, $\mathcal{A}_\sigma(E) = \pi jE$ for $E \le \rho_0^2$, $D_\sigma(z) = z/\sqrt{j}$ for $|z| \le \sqrt{j}\,\rho_0$, and in (iv) $f_\sigma(z) = F_\sigma(z/\sqrt{j})$ for $|z| \le \sqrt{j}\,\rho_0$.*

Proof. (i) In polar coordinates,

$$\begin{aligned}\mathcal{A}_\sigma(E) &= \int_0^{\sqrt{E}} \rho \int_0^{2\pi} J_\sigma \, d\psi \, d\rho \\ &= \int_0^{\sqrt{E}} 2\pi\rho \, \bar{J}_\sigma(\rho^2) \, d\rho = \pi \int_0^E \bar{J}_\sigma(t) \, dt.\end{aligned}$$

Since $J_\sigma > 0$ is continuous, $\mathcal{A}_\sigma$ is continuous and strictly increasing with $\mathcal{A}_\sigma(0) = 0$, and it increases to $\mathcal{A}_\sigma(\infty)$.

(ii) For $w \neq 0$ the real-linear map $\zeta \mapsto w\zeta$ has determinant $|w|^2$ and maps the unit disc onto $\{|x| \leq |w|\}$, so $\pi|w|^2\mathcal{G}_\sigma(w) = \int_{|x|\leq|w|} J_\sigma(x)\,dx = \mathcal{A}_\sigma(|w|^2)$; at $w = 0$, $\mathcal{G}_\sigma(0) = J_\sigma(0)$. With N_0 and ρ_1 as in the proof of Lemma 4.4, the integrand $J(\sigma, w\zeta)$ is C^∞ in $(\sigma, w, \zeta) \in N_0 \times B(0,\rho_1) \times \bar{B}(0,1)$ and positive, so $\mathcal{G}$ has a positive C^∞ extension $\tilde{\mathcal{G}}$ to $N_0 \times B(0,\rho_1)$.

(iii) In polar coordinates $w = \rho e^{i\psi}$, $\Psi_\sigma(w) = r(\rho)e^{i\psi}$ with $\pi r(\rho)^2 = \mathcal{A}_\sigma(\rho^2)$ by (ii). As ρ runs over $[0,\infty)$, πr^2 increases strictly and continuously from 0 to $\mathcal{A}_\sigma(\infty)$; hence Ψ_σ is a bijection of $\mathbb{R}^2$ onto $D(\mathcal{A}_\sigma(\infty))$, and $E_\sigma(\pi|\Psi_\sigma(w)|^2) = |w|^2$. On $w \neq 0$, by (i),

$$\Psi_\sigma^*\omega_{\mathbb{C}} = r\,dr \wedge d\psi = \tfrac{1}{2\pi}\, d(\pi r^2) \wedge d\psi = \tfrac{1}{2\pi}\,\pi\bar{J}_\sigma(\rho^2)\,2\rho\,d\rho \wedge d\psi = \bar{J}_\sigma(\rho^2)\,du \wedge dv,$$

so $\det D\Psi_\sigma(w) = \bar{J}_\sigma(|w|^2)$; by continuity this also holds at $w = 0$, where $D\Psi_\sigma(0) = \sqrt{\mathcal{G}_\sigma(0)}\,\mathrm{Id}$ and both sides equal $J_\sigma(0)$. Hence Ψ_σ is a diffeomorphism onto the open disc. Its inverse maps the circle $\{\pi|z|^2 = I\}$ radially onto $\{|w|^2 = E_\sigma(I)\}$, i.e. it is $z \mapsto \sqrt{\pi E_\sigma(I)/I}\,z = D_\sigma(z)$ (and $0 \mapsto 0$). This gives $|D_\sigma(z)|^2 = E_\sigma(\pi|z|^2)$, the pull-back identity, and $DD_\sigma(0) = D\Psi_\sigma(0)^{-1}$. Joint smoothness: let $(\sigma_0, z_0) \in \Omega$ and $w_0 = D_{\sigma_0}(z_0)$; choose $\rho_1 > |w_0|$, N_0 and $\tilde{\mathcal{G}}$ as in (ii). The map $(\sigma, w) \mapsto (\sigma, \sqrt{\tilde{\mathcal{G}}(\sigma,w)}\,w)$ is C^∞ on $N_0 \times B(0,\rho_1)$ and its derivative at (σ_0, w_0) has determinant $\bar{J}_{\sigma_0}(|w_0|^2) > 0$, so it has a C^∞ inverse $\mathcal{L}$ on an open neighbourhood N of (σ_0, z_0). For $(\sigma, z) \in N \cap \Omega$, $\mathcal{L}(\sigma, z) = (\sigma, w')$ with $\Psi_\sigma(w') = z$, so $w' = D_\sigma(z)$ by injectivity of Ψ_σ. Finally, from $\Psi_\sigma(D_\sigma(z)) = z$ we get $|z|^2 = \mathcal{G}_\sigma(D_\sigma(z))|D_\sigma(z)|^2$, i.e. $\pi E_\sigma(I)/I = |D_\sigma(z)|^2/|z|^2 = 1/\mathcal{G}_\sigma(D_\sigma(z))$, which is smooth in (σ, z) as a composite of smooth maps.

(iv) D_σ is a diffeomorphism of $D(\mathcal{A}_\sigma(\infty))$ onto $\mathbb{R}^2$, T_σ^{-1} a diffeomorphism of $\mathbb{R}^2$, and F_σ a diffeomorphism of $\mathbb{R}^2$ onto $\mathcal{V}_\sigma$. By Lemma 4.4(iii) and (iii) above,

$$f_\sigma^*(dx \wedge d\theta) = D_\sigma^*(T_\sigma^{-1})^*\big(J_\sigma\,du \wedge dv\big) = D_\sigma^*\big(\bar{J}_\sigma\,du \wedge dv\big) = \omega_{\mathbb{C}}.$$

Joint smoothness follows from that of the three factors. D_σ maps $\{\pi|z|^2 = I\}$ onto $\{|w|^2 = E_\sigma(I)\}$, which T_σ^{-1} preserves. Finally $f_\sigma(0) = F_\sigma(0)$, and $Df_\sigma(0) = DF_\sigma(0)\,DT_\sigma^{-1}(0)\,DD_\sigma(0) = J_\sigma(0)^{-1/2}DF_\sigma(0)$ by Lemma 4.4(iii).

(v) For $|w| \leq \rho_0$ and $|\zeta| \leq 1$, $|w\zeta| \leq \rho_0$, so $\mathcal{G}_\sigma(w) = j$ and $\Psi_\sigma(w) = \sqrt{j}\,w$ on the closed disc of radius ρ_0; inverting, $D_\sigma(z) = z/\sqrt{j}$ for $|z| \leq \sqrt{j}\rho_0$, and $\mathcal{A}_\sigma(E) = \pi j E$ for $E \leq \rho_0^2$ by (ii). The last claim follows from Lemma 4.4(iv). □

4.4. **The inner map.** The polar chart places the Morse rectangle into the disc of area h. For $y \in \mathcal{Y}$ let

$$C_y(x,\theta) = -\sqrt{\frac{p^*(y)+x}{\pi}}\,e^{2\pi i\theta}, \qquad (x,\theta) \in \mathcal{C}_y = \{p^*(y)+x > 0,\ |\theta| < \tfrac{1}{2}\}. \tag{3}$$

It has polar radius $\sqrt{(p^*+x)/\pi}$ and polar angle $2\pi\theta + \pi \in (0, 2\pi)$; so C_y is injective, $\pi|C_y(x,\theta)|^2 = p^*(y) + x$, the image of $\mathcal{C}_y$ is $\mathbb{C}$ minus the closed positive real axis, and $C_y^*(dp \wedge d\vartheta) = dx \wedge d\theta$, i.e. C_y is symplectic. Its centre line $\theta = 0$ is the negative real axis, and $C_y(0) = b_y := (-r_b(y), 0)$ with $r_b(y) = \sqrt{p^*(y)/\pi}$.

Definition 4.6 (The inner map)**.** Let (U,V) be an admissible potential pair over $\mathcal{Y}$ and $y\in\mathcal{Y}$. Apply the formulas of Definition 4.3 to the density $J_y = X_y'Y_y'$ of Lemma 4.2 (parameters $\sigma = y \in \mathcal{P} = \mathcal{Y}$), and write T_y, D_y, $\mathcal{A}_y$, E_y for its normalisations. In the slice case the claims about these maps are made on the disc $D(p^\dagger)$ of Definition 7.3 and are proved with the constant-tail extension above. Put

$$f_y = F_y\circ T_y^{-1}\circ D_y\colon D(\mathcal{A}_y(\infty))\to\mathcal{R}_y.$$

Let $E_{\mathrm{rect}}(y)\in(0,\infty]$ be the infimum of $U_y(p)-U_{\min}(y)+V_y(\theta)$ over $(I_U\times I_V)\setminus\big((0,h)\times(-\frac12,\frac12)\big)$ (and $+\infty$ if this set is empty), let $I_{\mathrm{rect}}(y)=\mathcal{A}_y(E_{\mathrm{rect}}(y))$ (with $\mathcal{A}_y(+\infty)=\mathcal{A}_y(\infty)$), and define the *inner map*

$$q_{\mathrm{in},y} = C_y\circ f_y\colon D\big(I_{\mathrm{rect}}(y)\big)\to\mathbb{C}.$$

We also put $c_x(y)=\sqrt{a_0(y)/b_0(y)}$ and $\mathbf{L}_y=\operatorname{diag}(c_x(y),c_x(y)^{-1})$.

By (P1), $U_y-U_{\min}$ is monotone on each side of p^* and V_y on each side of 0, so $E_{\mathrm{rect}}(y)$ is the least of the values $U_y(0)-U_{\min}$, $U_y(h)-U_{\min}$, $V_y(-\frac12)$, $V_y(\frac12)$ at those of the points $0,h\in\mathbb{R}$ and $\pm\frac12$ that lie in I_U, respectively I_V; it is positive since $p^*(y)\in(0,h)$. By Lemma 4.2(iv), $\mathcal{A}_y(E)$ is the area of the sublevel $\{U_y-U_{\min}+V_y\le E\}$ and $\mathcal{A}_y(\infty)=|I_U|\,|I_V|$. For $E<E_{\mathrm{rect}}(y)$ this sublevel lies in $(0,h)\times(-\frac12,\frac12)$, so $\mathcal{A}_y(E)\le h$; letting $E\uparrow E_{\mathrm{rect}}(y)$ gives $I_{\mathrm{rect}}(y)\le h$.

Theorem 4.7 (The inner map)**.** *Let (U,V) be an admissible potential pair. Put $\mathcal{D}_y^{\mathrm{src}}=\mathbb{C}$ for the equal-ball family and $\mathcal{D}_y^{\mathrm{src}}=D(p^\dagger)$, with $p^\dagger$ of Definition 7.3, for the slice family.*

(a) For each y, f_y is a symplectic embedding of $\mathcal{D}_y^{\mathrm{src}}$ into the Morse rectangle $\mathcal{R}_y$, and $(y,z)\mapsto f_y(z)$ is smooth on the corresponding source set. In the equal-ball case it is a diffeomorphism onto $\mathcal{R}_y=\mathbb{R}^2$. Moreover $f_y(0)=0$ and $Df_y(0)=\mathbf{L}_y$, a diagonal matrix of determinant 1.

(b) (Energy identity.) If $f_y(z)=(x,\theta)$, then

$$U_y\big(p^*(y)+x\big)+V_y(\theta)=U_{\min}(y)+E_y(\pi|z|^2),$$

so f_y maps each circle $\{\pi|z|^2=I\}$ onto the level curve of $U_y(p^*+\cdot)-U_{\min}+V_y$ which encloses area I.

(c) The inner map $q_{\mathrm{in},y}$ is a symplectic embedding of $D(I_{\mathrm{rect}}(y))\cap\mathcal{D}_y^{\mathrm{src}}$ into $D(h)\setminus[0,\infty)$, jointly smooth in (y,z), with $q_{\mathrm{in},y}(0)=b_y$, $Dq_{\mathrm{in},y}(0)=\mathbf{A}_y:=DC_y(0)\,\mathbf{L}_y$, and, for $f_y(z)=(x,\theta)$,

$$\pi|q_{\mathrm{in},y}(z)|^2=p^*(y)+x,\qquad U_y\big(\pi|q_{\mathrm{in},y}(z)|^2\big)+V_y(\theta)=U_{\min}(y)+E_y(\pi|z|^2).$$

In particular $U_y(\pi|q_{\mathrm{in},y}(z)|^2)\le U_{\min}(y)+E_y(\pi|z|^2)$, with equality on the centre line $\theta=0$.

(d) If the pair is exactly quadratic near the minimum, then $f_y(z)=\mathbf{L}_yz$ for $|z|\le\sqrt{j_0(y)}\,\rho_T(y)$.

Proof. (a) For the whole-plane family, apply Lemma 4.5(iv) in setting (N) with $\mathcal{P}=\mathcal{Y}$, $\mathcal{U}=\mathcal{Y}\times\mathbb{R}^2$, the density J_y and F_y (Lemma 4.2(iv)), whose image is $\mathcal{V}_y=\mathcal{R}_y$. For the slice family, use the constant-tail extension following Definition 4.3. Its area and rotation normalisations are smooth and have the stated determinant identities by Lemmas 4.5 and 4.4. On $D(p^\dagger)$ they agree with the original normalisations, and their Morse images stay where the extended density equals J_y. There $\det DF_y=J_y$ gives the same symplectic identity after composition. Thus the assertion holds on each source domain. Further $f_y(0)=F_y(0)=0$ and $Df_y(0)=j_0^{-1/2}\operatorname{diag}(a_0,b_0)=\operatorname{diag}(\sqrt{a_0/b_0},\sqrt{b_0/a_0})=\mathbf{L}_y$.

(b) $(u,v) = T_y^{-1} D_y(z)$ satisfies $u^2+v^2 = |D_y(z)|^2 = E_y(\pi|z|^2)$, since T_y^{-1} preserves $|w|$ (Lemmas 4.4(ii) and 4.5(iii)). Apply the level identity of Lemma 4.2(iv).

(c) If $\pi|z|^2 < I_{\mathrm{rect}}(y)$ then $E_y(\pi|z|^2) < E_{\mathrm{rect}}(y)$, so by (b) the point (p^*+x,θ) lies in a sublevel below E_{rect}, hence in $(0,h)\times(-\frac12,\frac12)$; thus $(x,\theta)\in\mathcal{C}_y$ and $0<\pi|q_{\mathrm{in},y}(z)|^2 = p^*+x<h$. The map $q_{\mathrm{in},y}$ is the composite of the symplectic embedding f_y and the symplectic embedding C_y, smooth in (y,z) because C_y depends smoothly on (y,x,θ) through $p^*(y)$. The value and derivative at 0 follow from (a), and the two displayed identities from the moment formula for C_y and (b). Equality on $\theta=0$ is $V_y(0)=0$.

(d) By Lemma 4.2(v), $J_y = j_0$ on $|w|\le\rho_T$, so by Lemma 4.5(v) $f_y(z) = F_y(z/\sqrt{j_0})$ for $|z|\le\sqrt{j_0}\rho_T$, and there $F_y(w) = (a_0u, b_0v)$; hence $f_y(z) = j_0^{-1/2}\operatorname{diag}(a_0,b_0)z = \mathbf{L}_y z$. □

A computation from (3) gives $DC_y(0) = \operatorname{diag}\big(-1/(2\pi r_b(y)),\, -2\pi r_b(y)\big)$, so

$$\mathbf{A}_y = \operatorname{diag}\Big(-\frac{c_x(y)}{2\pi r_b(y)},\, -\frac{2\pi r_b(y)}{c_x(y)}\Big), \qquad \det\mathbf{A}_y = 1, \tag{4}$$

a diagonal matrix with two negative entries. The comparison inequalities that turn the energy identity into a rearrangement estimate depend on the particular potentials and are proved in §5 and §7: one bounds $\mathcal{A}_y(E)$ from below by the area of a rectangle, which bounds $E_y(I)$ from above.

4.5. **Density interpolation and the charts C_λ.** The inner map is an embedding of a disc, not a map of $D(h)$ to itself, and it comes with no path to the identity. Each of its two nonlinear factors has an obvious linearisation at the origin: the Morse chart F_y is deformed to the linear map $\operatorname{diag}(a_0,b_0)$ by interpolating X_y, Y_y linearly with their linear parts, and the chart C_y is deformed to its affine part by the rescaling $w\mapsto b_y + [C_y(\lambda w)-b_y]/\lambda$, $\lambda\to 0$. The chart parameter λ must start at 0, so that the path starts at an affine map.

Definition 4.8 (Density family, rescaled charts and schedule)**.** Let (U,V) be an admissible potential pair over $\mathcal{Y}$.

(a) For $y\in\mathcal{Y}$ and $s\in\mathbb{R}$ put

$$X_{y,s}(u) = (1-s)a_0(y)u + sX_y(u), \quad Y_{y,s}(v) = (1-s)b_0(y)v + sY_y(v), \quad J_{y,s}(u,v) = X'_{y,s}(u)\,Y'_{y,s}(v),$$

$F_{y,s} = (X_{y,s}, Y_{y,s})$, and for $s\in[0,1]$ let $T_{y,s}$, $D_{y,s}$, $\mathcal{A}_{y,s}$, $E_{y,s}$ be the normalisations of $J_{y,s}$ (Definition 4.3 with parameters $(y,s)\in\mathcal{Y}\times[0,1]$), and $f_{y,s} = F_{y,s}\circ T_{y,s}^{-1}\circ D_{y,s}$ on $D(\mathcal{A}_{y,s}(\infty))$.

(b) For (y,λ,w) in the open set $\mathcal{W} = \{(y,\lambda,w)\in\mathcal{Y}\times\mathbb{R}\times\mathbb{R}^2 :\ \lambda w\in\mathcal{C}_y\}$ put

$$C_{y,\lambda}(w) = b_y + \int_0^1 DC_y(t\lambda w)\, w\, dt.$$

(c) For numbers $\lambda_0\in(0,\frac12]$ and $\delta_s\in(0,\frac14]$, the *schedule* is

$$\lambda(s) = \lambda_0\, S(2s) + (1-\lambda_0)\, S\Big(\frac{s-1+\delta_s}{\delta_s}\Big), \qquad s\in[0,1].$$

Since $\mathcal{C}_y$ is convex and contains 0, $\lambda w\in\mathcal{C}_y$ implies $t\lambda w\in\mathcal{C}_y$ for $t\in[0,1]$, so the integrand in (b) is defined and $C_{y,\lambda}$ is C^∞ on $\mathcal{W}$. The schedule satisfies $\lambda(0)=0$, $\lambda(1)=1$, $\lambda(s) = \lambda_0 S(2s)\le\lambda_0$ for $s\le 1-\delta_s$ (the *early range*), $\lambda=\lambda_0$ on $[\frac12, 1-\delta_s]$, and $\lambda(s)\in[\lambda_0,1]$ for $s\ge 1-\delta_s$ (the *late range*), because $\delta_s\le\frac14$; λ is smooth.

Lemma 4.9 (The rescaled charts)**.** *For $(y,\lambda,w)\in\mathcal{W}$: if $\lambda\neq 0$ then $C_{y,\lambda}(w) = b_y + [C_y(\lambda w) - b_y]/\lambda$; $C_{y,0}(w) = b_y + DC_y(0)w$; $DC_{y,\lambda}(w) = DC_y(\lambda w)$, so $\det DC_{y,\lambda} = 1$; and $C_{y,\lambda}$ is injective on $\{w : \lambda w\in\mathcal{C}_y\}$.*

Proof. For $\lambda \neq 0$ the fundamental theorem of calculus gives

$$C_y(\lambda w) - C_y(0) = \int_0^1 DC_y(t\lambda w)\lambda w \, dt.$$

Divide by λ. The quotient formula has w-derivative $DC_y(\lambda w)$, and at $\lambda = 0$ the formula is affine with linear part $DC_y(0)$. For $\lambda \neq 0$, $C_{y,\lambda}$ is $C_y \circ (\lambda\,\cdot)$ followed by an affine bijection, and C_y is injective on $\mathcal{C}_y$; $C_{y,0}$ is affine with $\det DC_y(0) = 1$. □

Equivalently, with $g(\zeta) = (e^\zeta - 1)/\zeta$ ($g(0) = 1$) and $w = (u, v)$,

$$C_{y,\lambda}(u,v) = b_y - 2\pi r_b(y)\, i\, v\, g(2\pi i\lambda v) - \frac{u\, e^{2\pi i\lambda v}}{\pi\big(r_b(y) + \sqrt{(p^*(y) + \lambda u)/\pi}\big)} \qquad \text{on } \mathcal{W}.$$

For $\lambda \neq 0$ this follows from $C_y(\lambda w) - b_y = r_b(1 - e^{2\pi i\lambda v}) + (r_b - \sqrt{(p^* + \lambda u)/\pi})\, e^{2\pi i\lambda v}$ and $r_b - \sqrt{(p^* + \lambda u)/\pi} = -\lambda u/\big(\pi(r_b + \sqrt{(p^* + \lambda u)/\pi})\big)$; for $\lambda = 0$ it is $b_y + DC_y(0)w$.

Lemma 4.10 (The density family and the uniform energy bound)**.** *Let (U, V) be an admissible potential pair over $\mathcal{Y}$, $y \in \mathcal{Y}$, $s \in [0,1]$, and abbreviate $a_0 = a_0(y)$, $b_0 = b_0(y)$, $j_0 = a_0 b_0$.*

(a) $X_{y,s}, Y_{y,s}$ are C^∞ and strictly increasing, and

$$J_{y,s} = (1-s)^2 j_0 + s(1-s)\big(a_0 Y_y' + b_0 X_y'\big) + s^2 J_y \;\geq\; (1-s)^2 j_0 + s^2 J_y > 0, \qquad J_{y,s}(0) = j_0.$$

(b) For $E \geq 0$,

$$\mathcal{A}_{y,s}(E) \;\geq\; \pi j_0 (1-s)^2 E + s^2 \mathcal{A}_y(E) \;\geq\; \mathcal{A}_y^-(E) := \frac{\pi j_0 E\, \mathcal{A}_y(E)}{\pi j_0 E + \mathcal{A}_y(E)} \quad (E > 0).$$

In particular $\mathcal{A}_{y,s}(\infty) = \infty$ for $s < 1$, and $\mathcal{A}_{y,1}(\infty) = \mathcal{A}_y(\infty)$.

(c) (Uniform energy bound.) $\mathcal{A}_y^-$ is continuous and strictly increasing. If $\bar{E} > 0$ and $\mathcal{A}_{y,s}(\bar{E}) \geq p^\dagger$ for every $s \in [0,1]$ (in both families this holds for the explicit $\bar{E}$ of Lemma 5.29(d), respectively Definition 7.3), then $E_{y,s}(I) \leq \bar{E}$ for $0 \leq I \leq p^\dagger$ and $s \in [0,1]$.

(d) $f_{y,s}$ is a symplectic embedding on the source domain $\mathcal{D}_y^{\mathrm{src}}$ of Theorem 4.7; the map $(y, s, z) \mapsto f_{y,s}(z)$ is smooth for $s \in [0,1]$ and $z \in \mathcal{D}_y^{\mathrm{src}}$. In the equal-ball case it is a diffeomorphism onto $\mathbb{R}^2$; $f_{y,s}(0) = 0$ and $Df_{y,s}(0) = \mathbf{L}_y$ for every s; $f_{y,0} = \mathbf{L}_y$ and $f_{y,1} = f_y$. If $(u, v) = T_{y,s}^{-1} D_{y,s}(z)$, then $f_{y,s}(z) = (X_{y,s}(u), Y_{y,s}(v))$ and $u^2 + v^2 = E_{y,s}(\pi|z|^2)$.

(e) If the pair is exactly quadratic near the minimum, then $J_{y,s} = j_0$ on $|w| \leq \rho_T(y)$ and $f_{y,s}(z) = \mathbf{L}_y z$ for $|z| \leq \sqrt{j_0}\, \rho_T(y)$, for every $s \in [0,1]$.

Proof. (a) $X_{y,s}' = (1-s)a_0 + sX_y'$ and $Y_{y,s}' = (1-s)b_0 + sY_y'$ are positive for $s \in [0,1]$ (one of the two terms is positive and the other nonnegative); expanding the product gives the formula, whose middle term is nonnegative. $X_y'(0) = a_0$ and $Y_y'(0) = b_0$ give $J_{y,s}(0) = j_0$.

(b) Integrate (a) over $\{|w|^2 \leq E\}$: the first term gives $\pi j_0 (1-s)^2 E$, the middle term is nonnegative, and the last gives $s^2 \mathcal{A}_y(E)$. For $\alpha, \beta > 0$ the minimum of $\alpha(1-s)^2 + \beta s^2$ over $s \in \mathbb{R}$ is $\alpha\beta/(\alpha + \beta)$. If $s < 1$, the first term tends to ∞ with E. For $s = 1$, $J_{y,1} = J_y$.

(c) For $E > 0$, $1/\mathcal{A}_y^-(E) = 1/(\pi j_0 E) + 1/\mathcal{A}_y(E)$, so positivity, continuity and strict increase of $\mathcal{A}_y$ give those of $\mathcal{A}_y^-$. Since $\mathcal{A}_{y,s}$ is strictly increasing, $\mathcal{A}_{y,s}(\bar{E}) \geq p^\dagger \geq I$ implies $E_{y,s}(I) \leq \bar{E}$. The gate is Lemma 5.29(d) for the equal-ball family and is proved in Lemma 7.4 for the slice family.

(d) The density $(y, s, w) \mapsto J_{y,s}(w)$ is C^∞ on $\mathcal{U} = \mathcal{Y} \times \mathbb{R} \times \mathbb{R}^2$ and positive on $\mathcal{P} \times \mathbb{R}^2$ with $\mathcal{P} = \mathcal{Y} \times [0,1]$, and has the common positive constant germ. In the slice case apply the constant-tail extension on the common energy disc of radius $\sqrt{\bar{E}}$ and use normalization locality on $D(p^\dagger)$; in the equal-ball case setting (N) applies directly. $F_{y,s}$ is C^∞ on $\mathcal{U}$, and for $s \in [0,1]$ it

is a diffeomorphism of $\mathbb{R}^2$ onto $X_{y,s}(\mathbb{R}) \times Y_{y,s}(\mathbb{R})$ with Jacobian $J_{y,s}$; for $s < 1$ both factors are onto $\mathbb{R}$ because the term $(1-s)a_0 u$ (resp. $(1-s)b_0 v$) is unbounded and X_y, Y_y are increasing. Lemma 4.5(iv), restricted by locality to the stated source domain, gives everything except the last three assertions, with $Df_{y,s}(0) = j_0^{-1/2}\,\mathrm{diag}(a_0, b_0) = \mathbf{L}_y$. For $s = 0$ the density is the constant j_0, so $\mathcal{M} = 0$, $T_{y,0} = \mathrm{id}$, $D_{y,0} = j_0^{-1/2}\mathrm{id}$ and $f_{y,0} = \mathbf{L}_y$; for $s = 1$, $F_{y,1} = F_y$ and $f_{y,1} = f_y$. The last assertion is the computation in the proof of Theorem 4.7(b).

(e) For $|w| \le \rho_T$, $X_y'(u) = a_0$ and $Y_y'(v) = b_0$ (Lemma 4.2(v)), hence $X_{y,s}'(u) = a_0$, $Y_{y,s}'(v) = b_0$ and $J_{y,s}(w) = j_0$; conclude as in Theorem 4.7(d). □

By (b) and (c), the energy needed to cover a disc of area $a < \mathcal{A}_y(\infty)$ is bounded uniformly in s, although $J_{y,s}$ is close to neither $J_{y,0}$ nor $J_{y,1}$ for intermediate s. The quantitative input of the isotopy is collected in the following definition. All entries are numbers, independent of y; each condition is required for every $y \in \mathcal{Y}$.

Definition 4.11 (Isotopy data)**.** Let (U, V) be an admissible potential pair over $\mathcal{Y}$ with disc area h, and consider positive numbers

$$p_{\mathrm{on}},\ p_{\mathrm{zero}},\ p^{\dagger},\ p^{*}_{\mathrm{lo}},\ p^{*}_{\mathrm{hi}},\ E_c,\ \Delta P,\ \Delta\theta,\ R_q,\ \delta_x,\ \delta_\theta,\ \delta_s,\ \bar{E},\ \bar{X},\ \bar{Y},\ \lambda_0,\ P_{\min},\ P_{\max},\ B_{\mathrm{lin}},\ B^{*},\ c_{\mathrm{sh}},$$

where the radial range bounds satisfy

$$P_{\min} \le \tfrac{1}{2}\min(p^{*}_{\mathrm{lo}}, \Delta P), \qquad P_{\max} \ge \max(\tfrac{3}{2}p^{*}_{\mathrm{hi}}, h).$$

From these numbers we form the *derived constants*

$$M_C = \frac{1}{2\sqrt{\pi P_{\min}}} + 2\sqrt{\pi P_{\max}},$$

$$R^{\dagger} = \sqrt{p^{\dagger}/\pi}, \quad R_h = \sqrt{h/\pi}, \quad r_b^{+} = \sqrt{p^{*}_{\mathrm{hi}}/\pi},$$

$$R_{\mathrm{supp}} = \max\big(R^{\dagger},\ r_b^{+} + c_{\mathrm{sh}}B_{\mathrm{lin}},\ r_b^{+} + c_{\mathrm{sh}}B^{*},\ R_q\big), \quad P_{\mathrm{supp}} = \pi R_{\mathrm{supp}}^2, \quad p_{\mathrm{bd}} = \tfrac{1}{2}(h + P_{\mathrm{supp}}).$$

The numbers are *isotopy data* for the pair if, for every $y \in \mathcal{Y}$:

(D1) $p_{\mathrm{on}} < p_{\mathrm{zero}} < p^{\dagger}$ and $p^{*}_{\mathrm{lo}} \le p^{*}(y) \le p^{*}_{\mathrm{hi}} < h$;

(D2) (endpoint margins) for all u, v with $u^2 < E_c$, $v^2 < E_c$:

$$\Delta P \le p^{*}(y) + X_y(u) \le h - \Delta P, \qquad p^{*}(y) + X_y(u) \le \pi R_q^2 < h, \qquad |Y_y(v)| \le \tfrac{1}{2} - \Delta\theta;$$

(D3) (interpolation error) $|a_0(y)u - X_y(u)| \le \delta_x$ and $|b_0(y)v - Y_y(v)| \le \delta_\theta$ for $u^2 < E_c$, $v^2 < E_c$;

(D4) (late range) $\delta_s \le \frac{1}{4}$, $\delta_s\delta_x \le \frac{1}{2}\Delta P$, $\delta_s\delta_\theta \le \frac{1}{2}\Delta\theta$ and either $(1-\delta_s)^2\mathcal{A}_y(E_c) > p^{\dagger}$, or the uniform-energy alternative $\mathcal{A}_{y,s}(E_c) > p^{\dagger}$ for all $s \in [0,1]$; the slice construction uses the latter with $E_c = \bar{E}$;

(D5) (uniform energy) $\mathcal{A}_{y,s}(\bar{E}) \ge p^{\dagger}$ for all $s \in [0,1]$, and $\max(a_0(y)|u|, |X_y(u)|) \le \bar{X}$, $\max(b_0(y)|v|, |Y_y(v)|) \le \bar{Y}$ for $u^2 \le \bar{E}$, $v^2 \le \bar{E}$;

(D6) (early range) $\lambda_0 \le \frac{1}{2}$, $\lambda_0\bar{X} \le \frac{1}{2}p^{*}_{\mathrm{lo}}$ and $\lambda_0\bar{Y} \le \frac{1}{4}$;

(D7) (ranges of the path)

$$R^{\dagger}\max\big(1, |(\mathbf{A}_y)_{11}|, |(\mathbf{A}_y)_{22}|\big) \le B_{\mathrm{lin}}, \qquad B^{*} \ge \min\Big(M_C\sqrt{\bar{X}^2 + \bar{Y}^2},\ \frac{\bar{X}}{\sqrt{\pi p^{*}_{\mathrm{lo}}}} + 2\pi r_b^{+}\bar{Y}\Big),$$

$$c_{\mathrm{sh}} \le \min\Big(\frac{1}{4},\ \frac{R_h - r_b^{+}}{8\max(R^{\dagger}, B_{\mathrm{lin}}, B^{*})}\Big).$$

The schedule $\lambda(s)$ of Definition 4.8(c) is formed with these λ_0 and δ_s.

The roles are as follows. The source disc of the inner family is $D(p^\dagger)$; the isotopy follows the inner family on the prescribed core $D(h-4e_q)\subset D(p_{\mathrm{on}})$ and is cut off between the areas p_{on} and p_{zero}; R_q bounds the radius of the endpoint image. E_c is an energy level whose Morse coordinates are mapped by the endpoint strictly inside the chart, with moment margin ΔP and angular margin $\Delta\theta$; δ_s is the length of the late range, short enough that the interpolation error is at most half of each margin and that the whole source disc $D(p^\dagger)$ stays below the level E_c; $\bar E,\bar X,\bar Y$ are crude bounds valid for all s, used only in the early range, where the chart parameter is at most λ_0; $[P_{\min},P_{\max}]$ is the range of radial coordinates met by the segments in Definition 4.8(b) and M_C bounds $\|DC_y\|$ there; B^* bounds the distance of the inner family from the centre b_y, B_{lin} bounds the linear segment, and c_{sh} shrinks all of these to at most $(R_h-r_b^+)/8$; $D(P_{\mathrm{supp}})$ is the disc outside of which nothing moves.

Lemma 4.12 (Existence and order of the data)**.** *(a) The two potential families have the explicit isotopy data given in §5.6 and Definition 7.3, respectively.*

(b) For any isotopy data: $0<\delta_s\le\frac14$, $p^\dagger<\mathcal{A}_y(E_c)<h$, $E_c\le E_{\mathrm{rect}}(y)$ and $p^\dagger<I_{\mathrm{rect}}(y)$ for every y, and

$$R_{\mathrm{supp}}<R_h,\qquad \textit{i.e.}\qquad p_{\mathrm{on}}<p^\dagger\le P_{\mathrm{supp}}<p_{\mathrm{bd}}<h.$$

Proof. (a) In the equal-ball family the positivity and source order are Lemma 5.28, the pointwise and area gates are Lemma 5.29, and the range and shrinking constants are Definition 5.32, assembled in Lemma 5.33. For the slice family the verification is Lemma 7.4.

(b) $\delta_s>0$ is part of the data. $\mathcal{A}_y(E_c)>p^\dagger$ follows from either alternative of (D4), taking $s=1$ in the uniform-energy case. If $|w|^2\le E_c'$ for some $E_c'<E_c$, then $u^2,v^2<E_c$, so by (D2) $F_y(w)$ lies in $[\Delta P-p^*,h-\Delta P-p^*]\times[-\frac12+\Delta\theta,\frac12-\Delta\theta]$; letting $E_c'\uparrow E_c$, $\mathcal{A}_y(E_c)\le(h-2\Delta P)(1-2\Delta\theta)<h$. The same inclusion shows that every point of $\{U_y-U_{\min}+V_y<E_c\}$ lies in $(0,h)\times(-\frac12,\frac12)$, so $E_c\le E_{\mathrm{rect}}(y)$, and then $I_{\mathrm{rect}}(y)\ge\mathcal{A}_y(E_c)>p^\dagger$. Consequently $R^\dagger<R_h$. By (D1), $r_b^+<R_h$, and (D7) gives $r_b^+ + c_{\mathrm{sh}}\max(B_{\mathrm{lin}},B^*)\le r_b^+ + \frac18(R_h-r_b^+)<R_h$. By (D2), $\pi R_q^2<h$. Hence $R_{\mathrm{supp}}<R_h$, $P_{\mathrm{supp}}<h$, and $P_{\mathrm{supp}}\ge\pi(R^\dagger)^2=p^\dagger$. □

Proposition 4.13 (The common inner family)**.** *Let (U,V) be an admissible potential pair over $\mathcal{Y}$ with isotopy data. For $y\in\mathcal{Y}$, $s\in[0,1]$ and $z\in\bar D(p^\dagger)$ write $(u,v)=T_{y,s}^{-1}D_{y,s}(z)$.*

(a) (A priori bounds.) $u^2+v^2=E_{y,s}(\pi|z|^2)\le\bar E$, $|X_{y,s}(u)|\le\bar X$ and $|Y_{y,s}(v)|\le\bar Y$.

(b) (Late margins.) If $s\ge1-\delta_s$, then $u^2+v^2<E_c$ and

$$\tfrac12\Delta P\le p^*(y)+X_{y,s}(u)\le h-\tfrac12\Delta P,\qquad |Y_{y,s}(v)|\le\tfrac12-\tfrac12\Delta\theta.$$

(c) (The family.) For $t\in[0,1]$ the point $t\lambda(s)f_{y,s}(z)$ lies in $\mathcal{C}_y$ and has radial coordinate $p^*(y)+t\lambda(s)X_{y,s}(u)\in[P_{\min},P_{\max}]$. Hence

$$q^{\mathrm{in}}_{y,s}=C_{y,\lambda(s)}\circ f_{y,s}\colon D(p^\dagger)\to\mathbb{C}$$

is defined, $(y,s,z)\mapsto q^{\mathrm{in}}_{y,s}(z)$ is smooth on $\mathcal{Y}\times[0,1]\times D(p^\dagger)$, and each $q^{\mathrm{in}}_{y,s}$ is a symplectic embedding with

$$q^{\mathrm{in}}_{y,0}(z)=b_y+\mathbf{A}_yz,\qquad q^{\mathrm{in}}_{y,1}=q_{\mathrm{in},y}|_{D(p^\dagger)},\qquad q^{\mathrm{in}}_{y,s}(0)=b_y,\qquad Dq^{\mathrm{in}}_{y,s}(0)=\mathbf{A}_y.$$

(d) (Ranges.) $|q^{\mathrm{in}}_{y,s}(z)-b_y|\le B^*$ and $|q^{\mathrm{in}}_{y,1}(z)|\le R_q$ for $z\in D(p^\dagger)$.

Proof. (a) By Lemma 4.10(d), $u^2+v^2=E_{y,s}(\pi|z|^2)$, and $E_{y,s}(\pi|z|^2)\le\bar E$ by (D5) and monotonicity of $\mathcal{A}_{y,s}$. $X_{y,s}(u)$ is a convex combination of a_0u and $X_y(u)$ with $u^2\le\bar E$, so $|X_{y,s}(u)|\le\bar X$ by (D5); likewise for $Y_{y,s}$.

(b) By Lemma 4.10(b) and the first alternative of (D4), or directly by its uniform-energy alternative, $\mathcal{A}_{y,s}(E_c)>p^\dagger\ge\pi|z|^2$, so $E_{y,s}(\pi|z|^2)<E_c$ and $u^2,v^2<E_c$. By (D2), $\Delta P\le$

$p^* + X_y(u) \le h - \Delta P$ and $|Y_y(v)| \le \frac{1}{2} - \Delta\theta$. By (D3) and (D4), $|X_{y,s}(u) - X_y(u)| = (1-s)|a_0 u - X_y(u)| \le \delta_s\delta_x \le \frac{1}{2}\Delta P$ and $|Y_{y,s}(v) - Y_y(v)| \le \delta_s\delta_\theta \le \frac{1}{2}\Delta\theta$.

(c) *Early range* $s \le 1-\delta_s$: $\lambda(s) \le \lambda_0$, so by (a) and (D6), $|t\lambda X_{y,s}(u)| \le \lambda_0\bar{X} \le \frac{1}{2}p^*_{\mathrm{lo}} \le \frac{1}{2}p^*(y)$ and $|t\lambda Y_{y,s}(v)| \le \lambda_0\bar{Y} \le \frac{1}{4}$; the radial coordinate lies in $[\frac{1}{2}p^*, \frac{3}{2}p^*] \subset [P_{\min}, P_{\max}]$. *Late range* $s \ge 1-\delta_s$: $\lambda \in [0,1]$, so $p^* + t\lambda X_{y,s}(u)$ lies between p^* and $p^* + X_{y,s}(u) \in [\frac{1}{2}\Delta P, h - \frac{1}{2}\Delta P]$, hence in $[\min(p^*, \frac{1}{2}\Delta P), \max(p^*, h)] \subset [P_{\min}, P_{\max}]$, and $|t\lambda Y_{y,s}(v)| \le \frac{1}{2} - \frac{1}{2}\Delta\theta$ by (b). In both cases the point lies in $\mathcal{C}_y$, so $(y, \lambda(s), f_{y,s}(z)) \in \mathcal{W}$.

Smoothness follows from Lemma 4.10(d), the smoothness of C on the open set $\mathcal{W}$ (Definition 4.8) and of λ. Symplecticity: $\det Dq^{\mathrm{in}}_{y,s} = \det DC_y(\lambda f_{y,s}) \det Df_{y,s} = 1$ (Lemma 4.9). Injectivity: $f_{y,s}$ is injective, and $C_{y,\lambda}$ is injective on $\{w : \lambda w \in \mathcal{C}_y\}$. At $s = 0$, $\lambda = 0$ and $f_{y,0} = \mathbf{L}_y$, so $q^{\mathrm{in}}_{y,0}(z) = b_y + DC_y(0)\mathbf{L}_y z$. At $s = 1$, $\lambda = 1$, $C_{y,1} = C_y$ and $f_{y,1} = f_y$; since $p^\dagger < I_{\mathrm{rect}}(y)$ (Lemma 4.12(b)), $q^{\mathrm{in}}_{y,1} = q_{\mathrm{in},y}$ on $D(p^\dagger)$. Since $f_{y,s}(0) = 0$, $C_{y,\lambda}(0) = b_y$ and $Df_{y,s}(0) = \mathbf{L}_y$, $q^{\mathrm{in}}_{y,s}(0) = b_y$ and $Dq^{\mathrm{in}}_{y,s}(0) = DC_y(0)\mathbf{L}_y = \mathbf{A}_y$.

(d) Let $w = f_{y,s}(z) = (x,\theta)$ and $\lambda = \lambda(s)$; by (a), $|x| \le \bar{X}$, $|\theta| \le \bar{Y}$. *First bound.* At a point with radial coordinate P the columns of DC_y are orthogonal of lengths $1/(2\sqrt{\pi P})$ and $2\sqrt{\pi P}$, so $\|DC_y\| \le M_C$ for $P \in [P_{\min}, P_{\max}]$; by (c) this applies along the segment in Definition 4.8(b), and $|C_{y,\lambda}(w) - b_y| \le M_C|w| \le M_C\sqrt{\bar{X}^2 + \bar{Y}^2}$. *Second bound.* For $\lambda > 0$ put $P = p^* + \lambda x > 0$ and $\varphi = 2\pi\lambda\theta$; then, with $r_b = r_b(y)$,

$$C_y(\lambda w) - b_y = r_b\big(1 - e^{i\varphi}\big) + \big(r_b - \sqrt{P/\pi}\big)e^{i\varphi}, \qquad \big|r_b - \sqrt{P/\pi}\big| = \frac{|p^* - P|}{\pi(r_b + \sqrt{P/\pi})} \le \frac{\lambda|x|}{\sqrt{\pi p^*}},$$

and $|1 - e^{i\varphi}| \le |\varphi|$. Dividing by λ, $|C_{y,\lambda}(w) - b_y| \le |x|/\sqrt{\pi p^*} + 2\pi r_b|\theta| \le \bar{X}/\sqrt{\pi p^*_{\mathrm{lo}}} + 2\pi r_b^+\bar{Y}$; for $\lambda = 0$, $|DC_y(0)w| \le |x|/(2\pi r_b) + 2\pi r_b|\theta|$ obeys the same bound, since $2\pi r_b = 2\sqrt{\pi p^*} \ge \sqrt{\pi p^*}$. Hence $|q^{\mathrm{in}}_{y,s}(z) - b_y| \le B^*$ by (D7). Finally $\pi|q^{\mathrm{in}}_{y,1}(z)|^2 = p^* + X_y(u) \le \pi R_q^2$ by (D2), because $u^2 < E_c$ by (b). □

4.6. **The five-segment ordinary path.** The images of the inner family stay within B^* of b_y, but B^* may exceed the distance $R_h - r_b$ from b_y to $\partial D(h)$; so $q^{\mathrm{in}}_{y,s}$ itself cannot be the isotopy. We give up area preservation for a moment: shrink by c_{sh}, run the inner family inside the shrunken picture, and expand back. The ambient flow and the area correction of the next subsections apply to any path with the following properties.

Definition 4.14 (Ordinary path)**.** Let $\mathcal{Y} \subset \mathbb{R}^d$ be open and let $p^\dagger > 0$, $0 < P_{\mathrm{supp}} < h$ and $0 < c_{\mathrm{sh}} \le 1$. An *ordinary path* with these constants is a smooth map $\mathfrak{e}\colon \mathcal{Y} \times \mathbb{R} \times D(p^\dagger) \to \mathbb{C}$, $(y,t,z) \mapsto \mathfrak{e}_{y,t}(z)$, together with a smooth function $\mu\colon \mathbb{R} \to (0,\infty)$, such that for all $y \in \mathcal{Y}$ and $t \in \mathbb{R}$:

(O1) $\mathfrak{e}_{y,t} = \mathrm{id}$ for $t \le 0$ and $\mathfrak{e}_{y,t} = \mathfrak{e}_{y,1}$ for $t \ge 1$;
(O2) $\mathfrak{e}_{y,t}$ is injective, $\det D\mathfrak{e}_{y,t}(z) = \mu_t$ for all z, $\mu_t \ge c_{\mathrm{sh}}^2$ and $\mu_1 = 1$;
(O3) $\pi|\mathfrak{e}_{y,t}(z)|^2 \le P_{\mathrm{supp}}$ for all $z \in D(p^\dagger)$.

By (O1), $\mu_t = 1$ for $t \le 0$ and $\mu_t = 1$ for $t \ge 1$, and $\partial_t\mathfrak{e}_{y,t} = 0$ for $t \notin (0,1)$.

Construction 4.15 (The five-segment path)**.** Let (U,V) be an admissible potential pair over $\mathcal{Y}$ with isotopy data, and c_{sh} the constant of the data. For $y \in \mathcal{Y}$ and $\sigma \in [0,1]$ let $\mathbf{M}_{y,\sigma} = \mathrm{Rot}_{\pi\sigma}\,\mathrm{diag}\big(|(\mathbf{A}_y)_{11}|^\sigma, |(\mathbf{A}_y)_{22}|^\sigma\big)$, and for $z \in D(p^\dagger)$ define the five segments

$$\mathfrak{e}^0_{y,\sigma}(z) = c_{\mathrm{sh}}^\sigma z, \qquad \mathfrak{e}^1_{y,\sigma}(z) = \sigma b_y + c_{\mathrm{sh}} z, \qquad \mathfrak{e}^2_{y,\sigma}(z) = b_y + c_{\mathrm{sh}}\,\mathbf{M}_{y,\sigma} z,$$
$$\mathfrak{e}^3_{y,\sigma}(z) = b_y + c_{\mathrm{sh}}\big(q^{\mathrm{in}}_{y,\sigma}(z) - b_y\big), \quad \mathfrak{e}^4_{y,\sigma}(z) = b_y + c_{\mathrm{sh}}^{1-\sigma}\big(q^{\mathrm{in}}_{y,1}(z) - b_y\big).$$

Put $\mathfrak{e}_{y,t} = \mathfrak{e}^{j}_{y,S(5t-j)}$ for $t \in [\frac{j}{5}, \frac{j+1}{5}]$, $j = 0, \dots, 4$, $\mathfrak{e}_{y,t} = \mathrm{id}$ for $t \le 0$ and $\mathfrak{e}_{y,t} = \mathfrak{e}_{y,1}$ for $t \ge 1$, and $\mu_t = c_{\mathrm{sh}}^{2S(5t)-2S(5t-4)}$.

By (4), $\mathbf{A}_y = \mathrm{Rot}_\pi \operatorname{diag}(\alpha_1, \alpha_2)$ with $\alpha_i = |(\mathbf{A}_y)_{ii}|$ and $\alpha_1\alpha_2 = 1$. Hence $\mathbf{M}_{y,0} = \mathrm{Id}$, $\mathbf{M}_{y,1} = \mathbf{A}_y$, $\det \mathbf{M}_{y,\sigma} = (\alpha_1\alpha_2)^\sigma = 1$, and $\|\mathbf{M}_{y,\sigma}\| \le \max(1, \alpha_1, \alpha_2)$.

Lemma 4.16 (The five-segment path is an ordinary path)**.** *The map $\mathfrak{e}$ of Construction 4.15, with μ, is an ordinary path with constants $p^\dagger$, P_{supp}, h, c_{sh}, and $\mathfrak{e}_{y,1} = q^{\mathrm{in}}_{y,1} = q_{\mathrm{in},y}$ on $D(p^\dagger)$.*

Proof. Matching. $\mathfrak{e}^0_{y,0} = \mathrm{id}$, $\mathfrak{e}^0_{y,1} = c_{\mathrm{sh}} z = \mathfrak{e}^1_{y,0}$, $\mathfrak{e}^1_{y,1} = b_y + c_{\mathrm{sh}} z = \mathfrak{e}^2_{y,0}$, $\mathfrak{e}^2_{y,1} = b_y + c_{\mathrm{sh}}\mathbf{A}_y z = b_y + c_{\mathrm{sh}}(q^{\mathrm{in}}_{y,0}(z) - b_y) = \mathfrak{e}^3_{y,0}$ (here $\lambda(0) = 0$ is used), $\mathfrak{e}^3_{y,1} = \mathfrak{e}^4_{y,0}$, and $\mathfrak{e}^4_{y,1} = q^{\mathrm{in}}_{y,1}$.

Smoothness. Each segment is smooth on $\mathcal{Y} \times [0,1] \times D(p^\dagger)$ (Proposition 4.13(c) for $j = 3, 4$). On each interval $[\frac{j}{5}, \frac{j+1}{5}]$ the clock $S(5t-j)$ of Lemma 2.5 takes values in $[0,1]$, and all its derivatives vanish at the endpoints. Hence all derivatives of $\mathfrak{e}$ containing at least one t-derivative vanish at $t \in \{0, \frac{1}{5}, \dots, 1\}$ from both sides, while the purely spatial derivatives agree there because consecutive segments match as maps. So the concatenation is smooth on $\mathcal{Y} \times \mathbb{R} \times D(p^\dagger)$ (Lemma 2.5), and (O1) holds.

Determinants and injectivity. The spatial Jacobian determinants of the segments are $c_{\mathrm{sh}}^{2\sigma}$, c_{sh}^2, $c_{\mathrm{sh}}^2 \det \mathbf{M}_{y,\sigma} = c_{\mathrm{sh}}^2$, $c_{\mathrm{sh}}^2 \det Dq^{\mathrm{in}}_{y,\sigma} = c_{\mathrm{sh}}^2$ and $c_{\mathrm{sh}}^{2(1-\sigma)}$; with $\sigma = S(5t-j)$ they equal μ_t, which lies in $[c_{\mathrm{sh}}^2, 1]$, and $\mu_1 = 1$. Segments $0, 1, 2$ are affine bijections; segments $3, 4$ are the embeddings $q^{\mathrm{in}}_{y,\sigma}$ followed by affine bijections. This is (O2).

Ranges. Let $|z| < R^\dagger$. Then $|\mathfrak{e}^0_{y,\sigma}(z)| < R^\dagger$, $|\mathfrak{e}^1_{y,\sigma}(z)| \le r_b^+ + c_{\mathrm{sh}} R^\dagger \le r_b^+ + c_{\mathrm{sh}} B_{\mathrm{lin}}$, $|\mathfrak{e}^2_{y,\sigma}(z)| \le r_b^+ + c_{\mathrm{sh}} B_{\mathrm{lin}}$ by (D7) and the norm bound for $\mathbf{M}_{y,\sigma}$, and $|\mathfrak{e}^3_{y,\sigma}(z)| \le r_b^+ + c_{\mathrm{sh}} B^*$ by Proposition 4.13(d). Finally $\mathfrak{e}^4_{y,\sigma}(z) = (1-\beta) b_y + \beta q^{\mathrm{in}}_{y,1}(z)$ with $\beta = c_{\mathrm{sh}}^{1-\sigma} \in [c_{\mathrm{sh}}, 1]$ is a convex combination of b_y and $q^{\mathrm{in}}_{y,1}(z)$, so its modulus is at most $\max(r_b^+, R_q)$ (Proposition 4.13(d)). All these are at most R_{supp}, which is (O3). The constants satisfy $P_{\mathrm{supp}} < h$ and $0 < c_{\mathrm{sh}} \le \frac{1}{4}$ by Lemma 4.12(b) and (D7). □

Segments 2 and 3 stay within $(R_h - r_b^+)/8$ of b_y; segment 0 stays in $\{|x| < R^\dagger\}$ and segment 1 in $\{|x| \le r_b^+ + c_{\mathrm{sh}} R^\dagger\}$; the last segment uses only convexity of the disc. Since μ_t does not depend on z and $\mu_0 = \mu_1 = 1$, the area correction of §4.8 acts on the core as the scaling $z \mapsto \mu_t^{-1/2} z$ and is the identity there at $t = 0$ and $t = 1$.

4.7. **The cut-off ambient flow.** An ordinary path consists of embeddings of the source disc only. Push its velocity forward to the moving image, cut it off in the *source* variable between the areas p_{on} and p_{zero}, and extend by zero. Cutting off in the source keeps the field inside the image $\mathfrak{e}_{y,t}(D(p^\dagger))$, where $\mathfrak{e}_{y,t}^{-1}$ is defined. Fix $0 < p_{\mathrm{on}} < p_{\mathrm{zero}} < p^\dagger$ and put $r_{\mathrm{on}} = \sqrt{p_{\mathrm{on}}/\pi}$, $r_{\mathrm{zero}} = \sqrt{p_{\mathrm{zero}}/\pi}$. The radial cutoff $\chi_{r_{\mathrm{on}}, r_{\mathrm{zero}}}$ of Definition 2.6 is $\chi_{r_{\mathrm{on}}, r_{\mathrm{zero}}}(z) = 1 - S\big((\pi|z|^2 - p_{\mathrm{on}})/(p_{\mathrm{zero}} - p_{\mathrm{on}})\big)$; by Lemma 2.7(a) it is C^∞, equals 1 for $\pi|z|^2 \le p_{\mathrm{on}}$ and 0 for $\pi|z|^2 \ge p_{\mathrm{zero}}$.

Lemma 4.17 (The ambient flow)**.** *Let $\mathfrak{e}$ be an ordinary path and $0 < p_{\mathrm{on}} < p_{\mathrm{zero}} < p^\dagger$. Let $\mathcal{O} = \{(y,t,x) : x \in \mathfrak{e}_{y,t}(D(p^\dagger))\}$ and define*

$$W_{y,t}(x) = \chi_{r_{\mathrm{on}}, r_{\mathrm{zero}}}\big(\mathfrak{e}_{y,t}^{-1}(x)\big)\, \big(\partial_t \mathfrak{e}_{y,t}\big)\big(\mathfrak{e}_{y,t}^{-1}(x)\big) \quad \text{on } \mathcal{O}, \qquad W_{y,t}(x) = 0 \quad \text{otherwise.}$$

Let $\phi_{y,t}$ be the flow of the time-dependent field $W_{y,t}$ with $\phi_{y,0} = \mathrm{id}$.

- *(a) W is C^∞ on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$ and vanishes for $t \notin (0,1)$ and for $\pi|x|^2 > P_{\mathrm{supp}}$. Hence $\phi_{y,t}$ is defined for all t, each $\phi_{y,t}$ is a diffeomorphism of $\mathbb{C}$, and $(y,t,x) \mapsto \phi_{y,t}(x)$ is C^∞.*
- *(b) $\phi_{y,t}(z) = \mathfrak{e}_{y,t}(z)$ for $\pi|z|^2 \le p_{\mathrm{on}}$, and $\det D\phi_{y,t}(z) = \mu_t$ there.*

(c) $\phi_{y,t} = \mathrm{id}$ *on* $\{\pi|x|^2 > P_{\mathrm{supp}}\}$*;* $\phi_{y,t}$ *maps* $\bar{D}(P)$ *onto itself for every* $P \geq P_{\mathrm{supp}}$ *and* $D(P)$ *onto itself for every* $P > P_{\mathrm{supp}}$*, in particular* $D(h)$ *onto* $D(h)$*; and* $\det D\phi_{y,t} > 0$ *everywhere. For* $t \leq 0$*,* $\phi_{y,t} = \mathrm{id}$*; for* $t \geq 1$*,* $\phi_{y,t} = \phi_{y,1}$*.*

Proof. (a) The map $\hat{\mathfrak{e}}(y,t,z) = (y,t,\mathfrak{e}_{y,t}(z))$ is smooth and injective on the open set $\mathcal{Y}\times\mathbb{R}\times D(p^\dagger)$, with Jacobian determinant $\mu_t > 0$; by the inverse function theorem it is a diffeomorphism onto the open set $\mathcal{O}$. Hence $(y,t,x) \mapsto \mathfrak{e}_{y,t}^{-1}(x)$ is smooth on $\mathcal{O}$, and so is W. W vanishes on $\mathcal{O}$ outside $K = \hat{\mathfrak{e}}(\mathcal{Y}\times[0,1]\times\bar{D}(p_{\mathrm{zero}}))$, because $\chi_{r_{\mathrm{on}},r_{\mathrm{zero}}}(z) = 0$ for $\pi|z|^2 \geq p_{\mathrm{zero}}$ and $\partial_t\mathfrak{e} = 0$ for $t \notin (0,1)$. The set K is closed in $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$: if $(y_n,t_n,\mathfrak{e}_{y_n,t_n}(z_n)) \to (y,t,x)$ with $y \in \mathcal{Y}$, $t_n \in [0,1]$, $z_n \in \bar{D}(p_{\mathrm{zero}})$, pass to a subsequence with $z_n \to z \in \bar{D}(p_{\mathrm{zero}})$; continuity gives $x = \mathfrak{e}_{y,t}(z)$. And $K \subset \mathcal{O}$ because $p_{\mathrm{zero}} < p^\dagger$. The open sets $\mathcal{O}$ and the complement of K cover $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$, and W is smooth on both, so W is smooth. By (O3), $W_{y,t}(x) = 0$ for $\pi|x|^2 > P_{\mathrm{supp}}$. A smooth time-dependent vector field depending smoothly on parameters and vanishing outside a fixed bounded set of x has a global flow by diffeomorphisms, smooth in (y,t,x).

(b) For $\pi|z|^2 \leq p_{\mathrm{on}}$, the curve $t \mapsto \mathfrak{e}_{y,t}(z)$ satisfies $\frac{d}{dt}\mathfrak{e}_{y,t}(z) = \chi_{r_{\mathrm{on}},r_{\mathrm{zero}}}(z)\,\partial_t\mathfrak{e}_{y,t}(z) = W_{y,t}(\mathfrak{e}_{y,t}(z))$ and $\mathfrak{e}_{y,0}(z) = z$; by uniqueness of solutions it is $t \mapsto \phi_{y,t}(z)$. On the open disc $D(p_{\mathrm{on}})$ the two maps agree, so their derivatives agree and $\det D\phi_{y,t} = \mu_t$; on its boundary circle this follows by continuity of $\det D\phi_{y,t}$.

(c) Where $W_{y,t}$ vanishes for all t the constant curve is a solution, so $\phi_{y,t} = \mathrm{id}$ on $\{\pi|x|^2 > P_{\mathrm{supp}}\}$. Since $\phi_{y,t}$ is a bijection of $\mathbb{C}$ fixing every point outside $\bar{D}(P_{\mathrm{supp}})$, it maps $\bar{D}(P_{\mathrm{supp}})$ onto itself, hence $\bar{D}(P)$ and $D(P)$ onto themselves for $P \geq P_{\mathrm{supp}}$, respectively $P > P_{\mathrm{supp}}$; $h > P_{\mathrm{supp}}$. The determinant $\det D\phi_{y,t}$ never vanishes, is continuous in t and equals 1 at $t = 0$. Since $W_{y,t} = 0$ for $t \notin (0,1)$, the flow is constant in t on $(-\infty,0]$ and on $[1,\infty)$. □

4.8. **Area normalisation of the ambient flow.** The flow $\phi_{y,t}$ has the right support and the right endpoint on the core, but it does not preserve area. We correct it with the normalisations of §4.3, applied to $G_{y,t}(w) = \phi_{y,t}(w/\sqrt{\mu_t})$, whose density is 1 near the origin for every t; this lets Lemma 4.5(v) apply with one radius for the whole family.

Lemma 4.18 (Normalisation of the whole path)**.** *Let* $\mathfrak{e}$ *be an ordinary path,* $0 < p_{\mathrm{on}} < p_{\mathrm{zero}} < p^\dagger$*, and* ϕ *the ambient flow of Lemma 4.17. For* $(y,t) \in \mathcal{Y}\times\mathbb{R}$ *put*

$$G_{y,t}(w) = \phi_{y,t}\big(w/\sqrt{\mu_t}\big), \qquad \hat{J}_{y,t}(w) = \det DG_{y,t}(w) = \frac{\det D\phi_{y,t}(w/\sqrt{\mu_t})}{\mu_t},$$

let $\hat{T}_{y,t}$*,* $\hat{D}_{y,t}$ *be the rotation and area normalisations of* $\hat{J}_{y,t}$ *(parameters* (y,t)*), and define*

$$\mathcal{N}_{y,t} = \mu_t^{-1/2}\,\hat{T}_{y,t}^{-1} \circ \hat{D}_{y,t}, \qquad \hat{q}_{y,t} = \phi_{y,t} \circ \mathcal{N}_{y,t} = G_{y,t} \circ \hat{T}_{y,t}^{-1} \circ \hat{D}_{y,t}.$$

(a) $\hat{J}$ *is* C^∞ *and positive on* $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$*;* $\hat{J}_{y,t}(w) = 1$ *for* $\pi|w|^2 \leq \mu_t p_{\mathrm{on}}$*, in particular for* $|w| \leq \rho_c = c_{\mathrm{sh}}\sqrt{p_{\mathrm{on}}/\pi}$*;* $\hat{J}_{y,t}(w) = 1/\mu_t$ *for* $\pi|w|^2 > \mu_t P_{\mathrm{supp}}$*; and* $\mathcal{A}_{\hat{J}_{y,t}}(\mu_t P/\pi) = \operatorname{area}\phi_{y,t}(\bar{D}(P))$*, which equals* $\mu_t P$ *for* $P \leq p_{\mathrm{on}}$ *and* P *for* $P \geq P_{\mathrm{supp}}$*. In particular* $\mathcal{A}_{\hat{J}_{y,t}}(\infty) = \infty$*.*

(b) Each $\hat{q}_{y,t}$ *is a symplectic diffeomorphism of* $\mathbb{C}$*; it maps* $\bar{D}(P)$ *onto itself for every* $P \geq P_{\mathrm{supp}}$ *and* $D(P)$ *onto itself for every* $P > P_{\mathrm{supp}}$*, in particular* $D(h)$ *and* $\bar{D}(h)$*.*

(c) (Core.) $\hat{q}_{y,t}(z) = \mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ *for* $\pi|z|^2 \leq \mu_t p_{\mathrm{on}}$*, in particular on the common core* $\pi|z|^2 \leq c_{\mathrm{sh}}^2 p_{\mathrm{on}}$*.*

(d) (Collar.) $\hat{q}_{y,t}(z) = z$ *for* $\pi|z|^2 > P_{\mathrm{supp}}$*.*

(e) (Smoothness.) $(y,t,z) \mapsto \hat{q}_{y,t}(z)$ *and* $(y,t,x) \mapsto \hat{q}_{y,t}^{-1}(x)$ *are* C^∞ *on* $\mathcal{Y}\times\mathbb{R}\times\mathbb{C}$*, including* $z = 0$*;* $\hat{q}_{y,t} = \mathrm{id}$ *for* $t \leq 0$ *and* $\hat{q}_{y,t} = \hat{q}_{y,1}$ *for* $t \geq 1$*.*

(f) (Endpoint.) $\hat{q}_{y,1} = \mathfrak{e}_{y,1}$ *on* $\bar{D}(p_{\mathrm{on}})$*.*

Proof. (a) $\det D\phi_{y,t} > 0$ is smooth in (y,t,x) (Lemma 4.17), and $\mu_t \geq c_{\mathrm{sh}}^2 > 0$ is smooth; the formula for $\hat{J}$ is the chain rule. If $\pi|w|^2 \leq \mu_t p_{\mathrm{on}}$, then $x = w/\sqrt{\mu_t}$ has $\pi|x|^2 \leq p_{\mathrm{on}}$, where $\det D\phi_{y,t} = \mu_t$ (Lemma 4.17(b)); so $\hat{J}_{y,t}(w) = 1$, and $\mu_t \geq c_{\mathrm{sh}}^2$ gives the uniform radius ρ_c. If $\pi|w|^2 > \mu_t P_{\mathrm{supp}}$, then $\phi_{y,t} = \mathrm{id}$ near $w/\sqrt{\mu_t}$, so $\hat{J}_{y,t}(w) = 1/\mu_t$. Since $G_{y,t}$ is a diffeomorphism with Jacobian $\hat{J}_{y,t}$, the change of variables formula gives $\mathcal{A}_{\hat{J}_{y,t}}(\mu_t P/\pi) = \operatorname{area} G_{y,t}(\{\pi|w|^2 \leq \mu_t P\}) = \operatorname{area} \phi_{y,t}(\bar{D}(P))$. For $P \leq p_{\mathrm{on}}$, $\phi_{y,t} = \mathfrak{e}_{y,t}$ on $\bar{D}(P)$ with constant Jacobian μ_t, so the area is $\mu_t P$; for $P \geq P_{\mathrm{supp}}$, $\phi_{y,t}$ maps $\bar{D}(P)$ onto itself (Lemma 4.17(c)), so the area is P.

Setting (N) applies with $\mathcal{P} = \mathcal{Y} \times \mathbb{R}$ (open), $\mathcal{U} = \mathcal{P} \times \mathbb{R}^2$ and $J = \hat{J}$, which equals 1 on $|w| \leq \rho_c$ for all parameters. By Lemmas 4.4 and 4.5, $\hat{T}$, $\hat{T}^{-1}$ and $\hat{D}$ are smooth in all variables, $\hat{T}_{y,t} = \hat{T}_{y,t}^{-1} = \mathrm{id}$ on $|w| \leq \rho_c$, $\hat{D}_{y,t}$ is a diffeomorphism of $\mathbb{C}$, and $\hat{q}_{y,t} = G_{y,t} \circ \hat{T}_{y,t}^{-1} \circ \hat{D}_{y,t}$ is a symplectic diffeomorphism of $\mathbb{C}$ onto $G_{y,t}(\mathbb{C}) = \mathbb{C}$ (Lemma 4.5(iv) with $F = G_{y,t}$).

(d) Let $I = \pi|z|^2 > P_{\mathrm{supp}}$. By (a), $\mathcal{A}_{\hat{J}_{y,t}}(\mu_t I/\pi) = I$, so $|\hat{D}_{y,t}(z)|^2 = \mu_t I/\pi$ and, $\hat{D}$ being radial, $\hat{D}_{y,t}(z) = \sqrt{\mu_t}\, z$. On the circle through $w = \sqrt{\mu_t} z$ the density $\hat{J}_{y,t}$ is the constant $1/\mu_t$, so $\mathcal{M}_{\hat{J}_{y,t}}(w) = 0$ and $\hat{T}_{y,t}^{-1}(w) = w$. Hence $\mathcal{N}_{y,t}(z) = z$ and $\hat{q}_{y,t}(z) = \phi_{y,t}(z) = z$.

(b) $\hat{q}_{y,t}$ is a bijection of $\mathbb{C}$ fixing every point outside $\bar{D}(P_{\mathrm{supp}})$ by (d), so it maps $\bar{D}(P_{\mathrm{supp}})$, hence $\bar{D}(P)$ for $P \geq P_{\mathrm{supp}}$ and $D(P)$ for $P > P_{\mathrm{supp}}$, onto itself.

(c) Let $\rho = \sqrt{\mu_t p_{\mathrm{on}}/\pi}$. By (a), $\hat{J}_{y,t} = 1$ on $|w| \leq \rho$, so Lemma 4.5(v) gives $\hat{q}_{y,t}(z) = G_{y,t}(z) = \phi_{y,t}(z/\sqrt{\mu_t})$ for $|z| \leq \rho$; since $\pi|z/\sqrt{\mu_t}|^2 \leq p_{\mathrm{on}}$, this equals $\mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ by Lemma 4.17(b).

(e) Joint smoothness of $\hat{q}$ was shown in the proof of (a) (all factors are smooth in $(y,t,\cdot)$). For $t \leq 0$, $\phi_{y,t} = \mathrm{id}$ and $\mu_t = 1$, so $\hat{J} = 1$, $\hat{T} = \hat{D} = \mathrm{id}$ and $\hat{q}_{y,t} = \mathrm{id}$; for $t \geq 1$, $\phi_{y,t} = \phi_{y,1}$ and $\mu_t = 1$, so $\hat{q}_{y,t} = \hat{q}_{y,1}$. The map $(y,t,z) \mapsto (y,t,\hat{q}_{y,t}(z))$ is a smooth bijection of $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$ with Jacobian determinant $\det D\hat{q}_{y,t} = 1$; by the inverse function theorem its inverse, $(y,t,x) \mapsto (y,t,\hat{q}_{y,t}^{-1}(x))$, is smooth.

(f) is (c) for $t = 1$, where $\mu_1 = 1$. □

4.9. **The Hamiltonian.** A symplectic isotopy of the plane is Hamiltonian: its velocity field ξ satisfies $d(\iota_\xi \omega_{\mathbb{C}}) = 0$, and a closed one-form on the plane is exact. The primitive is written down by the radial (Poincaré) formula in the source variable, and the additive constant is fixed by evaluation at a point of the collar, where the isotopy is the identity and the primitive is constant.

Construction 4.19 (The Hamiltonian)**.** For an ordinary path with normalised isotopy $\hat{q}$, $y \in \mathcal{Y}$, $t \in \mathbb{R}$ and $z, w \in \mathbb{C}$, put

$$\mathfrak{a}_{y,t}(z)[w] = \omega_{\mathbb{C}}\big(\partial_t \hat{q}_{y,t}(z),\, D\hat{q}_{y,t}(z)w\big), \qquad \mathfrak{h}_{y,t}(z) = \int_0^1 \mathfrak{a}_{y,t}(rz)[z]\, dr,$$

$$z_{\mathrm{bd}} = \big(\sqrt{p_{\mathrm{bd}}/\pi}, 0\big), \quad p_{\mathrm{bd}} = \tfrac{1}{2}(h + P_{\mathrm{supp}}), \qquad Q_{y,t}(x) = -\big[\mathfrak{h}_{y,t} - \mathfrak{h}_{y,t}(z_{\mathrm{bd}})\big]\big(\hat{q}_{y,t}^{-1}(x)\big).$$

Proposition 4.20 (Q generates $\hat{q}$)**.** *Q is C^∞ on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$; $Q_{y,t}$ and all its derivatives vanish on $\{\pi|x|^2 > P_{\mathrm{supp}}\}$ for every (y,t), and $Q_{y,t} = 0$ for $t \notin (0,1)$. With $\xi_{y,t} = (\partial_t \hat{q}_{y,t}) \circ \hat{q}_{y,t}^{-1}$,*

$$\iota_{\xi_{y,t}} \omega_{\mathbb{C}} = -dQ_{y,t}, \qquad \partial_t \hat{q}_{y,t} = \xi_{y,t} \circ \hat{q}_{y,t},$$

so $\hat{q}_{y,\cdot}$ is the isotopy generated by $Q_{y,\cdot}$ in the convention $\iota_X \omega_{\mathbb{C}} = -dQ$.

Proof. By definition $\mathfrak{a}_{y,t} = \hat{q}_{y,t}^*(\iota_{\xi_{y,t}} \omega_{\mathbb{C}})$. Differentiating $\hat{q}_{y,t}^* \omega_{\mathbb{C}} = \omega_{\mathbb{C}}$ in t gives $\hat{q}_{y,t}^* \mathcal{L}_{\xi_{y,t}} \omega_{\mathbb{C}} = 0$, so $d\iota_{\xi_{y,t}} \omega_{\mathbb{C}} = \mathcal{L}_{\xi_{y,t}} \omega_{\mathbb{C}} = 0$ and $\mathfrak{a}_{y,t}$ is closed. The plane is star-shaped about 0, so the radial primitive satisfies $d\mathfrak{h}_{y,t} = \mathfrak{a}_{y,t}$; it is smooth in (y,t,z) because the integrand is. Then $-dQ_{y,t} = (\hat{q}_{y,t}^{-1})^* d\mathfrak{h}_{y,t} = \iota_{\xi_{y,t}} \omega_{\mathbb{C}}$. On the connected set $\{\pi|z|^2 > P_{\mathrm{supp}}\}$, $\hat{q}_{y,t} = \mathrm{id}$ for all t (Lemma 4.18(d)), so $\partial_t \hat{q}_{y,t} = 0$ and $\mathfrak{a}_{y,t} = 0$ there; hence $\mathfrak{h}_{y,t}$ is constant on this set, equal to $\mathfrak{h}_{y,t}(z_{\mathrm{bd}})$ because $P_{\mathrm{supp}} < p_{\mathrm{bd}}$. Therefore $Q_{y,t}$ vanishes identically on $\hat{q}_{y,t}(\{\pi|z|^2 > P_{\mathrm{supp}}\}) = \{\pi|x|^2 > P_{\mathrm{supp}}\}$,

with all derivatives. For $t \notin (0,1)$, $\partial_t \hat{q}_{y,t} = 0$, so $\mathfrak{a}_{y,t} = 0$ and $Q_{y,t} = 0$. Smoothness of Q follows from that of $\mathfrak{h}$ and of $(y,t,x) \mapsto \hat{q}_{y,t}^{-1}(x)$ (Lemma 4.18(e)). □

The radial primitive gives $\hat{q}_{y,t}^{*}(\iota_{\xi_{y,t}}\omega_{\mathbb{C}}) = +d\mathfrak{h}_{y,t}$ in the source. The sign of Q matches the fibre convention $\iota_X\omega_{\mathbb{C}} = -dQ$ of §M3, which is the convention used for the suspension in §M4.

4.10. **The whole-disc isotopy.**

Theorem 4.21 (Whole-disc Hamiltonian isotopy with prescribed endpoint on the core)**.** [Lean: needed consequence verified.]

Let (U,V) be an admissible potential pair over $\mathcal{Y}$ with disc area h, and let isotopy data be given (Definition 4.11). Let $\hat{q}$ and Q be obtained from the five-segment path of Construction 4.15 by Lemma 4.18, with the levels $p_{\mathrm{on}} < p_{\mathrm{zero}}$ of the data, and Construction 4.19. Then $P_{\mathrm{supp}} < h$, and for every $y \in \mathcal{Y}$:

- *(a) each $\hat{q}_{y,t}$ is a symplectic diffeomorphism of $\mathbb{C}$ which is the identity on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ and maps $\bar{D}(P)$ onto itself for every $P \geq P_{\mathrm{supp}}$, in particular $\bar{D}(h)$ and $D(h)$;*
- *(b) $\hat{q}_{y,t} = \mathrm{id}$ for $t \leq 0$ and $\hat{q}_{y,t} = \hat{q}_{y,1}$ for $t \geq 1$;*
- *(c) (same endpoint on the core) $\hat{q}_{y,1}(z) = q_{\mathrm{in},y}(z)$ for $\pi|z|^2 \leq p_{\mathrm{on}}$;*
- *(d) $Q_{y,t}$ vanishes with all derivatives on $\{\pi|x|^2 > P_{\mathrm{supp}}\}$ and for $t \notin (0,1)$, and generates $\hat{q}_{y,t}$: $\partial_t \hat{q}_{y,t} = \xi_{y,t} \circ \hat{q}_{y,t}$ with $\iota_{\xi_{y,t}}\omega_{\mathbb{C}} = -dQ_{y,t}$;*
- *(e) (core formula) $\hat{q}_{y,t}(z) = \mathfrak{e}_{y,t}(z/\sqrt{\mu_t})$ for $\pi|z|^2 \leq \mu_t p_{\mathrm{on}}$.*

Proof. By Lemma 4.16 the five-segment path is an ordinary path with constants $p^{\dagger}$, P_{supp}, h, c_{sh}; $p_{\mathrm{on}} < p_{\mathrm{zero}} < p^{\dagger}$ by (D1), and $P_{\mathrm{supp}} < h$ by Lemma 4.12(b). Items (a), (b) and (e) are Lemma 4.18(b)–(e); (d) is Proposition 4.20. For (c), Lemma 4.18(f) gives $\hat{q}_{y,1} = \mathfrak{e}_{y,1}$ on $\bar{D}(p_{\mathrm{on}})$, and $\mathfrak{e}_{y,1} = q_{y,1}^{\mathrm{in}} = q_{\mathrm{in},y}$ on $D(p^{\dagger}) \supset \bar{D}(p_{\mathrm{on}})$ by Lemma 4.16. □

On the annulus $\{p_{\mathrm{on}} < \pi|z|^2 \leq h\}$ we use only $\hat{q}_{y,1}(\bar{D}(h)) = \bar{D}(h)$:

Corollary 4.22 (Comparison on the whole closed disc)**.** [Lean: needed consequence verified.]

In the situation of Theorem 4.21, let $y \in \mathcal{Y}$, let $\Upsilon\colon [0,h] \to \mathbb{R}$ be any function, $g\colon [0,h] \to \mathbb{R}$ nondecreasing and $\varepsilon \geq 0$, and suppose that $\Upsilon(\pi|q_{\mathrm{in},y}(z)|^2) \leq g(\pi|z|^2) + \varepsilon$ whenever $\pi|z|^2 \leq p_{\mathrm{on}}$, and that $\sup_{[0,h]} \Upsilon \leq g(p_{\mathrm{on}}) + \varepsilon$. Then

$$\Upsilon\big(\pi|\hat{q}_{y,1}(z)|^2\big) \leq g(\pi|z|^2) + \varepsilon \qquad \text{for all } z \in \bar{D}(h).$$

Proof. For $\pi|z|^2 \leq p_{\mathrm{on}}$ use Theorem 4.21(c). For $p_{\mathrm{on}} < \pi|z|^2 \leq h$, $\hat{q}_{y,1}(z) \in \bar{D}(h)$ by Theorem 4.21(a), so $\Upsilon(\pi|\hat{q}_{y,1}(z)|^2) \leq \sup \Upsilon \leq g(p_{\mathrm{on}}) + \varepsilon \leq g(\pi|z|^2) + \varepsilon$. □

All constructions above were made with y as a variable, and the data are numbers independent of y. The following theorem records the consequences used for families.

Theorem 4.23 (Parametrised version)**.** *In the situation of Theorem 4.21:*

- *(a) the maps $(y,t,z) \mapsto \hat{q}_{y,t}(z)$, $(y,t,x) \mapsto \hat{q}_{y,t}^{-1}(x)$ and $(y,t,x) \mapsto Q_{y,t}(x)$ are C^{∞} on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$, and $(y,s,z) \mapsto q_{y,s}^{\mathrm{in}}(z)$ is smooth on $\mathcal{Y} \times [0,1] \times D(p^{\dagger})$; the numbers p_{on}, P_{supp}, c_{sh}, and hence the collar $\{\pi|z|^2 > P_{\mathrm{supp}}\}$, the core $\bar{D}(p_{\mathrm{on}})$ and the common core $\bar{D}(c_{\mathrm{sh}}^2 p_{\mathrm{on}})$, do not depend on y;*
- *(b) if $\mathcal{Y}_0 \subset \mathcal{Y}$ is compact, then the derivatives of order ≥ 1 of $\hat{q}$ and $\hat{q}^{-1}$, and all derivatives (including order 0) of $\hat{q} - \mathrm{id}$, $\hat{q}^{-1} - \mathrm{id}$ and Q, are bounded on $\mathcal{Y}_0 \times \mathbb{R} \times \mathbb{C}$.*

Proof. (a) is Lemma 4.18(e), Proposition 4.20 and Proposition 4.13(c). (b) The maps are smooth on $\mathcal{Y} \times \mathbb{R} \times \mathbb{C}$ and constant in t outside $[0,1]$, and $\hat{q}_{y,t} - \mathrm{id}$, $\hat{q}_{y,t}^{-1} - \mathrm{id}$ and $Q_{y,t}$ vanish outside $\bar{D}(P_{\mathrm{supp}})$. Hence each of the listed derivatives is a continuous function on $\mathcal{Y}_0 \times \mathbb{R} \times \mathbb{C}$ which,

outside the compact set $\mathcal{Y}_0 \times [0,1] \times \bar{D}(P_{\mathrm{supp}})$, coincides with its value at $t = 0$ or $t = 1$, or equals 0, or (for first derivatives of $\hat{q}$ and $\hat{q}^{-1}$) the identity matrix; so it is bounded. □

Remark 4.24 (Explicitness). Every object of this section is a defined term in the sense of Definition M1.3: finite expressions in the potentials and their derivatives, integrals over compact domains ($\bar{J}$, $\mathcal{M}$, $\mathcal{G}$, the charts $C_{y,\lambda}$, the radial primitive), inverses of explicit bijections with nonvanishing derivative (X_y, Y_y, $\mathcal{A}_{y,s}$, Ψ_σ, T_σ, $(y,t,z) \mapsto (y,t,\mathfrak{e}_{y,t}(z))$, $\hat{q}_{y,t}$), and the flow of an explicit compactly supported vector field. The base point z_{bd}, the schedule $\lambda(s)$, the clocks $S(5t-j)$ of Lemma 2.5 and the time intervals $[\frac{j}{5}, \frac{j+1}{5}]$ are fixed. The isotopy data are the numbers of §5 and §7, given by finite expressions; square roots and compact extrema occur in both constructions.

Theorem 4.21, Corollary 4.22 and Theorem 4.23 supply exactly the three properties of the fibre isotopy used in §M4.

5. The explicit inner map for equal balls

We construct the fibre isotopy (q_t) used in Proposition M3.8: an explicit compactly supported Hamiltonian isotopy of the disc $\bar{D}(h)$ whose endpoint q satisfies $\mathcal{R}_{\mathrm{tent}}(\pi|q(z)|^2) \le \pi|z|^2 + 4e_q$ on $\bar{D}(h)$ (Theorem 5.34). The isotopy is the output of §4 for one potential pair $(\tilde{R}, \tilde{V})$: $\tilde{R}$ is a hyperbolic smoothing of the tent $\mathcal{R}_{\mathrm{tent}}$, and $\tilde{V}$ is a quadratic angular potential with a steep step at $\theta = \pm\frac{1}{2}$. Both potentials are made exactly quadratic near their minima. Consequently the Morse chart, its density and both normalisations of §4 are linear near the origin, and the inner map equals the linear symplectic map $\mathbf{L} = \mathrm{diag}(c_x, c_y)$ on an explicit disc (Proposition 5.18); this gives smoothness at the origin with no further argument. The error $4e_q$ has two sources: on the core, the smoothing of the tent and the mean of the angular potential (Proposition 5.24); outside it, an annulus of area $4e_q$ on which only $q(\bar{D}(h)) = \bar{D}(h)$ is used.

5.1. **Setting and conventions.** *Parameters.* Fix an integer $k \ge 10$ (so $\sqrt{k} > 3$) and $t_0 \in (\frac{7}{8}, 1)$ (in the equal-capacity branch $t_0 = \frac{1}{2}(1 + \rho\sqrt{k})$, Remark 13.25), and let

$$a = \frac{t_0}{\sqrt{k}}, \qquad h = \frac{t_0}{3}, \qquad A = \frac{h}{a}, \qquad B = \frac{h}{h-a}$$

be as in §M3.4. By §M3.4 and Lemma 3.9, $0 < a < h$, $A, B > 0$,

$$\frac{1}{A} + \frac{1}{B} = 1 \quad (\text{equivalently } A + B = AB), \qquad \frac{h}{A} = a, \qquad \frac{h}{B} = h - a.$$

The tent is $\mathcal{R}_{\mathrm{tent}}(x) = A(a - x)$ for $x \le a$ and $\mathcal{R}_{\mathrm{tent}}(x) = B(x - a)$ for $x \ge a$ ($x \in \mathbb{R}$); it is the function Λ of §M3.4, convex, with $\mathcal{R}_{\mathrm{tent}}(0) = \mathcal{R}_{\mathrm{tent}}(h) = h$ and $\mathcal{R}_{\mathrm{tent}}(a) = 0$. By Lemma M3.6, for every level $\beta \ge 0$,

$$\{x \in \mathbb{R} : \mathcal{R}_{\mathrm{tent}}(x) \le \beta\} = \Big[a - \frac{\beta}{A},\ a + \frac{\beta}{B}\Big], \tag{5}$$

an interval of length β, contained in $[0,h]$ if $\beta \le h$; in particular $\{\mathcal{R}_{\mathrm{tent}} < h\} = (0,h)$. Throughout, $e = e_q$ is the number of Definition 3.10. The estimates below use only $0 < a < h$, $A, B > 0$, Lemma 3.9 and $0 < e \le h/100$.

The fibre. The fibre is the plane $\mathbb{C}$ with the area form ω_2 and moment $p(\zeta) = \pi|\zeta|^2$. $D(P_0) = \{\pi|\zeta|^2 < P_0\}$ is the open disc of *area* P_0 and $\bar{D}(P_0)$ its closure. A map between open subsets of the plane is symplectic if its Jacobian determinant is identically 1. As in §4, three planes occur: the *source* plane with coordinate z; the *Morse plane* with coordinate $w = (u,v)$, in which the potentials become $u^2 + v^2$; and the *rectangle coordinates* (x,θ), where x is a shifted moment and θ an angle. In the notation of §4 the parameter domain is a point ($d = 0$), $I_U = I_V = \mathbb{R}$,

$U(p) = \tilde{R}(p-p^*)$ and $V = \tilde{V}$ (Construction 5.9), and we work in the shifted moment $x = p - p^*$. Fibre Hamiltonians Q use $\iota_\Xi \omega_2 = -dQ$. S is the flat step of Definition 2.3; by Lemma 2.4 it is C^∞ and nondecreasing, $S = 0$ on $(-\infty, 0]$, $S = 1$ on $[1, \infty)$, all its derivatives vanish at 0 and at 1, and $S(x) + S(1-x) = 1$, so $S(\frac{1}{2}) = \frac{1}{2}$. Rot_ϕ is the rotation by the angle ϕ.

5.2. **Constants and the hyperbola.** On $\mathbb{R}$ we have $\mathcal{R}_{\mathrm{tent}}(p) = m_-(p-a) + m_+|p-a|$ with the two constants $m_\pm$ below (Lemma 5.7). The corner at $p = a$ is smoothed by a hyperbola of width η_{w}, chosen so that the smoothing error is $\eta_{\mathrm{w}} m_+ = e/8$. The shift x^* moves the minimum of the smoothed function to $x = 0$, and p^* is the minimum point in the moment variable. α_0 and e are the second derivatives at 0 of the two potentials.

Definition 5.1 (Constants)**.**

$$m_- = \frac{B-A}{2}, \quad m_+ = \frac{A+B}{2}, \quad r = \sqrt{AB},$$
$$\eta_{\mathrm{w}} = \frac{e}{8m_+}, \quad x^* = -\frac{m_-\eta_{\mathrm{w}}}{r}, \quad p^* = a + x^*, \quad \delta = \frac{e}{8}.$$

Definition 5.2 (Curvatures and linear constants)**.**

$$\alpha_0 = \frac{r^3}{\eta_{\mathrm{w}} m_+^2}, \qquad a_0 = \sqrt{2/\alpha_0}, \qquad b_0 = \sqrt{2/e}, \qquad j_0 = a_0 b_0,$$
$$c_x = \sqrt{a_0/b_0}, \qquad c_y = 1/c_x, \qquad \mathbf{L} = \mathrm{diag}(c_x, c_y).$$

Lemma 5.3 (Relations between the constants)**.** *$r^2 = 2m_+ = m_+^2 - m_-^2$, hence $0 < r \le m_+$ and $|m_-| < m_+$; moreover $\alpha_0 = 8r^3/(em_+) = 16r/e$, $\eta_{\mathrm{w}} m_+ = \delta$, $a_0 = \sqrt{e/(8r)}$, $j_0 = 1/(2\sqrt{r})$ and $c_x^4 = e^2/(16r)$.*

Proof. By Lemma 3.9, $r^2 = AB = A + B = 2m_+$, and $m_+^2 - m_-^2 = AB = r^2$. As $A, B > 0$, $|m_-| < m_+$ and $r \le m_+$. Then $\alpha_0 = r^3 \cdot 8m_+/(em_+^2) = 8r^3/(em_+) = 8r \cdot r^2/(em_+) = 16r/e$, and $\eta_{\mathrm{w}} m_+ = e/8$. Hence $a_0^2 = 2/\alpha_0 = e/(8r)$, $j_0^2 = a_0^2 b_0^2 = (e/(8r))(2/e) = 1/(4r)$ and $c_x^4 = a_0^2/b_0^2 = (e/(8r))(e/2) = e^2/(16r)$. □

Definition 5.4 (The hyperbola and its inverse)**.** For real x and u put

$$R(x) = m_-(x^* + x) + m_+\sqrt{(x^*+x)^2 + \eta_{\mathrm{w}}^2} - \eta_{\mathrm{w}} r, \qquad X(u) = \frac{-m_- u^2 + m_+ u\sqrt{u^2 + 2\eta_{\mathrm{w}} r}}{r^2}.$$

Lemma 5.5 (The hyperbola)**.** *R is C^∞ and strictly convex, $R(0) = R'(0) = 0$, $R''(0) = \alpha_0 > 0$, $R \ge 0$, and $R(x) \to +\infty$ as $x \to \pm\infty$. In particular 0 is the unique minimum point of R, and it is nondegenerate.*

Proof. Write $\xi = x^* + x$. Since $(x^*)^2 + \eta_{\mathrm{w}}^2 = \eta_{\mathrm{w}}^2(m_-^2 + r^2)/r^2 = (\eta_{\mathrm{w}} m_+/r)^2$,

$$R(0) = -\frac{m_-^2\eta_{\mathrm{w}}}{r} + \frac{m_+^2\eta_{\mathrm{w}}}{r} - \eta_{\mathrm{w}} r = \frac{\eta_{\mathrm{w}}(m_+^2 - m_-^2)}{r} - \eta_{\mathrm{w}} r = 0.$$

$R'(x) = m_- + m_+\xi/\sqrt{\xi^2 + \eta_{\mathrm{w}}^2}$, so $R'(0) = m_- + m_+(-m_-\eta_{\mathrm{w}}/r)/(\eta_{\mathrm{w}} m_+/r) = 0$, and $R''(x) = m_+\eta_{\mathrm{w}}^2/(\xi^2 + \eta_{\mathrm{w}}^2)^{3/2} > 0$, with $R''(0) = m_+\eta_{\mathrm{w}}^2(r/(\eta_{\mathrm{w}} m_+))^3 = r^3/(\eta_{\mathrm{w}} m_+^2) = \alpha_0$. Nonnegativity also follows from Cauchy–Schwarz: $\eta_{\mathrm{w}} r - m_-\xi \le \sqrt{m_-^2 + r^2}\sqrt{\xi^2 + \eta_{\mathrm{w}}^2} = m_+\sqrt{\xi^2 + \eta_{\mathrm{w}}^2}$. Finally $R(x) \ge (m_+ - |m_-|)|\xi| - \eta_{\mathrm{w}} r$, and $m_+ - |m_-| > 0$. □

Lemma 5.6 (X inverts R)**.** *X is a C^∞ diffeomorphism of $\mathbb{R}$ with $X' > 0$, $\operatorname{sign} X(u) = \operatorname{sign} u$ and $R(X(u)) = u^2$. For $u \ne 0$, $X(u)$ is the only solution of $R(x) = u^2$ with the sign of u. Moreover $X'(0) = \sqrt{2/\alpha_0} = a_0$.*

Proof. By Lemma 5.5, for $u \neq 0$ the equation $R(x) = u^2$ has exactly two solutions, one positive and one negative. Any solution satisfies $m_+\sqrt{\xi^2+\eta_{\mathrm{w}}^2} = u^2 + \eta_{\mathrm{w}} r - m_-\xi$ ($\xi = x^* + x$); squaring and using $m_+^2 - m_-^2 = r^2$,

$$r^2\xi^2 + 2m_-(u^2+\eta_{\mathrm{w}}r)\xi + m_+^2\eta_{\mathrm{w}}^2 - (u^2+\eta_{\mathrm{w}}r)^2 = 0.$$

Since $m_-^2 + r^2 = m_+^2$ and $(u^2+\eta_{\mathrm{w}}r)^2 - r^2\eta_{\mathrm{w}}^2 = u^2(u^2+2\eta_{\mathrm{w}}r)$, the roots are $\xi = [-m_-(u^2+\eta_{\mathrm{w}}r) \pm m_+|u|\sqrt{u^2+2\eta_{\mathrm{w}}r}]/r^2$, that is, $x_\pm = [-m_-u^2 \pm m_+|u|\sqrt{u^2+2\eta_{\mathrm{w}}r}]/r^2$. As $m_+\sqrt{u^2+2\eta_{\mathrm{w}}r} > m_+|u| \geq |m_-||u|$, $x_+ > 0 > x_-$ for $u \neq 0$. Hence the positive solution is x_+, the negative one is x_-, both are given by $X(u)$ according to the sign of u, and $X(0) = 0$. Thus $R(X(u)) = u^2$ and $\operatorname{sign} X(u) = \operatorname{sign} u$. X is C^∞ because $2\eta_{\mathrm{w}}r > 0$. Differentiating, $R'(X(u))X'(u) = 2u$; for $u \neq 0$, $R'(X(u))$ has the sign of u, so $X'(u) > 0$; and $X'(0) = m_+\sqrt{2\eta_{\mathrm{w}}r}/r^2 > 0$ with $X'(0)^2 = 2m_+^2\eta_{\mathrm{w}}/r^3 = 2/\alpha_0$. Finally $X(u) \sim (m_+ - m_-)u^2/r^2 \to +\infty$ as $u \to +\infty$ and $X(u) \sim -(m_+ + m_-)u^2/r^2 \to -\infty$ as $u \to -\infty$, so X is a bijection of $\mathbb{R}$ with positive derivative, hence a diffeomorphism. □

Lemma 5.7 (Comparison of R with the tent)**.** $\mathcal{R}_{\mathrm{tent}}(p) = m_-(p-a) + m_+|p-a|$ *for all real* p, *and*

$$\big|R(p-p^*) - \mathcal{R}_{\mathrm{tent}}(p)\big| \leq \delta = \frac{e}{8} \qquad (p \in \mathbb{R}). \tag{6}$$

Proof. For $p \leq a$, $m_-(p-a) - m_+(p-a) = -A(p-a)$; for $p \geq a$, $(m_- + m_+)(p-a) = B(p-a)$. With $\xi = p - a = x^* + (p-p^*)$,

$$R(p-p^*) - \mathcal{R}_{\mathrm{tent}}(p) = m_+\big(\sqrt{\xi^2+\eta_{\mathrm{w}}^2} - |\xi|\big) - \eta_{\mathrm{w}}r \in [-\eta_{\mathrm{w}}r,\ \eta_{\mathrm{w}}(m_+ - r)],$$

because $0 \leq \sqrt{\xi^2+\eta_{\mathrm{w}}^2} - |\xi| \leq \eta_{\mathrm{w}}$. Since $0 < r \leq m_+$, the absolute value is at most $\eta_{\mathrm{w}}m_+ = \delta$. □

Lemma 5.8 (Position of the minimum)**.** $0 < p^* < h$.

Proof. Take $p = p^*$ in (6) and use $R(0) = 0$ and $\mathcal{R}_{\mathrm{tent}} \geq 0$: $\mathcal{R}_{\mathrm{tent}}(p^*) \leq e/8 \leq h/800 < h$. By (5), $\{\mathcal{R}_{\mathrm{tent}} < h\} = (0, h)$. □

5.3. **Potentials exactly quadratic near the origin.** Near its minimum, R is $\frac{1}{2}\alpha_0 x^2$ up to a cubic error g_R. We replace R by its quadratic part on $|x| \leq \varepsilon_R$ and interpolate with a cutoff on $\varepsilon_R \leq |x| \leq 2\varepsilon_R$. The scale ε_R is chosen so small that the change of R is far below e and the cutoff creates no new critical point (Lemmas 5.11 and 5.12). The angular potential $\tilde{V}$ is the quadratic $e\theta^2/2$ plus a smooth step of height $2h$ supported in a band of width w_V at $\theta = \pm\frac{1}{2}$. The step confines the level sets used below to $|\theta| < \frac{1}{2}$ (Lemma 5.14(c)) and adds at most $e/5$ to the mean of $\tilde{V}$ (Lemma 5.22).

Construction 5.9 (The potential pair)**.** *Radial potential.* Put

$$\psi_R(t) = 1 - S\Big(\frac{t^2-1}{3}\Big) \quad (t \in \mathbb{R}), \qquad C_\psi = \sup_{t\in\mathbb{R}} |\psi_R'(t)|, \qquad K_3 = \frac{m_+}{\eta_{\mathrm{w}}^2},$$

$$\varepsilon_R = \frac{e\, r^3}{256\, m_+^4\,(1+C_\psi)}, \qquad \kappa_R = \frac{e}{196608} = \frac{e}{3\cdot 2^{16}}, \qquad \delta' = \delta + \kappa_R = \frac{e}{8} + \kappa_R,$$

$g_R(x) = \frac{1}{2}\alpha_0 x^2 - R(x)$ and

$$\tilde{R}(x) = R(x) + \psi_R(x/\varepsilon_R)\, g_R(x) = \tfrac{1}{2}\alpha_0 x^2 - \big(1 - \psi_R(x/\varepsilon_R)\big) g_R(x) \qquad (x \in \mathbb{R}).$$

Let $\Sigma_R(x) = \operatorname{sign}(x)\sqrt{\tilde{R}(x)}$, let $\tilde{X} = \Sigma_R^{-1}$ (a bijection of $\mathbb{R}$ by Lemma 5.13), and put $u_\varepsilon = \sqrt{\alpha_0/2}\,\varepsilon_R$.

Angular potential. Put $w_V = e/(20h)$, $c_V = 1 - 2w_V$, and for real θ

$$\mathrm{st}(\theta) = S\Big(\frac{4\theta^2 - c_V^2}{1 - c_V^2}\Big), \qquad \tilde V(\theta) = \frac{e\,\theta^2}{2} + 2h\,\mathrm{st}(\theta), \qquad \theta_c = \frac{1}{2}\sqrt{c_V^2 + \frac{1 - c_V^2}{2}}\,.$$

Let $\sigma_V(\theta) = \mathrm{st}(\theta)/\theta^2$ for $\theta \neq 0$ and $\sigma_V(0) = 0$, put $\Sigma_V(\theta) = \theta\sqrt{e/2 + 2h\,\sigma_V(\theta)}$, let $\tilde Y = \Sigma_V^{-1}$ (a bijection of $\mathbb{R}$ by Lemma 5.14), and put $v_\varepsilon = \sqrt{e/2}\,(\frac{1}{2} - w_V)$ and $\rho_T = \min(u_\varepsilon, v_\varepsilon)$.

The function ψ_R is C^∞, even, nonincreasing in $|t|$, with values in $[0,1]$; $\psi_R(t) = 1$ for $|t| \le 1$ (there $(t^2-1)/3 \le 0$) and $\psi_R(t) = 0$ for $|t| \ge 2$ (there $(t^2-1)/3 \ge 1$). Since ψ_R' is continuous and vanishes outside $[-2,2]$, C_ψ is the maximum of $|\psi_R'|$ over $[-2,2]$, a real number in $[0,\infty)$. Only the inequality $|\psi_R'| \le C_\psi$ enters the proofs (Lemma 5.12); the value of C_ψ enters the construction through ε_R. Thus $\tilde R(x) = \frac{1}{2}\alpha_0 x^2$ for $|x| \le \varepsilon_R$ and $\tilde R(x) = R(x)$ for $|x| \ge 2\varepsilon_R$. Since $e \le h/100$, $w_V \le 1/2000$ and $c_V \in (0,1)$. The argument of S in st is ≤ 0 for $|\theta| \le \frac{1}{2} - w_V$ (i.e. $4\theta^2 \le c_V^2$) and ≥ 1 for $|\theta| \ge \frac{1}{2}$; so st $= 0$ on $|\theta| \le \frac{1}{2} - w_V$, st $= 1$ on $|\theta| \ge \frac{1}{2}$, and st is even and nondecreasing in $|\theta|$.

Lemma 5.10 (Third derivative and Taylor bounds)**.** *(i) With $\xi = x^* + x$,*

$$R'''(x) = -3m_+\eta_{\mathrm w}^2\xi\,(\xi^2 + \eta_{\mathrm w}^2)^{-5/2}, \qquad |R'''(x)| \le K_3 \quad (x \in \mathbb{R}).$$

(ii) $g_R(0) = g_R'(0) = g_R''(0) = 0$ and $|g_R'''| \le K_3$; hence $|g_R'(x)| \le K_3x^2/2$ and $|g_R(x)| \le K_3|x|^3/6$.

(iii) $K_3\,\varepsilon_R = \alpha_0/\big(32(1 + C_\psi)\big)$.

Proof. (i) Differentiate $R''(x) = m_+\eta_{\mathrm w}^2(\xi^2 + \eta_{\mathrm w}^2)^{-3/2}$. Substituting $\xi = \eta_{\mathrm w}\tau$ gives

$$|R'''(x)| = (3m_+/\eta_{\mathrm w}^2)\varphi(|\tau|), \qquad \varphi(\tau) = \tau(1+\tau^2)^{-5/2} \quad (\tau \ge 0).$$

Since

$$\varphi'(\tau) = (1+\tau^2)^{-7/2}(1 - 4\tau^2),$$

φ is maximal at $\tau = \frac{1}{2}$, where

$$\varphi(\tfrac{1}{2}) = \tfrac{1}{2}(\tfrac{4}{5})^{5/2} = \frac{16}{25\sqrt{5}}.$$

Hence $3\varphi \le \frac{48}{25\sqrt{5}} < 1$, because $48^2 = 2304 < 3125 = 25^2 \cdot 5$, and $|R'''| \le K_3$.

(ii) By Lemma 5.5, $R(0) = R'(0) = 0$ and $R''(0) = \alpha_0$, so $g_R(0) = g_R'(0) = g_R''(0) = 0$, and $g_R''' = -R'''$. Integrating three times from 0: $|g_R''(x)| \le K_3|x|$, $|g_R'(x)| \le K_3x^2/2$, $|g_R(x)| \le K_3|x|^3/6$.

(iii) $\eta_{\mathrm w} = e/(8m_+)$ gives $K_3 = 64m_+^3/e^2$, and $\alpha_0 = 8r^3/(em_+)$ by Lemma 5.3. So

$$K_3\varepsilon_R = \frac{64m_+^3}{e^2}\cdot\frac{e\,r^3}{256\,m_+^4(1+C_\psi)} = \frac{r^3}{4em_+(1+C_\psi)} = \frac{8r^3}{em_+}\cdot\frac{1}{32(1+C_\psi)} = \frac{\alpha_0}{32(1+C_\psi)}. \qquad \square$$

Lemma 5.11 (Closeness)**.** *For all real x and p,*

$$|\tilde R(x) - R(x)| \le \kappa_R, \qquad \big|\tilde R(p - p^*) - \mathcal{R}_{\mathrm{tent}}(p)\big| \le \delta' = \frac{e}{8} + \kappa_R. \tag{7}$$

Proof. $\tilde R - R = \psi_R(x/\varepsilon_R)\,g_R$ vanishes for $|x| \ge 2\varepsilon_R$, and $0 \le \psi_R \le 1$. For $|x| \le 2\varepsilon_R$, Lemma 5.10(ii),(iii) give

$$|\tilde R(x) - R(x)| \le |g_R(x)| \le \frac{K_3(2\varepsilon_R)^3}{6} = \frac{4}{3}K_3\varepsilon_R\cdot\varepsilon_R^2 = \frac{\alpha_0\varepsilon_R^2}{24(1+C_\psi)}.$$

With $\alpha_0 = 8r^3/(em_+)$ and $\varepsilon_R^2 = e^2r^6/(2^{16}m_+^8(1+C_\psi)^2)$ this equals

$$\frac{8\,e\,r^9}{24\cdot 2^{16}\,m_+^9(1+C_\psi)^3} = \frac{e}{3\cdot 2^{16}}\Big(\frac{r}{m_+}\Big)^9\frac{1}{(1+C_\psi)^3} \le \kappa_R,$$

since $0 < r \le m_+$ and $C_\psi \ge 0$. The second inequality follows from the first and (6). □

Lemma 5.12 (Monotonicity)**.** *$\tilde R$ is C^∞, and $\operatorname{sign}(x)\,\tilde R'(x) > 0$ for every $x \neq 0$; more precisely $\operatorname{sign}(x)\tilde R'(x) \ge \frac{91}{96}\alpha_0|x|$ for $0 < |x| \le 2\varepsilon_R$. Consequently $\tilde R$ is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0,\infty)$, $\tilde R(0) = 0$, $\tilde R(x) > 0$ for $x \neq 0$, and $\tilde R(x) \to +\infty$ as $x \to \pm\infty$.*

Proof. $\tilde R$ is C^∞ because R, ψ_R and g_R are, and

$$\tilde R'(x) = \alpha_0 x - \big(1-\psi_R(x/\varepsilon_R)\big)g_R'(x) + \varepsilon_R^{-1}\psi_R'(x/\varepsilon_R)\,g_R(x).$$

Let $0 < |x| \le 2\varepsilon_R$. By Lemma 5.10, $|g_R'(x)| \le K_3x^2/2 \le K_3\varepsilon_R|x| = \alpha_0|x|/(32(1+C_\psi)) \le \alpha_0|x|/32$, and

$$\big|\varepsilon_R^{-1}\psi_R'(x/\varepsilon_R)g_R(x)\big| \le \frac{C_\psi}{\varepsilon_R}\cdot\frac{K_3|x|(2\varepsilon_R)^2}{6} = \frac{2}{3}\cdot\frac{C_\psi}{1+C_\psi}\cdot\frac{\alpha_0|x|}{32} \le \frac{\alpha_0|x|}{48}.$$

Since $0 \le 1-\psi_R \le 1$, $\operatorname{sign}(x)\tilde R'(x) \ge \alpha_0|x|\big(1-\frac{1}{32}-\frac{1}{48}\big) = \frac{91}{96}\alpha_0|x|$. For $|x| > 2\varepsilon_R$, $\tilde R = R$ near x, and $\operatorname{sign}(x)R'(x) > 0$ because R is strictly convex with $R'(0) = 0$. Hence $\tilde R$ is strictly monotone on each half-line with minimum $\tilde R(0) = 0$, and $\tilde R = R \to \infty$ as $|x| \to \infty$. □

Near 0 the signed square root of $\tilde R$ is the linear function $\sqrt{\alpha_0/2}\,x$, so the inverse $\tilde X$ is obtained without differentiating the square root of a function that vanishes at 0.

Lemma 5.13 ($\tilde X$ inverts $\tilde R$ and is linear near 0)**.**
(i) Σ_R is a C^∞ diffeomorphism of $\mathbb{R}$ with $\Sigma_R' > 0$, and $\Sigma_R(x) = \sqrt{\alpha_0/2}\,x$ for $|x| \le \varepsilon_R$.
(ii) $\tilde X$ is a C^∞ diffeomorphism of $\mathbb{R}$ with $\tilde X' > 0$, $\operatorname{sign}\tilde X(u) = \operatorname{sign} u$ and $\tilde R(\tilde X(u)) = u^2$.
(iii) $\tilde X(u) = a_0u$ for $|u| \le u_\varepsilon$; in particular $\tilde X'(u) = a_0$ there.
(iv) If $|X(u)| \ge 2\varepsilon_R$ or $|\tilde X(u)| \ge 2\varepsilon_R$, then $\tilde X(u) = X(u)$. Hence

$$|\tilde X(u)| \le \max\big(|X(u)|, 2\varepsilon_R\big) \le |X(u)| + 2\varepsilon_R \qquad (u \in \mathbb{R}).$$

Proof. (i) The open sets $(-\varepsilon_R, \varepsilon_R)$, $(0,\infty)$, $(-\infty, 0)$ cover $\mathbb{R}$. On the first, $\tilde R = \alpha_0x^2/2$, so $\Sigma_R(x) = \sqrt{\alpha_0/2}\,x$ (also on the closed interval, by continuity). On the second, $\tilde R > 0$ (Lemma 5.12), so $\Sigma_R = \sqrt{\tilde R}$ is C^∞ with $\Sigma_R' = \tilde R'/(2\sqrt{\tilde R}) > 0$; on the third $\Sigma_R = -\sqrt{\tilde R}$ and $\Sigma_R' = -\tilde R'/(2\sqrt{\tilde R}) > 0$. Since $\Sigma_R^2 = \tilde R \to \infty$ at $\pm\infty$ and Σ_R has the sign of x, Σ_R is a strictly increasing bijection of $\mathbb{R}$, hence a C^∞ diffeomorphism by the inverse function theorem. (ii) $\tilde X' = 1/\Sigma_R'(\tilde X) > 0$, $\tilde X$ has the sign of u, and $\tilde R(\tilde X(u)) = \Sigma_R(\tilde X(u))^2 = u^2$. (iii) For $|u| \le u_\varepsilon$, $|a_0u| \le a_0u_\varepsilon = \varepsilon_R$, so $\Sigma_R(a_0u) = \sqrt{\alpha_0/2}\,a_0u = u$. (iv) If $|X(u)| \ge 2\varepsilon_R$, then $\tilde R(X(u)) = R(X(u)) = u^2$ with $X(u)$ of the sign of u (Lemma 5.6), so $\Sigma_R(X(u)) = u$ and $X(u) = \tilde X(u)$. If $|\tilde X(u)| \ge 2\varepsilon_R$, then $R(\tilde X(u)) = \tilde R(\tilde X(u)) = u^2$ with $\tilde X(u)$ of the sign of u, so $\tilde X(u) = X(u)$ by the uniqueness in Lemma 5.6 (for $u = 0$ both vanish). The bound follows by distinguishing $|\tilde X(u)| < 2\varepsilon_R$ and $|\tilde X(u)| \ge 2\varepsilon_R$. □

Lemma 5.14 (The angular wall and its signed inverse)**.**
(a) $\tilde V$ is C^∞, even and nonnegative, $\tilde V(\theta) = e\theta^2/2$ for $|\theta| \le \frac12 - w_V$, $\tilde V(\theta) \ge e\theta^2/2$ for all θ, and $\tilde V'(\theta) \ge e\theta > 0$ for $\theta > 0$; in particular $\tilde V$ is strictly increasing in $|\theta|$.
(b) $\tilde V(\pm\frac12) = e/8 + 2h > h$.
(c) $\frac12 - w_V < \theta_c < \frac12$, and if $\tilde V(\theta) \le h - 2e$, then $\operatorname{st}(\theta) < \frac12$ and $|\theta| < \theta_c$.

(d) Σ_V *is an odd* C^∞ *diffeomorphism of* $\mathbb{R}$ *with* $\Sigma_V' > 0$, $\Sigma_V^2 = \tilde V$, $|\Sigma_V(\theta)| \ge \sqrt{e/2}\,|\theta|$, *and* $\Sigma_V(\theta) = \sqrt{e/2}\,\theta$ *for* $|\theta| \le \frac12 - w_V$.

(e) $\tilde Y$ *is an odd* C^∞ *diffeomorphism of* $\mathbb{R}$, $\tilde Y' > 0$ *is even,* $\tilde V(\tilde Y(v)) = v^2$, $|\tilde Y(v)| \le b_0|v|$, *and* $\tilde Y(v) = b_0 v$ *for* $|v| \le v_\varepsilon$.

Proof. (a) st is C^∞ (a composite of S with a polynomial), even, zero on $|\theta| \le \frac12 - w_V$, and $\mathrm{st}'(\theta) \ge 0$ for $\theta > 0$. Hence $\tilde V'(\theta) = e\theta + 2h\,\mathrm{st}'(\theta) \ge e\theta$ for $\theta > 0$; the other claims are immediate. (b) $\mathrm{st}(\pm\frac12) = S(1) = 1$. (c) $\theta_c^2 = \frac14 \cdot \frac{c_V^2+1}{2}$ lies strictly between $c_V^2/4 = (\frac12 - w_V)^2$ and $\frac14$, since $0 < c_V < 1$. If $\tilde V(\theta) \le h - 2e$, then $2h\,\mathrm{st}(\theta) \le \tilde V(\theta) < h$, so $\mathrm{st}(\theta) < \frac12$. As S is nondecreasing and $S(\frac12) = \frac12$, the argument of S is $< \frac12$: $4\theta^2 < c_V^2 + (1 - c_V^2)/2 = 4\theta_c^2$. (d) σ_V is C^∞: it vanishes on $|\theta| < \frac12 - w_V$, and it is a quotient of C^∞ functions on $\theta \ne 0$; these open sets cover $\mathbb{R}$. So $\gamma_V = \sqrt{e/2 + 2h\sigma_V}$ is C^∞, even and $\ge \sqrt{e/2}$, and $\Sigma_V = \theta\gamma_V$ is odd and C^∞, with $|\Sigma_V(\theta)| \ge \sqrt{e/2}\,|\theta|$ and $\Sigma_V^2 = e\theta^2/2 + 2h\,\mathrm{st} = \tilde V$. On $|\theta| \le \frac12 - w_V$, $\gamma_V = \sqrt{e/2}$. For $\theta > 0$, $\Sigma_V = \sqrt{\tilde V}$ with $\tilde V > 0$, so $\Sigma_V' = \tilde V'/(2\sqrt{\tilde V}) > 0$ by (a); by oddness $\Sigma_V' > 0$ for $\theta < 0$, and $\Sigma_V'(0) = \sqrt{e/2}$. A strictly increasing proper C^∞ function with positive derivative is a C^∞ diffeomorphism of $\mathbb{R}$. (e) $\tilde Y$ is odd because Σ_V is, $\tilde Y' = 1/\Sigma_V'(\tilde Y) > 0$ is even as the derivative of an odd function, $\tilde V(\tilde Y(v)) = \Sigma_V(\tilde Y(v))^2 = v^2$, and $|v| = |\Sigma_V(\tilde Y(v))| \ge \sqrt{e/2}\,|\tilde Y(v)|$. For $|v| \le v_\varepsilon$, $|b_0 v| \le \frac12 - w_V$, so $\Sigma_V(b_0 v) = \sqrt{e/2}\,b_0 v = v$. □

Lemma 5.15 (The equal-ball potential pair)**.** *Let* $d = 0$, $I_U = I_V = \mathbb{R}$, $U(p) = \tilde R(p - p^*)$ *and* $V = \tilde V$. *Then* (U, V) *is an admissible potential pair with disc area* h *(Definition 4.1), with minimum point* p^* *and minimum value* $m = 0$, *and it is exactly quadratic near the minimum with* $\varepsilon_U = \varepsilon_R$ *and* $\varepsilon_V = \frac12 - w_V$. *Its objects in Lemma 4.2 are* $X_y = \tilde X$, $Y_y = \tilde Y$, $a_0(y) = a_0$, $b_0(y) = b_0$, $j_0(y) = j_0$, $\rho_T(y) = \rho_T$, *and* $\mathbf{L}_y = \mathbf{L}$ *in Definition 4.6.*

Proof. (P1): $p^* \in (0, h)$ by Lemma 5.8. By Lemma 5.12, $U'(p)(p - p^*) = \tilde R'(x)\,x > 0$ for $x = p - p^* \ne 0$, and $U''(p^*) = \tilde R''(0) = \alpha_0 > 0$ because $\tilde R = \frac12\alpha_0 x^2$ on $|x| \le \varepsilon_R$. By Lemma 5.14(a), $\tilde V(0) = 0$, $\tilde V''(0) = e > 0$, and $\tilde V'(\theta) \ge e\theta > 0$ for $\theta > 0$; since $\tilde V$ is even, $\tilde V'$ is odd, so $\tilde V'(\theta)\,\theta > 0$ for $\theta \ne 0$. (P2): $\tilde R(x) \to +\infty$ as $x \to \pm\infty$ (Lemma 5.12) and $\tilde V(\theta) \ge e\theta^2/2$. Further $m = U(p^*) = \tilde R(0) = 0$; $U(p) = \frac12\alpha_0(p - p^*)^2$ for $|p - p^*| \le \varepsilon_R$, and $\tilde V(\theta) = \frac{e}{2}\theta^2$ for $|\theta| \le \frac12 - w_V$.

The signed square root of Lemma 4.2(ii) is $\mathrm{sign}(x)\sqrt{U(p^* + x) - m} = \mathrm{sign}(x)\sqrt{\tilde R(x)} = \Sigma_R(x)$, so $X_y = \Sigma_R^{-1} = \tilde X$. By Lemma 5.14(d), Σ_V is odd, increasing and $\Sigma_V^2 = \tilde V$, so Σ_V is the signed square root of $\tilde V$ and $Y_y = \tilde Y$. Then $a_0(y) = \sqrt{2/U''(p^*)} = \sqrt{2/\alpha_0} = a_0$ and $b_0(y) = \sqrt{2/\tilde V''(0)} = \sqrt{2/e} = b_0$, hence $j_0(y) = j_0$ and $\mathbf{L}_y = \mathrm{diag}(c_x, 1/c_x) = \mathbf{L}$. Finally $\varepsilon_R/a_0 = \sqrt{\alpha_0/2}\,\varepsilon_R = u_\varepsilon$ and $(\frac12 - w_V)/b_0 = v_\varepsilon$, so $\rho_T(y) = \min(u_\varepsilon, v_\varepsilon) = \rho_T$. □

5.4. **The inner map.**

Lemma 5.16 (The Morse chart and its density)**.** *Put* $F_{\mathrm{mor}}(u, v) = (\tilde X(u), \tilde Y(v))$, $J(u, v) = \tilde X'(u)\tilde Y'(v)$ *and* $\mathcal{A}_J(E) = \int_{|w|^2 \le E} J(w)\,dw$ *for* $E \ge 0$. *Then* F_{mor} *is a* C^∞ *diffeomorphism of* $\mathbb{R}^2$ *with inverse* (Σ_R, Σ_V), $J > 0$ *is* C^∞, $J(w) = j_0$ *for* $|w| \le \rho_T$, $F_{\mathrm{mor}}^*(dx \wedge d\theta) = J\,du \wedge dv$, $\tilde R(\tilde X(u)) + \tilde V(\tilde Y(v)) = u^2 + v^2$, *and* $\mathcal{A}_J(E) = \mathrm{area}\{(x, \theta) : \tilde R(x) + \tilde V(\theta) \le E\} \to \infty$ *as* $E \to \infty$.

Proof. By Lemma 5.15, F_{mor} and J are the Morse chart F_y and its density J_y of Lemma 4.2(iv) for the pair (U, V), whose Morse rectangle is $\mathbb{R}^2$; that lemma gives all claims except the constant value near 0 and the limit, and (v) gives $J = j_0$ on $|w| \le \rho_T$. Directly: if $|w| \le \rho_T$ then $|u| \le u_\varepsilon$

and $|v| \le v_\varepsilon$, so $\tilde X'(u) = a_0$ and $\tilde Y'(v) = b_0$ by Lemmas 5.13(iii) and 5.14(e). Since $\tilde R$ and $\tilde V$ are proper and nonnegative, the sublevel sets exhaust $\mathbb{R}^2$, and $\mathcal{A}_J(E) \to \infty$. □

Let T_J, D_J, E_J be the rotation normalisation, the area normalisation and the inverse of $\mathcal{A}_J$ for the density J (Definition 4.3), and $\bar J$ the circle mean of J.

Definition 5.17 (The inner map)**.** $f = F_{\mathrm{mor}} \circ T_J^{-1} \circ D_J$. By Lemma 5.15 this is the map f_y of Definition 4.6 for the pair (U, V).

Proposition 5.18 (The inner map; exactly linear at the origin)**.**

(a) The map f is a smooth diffeomorphism of the plane, and

$$f^*(dx \wedge d\theta) = \omega_2.$$

If $f(z) = (x, \theta)$, then $\tilde R(x) + \tilde V(\theta) = E_J(\pi|z|^2)$.

(b) On $\{|w| \le \rho_T\}$, $\bar J = j_0$ and $T_J = T_J^{-1} = \mathrm{id}$; $\mathcal{A}_J(E) = \pi j_0 E$ for $E \le \rho_T^2$, $E_J(I) = I/(\pi j_0)$ for $I \le \pi j_0 \rho_T^2$, and $D_J(z) = z/\sqrt{j_0}$ for $|z| \le \sqrt{j_0}\,\rho_T$.

(c) For $|z| \le \sqrt{j_0}\,\rho_T$,

$$f(z) = \mathrm{diag}\big(a_0/\sqrt{j_0},\, b_0/\sqrt{j_0}\big) z = \mathrm{diag}(c_x, c_y)\, z = \mathbf{L} z,$$

a linear map with $\det \mathbf{L} = 1$.

Proof. By Lemma 5.15, (U, V) is an admissible potential pair, exactly quadratic near the minimum, with $\mathcal{A}_J(\infty) = \infty$ (Lemma 5.16) and Morse rectangle $\mathbb{R}^2$. (a) is Theorem 4.7(a),(b), with $m = 0$ and $U(p^* + x) = \tilde R(x)$. (b) Since $J = j_0$ on $\{|w| \le \rho_T\}$ (Lemma 5.16), the circle mean is $\bar J = j_0$ there, and Lemma 4.4(iv) gives $T_J = T_J^{-1} = \mathrm{id}$ there; Lemma 4.5(v) gives $\mathcal{A}_J(E) = \pi j_0 E$ for $E \le \rho_T^2$ and $D_J(z) = z/\sqrt{j_0}$ for $|z| \le \sqrt{j_0}\,\rho_T$, and inverting the first formula gives E_J. (c) is Theorem 4.7(d) with $\mathbf{L}_y = \mathbf{L}$ (Lemma 5.15); directly, $w = z/\sqrt{j_0}$ has $|w| \le \rho_T$, so $f(z) = F_{\mathrm{mor}}(w) = (a_0 u, b_0 v)$ by (b) and Lemmas 5.13(iii) and 5.14(e), and $a_0/\sqrt{j_0} = \sqrt{a_0/b_0} = c_x$, $b_0/\sqrt{j_0} = 1/c_x = c_y$. □

5.5. **The physical chart and the comparison.** The chart C_{ph} cuts the fibre disc along the positive real axis; its centre line $\theta = 0$ is the negative real axis. The comparison has two parts: the moment of $C_{\mathrm{ph}} f(z)$ is controlled by the area estimate $E_J(I) \le I + d_0$, and on the source disc of area $p_+ = h - 15e/4$ the sublevel sets stay strictly inside the chart, so $C_{\mathrm{ph}} f$ is defined there.

Definition 5.19 (Physical chart)**.** Let $\mathcal{U} = \{(x, \theta) \in \mathbb{R}^2 : p^* + x > 0,\ |\theta| < \tfrac12\}$ and

$$C_{\mathrm{ph}}(x, \theta) = -\sqrt{\frac{p^* + x}{\pi}}\, e^{2\pi i \theta} \qquad ((x, \theta) \in \mathcal{U}). \tag{8}$$

Put $r_b = \sqrt{p^*/\pi}$ and $b = C_{\mathrm{ph}}(0) = (-r_b, 0)$. For the pair of Lemma 5.15 these are the chart C_y, its domain $\mathcal{C}_y$, and $r_b(y)$, b_y of (3).

Lemma 5.20 (The physical chart)**.** *$\mathcal{U}$ is convex and contains 0. C_{ph} is a C^∞ injective map on $\mathcal{U}$ with $\det DC_{\mathrm{ph}} = 1$, i.e. $C_{\mathrm{ph}}^* \omega_2 = dx \wedge d\theta$, and $p(C_{\mathrm{ph}}(x, \theta)) = p^* + x$. In particular C_{ph} maps $\mathcal{U} \cap \{x < h - p^*\}$ into $D(h)$. At a point with $p^* + x = P$ the columns of DC_{ph} are orthogonal, of lengths $1/(2\sqrt{\pi P})$ and $2\sqrt{\pi P}$, and $DC_{\mathrm{ph}}(0) = \mathrm{diag}\big(-\frac{1}{2\pi r_b},\, -2\pi r_b\big)$.*

Proof. $0 \in \mathcal{U}$ by Lemma 5.8, and $\mathcal{U}$ is an intersection of half-planes. C_{ph} has polar radius $\rho_C = \sqrt{(p^* + x)/\pi}$ and polar angle $2\pi\theta + \pi \in (0, 2\pi)$, which gives injectivity and the moment. The area form is $\rho_C\, d\rho_C \wedge d(2\pi\theta) = \frac{1}{2\pi} d(p^* + x) \wedge 2\pi\, d\theta = dx \wedge d\theta$. With $P = p^* + x$, $\partial_x C_{\mathrm{ph}} = -\frac{1}{2\sqrt{\pi P}} e^{2\pi i\theta}$ and $\partial_\theta C_{\mathrm{ph}} = -2\pi i \sqrt{P/\pi}\, e^{2\pi i\theta}$, which are orthogonal with the stated lengths; at 0, $e^{2\pi i\theta} = 1$ and $\sqrt{\pi p^*} = \pi r_b$. □

Definition 5.21 (Comparison constants)**.** $\mu_V = \int_{-1/2}^{1/2} \tilde{V}(\theta)\,d\theta$, $d_0 = \delta' + \mu_V$, $p_+ = h - 15e/4$, and $E_c = h - 2e$.

Lemma 5.22 (Size of the comparison constants)**.** $\mu_V \le e/24 + e/5 = 29e/120 < 7e/24$. *Moreover* $d_0 < 5e/12$, $2\delta' + \mu_V < 13e/24$, $p_+ > 0$, *and* $E_c - d_0 > p_+$.

Proof. $\int_{-1/2}^{1/2} e\theta^2/2\,d\theta = e/24$. The function st takes values in $[0,1]$ and vanishes on $|\theta| \le \frac{1}{2} - w_V$, so $\int_{-1/2}^{1/2} 2h\,\mathrm{st} \le 2h \cdot 2w_V = e/5$. Hence $\mu_V \le 5e/120 + 24e/120 = 29e/120 < 7e/24$. Since $\kappa_R = e/196608 < e/40$,

$$d_0 \le \frac{15e}{120} + \frac{29e}{120} + \kappa_R < \frac{50e}{120} = \frac{5e}{12}, \qquad 2\delta' + \mu_V \le \frac{59e}{120} + 2\kappa_R < \frac{65e}{120} = \frac{13e}{24}.$$

As $e \le h/100$, $15e/4 < h$, so $p_+ > 0$; and $E_c - d_0 > h - 2e - 5e/12 = h - 29e/12 > h - 45e/12 = p_+$. □

The area lower bound is a slicing argument: for each angle θ, the x-slice of $\{\tilde{R} + \tilde{V} \le E\}$ contains the slice of $\{\mathcal{R}_{\mathrm{tent}}(p^* + x) \le E - \delta' - \tilde{V}(\theta)\}$, an interval of length $E - \delta' - \tilde{V}(\theta)$; integrating over θ loses the mean μ_V and the smoothing error δ'.

Lemma 5.23 (Area lower bound)**.** $\mathcal{A}_J(E) \ge E - d_0$ *for all* $E \ge 0$.

Proof. By Lemma 5.16, $\mathcal{A}_J(E)$ is the area of $\{\tilde{R}(x) + \tilde{V}(\theta) \le E\}$. For fixed θ with $|\theta| \le \frac{1}{2}$, (7) gives $\tilde{R}(x) \le \mathcal{R}_{\mathrm{tent}}(p^* + x) + \delta'$, so the x-section contains $\{x : \mathcal{R}_{\mathrm{tent}}(p^* + x) \le E - \delta' - \tilde{V}(\theta)\}$, an interval of length $(E - \delta' - \tilde{V}(\theta))_+$ by (5). Integrating over $\theta \in [-\frac{1}{2}, \frac{1}{2}]$, $\mathcal{A}_J(E) \ge \int_{-1/2}^{1/2} (E - \delta' - \tilde{V}(\theta))_+\,d\theta \ge \int_{-1/2}^{1/2} (E - \delta' - \tilde{V}(\theta))\,d\theta = E - d_0$. □

Proposition 5.24 (Comparison and chart containment)**.** *Let* $z \in \mathbb{C}$, $I = \pi|z|^2$, *and suppose that* $f(z) \in \mathcal{U}$. *Then* $E_J(I) \le I + d_0$ *and*

$$\mathcal{R}_{\mathrm{tent}}\big(p(C_{\mathrm{ph}} f(z))\big) \le I + 2\delta' + \mu_V < I + \frac{13e}{24}. \tag{9}$$

Moreover, if $I \le p_+$ *then* $E_J(I) \le p_+ + d_0 < h - \delta'$ *and* $f(z) \in \{(x,\theta) : -p^* < x < h - p^*,\ |\theta| < \frac{1}{2}\}$; *thus* $C_{\mathrm{ph}} f$ *is defined on* $\bar{D}(p_+)$, *takes values in* $\{0 < p < h\}$, *and satisfies* (9) *there.*

Proof. Let $f(z) = (x,\theta)$ and $E = E_J(I)$. By Lemma 5.23, $I = \mathcal{A}_J(E) \ge E - d_0$. By Proposition 5.18(a), $\tilde{R}(x) \le \tilde{R}(x) + \tilde{V}(\theta) = E$. By Lemma 5.20 and (7), $\mathcal{R}_{\mathrm{tent}}(p(C_{\mathrm{ph}} f(z))) = \mathcal{R}_{\mathrm{tent}}(p^* + x) \le \tilde{R}(x) + \delta' \le I + d_0 + \delta' = I + 2\delta' + \mu_V$, and $2\delta' + \mu_V < 13e/24$ by Lemma 5.22.

Let $I \le p_+$. Then $E \le p_+ + d_0 < h - 15e/4 + 5e/12 = h - 10e/3 < h - \delta'$. At $x = -p^*$ and $x = h - p^*$ (moments 0 and h) we have $\mathcal{R}_{\mathrm{tent}} = h$, hence $\tilde{R} \ge h - \delta' > E$ by (7). Since $\tilde{R}$ decreases strictly on $(-\infty, 0]$ and increases strictly on $[0, \infty)$ (Lemma 5.12) and $0 \in (-p^*, h - p^*)$ (Lemma 5.8), the sublevel set $\{\tilde{R} \le E\}$ is an interval contained in $(-p^*, h - p^*)$. At the angular boundary $\tilde{V}(\pm\frac{1}{2}) = e/8 + 2h > E$, and $\tilde{V}$ is even and increasing in $|\theta|$ (Lemma 5.14), so $\{\tilde{V} \le E\} \subset (-\frac{1}{2}, \frac{1}{2})$. Since $\tilde{R}(x) \le E$ and $\tilde{V}(\theta) \le E$, $f(z)$ lies in the stated rectangle, and the first part applies. □

The error $13e/24$ consists of the smoothing error δ', which occurs twice (in the area estimate and in the comparison of $\tilde{R}$ with $\mathcal{R}_{\mathrm{tent}}$), and the mean μ_V of the angular potential; the quadratic modification at the origin contributes only $2\kappa_R = e/98304$.

5.6. **The data of the whole-disc isotopy.** Theorem 4.21 turns $C_{\mathrm{ph}}f$ into a compactly supported Hamiltonian isotopy of $D(h)$ whose endpoint equals $C_{\mathrm{ph}}f$ on a core disc, once isotopy data (Definition 4.11) are supplied. This subsection supplies them.

Definition 5.25 (The density family)**.** For $s \in \mathbb{R}$ let

$$X_s(u) = (1-s)a_0u + s\tilde{X}(u), \qquad Y_s(v) = (1-s)b_0v + s\tilde{Y}(v), \qquad J_s(u,v) = X_s'(u)\,Y_s'(v),$$

$F_s = (X_s, Y_s)$, and let $\mathcal{A}_s$, $E_s = \mathcal{A}_s^{-1}$, T_{J_s}, D_{J_s} and $f_s = F_s \circ T_{J_s}^{-1} \circ D_{J_s}$ be the objects of Definition 4.8(a) for the pair of Lemma 5.15. Put $K_0 = \pi j_0$ and $\kappa_J = K_0/(1+K_0)$.

Lemma 5.26 (Positivity and area growth)**.** *For all $s \in [0,1]$:*

(a) $J_s = (1-s)^2 j_0 + s(1-s)\big(a_0\tilde{Y}'(v) + b_0\tilde{X}'(u)\big) + s^2 J > 0$;
(b) $J_s(w) = j_0$ for $|w| \le \rho_T$ (this holds for every $s \in \mathbb{R}$);
(c) for all $E \ge 0$, $\mathcal{A}_s(E) \ge [K_0(1-s)^2 + s^2]E - s^2 d_0 \ge \kappa_J E - d_0$.

In particular $\mathcal{A}_s$ is an increasing bijection of $[0,\infty)$, and for every $I_0 \ge 0$, $E_s(I) \le (I_0 + d_0)/\kappa_J$ whenever $0 \le I \le I_0$.

Proof. (a) is Lemma 4.10(a). (b) For $|w| \le \rho_T$, $\tilde{X}'(u) = a_0$ and $\tilde{Y}'(v) = b_0$ (Lemma 5.16), hence $X_s'(u) = a_0$, $Y_s'(v) = b_0$. (c) By Lemma 4.10(b), $\mathcal{A}_s(E) \ge K_0(1-s)^2E + s^2\mathcal{A}_J(E)$, and $\mathcal{A}_J(E) \ge E - d_0$ by Lemma 5.23. The minimum of $K_0(1-s)^2 + s^2$ over $s \in \mathbb{R}$ is $K_0/(1+K_0)$, and $s^2 d_0 \le d_0$. $\mathcal{A}_s$ is continuous and strictly increasing with $\mathcal{A}_s(0) = 0$ because $J_s > 0$, and $\mathcal{A}_s(E) \to \infty$ by (c). If $\mathcal{A}_s(E) = I \le I_0$, then $\kappa_J E - d_0 \le I_0$. □

$E_c = h - 2e$ is the energy level below which the Morse coordinates are mapped a fixed distance away from the slit and the boundary: points with $u^2, v^2 < E_c$ have moments $p^* + \tilde{X}(u)$ in $[c_P/A,\ h - c_P/B]$, where $c_P = 2e - \delta'$, and angles $|\tilde{Y}(v)| < \theta_c$ (Lemma 5.29). ΔP and $\Delta\theta$ are the resulting margins. $\tilde{M}_X(E)$ bounds $|\tilde{X}(u)|$ for $u^2 \le E$, and D_x, D_y bound the interpolation errors below the level E_c. The interpolation parameter $s \in [0,1]$ has an *early range* $s \le 1 - \delta_s$, where the chart parameter $\lambda(s)$ of Definition 4.8(c) is at most λ_0, and a *late range* $s \ge 1 - \delta_s$. The length δ_s of the late range is chosen so that the interpolation error is at most half of each margin there and the whole source disc $D(p^\dagger)$ lies below the level E_c. $\bar{E}$, $\bar{X}$, $\bar{Y}$ are crude bounds valid for all s.

Definition 5.27 (Constants of the common source disc)**.** Put $c_P = 2e - \delta'$ and, for $E \ge 0$,

$$\Delta P = \frac{c_P}{\max(A,B)}, \qquad \Delta\theta = \tfrac{1}{2} - \theta_c,$$

$$M_X(E) = \frac{|m_-|E + m_+\sqrt{E(E+2\eta_{\mathrm{w}}r)}}{r^2}, \quad \tilde{M}_X(E) = M_X(E) + 2\varepsilon_R,$$

$$D_x = a_0\sqrt{E_c} + \tilde{M}_X(E_c), \qquad D_y = b_0\sqrt{E_c} + \theta_c, \qquad R_q = \sqrt{\big(h - c_P/B\big)/\pi},$$

$$\delta_s = \min\Big(\frac{1}{4},\ \frac{1}{2}\Big[1 - \sqrt{\tfrac{p_+}{E_c - d_0}}\Big],\ \frac{\Delta P}{2D_x},\ \frac{\Delta\theta}{2D_y}\Big), \qquad p^\dagger = \tfrac{1}{2}\big[p_+ + (1-\delta_s)^2(E_c - d_0)\big],$$

$$\bar{E} = \frac{p^\dagger + d_0}{\kappa_J}, \qquad \bar{X} = \max\big(a_0\sqrt{\bar{E}},\ \tilde{M}_X(\bar{E})\big), \qquad \bar{Y} = b_0\sqrt{\bar{E}}.$$

Lemma 5.28 (Order of the constants)**.** *$0 < c_P < h$, all four entries of the minimum defining δ_s are positive, so $0 < \delta_s \le \frac{1}{4}$, and*

$$p_+ < p^\dagger < (1-\delta_s)^2(E_c - d_0) < h.$$

Proof. $c_P = 15e/8 - \kappa_R$ lies in $(0, h)$. $\theta_c < \frac{1}{2}$ by Lemma 5.14(c), so $\Delta\theta > 0$; $\Delta P > 0$; $D_x, D_y > 0$. By Lemma 5.22, $E_c - d_0 > p_+ > 0$, so $0 < \sqrt{p_+/(E_c - d_0)} < 1$ and the second entry is positive. Next $1 - \delta_s \geq \frac{1}{2}(1 + \sqrt{p_+/(E_c - d_0)}) > \sqrt{p_+/(E_c - d_0)}$, so $(1 - \delta_s)^2(E_c - d_0) > p_+$, and $p^\dagger$ is the midpoint of these two numbers. Finally $(1 - \delta_s)^2(E_c - d_0) < E_c < h$. □

Lemma 5.29 (Bounds for the instance)**.** *(a) For all real u, $|X(u)| \leq M_X(u^2)$ and $|\tilde{X}(u)| \leq \tilde{M}_X(u^2)$; $\tilde{M}_X$ is increasing. For all u, v with $u^2 \leq \bar{E}$, $v^2 \leq \bar{E}$:*

$$\max\big(a_0|u|, |\tilde{X}(u)|\big) \leq \bar{X}, \qquad \max\big(b_0|v|, |\tilde{Y}(v)|\big) \leq \bar{Y}.$$

(b) For all u, v with $u^2 < E_c$ and $v^2 < E_c$:

$$\Delta P \leq \frac{c_P}{A} \leq p^* + \tilde{X}(u) \leq h - \frac{c_P}{B} = \pi R_q^2 \leq h - \Delta P, \qquad |\tilde{Y}(v)| < \theta_c = \tfrac{1}{2} - \Delta\theta.$$

(c) For all u, v with $u^2 < E_c$ and $v^2 < E_c$: $|a_0 u - \tilde{X}(u)| \leq D_x$ and $|b_0 v - \tilde{Y}(v)| \leq D_y$.
(d) $(1 - \delta_s)^2 \mathcal{A}_J(E_c) > p^\dagger$, and $\mathcal{A}_s(\bar{E}) \geq p^\dagger$ for every $s \in [0, 1]$.

Proof. (a) From the formula for X, $|X(u)| \leq (|m_-|u^2 + m_+|u|\sqrt{u^2 + 2\eta_{\mathrm{w}} r})/r^2 = M_X(u^2)$, and $|\tilde{X}(u)| \leq |X(u)| + 2\varepsilon_R \leq \tilde{M}_X(u^2)$ by Lemma 5.13(iv); M_X is increasing. If $u^2 \leq \bar{E}$, then $a_0|u| \leq a_0\sqrt{\bar{E}} \leq \bar{X}$ and $|\tilde{X}(u)| \leq \tilde{M}_X(\bar{E}) \leq \bar{X}$. If $v^2 \leq \bar{E}$, then $|\tilde{Y}(v)| \leq b_0|v| \leq b_0\sqrt{\bar{E}} = \bar{Y}$ by Lemma 5.14(e).

(b) $\tilde{R}(\tilde{X}(u)) = u^2 < E_c$, so (7) gives $\mathcal{R}_{\mathrm{tent}}(p^* + \tilde{X}(u)) < E_c + \delta' = h - c_P$. As $0 < c_P < h$ (Lemma 5.28), (5) with $\beta = h - c_P$ and $h/A = a$, $h/B = h - a$ (Lemma 3.9) gives

$$p^* + \tilde{X}(u) \in \big[a - (h - c_P)/A,\ a + (h - c_P)/B\big] = \big[c_P/A,\ h - c_P/B\big].$$

Since $\Delta P = c_P/\max(A, B)$ is at most c_P/A and c_P/B, the outer inequalities follow, and $h - c_P/B = \pi R_q^2$ by definition. Next $\tilde{V}(\tilde{Y}(v)) = v^2 < E_c = h - 2e$, so $|\tilde{Y}(v)| < \theta_c$ by Lemma 5.14(c).

(c) $|a_0 u - \tilde{X}(u)| \leq a_0|u| + |\tilde{X}(u)| \leq a_0\sqrt{E_c} + \tilde{M}_X(E_c) = D_x$ by (a), and $|b_0 v - \tilde{Y}(v)| \leq b_0\sqrt{E_c} + \theta_c = D_y$ by (b).

(d) By Lemmas 5.23 and 5.28, $(1 - \delta_s)^2 \mathcal{A}_J(E_c) \geq (1 - \delta_s)^2(E_c - d_0) > p^\dagger$. By Lemma 5.26(c), $\mathcal{A}_s(\bar{E}) \geq \kappa_J \bar{E} - d_0 = p^\dagger$. □

The core $D(p_{\mathrm{on}})$, on which the isotopy ends at $C_{\mathrm{ph}} f$, must contain the region $\{I \leq h - 4e\}$ where the comparison is used, and the cutoff must be complete before the edge $p^\dagger$ of the source disc; the levels below satisfy $h - 4e < p_{\mathrm{on}} < p_{\mathrm{zero}} < p_+ < p^\dagger$.

Definition 5.30 (Cutoff levels)**.** Put $p_{\mathrm{on}} = h - \frac{31e}{8}$ and $p_{\mathrm{zero}} = h - \frac{61e}{16}$. The source cutoff of §4.7 is then $\chi(I) = 1 - S\big((I - p_{\mathrm{on}})/(p_{\mathrm{zero}} - p_{\mathrm{on}})\big)$.

Lemma 5.31 (Cutoff levels)**.** *$0 < h - 4e < p_{\mathrm{on}} < p_{\mathrm{zero}} < p_+ < p^\dagger$; $\chi = 1$ on $I \leq p_{\mathrm{on}}$ and $\chi = 0$ on $I \geq p_{\mathrm{zero}}$.*

Proof. $h - 4e > 0$ since $e \leq h/100$; $4 > \frac{31}{8} > \frac{61}{16} > \frac{15}{4}$, and $p_+ < p^\dagger$ by Lemma 5.28. The values of χ follow from $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$. □

Definition 5.32 (Constants of the path)**.** Put $P_{\min} = \frac{1}{2}\min(p^*, \Delta P)$ and $P_{\max} = \max(\frac{3}{2}p^*, h)$, and let M_C, $R^\dagger$, R_h, r_b^+, R_{supp}, P_{supp} and p_{bd} be the derived constants of Definition 4.11, formed with $p^*_{\mathrm{lo}} = p^*_{\mathrm{hi}} = p^*$ (so $r_b^+ = r_b$), the numbers ΔP, $p^\dagger$, R_q of Definition 5.27, and the numbers B_3, B^*, $c = c_{\mathrm{new}}$ below. Let $\mathbf{A}_0 = DC_{\mathrm{ph}}(0)\,\mathbf{L} = \mathrm{diag}\big(-\frac{c_x}{2\pi r_b},\ -2\pi r_b c_y\big)$ and put

$$\lambda_0 = \min\Big(\frac{1}{2},\ \frac{p^*}{2\bar{X}},\ \frac{1}{4\bar{Y}}\Big), \qquad B_3 = R^\dagger \max\big(1, |(\mathbf{A}_0)_{11}|, |(\mathbf{A}_0)_{22}|\big),$$

$$B^* = \min\Big(M_C\sqrt{\bar{X}^2 + \bar{Y}^2},\ \frac{\bar{X}}{\sqrt{\pi p^*}} + 2\pi r_b \bar{Y}\Big), \qquad c_{\mathrm{new}} = \min\Big(\frac{1}{4},\ \frac{R_h - r_b}{8\max(R^\dagger, B_3, B^*)}\Big).$$

Lemma 5.33 (Isotopy data for the equal-ball pair)**.** *For the admissible potential pair $U(p) = \tilde{R}(p-p^*)$, $V = \tilde{V}$ of Lemma 5.15, the numbers*

$$p_{\mathrm{on}},\ p_{\mathrm{zero}},\ p^{\dagger},\ p^*_{\mathrm{lo}} = p^*_{\mathrm{hi}} = p^*,\ E_c,\ \Delta P,\ \Delta\theta,\ R_q,\ D_x,\ D_y,\ \delta_s,\ \bar{E},\ \bar{X},\ \bar{Y},\ \lambda_0,\ B_3,\ B^*,\ c = c_{\mathrm{new}}$$

of Definitions 5.21, 5.27, 5.30 and 5.32 are isotopy data in the sense of Definition 4.11. In the notation of that definition, $X_y = \tilde{X}$, $Y_y = \tilde{Y}$, $\mathcal{A}_y = \mathcal{A}_J$, $\mathcal{A}_{y,s} = \mathcal{A}_s$ and $\mathbf{A}_y = \mathbf{A}_0$.

Proof. By Lemma 5.15, $X_y = \tilde{X}$, $Y_y = \tilde{Y}$, $a_0(y) = a_0$, $b_0(y) = b_0$ and $\mathbf{L}_y = \mathbf{L}$; hence the density $J_{y,s}$ of Definition 4.8(a) is J_s, so $\mathcal{A}_{y,s} = \mathcal{A}_s$ and $\mathcal{A}_{y,1} = \mathcal{A}_J$; and by (4) and Definition 5.19, $\mathbf{A}_y = DC_{\mathrm{ph}}(0)\mathbf{L} = \mathbf{A}_0$.

Positivity. $p_{\mathrm{on}} > h - 4e > 0$ (Lemma 5.31), and p_{zero}, $p^{\dagger}$ are larger; $p^* > 0$ (Lemma 5.8); $E_c = h - 2e > 0$; $\Delta P, \Delta\theta, D_x, D_y, \delta_s > 0$ (Lemma 5.28); $R_q > 0$ because $c_P < h$ and $B = h/(h-a) > 1$; $\bar{E} > 0$, hence $\bar{X} \geq a_0\sqrt{\bar{E}} > 0$ and $\bar{Y} > 0$; so $\lambda_0 > 0$, $B_3 \geq R^{\dagger} > 0$ and $B^* > 0$; and $c_{\mathrm{new}} > 0$ because $R_h > r_b$, i.e. $p^* < h$.

(D1): $p_{\mathrm{on}} < p_{\mathrm{zero}} < p^{\dagger}$ by Lemma 5.31, and $p^* < h$ by Lemma 5.8.

(D2): Lemma 5.29(b), since $|\tilde{Y}(v)| < \theta_c = \frac{1}{2} - \Delta\theta$.

(D3): Lemma 5.29(c).

(D4): $\delta_s \leq \frac{1}{4}$, $\delta_s D_x \leq \frac{1}{2}\Delta P$ and $\delta_s D_y \leq \frac{1}{2}\Delta\theta$ because δ_s is the minimum in Definition 5.27; $(1-\delta_s)^2\mathcal{A}_J(E_c) > p^{\dagger}$ is Lemma 5.29(d).

(D5): $\mathcal{A}_s(\bar{E}) \geq p^{\dagger}$ for $s \in [0,1]$ is Lemma 5.29(d), and the bounds for $u^2 \leq \bar{E}$, $v^2 \leq \bar{E}$ are Lemma 5.29(a).

(D6): $\lambda_0 \leq \frac{1}{2}$, $\lambda_0\bar{X} \leq \frac{1}{2}p^*$ and $\lambda_0\bar{Y} \leq \frac{1}{4}$ because λ_0 is the minimum in Definition 5.32.

(D7): with $p^*_{\mathrm{lo}} = p^*_{\mathrm{hi}} = p^*$, $r_b^+ = r_b$, $B_{\mathrm{lin}} = B_3$ and $\mathbf{A}_y = \mathbf{A}_0$, the three conditions hold with equality by the definitions of B_3, B^* and c_{new}; and $P_{\min}, P_{\max}$ satisfy the range conditions of Definition 4.11 with equality. □

5.7. **The isotopy and the comparison on the closed disc.** Let $(\hat{q}_t)_{t\in\mathbb{R}}$ and $(Q_t)_{t\in\mathbb{R}}$ be the isotopy and the Hamiltonian that Theorem 4.21 assigns to the pair of Lemma 5.15 with the data of Lemma 5.33 (Lemma 4.18 and Construction 4.19), and let P_{supp} be the derived constant of Definition 5.32.

Theorem 5.34 (The explicit inner map and its isotopy)**.** *Let $k \geq 10$ and $t_0 \in (\frac{7}{8}, 1)$, with h, $\mathcal{R}_{\mathrm{tent}}$ and e_q as defined in §5.1. Let $q_t = \hat{q}_t$ ($t \in \mathbb{R}$) and $q = q_1$. The family (q_t), its Hamiltonian (Q_t) and the number P_{supp} have the following properties.*

- *(a) Each q_t is a symplectic diffeomorphism of $\mathbb{C}$, and $(t,z) \mapsto q_t(z)$, $(t,\zeta) \mapsto q_t^{-1}(\zeta)$ are C^∞ on $\mathbb{R}\times\mathbb{C}$; $q_t = \mathrm{id}$ for $t \leq 0$ and $q_t = q_1$ for $t \geq 1$.*
- *(b) $P_{\mathrm{supp}} < h$; for every t, $q_t = \mathrm{id}$ on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$, and q_t maps $\bar{D}(P')$ onto itself for every $P' \geq P_{\mathrm{supp}}$, in particular $\bar{D}(h)$ and $D(h)$ onto themselves.*
- *(c) Q is C^∞ on $\mathbb{R}\times\mathbb{C}$, Q_t vanishes with all its derivatives on $\{\pi|\zeta|^2 > P_{\mathrm{supp}}\}$ for every t, $Q_t = 0$ for $t \notin (0,1)$, and q_t is the isotopy generated by Q_t: $\partial_t q_t = \Xi_t \circ q_t$ with $\iota_{\Xi_t}\omega_2 = -dQ_t$.*
- *(d) q equals $C_{\mathrm{ph}} \circ f$ on $\bar{D}(p_{\mathrm{on}})$, $p_{\mathrm{on}} = h - \frac{31}{8}e_q$; in particular on $\bar{D}(h - 4e_q)$.*
- *(e) For every $z \in \bar{D}(h)$, with $I = \pi|z|^2$,*

$$\mathcal{R}_{\mathrm{tent}}\big(p(q(z))\big) \leq I + 4e_q. \tag{10}$$

Proof. By Lemmas 5.15 and 5.33, Theorem 4.21 applies with $d = 0$.

(a) Symplecticity, bijectivity and the behaviour for $t \leq 0$ and $t \geq 1$ are Theorem 4.21(a),(b); joint smoothness of $(t,z) \mapsto q_t(z)$ and $(t,\zeta) \mapsto q_t^{-1}(\zeta)$ is Lemma 4.18(e) (equivalently Theorem 4.23(a)).

(b) $P_{\mathrm{supp}} < h$ is Lemma 4.12(b) (also part of Theorem 4.21); the rest is Theorem 4.21(a).

(c) Smoothness of Q on $\mathbb{R}\times\mathbb{C}$ is Proposition 4.20 (equivalently Theorem 4.23(a)); the support properties and the generating equation are Theorem 4.21(d), with $\Xi_t = \xi_{y,t}$.

(d) By Theorem 4.21(c), $q(z) = q_{\mathrm{in},y}(z)$ for $\pi|z|^2 \le p_{\mathrm{on}}$, and by Definition 4.6, Definition 5.17 and Definition 5.19, $q_{\mathrm{in},y} = C_{\mathrm{ph}} \circ f$ on $D(I_{\mathrm{rect}}(y))$, which contains $\bar{D}(p_{\mathrm{on}})$ because $p_{\mathrm{on}} < p^\dagger < I_{\mathrm{rect}}(y)$ (Lemmas 5.31 and 4.12(b)). Since $h - 4e_q < p_{\mathrm{on}}$ (Lemma 5.31), this is (d).

(e) Apply Corollary 4.22 with $\Upsilon = \mathcal{R}_{\mathrm{tent}}|_{[0,h]}$, $g(I) = I + 4e_q$ and $\varepsilon = 0$. If $\pi|z|^2 \le p_{\mathrm{on}}$, then $\pi|z|^2 < p_+$ (Lemma 5.31), so by (d) and Proposition 5.24, $\Upsilon(\pi|q_{\mathrm{in},y}(z)|^2) < \pi|z|^2 + 13e_q/24 \le g(\pi|z|^2)$. Since $\mathcal{R}_{\mathrm{tent}}$ is convex, $\sup_{[0,h]} \Upsilon = \max(\mathcal{R}_{\mathrm{tent}}(0), \mathcal{R}_{\mathrm{tent}}(h)) = h < p_{\mathrm{on}} + 4e_q = g(p_{\mathrm{on}})$. The corollary gives (10). □

The bound $4e_q$ is needed only on the annulus $\{p_{\mathrm{on}} < \pi|z|^2 \le h\}$, on which only $q(\bar{D}(h)) = \bar{D}(h)$ is used; on the core the error is below $13e_q/24$. With c_s and g as in §M3.4, the error costs $4c_s e_q \le g/32$ in Proposition M3.8 (Lemma 3.11): the gap drops from g to $\frac{31}{32}g$, a relative loss of at most $1/32$, uniformly in k and t_0.

6. The scalar comparison datum for the cubic model

On an interval the comparison construction acts directly on continuous convex functions, and no polyhedral bookkeeping is needed. This section gives the scalar iteration with its stopping rule, a fixed mollifier, the planar potentials and the Hamiltonian realiser, and then substitutes the parameters of the four-dimensional cubic model for arbitrary $k \ge 1$ and arbitrary (possibly unequal) capacities. The iteration is the scalar instance of [Ope26, §2]. We use $D(A) = \{z \in \mathbb{C} : \pi|z|^2 < A\}$, the standard area form $\omega_{\mathbb{C}}$, and the flat step S of Definition 2.3; fibre Hamiltonians satisfy $\iota_{X_Q}\omega_{\mathbb{C}} = -dQ$.

6.1. **Finite iteration and fixed mollification.** Let $h > 0$ and let $P\colon \mathbb{R} \to \mathbb{R}$ be continuous and concave on $[0,h]$. Suppose that

$$\int_0^h P(p)\,dp = \mu h, \qquad 0 < \tau < 1, \qquad P(p) \ge \beta > 0 \quad (\tau h \le p \le h), \tag{11}$$
$$0 < d < \min(\mu, \beta), \qquad P_{\min} \le P(p), \qquad |P(p)| \le \bar{P},$$
$$|P(p) - P(q)| \le L_P|p - q| \quad (p, q \in [0,h]),$$

where $\bar{P} > 0$ and $L_P \ge 0$. For a continuous convex f on $[0,h]$, write

$$(Rf)(I) = \min_{0 \le x \le h-I} \max\bigl(f(x), f(x+I)\bigr), \qquad 0 \le I \le h. \tag{12}$$

This is the coordinate rearrangement of §3 for $m = 1$ and $\Delta = [0,h]$ (whose proofs do not use rationality). By Lemmas 3.1, 3.2 and 3.7, Rf is the increasing rearrangement of f; it is continuous, convex and nondecreasing, $Rf(0) = \min f$, $Rf(h) = \max f$; $\int_0^h Rf = \int_0^h f$; $R(f + c) = Rf + c$; $f \le f'$ implies $Rf \le Rf'$; $\|Rf - Rf'\| \le \|f - f'\|$; and Rf is L-Lipschitz whenever f is.

Set

$$\eta_{\mathrm{c}} = (1-\tau)(\mu - d), \qquad B_P = \max(1, d - P_{\min} + 1), \qquad \lambda = 1 - \frac{\eta_{\mathrm{c}}}{2(B_P + \eta_{\mathrm{c}})}, \tag{13}$$
$$K_0 = 0, \qquad K_{r+1} = R(\lambda K_r - P), \qquad M = \frac{L_P}{1-\lambda},$$
$$N = \left\lfloor \frac{\log^+\bigl(16\bar{P}/[d(1-\lambda)]\bigr)}{1-\lambda} \right\rfloor + 1, \qquad \log^+ x = \max(0, \log x).$$

Since $\eta_c > 0$ and $B_P \geq 1$, $0 < 1-\lambda < \frac{1}{2}$. The integer N is explicit. When P is piecewise affine, as in §6.5, each K_r is a continuous convex piecewise affine function with finitely many knots, computed from K_{r-1} by Lemma 3.8 (whose proof holds verbatim for real knots and values).

Lemma 6.1 (Finite scalar gap)**.** *Each K_r is continuous, convex, nondecreasing and M-Lipschitz on $[0,h]$, and*

$$R(K_N - P)(I) \leq K_N(I) - \frac{7}{8}d \qquad (0 \leq I \leq h). \tag{14}$$

Proof. By induction, $\lambda K_r - P$ is continuous, convex and $(\lambda M + L_P)$-Lipschitz, and $\lambda M + L_P = M$; so $K_{r+1} = R(\lambda K_r - P)$ is continuous, convex, nondecreasing and M-Lipschitz. Nonexpansiveness gives $\|K_1 - K_0\| = \|R(-P) - R(0)\| \leq \bar{P}$ and $\|K_{r+1} - K_r\| \leq \lambda \|K_r - K_{r-1}\|$, hence

$$\|K_{r+1} - K_r\|_{[0,h]} \leq \lambda^r \bar{P}. \tag{15}$$

Integral preservation gives $\int_0^h K_{r+1} = \lambda \int_0^h K_r - \mu h$, so

$$\int_0^h K_r = -\frac{\mu h(1-\lambda^r)}{1-\lambda}, \qquad (1-\lambda)K_r(0) \leq -\mu(1-\lambda^r),$$

the second because the nondecreasing K_r is at most its mean at 0.

We next prove

$$(1-\lambda)K_r(h) \leq \lambda^r \bar{P} - d. \tag{16}$$

We have $\mu \leq \bar{P}$ by integration, and

$$\frac{\lambda \eta_c}{1-\lambda} = 2B_P + \eta_c > d - P_{\min}, \qquad \eta_c \leq \mu - d.$$

Consequently $(1-\lambda)(d - P_{\min}) \leq \lambda(\mu - d)$, or $\lambda\mu + (1-\lambda)P_{\min} \geq d$. The case $r = 0$ of (16) follows from $d < \mu \leq \bar{P}$. For the induction step, $Rf(h) = \max f$ and convexity of $\lambda K_r - P$ give

$$K_{r+1}(h) = \max\{\lambda K_r(0) - P(0), \lambda K_r(h) - P(h)\}.$$

After multiplication by $1-\lambda$, the first entry is bounded above by $-\lambda\mu + \lambda^{r+1}\mu - (1-\lambda)P_{\min} \leq \lambda^{r+1}\bar{P} - d$. The second is bounded above by $\lambda^{r+1}\bar{P} - \lambda d - (1-\lambda)P(h) \leq \lambda^{r+1}\bar{P} - d$, since $P(h) \geq \beta > d$. This proves (16).

Since $\lambda \leq e^{-(1-\lambda)}$ and $(1-\lambda)N > \log^+\big(16\bar{P}/[d(1-\lambda)]\big)$,

$$\lambda^N \bar{P} < \bar{P} \exp\Big(-\log^+ \frac{16\bar{P}}{d(1-\lambda)}\Big) \leq \frac{d(1-\lambda)}{16} \leq \frac{d}{16}.$$

Finally, $(1-\lambda)K_N \leq (1-\lambda)K_N(h)$, order preservation and translation invariance give

$$\begin{aligned} R(K_N - P) &\leq R(\lambda K_N - P) + (1-\lambda)K_N(h) \\ &= K_{N+1} + (1-\lambda)K_N(h) \leq K_N + 2\lambda^N \bar{P} - d \leq K_N - \frac{7}{8}d, \end{aligned}$$

using (15) and (16). □

Define the extension and fixed mollifier by

$$K^{\mathrm{ext}}(p) = \min_{0 \leq x \leq h} \big(K_N(x) + M|p-x|\big), \qquad b(t) = \begin{cases} \exp\big(-1/(1-t^2)\big), & |t| < 1, \\ 0, & |t| \geq 1, \end{cases} \tag{17}$$

$$b_0 = \int_{-1}^{1} b(t)\,dt, \qquad w = \frac{d}{32(M+1)}, \qquad K(p) = \frac{1}{b_0}\int_{-1}^{1} K^{\mathrm{ext}}(p - wt)b(t)\,dt.$$

Proposition 6.2 (Smooth scalar comparator)**.** *The function K of* (17) *is smooth, convex and M-Lipschitz on $\mathbb{R}$. Moreover,*

$$\|K - K_N\|_{[0,h]} \le \frac{d}{32}, \qquad R(K-P)(I) \le K(I) - \frac{13}{16}d \quad (0 \le I \le h). \tag{18}$$

Proof. For $p \in [0,h]$, $K_N(x) + M|p-x| \ge K_N(p)$ for all $x \in [0,h]$ because K_N is M-Lipschitz, with equality at $x = p$; so $K^{\mathrm{ext}} = K_N$ on $[0,h]$. As an infimum of M-Lipschitz functions of p, K^{ext} is M-Lipschitz. It is convex, being the partial minimum over the compact convex set $[0,h]$ of the jointly convex function $(p,x) \mapsto K_N(x) + M|p-x|$. Convolution with the smooth compactly supported probability density $t \mapsto b(t)/b_0$ preserves convexity and the Lipschitz constant and produces a smooth function. Since $|t| \le 1$ on the support, $|K(p) - K^{\mathrm{ext}}(p)| \le Mw < d/32$. Nonexpansiveness and Lemma 6.1 now give

$$R(K-P) \le R(K_N - P) + d/32 \le K_N - 7d/8 + d/32 \le K - 13d/16.$$

□

6.2. **The scalar planar potentials.** Let F be continuous, convex and L-Lipschitz on $[0,h]$, where $L \ge 0$, and let $e > 0$. Suppose that $U_0 \colon \mathbb{R} \to \mathbb{R}$ is smooth and convex, with

$$F \le U_0 \le F + \sigma_{\mathrm{in}} \quad \text{on } [0,h], \qquad 0 \le \sigma_{\mathrm{in}} \le e/16. \tag{19}$$

Define

(20)

$$\eta = \frac{1}{2}\min\left(\frac{h}{16}, \frac{e}{32(L+1)}\right), \qquad \beta_{\mathrm{a}} = \frac{\eta}{8h}, \qquad \sigma = \frac{e}{8}, \qquad \nu = 4\sigma\beta_{\mathrm{a}}(1-\beta_{\mathrm{a}}), \qquad \kappa = \frac{\sigma\eta}{8},$$

$$U(p) = U_0(p) + \kappa\left(\frac{1}{p} + \frac{1}{h-p}\right) \quad (0 < p < h), \qquad V(y) = \frac{\nu y^2}{\frac{1}{4} - y^2} \quad (|y| < \tfrac{1}{2}).$$

Then $0 < \eta \le h/32$, $L\eta \le e/64$, $0 < \beta_{\mathrm{a}} \le 1/256$, and $\kappa, \nu > 0$. Here p is the moment and y the angle of the rectangle $(0,h) \times (-\frac{1}{2}, \frac{1}{2})$.

Lemma 6.3 (Area below a scalar potential level)**.** *U and V are smooth and strictly convex on their intervals and tend to $+\infty$ at both endpoints. Hence U has a unique critical point $p^* \in (0,h)$, its minimum, with $U''(p^*) > 0$; we write $m_* = U(p^*)$. Also $V(0) = 0$ is the minimum of V and $V''(0) = 8\nu$. For $0 \le I \le h - 4\eta$, put $b_I = RF(I + 3\eta) + 2\sigma$. Then*

$$\mathrm{area}\{(p,y) \in (0,h) \times (-\tfrac{1}{2}, \tfrac{1}{2}) : U(p) + V(y) \le b_I\} > I. \tag{21}$$

Proof. On $(0,h)$, $U'' = U_0'' + 2\kappa(p^{-3} + (h-p)^{-3}) > 0$, and $U \to +\infty$ at 0 and h because U_0 is bounded on $[0,h]$; so U' is strictly increasing from $-\infty$ to $+\infty$ and has a unique zero p^*. Writing

$$V(y) = \frac{\nu}{4}\left(\frac{1}{y + 1/2} + \frac{1}{1/2 - y}\right) - \nu$$

shows that V is strictly convex and proper, and $V'(y) = (\nu/2)y/(\frac{1}{4} - y^2)^2$ gives $V''(0) = 8\nu$.

On $[\eta, h-\eta]$ the barrier is at most $2\kappa/\eta = \sigma/4$, so $U \le F + \sigma_{\mathrm{in}} + \sigma/4 \le F + 3\sigma/4$ there. For $|y| \le \frac{1}{2} - \beta_{\mathrm{a}}$, $\frac{1}{4} - y^2 \ge \beta_{\mathrm{a}}(1-\beta_{\mathrm{a}})$ and $V(y) \le \nu/(4\beta_{\mathrm{a}}(1-\beta_{\mathrm{a}})) = \sigma$.

Since $I + 3\eta \le h$, Lemma 3.1(a) gives an interval $[a, a + I + 3\eta] \subset [0,h]$ on which $F \le RF(I + 3\eta)$. Its intersection with $[\eta, h-\eta]$ has length at least $I + \eta$, and its product with $[-\frac{1}{2} + \beta_{\mathrm{a}}, \frac{1}{2} - \beta_{\mathrm{a}}]$ lies in the set of (21), because $\frac{3}{4}\sigma + \sigma < 2\sigma$. This rectangle has area at least

$$(I + \eta)(1 - 2\beta_{\mathrm{a}}) = I + \eta - \frac{\eta(I+\eta)}{4h} > I,$$

since $I + \eta < 4h$. □

6.3. **Quadratic wells and the endpoint map.** Both potentials are modified near their minima to be exactly quadratic. Let $\psi(t) = 1 - S((t^2-1)/3)$; it is smooth and even, $\psi = 1$ on $[-1,1]$, $\psi = 0$ outside $(-2,2)$, $0 \le \psi \le 1$, and $C_\psi = \max_{[-2,2]} |\psi'|$ is finite. Let f be a smooth function on an interval (A,B) and $c \in (A,B)$ with $f'(c) = 0$, $\alpha = f''(c) > 0$, $(x-c)f'(x) > 0$ for $x \ne c$, and $f(x) \to +\infty$ as $x \to A$ and as $x \to B$. Put

(22)
$$\delta_{\mathrm{w}} = e/64, \qquad g_f(x) = f(c+x) - f(c), \qquad r_f = \tfrac{1}{4}\min(c-A, B-c), \qquad \chi_f(x) = \psi(x/r_f),$$
$$K_{3,f} = 1 + \max_{x\in[-2r_f,2r_f]} |(\chi_f g_f)'''(x)|, \qquad \varepsilon_f = \min\left(\frac{r_f}{4}, \frac{\alpha}{32K_{3,f}(1+C_\psi)}, \sqrt{\frac{\delta_{\mathrm{w}}}{\alpha}}\right),$$
$$\widetilde{f}(x) = g_f(x) + \psi(x/\varepsilon_f)\big(\alpha x^2/2 - \chi_f(x)g_f(x)\big) \qquad (A-c < x < B-c).$$

Here $\chi_f g_f$ is extended by zero outside $[-2r_f, 2r_f] \subset (A-c, B-c)$; the extension is smooth. Since $2\varepsilon_f \le r_f/2$, $\chi_f = 1$ on the support of $\psi(x/\varepsilon_f)$, so $\widetilde{f} = g_f + \psi(x/\varepsilon_f)(\alpha x^2/2 - g_f)$, and $K_{3,f} > \max_{|x|\le 2\varepsilon_f} |g_f'''(x)|$.

Lemma 6.4 (Quadratic modification of a well)**.** *Let f, c, α be as above, and let $\varepsilon > 0$, $K > 0$, $\delta > 0$ satisfy $2\varepsilon < \min(c-A, B-c)$, $|g_f'''| \le K$ on $[-2\varepsilon, 2\varepsilon]$, $\varepsilon \le \alpha/(32K(1+C_\psi))$ and $\varepsilon \le \sqrt{\delta/\alpha}$. Put $\widetilde{f}(x) = g_f(x) + \psi(x/\varepsilon)(\alpha x^2/2 - g_f(x))$ on $(A-c, B-c)$. Then $\widetilde{f}$ is smooth and*

$$|\widetilde{f} - g_f| \le \delta/24, \qquad \widetilde{f}(x) = \alpha x^2/2 \quad (|x| \le \varepsilon), \qquad x\widetilde{f}'(x) > 0 \quad (x \ne 0), \qquad \widetilde{f} = g_f \quad (|x| \ge 2\varepsilon).$$

The map $\Sigma_f(x) = \operatorname{sign}(x)\sqrt{\widetilde{f}(x)}$ is an increasing smooth diffeomorphism of $(A-c, B-c)$ onto $\mathbb{R}$, and its inverse X_f satisfies

$$\widetilde{f}(X_f(u)) = u^2, \qquad X_f'(u) > 0, \qquad X_f(u) = u/\sqrt{\alpha/2} \quad (|u| \le \sqrt{\alpha/2}\,\varepsilon).$$

In particular this applies to (22) with $(\varepsilon, K, \delta) = (\varepsilon_f, K_{3,f}, \delta_{\mathrm{w}})$, and to $(f, A, B, c) = (U, 0, h, p^*)$ and $(V, -\frac{1}{2}, \frac{1}{2}, 0)$; we write $\widetilde{U}$, $\widetilde{V}$ for the two modifications.

Proof. Since $g_f(0) = g_f'(0) = 0$ and $g_f''(0) = \alpha$, Taylor's theorem gives, for $|x| \le 2\varepsilon$,

$$|g_f(x) - \alpha x^2/2| \le K|x|^3/6, \qquad |g_f'(x) - \alpha x| \le Kx^2/2.$$

Hence $|\widetilde{f} - g_f| \le K(2\varepsilon)^3/6 = \frac{4}{3}K\varepsilon\cdot\varepsilon^2 \le \frac{4}{3}\cdot\frac{\alpha}{32}\cdot\frac{\delta}{\alpha} = \delta/24$. Differentiating, $\widetilde{f}'(x) - \alpha x = (1-\psi(x/\varepsilon))(g_f'(x) - \alpha x) + \varepsilon^{-1}\psi'(x/\varepsilon)(\alpha x^2/2 - g_f(x))$, and for $0 < |x| < 2\varepsilon$ (where $|x|/\varepsilon < 2$)

$$|\widetilde{f}'(x) - \alpha x| \le Kx^2\Big(\frac{1}{2} + \frac{C_\psi}{3}\Big) \le K|x|\cdot 2\varepsilon(1+C_\psi) \le \frac{\alpha|x|}{16}.$$

So $x\widetilde{f}'(x) > 0$ there; for $|x| \ge 2\varepsilon$, $\widetilde{f} = g_f$ and $xg_f'(x) > 0$ by hypothesis. As $\widetilde{f}(0) = 0$, $\widetilde{f} > 0$ for $x \ne 0$. On $|x| < \varepsilon$, $\Sigma_f(x) = \sqrt{\alpha/2}\,x$; on $x \ne 0$, Σ_f is smooth with $\Sigma_f' = \widetilde{f}'/(2\Sigma_f) > 0$. Since $\widetilde{f} = g_f \to +\infty$ at both ends, Σ_f is an increasing bijection onto $\mathbb{R}$ with positive derivative, hence a diffeomorphism; the identities for $X_f = \Sigma_f^{-1}$ follow. For (22), $2\varepsilon_f \le r_f/2$, and the hypotheses on U and V hold by Lemma 6.3: strict convexity gives $(x-c)f'(x) > 0$. □

Let X_U, X_V be the inverses X_f of Lemma 6.4 for $f = U$ and $f = V$, with widths $\varepsilon_U, \varepsilon_V$, so that $\widetilde{U}(X_U(u)) = u^2$ and $\widetilde{V}(X_V(v)) = v^2$, X_U maps $\mathbb{R}$ onto $(-p^*, h-p^*)$ and X_V maps $\mathbb{R}$ onto

$(-\frac12,\frac12)$. Put

$$b_{\rm top}=RF(h-\eta)+2\sigma,\qquad \bar E=b_{\rm top}-m_*+2\delta_{\rm w},\qquad u_1=\sqrt{\bar E}+1,\tag{23}$$
$$\varphi(t)=tS(t),\qquad Z(u)=\varphi(u-u_1)-\varphi(-u-u_1),$$
$$X(u)=X_U(u)+Z(u),\qquad Y(v)=X_V(v)+Z(v),\qquad a=\sqrt{2/U''(p^*)},\qquad b=\sqrt{2/V''(0)},$$
$$\varepsilon_{\rm M}=\min\bigl(\sqrt{U''(p^*)/2}\,\varepsilon_U,\sqrt{V''(0)/2}\,\varepsilon_V,1\bigr).$$

By (21) with $I=h-4\eta>0$ the sublevel $\{U+V\le b_{\rm top}\}$ is nonempty, so $b_{\rm top}\ge m_*$ and $\bar E>0$. The function φ is smooth, vanishes on $(-\infty,0]$, equals t on $[1,\infty)$ and has $\varphi'=S+tS'\ge0$; so Z is smooth, nondecreasing, zero on $[-u_1,u_1]$ and unbounded in both directions. Consequently X,Y are smooth increasing diffeomorphisms of $\mathbb R$ with positive derivatives, $X=X_U$ and $Y=X_V$ on $[-u_1,u_1]$, and $X(u)=au$, $Y(v)=bv$ on $[-\varepsilon_{\rm M},\varepsilon_{\rm M}]$.

We use the rotation normalisation T_J and the area normalisation D_J of a density J from Definition 4.3: with $\mathrm{Rot}_\theta\, w=e^{i\theta}w$,

$$\bar J(t)=\frac{1}{2\pi}\int_0^{2\pi}J(\mathrm{Rot}_\theta\sqrt{t})\,d\theta,\qquad B_J(w)=\frac{1}{2\pi}\int_0^{2\pi}(\theta-\pi)J(\mathrm{Rot}_\theta\, w)\,d\theta,\tag{24}$$
$$T_J(w)=\mathrm{Rot}_{B_J(w)/\bar J(|w|^2)}\,w,\qquad \mathcal A_J(E)=\pi\int_0^E\bar J(t)\,dt,\qquad E_J=\mathcal A_J^{-1},$$
$$D_J(z)=\sqrt{\frac{\pi E_J(I)}{I}}\,z\quad(I=\pi|z|^2>0),\qquad D_J(0)=0.$$

If J is smooth and positive on $\mathbb R^2$, constant near 0, and $\mathcal A_J(E)\to\infty$, then by Lemmas 4.4 and 4.5 T_J is a diffeomorphism of $\mathbb R^2$ preserving each circle about 0 with $T_J^*(\bar J(|w|^2)\,du\wedge dv)=J\,du\wedge dv$, D_J is a diffeomorphism of $\mathbb C$ with $|D_J(z)|^2=E_J(\pi|z|^2)$ and $D_J^*(\bar J(|w|^2)\,du\wedge dv)=\omega_{\mathbb C}$, and both maps are linear near 0.

Define

$$F_{\rm M}(u+iv)=X(u)+iY(v),\qquad J(u+iv)=X'(u)Y'(v),\qquad f=F_{\rm M}\circ T_J^{-1}\circ D_J.\tag{25}$$

Here $J>0$ is smooth and equals ab for $|u|,|v|\le\varepsilon_{\rm M}$, and $F_{\rm M}$ is a diffeomorphism of $\mathbb R^2$ with $F_{\rm M}^*(dx\wedge dy)=J\,du\wedge dv$. Writing $\varrho_X(x)=(X^{-1}(x))^2$ and $\varrho_Y(y)=(Y^{-1}(y))^2$, $F_{\rm M}$ maps $\{u^2+v^2\le E\}$ onto $\{\varrho_X(x)+\varrho_Y(y)\le E\}$, so

$$\mathcal A_J(E)=\mathrm{area}\{\varrho_X(x)+\varrho_Y(y)\le E\}\longrightarrow\infty,\qquad \varrho_X(\mathrm{Re}\,f(z))+\varrho_Y(\mathrm{Im}\,f(z))=E_J(\pi|z|^2),\tag{26}$$

the second because T_J^{-1} preserves $|w|$. Hence f is an area-preserving diffeomorphism of $\mathbb C$ (as in Lemma 4.5(iv)). If $\varrho_X(x)<u_1^2$, then $X^{-1}(x)\in(-u_1,u_1)$, so $X^{-1}(x)=X_U^{-1}(x)$ and $\varrho_X(x)=\widetilde U(x)$; likewise $\varrho_Y(y)<u_1^2$ implies $\varrho_Y(y)=\widetilde V(y)$. Conversely $\widetilde U(x)<u_1^2$ implies $|X_U^{-1}(x)|<u_1$, hence $\varrho_X(x)=\widetilde U(x)$, and likewise for $\widetilde V$.

The endpoint is

$$P_-=p^*+X_U(-u_1),\qquad P_+=p^*+X_U(u_1),\qquad a_{\rm cut}=P_-/4,\tag{27}$$
$$s_a(p)=p\,S((p-a)/a)+\tfrac a2\bigl(1-S((p-a)/a)\bigr),$$
$$C_a(x+iy)=-\sqrt{s_a(p^*+x)/\pi}\,e^{2\pi iy},\qquad \Phi=C_{a_{\rm cut}}\circ f.$$

Here $0<P_-<p^*<P_+<h$; $s_a(p)=a/2$ for $p\le a$, $s_a(p)=p$ for $p\ge2a$, and in between $s_a(p)$ is a convex combination of $p>a$ and $a/2$; so $s_a>0$ and C_a is smooth on $\mathbb C$.

Lemma 6.5 (Scalar endpoint and its error)**.** *The map Φ is smooth on $\mathbb{C}$, is injective and area preserving on $D(h-4\eta)$, and satisfies*

$$F(\pi|\Phi(z)|^2) < RF(\pi|z|^2) + e \qquad (\pi|z|^2 \le h - 4\eta). \tag{28}$$

On the latter closed disc, $P_- < \pi|\Phi(z)|^2 < P_+$.

Proof. Let $0 \le I \le h-4\eta$, $b_I = RF(I+3\eta) + 2\sigma \le b_{\mathrm{top}}$ (as RF is nondecreasing) and $E' = b_I - m_* + 2\delta_{\mathrm{w}} \le \bar{E}$. Write $g_U(x) = U(p^*+x) - m_*$, so $g_V = V$. If $U(p)+V(y) \le b_I$ and $x = p - p^*$, then $g_U(x) + V(y) \le b_I - m_*$, and Lemma 6.4 gives $\widetilde{U}(x) + \widetilde{V}(y) \le E' < u_1^2$; as both terms are nonnegative, $\varrho_X(x) = \widetilde{U}(x)$, $\varrho_Y(y) = \widetilde{V}(y)$, and $\varrho_X(x) + \varrho_Y(y) \le E'$. Translation preserves area, so by Lemma 6.3 and (26), $\mathcal{A}_J(E') > I$, that is,

$$E_J(I) < b_I - m_* + 2\delta_{\mathrm{w}} \le \bar{E} < u_1^2. \tag{29}$$

Now let $\pi|z|^2 = I$ and $(x,y) = f(z)$. By (26), $\varrho_X(x), \varrho_Y(y) < u_1^2$, so $|X^{-1}(x)|, |Y^{-1}(y)| < u_1$; hence $p^* + x \in (P_-, P_+) \subset (0,h)$, $|y| < \frac{1}{2}$, and $\widetilde{U}(x) + \widetilde{V}(y) = E_J(I)$. By Lemma 6.4, $U(p^*+x) + V(y) \le m_* + E_J(I) + 2\delta_{\mathrm{w}} < b_I + 4\delta_{\mathrm{w}}$. Since $F \le U_0 < U$ on $(0,h)$, $V \ge 0$ and RF is L-Lipschitz (Lemma 3.7),

$$F(p^*+x) < RF(I+3\eta) + 2\sigma + 4\delta_{\mathrm{w}} \le RF(I) + 3L\eta + 2\sigma + 4\delta_{\mathrm{w}} \le RF(I) + \frac{23}{64}e < RF(I) + e.$$

As $p^* + x > P_- = 4a_{\mathrm{cut}}$, $s_{a_{\mathrm{cut}}}(p^*+x) = p^* + x$, so $\pi|\Phi(z)|^2 = p^* + x$ and (28) follows. Thus f maps $D(h-4\eta)$ into the open set $\{(x,y) : p^* + x > 2a_{\mathrm{cut}},\ |y| < \frac{1}{2}\}$, on which $C_{a_{\mathrm{cut}}}$ is the polar chart $(x,y) \mapsto -\sqrt{(p^*+x)/\pi}\, e^{2\pi i y}$; this chart is injective (its polar angle $2\pi y + \pi$ lies in $(0, 2\pi)$) and pulls $\omega_{\mathbb{C}}$ back to $dx \wedge dy$ (Lemma 5.20). Since f is an area-preserving diffeomorphism, Φ is injective and area preserving on $D(h-4\eta)$. Smoothness of Φ on $\mathbb{C}$ follows from that of f and $C_{a_{\mathrm{cut}}}$. □

6.4. **The Alexander path and its Hamiltonian.** The endpoint is joined to its linearisation by the integral Alexander family

$$b_* := \Phi(0) = -r_b, \qquad r_b = \sqrt{p^*/\pi}, \qquad q_\alpha(z) = b_* + \int_0^1 D\Phi(t\alpha z)[z]\, dt \quad (0 \le \alpha \le 1). \tag{30}$$

Indeed $f(0) = F_{\mathrm{M}}(0) = 0$ and $s_{a_{\mathrm{cut}}}(p^*) = p^*$, so $\Phi(0) = -r_b$. Thus $q_\alpha = b_* + (\Phi(\alpha z) - b_*)/\alpha$ for $\alpha > 0$, $q_1 = \Phi$ and $q_0 = b_* + A_0 z$ with $A_0 = D\Phi(0)$. Near 0 the density J is the constant ab, so $T_J = \mathrm{id}$ and $D_J = (ab)^{-1/2}\mathrm{id}$ there (Lemmas 4.4(iv) and 4.5(v)), and $Df(0) = \mathrm{diag}(c_x, c_y)$ with $c_x = a/\sqrt{ab}$, $c_y = b/\sqrt{ab}$, $c_x c_y = 1$. The polar chart has $DC_{a_{\mathrm{cut}}}(0) = \mathrm{diag}(-1/(2\pi r_b), -2\pi r_b)$ (Lemma 5.20), so

$$A_0 = \mathrm{diag}(A_{11}, A_{22}), \qquad A_{11} = -\frac{c_x}{2\pi r_b}, \qquad A_{22} = -2\pi r_b c_y, \qquad A_{11}A_{22} = 1.$$

Let

$$p^\dagger = h - 5\eta, \qquad R^\dagger = \sqrt{p^\dagger/\pi}, \qquad R_h = \sqrt{h/\pi}, \qquad R_q = \sqrt{P_+/\pi}, \tag{31}$$

$$M_\Phi = \max_{\pi|z|^2 \le p^\dagger} \|D\Phi(z)\|, \qquad B_* = M_\Phi R^\dagger, \qquad B_3 = R^\dagger \max(1, |A_{11}|, |A_{22}|),$$

$$c_{\mathrm{sh}} = \min\left(\frac{1}{4}, \frac{R_h - r_b}{8\max(R^\dagger, B_3, B_*)}\right),$$

$$R_{\mathrm{supp}} = \max(R^\dagger, r_b + c_{\mathrm{sh}}B_3, r_b + c_{\mathrm{sh}}B_*, R_q), \qquad P_{\mathrm{supp}} = \pi R_{\mathrm{supp}}^2.$$

M_Φ is the maximum of a continuous function on a compact disc. With

$$M_\alpha = \mathrm{Rot}_{\pi\alpha}\, \mathrm{diag}(|A_{11}|^\alpha, |A_{22}|^\alpha),$$

so that $M_0 = \mathrm{Id}$, $M_1 = A_0$ (both entries of A_0 are negative), $\det M_\alpha = 1$ and $\|M_\alpha\| \le \max(1, |A_{11}|, |A_{22}|)$, the five segments on $D(p^\dagger)$ are

$$\begin{aligned} &\mathfrak{e}^0_\alpha(z) = c_{\mathrm{sh}}^{\alpha} z, && \mathfrak{e}^1_\alpha(z) = \alpha b_* + c_{\mathrm{sh}} z, \\ &\mathfrak{e}^2_\alpha(z) = b_* + c_{\mathrm{sh}} M_\alpha z, \\ &\mathfrak{e}^3_\alpha(z) = b_* + c_{\mathrm{sh}}(q_\alpha(z) - b_*), \\ &\mathfrak{e}^4_\alpha(z) = b_* + c_{\mathrm{sh}}^{1-\alpha}(\Phi(z) - b_*). \end{aligned} \tag{32}$$

Put $\mathfrak{e}_t = \mathfrak{e}^j_{S(5t-j)}$ for $j/5 \le t \le (j+1)/5$, $j = 0, \dots, 4$, $\mathfrak{e}_t = \mathrm{id}$ for $t \le 0$ and $\mathfrak{e}_t = \Phi$ for $t \ge 1$. This is Construction 4.15 with the Alexander family q_α in place of the inner family.

Lemma 6.6 (The scalar ordinary path)**.** *$(\mathfrak{e}_t)$ is an ordinary path in the sense of Definition 4.14 (with a one-point parameter space), with constants $p^\dagger$, P_{supp}, h, c_{sh} and*

$$\gamma_t = c_{\mathrm{sh}}^{2S(5t)-2S(5t-4)}, \qquad c_{\mathrm{sh}}^2 \le \gamma_t \le 1, \qquad \gamma_1 = 1, \qquad P_{\mathrm{supp}} < h.$$

Proof. The integrand in (30) is smooth in (t, α, z), so $(\alpha, z) \mapsto q_\alpha(z)$ is smooth on $[0,1] \times D(p^\dagger)$, including $\alpha = 0$. For $\alpha > 0$, q_α is $\Phi \circ (\alpha\,\cdot)$ followed by an affine bijection, and Φ is injective with $\det D\Phi = 1$ on $D(p^\dagger) \subset D(h - 4\eta)$ (Lemma 6.5); so q_α is injective with $\det Dq_\alpha = 1$; q_0 is affine with $\det A_0 = 1$. Also $|q_\alpha(z) - b_*| \le M_\Phi |z| \le B_*$.

Seams and smoothness. $\mathfrak{e}^0_1 = \mathfrak{e}^1_0$, $\mathfrak{e}^1_1 = \mathfrak{e}^2_0$ (as $M_0 = \mathrm{Id}$), $\mathfrak{e}^2_1 = \mathfrak{e}^3_0$ (as $M_1 = A_0$ and $q_0 - b_* = A_0 z$), $\mathfrak{e}^3_1 = \mathfrak{e}^4_0$ and $\mathfrak{e}^4_1 = \Phi$. All derivatives of S vanish at 0 and 1, so the concatenation is smooth on $\mathbb{R} \times D(p^\dagger)$ (Lemma 2.5); this gives (O1).

Determinants. The segments are injective with constant Jacobian determinants $c_{\mathrm{sh}}^{2\alpha}$, c_{sh}^2, c_{sh}^2, c_{sh}^2 and $c_{\mathrm{sh}}^{2(1-\alpha)}$; with $\alpha = S(5t-j)$ these equal $\gamma_t \in [c_{\mathrm{sh}}^2, 1]$, and $\gamma_1 = 1$. This is (O2).

Ranges. Let $|z| < R^\dagger$. Then $|\mathfrak{e}^0_\alpha(z)| < R^\dagger$; $|\mathfrak{e}^1_\alpha(z)| \le r_b + c_{\mathrm{sh}} R^\dagger$ and $|\mathfrak{e}^2_\alpha(z)| \le r_b + c_{\mathrm{sh}}\|M_\alpha\| R^\dagger$ are at most $r_b + c_{\mathrm{sh}} B_3$; $|\mathfrak{e}^3_\alpha(z)| \le r_b + c_{\mathrm{sh}} B_*$; and $\mathfrak{e}^4_\alpha(z)$ is a convex combination of b_* and $\Phi(z)$, both of modulus at most R_q (Lemma 6.5 and $p^* < P_+$). So all images lie in $\bar{D}(P_{\mathrm{supp}})$, which is (O3). Finally $R^\dagger < R_h$, $R_q < R_h$ (as $P_+ < h$), and $r_b + c_{\mathrm{sh}} \max(B_3, B_*) \le r_b + \frac{1}{8}(R_h - r_b) < R_h$ (as $p^* < h$); hence $P_{\mathrm{supp}} < h$. □

The ambient realiser is that of §§4.7–4.9 for this path. With $e_q = 2\eta$ put

$$\begin{aligned} &p_{\mathrm{on}} = h - 31e_q/8, \qquad p_{\mathrm{zero}} = h - 61e_q/16, \qquad \chi(I) = 1 - S\!\left(\frac{I - p_{\mathrm{on}}}{p_{\mathrm{zero}} - p_{\mathrm{on}}}\right), \\ &W_t(y) = \chi(\pi|\mathfrak{e}_t^{-1}(y)|^2)\, \partial_t \mathfrak{e}_t(\mathfrak{e}_t^{-1}(y)) \quad \text{on } \mathfrak{e}_t(D(p^\dagger)), \qquad W_t = 0 \quad \text{elsewhere}, \end{aligned} \tag{33}$$

where $\mathfrak{e}_t^{-1}(y)$ is the unique preimage in $D(p^\dagger)$. Since $h - 4e_q < p_{\mathrm{on}} < p_{\mathrm{zero}} < h - 15e_q/4 < p^\dagger$, these are the field and cutoff of Lemma 4.17. Let ϕ_t be the flow of W_t with $\phi_0 = \mathrm{id}$, and put

$$G_t(w) = \phi_t(w/\sqrt{\gamma_t}), \qquad \widehat{J}_t = \det DG_t, \qquad \widehat{q}_t = G_t \circ T^{-1}_{\widehat{J}_t} \circ D_{\widehat{J}_t}, \tag{34}$$

with the normalisations (24); this is $\hat{q}$ of Lemma 4.18. Finally define, as in Construction 4.19,

$$\begin{aligned} &a^q_t(z)[v] = \omega_{\mathbb{C}}\big(\partial_t \widehat{q}_t(z), D\widehat{q}_t(z)[v]\big), \qquad h^q_t(z) = \int_0^1 a^q_t(rz)[z]\, dr, \\ &z_{\mathrm{bd}} = \sqrt{(h + P_{\mathrm{supp}})/(2\pi)}, \qquad Q_t(y) = -h^q_t(\widehat{q}_t^{-1}(y)) + h^q_t(z_{\mathrm{bd}}). \end{aligned} \tag{35}$$

Proposition 6.7 (Scalar Hamiltonian realisation)**.** *The maps $\widehat{q}_t$ ($t \in \mathbb{R}$) are symplectic diffeomorphisms of $\mathbb{C}$, smooth in (t, z) with smooth inverses; $\widehat{q}_t = \mathrm{id}$ for $t \le 0$ and $\widehat{q}_t = \widehat{q}_1$ for $t \ge 1$; each $\widehat{q}_t$ is the identity on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ and maps $\bar{D}(h)$ onto itself. The function Q of (35) is*

smooth on $\mathbb{R}\times\mathbb{C}$, vanishes for $t\notin(0,1)$ and on $\{\pi|y|^2>P_{\mathrm{supp}}\}$, and generates $\widehat{q}$: $\partial_t\widehat{q}_t=\Xi_t\circ\widehat{q}_t$ with $\iota_{\Xi_t}\omega_{\mathbb{C}}=-dQ_t$. The endpoint satisfies

$$F(\pi|\widehat{q}_1(z)|^2)\le RF(\pi|z|^2)+e\qquad(\pi|z|^2\le h). \tag{36}$$

Proof. By Lemma 6.6, Lemmas 4.17 and 4.18 and Proposition 4.20 apply to $(\mathfrak{e}_t)$ with the levels $p_{\mathrm{on}}<p_{\mathrm{zero}}<p^{\dagger}$. They give all assertions except (36), together with $\widehat{q}_1=\mathfrak{e}_1=\Phi$ on $\bar{D}(p_{\mathrm{on}})\supset\bar{D}(h-4e_q)=\bar{D}(h-8\eta)$ (Lemma 4.18(f)).

For $I=\pi|z|^2\le h-8\eta$, (36) is Lemma 6.5. For $h-8\eta<I\le h$, $\pi|\widehat{q}_1(z)|^2\in[0,h]$, so, using $RF(h)=\max F$ and the L-Lipschitz bound for RF,

$$F(\pi|\widehat{q}_1(z)|^2)\le\max_{[0,h]}F=RF(h)\le RF(I)+L(h-I)\le RF(I)+8L\eta\le RF(I)+e/8.\qquad\square$$

6.5. **Substitution of the cubic parameters.** Let $k\ge1$, $0<\rho_i<1/3$ and $\sum_i\rho_i^2<1$. The auxiliary quantities of Lemma 8.5 are

(37)

$$D_\rho=1-\sum_i\rho_i^2,\quad S_\rho=\sum_i\rho_i,\quad\rho_{\max}=\max_i\rho_i,\qquad\delta_a=\min\left(\frac{1/3-\rho_{\max}}{8},\frac{D_\rho}{8(S_\rho+k)},\frac{1}{8}\right),$$

$$a_i=\rho_i+\delta_a,\qquad a_{\max}=\rho_{\max}+\delta_a,\qquad D_a=1-\sum_i a_i^2,\qquad\varsigma=\min\left(\frac{1/3-a_{\max}}{2},\frac{D_a}{12}\right),$$

$$h=\frac{1}{3}-\varsigma,\qquad\tau=\frac{a_{\max}+h}{2h},\qquad H_{\mathrm{b}}(p)=3(1-3p),\qquad P_{\mathrm{b}}(p)=H_{\mathrm{b}}(p)-\sum_i(a_i-p)_+,$$

$$\beta=3(1-3h),\qquad\mu=\frac{1-\sum_i a_i^2-(1-3h)^2}{2h},\qquad d=\tfrac12\min(\mu,\beta),\qquad d_n=\tfrac12(h+1/3).$$

For (11) take

$$P=P_{\mathrm{b}},\qquad P_{\min}=-\sum_i a_i,\qquad\bar{P}=3+\sum_i a_i,\qquad L_P=9+k. \tag{38}$$

Construct K by (13) and (17). For the planar input, set

$$F(p)=K(p)-P_{\mathrm{b}}(p),\qquad e=d/16,\qquad L=M+9+k,\qquad\varepsilon_{\mathrm{cap}}=d/(128k), \tag{39}$$

$$U_0(p)=K(p)-H_{\mathrm{b}}(p)+\sum_i\frac{a_i-p+\sqrt{(a_i-p)^2+\varepsilon_{\mathrm{cap}}^2}}{2},\qquad\sigma_{\mathrm{in}}=d/256.$$

Let $\widehat{q}$ and Q be (34) and (35) for these data, and put $g=3d/4$.

Proposition 6.8 (The cubic comparison datum)**.** *For every $k\ge1$ and all $\rho_1,\dots,\rho_k\in(0,\frac13)$ with $\sum_i\rho_i^2<1$, the data* (37)–(39) *satisfy the hypotheses* (11) *of Lemma 6.1 and* (19) *of §6.2, and*

$$K(I)-F(\pi|\widehat{q}_1(z)|^2)\ge g\ge\tfrac{31}{32}g\qquad(I=\pi|z|^2\le h). \tag{40}$$

Consequently $(k,\rho_i,a_i,h,d_n,g,K,\widehat{q},Q)$ are valid cubic data in the sense of Definition 10.13.

Proof. Auxiliary capacities. Since $\delta_a\le\frac18$ and $\delta_a\le D_\rho/(8(S_\rho+k))$, $\sum_i a_i^2-\sum_i\rho_i^2=2\delta_aS_\rho+k\delta_a^2\le2\delta_a(S_\rho+k)\le D_\rho/4$, so $0<\frac34D_\rho\le D_a\le1$. Also $a_{\max}=\rho_{\max}+\delta_a<\frac13$, so $\varsigma>0$ and $a_{\max}<h<\frac13$; hence $0<\tau<1$ and $\tau h=\frac12(a_{\max}+h)>a_{\max}$. Moreover $(1-3h)^2=9\varsigma^2\le D_a^2/16<D_a$.

Hypotheses (11). Since $\int_0^h H_{\mathrm{b}}=\frac12[1-(1-3h)^2]$ and $\int_0^h(a_i-p)_+\,dp=\frac12a_i^2$, $\int_0^h P_{\mathrm{b}}=\frac12[D_a-(1-3h)^2]=\mu h>0$. For $p\ge\tau h>a_{\max}$ the caps vanish and $P_{\mathrm{b}}(p)=H_{\mathrm{b}}(p)\ge H_{\mathrm{b}}(h)=\beta>0$.

So $0 < d < \min(\mu, \beta)$. The function P_{b} is continuous, concave and $(9+k)$-Lipschitz, and on $[0,h]$ we have $H_{\mathrm{b}} > 0$, hence $-\sum_i a_i \le P_{\mathrm{b}} \le 3$; this gives $P_{\min}$, $\bar{P}$ and L_P.

Hypotheses (19). By Proposition 6.2, K is smooth, convex and M-Lipschitz, so $F = K - H_{\mathrm{b}} + \sum_i (a_i - p)_+$ is convex and L-Lipschitz on $[0,h]$ with $L = M + 9 + k$. The soft cap $s_\varepsilon(x) = \frac{1}{2}(x + \sqrt{x^2+\varepsilon^2})$, $\varepsilon > 0$, is smooth and convex with $0 \le s_\varepsilon(x) - x_+ \le \varepsilon/2$. Hence $U_0 = K - H_{\mathrm{b}} + \sum_i s_{\varepsilon_{\mathrm{cap}}}(a_i - p)$ is smooth and convex on $\mathbb{R}$ and $F \le U_0 \le F + k\varepsilon_{\mathrm{cap}}/2 = F + d/256$, with $\sigma_{\mathrm{in}} = d/256 = e/16$.

The gap. By (36) and (18), for $\pi|z|^2 = I \le h$,

$$F(\pi|\widehat{q}_1(z)|^2) \le RF(I) + \frac{d}{16} \le K(I) - \frac{13d}{16} + \frac{d}{16} = K(I) - \frac{3d}{4} = K(I) - g.$$

Validity. Definition 10.13(a): $0 < \rho_i < a_i \le a_{\max} < h < d_n < \frac{1}{3}$, and $0 < g = \frac{3}{4}d < \beta = H_{\mathrm{b}}(h)$. Item (b) is Proposition 6.7 ($\widehat{q}_0 = \mathrm{id}$, $Q_t = 0$ for $t \notin (0,1)$, and $\widehat{q}_t = \mathrm{id}$, $Q_t = 0$ on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ with $P_{\mathrm{supp}} < h$). Item (c) is (40). □

The handle construction of §10.9 uses, with these data, the cap and the lower base profile

$$e_h = g/16, \qquad H^* = H_{\mathrm{b}} - e_h, \qquad \mu_h = g/(16k), \qquad J = K - H^*,$$
$$h_i(p) = \tfrac{1}{2}\big(a_i - p + \sqrt{(a_i - p)^2 + \mu_h^2}\big);$$

the smoothing width μ_h of the handle cap is distinct from the width $\varepsilon_{\mathrm{cap}}$ of the planar majorant.

The source capacities of the general cubic embeddings are

$$\bar{\rho}_i = \frac{\rho_i + a_i}{2}, \qquad \delta_i = a_i - \bar{\rho}_i, \qquad \hat{\rho}_i = \bar{\rho}_i + \frac{\delta_i}{4}, \tag{41}$$

so $\rho_i < \bar{\rho}_i < \hat{\rho}_i < a_i$. With the cubic data of Proposition 6.8, the maps $f_i^{\mathrm{cap}} = b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N} \circ f_i^N$ of Remark 10.30 are smooth symplectic embeddings of $B^\circ(\hat{\rho}_i)$ into $B^\circ(1)$ with pairwise disjoint images (Proposition 10.29, Theorem 13.21 and Theorem 13.23). For a target capacity $C > 0$ and given capacities c_i with $3c_i < C$ and $\sum_i c_i^2 < C^2$, put $\rho_i = c_i/C$ and

$$f_i(u) = \sqrt{C}\, f_i^{\mathrm{cap}}(u/\sqrt{C}), \qquad c_i' = C\bar{\rho}_i, \qquad \bar{c}_i = C\hat{\rho}_i. \tag{42}$$

The homothety $u \mapsto u/\sqrt{C}$ maps $B^\circ(\bar{c}_i)$ onto $B^\circ(\hat{\rho}_i)$ and multiplies ω_0 by $1/C$, so f_i is a smooth symplectic embedding of $B^\circ(\bar{c}_i)$ into $B^\circ(C)$; the images $f_i(B^\circ(\bar{c}_i))$, and in particular the compact sets $f_i(B(c_i'))$, are pairwise disjoint, and $c_i < c_i' < \bar{c}_i$, as required in Definition M6.2.

7. Rearranging convex functions by explicit isotopies

This section proves Theorem M3.4: one coordinate rearrangement R_j of a convex polyhedral function with real coefficients is realised, up to an arbitrarily small error, by an explicit Hamiltonian isotopy of the toric domain V_Δ (Theorem 7.13). Theorem 7.7 is the parameter-free planar core. Outside the equal-capacity case the four-dimensional branch uses instead the scalar construction of §6 (Remark 7.8).

The construction has three layers. On one slice, rescaled to $[0,1]$, the convex function f is replaced by a proper strictly convex potential U (log-sum-exp smoothing plus a reciprocal barrier), and the angle receives a proper wall V. Both centred wells are then modified on explicit small intervals to be exactly quadratic. The construction of §4 turns this modified pair into a compactly supported Hamiltonian isotopy of the disc of area 1 whose endpoint carries, on a core disc, the centred circle of area I onto the level set of $\widetilde{U}_y(p - p^*(y)) + \widetilde{V}(\theta)$ enclosing area I. The only estimate specific to convex f is an area estimate (Lemma 7.5): that level lies below $g(I)$ plus a small error, where g is the increasing rearrangement of f. In the second layer every constant is chosen uniformly in the other moments $y = (p_i)_{i \ne j}$ on an explicit open

parameter domain, which gives one jointly smooth family (Theorem 7.9). In the third layer the family is lifted to $\mathbb{C}^m$ by the rescaling of [Ope26, §2.3], with an explicit smooth slice length $\ell(y) < L(y)$, an explicit cutoff $\chi(y)$ near degenerate slices, and two open sets which define the lifted Hamiltonian on all of $\mathbb{C}^m$ (Definition 7.10). In [Ope26, §2.3] the planar maps are obtained from an existence lemma on nested discs (compare [Sch99, Lemma 3.2]). Here they are given by closed formulas: every constant is an explicit function of the data, and the resulting isotopy is one jointly smooth function of the parameter, the time and the point.

Fibre Hamiltonians follow $\iota_{X_Q}\omega = -dQ$ (§2.1). We use the conventions of §4: $D(A) = \{z \in \mathbb{C} : \pi|z|^2 < A\}$, $\bar{D}(A)$ its closure, and the standard area form on every plane. The notation of §3 is in force: $\Delta = \{p \in \mathbb{R}^m_{\geq 0} : \langle b_\nu, p\rangle \leq \ell_\nu\}$, the toric domain $V_\Delta = \{w \in \mathbb{C}^m : p(w) \in \Delta\}$ with moments $p(w) = (\pi|w_i|^2)_i$ and form $\omega_V = \sum_i dp_i \wedge d\theta_i$, the sum of the standard area forms of the coordinate planes, the slice lengths L_ν, L, the projection Δ_j, the coordinate rearrangement R_j (Definition M3.1) and the slice description (2).

7.1. **Planar data and potentials.**

Planar data. Let $d \geq 0$ and let $\mathcal{O} \subset \mathbb{R}^d$ be open ($d = 0$ means that $\mathcal{O}$ is one point and the parameter is absent). Let $\mathcal{I}$ be a finite nonempty index set and $a_r, b_r \in C^\infty(\mathcal{O})$, $r \in \mathcal{I}$, and put

$$f_y(p) = \max_{r\in\mathcal{I}}\big(a_r(y) + b_r(y)p\big) \qquad (y \in \mathcal{O},\ p \in \mathbb{R}). \tag{43}$$

Let $\Lambda \geq 1$ and M_0 be real numbers with

$$|b_r(y)| \leq \Lambda \quad \text{and} \quad |a_r(y)| + |b_r(y)| \leq M_0 \qquad (r \in \mathcal{I},\ y \in \mathcal{O}).$$

We call $(\mathcal{O}, (a_r, b_r)_{r\in\mathcal{I}}, \Lambda, M_0)$ *planar data*; the families used below are the rescaled polytope slices of Definition 7.10, and $d = 0$ gives the parameter-free case. The second bound implies $|f_y| \leq M_0$ on $[0,1]$ and $|b_r(y)| \leq M_0$. For $y \in \mathcal{O}$, g_y denotes the increasing rearrangement of $f_y|_{[0,1]}$,

$$g_y(x) = \min_{0\leq a\leq 1-x} \max\big(f_y(a), f_y(a+x)\big) \qquad (0 \leq x \leq 1),$$

which is nondecreasing with $g_y(0) = \min_{[0,1]} f_y$ and $g_y(1) = \max_{[0,1]} f_y$ (Lemma 3.1). Since f_y is Λ-Lipschitz on $[0,1]$, Lemma 3.7 gives

$$0 \leq g_y(x') - g_y(x) \leq \Lambda(x' - x) \qquad (0 \leq x \leq x' \leq 1). \tag{44}$$

The scale η below is the width of the boundary layer of $[0,1]$ that is sacrificed, β_0 the width of the corresponding angular strips, σ_0 the size of each smoothing error, κ_0 and ν_0 the strengths of the barrier and of the wall, c_0 the sharpness of the log-sum-exp smoothing, r_0 a bracket for the minimum point of U_y, and Λ_2 a curvature bound. They are chosen so that all errors add up to less than $\frac{11}{32}\xi$ (Lemma 7.5).

Definition 7.1 (Planar constants and potentials)**.** Given planar data and a real $\xi > 0$, put

$$\eta = \frac{1}{2}\min\Big(\frac{1}{16}, \frac{\xi}{32(\Lambda+1)}\Big), \quad \beta_0 = \frac{\eta}{8}, \quad \sigma_0 = \frac{\xi}{8}, \quad \nu_0 = 4\sigma_0\beta_0(1-\beta_0), \quad \kappa_0 = \frac{\sigma_0\eta}{8},$$
$$c_0 = \frac{8(\#\mathcal{I}+1)}{\sigma_0}, \qquad r_0 = \min\Big(\frac{1}{8}, \sqrt{\frac{\kappa_0}{4(\Lambda+2\kappa_0+1)}}\Big), \qquad \Lambda_2 = c_0\Lambda^2,$$

and define, for $y \in \mathcal{O}$,

$$\tilde{f}_y(p) = \frac{1}{c_0} \log \sum_{r \in \mathcal{I}} \exp\big(c_0(a_r(y) + b_r(y)p)\big) \quad (p \in \mathbb{R}),$$

$$U_y(p) = \tilde{f}_y(p) + \kappa_0\Big(\frac{1}{p} + \frac{1}{1-p}\Big) \quad (0 < p < 1),$$

$$V(\theta) = \frac{\nu_0 \theta^2}{\frac{1}{4} - \theta^2} \quad \big(-\tfrac{1}{2} < \theta < \tfrac{1}{2}\big).$$

The barrier $\kappa_0(1/p + 1/(1-p))$ makes U_y proper on $(0,1)$ even where f_y is affine; the wall V plays the same role in the angle. Both perturbations are small away from the boundary: the barrier is at most $\sigma_0/4$ on $[\eta, 1-\eta]$, and $V \le \sigma_0$ outside the strips of width β_0 at the two angular ends.

Lemma 7.2 (The potentials)**.** *For planar data and $\xi > 0$, with the constants of Definition 7.1:*

(a) $\tilde{f}_y(p)$ is C^∞ in $(y,p) \in \mathcal{O} \times \mathbb{R}$ and real-analytic and convex in p; U is C^∞ on $\mathcal{O} \times (0,1)$ and real-analytic in p; V is even and real-analytic on $(-\frac{1}{2}, \frac{1}{2})$ with $V(0) = 0$.

(b) (Approximation.) $f_y \le \tilde{f}_y < f_y + \sigma_0/4$ on $\mathbb{R}$; $U_y > f_y$ on $(0,1)$; $0 \le U_y - f_y < \sigma_0/2$ on $[\eta, 1-\eta]$; $0 \le V \le \sigma_0$ on $[-\frac{1}{2} + \beta_0, \frac{1}{2} - \beta_0]$.

(c) (Convexity and properness.) $|\tilde{f}_y'| \le \Lambda$ and $0 \le \tilde{f}_y'' \le \Lambda_2$ on $\mathbb{R}$; $U_y'' \ge 32\kappa_0$ on $(0,1)$; $U_y(p) \to +\infty$ as $p \downarrow 0$ or $p \uparrow 1$; $V'' > 0$ and $V(\theta) \to +\infty$ as $|\theta| \uparrow \frac{1}{2}$.

(d) (Minimum.) The minimum point $p^*(y)$ of U_y satisfies $r_0 < p^*(y) < 1 - r_0$; with $U_{\min}(y) = U_y(p^*(y))$,

$$32\kappa_0 \le U_y''(p^*(y)) \le \Lambda_2 + \frac{4\kappa_0}{r_0^3}, \qquad a_{\min} := \Big(\frac{2}{\Lambda_2 + 4\kappa_0/r_0^3}\Big)^{1/2} \le a_0(y) \le a_{\max} := \frac{1}{4\sqrt{\kappa_0}},$$

where $a_0(y) = \sqrt{2/U_y''(p^*(y))}$; and $b_0 = \sqrt{2/V''(0)} = 1/(2\sqrt{\nu_0})$.

(e) (Angular inverse.) The inverse Y of the signed square root of V is $Y(v) = v/(2\sqrt{v^2 + \nu_0})$.

(f) (Energy.) Let $\delta_E = \eta/8$ and $E_{\mathrm{box}} = \Lambda + \sigma_0/4 + 16\kappa_0/\eta + \nu_0/(4\delta_E(1-\delta_E))$. For $E \ge E_{\mathrm{box}}$, the area $\mathcal{A}_y(E)$ of $\{(p,\theta) \in (0,1) \times (-\frac{1}{2}, \frac{1}{2}) : U_y(p) - U_{\min}(y) + V(\theta) \le E\}$ is at least $A_{\min} = (1 - 2\delta_E)^2$, and $A_{\min} > 1 - 2\eta$.

(g) (Margins.) Let $E > 0$. If $0 < p < 1$ and $U_y(p) - U_{\min}(y) < E$, then $\min(p, 1-p) > \kappa_0/(M_0 + 4\kappa_0 + E)$. If $|\theta| < \frac{1}{2}$ and $V(\theta) < E$, then $|\theta| < \frac{1}{2}\sqrt{E/(E + \nu_0)}$.

Proof. (a) For fixed y, $\tilde{f}_y$ is the log-sum-exp of the affine functions $p \mapsto a_r(y) + b_r(y)p$, hence real-analytic and convex (Lemma 3.5(b)); in (y,p) it is the composite of the smooth map $(y,p) \mapsto (a_r(y) + b_r(y)p)_r$ with the real-analytic function $x \mapsto c_0^{-1} \log \sum_r e^{c_0 x_r}$. The barrier and V are rational functions without poles on the stated intervals.

(b) By Lemma 3.5(a), $f_y \le \tilde{f}_y \le f_y + \log(\#\mathcal{I})/c_0$, and $\log(\#\mathcal{I})/c_0 < \#\mathcal{I}/c_0 < \sigma_0/4$. The barrier is positive, so $U_y > f_y$, and on $[\eta, 1-\eta]$ it is at most $2\kappa_0/\eta = \sigma_0/4$. The function V is nondecreasing in $|\theta|$; at $|\theta| = \frac{1}{2} - \beta_0$, $\theta^2 \le \frac{1}{4}$ and $\frac{1}{4} - \theta^2 = (\frac{1}{2} - |\theta|)(\frac{1}{2} + |\theta|) = \beta_0(1 - \beta_0)$, so $V \le \nu_0/(4\beta_0(1-\beta_0)) = \sigma_0$.

(c) By Lemma 3.5(c), $\tilde{f}_y' = \sum_r \varpi_r b_r(y)$ with convex weights ϖ_r, so $|\tilde{f}_y'| \le \Lambda$, and $\tilde{f}_y''$ is c_0 times the variance of the slopes $b_r(y)$ under these weights (proof of Lemma 3.5(b)), which lies in $[0, c_0\Lambda^2]$. The barrier has second derivative $2\kappa_0(p^{-3} + (1-p)^{-3})$, a convex function symmetric about $\frac{1}{2}$ with minimum $32\kappa_0$ there. $\tilde{f}_y$ is bounded on $[0,1]$, so U_y is proper. Writing $V(\theta) = \nu_0\big(\frac{1}{4}(\frac{1}{4} - \theta^2)^{-1} - 1\big)$ one finds $V''(\theta) = \nu_0(1 + 12\theta^2)/(8(\frac{1}{4} - \theta^2)^3) > 0$, and V is proper. For the unmodified wells, U_y' is strictly increasing and tends to $\mp\infty$ at 0 and 1, so it has a unique

zero $p^*(y) \in (0,1)$ and changes sign there, and $U_y''(p^*) > 0$; V is even and strictly convex with $V(0) = 0$, so $V'(\theta)\theta > 0$ for $\theta \neq 0$, and $V''(0) = 8\nu_0 > 0$.

(d) Since $r_0 < \frac{1}{4}$ and $|\tilde f_y'| \le \Lambda$,

$$U_y'(r_0) \le \Lambda - \frac{\kappa_0}{r_0^2} + \frac{\kappa_0}{(1-r_0)^2} < \Lambda + 2\kappa_0 - \frac{\kappa_0}{r_0^2} < 0,$$

because $\kappa_0/r_0^2 \ge 4(\Lambda + 2\kappa_0 + 1) > \Lambda + 2\kappa_0$; symmetrically $U_y'(1-r_0) > 0$. Hence $r_0 < p^*(y) < 1 - r_0$. The lower bound for $U_y''(p^*)$ is (c); the upper bound is $\tilde f_y'' \le \Lambda_2$ plus $2\kappa_0((p^*)^{-3} + (1-p^*)^{-3}) \le 4\kappa_0/r_0^3$. The bounds for a_0 follow, and $V''(0) = 8\nu_0$ gives b_0.

(e) Y is odd, real-analytic and increasing, with limits $\pm\frac{1}{2}$. With $\vartheta = Y(v)$, $\vartheta^2 = v^2/(4(v^2+\nu_0))$ and $\frac{1}{4} - \vartheta^2 = \nu_0/(4(v^2+\nu_0))$, so $V(\vartheta) = v^2$; since Y has the sign of v, Y inverts the signed square root $\operatorname{sign}(\theta)\sqrt{V(\theta)}$.

(f) On $Q_E = [\delta_E, 1-\delta_E] \times [-\frac{1}{2} + \delta_E, \frac{1}{2} - \delta_E]$, by (b), $U_y(p) \le \max_{[0,1]} f_y + \sigma_0/4 + 2\kappa_0/\delta_E$, while $U_{\min}(y) \ge \min_{[0,1]} f_y$ and $\max_{[0,1]} f_y - \min_{[0,1]} f_y \le \Lambda$; and $V \le \nu_0/(4\delta_E(1-\delta_E))$ on Q_E, as in (b). So $U_y - U_{\min}(y) + V \le E_{\text{box}}$ on Q_E, and the sublevel at $E \ge E_{\text{box}}$ contains Q_E, of area $A_{\min}$. Finally $A_{\min} = 1 - \eta/2 + \eta^2/16 > 1 - 2\eta$.

(g) By (c), $|\tilde f_y'| \le \max_r |b_r(y)| \le M_0$, so $|\tilde f_y(\frac{1}{2}) - \tilde f_y(p)| \le M_0/2$ for $p \in (0,1)$; and $U_{\min}(y) \le U_y(\frac{1}{2}) = \tilde f_y(\frac{1}{2}) + 4\kappa_0$. If $U_y(p) - U_{\min}(y) < E$, then

$$\frac{\kappa_0}{p} < U_y(p) - \tilde f_y(p) < U_{\min}(y) + E - \tilde f_y(p) \le \tilde f_y(\tfrac{1}{2}) - \tilde f_y(p) + 4\kappa_0 + E \le M_0 + 4\kappa_0 + E,$$

and likewise for $1-p$. If $V(\theta) < E$, then $\nu_0\theta^2 < E(\frac{1}{4} - \theta^2)$, so $\theta^2 < E/(4(E+\nu_0))$. □

7.2. **Isotopy data.** Theorem 4.21 needs an admissible potential pair and numerical data controlling the isotopy uniformly in the parameter (Definition 4.11). For the convex potentials both are given by closed formulas in the planar constants. The roles of the data are those listed after Definition 4.11; in addition $r_B < 1$ bounds the support, and $K_{\min}$, $A_{\max}$ are auxiliary.

Definition 7.3 (Modified wells and isotopy data of the convex pair)**.** Given planar data and $\xi > 0$, with the constants of Definition 7.1, let ψ and C_ψ be the flat cutoff and its derivative bound of §6.3, and put $\delta = \xi/64$, $\alpha_y = U_y''(p^*(y))$, $\alpha_{\min} = 32\kappa_0$, $\alpha_{\max} = c_0\Lambda^2 + 4\kappa_0/r_0^3$,

$$K_3 = 8c_0^2\Lambda^3 + 192\kappa_0/r_0^4 + 1, \qquad \epsilon_U = \min\left(\frac{r_0}{8}, \frac{\alpha_{\min}}{32K_3(1+C_\psi)}, \sqrt{\frac{\delta}{\alpha_{\max}}}\right),$$

$$g_y(x) = U_y(p^*(y)+x) - U_y(p^*(y)),$$

$$\widetilde U_y(x) = g_y(x) + \psi(x/\epsilon_U)\left(\frac{\alpha_y x^2}{2} - g_y(x)\right) \qquad (-p^*(y) < x < 1 - p^*(y)).$$

For the angle, apply (22) to $(f, A, B, c) = (V, -\frac{1}{2}, \frac{1}{2}, 0)$ with δ in place of δ_{w}: $r_V = \frac{1}{8}$, $\alpha_V = V''(0) = 8\nu_0$, $G_V = \psi(x/r_V)V$, $K_V = 1 + \max_x |G_V'''(x)|$,

$$\epsilon_V = \min\left(\frac{r_V}{4}, \frac{\alpha_V}{32K_V(1+C_\psi)}, \sqrt{\delta/\alpha_V}\right),$$

$$\widetilde V(x) = V(x) + \psi(x/\epsilon_V)\left(\frac{\alpha_V x^2}{2} - G_V(x)\right) \qquad (|x| < \tfrac{1}{2}).$$

The *modified convex pair* is $U_y(p) := \widetilde U_y(p - p^*(y))$ on $I_U = (0,1)$ and $V := \widetilde V$ on $I_V = (-\frac{1}{2}, \frac{1}{2})$, with disc area $h = 1$; X_y and Y denote its Morse inverses (Lemma 4.2). Its slopes at zero are $a_0(y) = \sqrt{2/\alpha_y}$ and $b_0 = 1/(2\sqrt{\nu_0})$.

The isotopy data are $p^*_{\mathrm{lo}} = r_0$, $p^*_{\mathrm{hi}} = 1 - r_0$ and

$$\delta_E = \eta/8, \quad A_{\min} = (1-2\delta_E)^2, \quad p^\dagger = 1 - 5\eta, \quad e_q = 2\eta,$$
$$p_{\mathrm{on}} = 1 - 31\eta/4, \quad p_{\mathrm{zero}} = 1 - 61\eta/8,$$
$$E_{\mathrm{rect}} = 2M_0 + 2 + 2\kappa_0/\delta_E + \nu_0/(2\delta_E) + 2\delta,$$
$$a_{\min} = \sqrt{2/\alpha_{\max}}, \quad a_{\max} = \sqrt{2/\alpha_{\min}},$$
$$K_{\min} = \pi a_{\min} b_0,$$
$$\bar{E} = E_{\mathrm{rect}} + \frac{p^\dagger A_{\min}}{K_{\min}(A_{\min} - p^\dagger)} + 1, \quad E_c = \bar{E},$$
$$\bar{X} = a_{\max}\sqrt{\bar{E}} + 1, \qquad \bar{Y} = b_0\sqrt{\bar{E}} + \tfrac{1}{2},$$
$$\Delta P = \frac{\kappa_0}{2M_0 + 2 + 4\kappa_0 + \bar{E} + \delta},$$
$$\Delta\theta = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{E}+\delta+1}{\bar{E}+\delta+1+\nu_0}}\right),$$
$$\delta_x = \bar{X}, \quad \delta_\theta = \bar{Y}, \quad \delta_s = \min\left(\frac{1}{4}, \frac{\Delta P}{2\bar{X}}, \frac{\Delta\theta}{2\bar{Y}}\right),$$
$$\lambda_0 = \min\left(\frac{1}{2}, \frac{r_0}{2\bar{X}}, \frac{1}{4\bar{Y}}\right),$$
$$P_{\min} = \tfrac{1}{4}\min(r_0, \Delta P), \quad P_{\max} = \tfrac{3}{2}, \quad M_C = \frac{1}{2\sqrt{\pi P_{\min}}} + 2\sqrt{\pi P_{\max}},$$
$$R^\dagger = \sqrt{p^\dagger/\pi}, \quad R_h = \sqrt{1/\pi}, \quad r_b^+ = \sqrt{(1-r_0)/\pi},$$
$$A_{\max} = \frac{\sqrt{a_{\max}/b_0}}{2\sqrt{\pi r_0}} + 2\sqrt{\pi}\sqrt{b_0/a_{\min}},$$
$$B_{\mathrm{lin}} = R^\dagger(1 + A_{\max}), \quad B^* = M_C\sqrt{\bar{X}^2 + \bar{Y}^2}, \quad R_q = \sqrt{(1 - \Delta P/2)/\pi},$$
$$c_{\mathrm{sh}} = \min\left(\frac{1}{4}, \frac{R_h - r_b^+}{8\max(R^\dagger, B_{\mathrm{lin}}, B^*)}\right),$$
$$R_{\mathrm{supp}} = \max(R^\dagger, r_b^+ + c_{\mathrm{sh}}B_{\mathrm{lin}}, r_b^+ + c_{\mathrm{sh}}B^*, R_q),$$
$$r_B = P_{\mathrm{supp}} = \pi R_{\mathrm{supp}}^2.$$

All entries are independent of y. Apart from K_V, the maximum of an explicit smooth compactly supported function, they are finite expressions in Λ, M_0, $\#\mathcal{I}$, ξ and C_ψ.

Lemma 7.4 (The modified convex pair and its isotopy data)**.** *For planar data and $\xi > 0$:*

(a) $|\widetilde{U}_y(x) - g_y(x)| \le \delta$ and $|\widetilde{V} - V| \le \delta$; $\widetilde{U}_y(x) = \frac{1}{2}\alpha_y x^2$ for $|x| \le \epsilon_U$ and $\widetilde{V}(x) = \frac{1}{2}\alpha_V x^2$ for $|x| \le \epsilon_V$; $x\widetilde{U}_y'(x) > 0$ and $x\widetilde{V}'(x) > 0$ for $x \ne 0$; $\widetilde{U}_y \ge 0$, $\widetilde{V} \ge 0$.

(b) The modified convex pair is an admissible potential pair over $\mathcal{Y} = \mathcal{O}$ with $h = 1$, exactly quadratic near the minimum, and $r_0 < p^*(y) < 1 - r_0$.

(c) The numbers of Definition 7.3 are isotopy data for this pair (using the uniform-energy alternative in (D4)); in particular $P_{\mathrm{supp}} = r_B < 1$.

Proof. (a) By Lemma 7.2(c),(d), U_y is smooth and strictly convex on $(0,1)$, proper, with unique critical point $p^*(y) \in (r_0, 1-r_0)$ and $\alpha_{\min} \le \alpha_y \le \alpha_{\max}$. We apply Lemma 6.4 to $f = U_y$, $c = p^*(y)$, $\varepsilon = \epsilon_U$, $K = K_3$. First $2\epsilon_U \le r_0/4 < \min(p^*, 1-p^*)$. For $|x| \le 2\epsilon_U$ the point

$p = p^* + x$ lies in $[\frac{3}{4}r_0, 1 - \frac{3}{4}r_0]$. There $g_y'''(x) = \tilde{f}_y'''(p) + 6\kappa_0((1-p)^{-4} - p^{-4})$, and $\tilde{f}_y'''$ is c_0^2 times the third central moment of the slopes $b_r(y)$ under the weights of Lemma 3.5(c), so $|\tilde{f}_y'''| \le c_0^2(2\Lambda)^3$; and $6\kappa_0 \max(p^{-4}, (1-p)^{-4}) \le 6\kappa_0(4/(3r_0))^4 < 192\kappa_0/r_0^4$. Hence $|g_y'''| \le K_3$ there. Finally $\epsilon_U \le \alpha_{\min}/(32K_3(1+C_\psi)) \le \alpha_y/(32K_3(1+C_\psi))$ and $\epsilon_U \le \sqrt{\delta/\alpha_{\max}} \le \sqrt{\delta/\alpha_y}$. Lemma 6.4 gives the assertions for $\widetilde{U}_y$ (with error $\delta/24$). For $\widetilde{V}$, V is smooth, even, strictly convex and proper with $V(0) = 0$ (Lemma 7.2(c)), $\chi_V g_V = G_V$, and ϵ_V is the width of (22); Lemma 6.4 applies again. Nonnegativity follows from the sign of the derivatives and the value 0 at 0.

(b) p^* is smooth on $\mathcal{O}$ by the implicit function theorem ($U_y'(p^*(y)) = 0$, $U_y'' > 0$), so α_y and $(y,x) \mapsto g_y(x)$ are smooth, and hence so is $(y,p) \mapsto \widetilde{U}_y(p - p^*(y))$; $\widetilde{V}$ does not depend on y. (P1) follows from (a): the minimum of U_y is at $p^*(y) \in (0,1)$ with second derivative $\alpha_y > 0$, and V has minimum 0 at 0 with $V''(0) = 8\nu_0 > 0$. (P2): near the ends of their intervals the modified wells equal g_y and V, which are proper, and they are exactly quadratic on the y-independent neighbourhoods $|x| \le \epsilon_U$, $|x| \le \epsilon_V$, with quadratic coefficients $\frac{1}{2}\alpha_y$ and $4\nu_0$.

(c) Write $\mathcal{A}_y$, $\mathcal{A}_{y,s}$ for the area functions of Lemma 4.10 for this pair. All numbers are positive: $A_{\min} = 1 - \frac{\eta}{2} + \frac{\eta^2}{16} > 1 - 5\eta = p^\dagger > 0$ since $\eta \le \frac{1}{32}$, and the remaining positivity is clear.

Area gate. Let $Q_E = [\delta_E, 1-\delta_E] \times [-\frac{1}{2} + \delta_E, \frac{1}{2} - \delta_E]$. For $(p,\theta) \in Q_E$, $|\tilde{f}_y(p) - \tilde{f}_y(p^*)| \le M_0$ (since $|\tilde{f}_y'| \le M_0$) and the barrier at p is at most $2\kappa_0/\delta_E$, so $g_y(p - p^*) = U_y(p) - U_{\min}(y) \le M_0 + 2\kappa_0/\delta_E$; and $V(\theta) \le \nu_0/(4\delta_E(1-\delta_E)) \le \nu_0/(2\delta_E)$. By (a), $\widetilde{U}_y(p-p^*) + \widetilde{V}(\theta) \le M_0 + 2\kappa_0/\delta_E + \nu_0/(2\delta_E) + 2\delta \le E_{\text{rect}}$ on Q_E. By Lemma 4.2(iv), $\mathcal{A}_y(E) \ge \operatorname{area} Q_E = A_{\min}$ for $E \ge E_{\text{rect}}$. Since $j_0(y) = a_0(y)b_0 \ge a_{\min}b_0$, Lemma 4.10(b) gives, for $s \in [0,1]$,

$$\mathcal{A}_{y,s}(\bar{E}) \ge \alpha(1-s)^2 + \beta s^2 \ge \frac{\alpha\beta}{\alpha+\beta}, \qquad \alpha = K_{\min}\bar{E}, \quad \beta = A_{\min}.$$

As $\alpha > p^\dagger\beta/(\beta - p^\dagger)$, we get $\alpha(\beta - p^\dagger) > p^\dagger\beta$, i.e. $\alpha\beta/(\alpha+\beta) > p^\dagger$. Hence $\mathcal{A}_{y,s}(E_c) > p^\dagger$ for all $s \in [0,1]$ (the uniform-energy alternative of (D4), $E_c = \bar{E}$), and (D5) holds for the areas.

(D1). $1 - \frac{31}{4}\eta < 1 - \frac{61}{8}\eta < 1 - 5\eta$, and $r_0 < p^*(y) < 1 - r_0 < 1$ by Lemma 7.2(d).

(D2). Let $u^2, v^2 < E_c$, $x = X_y(u)$, $\theta = Y(v)$ and $p = p^* + x$. Then $\widetilde{U}_y(x) = u^2 < \bar{E}$, so $U_y(p) - U_{\min}(y) = g_y(x) < \bar{E} + \delta$, and Lemma 7.2(g) gives $\min(p, 1-p) > \kappa_0/(M_0 + 4\kappa_0 + \bar{E} + \delta) > \Delta P$. Thus $\Delta P < p < 1 - \Delta P \le \pi R_q^2 < 1$. Likewise $V(\theta) \le \widetilde{V}(\theta) + \delta < \bar{E} + \delta$, so $|\theta| < \frac{1}{2}\sqrt{(\bar{E}+\delta)/(\bar{E}+\delta+\nu_0)} \le \frac{1}{2} - \Delta\theta$, the function $E \mapsto E/(E+\nu_0)$ being increasing.

(D3) and (D5). For $u^2 \le \bar{E}$: $a_0(y)|u| \le a_{\max}\sqrt{\bar{E}}$ and $|X_y(u)| < 1$ (as $X_y(u) \in (-p^*, 1-p^*)$), so $\max(a_0|u|, |X_y(u)|) \le \bar{X}$ and $|a_0 u - X_y(u)| \le \bar{X} = \delta_x$. For $v^2 \le \bar{E}$: $b_0|v| \le b_0\sqrt{\bar{E}}$ and $|Y(v)| < \frac{1}{2}$, so similarly with $\bar{Y} = \delta_\theta$.

(D4). $\delta_s \le \frac{1}{4}$, $\delta_s\delta_x \le \frac{1}{2}\Delta P$ and $\delta_s\delta_\theta \le \frac{1}{2}\Delta\theta$ by the definition of δ_s; the energy condition was shown above.

(D6). By the definition of λ_0, with $p^*_{\text{lo}} = r_0$.

(D7). $P_{\min} = \frac{1}{4}\min(r_0, \Delta P) \le \frac{1}{2}\min(p^*_{\text{lo}}, \Delta P)$ and $P_{\max} = \frac{3}{2} \ge \max(\frac{3}{2}p^*_{\text{hi}}, 1)$. By (4), with $c_x = \sqrt{a_0/b_0}$ and $r_b = \sqrt{p^*/\pi}$,

$$|(\mathbf{A}_y)_{11}| = \frac{\sqrt{a_0/b_0}}{2\sqrt{\pi p^*}} \le \frac{\sqrt{a_{\max}/b_0}}{2\sqrt{\pi r_0}}, \qquad |(\mathbf{A}_y)_{22}| = 2\sqrt{\pi p^*}\sqrt{b_0/a_0} \le 2\sqrt{\pi}\sqrt{b_0/a_{\min}};$$

so both are at most $A_{\max}$ and $R^\dagger \max(1, |(\mathbf{A}_y)_{11}|, |(\mathbf{A}_y)_{22}|) \le B_{\text{lin}}$. The bounds for B^* and c_{sh} hold by definition.

Lemma 4.12(b) now gives $P_{\text{supp}} < h = 1$. □

7.3. **The area estimate.**

Lemma 7.5 (Area estimate)**.** *Assume planar data, $\xi > 0$ and the constants of Definition 7.1, and write $\check{\mathcal{A}}_y(E)$ for the area of $\{(p,\theta) \in (0,1) \times (-\frac{1}{2},\frac{1}{2}) : U_y(p) - U_{\min}(y) + V(\theta) \le E\}$. Let $y \in \mathcal{O}$ and $0 \le I \le 1 - 3\eta$. Then*

$$\text{area}\big\{(p,\theta) \in (0,1) \times (-\tfrac{1}{2},\tfrac{1}{2}) : U_y(p) + V(\theta) \le g_y(I+3\eta) + 2\sigma_0\big\} \ge (I+\eta)(1-2\beta_0) > I. \tag{45}$$

Consequently, if $E \ge 0$ and $\check{\mathcal{A}}_y(E) \le I$, then $U_{\min}(y) + E < g_y(I+3\eta) + 2\sigma_0$, and every (p,θ) with $U_y(p) + V(\theta) \le U_{\min}(y) + E$ satisfies

$$f_y(p) < g_y(I) + 3\Lambda\eta + 2\sigma_0 < g_y(I) + \tfrac{11}{32}\xi. \tag{46}$$

Proof. Put $b = g_y(I+3\eta)$; note $I+3\eta \le 1$. By Lemma 3.1(a) the sublevel set $\{p \in [0,1] : f_y(p) \le b\}$ is a closed interval J of length at least $I+3\eta$, and $J' = J \cap [\eta, 1-\eta]$ has length at least $I+\eta$. For $p \in J'$, Lemma 7.2(b) gives $U_y(p) < f_y(p) + \sigma_0/2 \le b + \sigma_0/2$, and for $|\theta| \le \frac{1}{2} - \beta_0$ it gives $V(\theta) \le \sigma_0$. So the rectangle $J' \times [-\frac{1}{2}+\beta_0, \frac{1}{2}-\beta_0]$ lies in the set of (45), whose area is therefore at least $(I+\eta)(1-2\beta_0)$. Since $I+\eta \le 1$ and $2\beta_0 = \eta/4$, $(I+\eta)(1-2\beta_0) \ge I+\eta-2\beta_0 = I+\frac{3}{4}\eta > I$.

If $\check{\mathcal{A}}_y(E) \le I$ but $U_{\min}(y) + E \ge b + 2\sigma_0$, the sublevel set at level $U_{\min}(y) + E$ contains the set of (45), so $\check{\mathcal{A}}_y(E) > I$, a contradiction. Hence $U_{\min}(y) + E < b + 2\sigma_0$. For (p,θ) with $U_y(p) + V(\theta) \le U_{\min}(y) + E$, Lemma 7.2(b) and $V \ge 0$ give $f_y(p) < U_y(p) + V(\theta) < b + 2\sigma_0$, and $b \le g_y(I) + 3\Lambda\eta$ by (44). Finally $3\Lambda\eta \le 3\xi/64$ and $2\sigma_0 = \xi/4$, whose sum is below $11\xi/32$. □

The estimate includes $I = 0$: the centre of the disc goes to the minimum of $U_y + V$, where the lemma bounds f_y by $\min f_y$ plus the error (smoothness of the inner map at the centre is Theorem 4.7).

Lemma 7.6 (Area estimate after quadratic modification)**.** *In the setting of Lemma 7.5, put $H_0(p,\theta) = U_y(p) - U_{\min}(y) + V(\theta)$ and $\widetilde{H}(p,\theta) = \widetilde{U}_y(p - p^*(y)) + \widetilde{V}(\theta)$ on $(0,1) \times (-\frac{1}{2},\frac{1}{2})$, with the modified wells of Definition 7.3, so that $|\widetilde{H} - H_0| \le 2\delta$, $\delta = \xi/64$ (Lemma 7.4(a)). Let $\widetilde{\mathcal{A}}_y(E)$ be the area of $\{\widetilde{H} \le E\}$. If $0 \le I \le 1 - 3\eta$, $E \ge 0$ and $\widetilde{\mathcal{A}}_y(E) \le I$, then every point with $\widetilde{H}(p,\theta) \le E$ satisfies*

$$f_y(p) < g_y(I) + 3\Lambda\eta + 2\sigma_0 + 5\delta \le g_y(I) + \frac{3}{8}\xi < g_y(I) + \xi.$$

Proof. Set $T = g_y(I+3\eta) + 2\sigma_0 - U_{\min}(y)$. The set $\{H_0 \le T\}$ has area greater than I by Lemma 7.5, and it is contained in $\{\widetilde{H} \le T + 2\delta\}$. Monotonicity of sublevel area and $\widetilde{\mathcal{A}}_y(E) \le I$ therefore imply $E < T + 2\delta$. At a point with $\widetilde{H} \le E$ we have

$$f_y(p) \le U_y(p) + V(\theta) = U_{\min}(y) + H_0(p,\theta) \le U_{\min}(y) + E + 2\delta < g_y(I+3\eta) + 2\sigma_0 + 4\delta.$$

By (44), $g_y(I+3\eta) \le g_y(I) + 3\Lambda\eta$, and $3\Lambda\eta + 2\sigma_0 + 5\delta \le \frac{3\xi}{64} + \frac{16\xi}{64} + \frac{5\xi}{64} = \frac{3}{8}\xi$. □

7.4. **One convex function.**

Theorem 7.7 (Planar rearrangement of a convex function)**.** *Let $f(p) = \max_{r\in\mathcal{I}}(a_r + b_r p)$ with finitely many real a_r, b_r, let $\Lambda \ge 1$ and M_0 be real with $|b_r| \le \Lambda$ and $|a_r| + |b_r| \le M_0$ for every r, let g be the increasing rearrangement of f on $[0,1]$, and let $\xi > 0$ be real. Let $Q^{\mathrm{pl}} = Q$ and $q_t = \hat{q}_t$ be the Hamiltonian and the isotopy of Theorem 4.21 for the modified pair of Definition 7.3 ($d = 0$) and the isotopy data of Definition 7.3, and let r_B be the number defined there. Then:*

- *(a) Q^{pl} is C^∞ on $\mathbb{R} \times \mathbb{C}$, and $Q^{\mathrm{pl}}(t,u) = 0$ if $\pi|u|^2 \ge r_B$ or $t \notin (0,1)$;*
- *(b) q_t is generated by Q^{pl} ($\iota_X\omega = -dQ^{\mathrm{pl}}_t$), $q_t = \mathrm{id}$ for $t \le 0$, $q_t = q_1$ for $t \ge 1$, $q_t = \mathrm{id}$ on $\{\pi|u|^2 \ge r_B\}$, and $q_t(\bar{D}(A)) = \bar{D}(A)$ for every $A \ge r_B$, in particular for $A = 1$;*

(c) for every $z \in \bar{D}(1)$,

$$f\big(\pi|q_1(z)|^2\big) \le g(\pi|z|^2) + \xi.$$

Proof. By Lemma 7.4, Theorem 4.21 applies with $h = 1$ and a one-point parameter space. Its items (a), (b), (d) give (a) and (b), because $P_{\mathrm{supp}} = r_B$ and Q, which vanishes with all derivatives on $\{\pi|u|^2 > P_{\mathrm{supp}}\}$, vanishes by continuity on $\{\pi|u|^2 \ge P_{\mathrm{supp}}\}$; likewise $q_t = \mathrm{id}$ there.

(c) Put $I = \pi|z|^2$. If $I \le 1 - 8\eta < p_{\mathrm{on}}$, Theorem 4.21(c) gives $q_1(z) = q_{\mathrm{in}}(z) = C(f_{\mathrm{in}}(z))$, where f_{in} is the map f_y of Definition 4.6 and C the polar chart (3). By Theorem 4.7(b),(c), with $f_{\mathrm{in}}(z) = (x, \theta)$ and $p = p^* + x = \pi|q_1(z)|^2$, we have $\widetilde{H}(p, \theta) = E(I)$, where $E(I)$ is the level whose closed sublevel $\{\widetilde{H} \le E(I)\}$ has area exactly I (Lemma 4.2(iv)). Since $I \le 1 - 3\eta$, Lemma 7.6 gives $f(\pi|q_1(z)|^2) < g(I) + \frac{3}{8}\xi$. If $1 - 8\eta < I \le 1$, then $\pi|q_1(z)|^2 \in [0, 1]$ by (b), so by (44)

$$f(\pi|q_1(z)|^2) \le \max_{[0,1]} f = g(1) \le g(I) + \Lambda(1 - I) < g(I) + 8\Lambda\eta \le g(I) + \tfrac{1}{8}\xi. \qquad \square$$

Remark 7.8 (Dimension four). For $m = 1$ there is no parameter: $\Delta = [0, h]$ with $h = L$ a positive real, ℓ is a constant in $[h - \zeta/2, h - \delta_1]$, and χ is a constant, equal to 1 when $h \ge \frac{7}{4}\eta_1$ (Definition 7.10, Lemma 7.11). If $\chi \ne 0$, Theorem 7.13 is Theorem 7.7 run with the clock $\chi S(t)$ and rescaled to the disc of area ℓ; if $\chi = 0$, the isotopy is the identity. The four-dimensional branch does not use this route: for $m = 1$ the iteration can act directly on continuous convex functions on $[0, h]$, without the polyhedral bookkeeping of §3 and without the slice length ℓ and the cutoff χ, and §6 carries this out with a fixed mollifier and one planar path on the whole disc of area h, with constants written for the cubic model.

7.5. **Families.**

Theorem 7.9 (Planar rearrangement with parameters)**.** *Let* $(\mathcal{O}, (a_r, b_r)_{r\in\mathcal{I}}, \Lambda, M_0)$ *be planar data and* $\xi > 0$ *real. Let* $Q^{\mathrm{pl}}(y, t, u) = Q_{y,t}(u)$ *and* $q_{y,t} = \hat{q}_{y,t}$ *be the Hamiltonian and the isotopy of Theorem 4.21 for the modified potential pair of Definition 7.3 over* $\mathcal{O}$ *and the isotopy data of Definition 7.3. Then* Q^{pl} *and* $(y, t, u) \mapsto q_{y,t}(u)$ *are* C^∞ *on* $\mathcal{O} \times \mathbb{R} \times \mathbb{C}$*; for every* $y \in \mathcal{O}$*,* $Q^{\mathrm{pl}}(y, \cdot, \cdot)$ *and* $q_{y,\cdot}$ *have properties (a) and (b) of Theorem 7.7 with the same* r_B*; and for every* $y \in \mathcal{O}$ *and* $z \in \bar{D}(1)$*,*

$$f_y\big(\pi|q_{y,1}(z)|^2\big) \le g_y(\pi|z|^2) + \xi.$$

Proof. The constants of Definitions 7.1 and 7.3 do not depend on y, and Lemmas 7.2 and 7.4 hold for every $y \in \mathcal{O}$; so Theorem 4.21 applies over the open set $\mathcal{Y} = \mathcal{O}$, and Theorem 4.23(a) gives joint smoothness. For fixed y the proof of Theorem 7.7 applies verbatim with $f_y, g_y, U_y, \mathcal{A}_y$. $\square$

7.6. **Slice lengths, cutoffs and the parameter domain.** We return to the polytope. Fix a convex polyhedral function with real coefficients $F = \max_{r\in\mathcal{I}} \phi_r$ (ϕ_r real affine on $\mathbb{R}^m$, $\mathcal{I}$ finite), a coordinate j and a real $e > 0$. Write $p = (y, x)$ with $x = p_j$, $y \in \mathbb{R}^{m-1}$, and let $\mathcal{N}_j = \{\nu : b_{\nu j} > 0\}$, which is nonempty (§3.1), $N_j = \#\mathcal{N}_j$. Recall that $N_\Delta \ge N_j$ is the number of faces of Δ (§3.1). For $m = 1$ the variable y is absent and every L_ν is a positive constant. The coordinate-box conditions below are then void; the face-length conditions remain the displayed tests on these constants.

Definition 7.10 (Slice data and the lifted Hamiltonian)**.** (1) (Constants.)

$$\Lambda_F = 1 + \sum_r |\partial_j \phi_r|, \quad \eta_1 = \frac{e}{64\Lambda_F}, \quad \zeta = \frac{\eta_1}{8}, \quad \xi = \frac{e}{4}, \quad \delta_1 = \frac{\zeta}{4}, \quad s_1 = \frac{\zeta}{4(N_\Delta + 1)},$$

and

$$\varrho = 2 + \sum_i \sum_{\nu:\, b_{\nu i} > 0} \frac{\ell_\nu}{b_{\nu i}}.$$

(2) (Slice length.) For $y \in \mathbb{R}^{m-1}$,

$$\ell(y) = -s_1 \log \sum_{\nu \in \mathcal{N}_j} \exp\bigl(-L_\nu(y)/s_1\bigr) - \delta_1.$$

(3) (Cutoffs.) With the flat step S of Definition 2.3,

$$\chi_L(y) = \prod_{\nu \in \mathcal{N}_j} S\Bigl(\frac{L_\nu(y) - 5\eta_1/4}{\eta_1/2}\Bigr), \quad \rho_{\mathrm{box}}(y) = \prod_{i \neq j}\Bigl(1 - S\Bigl(\frac{y_i^2 - \varrho^2}{3\varrho^2}\Bigr)\Bigr), \quad \chi = \rho_{\mathrm{box}}\,\chi_L.$$

(4) (Parameter domain.) $\mathcal{O}_j = \{y \in \mathbb{R}^{m-1} : \|y\|_\infty < 3\varrho,\ L_\nu(y) > \frac{3}{4}\eta_1\ (\nu \in \mathcal{N}_j)\}$, an open polyhedron containing $\{y \in \Delta_j : L(y) > \frac{3}{4}\eta_1\}$ (Lemma 7.11(c)).

(5) (Planar data on slices.) $a_r(y) = \phi_r(y, 0)$ and $b_r(y) = \ell(y)\,\partial_j\phi_r$, so that $f_y(p) = F(y, \ell(y)p)$ in (43);

$$\bar\ell = 1 + \sum_{\nu \in \mathcal{N}_j} \frac{|\ell_\nu| + 4\varrho\sum_i |b_{\nu i}|}{b_{\nu j}}, \qquad \Lambda = 1 + \bar\ell \sum_r |\partial_j \phi_r|,$$

$$M_0 = 1 + \sum_r \left(|\phi_r(0)| + 4\varrho \sum_i |\partial_i \phi_r| + |\partial_j \phi_r| \bar\ell \right).$$

Let Q^{pl}, $q_{y,t}$ and r_B be those of Theorem 7.9 for the planar data $(\mathcal{O}_j, (a_r, b_r), \Lambda, M_0)$ and ξ.

(6) (Formula and zero sets.) Let $\mathcal{Z}_j, \mathcal{F}_j \subset \mathbb{R}^{m-1}$ be the open sets

$$\mathcal{Z}_j = \bigl\{y : |y_i| > \tfrac{9}{4}\varrho \text{ for some } i \neq j, \text{ or } L_\nu(y) < \tfrac{9}{8}\eta_1 \text{ for some } \nu \in \mathcal{N}_j\bigr\},$$
$$\mathcal{F}_j = \bigl\{y : |y_i| < \tfrac{5}{2}\varrho \text{ for all } i \neq j, \text{ and } L_\nu(y) > \eta_1 \text{ for all } \nu \in \mathcal{N}_j\bigr\}.$$

(7) (Lifted Hamiltonian.) For $t \in \mathbb{R}$ and $w \in \mathbb{C}^m$, with $y = y(w) = (\pi|w_i|^2)_{i \neq j}$,

$$Q^{(j)}(t, w) = \begin{cases} \chi(y) S'(t)\, \ell(y)\, Q^{\mathrm{pl}}\bigl(y, \chi(y)S(t),\, w_j/\sqrt{\ell(y)}\bigr) & \text{if } y \in \mathcal{F}_j, \\ 0 & \text{if } y \in \mathcal{Z}_j. \end{cases} \tag{47}$$

The sets $\mathcal{F}_j$ and $\mathcal{Z}_j$ make (47) a definition on all of $\mathbb{C}^m$: the planar family exists only over $\mathcal{O}_j$, the formula is used only on $\mathcal{F}_j \subset \mathcal{O}_j$, and by Lemma 7.11(e) the two cases cover $\mathbb{R}^{m-1}$ and agree on their overlap. If $\mathcal{O}_j = \emptyset$, then $\mathcal{F}_j = \emptyset$ and $Q^{(j)} = 0$.

Lemma 7.11 (Properties of the slice data)**.** *In the situation of Definition 7.10:*

(a) ℓ is real-analytic on $\mathbb{R}^{m-1}$ and $\delta_1 \le L(y) - \ell(y) \le \zeta/2$ for all y.

(b) χ is C^∞ with values in $[0,1]$. $\rho_{\mathrm{box}}(y) = 1$ if $\|y\|_\infty \le \varrho$, and $\rho_{\mathrm{box}} = 0$ on the open set $\{|y_i| > 2\varrho$ for some $i\}$; $\chi_L = 0$ on the open set $\{L_\nu < \frac{5}{4}\eta_1$ for some $\nu \in \mathcal{N}_j\}$, and $\chi_L(y) = 1$ if $L(y) \ge \frac{7}{4}\eta_1$.

(c) $\Delta_j \subset \{\|y\|_\infty \le \varrho - 2\}$; hence $\{y \in \Delta_j : L(y) > \frac{3}{4}\eta_1\} \subset \mathcal{O}_j$, and $\chi(y) = 1$ for $y \in \Delta_j$ with $L(y) \ge 2\eta_1$.

(d) On $\mathcal{O}_j$: $\ell > \frac{5}{8}\eta_1$, $|b_r| \le \Lambda$, and $|f_y(p)| \le M_0$ for $0 \le p \le 1$; thus $(\mathcal{O}_j, (a_r, b_r), \Lambda, M_0)$ are planar data. Moreover $\ell \le \bar\ell$ on $\mathcal{O}_j$, and f_y is $\Lambda_F \ell(y)$-Lipschitz on $[0,1]$.

(e) $\mathcal{Z}_j \cup \mathcal{F}_j = \mathbb{R}^{m-1}$, $\mathcal{F}_j \subset \mathcal{O}_j$, $\ell > \frac{7}{8}\eta_1$ on $\mathcal{F}_j$, and $\chi = 0$ on the open set $\mathcal{Z}_j$.

(f) If $\chi(y) \neq 0$, then $|y_i| < 2\varrho$ for all $i \neq j$, $L_\nu(y) > \frac{5}{4}\eta_1$ for all $\nu \in \mathcal{N}_j$, $y \in \mathcal{F}_j$, and $\ell(y) > \eta_1$.

Proof. (a) $\ell + \delta_1$ is the soft minimum $-s_1 \log \sum_\nu e^{-L_\nu/s_1}$ of the affine functions L_ν, $\nu \in \mathcal{N}_j$; Lemma 3.5(a),(b), applied to the functions $-L_\nu$, shows that it is real-analytic and lies in $[L -$

$s_1 \log N_j,\, L]$. Since $\log N_j < N_j \le N_\Delta < N_\Delta + 1$, $s_1 \log N_j < \zeta/4$, and $L - \ell \in [\delta_1, \delta_1 + \zeta/4] \subset [\delta_1, \zeta/2]$.

(b) Smoothness and range follow from Lemma 2.4. If $|y_i| \le \varrho$, the argument of S in the i-th factor of ρ_{box} is ≤ 0 and the factor is 1; if $|y_i| \ge 2\varrho$, the argument is ≥ 1 and the factor is 0. If $L_\nu < \frac{5}{4}\eta_1$, the ν-th factor of χ_L vanishes; if $L \ge \frac{7}{4}\eta_1$, every $L_\nu \ge L$ gives argument ≥ 1.

(c) Let $y \in \Delta_j$ and $i \ne j$. Some ν has $b_{\nu i} > 0$ (by compactness of Δ, as for j in §3.1), and $b_{\nu i} y_i \le \langle b_\nu, (y,0)\rangle \le \ell_\nu$ with $y_i \ge 0$; so $0 \le y_i \le \varrho - 2$. Then $\rho_{\mathrm{box}}(y) = 1$ and $\|y\|_\infty < 3\varrho$. If $L(y) > \frac{3}{4}\eta_1$, then $L_\nu(y) \ge L(y) > \frac{3}{4}\eta_1$ for all $\nu \in \mathcal{N}_j$, so $y \in \mathcal{O}_j$; if $L(y) \ge 2\eta_1$, moreover $\chi_L(y) = 1$ by (b).

(d) On $\mathcal{O}_j$ every $L_\nu > \frac{3}{4}\eta_1$, so $L > \frac{3}{4}\eta_1$ and $\ell \ge L - \zeta/2 > \frac{3}{4}\eta_1 - \frac{1}{16}\eta_1 > \frac{5}{8}\eta_1$. For $\nu \in \mathcal{N}_j$ and $\|y\|_\infty < 3\varrho$, $L_\nu(y) \le (\ell_\nu + 3\varrho \sum_{i\ne j} b_{\nu i})/b_{\nu j}$, so $0 < \ell < L \le \bar\ell$ and $|b_r| = \ell |\partial_j \phi_r| \le \bar\ell \sum_r |\partial_j \phi_r| < \Lambda$. For $0 \le p \le 1$, $|a_r(y)| + |b_r(y)| \le M_0$ and therefore $|f_y(p)| \le M_0$, since $|\phi_r(y,0)| \le |\phi_r(0)| + 3\varrho \sum_{i\ne j} |\partial_i \phi_r|$. The functions a_r are affine and b_r real-analytic. The Lipschitz bound is the slope bound $\max_r |b_r(y)| \le \Lambda_F \ell(y)$.

(e) If $y \notin \mathcal{Z}_j$, then $|y_i| \le \frac{9}{4}\varrho < \frac{5}{2}\varrho$ and $L_\nu(y) \ge \frac{9}{8}\eta_1 > \eta_1$, so $y \in \mathcal{F}_j$. If $y \in \mathcal{F}_j$, then $\|y\|_\infty < \frac{5}{2}\varrho < 3\varrho$ and $L_\nu > \eta_1 > \frac{3}{4}\eta_1$, so $y \in \mathcal{O}_j$, and $\ell \ge L - \zeta/2 > \eta_1 - \frac{1}{16}\eta_1$. If $y \in \mathcal{Z}_j$, either some $|y_i| > \frac{9}{4}\varrho > 2\varrho$, and $\rho_{\mathrm{box}}(y) = 0$ by (b), or some $L_\nu(y) < \frac{9}{8}\eta_1 < \frac{5}{4}\eta_1$, and $\chi_L(y) = 0$ by (b).

(f) If $\chi(y) \ne 0$, every factor of $\chi_L(y)$ is nonzero, so $(L_\nu(y) - \frac{5}{4}\eta_1)/(\eta_1/2) > 0$, and every factor of $\rho_{\mathrm{box}}(y)$ is nonzero, so $y_i^2 < 4\varrho^2$. Hence $|y_i| < 2\varrho < \frac{5}{2}\varrho$ and $L_\nu(y) > \frac{5}{4}\eta_1 > \eta_1$, so $y \in \mathcal{F}_j$, and $\ell \ge L - \zeta/2 > \frac{5}{4}\eta_1 - \frac{1}{16}\eta_1 > \eta_1$. □

7.7. **The lift.**

Lemma 7.12 (Support and outer faces)**.** *Let $Q^{(j)}$ be as in Definition 7.10 and put*

$$c_j = \tfrac{1}{2}(1 - r_B)\, \eta_1 \min_{\nu \in \mathcal{N}_j} b_{\nu j} > 0.$$

- *(a) If $Q^{(j)}(t,w) \ne 0$, then, with $y = y(w)$, $\chi(y) \ne 0$, $\|y\|_\infty < 2\varrho$ and $p_j(w) < r_B\, \ell(y)$. In particular $Q^{(j)}$ vanishes outside $\mathbb{R} \times K_j$, $K_j = \{w : \|y(w)\|_\infty \le 2\varrho,\ p_j(w) \le r_B \bar\ell\}$, a compact set.*
- *(b) For $\nu \in \mathcal{N}_j$ and every t: if $Q^{(j)}(t,w) \ne 0$, then*

$$\ell_\nu - \langle b_\nu, p(w)\rangle = b_{\nu j}\big(L_\nu(y) - p_j\big) \ge b_{\nu j}\big((L(y) - \ell(y)) + (1 - r_B)\ell(y)\big) > b_{\nu j}(1 - r_B)\eta_1. \tag{48}$$

  Hence $Q_t^{(j)}$ vanishes on the open set $\{w : \langle b_\nu, p(w)\rangle > \ell_\nu - c_j\}$.
- *(c) For $\nu \notin \mathcal{N}_j$ and every t, $\{Q_t^{(j)}, \langle b_\nu, p\rangle\} = 0$ on $\mathbb{C}^m$.*

Proof. (a) By (47), $Q^{(j)}(t,w) \ne 0$ forces $y \in \mathcal{F}_j$, $\chi(y) \ne 0$ and $Q^{\mathrm{pl}}(y, \chi(y) S(t), w_j/\sqrt{\ell(y)}) \ne 0$. Lemma 7.11(f) gives $\|y\|_\infty < 2\varrho$, and, since $Q^{\mathrm{pl}}(y,\cdot,u) = 0$ for $\pi|u|^2 \ge r_B$ (Theorem 7.9), $\pi|w_j|^2/\ell(y) < r_B$. By Lemma 7.11(d),(e), $\ell(y) \le \bar\ell$. K_j is closed and bounded, since $|w_i|^2 \le 2\varrho/\pi$ for $i \ne j$ and $|w_j|^2 \le r_B \bar\ell/\pi$.

(b) $\langle b_\nu, p\rangle = \sum_{i\ne j} b_{\nu i} y_i + b_{\nu j} p_j$ and $L_\nu = (\ell_\nu - \sum_{i\ne j} b_{\nu i} y_i)/b_{\nu j}$ give the identity. By (a) $p_j < r_B \ell$, and $L_\nu \ge L$; this gives the first inequality. By Lemma 7.11(a) $L - \ell \ge 0$, and by Lemma 7.11(f) $\ell > \eta_1$; this gives the second. Since $c_j < b_{\nu j}(1 - r_B)\eta_1$, no point of $\{\ell_\nu - \langle b_\nu, p\rangle < c_j\}$ has $Q_t^{(j)} \ne 0$.

(c) $\langle b_\nu, p\rangle = \sum_{i\ne j} b_{\nu i} p_i$. The Hamiltonian field of $p_i = \pi|w_i|^2$ rotates the i-th coordinate plane and fixes the other coordinates, and $Q_t^{(j)}$ depends on w_i, $i \ne j$, only through p_i; so it is invariant under these rotations, and $\{Q_t^{(j)}, p_i\} = \pm dQ_t^{(j)}(X_{p_i}) = 0$. □

Theorem 7.13 (Realisation of one coordinate rearrangement)**.** [Lean: needed consequence verified.]
Let $F = \max_{r\in\mathcal{I}} \phi_r$ be convex polyhedral with real coefficients on Δ, $j \in \{1,\dots,m\}$ and $e > 0$ real, and let $Q^{(j)}$ be the function of Definition 7.10. Then $Q^{(j)}$ is C^∞ on $\mathbb{R}\times\mathbb{C}^m$, vanishes outside $\mathbb{R}\times K_j$ with K_j compact, and $Q^{(j)}(t,\cdot) = 0$ for $t \le 0$ and $t \ge 1$. Its flow $(q_t^{(j)})_{t\in[0,1]}$ ($\iota_X\omega_V = -dQ_t^{(j)}$) is a smooth Hamiltonian isotopy of $\mathbb{C}^m$ with:

- *(a) $q_0^{(j)} = \mathrm{id}$; each $q_t^{(j)}$ preserves the moments p_i, $i \ne j$, and maps V_Δ onto V_Δ; explicitly, for w with $\chi(y(w)) \ne 0$ the j-th coordinate of $q_t^{(j)}(w)$ is $\sqrt{\ell(y)}\, q_{y,\chi(y)S(t)}(w_j/\sqrt{\ell(y)})$, $y = y(w)$, and for $\chi(y(w)) = 0$ it is w_j; the coordinates w_i, $i \ne j$, are rotated in their planes, so their moduli are preserved;*
- *(b) with the positive c_j of Lemma 7.12, for every $\nu \in \mathcal{N}_j$ and every t, $Q_t^{(j)}$ vanishes on $\{\langle b_\nu, p\rangle > \ell_\nu - c_j\}$, and for every $\nu \notin \mathcal{N}_j$, $Q_t^{(j)}$ Poisson-commutes with $\langle b_\nu, p\rangle$;*
- *(c) $F\big(p(q_1^{(j)}(w))\big) \le (R_jF)(p(w)) + e$ for every $w \in V_\Delta$.*

In particular Theorem M3.4 holds, with $Q_t^{(j)} = Q^{(j)}(t,\cdot)$.

Proof. Smoothness. The sets $\Omega_Z = \{w : y(w) \in \mathcal{Z}_j\}$ and $\Omega_F = \{w : y(w) \in \mathcal{F}_j\}$ are open in $\mathbb{C}^m$, since $w \mapsto y(w)$ is continuous, and they cover $\mathbb{C}^m$ by Lemma 7.11(e). On $\mathbb{R}\times\Omega_Z$, $Q^{(j)} = 0$. On $\mathbb{R}\times\Omega_F$, $y(w) \in \mathcal{O}_j$ and $\ell(y(w)) > \frac{7}{8}\eta_1$ (Lemma 7.11(e)), so the formula in (47) is a composite of C^∞ maps: $w \mapsto y(w)$ is polynomial, χ and ℓ are smooth, $\ell^{-1/2}$ is smooth where $\ell > 0$, $w \mapsto w_j/\sqrt{\ell(y(w))}$ is smooth including at $w_j = 0$ and on every axis $w_i = 0$, and Q^{pl} is smooth on $\mathcal{O}_j\times\mathbb{R}\times\mathbb{C}$ (Theorem 7.9). On the overlap the formula vanishes, because $\chi(y) = 0$ there by Lemma 7.11(e). Hence $Q^{(j)}$ is well defined and C^∞. By Lemma 7.12(a) it vanishes outside $\mathbb{R}\times K_j$. For $t \le 0$ and $t \ge 1$, $S'(t) = 0$, so $Q^{(j)}(t,\cdot) = 0$. The flow of a smooth time-dependent field supported in $\mathbb{R}\times K_j$ exists for all times and is a smooth isotopy of $\mathbb{C}^m$.

The flow (a). Since ω_V is the sum of the area forms of the coordinate planes, the w_i-component of the field at w is the planar Hamiltonian field of $w_i \mapsto Q^{(j)}(t,w)$, the other coordinates being fixed. For $i \ne j$ this function depends on w_i only through p_i, so the component is $\partial Q_t^{(j)}/\partial p_i$ times the field of p_i, which rotates the i-th plane; hence $|w_i|$ and p_i are constant along the flow (equivalently $\{Q_t^{(j)}, p_i\} = 0$, as in Lemma 7.12(c)), and so is y. If $\chi(y) = 0$, the function $w_j \mapsto Q^{(j)}(t,w)$ vanishes identically (for $y \in \mathcal{Z}_j$ by definition, for $y \in \mathcal{F}_j$ through the factor $\chi(y)$), so w_j is constant. If $\chi(y) \ne 0$, then $y \in \mathcal{F}_j$ and $Q^{(j)}(t,w) = \ell\, G_t(w_j/\sqrt{\ell})$ with $G_t(u) = \chi S'(t) Q^{\mathrm{pl}}(y, \chi S(t), u)$, $\ell = \ell(y)$, $\chi = \chi(y)$. For a planar Hamiltonian $\ell\, G(w/\sqrt{\ell})$ the field at w is $\sqrt{\ell}\, X_G(w/\sqrt{\ell})$, because the gradient of $\ell\, G(w/\sqrt{\ell})$ is $\sqrt{\ell}(\nabla G)(w/\sqrt{\ell})$ and the field is a fixed linear map applied to the gradient. Hence $u = w_j/\sqrt{\ell}$ solves $\dot u = \chi S'(t) X_{Q^{\mathrm{pl}}(y,\chi S(t),\cdot)}(u)$. The curve $t \mapsto q_{y,\chi S(t)}(u_0)$ solves the same equation with the same initial value, so $u(t) = q_{y,\chi S(t)}(u_0)$. This is the description in (a), and $q_0^{(j)} = \mathrm{id}$.

By (2), $w \in V_\Delta$ if and only if $y(w) \in \Delta_j$ and $p_j(w) \le L(y(w))$. The flow preserves y. If $\chi(y) = 0$ it fixes w_j. If $\chi(y) \ne 0$, the planar map $q_{y,\chi S(t)}$ maps $\bar D(1)$ onto itself and is the identity on $\{\pi|u|^2 \ge r_B\}$ (Theorem 7.9); in the variable $w_j = \sqrt{\ell}u$ it maps $\{p_j \le \ell\}$ onto itself and fixes every point with $p_j \ge r_B\ell$. Since $\ell < L$ (Lemma 7.11(a)), it maps $\{p_j \le L(y)\}$ into itself and fixes every point with $p_j > L(y)$. Thus $q_t^{(j)}$ maps V_Δ into V_Δ and $\mathbb{C}^m\setminus V_\Delta$ into $\mathbb{C}^m\setminus V_\Delta$; being a bijection of $\mathbb{C}^m$, it maps V_Δ onto V_Δ.

Faces (b). This is Lemma 7.12(b),(c).

The estimate (c). Let $w \in V_\Delta$, $p(w) = (y,x)$, so $y \in \Delta_j$ and $0 \le x \le L(y)$, and let $x' = p_j(q_1^{(j)}(w))$. By (a), $p(q_1^{(j)}(w)) = (y,x')$ with $0 \le x' \le L(y)$. Along the slice, $F(y,\cdot)$ is

Λ_F-Lipschitz, and by Lemma 3.2(i) $(R_jF)(y,x)$ is the increasing rearrangement of $F(y,\cdot)$ on $[0,L(y)]$ at x; in particular $(R_jF)(y,x) \geq \min_{[0,L(y)]} F(y,\cdot)$.

Case $L(y) < 2\eta_1$. Then

$$F(y,x') \leq \max_{[0,L(y)]} F(y,\cdot) \leq \min_{[0,L(y)]} F(y,\cdot) + \Lambda_F L(y) < (R_jF)(y,x) + 2\Lambda_F\eta_1,$$

and $2\Lambda_F\eta_1 = e/32$. This case covers all slices on which χ may lie strictly between 0 and 1 and all degenerate slices.

Case $L(y) \geq 2\eta_1$. Then $\chi(y) = 1$ and $y \in \mathcal{O}_j$ (Lemma 7.11(c)). Write $\ell = \ell(y)$, $L = L(y)$, and let g_y be the increasing rearrangement of $f_y(p) = F(y,\ell p)$ on $[0,1]$. For $0 \leq x'' \leq \ell$, the substitution $a \mapsto \ell a$ gives

$$g_y(x''/\ell) = \min_{0\leq a\leq \ell - x''} \max\big(F(y,a), F(y,a+x'')\big), \tag{49}$$

the rearrangement of $F(y,\cdot)$ on the shortened slice $[0,\ell]$.

If $x \leq \ell$: by (a), $x' = \ell\,\pi|q_{y,1}(u)|^2$ with $u = w_j/\sqrt{\ell} \in \bar{D}(1)$, $\pi|u|^2 = x/\ell$. Theorem 7.9 gives $F(y,x') = f_y(\pi|q_{y,1}(u)|^2) \leq g_y(x/\ell) + \xi$. Let $[a, a+x] \subset [0,L]$ be a minimising interval for $(R_jF)(y,x)$ and $a' = \min(a, \ell - x) \geq 0$. Then $[a', a'+x] \subset [0,\ell]$ and

$$0 \leq a - a' = \max(0, a + x - \ell) \leq L - \ell,$$

so both endpoints move by at most $L - \ell$, and by (49)

$$g_y(x/\ell) \leq \max\big(F(y,a'), F(y,a'+x)\big) \leq (R_jF)(y,x) + \Lambda_F(L-\ell).$$

If $\ell < x \leq L$: then $\pi|u|^2 = x/\ell > 1 \geq r_B$ for $u = w_j/\sqrt{\ell}$, so $x' = x$ by (a). For a minimising interval $[a, a+x]$ of $(R_jF)(y,x)$, $0 \leq a \leq L - x < L - \ell$, hence $F(y,x) \leq F(y,a+x) + \Lambda_F a \leq (R_jF)(y,x) + \Lambda_F(L-\ell)$.

In both subcases, by Lemma 7.11(a),

$$F(y,x') \leq (R_jF)(y,x) + \xi + \Lambda_F\frac{\zeta}{2} = (R_jF)(y,x) + \frac{e}{4} + \frac{e}{1024} < (R_jF)(y,x) + e.$$

This proves (c). Items (a)–(c) are the conclusions of Theorem M3.4: the isotopy is defined on all of $\mathbb{C}^m$, hence on a neighbourhood of V_Δ, and $c_j > 0$. A positive dyadic number below c_j gives a rational collar if required, without changing the generator. □

8. Dimension four: the cubic model

Here $n = 2$ and the target is $B^\circ(1)$. We construct the normal disc-bundle model associated with a smooth cubic in $\mathbb{CP}^2$, as in Biran's decomposition [Bir01], and a transfer of its truncated regions to the ball. The bundle has degree 9 over the cubic. The base is a fixed hexagonal torus with an explicit periodic area density of integral three. The model is written on its cover $\mathbb{R}^2 \times \mathbb{C}$, with two explicit deck maps. Its horizontal coefficient is $1 - 3p$, so the fibre interval stops below $p = 1/3$; this gives the condition $3c_i < C$. The integrated profile tends to $1/2$ as the height tends to $1/3$. An explicit unit lift satisfying the Fermat equation supplies the transport and return. A cube-root symmetry, together with injectivity modulo phase, ensures that the final ball chart is defined.

8.1. **The model.** Let $\zeta = e^{2\pi i/3}$ and let $\Gamma = (\mathbb{R}/\mathbb{Z})^2$, identified with $\mathbb{C}/(\mathbb{Z} + \zeta\mathbb{Z})$ by $(u,v) \mapsto u + \zeta v$. The theta formulas in §9.2 give a smooth map $Z : \mathbb{R}^2 \to \mathbb{C}^3$ with

$$|Z| = 1, \quad Z_0^3 + Z_1^3 + Z_2^3 = 0, \quad Z(u+1,v) = -Z(u,v), \quad Z(u,v+1) = \phi(u,v)Z(u,v), \quad |\phi| = 1.$$

Representatives related by $(Z,U) \mapsto (e^{i\chi}Z, e^{3i\chi}U)$ are called gauge equivalent. Forms are evaluated on the cover and on horizontal unit representatives.

Definition 8.1 (The cubic cover model)**.** Use the deck maps

$$T_u(u,v,w) = (u+1, v, -w), \qquad T_v(u,v,w) = (u, v+1, \phi(u,v)^3 w)$$

on $\widetilde{N} = \mathbb{R}^2 \times \mathbb{C}$; N denotes these cover data with their deck identification. Put

$$p = \pi|w|^2, \quad \eta = \frac{1}{2\pi} \operatorname{Im} \sum_j \overline{Z_j(u,v)}\, dZ_j(u,v), \quad \sigma = d\eta, \quad \alpha = \frac{d \arg w}{2\pi} - 3\eta,$$

$$\Omega_0 = \sigma + d(p\alpha) = \omega_V + (1-3p)\sigma - 3dp \wedge \eta, \quad \Delta = [0,h], \quad 0 < h < 1/3, \quad H_{\mathrm{b}}(p) = 3(1-3p).$$

The expression $p\, d\arg w/(2\pi) = \frac{1}{2}\operatorname{Im}(\bar{w}\, dw)$ makes the form smooth at $w = 0$. The marked set $\mathcal{Z}$ consists of the three torus points at which $Z_0 = 0$; its inverse image in the cover is $\widetilde{\mathcal{Z}} = \mathcal{Z} + \mathbb{Z}^2$, using the three representatives given below. The cover-to-representative map is $\mathsf{N}(u,v,w) = (Z(u,v), w)$.

The deck transformations preserve p and Ω_0 and map N to gauge-equivalent pairs. The density $\sigma = \sigma_\Gamma(u,v)du \wedge dv$ is smooth, periodic and strictly positive, with

$$\int_0^1 \int_0^1 \sigma_\Gamma(u,v)\, du\, dv = 3$$

(Lemma 9.10). In particular Ω_0 is nondegenerate for $p < 1/3$. The construction uses these cover identities directly.

Lemma 8.2 (Integrated profile)**.** *For $0 < h < 1/3$,*

$$\int_0^h H_{\mathrm{b}}(p)\, dp = 3h - \frac{9}{2}h^2, \qquad \frac{1}{2} - \int_0^h H_{\mathrm{b}}(p)\, dp = \frac{1}{2}(1-3h)^2.$$

Thus this scalar budget tends to $1/2$ as $h \uparrow 1/3$.

Proof. Integrate the affine function $3 - 9p$ and complete the square. □

8.2. **Handle data.** *Torus coordinates.* Let $\zeta = e^{2\pi i/3}$ and $\Lambda_0 = \mathbb{Z} + \zeta\mathbb{Z}$. The Weierstrass parametrisation

$$\Phi_\Gamma(z) = \big[\, 2c_F\, \wp(z) : -(1 + d_F\, \wp'(z)) : -(1 - d_F\, \wp'(z)) \,\big], \qquad z = u + \zeta v,\ (u,v) \in \mathbb{R}^2,$$

with $\Phi_\Gamma(0) = [0:1:-1]$, $\wp = \wp_{\Lambda_0}$, $g_3 = g_3(\Lambda_0) \neq 0$, $c_F = g_3^{-1/3}$ and $d_F = (3g_3)^{-1/2}$, is injective modulo the lattice as a map to gauge classes of Fermat vectors. An explicit theta lift $Z(u,v)$ without poles makes σ a smooth form on $(\mathbb{R}/\mathbb{Z})^2$ and $Z^*\eta_H$ a smooth form on $\mathbb{R}^2$ with explicit lattice periods; the base data below are smooth (§9). In these coordinates $\nu = du \wedge dv$ is a positive area form with $\int \nu = 1$, and

$$\mathcal{Z} = \{b_1, b_2, b_3\}, \qquad b_1 = (0,0), \quad b_2 = (\tfrac{1}{3}, \tfrac{2}{3}), \quad b_3 = (\tfrac{2}{3}, \tfrac{1}{3}).$$

(The point b_1 is $\Phi_\Gamma(0) = [0:1:-1]$; b_2, b_3 are the two zeros of $\wp$, which are the non-lattice fixed points of $z \mapsto \zeta z$.)

Proposition 8.3 (Handle data on the cubic)**.** *In the torus coordinates (u,v) the following are handle data for the cubic model $(\Gamma, N, [0,h], \Omega_0, \mathcal{Z})$ (Definition M4.1):*

- *(a) the annuli $\mathcal{A} = \{\frac{1}{8} \le u \le \frac{5}{24}\} \subset \mathcal{A}^+ = \{\frac{1}{12} \le u \le \frac{1}{4}\}$ with handle coordinates $(s,v) = (12u - \frac{3}{2},\, v)$, so that $\mathcal{A} = [0,1] \times \mathbb{R}/\mathbb{Z}$ and $\mathcal{A}^+ = [-\frac{1}{2}, \frac{3}{2}] \times \mathbb{R}/\mathbb{Z}$;*
- *(b) the dual band $\mathcal{B} = \{\frac{1}{9} < v < \frac{2}{9}\}$, containing the closed clock band $[\frac{1}{8}, \frac{5}{24}]$; its active intersection with $\mathcal{A}^+$ is $[\frac{1}{12}, \frac{1}{4}] \times [\frac{1}{8}, \frac{5}{24}]$, with $v_- = \frac{1}{8}$, $v_+ = \frac{5}{24}$;*
- *(c) the closed one-form $\beta = b(v)\, dv$, $b = \varphi / \int_0^1 \varphi$, where $\varphi(v) = a_{\mathrm{flat}}(v - \frac{1}{8})\, a_{\mathrm{flat}}(\frac{5}{24} - v)$ on $[0,1]$, extended 1-periodically, with $a_{\mathrm{flat}}(x) = e^{-1/x}$ for $x > 0$ and $a_{\mathrm{flat}}(x) = 0$ for $x \le 0$;*

(d) the area form $\nu = du \wedge dv = \frac{1}{12}\, ds \wedge dv$*, so* $c_\nu = \frac{1}{12}$*;*

(e) the primitive operator P_Γ*: for a smooth function* $\mathfrak{b}$ *on* $(\mathbb{R}/\mathbb{Z})^2$ *with* $\int \mathfrak{b}\,\nu = 0$*,*

$$P_\Gamma(\mathfrak{b}\,\nu) = A_0\, dv - B_0\, du,$$
$$A_0(u,v) = \int_0^u \big(\mathfrak{b}(s',v) - \mathfrak{b}_0(v)\big) ds',$$
$$\mathfrak{b}_0(v) = \int_0^1 \mathfrak{b}(s',v)\, ds', \qquad B_0(v) = \int_0^v \mathfrak{b}_0(t)\, dt;$$

(f) the unitary frame $(Z_0/|Z_0|)^3$ *of* N *over* $\mathcal{A}^+$*, with fibre coordinate* $w_\mathcal{A}$ *defined by* $U = (Z_0/|Z_0|)^3 w_\mathcal{A}$*, angle* $\theta_\mathcal{A} = \arg w_\mathcal{A}/2\pi$*, and the toric one-form*

$$\Gamma_\mathcal{A}(p) = (1-3p)\,\lambda_\Gamma, \qquad \lambda_\Gamma = \eta_H - \frac{1}{2\pi}\, d\arg Z_0,$$

so that $\Omega_0 = \omega_V + d\Gamma_\mathcal{A}$ *on* $N|_{\mathcal{A}^+}$*, with* $\omega_V = dp \wedge d\theta_\mathcal{A}$*;*

(g) the coordinate discs U_j *of radius* $\frac{1}{100}$ *about* b_j *in the centred coordinates* $(u - u(b_j), v - v(b_j))$*.*

Proof. (a), (b) The maps $u \mapsto 12u - \frac{3}{2}$ and $v \mapsto v$ identify the closed bands as stated, and $ds \wedge dv = 12\, du \wedge dv$ is positive. Modulo 1, the u-coordinates $0, \frac{1}{3}, \frac{2}{3}$ of the b_j have distance at least $\frac{1}{12}$ from $[\frac{1}{12}, \frac{1}{4}]$, and their v-coordinates $0, \frac{2}{3}, \frac{1}{3}$ have positive distance from $[\frac{1}{9}, \frac{2}{9}]$; so $\mathcal{A}^+$ and $\mathcal{B}$ avoid $\mathcal{Z}$. The base is closed, so there is no boundary collar. (c) φ is smooth on $\mathbb{R}$ because a_{flat} is flat at 0; it vanishes outside $[\frac{1}{8}, \frac{5}{24}] \subset (0,1)$ and is positive exactly on $(\frac{1}{8}, \frac{5}{24})$. Hence $b \geq 0$ is smooth and 1-periodic, $b > 0$ exactly on (v_-, v_+), $\int_0^1 b = 1$, $\operatorname{supp}\beta \subset \mathcal{B}$, β is closed because it depends on v only, and $\int_{\{s\}\times\mathbb{R}/\mathbb{Z}} \beta = 1$. (d) is the positivity of ν (§9). (e) A_0 and B_0 are 1-periodic because $\int_0^1 (\mathfrak{b} - \mathfrak{b}_0)\, ds' = 0$ and $\int_0^1 \mathfrak{b}_0 = \int \mathfrak{b}\,\nu = 0$, and $d(A_0\, dv - B_0\, du) = (\mathfrak{b} - \mathfrak{b}_0)\, du \wedge dv + \mathfrak{b}_0\, du \wedge dv = \mathfrak{b}\,\nu$; the operator is linear and preserves smooth dependence on parameters, with all jets, by differentiation under the integral sign. The support condition of Definition M4.1(e) is void on a closed base. (f) $\mathcal{Z} = \{b_1, b_2, b_3\}$ is disjoint from $\mathcal{A}^+$ by (a), so $Z_0 \neq 0$ on $\mathcal{A}^+$ and the frame is defined there. The identity $\Omega_0 = \omega_V + d\Gamma_\mathcal{A}$ is Lemma 9.13: λ_Γ is basic with $d\lambda_\Gamma = \sigma$, and $\arg U = 3\arg Z_0 + \arg w_\mathcal{A}$ gives $\alpha = d\theta_\mathcal{A} - 3\lambda_\Gamma$. The form $\Gamma_\mathcal{A}$ is a smooth family of base one-forms depending on p, hence toric. (g) The b_j have mutual distance at least $\frac{1}{3}$ in the flat metric of the torus, and by (a), (b) distance at least $\frac{1}{12}$ from $\mathcal{A}^+ \cup \mathcal{B}$; both bounds exceed $\frac{2}{100}$. So the closed discs $\overline{U}_j$ are disjoint from each other and from $\mathcal{A}^+ \cup \mathcal{B}$. □

The base is closed, so there is no collar ($\chi_\partial = 0$) and no finite atlas: the primitive operator is the global operator P_Γ of (e), and the correction region of Definition M4.1(d) is the whole torus, on which $\kappa(p) = (1-3p)\sigma \geq (1-3h)\sigma$ for $p \leq h$. The background is $e_h\nu$ (Definition 9.17). The twist, the three paths, the Moser map and the layer maps over these data are constructed in §10.9.

8.3. **Transfer into the ball.** The transfer is constructed by a deformation to the normal cone, following the scheme of [Ope26]. It embeds each working region $\{p < h\}$, away from the marked fibres and modulo deck identification, into the ball. Let $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$ and consider the family $X_T = \{\mathcal{F}(Z) = TW\}$, where W has weight three. For $T = 1$ the equation says $W = \mathcal{F}(Z)$, so X_1 is a graph over $\mathbb{CP}^2$. For $T = 0$ it says $\mathcal{F}(Z) = 0$ with W free, so X_0 is the total space of the normal bundle $\mathcal{O}(3)|_E$ of the cubic E, which is the disc-bundle model. One Kähler potential Φ_E restricts to all fibres X_T. The weight-three injection places the model in X_0 with its own form; the transport is the Ω_E-orthogonal lift of ∂_T on a bounded region, and carries its points from X_0 to X_1 preserving the restricted forms; projection identifies X_1 with $\mathbb{CP}^2$, where

the restricted form is ω_{FS} plus the exact term $d_n\,dd^c\log(1+\mathfrak{u})$; and the Moser return removes that term. Both vector fields are given in closed form, with denominators bounded below for $0\le T\le 1$ and $0\le s\le 1$. To land in $B^\circ(1)=\mathbb{CP}^2\setminus\{Z_0=0\}$ the image must avoid the line $\{Z_0=0\}$. Every step commutes with the symmetry $\mathsf{r}\colon Z_0\mapsto\zeta Z_0$, whose fixed points in $\mathbb{CP}^2$ are that line and the point $[1:0:0]$. A map that commutes with r and is injective modulo gauge sends non-fixed points to non-fixed points, so the model over $E\setminus\{Z_0=0\}$ is carried off the line (Proposition 13.19).

Construction 8.4 (The cubic transfer)**.** Fix $h<d_n<1/3$. Use pairs (Z,W) with $|Z|=1$, with the gauge relation $(Z,W)\sim(e^{i\chi}Z,e^{3i\chi}W)$. The ambient form is defined on horizontal representatives by the potential

$$\Phi_E(Z,W)=\log|Z|^2+d_n\log(1+|W|^2/|Z|^6),\qquad \Omega_E=dd^c\Phi_E.$$

The constraint is $X_T=\{(Z,W):|Z|=1,\ \mathcal{F}(Z)=TW\}$. In unit representatives put $p_W=d_n|W|^2/(1+|W|^2)$, $A=1-3p_W$ and $B=d_n/(1+|W|^2)^2$. For $\|Z\|=1$ put $f=\mathcal{F}(Z)$, $\mathbf{n}=3\bar Z^2-3\bar f Z$ (with $\bar Z^2=(\bar Z_0^2,\bar Z_1^2,\bar Z_2^2)$), $G=\|\mathbf{n}\|^2$, $N_{\mathrm{den}}=G/A+T^2/B$ and $\mathfrak{u}=|\mathcal{F}(Z)|^2/\|Z\|^6$. The transfer is the composite of the following four maps.

(1) The *weight-three injection* $\iota[Z,U]=[Z,\sqrt{\pi}\,U/\sqrt{d_n-p}]$ for unit pairs with $p<d_n$.

(2) The *transport* τ_1, the time-one map, from $T=0$ to $T=1$, of the field

$$\dot Z=\frac{W\,\mathbf{n}}{A\,N_{\mathrm{den}}},\qquad \dot W=-\frac{T\,W}{B\,N_{\mathrm{den}}},\qquad \dot T=1$$

on unit representatives in the gauge $Z^\dagger\dot Z=0$.

(3) The projection $\mathrm{pr}(Z,W)=Z$ from the graph constraint at $T=1$.

(4) The *Moser return* ψ_1^{-1}, where ψ_s is the unit-representative flow of

$$Y_s=-\frac{d_n}{2}\,\frac{f\,\mathbf{n}}{(1+\mathfrak{u})(A_s+B_sG)},\qquad A_s=1-3sd_n\frac{\mathfrak{u}}{1+\mathfrak{u}},\quad B_s=\frac{sd_n}{(1+\mathfrak{u})^2},$$

with the graph forms $\omega_s^{\mathrm{gr}}=\omega_{\mathrm{FS}}+s\,d_n\,dd^c\log(1+\mathfrak{u})$, $0\le s\le 1$.

Finally set $\mathsf{r}Z=(\zeta Z_0,Z_1,Z_2)$ and $\mathsf{r}(Z,W)=(\mathsf{r}Z,W)$. This preserves the central constraint and the gauge relation.

The ball chart b_{ball} (Lemma 13.20) is applied to the output of ψ_1^{-1}. The resulting cover transfer $\mathcal{T}=b_{\mathrm{ball}}\circ\psi_1^{-1}\circ\mathrm{pr}\circ\tau_1\circ\iota\circ\mathsf{N}$ is defined on $\{p<h\}$ off the marked fibres, pulls ω_0 back to Ω_0 and identifies exactly the deck-equivalent points (Theorem 13.21 in §13.6); the avoidance of the line $\{Z_0=0\}$ is Proposition 13.19.

8.4. **The packing.** The balls of capacities ρ_i have total volume $\frac12\sum_i\rho_i^2$. The integrated profile is $\frac12-\frac12(1-3h)^2$ (Lemma 8.2). We enlarge each ρ_i by a slack δ to an auxiliary capacity a_i and choose h so that the scalar budget after subtracting the enlarged ball capacities stays positive; this budget is $\int_0^h P_{\mathrm{b}}$. As $\sum_i\rho_i^2\uparrow 1$, the slack δ, the defect $\frac13-h$ and the mean of P_{b} tend to zero.

Lemma 8.5 (Auxiliary capacities)**.** *Let $k\ge 1$ and $0<\rho_1,\dots,\rho_k<\frac13$ with $\sum_i\rho_i^2<1$. Put $S_\rho=\sum_i\rho_i$, $D=1-\sum_i\rho_i^2$, $\varpi=\frac13-\max_i\rho_i$,*

$$\delta=\min\Big(\frac{\varpi}{8},\ \frac{D}{8(S_\rho+k)},\ \frac18\Big),\qquad a_i=\rho_i+\delta,\qquad D_a=1-\sum_i a_i^2,$$

$$\varsigma=\min\Big(\frac{\frac13-a_{\max}}{2},\ \frac{D_a}{12}\Big),\qquad h=\frac13-\varsigma,\qquad \tau_\Delta=\frac{a_{\max}+h}{2h},$$

with $a_{\max}=\max_i a_i$, and let $P_{\mathrm{b}}(p)=H_{\mathrm{b}}(p)-\sum_i(a_i-p)_+$ on $\Delta=[0,h]$. Then:

(i) $\rho_i < a_i \le a_{\max} < \tau_\Delta h < h < \frac{1}{3}$, *and* $0 < \tau_\Delta < 1$;
(ii) $\sum_i a_i^2 + (1-3h)^2 < 1$;
(iii) P_{b} *is continuous, concave and piecewise affine with breakpoints in* $\{a_1, \dots, a_k\}$, *and*

$$\int_0^h P_{\mathrm{b}}(p)\,dp = \frac{1 - \sum_i a_i^2 - (1-3h)^2}{2} > 0;$$

(iv) $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) \ge 3(1-3h) > 0$ *for* $p \in [\tau_\Delta h, h]$.

If the ρ_i *are rational, so are* δ, a_i, ς, h, τ_Δ *and the breakpoints and values of* P_{b}.

Proof. (ii) and positivity of D_a. Since $\delta \le \frac{1}{8}$, $\sum_i a_i^2 - \sum_i \rho_i^2 = 2S_\rho\delta + k\delta^2 \le \delta(2S_\rho + k) \le 2\delta(S_\rho + k) \le D/4$, so $D_a \ge \frac{3}{4}D > 0$; also $D_a \le 1$. Then $(1-3h)^2 = 9\varsigma^2 \le 9D_a^2/144 = D_a^2/16 < D_a$, i.e. $\sum_i a_i^2 + (1-3h)^2 < 1$.

(i) $\delta > 0$ gives $a_i > \rho_i$, and $a_{\max} \le \max_i \rho_i + \varpi/8 < \frac{1}{3}$, so $\varsigma > 0$ and $h < \frac{1}{3}$. Further $h \ge \frac{1}{3} - \frac{1}{2}(\frac{1}{3} - a_{\max}) = \frac{1}{2}(\frac{1}{3} + a_{\max}) > a_{\max}$. Hence $\tau_\Delta h = \frac{1}{2}(a_{\max} + h)$ lies strictly between $a_{\max}$ and h, and $0 < \tau_\Delta < 1$.

(iii) H_{b} is affine and each $-(a_i - p)_+$ is concave and piecewise affine with the single breakpoint a_i. Since $a_i < h$, $\int_0^h (a_i - p)_+\,dp = a_i^2/2$, and $\int_0^h H_{\mathrm{b}} = 3h - \frac{9}{2}h^2 = \frac{1}{2}\big(1 - (1-3h)^2\big)$ by Lemma 8.2. Positivity is (ii).

(iv) For $p \ge \tau_\Delta h > a_{\max}$ every $(a_i - p)_+$ vanishes, and $H_{\mathrm{b}}(p) = 3(1-3p) \ge 3(1-3h)$.

Rationality is clear from the formulas. □

With these data the packing in the model and the transfer give the four-dimensional embeddings, Theorem 13.23 in §13.7; the parameters of the equal-capacity branch are in Remark 13.25.

9. The Fermat torus cover

The base Γ is the hexagonal torus $(\mathbb{R}/\mathbb{Z})^2$, with $z = u + \zeta v$ and $\zeta = e^{2\pi i/3}$. This section constructs the unit Fermat lift $Z(u,v)$, the two deck transformations of $\mathbb{R}^2 \times \mathbb{C}$, the model form and the handle data used in §8, and the first Moser primitive α_1, which vanishes as an ambient covector on the three marked fibres. The base area three is computed from the quasi-periods of the lift. We write $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$, $S_\Gamma = \{|Z| = 1,\ \mathcal{F}(Z) = 0\} \subset \mathbb{C}^3$, and $\eta_H = \mathrm{Im}(Z^\dagger dZ)/(2\pi)$ on the unit sphere.

9.1. The representative and cover data.

Lemma 9.1 (Unit Fermat representatives)**.** *The theta lift defined below is smooth, has unit norm and satisfies* $Z_0(u,v)^3 + Z_1(u,v)^3 + Z_2(u,v)^3 = 0$ *for every* $(u,v) \in \mathbb{R}^2$.

Proof. Away from the lattice the holomorphic lift $\hat{Z}$ is a nonzero scalar multiple of the Weierstrass vector of Lemma 9.5, so it satisfies the homogeneous cubic equation. At the lattice it is a nonzero multiple of $(0, 1, -1)$. Normalization preserves that equation, gives unit norm, and is smooth because $\hat{Z}$ never vanishes (Lemma 9.7). □

We use the cover model of Definition 8.1: $\widetilde{N} = \mathbb{R}^2 \times \mathbb{C}$ with the deck maps T_u, T_v, the map $\mathsf{N}(u,v,w) = (Z(u,v), w)$, the gauge relation $(Z,U) \sim (e^{i\chi}Z, e^{3i\chi}U)$, $p = \pi|w|^2$, $\eta = \mathrm{Im}\sum_j \bar{Z}_j dZ_j/(2\pi)$ and $\sigma = d\eta$; here $Z(u+1,v) = -Z(u,v)$, $Z(u,v+1) = \phi(u,v)Z(u,v)$ and $|\phi| = 1$.

Lemma 9.2 (Angular and model forms)**.** *On* $w \neq 0$, *the form* $\alpha = d\arg w/(2\pi) - 3\eta$ *is deck invariant and* $d\alpha = -3\sigma$. *It has integral one on each fibre circle. The product* $p\alpha$ *extends smoothly across* $w = 0$.

Proof. Under a variable phase $Z \mapsto e^{i\chi}Z$, η changes by $d\chi/(2\pi)$. The fibre coordinate changes by $e^{3i\chi}$, whose angular form changes by $3d\chi/(2\pi)$. These changes cancel. The remaining claims follow from $d\eta = \sigma$ and $p\,d\arg w/(2\pi) = \frac{1}{2}\operatorname{Im}(\bar{w}\,dw)$. □

Lemma 9.3 (The coordinate model form)**.** *On the cover,*

$$\Omega_0 = \sigma + d(p\alpha) = \omega_V + (1-3p)\sigma - 3dp \wedge \eta. \tag{50}$$

It is closed, deck invariant and smooth at the fibre axis. Its square is $2(1-3p)\omega_V \wedge \sigma$*, so it is nondegenerate for* $p < 1/3$*.*

Proof. Differentiate the smooth primitive $\frac{1}{2}\operatorname{Im}(\bar{w}\,dw) + (1-3p)\eta$. In its square, every additional mixed term has a repeated vertical differential or three base differentials. Positivity of σ is Lemma 9.10. □

9.2. **An explicit parametrisation and atlas.** We fix an explicit uniformisation of Γ by the torus $\mathbb{C}/\Lambda_0$, $\Lambda_0 = \mathbb{Z} + \zeta\mathbb{Z}$, $\zeta = e^{2\pi i/3}$, with real coordinates $(u, v) \in (\mathbb{R}/\mathbb{Z})^2$, $z = u + \zeta v$, and a lift of the curve to the unit sphere without poles. In these coordinates σ, η_H and all base data are explicit functions of (u, v), smooth, with the flat cutoffs of §9.3. Since $\zeta_6 := e^{i\pi/3} = 1 + \zeta$, also $\Lambda_0 = \mathbb{Z} + \zeta_6\mathbb{Z}$, and $\zeta_6\Lambda_0 = \Lambda_0$ because $\zeta_6^2 = \zeta_6 - 1$; likewise $\zeta\Lambda_0 = \Lambda_0$.

Definition 9.4 (Torus parametrisation)**.** Let $\wp = \wp_{\Lambda_0}$ be the Weierstrass function of Λ_0, $w_\wp = \wp'(z)$, and g_2, g_3 the invariants of Λ_0, so that $w_\wp^2 = 4\wp^3 - g_2\wp - g_3$. Put

$$c_F = g_3^{-1/3}, \qquad d_F = (3g_3)^{-1/2} \qquad \text{(principal powers)},$$

so that $c_F^3 g_3 = 1$ and $3d_F^2 g_3 = 1$ ($g_3 \neq 0$ by Lemma 9.5). Define Φ_Γ on the quotient by the lattice, with brackets denoting nonzero complex vectors modulo a scalar by

$$\Phi_\Gamma(z) = \big[\, 2c_F\,\wp(z) : -(1 + d_F\,w_\wp) : -(1 - d_F\,w_\wp)\,\big] \quad (z \notin \Lambda_0), \qquad \Phi_\Gamma(0) = [0:1:-1].$$

The torus is oriented by $\nu = du \wedge dv$, so $\int \nu = 1$.

The orientation of ν is the complex one: $d\operatorname{Re} z \wedge d\operatorname{Im} z = \operatorname{Im}\zeta\, du \wedge dv$ and $\operatorname{Im}\zeta = \sqrt{3}/2 > 0$. The constants c_F, d_F normalise the map rather than the lattice.

Lemma 9.5 (Injectivity of the Fermat representative map)**.** *The invariants of* Λ_0 *are* $g_2 = 0$ *and* $g_3 \neq 0$*, so* $w_\wp^2 = 4\wp^3 - g_3$*. The map* Φ_Γ *takes values in gauge classes of nonzero Fermat vectors, and it is injective on* $\mathbb{C}/\Lambda_0$*.*

Proof. Invariants. The Eisenstein series satisfy $G_k(\lambda\Lambda) = \lambda^{-k}G_k(\Lambda)$. With $\lambda = \zeta_6$ and $\zeta_6\Lambda_0 = \Lambda_0$ this gives $G_4 = \zeta_6^{-4}G_4$, and $\zeta_6^4 \neq 1$, so $G_4 = 0$ and $g_2 = 60G_4 = 0$. Suppose $g_3 = 0$, so $w_\wp^2 = 4\wp^3$. At the half period $\omega_1 = \frac{1}{2}$, $\wp'(\omega_1) = 0$ ($\wp'$ is odd and periodic), hence $\wp(\omega_1) = 0$. If $\wp - \wp(\omega_1)$ vanishes to order $n \geq 1$ at ω_1, then $\wp'$ vanishes to order $n-1$, and $w_\wp^2 = 4\wp^3$ gives $2(n-1) = 3n$, which is impossible. So $g_3 \neq 0$, and c_F, d_F are defined and nonzero.

Image. Off Λ_0,

$$(2c_F\wp)^3 - (1 + d_F w_\wp)^3 - (1 - d_F w_\wp)^3 = 8c_F^3\wp^3 - 2 - 6d_F^2 w_\wp^2 = \frac{8\wp^3 - 2g_3 - 2w_\wp^2}{g_3} = 0.$$

At the lattice point, $0^3 + 1^3 + (-1)^3 = 0$. At the lattice point the formula takes this value.

Injectivity. For $z \notin \Lambda_0$ the last two homogeneous coordinates of $\Phi_\Gamma(z)$ sum to $-2 \neq 0$, while for $[0:1:-1]$ they sum to 0; so the lattice point is the only preimage of $[0:1:-1]$. If $\Phi_\Gamma(z) = \Phi_\Gamma(z')$ with $z, z' \notin \Lambda_0$, the representatives agree up to a scalar, which is 1 by the sums of the last two coordinates; since $c_F, d_F \neq 0$, $\wp(z) = \wp(z')$ and $\wp'(z) = \wp'(z')$. The elliptic function $\wp - \wp(z)$ has a single pole in $\mathbb{C}/\Lambda_0$, of order two, so it has exactly two zeros counted with multiplicity; since $\wp$ is even, these are z and $-z$. Hence $z' \equiv \pm z$ modulo Λ_0. If $z' \equiv -z$,

then $\wp'(z) = \wp'(z') = -\wp'(z)$, so $\wp'(z) = 0$. The odd elliptic function $\wp'$ has a single pole, of order three, so it has exactly three zeros counted with multiplicity, and it vanishes at the three half periods; so z is a half period and $-z \equiv z$. In either case $z' \equiv z$. □

Definition 9.6 (Theta lift)**.** Let $x = \pi z = \pi(u + \zeta v)$ and

$$\vartheta(x) = 2\sum_{n\geq 0}(-1)^n e^{i\pi\zeta(n+1/2)^2}\sin\big((2n+1)x\big),$$

the Jacobi function $\vartheta_1(x\,|\,\zeta)$; primes denote x-derivatives. Put $r_\vartheta = \vartheta'''(0)/(3\vartheta'(0))$,

$$P_\vartheta = \vartheta'^2\vartheta - \vartheta''\vartheta^2 + r_\vartheta\vartheta^3, \qquad Q_\vartheta = -\vartheta'''\vartheta^2 + 3\vartheta'\vartheta''\vartheta - 2\vartheta'^3,$$

$$\hat{Z} = \big(2c_F\pi^2 P_\vartheta,\ -\vartheta^3 - d_F\pi^3 Q_\vartheta,\ -\vartheta^3 + d_F\pi^3 Q_\vartheta\big), \qquad Z = \frac{\hat{Z}}{\|\hat{Z}\|},$$

all evaluated at $x = \pi z$.

Equivalently, $\vartheta(x) = -i\,e^{i\pi\zeta/4+ix}\,\Theta(x/\pi + \frac{1}{2} + \frac{\zeta}{2}, \zeta)$ with $\Theta(y,\zeta) = \sum_{n\in\mathbb{Z}} e^{2\pi i n y + i\pi n^2\zeta}$.

Lemma 9.7 (The lift has no zero)**.** *The zeros of ϑ are exactly the points of $\pi\Lambda_0$, and they are simple. Off Λ_0,*

$$\wp(z) = -\pi^2(\log\vartheta)''(\pi z) + \pi^2 r_\vartheta,$$

and $\hat{Z} = \vartheta^3\cdot(2c_F\wp,\ -1 - d_F w_\wp,\ -1 + d_F w_\wp)$ wherever $\vartheta \neq 0$. $\hat{Z}$ is an entire function of z without zeros, so Z is a smooth map $\mathbb{R}^2 \to \mathbb{C}^3$ with $\|Z\| = 1$. Under $(u,v) \mapsto (u+1, v)$ and $(u,v) \mapsto (u, v+1)$, $\hat{Z}$ is multiplied by -1 and by $\mathfrak{m}(\pi z)^3$, where $\mathfrak{m}(x) = -e^{-i\pi\zeta}e^{-2ix}$.

Proof. Quasi-periodicity. From the series, ϑ is entire and odd, so $\vartheta(0) = 0$, and $\vartheta(x+\pi) = -\vartheta(x)$, $\vartheta(x+\pi\zeta) = \mathfrak{m}(x)\vartheta(x)$ (these follow from the corresponding identities of Θ). Since $\mathfrak{m}'/\mathfrak{m} = -2i$, the logarithmic derivative $\ell = \vartheta'/\vartheta$ satisfies $\ell(x+\pi) = \ell(x)$ and $\ell(x+\pi\zeta) = \ell(x) - 2i$.

Zeros. For every $q \in \mathbb{C}$ there is a parallelogram $P = a + [0,1]\pi + [0,1]\pi\zeta$ containing q in its interior with no zero of ϑ on its boundary. By the argument principle the number of zeros in P is $(2\pi i)^{-1}\oint_{\partial P}\ell\,dx$. The two edges parallel to $\pi\zeta$ cancel, and the two edges parallel to π contribute $\int_a^{a+\pi}(\ell(x) - \ell(x+\pi\zeta))\,dx = 2\pi i$. So every such P contains exactly one zero, and it is simple. Since $\vartheta(0) = 0$ and the zero set is $\pi\Lambda_0$-invariant, every point of $\pi\Lambda_0$ is a zero. Conversely, let q be a zero and $P \ni q$ as above. Then ∂P contains no point of $\pi\Lambda_0$, so the translate P of a fundamental parallelogram contains exactly one point of $\pi\Lambda_0$, in its interior; this point and q are zeros in P, so they coincide. Thus the zeros are exactly $\pi\Lambda_0$; in particular $\vartheta' \neq 0$ there.

The formula for $\wp$. Put $F(z) = -\pi^2(\log\vartheta)''(\pi z) = \pi^2(\vartheta'^2 - \vartheta\vartheta'')/\vartheta^2$. Since ℓ changes only by additive constants under $\pi\Lambda_0$, F is Λ_0-periodic. From $\vartheta(x) = \vartheta'(0)x\big(1 + \frac{\vartheta'''(0)}{6\vartheta'(0)}x^2 + O(x^4)\big)$ we get $(\log\vartheta)'' = -x^{-2} + \frac{\vartheta'''(0)}{3\vartheta'(0)} + O(x^2)$, so $F(z) = z^{-2} - \pi^2 r_\vartheta + O(z^2)$. Thus $\wp - F$ is periodic, holomorphic off Λ_0 and bounded near the lattice, hence constant by Liouville's theorem; the constant is $\pi^2 r_\vartheta$ because $\wp$ has zero constant Laurent term at 0. So $\vartheta^3\wp = \pi^2 P_\vartheta$. Differentiating in z (one more factor π), $w_\wp = \pi^3\frac{d}{dx}\frac{\vartheta'^2 - \vartheta\vartheta''}{\vartheta^2} = \pi^3 Q_\vartheta/\vartheta^3$, which gives the formula for $\hat{Z}$.

No zero. Where $\vartheta \neq 0$ the sum of the last two components is $-2\vartheta^3 \neq 0$. Where $\vartheta = 0$, $\vartheta' \neq 0$ and $Q_\vartheta = -2\vartheta'^3$, so the second component is $2d_F\pi^3\vartheta'^3 \neq 0$. $\hat{Z}$ is entire, being a polynomial in $\vartheta, \dots, \vartheta'''$ at πz.

Automorphy. Under $x \mapsto x + \pi$ all of $\vartheta, \dots, \vartheta'''$ change sign, and P_ϑ, Q_ϑ, ϑ^3 are homogeneous of degree three in them, so $\hat{Z} \mapsto -\hat{Z}$. Under $x \mapsto x + \pi\zeta$, off $\pi\Lambda_0$ we have $P_\vartheta = \vartheta^3\wp/\pi^2$ and $Q_\vartheta = \vartheta^3 w_\wp/\pi^3$ by the previous step; since $\wp$ and $w_\wp$ are Λ_0-periodic, P_ϑ, Q_ϑ and ϑ^3 are all multiplied by $\mathfrak{m}(x)^3$. By continuity the same holds at the points of $\pi\Lambda_0$. □

At the lattice points $P_\vartheta = 0$ and $Q_\vartheta = -2\vartheta'^3$, so $[\hat{Z}] = [0 : 1 : -1]$; off the lattice $[\hat{Z}] = \Phi_\Gamma(z)$. Hence Z is a smooth unit Fermat lift on $\mathbb{R}^2$ with that gauge class.

Construction 9.8 (The atlas of N)**.** (a) *Base.* The chart $(u, v) \mapsto \Phi_\Gamma(u + \zeta v)$ is the injective Fermat representative map of the base torus $\Gamma = (\mathbb{R}/\mathbb{Z})^2$; all base objects are written in (u, v), and $\nu = du \wedge dv$.

(b) *The torus cover of N.* The map $\mathbb{R}^2 \times \mathbb{C} \to N$, $(u, v, w) \mapsto [Z(u, v), w]$, with $w = \xi + i\eta$, is the cover-to-representative map. Let $\phi(u, v) = \mathfrak{m}(\pi z)^3/|\mathfrak{m}(\pi z)|^3$, so that $Z(u+1, v) = -Z(u, v)$ and $Z(u, v + 1) = \phi(u, v)Z(u, v)$ (Lemma 9.7). The deck maps

$$T_u(u, v, w) = (u + 1, v, -w), \qquad T_v(u, v, w) = (u, v + 1, \phi(u, v)^3 w)$$

generate a free $\mathbb{Z}^2$-action, and $(u, v, w) \mapsto [Z(u, v), w]$ is exactly the deck relation defining N.

(c) *Algebraic frames.* For $j = 0, 1, 2$, $Z\bar{Z}_j/|Z_j|$ is a local unitary frame over $\Gamma \cap \{Z_j \neq 0\}$ in the sense of Definition 8.1. Its fibre coordinate w_j is given by $U = (Z_j/|Z_j|)^3 w_j$, so its unit section is $[Z] \mapsto [Z, (Z_j/|Z_j|)^3]$, and its connection form is $-3\eta_H + \frac{3}{2\pi} d \arg Z_j$. These three sets cover Γ, and the transition functions $w_j = (Z_k|Z_j|/(|Z_k|Z_j))^3 w_k$ have weight three.

(d) *The model form in the torus frame.* In the frame (b), $p = \pi|w|^2$, $\alpha = d\theta - 3Z^*\eta_H$, and

$$\Omega_0 = d\Big((1 - 3p)\, Z^*\eta_H + \tfrac{1}{2}(\xi\, d\eta - \eta\, d\xi)\Big) = \omega_V + (1 - 3p)\sigma + dp \wedge (-3Z^*\eta_H).$$

Verification. (b) $[Z(u + 1, v), -w] = [-Z(u, v), -w] = [Z(u, v), (-1)^{-3}(-w)] = [Z(u, v), w]$ by $(e^{i\pi} Z, e^{3i\pi} U) \sim (Z, U)$, and similarly $[Z(u, v + 1), \phi^3 w] = [\phi Z(u, v), \phi^3 w] = [Z(u, v), w]$; so the map is invariant. Since $\mathfrak{m}(x + \pi) = \mathfrak{m}(x)$, we have $\phi(u + 1, v) = \phi(u, v)$, so $T_u T_v(u, v, w) = (u + 1, v + 1, -\phi(u, v)^3 w) = T_v T_u(u, v, w)$. If two unit representative pairs from the cover are gauge equivalent, projective injectivity of the explicit torus parametrization makes their base coordinates differ by an integral lattice vector. Apply the corresponding deck map; the base coordinates then agree and unit gauge equivalence fixes the fibre coordinate. Thus gauge equivalence on this cover is exactly the stated deck relation. (c) $Z\bar{Z}_j/|Z_j|$ is unchanged under $Z \mapsto e^{i\psi} Z$, so it is a gauge-invariant normalized lift on the cover patch $\{Z_j \neq 0\}$, and $[Z, (Z_j/|Z_j|)^3 w_j] = [e^{-i \arg Z_j} Z, w_j]$ by the weight-three relation. Since $\arg U = 3 \arg Z_j + \arg w_j$, the angular form of Lemma 9.2 is $\alpha = d \arg w_j/(2\pi) + \frac{3}{2\pi} d \arg Z_j - 3\eta_H$, so the connection form of the frame is $-3\eta_H + \frac{3}{2\pi} d \arg Z_j$. No point of Γ has $Z_0 = Z_1 = Z_2 = 0$. On overlaps, $[Z, (Z_k/|Z_k|)^3 w_k] = [Z, (Z_j/|Z_j|)^3 w_j]$ gives $w_j = (Z_k|Z_j|/(|Z_k|Z_j))^3 w_k$. (d) Lemma 9.2 and Lemma 9.3 with $a_{\mathrm{loc}} = -3Z^*\eta_H$, and $d(Z^*\eta_H) = \sigma$; the primitive is $Z^*\eta_H + p\alpha$. □

Lemma 9.9 (The deck phase)**.** *The two fibre factors are -1 and ϕ^3. They commute with the lattice translations. On $v = 0$, $\phi(u, 0) = \phi_0 e^{-6\pi i u}$ for a constant $|\phi_0| = 1$.*

Proof. The theta multiplier is periodic under $u \mapsto u + 1$, so the two deck maps commute. On $v = 0$, its formula is $\mathfrak{m}(\pi u) = -e^{-i\pi\zeta} e^{-2\pi i u}$. Taking its unit phase and cubing gives the displayed identity. □

Lemma 9.10 (The area of the base)**.** *Write $Z^*\eta_H = \mathsf{h}_u\, du + \mathsf{h}_v\, dv$ and $\sigma = d(Z^*\eta_H) = \sigma_\Gamma(u, v)\, du \wedge dv$ on $\mathbb{R}^2$. Then σ_Γ is smooth, 1-periodic in u and in v, strictly positive, and $\int_0^1 \int_0^1 \sigma_\Gamma\, du\, dv = 3$.*

Proof. Positivity. σ_Γ is the pull-back of $\omega_{\mathrm{FS}} = dd^c \log \|Z\|^2$ by $z \mapsto [\hat{Z}(z)]$:

$$\sigma_\Gamma = \operatorname{Im} \zeta \cdot \frac{\|\hat{Z}\|^2 \|\hat{Z}'\|^2 - |\hat{Z}^\dagger \hat{Z}'|^2}{\pi \|\hat{Z}\|^4} \qquad (\hat{Z}' = d\hat{Z}/dz),$$

which by the Cauchy–Schwarz equality case is positive unless $\hat{Z}'$ is proportional to $\hat{Z}$. Off the lattice, $\hat{Z} = \vartheta^3 R$ with $R = (2c_F\wp, -1 - d_F w_\wp, -1 + d_F w_\wp)$, and $\hat{Z}' \parallel \hat{Z}$ iff $R' \parallel R$. Now $R' = (2c_F w_\wp, -d_F\wp'', d_F\wp'')$; if $R' = aR$, adding the last two components gives $0 = -2a$, so $R' = 0$, i.e. $w_\wp = 0$ and $\wp'' = 0$. This is impossible: differentiating $w_\wp^2 = 4\wp^3 - g_3$ gives $\wp'' = 6\wp^2$, and $w_\wp = 0$ forces $4\wp^3 = g_3 \neq 0$. At a lattice point the first component of $\hat{Z}$ vanishes while that of $\hat{Z}'$ is $2c_F\pi^3 P'_\vartheta = 2c_F\pi^3\vartheta'^3 \neq 0$, so again $\hat{Z}' \nparallel \hat{Z} \neq 0$.

Periodicity. By Lemma 9.7, Z changes by unit phases ϕ under the deck translations, and η_H changes by the closed form $(2\pi i)^{-1}\bar{\phi}\, d\phi$; so $\sigma = d(Z^*\eta_H)$ is periodic.

Integral. By Stokes on the unit square, $\iint \sigma_\Gamma = \iint(\partial_u \mathsf{h}_v - \partial_v \mathsf{h}_u)$. Under $u \mapsto u+1$ the lift changes by the constant phase -1, so $\mathsf{h}_v(1,v) = \mathsf{h}_v(0,v)$ and the first term integrates to 0. Under $v \mapsto v+1$ it changes by $\phi(u,v)$, and on the edge $v = 0$, $\phi = \phi_0 e^{-6\pi i u}$ (Lemma 9.9); hence $\mathsf{h}_u(u,1) = \mathsf{h}_u(u,0) + (2\pi i)^{-1}(-6\pi i) = \mathsf{h}_u(u,0) - 3$. So $\iint \partial_v \mathsf{h}_u = \int_0^1 (\mathsf{h}_u(u,1) - \mathsf{h}_u(u,0))\, du = -3$ and $\iint \sigma_\Gamma = 3$. □

Lemma 9.11 (The three points on L)**.** *The zero set $\mathcal{Z} = \{(u,v) \in \Gamma : Z_0(u,v) = 0\}$, the set of base points lying on the line $L = \{Z_0 = 0\}$, consists of the three points*

$$b_1 = (0,0), \qquad b_2 = (\tfrac{1}{3}, \tfrac{2}{3}), \qquad b_3 = (\tfrac{2}{3}, \tfrac{1}{3}).$$

Proof. $Z_0 = 0$ iff $P_\vartheta = 0$, i.e. iff $\vartheta = 0$ (the point b_1, where $\Phi_\Gamma = [0:1:-1]$) or $\wp = 0$. Since $\zeta\Lambda_0 = \Lambda_0$, $\wp(\zeta z) = \zeta^{-2}\wp(z) = \zeta\wp(z)$; at a non-lattice fixed point of $z \mapsto \zeta z$ modulo Λ_0 this forces $\wp(z) = 0$. For $z_2 = \frac{1}{3} + \frac{2}{3}\zeta$, $\zeta z_2 = -\frac{2}{3} - \frac{1}{3}\zeta \equiv z_2$ (using $\zeta^2 = -1 - \zeta$), and similarly for $z_3 = \frac{2}{3} + \frac{1}{3}\zeta \equiv -z_2$. The elliptic function $\wp$ has a single pole, of order two, in $\mathbb{C}/\Lambda_0$, hence exactly two zeros counted with multiplicity; since $z_2 \not\equiv z_3$, these are z_2 and z_3. □

9.3. **Handle data.** The handle data of the cubic model are stated in Proposition 8.3. Picture the torus as the unit square in (u,v). The annulus $\mathcal{A}$ is the vertical band $\frac{1}{8} \le u \le \frac{5}{24}$ inside the wider band $\mathcal{A}^+ = \{\frac{1}{12} \le u \le \frac{1}{4}\}$; the dual band $\mathcal{B} = \{\frac{1}{9} < v < \frac{2}{9}\}$ contains the closed clock band $\frac{1}{8} \le v \le \frac{5}{24}$ and carries the closed one-form $\beta = b(v)\, dv$; the two bands meet in one square. The three points b_j lie far from both bands and carry small discs U_j. This subsection gives the frame over $\mathcal{A}^+$, the primitive operator and the cutoffs used in §9.4.

Definition 9.12 (Frame over $\mathcal{A}^+$)**.** On $\Gamma \cap \{Z_0 \neq 0\}$ let $w_\mathcal{A}$ be the fibre coordinate of the local unitary frame $Z\bar{Z}_0/|Z_0|$ of Construction 9.8(c), i.e. $U = (Z_0/|Z_0|)^3 w_\mathcal{A}$; we also call it the frame $(Z_0/|Z_0|)^3$, after its unit section. Let $\theta_\mathcal{A} = \arg w_\mathcal{A}/2\pi$, and put

$$\lambda_\Gamma = \eta_H - \frac{1}{2\pi}\operatorname{Im}\frac{dZ_0}{Z_0}, \qquad a_\mathcal{A} = -3\lambda_\Gamma, \qquad \Gamma_\mathcal{A}(p) = (1-3p)\,\lambda_\Gamma,$$

so that $a_\mathcal{A}$ is the connection form a_{loc} of this frame.

Lemma 9.13 (The frame over the annulus)**.** *λ_Γ is a smooth one-form on $\Gamma \cap \{Z_0 \neq 0\}$ (no branch of* arg *is needed), $d\lambda_\Gamma = \sigma$, and on $N|_{\Gamma\cap\{Z_0\neq 0\}}$*

$$\alpha = d\theta_\mathcal{A} + a_\mathcal{A}, \qquad \Omega_0 = \omega_V + d\Gamma_\mathcal{A}, \quad \omega_V = dp \wedge d\theta_\mathcal{A}.$$

Proof. $\operatorname{Im}(dZ_0/Z_0) = d\arg Z_0$ is a smooth closed form where $Z_0 \neq 0$. The form λ_Γ on S_Γ is invariant and vanishes on the generator iZ $(\frac{1}{2\pi} - \frac{1}{2\pi})$, so it is basic; $d\lambda_\Gamma = d\eta_H = \sigma$. Since $\arg U = 3\arg Z_0 + \arg w_\mathcal{A}$, $\alpha = d\theta_\mathcal{A} + \frac{3}{2\pi} d\arg Z_0 - 3\eta_H = d\theta_\mathcal{A} - 3\lambda_\Gamma$. Then $\Omega_0 = d\lambda_\Gamma + d(p\, d\theta_\mathcal{A}) - 3\, d(p\lambda_\Gamma) = \omega_V + d\big((1-3p)\lambda_\Gamma\big)$. □

Definition 9.14 (Torus primitive operator)**.** This is the operator of Proposition 8.3(e). For a smooth function $\mathfrak{b}$ on $(\mathbb{R}/\mathbb{Z})^2$ with $\int \mathfrak{b}\,\nu = 0$ put

$$\mathfrak{b}_0(v) = \int_0^1 \mathfrak{b}(s', v)\, ds', \quad A_0(u, v) = \int_0^u \big(\mathfrak{b}(s', v) - \mathfrak{b}_0(v)\big)\, ds', \quad B_0(v) = \int_0^v \mathfrak{b}_0(t)\, dt,$$

and $P_\Gamma(\mathfrak{b}\,\nu) = A_0\, dv - B_0\, du$.

Lemma 9.15. *A_0 and B_0 are 1-periodic in both variables, so $P_\Gamma(\mathfrak{b}\nu)$ is a smooth one-form on the torus, and $d_\Sigma P_\Gamma(\mathfrak{b}\nu) = \mathfrak{b}\nu$. P_Γ is linear, and if $\mathfrak{b}$ depends smoothly on parameters then so does $P_\Gamma(\mathfrak{b}\nu)$, with all jets; in particular it preserves affine dependence on p.*

Proof. $A_0(u+1, v) - A_0(u, v) = \int_u^{u+1} (\mathfrak{b} - \mathfrak{b}_0)\, ds' = 0$, and A_0 is periodic in v because $\mathfrak{b}$ and $\mathfrak{b}_0$ are; $B_0(v+1) - B_0(v) = \int_0^1 \mathfrak{b}_0 = \int \mathfrak{b}\,\nu = 0$. Then $d(A_0\, dv - B_0\, du) = \partial_u A_0\, du \wedge dv - \mathfrak{b}_0\, dv \wedge du = (\mathfrak{b} - \mathfrak{b}_0 + \mathfrak{b}_0)\,\nu$. Smooth dependence on parameters follows by differentiation under the integral sign. □

Proposition 9.16 (Cutoffs on the cubic handle)**.** *Let $\mathcal{A} \subset \mathcal{A}^+$, $\mathcal{B}$ and β be the handle data of Proposition 8.3, and take the discs of its item (g) closed: $U_j = \{x_j^2 + y_j^2 \le 10^{-4}\}$, of radius $\frac{1}{100}$ about b_j in the centred coordinates $(x_j, y_j) = (u - u(b_j), v - v(b_j))$, reduced to $[-\frac{1}{2}, \frac{1}{2})^2$. With the flat step S and the cutoffs of Definition 2.6, the cutoffs*

$$\zeta_{\mathcal{A}}(u) = \chi^{\mathrm{I}}_{9/96,\,23/96,\,2/96}(u), \qquad \zeta_j = \chi_{1/200,\,1/100}(x_j, y_j)$$

(extended by zero) satisfy: $\zeta_{\mathcal{A}} = 1$ on $\{\frac{11}{96} \le u \le \frac{21}{96}\}$, a neighbourhood of $\mathcal{A}$, and $\operatorname{supp}\zeta_{\mathcal{A}} = \{\frac{9}{96} \le u \le \frac{23}{96}\} \subset \operatorname{int}\mathcal{A}^+$; $\zeta_j = 1$ on the disc of radius $\frac{1}{200}$ about b_j, and $\operatorname{supp}\zeta_j = U_j$. The discs U_j are pairwise disjoint and disjoint from $\mathcal{A}^+ \cup \mathcal{B}$, and $\operatorname{supp}\beta \cap U_j = \varnothing$. All these data are independent of the input.

Proof. By Lemma 2.7(b), $\chi^{\mathrm{I}}_{9/96,23/96,2/96} = 1$ on $[\frac{11}{96}, \frac{21}{96}] \supset [\frac{12}{96}, \frac{20}{96}] = \mathcal{A}$, and it is positive exactly on $(\frac{9}{96}, \frac{23}{96})$, so its support is $[\frac{9}{96}, \frac{23}{96}]$; moreover $\frac{8}{96} < \frac{9}{96}$ and $\frac{23}{96} < \frac{24}{96}$, so this interval lies in the interior of $\mathcal{A}^+ = [\frac{8}{96}, \frac{24}{96}]$. By Lemma 2.7(a), $\chi_{1/200,1/100} = 1$ on the closed disc of radius $\frac{1}{200}$, vanishes outside the open disc of radius $\frac{1}{100}$, and is positive on that open disc, since $S < 1$ on $(-\infty, 1)$; so its support is the closed disc U_j. The b_j have mutual distance at least $\frac{1}{3}$ in the flat metric of the torus, and distance at least $\frac{1}{12}$ from $\mathcal{A}^+ \cup \mathcal{B}$ (proof of Proposition 8.3(a),(b)); both bounds exceed $\frac{2}{100}$, which gives the disjointness. Finally $\operatorname{supp}\beta \subset \mathcal{B}$. □

9.4. **A global primitive vanishing on three whole fibres.** We now write the first path of Proposition M4.3 for the cubic in closed form. We start from an explicit primitive α^0 of the exact base form $\kappa_1 - \kappa$ and correct it twice by d_Σ-closed forms, which leaves the base curvature unchanged: over $\mathcal{A}^+$ we remove the difference between α^0 and the desired local primitive (a closed form on an annulus, exact after subtracting its period), and near the three points b_j we remove the value of the primitive. The supports of the two corrections are disjoint (Proposition 9.16).

The scalar data in this subsection are the concrete formulas below. They apply to both branches of the four-dimensional comparator.

Definition 9.17 (Admissible auxiliary data and base area forms)**.** Let $0 < h < 1/3$ and $0 < g \le 3(1-3h)$. Set

$$e_h = g/16, \qquad H^*(p) = 3(1-3p) - e_h, \qquad f(s) = S(2s - 1/2), \qquad \Gamma_{\mathrm{sm}} = e_h(u - 1/6)\, dv.$$

Thus $0 < e_h \le H_{\mathrm{b}}(h)/2$, f is nondecreasing, zero for $s \le 1/4$ and one for $s \ge 3/4$, and $d\Gamma_{\mathrm{sm}} = e_h\nu$. Put $\chi = f'(s)\, ds \wedge \beta$ (on $\mathcal{A}$, extended by zero) and

$$\kappa(p) = (1-3p)\,\sigma, \qquad \kappa_1(p) = e_h\nu + H^*(p)\chi.$$

Lemma 9.18 (Base area forms)**.** $\chi \geq 0$, $\int \chi = 1$, *and* $\int \kappa(p) = \int \kappa_1(p) = H_{\mathrm{b}}(p)$. *On* $[0,h]$, $H^* \geq \frac{1}{2}H_{\mathrm{b}}(h) > 0$. *For every* $p < \frac{1}{3}$ *with* $H^*(p) \geq 0$, *both* $\kappa(p)$ *and* $\kappa_1(p)$ *are positive area forms, and* $\kappa_1(p) \geq e_h\nu$.

Proof. $\chi = f'(s)b(v)\,ds \wedge dv \geq 0$ since $f' \geq 0$, $b \geq 0$, and $\int \chi = \int_0^1 f'\,ds \cdot \int_0^1 b\,dv = 1$. By Lemma 9.10, $\int \kappa = 3(1-3p)$, and $\int \kappa_1 = e_h + H^* = H_{\mathrm{b}}$. On $[0,h]$, $H_{\mathrm{b}} \geq H_{\mathrm{b}}(h)$ and $e_h \leq \frac{1}{2}H_{\mathrm{b}}(h)$. Finally $\sigma > 0$ (Lemma 9.10), and $\kappa_1 - e_h\nu = H^*\chi \geq 0$. □

Definition 9.19 (The annular difference and its period)**.** Let $\alpha^0(p) = P_\Gamma\big(\kappa_1(p) - \kappa(p)\big)$, a base one-form with coefficients depending on the base point and affinely on p, with $d_\Sigma\alpha^0 = \kappa_1 - \kappa$ (Lemmas 9.15 and 9.18). On $\mathcal{A}^+$ put

$$\Delta_0(p) = \Gamma_{\mathcal{A}}(p) + \alpha^0(p) - \Gamma_{\mathrm{sm}} - f(s)H^*(p)\beta,$$

and let $c(p) = \oint_{\{u=\mathrm{const}\}} \Delta_0(p)$ be its period over a v-circle in $\mathcal{A}^+$.

Lemma 9.20. *For each* p, $\Delta_0(p)$ *is* d_Σ*-closed on* $\mathcal{A}^+$*; hence* $c(p)$ *does not depend on the circle, and* c *is affine in* p.

Proof. $d_\Sigma\Gamma_{\mathcal{A}} = (1-3p)\sigma = \kappa$ by Lemma 9.13, $d_\Sigma\alpha^0 = \kappa_1 - \kappa$, $d_\Sigma\Gamma_{\mathrm{sm}} = e_h\nu$ and $d_\Sigma(fH^*\beta) = H^*f'\,ds \wedge \beta = H^*\chi$; so $d_\Sigma\Delta_0 = \kappa + \kappa_1 - \kappa - e_h\nu - H^*\chi = 0$. Two v-circles bound an annulus in $\mathcal{A}^+$, so by Stokes the period does not depend on the circle. Every term of Δ_0 is affine in p (H_{b} is affine, e_h and Γ_{sm} are constant in p, and P_Γ is linear). □

The period $c(p)$ records the holonomy of the frame $(Z_0/|Z_0|)^3$ around the core circle together with the periods of the chosen primitives; in general $c(p) \neq 0$. Since $\oint \beta = 1$, $\Delta_0 - c(p)\beta$ has period zero and is d_Σ-exact on $\mathcal{A}^+$.

Definition 9.21 (Annular primitive)**.** Write $\Delta_0(p) - c(p)\beta = D_u\,du + D_v\,dv$ on $\mathcal{A}^+$ and put

$$F_{\mathcal{A}}(u,v,p) = \int_{1/6}^{u} D_u(t,0,p)\,dt + \int_0^v D_v(u,t,p)\,dt \qquad (\tfrac{1}{12} \leq u \leq \tfrac{1}{4},\ v \in \mathbb{R}),$$

the integral of $\Delta_0 - c\beta$ along the path from $(\frac{1}{6},0)$ horizontally and then vertically.

Lemma 9.22. $F_{\mathcal{A}}$ *is a smooth function on* $\mathcal{A}^+$, 1*-periodic in* v, *affine in* p, *and* $d_\Sigma F_{\mathcal{A}} = \Delta_0 - c(p)\beta$.

Proof. $\partial_v F_{\mathcal{A}} = D_v$. By closedness $\partial_u D_v = \partial_v D_u$, so $\partial_u F_{\mathcal{A}} = D_u(u,0) + \int_0^v \partial_t D_u(u,t)\,dt = D_u(u,v)$. Periodicity in v: $\int_0^1 D_v(u,t)\,dt$ is the period of $\Delta_0 - c\beta$, which is $c(p) - c(p)\cdot 1 = 0$ by Proposition 8.3(c). Smoothness and affine dependence follow from those of D_u, D_v. □

Construction 9.23 (The global primitive α_1)**.** Define the base one-form, with $\zeta_{\mathcal{A}}F_{\mathcal{A}}$ extended by zero outside $\mathcal{A}^+$,

$$\alpha_{\mathrm{pre}}(p) = \alpha^0(p) - c(p)\beta - d_\Sigma\big(\zeta_{\mathcal{A}}F_{\mathcal{A}}(\cdot,p)\big),$$

write its value at b_j as $A_j(p)\,dx_j + B_j(p)\,dy_j$ (A_j, B_j affine in p), and put

$$\alpha_1(p) = \alpha_{\mathrm{pre}}(p) - \sum_{j=1}^{3} d_\Sigma\Big[\zeta_j\big(A_j(p)\,x_j + B_j(p)\,y_j\big)\Big]. \tag{51}$$

We use α_1 as the toric one-form $\pi_N^*\alpha_1(p(\cdot))$ on N. For the data of §M4 this is the one-form (M15), with the base point $(\frac{1}{6},0)$ for $F_{\mathcal{A}}$ and the cutoffs $\zeta_{\mathcal{A}}$, ζ_j of Proposition 9.16.

Lemma 9.24 (Properties of α_1)**.** *(i)* α_1 *is a smooth one-form on* N, *horizontal (no* dp *and no fibre-angle component), affine in* p, *and* $d_\Sigma\alpha_1 = \kappa_1 - \kappa$.

(ii) Over $\mathcal{A}$, in the frame $(Z_0/|Z_0|)^3$,

$$\Gamma_{\mathcal{A}} + \alpha_1 = \Gamma_{\mathrm{sm}} + f(s)H^*(p)\beta, \qquad \textit{hence} \qquad \Omega_0 + d\alpha_1 = \omega_V + e_h\nu + d\big(f(s)H^*(p)\beta\big), \tag{52}$$

with the full exterior derivative, which contains the mixed term $f(s)H^{*\prime}(p)\,dp\wedge\beta = -9f(s)\,dp\wedge\beta$.

Proof. (i) Each term is a base one-form whose coefficients are smooth functions of the base point and of p, affine in p; as p is smooth on N, the pull-back is a smooth horizontal one-form. The terms $c(p)\beta$, $d_\Sigma(\zeta_{\mathcal{A}}F_{\mathcal{A}})$ and the corrections at the b_j are d_Σ-closed, so $d_\Sigma\alpha_1 = d_\Sigma\alpha^0 = \kappa_1 - \kappa$. (ii) On $\mathcal{A}$, $\zeta_{\mathcal{A}} = 1$ on a neighbourhood and all $\zeta_j = 0$ (Proposition 9.16), so $\alpha_1 = \alpha^0 - c\beta - (\Delta_0 - c\beta) = \alpha^0 - \Delta_0$, which is the first identity by the definition of Δ_0. By Lemma 9.13, $\Omega_0 + d\alpha_1 = \omega_V + d(\Gamma_{\mathcal{A}} + \alpha_1) = \omega_V + d\Gamma_{\mathrm{sm}} + d(fH^*\beta)$, and $d\Gamma_{\mathrm{sm}} = e_h\nu$ because Γ_{sm} does not depend on p. □

Remark 9.25 (The period is subtracted in α_1). The term $-c(p)\beta$ is part of α_1, not only of $F_{\mathcal{A}}$: otherwise the left side of (52) gains $c(p)\beta$, whose ambient differential $c'(p)\,dp\wedge\beta$ is in general nonzero. Since β is closed on the torus and $\operatorname{supp}\beta\cap U_j = \varnothing$, the subtraction changes neither the base curvature nor the values at the b_j; it only changes the mixed term, which does not enter (53).

Lemma 9.26 (Vanishing on three whole fibres)**.** *At every point of each of the three fibres $N_{b_1}, N_{b_2}, N_{b_3}$, α_1 is the zero covector of T^*N.*

Proof. On the disc of radius $\frac{1}{200}$ about b_j we have $\zeta_j = 1$, and the cutoffs ζ_i, $i \neq j$, vanish on U_j (Proposition 9.16). Hence near b_j the j-th correction equals $d_\Sigma(A_j(p)x_j + B_j(p)y_j) = A_j(p)\,dx_j + B_j(p)\,dy_j$, and the base covector $\alpha_1(p)_{b_j} = \alpha_{\mathrm{pre}}(p)_{b_j} - A_j(p)\,dx_j - B_j(p)\,dy_j$ vanishes for every p. At a point of N_{b_j} with moment p_0, the toric form α_1 is the pull-back of $\alpha_1(p_0)_{b_j}$ by $d\pi_N$, hence zero. □

Remark 9.27 (Ambient covector versus pull-back). The pull-back of α_1 to a fibre vanishes automatically, since α_1 is horizontal. That weaker statement would only make the Moser field Y of the path ($\iota_Y\Omega_\lambda = -\alpha_1$) Ω_λ-orthogonal to the fibre, and such a field is in general transverse to it. Lemma 9.26 gives $Y = 0$ at every point of N_{b_j}, so the Moser flow fixes these fibres pointwise and preserves their complement, which the $\mathbb{Z}/3$-equivariant transfer maps into $\{Z_0 \neq 0\}$ (Lemma 9.26 and Proposition 13.19).

9.5. **Positivity including the mixed terms.** The corrections above change α_1 by base forms whose coefficients depend on p, so their ambient differentials contain mixed terms $dp\wedge(\cdots)$ that do not appear in the base curvature. They do not affect nondegeneracy: locally the space is a two-dimensional base times a fibre, and a mixed term $dp\wedge a$ with a a base one-form wedges to zero with every base two-form (three base factors), with $\omega_V = dp\wedge d\theta$ (two factors dp), and with itself.

Lemma 9.28 (Positivity)**.** *For $\lambda\in[0,1]$ let $\Omega_\lambda = \Omega_0 + \lambda\,d\alpha_1$ and $\kappa_\lambda = (1-\lambda)\kappa + \lambda\kappa_1$. In every local unitary frame,*

$$\Omega_\lambda = \omega_V + \kappa_\lambda + dp\wedge a_\lambda, \qquad a_\lambda = a_{\mathrm{loc}} + \lambda\,\partial_p\alpha_1,$$

with a_λ a base one-form, and

$$\Omega_\lambda^2 = 2\,\omega_V\wedge\kappa_\lambda, \tag{53}$$

smoothly across the zero section. Hence Ω_λ is a circle-invariant symplectic form on the open set $\{p < \frac{1}{3},\ H^*(p) > 0\} = \{p < \frac{1}{3} - \frac{e_h}{9}\}$, which contains $N_{p\le h}$; moreover $\iota_{X_p}\Omega_\lambda = -dp$ for the generator X_p of the circle action, and $\partial_\lambda\Omega_\lambda = d\alpha_1$.

Proof. By Lemma 9.24(i), $d\alpha_1 = d_\Sigma\alpha_1 + dp\wedge\partial_p\alpha_1 = \kappa_1 - \kappa + dp\wedge\partial_p\alpha_1$, and with (50) this gives the local form. By the paragraph preceding the lemma (Lemma 2.8(a) with $m=1$), $\Omega_\lambda^2 = 2\,\omega_V\wedge\kappa_\lambda$. Write $\kappa_\lambda = k_\lambda\nu$. Where $p<\frac13$ and $H^*(p)\ge 0$, $k_\lambda>0$ by Lemma 9.18, and $\omega_V\wedge\nu$ is a positive volume form; $H^*(p)>0$ iff $p<\frac13-\frac{e_h}{9}$, and $h<\frac13-\frac{e_h}{9}$ because $e_h\le\frac32(1-3h)<9(\frac13-h)$. In Cartesian fibre coordinates $\omega_V = d\xi\wedge d\eta$ and $dp = 2\pi(\xi\,d\xi+\eta\,d\eta)$ are smooth, so the decomposition and (53) hold across $w=0$. The forms Ω_0 and α_1 are invariant under the circle action; $\iota_{X_p}\Omega_0 = -dp$ (Lemma 2.9(c)) and $\iota_{X_p}d\alpha_1 = \mathcal{L}_{X_p}\alpha_1 - d(\alpha_1(X_p)) = 0$ because α_1 is invariant and horizontal. □

Lemmas 9.24, 9.26 and 9.28 give the first path and the full marked-fibre vanishing for the cubic model; (53) is the four-dimensional case of the toric top-power identity. The twist, the second and third paths, their primitives and the Moser map are constructed in §10.9; all three variation primitives vanish on the marked fibres (Lemma 10.26), so the Moser map fixes them pointwise (Proposition 10.28).

Remark 9.29 (Scalar data for both four-dimensional branches). The equal branch uses $g = 3(1-t_0)$ and $h = t_0/3$, so $g = H_{\mathrm{b}}(h)$; the general branch uses $g = 3d/4$ with the scalar comparison datum of Proposition 6.8. In both cases the data are valid in the sense of Definition 10.13: $g\le H_{\mathrm{b}}(h)$, and (63) holds with constant $\frac{31}{32}g$. With $e_h = g/16$ and the cap smoothing scale $\mu_h = g/(16k)$, the smoothing loss is at most $g/32$ and the residual gap is

$$\frac{31g}{32} - \frac{g}{16} - \frac{g}{32} = \frac{7g}{8} > 0.$$

This is Lemma 10.15(a),(c). The cover primitives, the marked-fibre vanishing and the positivity proof above therefore apply to both branches.

10. The handle construction

The construction is an explicit version of the surface packing lemma of [Ope26, §3.2]. For the higher-dimensional model inputs of Definition M2.3 the choices are fixed in §12; this section supplies the finite primitive used there, the handle data of the diagonal-quadric presentation, and the nondegeneracy and flow statements behind §M4. The periodic four-dimensional instance is treated in §10.9. Several objects of §§10.1–10.8 (the bump forms, the tree primitives and the normalisation) are constructed in §12, which comes after §11 because it also fixes the Moser fields and the maps ψ_i; §10.9 does not use §12.

10.1. **The finite primitive.**

Construction 10.1 (Coordinate primitive)**.** Let $r\,dx\wedge dy$ be a smooth two-form with compact support in an open coordinate rectangle $I\times I'$, put $M(x) = \int r(x,y)\,dy$ and fix a smooth bump b with compact support in I' and $\int b = 1$. With integrals starting at the lower endpoints of I and I', put

$$P(r) = -\left[\int^y \big(r(x,t) - b(t)M(x)\big)\,dt\right]dx + b(y)\left[\int^x M(t)\,dt\right]dy. \tag{54}$$

For a finite presentation (Definition 10.3) with partition weights ζ_i, unit-mass bumps η_i and tree primitives T_i (§12), the operator is

$$\mathcal{P}\omega = \sum_i \left[P_i(\zeta_i\omega - m_i(\omega)\eta_i) + m_i(\omega)T_i\right], \qquad m_i(\omega) = \int \zeta_i\omega. \tag{55}$$

Each local coefficient is extended by zero only after multiplication by its compactly supported weight.

Lemma 10.2 (Primitive identity)**.** *If $\int_{I\times I'} r\,dx\wedge dy = 0$, then $P(r)$ has compact support in $I\times I'$ and $dP(r) = r\,dx\wedge dy$. Let $\omega(p)$ be a smooth family of two-forms supported in the correction core with $m(\omega(p)) = \sum_i m_i(\omega(p)) = 0$. Then $d_\Sigma \mathcal{P}\omega(p) = \omega(p)$, and $\mathcal{P}\omega(p)$ has fixed compact support in the interior of the model base and depends smoothly on (p, x).*

Proof. The differential of the first term of (54) is $(r - bM)\,dx\wedge dy$, that of the second is $bM\,dx\wedge dy$. The dx-coefficient vanishes for y below $\operatorname{supp} r \cup \operatorname{supp} b$, and above it because $\int (r(x,t) - b(t)M(x))\,dt = 0$; it vanishes for x outside the x-projection of $\operatorname{supp} r$. The dy-coefficient vanishes outside $\operatorname{supp} b$ in y, below $\operatorname{supp} M$ in x, and above it because $\int M = 0$. For the operator, every argument of P_i has zero mass, so $dP_i(\cdot)$ returns it; the tree primitives satisfy $dT_i = \eta_i - \eta_0$ (§12); hence $d\mathcal{P}\omega = \omega - m(\omega)\eta_0 = \omega$. All integrals are over fixed compact coordinate domains with smooth integrands, which gives the support and smoothness statements. □

10.2. **The handle data of the diagonal quadric.**

Definition 10.3 (Finite presentation)**.** A finite presentation consists of coordinate maps with explicit inverses, compact inner and outer coordinate boxes, partition weights ζ_i (ζ_i supported in outer box i, $\sum_i \zeta_i = 1$ on the union of the inner boxes), unit-mass bumps in the boxes and in the overlaps of parent and child, and a rooted tree on the boxes in which every box $i \neq 0$ has a parent $j < i$ whose overlap with box i contains a coordinate box.

Construction 10.4 (Handle and normalisation data)**.** For the diagonal affine curve Σ of Definition 14.1 take the transverse cylinder $e(s,\theta)$, the dual form $\beta = \beta_0 - d(e_*(\chi F))$ and the clock profile b of Proposition 15.4. The annular coordinate of Definition M4.1(a) is $s_{\mathrm{a}} = s$ on $|s| < w = \frac{1}{8}$, with working half-width $b = \frac{1}{16}$; on $|s| < \frac{5}{32}$, which contains this annulus, $e^*\beta = b(\theta)\,d\theta$. The reference primitive is $\Gamma_{\mathrm{ref}} = \lambda = d^c \log(1 + |z|^2)$, with $\Lambda = \sigma$, and $\Gamma_{\mathrm{phys}}(p) = (1 - 2|p|)\lambda$, so $\kappa(p) = (1 - 2|p|)\sigma$. On Σ_ℓ with $R_\ell \geq 5$ the correction atlas is the family of Lemma 15.3 with $R = R_\ell - 1$, and the flow atlas is the same family with $R = R_\ell$, restricted to a neighbourhood of Σ_ℓ. The collar cutoff is $\chi_\partial = S\big((|t|^2 - (R_\ell - 2)^2)/((R_\ell - 1)^2 - (R_\ell - 2)^2)\big)$. The mixed model of §17.4 uses the same base data, and the six-dimensional model uses them with its physical primitive $\Gamma(p)$ of Definition 18.3.

Lemma 10.5 (Handle certificates)**.** *The form β is smooth, closed and supported in $\{|t| \leq 3\} \subset \operatorname{int}\Sigma_\ell$; on the annulus it equals $b(\theta)\,d\theta$ with $b \geq 0$, $b > 0$ on $(\frac{1}{4}, \frac{3}{4})$ and $\int_0^1 b = 1$. The inner boxes of the correction atlas cover $\{|t| \leq R_\ell - 1\}$, which contains the support of $1 - \chi_\partial$ and the handle, and all its outer boxes lie in $\operatorname{int}\Sigma_\ell$. For $p \in \Delta_h$, $\kappa(p) \geq (1 - 2h)\Lambda$ on all of Σ_ℓ, so the correction region of Definition M4.1(d) is Σ_ℓ with $\kappa_- = 1 - 2h$. The inner boxes of the flow atlas cover Σ_ℓ.*

Proof. The statements on β and b are Proposition 15.4; the annuli A, C lie in $|t| \leq \frac{5}{3}$ (proof of Lemma 15.5). The covering and containment statements are Lemma 15.3 with $R = R_\ell - 1$ and $R = R_\ell$: its branch-square condition $i + \frac{1}{16} < (R+1)^2$ holds because $i \leq m < R_\ell - 1$. The cutoff χ_∂ vanishes for $|t| \leq R_\ell - 2$, so $\operatorname{supp}(1 - \chi_\partial) \subset \{|t| \leq R_\ell - 1\}$, and the handle lies in $|t| \leq 3 < R_\ell - 1$. Finally $1 - 2|p| \geq 1 - 2h > 0$ on Δ_h. □

10.3. **Normalisation and the budget.**

Lemma 10.6 (Normalised datum)**.** *Let γ_* be the reserve of the comparison datum. The primitive Γ_n of Construction 12.2 agrees with Γ_{phys} near $\partial\Sigma_\ell$, its base curvature $d_\Sigma\Gamma_n(p) = \kappa_{\mathrm{sm}}(p) + H^*(p)\eta$ is positive for $p \in \Delta$, and on the annulus*

$$\Gamma_n(p) = e_\Lambda \Gamma_{\mathrm{ref}} + f(s)H^*(p)\beta. \tag{56}$$

The mass H^* is smooth, nonnegative on Δ, and $H^* \geq H_{\mathrm{b}} - \gamma_*/16$.

Proof. This is Proposition 12.3. Agreement near the boundary holds because the corrections $\mathcal{P}(\cdot)$, $c_p\beta$ and $d_\Sigma(\zeta u_p)$ are supported in the correction core, in $\operatorname{supp}\beta$ and in the annulus, all compact in $\operatorname{int}\Sigma_\ell$ (Lemma 10.5). □

For given capacities $\rho_i < a_i$ the layers use

$$\rho_i^\sharp = \frac{3a_i+\rho_i}{4}, \qquad \varepsilon_i = \frac{a_i-\rho_i^\sharp}{4}. \tag{57}$$

The model estimates are

$$\int_{\Sigma_\ell}\chi_\partial\kappa(p) \le \gamma_*/32, \qquad e_\Lambda\int_{\Sigma_\ell}\Lambda \le \gamma_*/32 \qquad (p\in\Delta), \tag{58}$$

and with

$$\kappa_{\mathrm{sm}} = \chi_\partial\kappa + (1-\chi_\partial)e_\Lambda\Lambda, \qquad H^* = H_{\mathrm{b}} - \int_{\Sigma_\ell}\kappa_{\mathrm{sm}} \tag{59}$$

they give, for $w\in V_\Delta$, by Proposition M4.5(ii),

$$H^* \ge H_{\mathrm{b}} - \gamma_*/16, \tag{60}$$
$$R^*(w) = K(p(q^{-1}w)) - K(p(w)) + H^*(p(w)) - \sum_i h_i(p(w)) \ge \gamma - \gamma_*/16 > 0.$$

Lemma 10.7 (Nondegeneracy and the Gram cutoffs)**.** *Let $K = V_\Delta(E)$. For $j=0,1,2$ and $t\in[0,1]$ the forms $\Omega_{j,t}$ of §12 are symplectic at every point of K, hence on an open neighbourhood of $[0,1]\times K$ in $[0,1]\times E$. The cutoff χ_j of §12 equals one on a neighbourhood of $[0,1]\times K$, and $\chi_j(t,\cdot)$ has compact support contained in the set where $\Omega_{j,t}$ is nondegenerate.*

Proof. All three paths are closed, being exact. By Lemma M4.2(a), for a toric primitive Q the top power of $\omega_V + dQ$ is $n\,\omega_V^m \wedge d_\Sigma Q$; mixed terms $dp_j\wedge(\cdot)$ do not contribute.

Path 0. $d_\Sigma Q_{0,t} = (1-t)\kappa(p) + t(\kappa_{\mathrm{sm}}(p) + H^*\eta)$. For $p\in\Delta$, $\kappa(p)>0$ (Definition M2.3), $\kappa_{\mathrm{sm}}(p)$ is a convex combination of $\kappa(p)$ and $e_\Lambda\Lambda>0$, and $H^*\eta \ge 0$ since $H^*\ge 0$ and $\eta = f'b\,ds\wedge dv \ge 0$. So the base form is positive.

Path 1. Off $E|_{\mathcal{A}}$, $\Omega_{1,t} = \Omega_{0,1}$. On $E|_{\mathcal{A}}$, $\Omega_{1,t} = \mathcal{D}_t^*(\omega_V + e_\Lambda\Lambda + d(G_t\beta))$ (Lemma M4.4(ii)), whose top power at $\mathcal{D}_t(x)$ is, by Lemma M4.2(b), $n(e_\Lambda a_\Lambda + b\,f'(s)H^*(p(q_t^{-1}w')))\,\omega_V^m\wedge ds\wedge dv$ with w' the fibre point of $\mathcal{D}_t(x)$. Since $q_\tau(V_\Delta)=V_\Delta$ for all τ (Definition M2.4), $x\in K$ gives $w'\in V_\Delta$ and $q_t^{-1}w'\in V_\Delta$, so $H^*\ge 0$ there; $e_\Lambda a_\Lambda>0$, $b\ge 0$, $f'\ge 0$, and $\mathcal{D}_t$ is a diffeomorphism.

Path 2. On $E|_{\mathcal{A}}$, $\Omega_{2,t} = \mathcal{D}^*(\omega_V + e_\Lambda\Lambda + d(\hat G_t\beta))$ by (M19), with top-power coefficient $e_\Lambda a_\Lambda + b\,\partial_s\hat G_t$ at $\mathcal{D}(x)$; here $\partial_s\hat G_t = (1-t)f'H^*(p(q^{-1}w')) + t\,\partial_s G_{\mathrm{f}} \ge 0$ on $\mathcal{A}\times V_\Delta$ by the previous case and Proposition M4.5(iii). Off $E|_{\mathcal{A}}$, $\Omega_{2,t} = \Omega_{0,1} + t\,dJ\wedge\beta$ differs from $\Omega_{0,1}$ by a mixed term and has the same top power.

Thus each $\Omega_{j,t}^n \ne 0$ on the compact set $[0,1]\times K$; since nondegeneracy is an open condition and the forms depend smoothly on (t,x), it holds on an open neighbourhood. For the cutoff, at each point of K some flow-atlas bump w_i equals one and the determinant of $\Omega_{j,t}$ in chart i is nonzero, so $g_j>0$ on $[0,1]\times K$ and $\mu_j>0$. The factor $S(4g_j/\mu_j - 1)$ equals one where $g_j \ge \mu_j/2$, and $S(2(R+1-|w|^2))$ equals one where $|w|^2 \le R+\frac12$; both sets contain an open neighbourhood of $[0,1]\times K$. Conversely $\chi_j(t,z)\ne 0$ forces $g_j(t,z) > \mu_j/4 > 0$, so some $w_i(z)\ne 0$ with nonzero determinant in chart i, and $|w|^2 < R+1$; the set $\{g_j(t,\cdot)\ge\mu_j/4,\ |w|^2\le R+1\}$ is closed, its base projection lies in the compact union of the supports of the w_i, and it is therefore compact. □

10.4. **The three paths.**

Proposition 10.8 (First path)**.** *The path $\Omega_{0,t} = \omega_V + d((1-t)\Gamma_{\mathrm{phys}} + t\Gamma_n)$ has the curvature, positivity and annular endpoint stated in Proposition M4.3. Its variation primitive is*

$$\alpha_1 = \Gamma_n - \Gamma_{\mathrm{phys}}, \tag{61}$$

which is supported in the union of the correction core, $\operatorname{supp}\beta$ *and the annulus.*

Proof. The base curvature and positivity are Path 0 of Lemma 10.7; the endpoint is (56); the support statement is the proof of Lemma 10.6. □

10.5. **The suspension path.**

Proposition 10.9 (Second path)**.** *The path $\Omega_{1,t} = \omega_V + d(\Gamma_n + \eta_t)$, with η_t the suspension primitive of Lemma M4.4, is closed, equals $\Omega_{0,1}$ at $t = 0$, and is symplectic on a neighbourhood of $V_\Delta(E)$. Over $\mathcal{A}$ it is the pullback by $\mathcal{D}_t$ of $\omega_V + e_\Lambda\Lambda + d(G_t\beta)$, $G_t = f(s)H^*(p(q_t^{-1}w))$.*

Proof. The pullback identity is Lemma M4.4(ii). The coefficient correction and the suspension term,

$$f(s)\big[H^*(p(q_t^{-1}q_{tf(s)}w)) - H^*(p(w))\big] \qquad \text{and} \qquad (Q_s^t \circ q_{tf(s)})\,ds,$$

vanish for $s \le 0$ and $s \ge 1$ (Lemma M4.4(iii)), so extension by zero is smooth and $\eta_0 = 0$. Nondegeneracy is Path 1 of Lemma 10.7. □

10.6. **The layer path.**

Proposition 10.10 (Third path)**.** *The path $\Omega_{2,t} = \omega_V + d(\Gamma_n + \eta_1 + t\xi)$ is closed and symplectic on a neighbourhood of $V_\Delta(E)$. Over the annulus it is the pullback by $\mathcal{D}$ of the interpolation from $G_1 = fH^* \circ p \circ q^{-1}$ to*

$$G_{\mathrm{f}} = J \circ p + \sum_i \varsigma_i h_i \circ p + \varsigma_* R^*, \qquad J = K - H^*, \tag{62}$$

whose s-derivative is nonnegative on $\mathcal{A} \times V_\Delta$. On ball layer i only the toric coefficient $A_i(p) + \varsigma_i(s)h_i(p)$ remains. Off the annulus the correction is the toric form $J(p)\beta$.

Proof. The clocks have disjoint ordered transition intervals, and $h_i \ge 0$, $R^* > 0$ on V_Δ by (60); this gives Proposition M4.5(iii). On both end collars of $\mathcal{A}$ the annular correction equals $J(p)\beta$, so ξ is smooth (M18). Nondegeneracy is Path 2 of Lemma 10.7. □

10.7. **Moser endpoints.**

Proposition 10.11 (The Moser maps)**.** *For $j = 0, 1, 2$ the cut field Y_j, defined by $\iota_{Y_{j,t}}\Omega_{j,t} = -\chi_j\,\partial_t Q_{j,t}$ on the nondegenerate locus and by zero elsewhere, is smooth on $[0,1]\times E$ with compact spatial support. Its time-one map Φ_j is a diffeomorphism of E with inverse the backward flow, $\Phi_j^{\pm1}(K) = K$, $\Phi_j^{\pm1}(\operatorname{int} K) = \operatorname{int} K$, and $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ on K. These are the identities of Proposition M4.6.*

Proof. By Lemma 10.7, $\chi_j(t,\cdot)$ is supported in a compact subset of the nondegenerate locus, where the defining equation has the smooth solution $-\Omega_{j,t}^{-1}(\chi_j\partial_t Q_{j,t})$ (Cramer's rule in each chart); this solution vanishes near the boundary of the support, so the extension by zero is smooth. A smooth field with compact spatial support on $[0,1]\times E$ has a complete flow, and its time-one map is a diffeomorphism whose inverse is the backward flow.

On a neighbourhood U of $[0,1]\times K$ we have $\chi_j = 1$, so Y_j is there the Moser field of the pair $(\Omega_{j,t}, \partial_t Q_{j,t})$, which satisfies $\partial_t\Omega_{j,t} = d\,\partial_t Q_{j,t}$. Lemma 12.5 shows that every integral curve of Y_j on a subinterval of $[0,1]$ which meets K (resp. $\operatorname{int} K$) stays in K (resp. $\operatorname{int} K$): near $\partial\Sigma_\ell$ the variation primitives vanish, so $Y_j = 0$ there; and near each outer face $\{\mu_\nu = \ell_\nu\}$ the face

moment satisfies $\iota_{X_\nu}\Omega_{j,t} = -d\mu_\nu$ and $\partial_t Q_{j,t}(X_\nu) = 0$, so $d\mu_\nu(Y_{j,t}) = 0$ by Lemma 2.10(d). The same holds in reverse time. Lemma 2.10(e) applied to $\mathcal{K} = K$ then gives $\Phi_j(K) = K$ and $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ on K; applied to the backward flow it gives $\Phi_j^{-1}(K) = K$, and the statement for $\operatorname{int} K$ follows in the same way, since every integral curve through a point of $\operatorname{int} K$ stays in it. □

10.8. **Assembly.**

Theorem 10.12 (Packing in the model)**.** *For the higher-dimensional model inputs, the maps ψ_i of Construction 12.4 satisfy Theorem M2.6.*

Proof. By Proposition 10.11, $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ satisfies $\Phi^*\Omega_{\mathrm{final}} = \Omega$ on K and Φ^{-1} maps $\operatorname{int} K$ onto itself. The layer maps E_i satisfy $E_i^*\Omega_{\mathrm{pre}} = \omega_0$ and have pairwise disjoint images in $\operatorname{int} K$ (Proposition 11.15), and $\mathcal{D}^{-1}$ pulls Ω_{final} back to Ω_{pre} (M19). The composite $\psi_i = \Phi^{-1} \circ \mathcal{D}^{-1} \circ E_i$ is therefore a smooth symplectic embedding of $B^\circ(\rho_i^\sharp)$, a neighbourhood of $B(\rho_i)$, and the images are pairwise disjoint because $\mathcal{D}^{-1}$ and Φ^{-1} are single injective maps (Proposition 11.17). □

10.9. **The four-dimensional handle.** For $n = 2$ the base is the closed torus Γ of §8, and there is neither a boundary collar nor a finite atlas. The torus coordinates (u, v) are global on the cover $\widetilde{N} = \mathbb{R}^2 \times \mathbb{C}$ of Definition 8.1, with fibre coordinate w and $p = \pi|w|^2$, and invariance under the deck maps replaces compactness of the base. A *deck map* is an element of the group generated by T_u and T_v; this group is free abelian, its element $T_u^m T_v^n$ shifts (u, v) by (m, n), and every deck map preserves p (Construction 9.8(b)). We write

$$\Omega_t = \Omega_0 + t\, d\alpha_1 \quad (t \in \mathbb{R}), \qquad \nu = du \wedge dv, \qquad \beta = b(v)\, dv,$$

with α_1 the toric one-form of Construction 9.23, β the dual form of Proposition 8.3(c) and S the flat step of Definition 2.3. The subsection constructs the twist, the second and third paths, the Moser map and the layer maps, and proves that every variation primitive vanishes on the marked fibres.

Definition 10.13 (Cubic data)**.** [Lean: needed consequence verified.]

Valid cubic data consist of an integer $k \geq 1$, numbers ρ_i, a_i $(1 \leq i \leq k)$, h, d_n, g, a smooth function K on $\mathbb{R}$, and a smooth isotopy $(q_t)_{t\in\mathbb{R}}$ of $(\mathbb{C}, \omega_{\mathbb{C}})$ with smooth Hamiltonian (Q_t), $\partial_t q_t = \Xi_t \circ q_t$ and $\iota_{\Xi_t}\omega_{\mathbb{C}} = -dQ_t$, such that:

(a) $0 < \rho_i < a_i < h < d_n < \frac{1}{3}$ and $0 < g \leq H_{\mathrm{b}}(h) = 3(1 - 3h)$;
(b) $q_0 = \mathrm{id}$, $Q_t = 0$ for $t \notin (0, 1)$, and for a number $P_{\mathrm{supp}} < h$ one has $q_t = \mathrm{id}$ and $Q_t = 0$ on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ for every t;
(c) with $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - p)_+$ and $F = K - P_{\mathrm{b}}$,

$$K(\pi|z|^2) - F(\pi|q_1(z)|^2) \geq \tfrac{31}{32}\, g \qquad (\pi|z|^2 \leq h). \tag{63}$$

Both branches take for (q_t) the whole-disc isotopy $\hat{q}$ of an ordinary path of disc area h (Definition 4.14, Lemma 4.18, Construction 4.19), which satisfies (b) by Theorem 4.21.

Since each q_t is bijective and is the identity outside $\bar{D}(P_{\mathrm{supp}})$, it maps $\bar{D}(c)$ onto itself for $c \geq P_{\mathrm{supp}}$ and $D(c)$ onto itself for $c > P_{\mathrm{supp}}$; so does q_t^{-1}. The two four-dimensional branches supply valid cubic data: the equal branch through the linear datum of Proposition M3.8 with the parameters of Remark 13.25, and the general branch through Proposition 6.8.

Definition 10.14 (Derived scalars)**.** For valid cubic data put $q = q_1$ and

$$e_h = \frac{g}{16}, \qquad H^* = H_{\mathrm{b}} - e_h, \qquad J = K - H^*, \qquad \mu_h = \frac{g}{16k}, \qquad \delta_m = \min\Big(\frac{d_n - h}{4}, \frac{g}{100}\Big),$$

$$h_i(p) = \tfrac{1}{2}\Big(a_i - p + \sqrt{(a_i - p)^2 + \mu_h^2}\Big), \qquad R^*(w) = K\big(p(q^{-1}w)\big) - J\big(p(w)\big) - \sum_{i=1}^{k} h_i\big(p(w)\big).$$

The caps h_i smooth the tents $(a_i - p)_+$ from above. Their total excess is the second item of the budget of Remark 9.29, the background e_h is the first.

Lemma 10.15 (Upper caps and the residual budget)**.** *For valid cubic data:*

(a) each h_i is smooth on $\mathbb{R}$, $(a_i - p)_+ < h_i(p) \le (a_i - p)_+ + \mu_h/2$ for all p, and $\sum_i h_i \le \sum_i (a_i - p)_+ + g/32$;
(b) $H^*(p) \ge 15g/16$ for $p \le h$;
(c) $R^*(w) \ge 7g/8$ whenever $\pi|w|^2 \le h$;
(d) if $p < h + 2\delta_m$, then $p < d_n$ and $H^*(p) \ge 15g/16 - 9(p-h)_+ > 15g/16 - 18\delta_m > 0$; if $\pi|w|^2 < h + 2\delta_m$, then $R^*(w) > 0$.

Proof. (a) Put $x = a_i - p$. Then $\mu_h > 0$ gives $|x| < \sqrt{x^2 + \mu_h^2} \le |x| + \mu_h$, and $h_i - x_+ = \frac{1}{2}(\sqrt{x^2 + \mu_h^2} - |x|) \in (0, \mu_h/2]$. Summing over k indices gives the excess $k\mu_h/2 = g/32$. (b) $H^*(p) \ge H_{\mathrm{b}}(h) - g/16 \ge 15g/16$. (c) Put $z = q^{-1}(w)$; then $\pi|z|^2 \le h$ because q^{-1} preserves $\bar{D}(h)$. By (a), $H^* - \sum_i h_i \ge P_{\mathrm{b}} - e_h - g/32$, so

$$R^*(w) \ge K(\pi|z|^2) - K(p(w)) + P_{\mathrm{b}}(p(w)) - \tfrac{3g}{32} = K(\pi|z|^2) - F(\pi|q(z)|^2) - \tfrac{3g}{32} \ge \tfrac{7g}{8}$$

by (63). (d) $p < h + (d_n - h)/2 < d_n$. For $p \le h$ use (b); for $p > h$, $H^*(p) = H^*(h) - 9(p-h)$. Finally $18\delta_m \le 18g/100 < 15g/16$. For the residual, the case $\pi|w|^2 \le h$ is (c). If $h < \pi|w|^2 < h + 2\delta_m$, then $q^{-1}w = w$, every $a_i < p$ gives $h_i(p) \le \mu_h/2$, and $R^*(w) = H^*(p) - \sum_i h_i(p) \ge 29g/32 - 9(p-h) > 29g/32 - 18g/100 > 0$. □

Definition 10.16 (Frame chart, clock and layers)**.** On the cover let $\mathcal{A} = \{\frac{1}{8} < u < \frac{5}{24}\} \subset \mathcal{A}^+ = \{\frac{1}{12} < u < \frac{1}{4}\}$ (single strips, not taken modulo one); in this subsection $\mathcal{A}$ and $\mathcal{A}^+$ denote these interiors of the closed annuli of Proposition 8.3(a). On $\mathcal{A}^+$ we have $Z_0 \neq 0$ (Proposition 8.3(a),(f)); put $\mathsf{F} = (Z_0/|Z_0|)^3$ and define the *frame chart*

$$\chi_{\mathcal{A}}(u, v, w_{\mathcal{A}}) = \big(u, v, \mathsf{F}(u, v)\, w_{\mathcal{A}}\big) \qquad (u \in \mathcal{A}^+).$$

The *clock* is $f(u) = S\big((u - u_0)/(u_1 - u_0)\big)$ with $u_0 = \frac{7}{48}$, $u_1 = \frac{3}{16}$. The *layers* are

$$N_\ell = k + 1, \quad \ell_s = \frac{u_1 - u_0}{N_\ell}, \quad l_j = u_0 + (j-1)\ell_s + \frac{\ell_s}{4}, \quad r_j = l_j + \frac{\ell_s}{2}, \quad \varsigma_j(u) = S\Big(\frac{u - l_j}{r_j - l_j}\Big)$$

for $1 \le j \le N_\ell$; the last index $j = N_\ell$ is denoted by $*$.

The frame chart is a diffeomorphism of $\mathcal{A}^+ \times \mathbb{C}$ onto itself which preserves base points and p. Since $Z_0(u, v+1) = \phi(u, v) Z_0(u, v)$ with $|\phi| = 1$, $\mathsf{F}(u, v+1) = \phi(u, v)^3 \mathsf{F}(u, v)$, so $\chi_{\mathcal{A}}(u, v+1, w_{\mathcal{A}}) = T_v \chi_{\mathcal{A}}(u, v, w_{\mathcal{A}})$. In terms of the handle coordinate $s = 12u - \frac{3}{2}$ of Proposition 8.3(a), f is the clock $S(2s - \frac{1}{2})$ of Definition 9.17; it vanishes for $u \le u_0$ and equals one for $u \ge u_1$. The layer intervals satisfy $u_0 < l_1 < r_1 < l_2 < \cdots < r_* = u_1 - \ell_s/4 < u_1$, and $\varsigma_j = 0$ for $u \le l_j$, $\varsigma_j = 1$ for $u \ge r_j$, $\varsigma_j' \ge 0$. Over $\mathcal{A}^+$ all formulas below are written in the frame coordinates $(u, v, w_{\mathcal{A}})$, and q_t acts on $w_{\mathcal{A}}$. Figure 1 shows the base sets.

Lemma 10.17 (The first path in the frame)**.** *Over $\mathcal{A}^+$, $\chi_{\mathcal{A}}^* \Omega_t = \omega_V + \kappa_t + dp \wedge (a_{\mathcal{A}} + t\, \partial_p \alpha_1)$ with the positive forms κ_t of Lemma 9.28. Over $\mathcal{A}$,*

$$\chi_{\mathcal{A}}^* \Omega_1 = \omega_V + e_h \nu + d\big(f(u) H^*(p) \beta\big).$$

Proof. The first identity is Lemma 9.28, written in the frame of Lemma 9.13. The second is Lemma 9.24(ii), with $d\Gamma_{\mathrm{sm}} = e_h \nu$. □

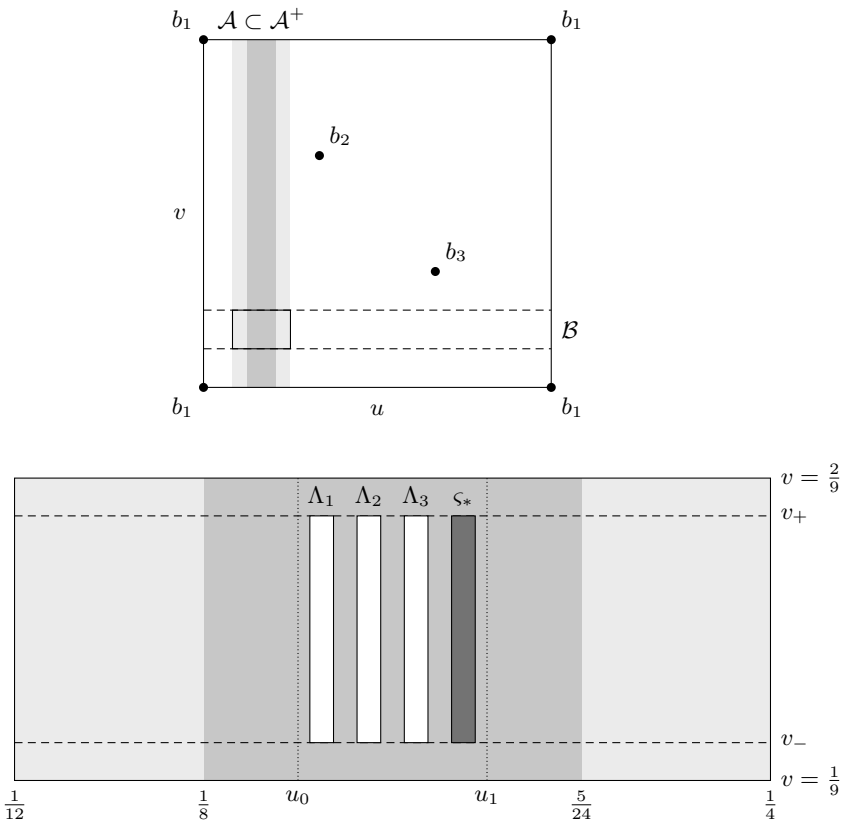


FIGURE 1. The handle data on the base torus $\Gamma = (\mathbb{R}/\mathbb{Z})^2$ in the coordinates (u, v) (Proposition 8.3, Definition 10.16). Top: the annuli $\mathcal{A} \subset \mathcal{A}^+$ (dark and light grey), the dual band $\mathcal{B}$ (between the dashed lines $v = \frac{1}{9}$, $v = \frac{2}{9}$) and the marked points b_1, b_2, b_3, at which $Z_0 = 0$. Bottom: the rectangle $\mathcal{A}^+ \cap \mathcal{B}$ enlarged, with dashed lines at $v = v_\pm$; the clock f rises on $[u_0, u_1]$, and the layers Λ_i (white) and the residual layer, where ς_* rises (black), lie over $(l_j, r_j) \times (v_-, v_+)$, where $b > 0$. Drawn for $k = 3$.

The twist. For $\lambda \in \mathbb{R}$ define on $\mathcal{A} \times \mathbb{C}$

$$\begin{aligned} \mathcal{D}_\lambda(u, v, w_\mathcal{A}) &= \big(u, v, q_{\lambda f(u)}(w_\mathcal{A})\big), \qquad \mathcal{D}_\lambda^{-1}(u, v, w_\mathcal{A}) = \big(u, v, q_{\lambda f(u)}^{-1}(w_\mathcal{A})\big), \\ Q_u^\lambda &= \lambda f'(u)\, Q_{\lambda f(u)}, \qquad G_\lambda(u, w_\mathcal{A}) = f(u)\, H^*\big(p(q_\lambda^{-1} w_\mathcal{A})\big), \end{aligned} \tag{64}$$

and $\mathcal{D} = \mathcal{D}_1$. These maps are smooth in $(\lambda, u, v, w_\mathcal{A})$ and mutually inverse, fix base points and preserve $\mathcal{A} \times D(c)$ for every $c > P_{\mathrm{supp}}$.

Lemma 10.18 (Suspension)**.** *On $\mathcal{A} \times \mathbb{C}$, $\mathcal{D}_\lambda^* \omega_V = \omega_V + d\big[(Q_u^\lambda \circ q_{\lambda f(u)})\, du\big]$ for every λ.*

Proof. This is Lemma M4.4(i) with $m = 1$ and s replaced by u: the differential of $\mathcal{D}_\lambda$ fixes ∂_v, maps a vertical vector ξ to $Dq\,\xi$ and ∂_u to $\partial_u + X_{Q_u^\lambda}$, and $q_{\lambda f(u)}$ is symplectic. □

Periodisation. For a one-form α on $\widetilde{N}$ supported over $\{u_0 \le u \le u_1\}$ put $\mathrm{per}(\alpha) = \sum_{m \in \mathbb{Z}} (T_u^m)^* \alpha$. The supports of the summands lie over the disjoint intervals $[u_0 - m, u_1 - m]$, so the sum is locally finite; $\mathrm{per}(\alpha)$ is T_u-invariant, agrees with α over $\{u_1 - 1 < u < 1 + u_0\}$, and vanishes near every point whose u-coordinate is not in $[u_0, u_1] + \mathbb{Z}$. If α is horizontal, then at a point with $\lfloor u \rfloor = m$ it is the covector of α at T_u^{-m} of that point.

Construction 10.19 (The second-path primitive)**.** On $\mathcal{A}\times\mathbb{C}$ let

$$\upsilon_\lambda^{\mathcal{A}} = \big(Q_u^\lambda \circ q_{\lambda f(u)}\big)\,du + f(u)\Big[H^*\big(p(q_\lambda^{-1}q_{\lambda f(u)}w_{\mathcal{A}})\big) - H^*\big(p(w_{\mathcal{A}})\big)\Big]\beta.$$

Let υ_λ be the one-form on $\widetilde{N}$ equal to $(\chi_{\mathcal{A}}^{-1})^*\upsilon_\lambda^{\mathcal{A}}$ over $\mathcal{A}$ and to zero elsewhere, let $\upsilon_\lambda^{\mathrm{per}} = \mathrm{per}(\upsilon_\lambda)$, and put

$$\Omega_\lambda^{(1)} = \Omega_1 + d\upsilon_\lambda^{\mathrm{per}}, \qquad \dot\upsilon_\lambda = \partial_\lambda \upsilon_\lambda^{\mathrm{per}}.$$

Lemma 10.20 (Properties of the second-path primitive)**.** *(a) $\upsilon_\lambda^{\mathcal{A}}$ vanishes for $u\le u_0$, for $u\ge u_1$, and for $p>P_{\mathrm{supp}}$; $\upsilon_0^{\mathcal{A}}=0$.*

(b) υ_λ and $\upsilon_\lambda^{\mathrm{per}}$ are horizontal one-forms on $\widetilde{N}$, smooth in (λ,x); υ_λ is T_v-invariant, $\upsilon_\lambda^{\mathrm{per}}$ is invariant under all deck maps, and $\upsilon_0^{\mathrm{per}}=0$. Both $\upsilon_\lambda^{\mathrm{per}}$ and $\dot\upsilon_\lambda$ vanish near every point with $p>P_{\mathrm{supp}}$ or with $u\notin[u_0,u_1]+\mathbb{Z}$.

(c) Over $\mathcal{A}$, $\chi_{\mathcal{A}}^*\Omega_\lambda^{(1)} = \mathcal{D}_\lambda^*\big(\omega_V + e_h\nu + d(G_\lambda\beta)\big)$.

Proof. (a) For $u\le u_0$, $f=f'=0$. For $u\ge u_1$, $f'=0$ and $q_\lambda^{-1}q_{\lambda f(u)} = q_\lambda^{-1}q_\lambda = \mathrm{id}$, so the bracket vanishes. For $p>P_{\mathrm{supp}}$, $Q_t=0$ near $w_{\mathcal{A}}$ and all q_t, q_t^{-1} fix $w_{\mathcal{A}}$. At $\lambda=0$ both terms vanish.

(b) By (a), $\upsilon_\lambda^{\mathcal{A}}$ vanishes over collars of both ends of $\mathcal{A}$, so the extension by zero is smooth. The form $\upsilon_\lambda^{\mathcal{A}}$ has only du and dv components, and $\chi_{\mathcal{A}}$ preserves base points, so υ_λ is horizontal. The coefficients of $\upsilon_\lambda^{\mathcal{A}}$ are 1-periodic in v, since b is periodic and $\mathcal{D}_\lambda$, G_λ do not depend on v; with $\chi_{\mathcal{A}}(u,v+1,\cdot) = T_v\chi_{\mathcal{A}}(u,v,\cdot)$ this gives $T_v^*\upsilon_\lambda=\upsilon_\lambda$. The periodisation is T_u-invariant, and T_v-invariant because T_u and T_v commute. The support statement follows from (a) and the properties of per, and it passes to the λ-derivative.

(c) Over $\mathcal{A}$ the form $\upsilon_\lambda^{\mathrm{per}}$ is υ_λ. The map $\mathcal{D}_\lambda$ fixes base points, so it preserves ν and β and $\mathcal{D}_\lambda^*(G_\lambda\beta) = (G_\lambda\circ\mathcal{D}_\lambda)\beta$ with $G_\lambda\circ\mathcal{D}_\lambda = f(u)H^*(p(q_\lambda^{-1}q_{\lambda f(u)}w_{\mathcal{A}}))$. Add Lemma 10.18 to Lemma 10.17. □

Proposition 10.21 (The second path)**.** *For valid cubic data and $\lambda\in[0,1]$, the form $\Omega_\lambda^{(1)}$ is closed, invariant under all deck maps, equal to Ω_1 for $\lambda=0$, and nondegenerate at every point of $\{p<h+2\delta_m\}$.*

Proof. Closedness, invariance and $\Omega_0^{(1)}=\Omega_1$ follow from Lemma 10.20(b) and the invariance of Ω_1 (Lemma 9.28). Since deck maps preserve p and the form, it suffices to check nondegeneracy over $\{u_1-1<u<1+u_0\}$. Near points with $u\notin[u_0,u_1]$ the form is Ω_1, which is symplectic on $\{p<\frac13-e_h/9\}\supset\{p<h+2\delta_m\}$ (Lemmas 9.28 and 10.15(d)). Over $\mathcal{A}$ use Lemma 10.20(c). By Lemma M4.2(b) with $n=2$, $\sigma_\Sigma=\nu$ and $e=e_h$,

$$\big(\omega_V + e_h\nu + d(G_\lambda\beta)\big)^2 = 2\big(e_h + b(v)\,\partial_u G_\lambda\big)\,\omega_V\wedge\nu, \qquad \partial_u G_\lambda(u,w') = f'(u)\,H^*\big(p(q_\lambda^{-1}w')\big).$$

If $p(w_{\mathcal{A}})<h+2\delta_m$, then $w'=q_{\lambda f(u)}(w_{\mathcal{A}})$ and $q_\lambda^{-1}w'$ lie in $D(h+2\delta_m)$, because $h+2\delta_m>P_{\mathrm{supp}}$; Lemma 10.15(d) gives $H^*>0$ there. Since $f'\ge0$, $b\ge0$ and $e_h>0$, the coefficient is positive at $\mathcal{D}_\lambda(x)$, and $\mathcal{D}_\lambda$ and $\chi_{\mathcal{A}}$ are diffeomorphisms. □

The final coefficient. With the residual of Definition 10.14 put

$$G_{\mathrm{f}}(u,w) = J\big(p(w)\big) + \sum_{i=1}^{k}\varsigma_i(u)\,h_i\big(p(w)\big) + \varsigma_*(u)\,R^*(w), \qquad A_i = J + \sum_{j<i}h_j.$$

Lemma 10.22 (The final coefficient)**.** *For valid cubic data, G_{f} is smooth on $\mathbb{R}\times\mathbb{C}$ and*

(a) $\partial_u G_{\mathrm{f}} = \sum_i \varsigma_i' h_i\circ p + \varsigma_*' R^* \ge 0$ at every (u,w) with $R^*(w)\ge0$;

(b) $G_{\mathrm{f}} = J\circ p$ for $u\le l_1$, and $G_{\mathrm{f}} = K\circ p\circ q^{-1}$ for $u\ge r_*$;

(c) $G_{\mathrm{f}}(u,w) = A_i(p(w)) + \varsigma_i(u)h_i(p(w))$ *for* $l_i < u < r_i$, $1 \le i \le k$.

Proof. (a) Differentiate; $h_i > 0$ by Lemma 10.15(a), and $\varsigma_j' \ge 0$. (b) For $u \le l_1$ all clocks vanish. For $u \ge r_*$ all equal one, and $J + \sum_i h_i + R^* = K \circ p \circ q^{-1}$. (c) On (l_i, r_i), $\varsigma_j = 1$ for $j < i$ and $\varsigma_j = 0$ for $j > i$, including $j = *$. □

Construction 10.23 (The third-path primitive)**.** On $\mathcal{A} \times \mathbb{C}$ let $\xi^{\mathcal{A}} = \big((G_{\mathrm{f}} - G_1) \circ \mathcal{D}\big)\beta$. Let ξ be the one-form on $\widetilde{N}$ equal to $J(p)\beta + (\chi_{\mathcal{A}}^{-1})^*\big(\xi^{\mathcal{A}} - J(p)\beta\big)$ over $\mathcal{A}$ and to $J(p)\beta$ elsewhere, and put

$$\xi^{\mathrm{per}} = J(p)\beta + \mathrm{per}\big(\xi - J(p)\beta\big), \qquad \Omega_\tau^{(2)} = \Omega_1^{(1)} + \tau\, d\xi^{\mathrm{per}},$$
$$\Omega_{\mathrm{fin}} = \Omega_1^{(2)}, \qquad \hat{G}_\tau = (1-\tau)G_1 + \tau G_{\mathrm{f}}.$$

Lemma 10.24 (Properties of the third-path primitive)**.** *(a)* $\xi^{\mathcal{A}} = J(p)\beta$ *for* $u \le u_0$ *and for* $u \ge u_1$.

(b) ξ^{per} *is a smooth one-form on* $\widetilde{N}$*, invariant under all deck maps, of the form* $\Xi\,\beta$ *with* Ξ *smooth; on* $\{p > P_{\mathrm{supp}}\}$*,* Ξ *is a function of* $(u \bmod 1, p)$.

(c) Over $\mathcal{A}$, $\chi_{\mathcal{A}}^*\Omega_\tau^{(2)} = \mathcal{D}^*\big(\omega_V + e_h\nu + d(\hat{G}_\tau\beta)\big)$*; near points with* $u \notin [u_0,u_1] + \mathbb{Z}$, $\Omega_\tau^{(2)} = \Omega_1 + \tau J'(p)\, b(v)\, dp \wedge dv$.

Proof. (a) For $u \le u_0$, $\mathcal{D} = \mathrm{id}$, $G_1 = 0$ and $G_{\mathrm{f}} = J \circ p$, since $u_0 < l_1$. For $u \ge u_1 > r_*$, $G_1 \circ \mathcal{D} = H^*(p(q^{-1}qw_{\mathcal{A}})) = H^*(p)$ and $G_{\mathrm{f}} \circ \mathcal{D} = K(p(q^{-1}qw_{\mathcal{A}})) = K(p)$, whose difference is $J(p)$.

(b) By (a), $\xi - J(p)\beta$ vanishes over collars of both ends of $\mathcal{A}$ and is supported over $[u_0,u_1]$, so ξ and the periodisation are smooth. Both summands are multiples of β: $\chi_{\mathcal{A}}$ preserves base points. Invariance follows as in Lemma 10.20(b), and $J(p)\beta$ is invariant because deck maps preserve p and v mod 1. On $\{p > P_{\mathrm{supp}}\}$ all q_t are the identity, $\mathcal{D} = \mathrm{id}$, and G_1, G_{f} are functions of (u,p).

(c) Over $\mathcal{A}$, $\xi^{\mathrm{per}} = \xi$ and $\chi_{\mathcal{A}}^*\xi = \xi^{\mathcal{A}} = \mathcal{D}^*((G_{\mathrm{f}} - G_1)\beta)$; add Lemma 10.20(c) at $\lambda = 1$. Near points with $u \notin [u_0,u_1] + \mathbb{Z}$, $v_1^{\mathrm{per}} = 0$ and $\xi^{\mathrm{per}} = J(p)\beta$, and $d(J(p)\beta) = J'(p)b(v)\, dp \wedge dv$. □

Proposition 10.25 (The third path and the final form)**.** *For valid cubic data and* $\tau \in [0,1]$, $\Omega_\tau^{(2)}$ *is closed, invariant under all deck maps and nondegenerate at every point of* $\{p < h + 2\delta_m\}$. *Over* $\mathcal{A}$,

$$\chi_{\mathcal{A}}^*\Omega_{\mathrm{fin}} = \mathcal{D}^*\Omega_{\mathrm{pre}}, \qquad (\chi_{\mathcal{A}} \circ \mathcal{D}^{-1})^*\Omega_{\mathrm{fin}} = \Omega_{\mathrm{pre}}, \qquad \Omega_{\mathrm{pre}} = \omega_V + e_h\nu + d(G_{\mathrm{f}}\beta),$$

and $\chi_{\mathcal{A}} \circ \mathcal{D}^{-1}$ *maps* $\mathcal{A} \times D(h)$ *diffeomorphically onto* $\mathcal{A} \times D(h)$.

Proof. Closedness and invariance follow from Proposition 10.21 and Lemma 10.24(b). As before it suffices to work over $\{u_1 - 1 < u < 1 + u_0\}$. Over $\mathcal{A}$, by Lemma 10.24(c) and Lemma M4.2(b), the square is $2(e_h + b\,\partial_u\hat{G}_\tau)\,\omega_V \wedge \nu$ at the image point $\mathcal{D}(x)$, and $\partial_u\hat{G}_\tau = (1-\tau)\partial_u G_1 + \tau\partial_u G_{\mathrm{f}}$. For $p(x) < h + 2\delta_m$ the image fibre point w' has $p(w') < h + 2\delta_m$; $\partial_u G_1(u,w') \ge 0$ as in Proposition 10.21, and $\partial_u G_{\mathrm{f}}(u,w') \ge 0$ by Lemmas 10.22(a) and 10.15(d). Near points with $u \notin [u_0,u_1]$, Lemma 10.24(c) adds to Ω_1 a term $dp \wedge (\cdot)$ with a base one-form; in the local form of Lemma 9.28 it only changes the mixed part, so the square is still $2\omega_V \wedge \kappa_1 > 0$ (Lemma 2.8). The displayed identities are Lemma 10.24(c) at $\tau = 1$. The last statement holds because $\mathcal{D}^{\pm 1}$ fix base points and $q_t^{\pm 1}$ preserve $D(h)$, while $\chi_{\mathcal{A}}$ preserves p. □

Lemma 10.26 (Vanishing on the marked fibres)**.** *Let* $\widetilde{\mathcal{F}}$ *be the union of all lifts of the fibres* $N_{b_1}, N_{b_2}, N_{b_3}$*, that is, the set of points* (u,v,w) *with* $(u,v) \in b_j + \mathbb{Z}^2$ *for some* j. *At every point of* $\widetilde{\mathcal{F}}$ *the covectors* α_1, $\dot{v}_\lambda$ $(\lambda \in \mathbb{R})$ *and* ξ^{per} *of* $T^*\widetilde{N}$ *are zero.*

Proof. For α_1 this is Lemma 9.26, lifted to the cover, where α_1 is periodic. The u-coordinates of the b_j are $0, \frac{1}{3}, \frac{2}{3}$, which are not in $[u_0, u_1] = [\frac{7}{48}, \frac{9}{48}]$ modulo one, so $\dot{v}_\lambda$ vanishes near $\widetilde{\mathcal{F}}$ by Lemma 10.20(b). By Lemma 10.24(b), $\xi^{\mathrm{per}} = \Xi\, b(v)\, dv$; the v-coordinates of the b_j are $0, \frac{2}{3}, \frac{1}{3}$, and b vanishes outside $[\frac{1}{8}, \frac{5}{24}] + \mathbb{Z}$ (Proposition 8.3(c)), so β vanishes at every point of $\widetilde{\mathcal{F}}$. □

Construction 10.27 (The handle Moser map)**.** The three stages are

$$(\Omega_s^{(0)}, \mathfrak{v}_s^{(0)}) = (\Omega_s, \alpha_1), \qquad (\Omega_s^{(1)}, \mathfrak{v}_s^{(1)}) = (\Omega_s^{(1)}, \dot{v}_s), \qquad (\Omega_s^{(2)}, \mathfrak{v}_s^{(2)}) = (\Omega_s^{(2)}, \xi^{\mathrm{per}}),$$

with matching endpoints $\Omega_1^{(0)} = \Omega_0^{(1)}$ and $\Omega_1^{(1)} = \Omega_0^{(2)}$. With the clocks $\sigma_i(t) = S(9t - 3i - 1)$, $i = 0, 1, 2$, which move only on $[\frac{3i+1}{9}, \frac{3i+2}{9}]$, put

$$\Omega_t^{\mathrm{M}} = \Omega_{\sigma_0(t)}^{(0)} + \sum_{i=1,2} \big(\Omega_{\sigma_i(t)}^{(i)} - \Omega_0^{(i)}\big), \qquad \mathfrak{v}_t = \sum_{i=0}^{2} \sigma_i'(t)\, \mathfrak{v}_{\sigma_i(t)}^{(i)}.$$

On $U = \{p < h + 2\delta_m\}$ let Y_t be the field with $\iota_{Y_t}\Omega_t^{\mathrm{M}} = -\mathfrak{v}_t$; in the coordinates $(\operatorname{Re} w, \operatorname{Im} w, u, v)$ it is the adjugate solution of a 4×4 skew system. With the moment cutoff

$$\mathfrak{c}(p) = S\Big(\frac{h + \frac{7}{4}\delta_m - p}{\delta_m/4}\Big),$$

which equals one for $p \le h + \frac{3}{2}\delta_m$ and zero for $p \ge h + \frac{7}{4}\delta_m$, let $\widehat{Y}_t = \mathfrak{c}(p) Y_t$, extended by zero. The handle Moser map Φ is the time-one map of $\widehat{Y}$ from time 0, and Φ^{-1} is its inverse.

The bordered construction of §M4 cuts each path off by a whole-path Gram cutoff of a finite atlas, because nondegeneracy is known only near the compact model region. Here the cover has one global chart and every stage form is nondegenerate on the whole set U for parameters in $[0, 1]$; the moment cutoff $\mathfrak{c}(p)$ is the only cutoff, and completeness comes from deck invariance.

Proposition 10.28 (The handle Moser map)**.** [Lean: needed consequence verified.]

For valid cubic data, Φ and Φ^{-1} are mutually inverse C^∞ diffeomorphisms of $\widetilde{N}$, and:

(a) $\Phi^*\Omega_{\mathrm{fin}} = \Omega_0$ and $(\Phi^{-1})^*\Omega_0 = \Omega_{\mathrm{fin}}$ at every point of $\{p \le h + \delta_m\}$;
(b) for every $c > P_{\mathrm{supp}}$, Φ and Φ^{-1} map each of $\{p \le c\}$, $\{p < c\}$, $\{p = c\}$ onto itself;
(c) Φ and Φ^{-1} fix every point of $\widetilde{\mathcal{F}}$;
(d) Φ and Φ^{-1} commute with every deck map.

Moreover $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$, where Φ_i is the time-one map of the cut stage field $\mathfrak{c}(p) Y_s^{(i)}$, $\iota_{Y_s^{(i)}}\Omega_s^{(i)} = -\mathfrak{v}_s^{(i)}$.

Proof. Hypotheses. On $[\frac{3i+1}{9}, \frac{3i+2}{9}]$ only σ_i moves, so by the matching endpoints $\Omega_t^{\mathrm{M}} = \Omega_{\sigma_i(t)}^{(i)}$ there, and Ω_t^{M} is Ω_0 for $t \le \frac{1}{9}$ and Ω_{fin} for $t \ge \frac{8}{9}$. Hence Ω_t^{M} is closed and nondegenerate on U for every t (Lemma 9.28 with Lemma 10.15(d), and Propositions 10.21, 10.25). The pair is smooth on $\mathbb{R} \times \widetilde{N}$, and $\partial_t \Omega_t^{\mathrm{M}} = d\mathfrak{v}_t$ because $\partial_s \Omega_s^{(i)} = d\mathfrak{v}_s^{(i)}$ for each stage; $\mathfrak{v}_t = 0$ for $t \notin (0, 1)$. By Lemma 2.10(a), Y is smooth on $\mathbb{R} \times U$, so $\widehat{Y}$ is smooth on $\mathbb{R} \times \widetilde{N}$. All stage forms and primitives are deck invariant, and so is $\mathfrak{c}(p)$, since the deck maps preserve p.

Toric band. Let $R = 2\pi(-y\partial_x + x\partial_y)$, $w = x + iy$, generate the fibre rotation, so that $\iota_R\Omega_0 = -dp$ (Lemma 9.28). On $U' = \{P_{\mathrm{supp}} < p < h + 2\delta_m\}$ one has $\iota_R\Omega_t^{\mathrm{M}} = -dp$ and $\mathfrak{v}_t(R) = 0$: α_1 is horizontal and $\iota_R d\alpha_1 = 0$; v_s^{per} and $\dot{v}_s$ vanish near U' (Lemma 10.20(b)); and on U', $\xi^{\mathrm{per}} = \Xi(u \bmod 1, p)\, b(v)\, dv$, whose differential has no fibre-angle component, so $\iota_R d\xi^{\mathrm{per}} = 0 = \xi^{\mathrm{per}}(R)$. Lemma 2.10(d) gives $dp(\widehat{Y}_t) = 0$ on $\{p > P_{\mathrm{supp}}\}$, where $\widehat{Y}$ vanishes for $p \ge h + \frac{7}{4}\delta_m$.

Completeness. The functions $du(\widehat{Y}_t)$, $dv(\widehat{Y}_t)$ are deck invariant (Lemma 2.10(g), since the stage data and $\mathfrak{c}(p)$ are deck invariant), and deck maps act on (u,v) by integral translations; hence $|du(\widehat{Y}_t)|+|dv(\widehat{Y}_t)| \le C_b$, the maximum over the compact set $[0,1]\times[0,1]^2\times\bar{D}(h+\frac{7}{4}\delta_m)$, and $\widehat{Y}_t = 0$ for $t \notin [0,1]$. By the toric band, an integral curve through x stays in $\{p \le \max(p(x),h)\}$, and its base coordinates move by at most C_b in unit time. Integral curves therefore remain in compact sets, and the flow and its inverse are defined on $\mathbb{R}\times\widetilde{N}$ and smooth.

Conclusions. (b) follows from $dp(\widehat{Y}_t) = 0$ on $\{p > P_{\mathrm{supp}}\}$, in both time directions. (a) For $p(x) \le h+\delta_m$ the integral curve stays in $\{p \le h+\delta_m\}$, because $dp(\widehat{Y}_t) = 0$ on $\{p > P_{\mathrm{supp}}\}$ (toric band) and $P_{\mathrm{supp}} < h$. For the same reason the open set $\{p < h + \frac{3}{2}\delta_m\}$ is invariant under the flow of $\widehat{Y}$, and $\mathfrak{c} = 1$ there, so $\widehat{Y} = Y$ on it and Φ agrees with the time-one map of the uncut field Y on a neighbourhood of x. Lemma 2.10(b) gives $(\Phi^*\Omega_1^{\mathrm{M}})_x = (\Omega_0^{\mathrm{M}})_x$. The inverse identity follows because Φ^{-1} preserves $\{p \le h+\delta_m\}$. (c) By Lemma 10.26, $\mathfrak{v}_t = 0$ on $\widetilde{\mathcal{F}}$ for every t, so $\widehat{Y}_t = 0$ there, and Lemma 2.10(c) applies in both time directions. (d) is Lemma 2.10(g), since all stage forms and primitives and $\mathfrak{c}(p)$ are deck invariant.

Decomposition. Each stage pair $(\Omega_s^{(i)}, \mathfrak{v}_s^{(i)})$ is deck invariant and satisfies the same rotation identity $\iota_R\Omega_s^{(i)} = -dp$, $\mathfrak{v}_s^{(i)}(R) = 0$ on U', so the completeness argument above applies to each cut stage field $\mathfrak{c}(p)Y_s^{(i)}$; write $\Phi_s^{(i)}$ for its flow from time 0 and $\Phi_i = \Phi_1^{(i)}$. On $[\frac{3i+1}{9}, \frac{3i+2}{9}]$ the cut field is $\widehat{Y}_t = \sigma_i'(t)\mathfrak{c}(p)Y^{(i)}_{\sigma_i(t)}$, so $t \mapsto \Phi^{(i)}_{\sigma_i(t)}(x)$ is an integral curve of $\widehat{Y}$ on that interval; on the gaps between these intervals and for $t \notin [\frac{1}{9}, \frac{8}{9}]$ one has $\mathfrak{v}_t = 0$, hence $\widehat{Y}_t = 0$. By uniqueness of integral curves, the integral curve of $\widehat{Y}$ through x is obtained by running $\Phi^{(0)}$, $\Phi^{(1)}$, $\Phi^{(2)}$ in turn, which gives $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ (Lemma 2.10(f)). □

Equivalently, Φ is defined through compactly supported fields: for $A \ge 0$ multiply $\widehat{Y}$ by a product cutoff in (u,v) equal to one on $\{|u|, |v| \le A + C_b + 2\}$, let Φ_A be the time-one map of the cut field, and set $\Phi(x) = \Phi_A(x)$ with $A = |u(x)| + |v(x)|$. The base speed bound shows that the integral curve through x never meets the region where the box cutoff differs from one, so this is the time-one map above.

Layer maps. On the layer $\Lambda_i = \{l_i < u < r_i,\ \frac{1}{8} < v < \frac{5}{24},\ p < h\}$ (frame coordinates) let $T(v) = \int_{1/8}^{v} b$, a diffeomorphism of $(\frac{1}{8}, \frac{5}{24})$ onto $(0,1)$ with $dT = \beta$ and inverse $v(T)$, and

$$\ell(T) = \frac{e_h}{b(v(T))}, \qquad x_i(u,T,p) = (u - l_i)\,\ell(T) + \varsigma_i(u)\,h_i(p).$$

Then $\partial_u x_i = \ell + \varsigma_i' h_i > 0$, $x_i = 0$ at $u = l_i$ and $x_i > h_i(p)$ at $u = r_i$, so for $0 < x < h_i(p)$ and $0 < T < 1$ there is a unique root $u = \sigma_i(x,T,p) \in (l_i, r_i)$. For $1 \le i \le k$ put $\bar{\rho}_i = (\rho_i + a_i)/2$, $\delta_i = a_i - \bar{\rho}_i$, $\hat{\rho}_i = \bar{\rho}_i + \delta_i/4$, and for $(z_{\mathrm{b}}, z_{\mathrm{f}}) \in B^\circ(\hat{\rho}_i) \subset \mathbb{C}^2$ with $(x,T) = \Psi_{\delta_i}(z_{\mathrm{b}})$ (Proposition 10.32) and $p = \pi|z_{\mathrm{f}}|^2$,

$$E_i(z_{\mathrm{b}}, z_{\mathrm{f}}) = \Big(\sigma_i(x,T,p),\ v(T),\ e^{-2\pi i T A_i'(p)} z_{\mathrm{f}}\Big), \qquad f_i^N = \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i.$$

Proposition 10.29 (The four-dimensional embeddings into the cover)**.** [Lean: needed consequence verified.]

Let the cubic data be valid and $1 \le i, j \le k$. Then $\rho_i < \bar{\rho}_i < \hat{\rho}_i < a_i$, and:

(a) f_i^N is defined and smooth on $B^\circ(\hat{\rho}_i)$, with values in $\{p < h\} \setminus \widetilde{\mathcal{F}}$, and $(f_i^N)^*\Omega_0 = \omega_0$;
(b) f_i^N is injective, $f_i^N(B^\circ(\hat{\rho}_i)) \cap f_j^N(B^\circ(\hat{\rho}_j)) = \varnothing$ for $i \ne j$, and $f_i^N(B(\bar{\rho}_i))$ is compact;
(c) if $f_i^N(X) = T\, f_j^N(X')$ for a deck map T, then $T = \mathrm{id}$, $i = j$ and $X = X'$.

Proof. Let $\varpi = \pi|z_{\mathrm{b}}|^2$ and $\varpi + p < \hat\rho_i$. By Proposition 10.32 and Lemma 10.15(a),

$$0 < x < \varpi + \frac{\delta_i}{2} < \hat\rho_i - p + \frac{\delta_i}{2} = a_i - p - \frac{\delta_i}{4} < h_i(p), \qquad p < a_i < h,$$

so the root exists, and E_i maps $B^\circ(\hat\rho_i)$ into Λ_i. Since $\ell\, b = e_h$, $e_h\nu = \ell\, du \wedge dT$; with Lemma 10.22(c), $d(G_{\mathrm{f}}\beta) = (\varsigma_i' h_i\, du + dA_i + \varsigma_i\, dh_i) \wedge dT$ on Λ_i, hence $\Omega_{\mathrm{pre}} = \omega_V + dx_i \wedge dT + dA_i(p) \wedge dT$ there. E_i is the composite of $\Psi_{\delta_i} \times \mathrm{id}$, the shear of Lemma 10.31 with $A = A_i$ and the inverse of the layer coordinates $(u, v, w_{\mathcal{A}}) \mapsto (x_i, T, w_{\mathcal{A}})$ (the periodic counterpart of the layer maps of Proposition 11.15); by Proposition 10.32 and Lemma 10.31 it is a diffeomorphism onto an open subset of Λ_i with $E_i^*\Omega_{\mathrm{pre}} = \omega_0$.

(a) By Proposition 10.25, $\chi_{\mathcal{A}}\mathcal{D}^{-1}$ maps Λ_i into $\mathcal{A} \times D(h)$ and pulls Ω_{fin} back to Ω_{pre}; by Proposition 10.28(a),(b), Φ^{-1} preserves $\{p < h\}$ and pulls Ω_0 back to Ω_{fin} there. Composing gives $(f_i^N)^*\Omega_0 = \omega_0$. The point $\chi_{\mathcal{A}}\mathcal{D}^{-1}E_i(X)$ lies over $\mathcal{A}$, which misses $\widetilde{\mathcal{F}}$; if $\Phi^{-1}(y) \in \widetilde{\mathcal{F}}$, then $y = \Phi(\Phi^{-1}y) = \Phi^{-1}y \in \widetilde{\mathcal{F}}$ by Proposition 10.28(c).

(b) E_i is injective, the images of different E_i lie over the disjoint intervals (l_i, r_i), and $\chi_{\mathcal{A}}\mathcal{D}^{-1}$ and Φ^{-1} are injective maps common to all i. The closed ball is compact.

(c) By Proposition 10.28(d), $\Phi^{-1}(y) = T\Phi^{-1}(y') = \Phi^{-1}(Ty')$ with $y = \chi_{\mathcal{A}}\mathcal{D}^{-1}E_i(X)$ and $y' = \chi_{\mathcal{A}}\mathcal{D}^{-1}E_j(X')$, so $y = Ty'$. Both points lie over the square $(\frac{1}{8}, \frac{5}{24})^2$, so the shift of T is zero and $T = \mathrm{id}$; then (b) gives $i = j$ and $X = X'$. □

Remark 10.30 (The four-dimensional embeddings). By Proposition 10.29(a),(c) the maps f_i^N satisfy the hypotheses of Theorem 13.21, and the four-dimensional embeddings are $f_i^{\mathrm{cap}} = b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N} \circ f_i^N$ on $B^\circ(\hat\rho_i)$. The general branch uses the data of Proposition 6.8 and the capacities of (41). The equal branch uses the same maps for the data of Remark 13.25: $\rho_i = \rho$, $a_i = a$, $K(p) = (3\sqrt{k}-9)p$, and (q_t) the isotopy of Theorem 5.34, with the recorded capacities of Remark 13.24. The strip constants are the four-dimensional values recorded in Definition 11.9: $\delta_i = a_i - \bar\rho_i = (a_i - \rho_i)/2$ and $\hat\rho_i = a_i - \frac{3}{4}\delta_i$; only the disc-to-strip map of Proposition 11.7 is taken from §11, and the layer identities are proved in Proposition 10.29.

10.10. **The shear and the strip map.** The two explicit maps used in the layers of §M4.4 are the following.

Lemma 10.31 (Multi-angle shear)**.** *Let A be a smooth function on an open set $W \subset \mathbb{R}^m$ and $\mathrm{Sh}_A(x, T, w) = (x, T, \mathrm{Sh}_{A,T}(w))$, where*

$$\mathrm{Sh}_{A,T}(w)_j = \exp\big(-2\pi i\, T\, \partial_j A(p(w))\big)\, w_j \qquad (1 \le j \le m),$$

on $\{(x, T, w) : p(w) \in W\} \subset \mathbb{R}^2 \times \mathbb{C}^m$. Then Sh_A is a diffeomorphism of this set onto itself, preserves p, x and T, and

$$\mathrm{Sh}_A^*\big(\omega_V + dx \wedge dT + dA(p) \wedge dT\big) = \omega_V + dx \wedge dT.$$

Proof. $\mathrm{Sh}_{A,T}$ multiplies each w_j by a unimodular factor that is a smooth function of $(T, p(w))$; it preserves p, and the same formula with $+2\pi i$ is its inverse. Both are smooth in Cartesian coordinates, also on the coordinate axes. Where all $w_j \neq 0$, write $w_j = \sqrt{p_j/\pi}\, e^{2\pi i\theta_j}$, so $\mathrm{Sh}_{A,T}$ is $\theta_j \mapsto \theta_j - T\partial_j A(p)$. Then

$$\mathrm{Sh}_A^*\omega_V = \sum_j dp_j \wedge d\big(\theta_j - T\partial_j A\big) = \omega_V - dA \wedge dT - T\sum_{j,k} \partial_{jk}A\, dp_j \wedge dp_k = \omega_V - dA \wedge dT,$$

the Hessian term vanishing by symmetry; $dx \wedge dT$ and $dA(p) \wedge dT$ are unchanged since x, T, p are. The identity extends to the coordinate axes by continuity. □

Proposition 10.32 (Disc-to-strip map)**.** *For every $\delta > 0$ there is an explicit smooth diffeomorphism $\Psi_\delta \colon \mathbb{C} \to (0,\infty)\times(0,1)$, $z \mapsto (x,T)$, with $\Psi_\delta^*(dx \wedge dT) = \omega_{\mathbb{C}}$ and*

$$0 < x < \pi|z|^2 + \frac{\delta}{2} \qquad \text{for all } z \in \mathbb{C}.$$

The map is $\Psi_\delta = F \circ \mathsf{T}^{-1} \circ \mathsf{D}$, where $F(u,v) = (X(u), Y(v))$ inverts the square roots of the Morse functions $x + \varepsilon^2/(4x) - \varepsilon$ on $(0,\infty)$ and $\varepsilon\,[(2\sqrt{T(1-T)})^{-1} - 1]$ on $(0,1)$, $\varepsilon = \delta/\pi$, and T, D are the explicit rotation and radial area normalisations of the density $X'(u)Y'(v)$. The estimate holds because the level set of energy $\mathcal{E}$ encloses area at least $\mathcal{E} - \varepsilon(\frac{\pi}{2} - 1)$ and lies in $\{x < \mathcal{E} + \varepsilon\}$ (Proposition 11.7). A compact subset of an open set of area a has area less than a, so no area-preserving map takes the closed disc of area a into $\{0 < x < a\} \times (0,1)$; the excess $\delta/2$ is paid once, for all concentric discs simultaneously.

11. The disc-to-strip map and the layer maps

This section constructs the planar disc-to-strip map and proves its estimate, which is Proposition 10.32. It then proves the layer step used in §M4.5: one ball is placed in each layer of the handle, and the maps $\psi_i = \Phi^{-1} \circ \mathcal{D}^{-1} \circ E_i$ are symplectic embeddings with pairwise disjoint images in the interior of the model (Propositions 11.15 and 11.17).

Conventions. On $\mathbb{C} \ni z$ we write $\omega_{\mathbb{C}} = dx \wedge dy$ for $z = x + iy$ and $I = \pi|z|^2$; recall $dx \wedge dy = dI \wedge d\theta$ with $\theta = \arg z / 2\pi$. The strip $(0,\infty)\times(0,1)$ carries coordinates (x,T) and the area form $dx \wedge dT$. An auxiliary plane $\mathbb{R}^2 \ni \xi = (\xi_1, \xi_2)$, identified with $\mathbb{C}$ by $\xi \mapsto \xi_1 + i\xi_2$, carries $d\xi = d\xi_1 \wedge d\xi_2$, and Rot_ϕ is the rotation by the angle ϕ. In §§11.3–11.6 we write $\mathbb{C}^n = \mathbb{C} \times \mathbb{C}^m \ni (z_{\mathrm{b}}, z_{\mathrm{f}})$, $\varpi = \pi|z_{\mathrm{b}}|^2$ and $p = p(z_{\mathrm{f}})$, so that $\omega_0 = \omega_{\mathbb{C}}(z_{\mathrm{b}}) + \omega_V(z_{\mathrm{f}})$ and $B(c) = \{\varpi + |p| \le c\}$, $B^\circ(c) = \{\varpi + |p| < c\}$.

11.1. **The strip Morse functions.** Fix $\delta > 0$ and put $b_{\mathrm{st}} = \delta/\pi$. The strip map Ψ_δ sends the circle of area I about 0 onto a level set $\{R_{\mathrm{st}}(x) + V_{\mathrm{st}}(T) = \mathcal{E}\}$ in the strip, where R_{st} and V_{st} are Morse functions on $(0,\infty)$ and $(0,1)$ with minimum 0. The level set of energy $\mathcal{E}$ lies in $\{x < \mathcal{E} + b_{\mathrm{st}}\}$ and encloses area at least $\mathcal{E} - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$, so the circle of area I lands in $\{x < I + b_{\mathrm{st}}\frac{\pi}{2}\} = \{x < I + \frac{\delta}{2}\}$. Since a ball meets every base disc $\{\varpi \le c\}$ at some fibre level, this excess must be uniform in the radius, and it is (Figure 2). The map is the product F_{st} of signed square-root inverses of R_{st} and V_{st}, followed by two normalisations of the area density that fix every circle: $X_{\mathrm{st}}(\xi_1)$ is the solution of $R_{\mathrm{st}}(x) = \xi_1^2$ on the side of the minimum $b_{\mathrm{st}}/2$ given by the sign of ξ_1, and $Y_{\mathrm{st}}(\xi_2)$ the solution of $V_{\mathrm{st}}(T) = \xi_2^2$ on the side of $\frac{1}{2}$ given by the sign of ξ_2.

Construction 11.1 (Strip Morse functions)**.** Let $\delta > 0$ and $b_{\mathrm{st}} = \delta/\pi$. For $x > 0$, $0 < T < 1$ and $\xi = (\xi_1, \xi_2) \in \mathbb{R}^2$ put

$$R_{\mathrm{st}}(x) = x + \frac{b_{\mathrm{st}}^2}{4x} - b_{\mathrm{st}}, \qquad V_{\mathrm{st}}(T) = b_{\mathrm{st}}\Big[\frac{1}{2\sqrt{T(1-T)}} - 1\Big],$$

$$X_{\mathrm{st}}(\xi_1) = \tfrac{1}{2}\Big[\xi_1^2 + b_{\mathrm{st}} + \xi_1\sqrt{\xi_1^2 + 2b_{\mathrm{st}}}\Big], \qquad Y_{\mathrm{st}}(\xi_2) = \frac{1}{2} + \frac{\xi_2\sqrt{\xi_2^2 + 2b_{\mathrm{st}}}}{2(\xi_2^2 + b_{\mathrm{st}})},$$

and $F_{\mathrm{st}}(\xi) = \big(X_{\mathrm{st}}(\xi_1), Y_{\mathrm{st}}(\xi_2)\big)$, $\mathcal{J}_{\mathrm{st}}(\xi) = X_{\mathrm{st}}'(\xi_1)\,Y_{\mathrm{st}}'(\xi_2)$.

Lemma 11.2 (Smooth Morse inverses)**.** *(i) $R_{\mathrm{st}} \ge 0$ on $(0,\infty)$ with unique zero $x = b_{\mathrm{st}}/2$, and $V_{\mathrm{st}} \ge 0$ on $(0,1)$ with unique zero $T = \frac{1}{2}$.*
(ii) $R_{\mathrm{st}}(X_{\mathrm{st}}(\xi_1)) = \xi_1^2$ and $V_{\mathrm{st}}(Y_{\mathrm{st}}(\xi_2)) = \xi_2^2$ for all $\xi_1, \xi_2 \in \mathbb{R}$.

(iii) X_{st} *and* Y_{st} *are smooth on* $\mathbb{R}$*, with*

$$X'_{\mathrm{st}}(\xi_1) = \frac{\big(\xi_1 + \sqrt{\xi_1^2 + 2b_{\mathrm{st}}}\big)^2}{2\sqrt{\xi_1^2 + 2b_{\mathrm{st}}}} > 0, \qquad Y'_{\mathrm{st}}(\xi_2) = \frac{b_{\mathrm{st}}^2}{\sqrt{\xi_2^2 + 2b_{\mathrm{st}}}\,(\xi_2^2 + b_{\mathrm{st}})^2} > 0.$$

(iv) X_{st} *is an increasing bijection* $\mathbb{R} \to (0,\infty)$ *and* Y_{st} *an increasing bijection* $\mathbb{R} \to (0,1)$*.* F_{st} *is a smooth diffeomorphism of* $\mathbb{R}^2$ *onto* $(0,\infty)\times(0,1)$ *with* $F_{\mathrm{st}}^*(dx \wedge dT) = \mathcal{J}_{\mathrm{st}}\, d\xi$ *and* $R_{\mathrm{st}} \circ X_{\mathrm{st}} + V_{\mathrm{st}} \circ Y_{\mathrm{st}} = |\xi|^2$*.*

Proof. (i) $R_{\mathrm{st}}(x) = (\sqrt{x} - b_{\mathrm{st}}/(2\sqrt{x}))^2$, which vanishes only at $x = b_{\mathrm{st}}/2$. Since $T(1-T) \le \frac{1}{4}$ with equality only at $T = \frac{1}{2}$, $V_{\mathrm{st}} \ge 0$ with unique zero $T = \frac{1}{2}$.

(ii) For $x > 0$, $R_{\mathrm{st}}(x) = \xi_1^2$ is equivalent to $x^2 - (\xi_1^2 + b_{\mathrm{st}})x + b_{\mathrm{st}}^2/4 = 0$, with discriminant $(\xi_1^2+b_{\mathrm{st}})^2 - b_{\mathrm{st}}^2 = \xi_1^2(\xi_1^2+2b_{\mathrm{st}})$ and roots $x_\pm = \frac{1}{2}[\xi_1^2 + b_{\mathrm{st}} \pm |\xi_1|\sqrt{\xi_1^2+2b_{\mathrm{st}}}]$, both positive because $x_+x_- = b_{\mathrm{st}}^2/4$. $X_{\mathrm{st}}(\xi_1)$ is x_+ for $\xi_1 \ge 0$ and x_- for $\xi_1 \le 0$. For the second identity write $Y_{\mathrm{st}} = \frac{1}{2} + \tau$ with $\tau = \xi_2\sqrt{\xi_2^2+2b_{\mathrm{st}}}/(2(\xi_2^2+b_{\mathrm{st}}))$. Then $\tau^2 = \frac{1}{4} - \frac{b_{\mathrm{st}}^2}{4(\xi_2^2+b_{\mathrm{st}})^2} < \frac{1}{4}$, so $0 < Y_{\mathrm{st}} < 1$, $T(1-T) = \frac{1}{4} - \tau^2 = b_{\mathrm{st}}^2/(4(\xi_2^2+b_{\mathrm{st}})^2)$ for $T = Y_{\mathrm{st}}(\xi_2)$, hence $1/(2\sqrt{T(1-T)}) = (\xi_2^2+b_{\mathrm{st}})/b_{\mathrm{st}}$ and $V_{\mathrm{st}}(T) = \xi_2^2$.

(iii) Smoothness holds because $\xi_1^2 + 2b_{\mathrm{st}} > 0$ and $\xi_2^2 + b_{\mathrm{st}} > 0$. With $r = \sqrt{\xi_1^2+2b_{\mathrm{st}}}$, $2X'_{\mathrm{st}} = 2\xi_1 + r + \xi_1^2/r = (\xi_1+r)^2/r$, and $\xi_1 + r > 0$ since $r > |\xi_1|$. With $\varphi(\xi_2) = \xi_2\sqrt{\xi_2^2+2b_{\mathrm{st}}}/(\xi_2^2+b_{\mathrm{st}})$,

$$\varphi'(\xi_2) = \frac{(2\xi_2^2+2b_{\mathrm{st}})(\xi_2^2+b_{\mathrm{st}}) - 2\xi_2^2(\xi_2^2+2b_{\mathrm{st}})}{\sqrt{\xi_2^2+2b_{\mathrm{st}}}\,(\xi_2^2+b_{\mathrm{st}})^2} = \frac{2b_{\mathrm{st}}^2}{\sqrt{\xi_2^2+2b_{\mathrm{st}}}\,(\xi_2^2+b_{\mathrm{st}})^2},$$

and $Y'_{\mathrm{st}} = \varphi'/2$.

(iv) Both maps are continuous and strictly increasing by (iii). As $\xi_1 \to +\infty$, $X_{\mathrm{st}} \to \infty$; for $\xi_1 < 0$, $X_{\mathrm{st}}(\xi_1) = x_- = b_{\mathrm{st}}^2/(4x_+)$ with $x_+ \to \infty$ as $\xi_1 \to -\infty$, so $X_{\mathrm{st}} \to 0^+$. Since φ is odd and $\varphi^2 = 1 - b_{\mathrm{st}}^2/(\xi_2^2+b_{\mathrm{st}})^2 \to 1$, Y_{st} increases from 0^+ to 1^-. Thus F_{st} is a smooth bijection with Jacobian $\mathcal{J}_{\mathrm{st}} > 0$, hence a smooth diffeomorphism by the inverse function theorem; the last identity is (ii). □

11.2. **The strip map.** F_{st} sends circles $|\xi|^2 = \mathcal{E}$ to level sets of $R_{\mathrm{st}} + V_{\mathrm{st}}$ but does not preserve area. Two normalisations correct the density without moving the circles: a rotation of each circle that makes the density constant along it, and a radial map that adjusts the enclosed area. Both are written in Cartesian coordinates, so that smoothness at the origin is visible. They are the normalisations of Definition 4.3 for a single density; the proofs are repeated here because the strip density $\mathcal{J}_{\mathrm{st}}$ is not constant near the origin, as setting (N) of §4.3 assumes.

Definition 11.3 (Rotation and area normalisation)**.** Let $\mathcal{J}$ be a positive smooth function on $\mathbb{R}^2$. Put

$$\begin{aligned}
\bar{\mathcal{J}}(|\xi|^2) &= \frac{1}{2\pi}\int_0^{2\pi} \mathcal{J}(\mathrm{Rot}_\phi\xi)\, d\phi,\\
\mathcal{M}(\xi) &= \frac{1}{2\pi}\int_0^{2\pi} (\phi-\pi)\, \mathcal{J}(\mathrm{Rot}_\phi\xi)\, d\phi,\\
\mathsf{T}_{\mathcal{J}}(\xi) &= \mathrm{Rot}_{\mathcal{M}(\xi)/\bar{\mathcal{J}}(|\xi|^2)}\,\xi.
\end{aligned}$$

and $\mathfrak{a}_{\mathcal{J}}(\mathcal{E}) = \int_{|\xi|^2\le\mathcal{E}} \mathcal{J}\, d\xi$ for $\mathcal{E} \ge 0$. In polar coordinates $\mathfrak{a}_{\mathcal{J}}(\mathcal{E}) = \int_0^{\sqrt{\mathcal{E}}}\int_0^{2\pi} \mathcal{J}\, d\psi\, r\, dr = \int_0^{\sqrt{\mathcal{E}}} 2\pi r\bar{\mathcal{J}}(r^2)\, dr = \pi\int_0^{\mathcal{E}} \bar{\mathcal{J}}(t)\, dt$ with $\bar{\mathcal{J}} > 0$, so $\mathfrak{a}_{\mathcal{J}}$ is continuous and strictly increasing with $\mathfrak{a}_{\mathcal{J}}(0) = 0$. If $\mathfrak{a}_{\mathcal{J}}(\mathcal{E}) \to \infty$ as $\mathcal{E} \to \infty$, it is a bijection of $[0,\infty)$; then let $\mathcal{E}_{\mathcal{J}} = \mathfrak{a}_{\mathcal{J}}^{-1}$ and, for $z \in \mathbb{C}$ with $I = \pi|z|^2$,

$$\mathsf{D}_{\mathcal{J}}(z) = \sqrt{\pi\,\mathcal{E}_{\mathcal{J}}(I)/I}\; z \quad (z \ne 0), \qquad \mathsf{D}_{\mathcal{J}}(0) = 0.$$

Lemma 11.4 (Normalisation)**.** *Let $\mathcal{J}$ be a positive smooth function on $\mathbb{R}^2$.*

(i) $\mathsf{T}_{\mathcal{J}}$ is a smooth diffeomorphism of $\mathbb{R}^2$ which maps every circle $\{|\xi| = r\}$ onto itself, and $\mathsf{T}_{\mathcal{J}}^*\big(\bar{\mathcal{J}}(|\xi|^2)\,d\xi\big) = \mathcal{J}\,d\xi$.

(ii) If $\mathfrak{a}_{\mathcal{J}}(\mathcal{E}) \to \infty$, then $\mathsf{D}_{\mathcal{J}}$ is a smooth diffeomorphism $\mathbb{C} \to \mathbb{R}^2$ with $|\mathsf{D}_{\mathcal{J}}(z)|^2 = \mathcal{E}_{\mathcal{J}}(\pi|z|^2)$ and $\mathsf{D}_{\mathcal{J}}^*\big(\bar{\mathcal{J}}(|\xi|^2)\,d\xi\big) = \omega_{\mathbb{C}}$.

(iii) If $\mathfrak{a}_{\mathcal{J}}(\mathcal{E}) \to \infty$ and $F\colon \mathbb{R}^2 \to U$ is a smooth diffeomorphism onto an open set $U \subset \mathbb{R}^2$ with $F^*(dx \wedge dT) = \mathcal{J}\,d\xi$, then $F \circ \mathsf{T}_{\mathcal{J}}^{-1} \circ \mathsf{D}_{\mathcal{J}}\colon \mathbb{C} \to U$ is a smooth diffeomorphism which pulls $dx \wedge dT$ back to $\omega_{\mathbb{C}}$ and maps the circle $\{\pi|z|^2 = I\}$ onto $F(\{|\xi|^2 = \mathcal{E}_{\mathcal{J}}(I)\})$.

Proof. If $g(\phi,\xi)$ is smooth on $\mathbb{R}\times\mathbb{R}^2$, then $\xi \mapsto \int_0^{2\pi} g(\phi,\xi)\,d\phi$ is smooth, by differentiation under the integral sign. Thus $\mathcal{M}$ and $\xi \mapsto \bar{\mathcal{J}}(|\xi|^2)$ are smooth, and $\bar{\mathcal{J}} > 0$. A smooth bijection with nowhere vanishing Jacobian has a smooth inverse by the inverse function theorem.

(i) Fix $r > 0$ and put $\mathcal{J}_r(\psi) = \mathcal{J}(re^{i\psi})$, a 2π-periodic function. Then $\mathcal{M}(re^{i\psi}) = \frac{1}{2\pi}\int_0^{2\pi}(\phi - \pi)\mathcal{J}_r(\psi+\phi)\,d\phi$, and integration by parts gives

$$\partial_\psi \mathcal{M}(re^{i\psi}) = \frac{1}{2\pi}\Big[(\phi-\pi)\mathcal{J}_r(\psi+\phi)\Big]_{\phi=0}^{2\pi} - \frac{1}{2\pi}\int_0^{2\pi}\mathcal{J}_r(\psi+\phi)\,d\phi = \mathcal{J}_r(\psi) - \bar{\mathcal{J}}(r^2).$$

So $\mathsf{T}_{\mathcal{J}}(re^{i\psi}) = re^{i\Theta(\psi)}$ with $\Theta = \psi + \mathcal{M}/\bar{\mathcal{J}}$, $\Theta' = \mathcal{J}/\bar{\mathcal{J}} > 0$ and $\Theta(\psi+2\pi) = \Theta(\psi) + 2\pi$: $\mathsf{T}_{\mathcal{J}}$ maps each circle diffeomorphically onto itself, $\mathsf{T}_{\mathcal{J}}(0) = 0$, and $\mathsf{T}_{\mathcal{J}}$ is a bijection of $\mathbb{R}^2$. The angle $\vartheta = \mathcal{M}/\bar{\mathcal{J}}(|\xi|^2)$ is smooth, so $\mathsf{T}_{\mathcal{J}}(\xi) = \mathrm{Rot}_{\vartheta(\xi)}\xi$ is smooth. In polar coordinates, since $\bar{\mathcal{J}}$ is radial and $\mathsf{T}_{\mathcal{J}}$ preserves r, $\mathsf{T}_{\mathcal{J}}^*(\bar{\mathcal{J}}\,r\,dr\wedge d\psi) = \bar{\mathcal{J}}\,r\,dr\wedge(\partial_r\Theta\,dr + \Theta'\,d\psi) = \mathcal{J}\,r\,dr\wedge d\psi$ off the origin. Hence $\det D\mathsf{T}_{\mathcal{J}} = \mathcal{J}/\bar{\mathcal{J}} > 0$ off the origin, and at the origin by continuity, so $\mathsf{T}_{\mathcal{J}}$ is a smooth diffeomorphism.

(ii) Put

$$G(\xi) = \int_0^1 \bar{\mathcal{J}}\big(t|\xi|^2\big)\,dt = \frac{1}{2\pi}\int_0^1\int_0^{2\pi} 2r\,\mathcal{J}\big(\mathrm{Rot}_\phi(r\xi)\big)\,d\phi\,dr,$$

a smooth positive function of ξ (the same argument as above, over the compact set $[0,1]\times[0,2\pi]$), with $\pi|\xi|^2 G(\xi) = \mathfrak{a}_{\mathcal{J}}(|\xi|^2)$ by the polar formula of Definition 11.3. The smooth map $\mathsf{D}'(\xi) = \sqrt{G(\xi)}\,\xi$ preserves the argument and satisfies $\pi|\mathsf{D}'(\xi)|^2 = \mathfrak{a}_{\mathcal{J}}(|\xi|^2)$; since $\mathfrak{a}_{\mathcal{J}}$ is a bijection of $[0,\infty)$, D' is a bijection of $\mathbb{R}^2$, and it is the inverse of $\mathsf{D}_{\mathcal{J}}$: if $I = \pi|\mathsf{D}'(\xi)|^2$ then $\mathcal{E}_{\mathcal{J}}(I) = |\xi|^2$ and $\mathsf{D}_{\mathcal{J}}(\mathsf{D}'(\xi)) = \sqrt{\pi|\xi|^2/I}\,\sqrt{G(\xi)}\,\xi = \xi$. In polar coordinates, with $\mathcal{E} = |\xi|^2$, D' preserves the angle ψ and sends $\mathcal{E}$ to $I = \mathfrak{a}_{\mathcal{J}}(\mathcal{E})$, so off the origin

$$(\mathsf{D}')^*\omega_{\mathbb{C}} = (\mathsf{D}')^*\Big(\frac{dI\wedge d\psi}{2\pi}\Big) = \frac{\pi\bar{\mathcal{J}}(\mathcal{E})\,d\mathcal{E}\wedge d\psi}{2\pi} = \bar{\mathcal{J}}(|\xi|^2)\,r\,dr\wedge d\psi = \bar{\mathcal{J}}(|\xi|^2)\,d\xi.$$

Thus $\det D\mathsf{D}' = \bar{\mathcal{J}}(|\xi|^2) > 0$, also at the origin by continuity, and D' is a smooth diffeomorphism. Hence so is $\mathsf{D}_{\mathcal{J}} = (\mathsf{D}')^{-1}$, and $\mathsf{D}_{\mathcal{J}}^*(\bar{\mathcal{J}}\,d\xi) = \omega_{\mathbb{C}}$. Finally $|\mathsf{D}_{\mathcal{J}}(z)|^2 = (\pi\mathcal{E}_{\mathcal{J}}(I)/I)|z|^2 = \mathcal{E}_{\mathcal{J}}(I)$.

(iii) By (i) and (ii) the composite is a smooth diffeomorphism onto $F(\mathbb{R}^2) = U$, and $(F\circ\mathsf{T}_{\mathcal{J}}^{-1}\circ\mathsf{D}_{\mathcal{J}})^*(dx\wedge dT) = \mathsf{D}_{\mathcal{J}}^*(\mathsf{T}_{\mathcal{J}}^{-1})^*(\mathcal{J}\,d\xi) = \mathsf{D}_{\mathcal{J}}^*(\bar{\mathcal{J}}\,d\xi) = \omega_{\mathbb{C}}$. $\mathsf{D}_{\mathcal{J}}$ maps $\{\pi|z|^2 = I\}$ onto $\{|\xi|^2 = \mathcal{E}_{\mathcal{J}}(I)\}$, which $\mathsf{T}_{\mathcal{J}}^{-1}$ preserves. □

We apply this to $\mathcal{J} = \mathcal{J}_{\mathrm{st}}$ and write $\bar{\mathcal{J}}_{\mathrm{st}}$ and $\mathfrak{a}_{\mathrm{st}} = \mathfrak{a}_{\mathcal{J}_{\mathrm{st}}}$. The next lemma contains the area facts behind the strip estimate.

Lemma 11.5 (Sublevel sets and area)**.** *Let $b_{\mathrm{st}} = \delta/\pi$.*

(i) For $c \ge 0$, $\{x > 0 : R_{\mathrm{st}}(x) \le c\} = [x_-(c), x_+(c)]$ with $x_\pm(c) = \frac{1}{2}\big[c + b_{\mathrm{st}} \pm \sqrt{c(c+2b_{\mathrm{st}})}\big]$. Its length is $\sqrt{c(c+2b_{\mathrm{st}})} \ge c$, and $0 < x_-(c) \le x_+(c) < c + b_{\mathrm{st}}$. For $c < 0$ the set is empty.

(ii) $\int_0^1 V_{\rm st}(T)\,dT = b_{\rm st}(\frac{\pi}{2}-1)$.
(iii) $\mathfrak{a}_{\rm st}(\mathcal{E})$ *is the* $dx\,dT$*-area of* $\{(x,T) : R_{\rm st}(x)+V_{\rm st}(T)\le\mathcal{E}\}$, *and* $\mathfrak{a}_{\rm st}(\mathcal{E}) \ge \mathcal{E} - b_{\rm st}(\frac{\pi}{2}-1)$.
(iv) $\mathfrak{a}_{\rm st}$ *is a continuous strictly increasing bijection of* $[0,\infty)$, *with* $\mathfrak{a}'_{\rm st}(\mathcal{E}) = \pi\bar{\mathcal{J}}_{\rm st}(\mathcal{E}) > 0$.

Proof. (i) This is the computation in the proof of Lemma 11.2(ii) with $\xi_1^2 = c$: $x_+ - x_- = \sqrt{c(c+2b_{\rm st})} \ge c$, and $\sqrt{c(c+2b_{\rm st})} < c + b_{\rm st}$ because $(c+b_{\rm st})^2 - c(c+2b_{\rm st}) = b_{\rm st}^2 > 0$; this gives $x_- > 0$ and $x_+ < c + b_{\rm st}$. For $c < 0$ the set is empty since $R_{\rm st} \ge 0$. (ii) $\int_0^1 dT/\sqrt{T(1-T)} = \pi$ (substitute $T = \sin^2\phi$). (iii) By Lemma 11.2(iv), $F_{\rm st}$ is a diffeomorphism onto the strip with Jacobian $\mathcal{J}_{\rm st}$ which maps $\{|\xi|^2 \le \mathcal{E}\}$ onto $\{R_{\rm st}+V_{\rm st}\le\mathcal{E}\}$; the change of variables formula gives the first claim. By Fubini and (i),

$$\mathfrak{a}_{\rm st}(\mathcal{E}) = \int_0^1 \big|\{x>0 : R_{\rm st}(x) \le \mathcal{E} - V_{\rm st}(T)\}\big|\,dT \ge \int_0^1 \big(\mathcal{E} - V_{\rm st}(T)\big)_+\,dT \ge \mathcal{E} - b_{\rm st}\big(\tfrac{\pi}{2}-1\big).$$

(iv) By Definition 11.3, $\mathfrak{a}_{\rm st}(\mathcal{E}) = \pi\int_0^{\mathcal{E}} \bar{\mathcal{J}}_{\rm st}$ is continuous and strictly increasing from 0, with the stated derivative; by (iii) it tends to ∞. □

Construction 11.6 (The strip map)**.** For $\delta > 0$ let $\mathsf{T}_{\rm st} = \mathsf{T}_{\mathcal{J}_{\rm st}}$, $\mathcal{E}_{\rm st} = \mathfrak{a}_{\rm st}^{-1}$ (defined on $[0,\infty)$ by Lemma 11.5(iv)) and $\mathsf{D}_{\rm st} = \mathsf{D}_{\mathcal{J}_{\rm st}}$, and define

$$\Psi_\delta = F_{\rm st} \circ \mathsf{T}_{\rm st}^{-1} \circ \mathsf{D}_{\rm st} \colon \mathbb{C} \to (0,\infty)\times(0,1), \qquad \Psi_\delta(z) = (x,T).$$

Proposition 11.7 (Disc-to-strip map)**.** *Let* $\delta > 0$ *and* $b_{\rm st} = \delta/\pi$. *Then:*
(i) Ψ_δ *is a smooth diffeomorphism of* $\mathbb{C}$ *onto* $(0,\infty)\times(0,1)$ *with* $\Psi_\delta^*(dx\wedge dT) = \omega_{\mathbb{C}}$;
(ii) $R_{\rm st}(x) + V_{\rm st}(T) = \mathcal{E}_{\rm st}(\pi|z|^2)$ *for* $(x,T) = \Psi_\delta(z)$;
(iii) for every $z\in\mathbb{C}$, *with* $(x,T) = \Psi_\delta(z)$,

$$0 < x < \pi|z|^2 + \frac{\delta}{2}, \qquad 0 < T < 1; \tag{65}$$

(iv) for $I_{\max} \ge 0$, $\mathcal{E}_{\max} = I_{\max} + b_{\rm st}(\frac{\pi}{2}-1)$ *and* $\pi|z|^2 \le I_{\max}$,

$$x \ge \tfrac{1}{2}\Big[\mathcal{E}_{\max} + b_{\rm st} - \sqrt{\mathcal{E}_{\max}(\mathcal{E}_{\max}+2b_{\rm st})}\Big] > 0, \qquad \big|T-\tfrac{1}{2}\big| \le \tfrac{1}{2}\sqrt{1-(1+\mathcal{E}_{\max}/b_{\rm st})^{-2}} < \tfrac{1}{2}.$$

In particular Ψ_δ *has the properties stated in Proposition 10.32.*

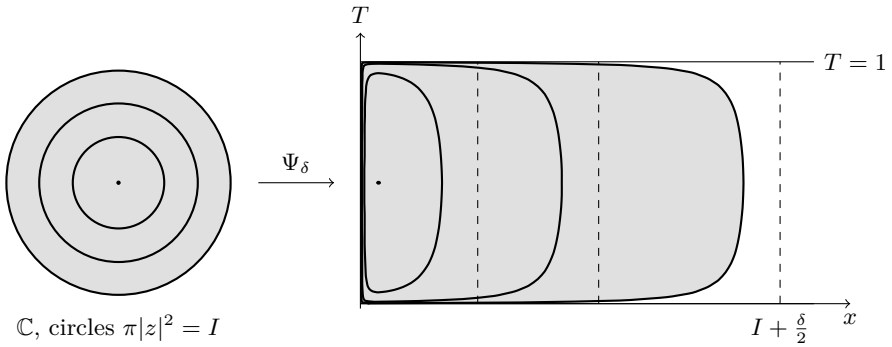


FIGURE 2. The disc-to-strip map Ψ_δ of Proposition 11.7: computed level sets for $b_{\rm st} = \delta/\pi = 0.12$ and $I = 0.2, 0.6, 1.2$. The circle of area I goes to the level set $\{R_{\rm st}+V_{\rm st} = \mathcal{E}_{\rm st}(I)\}$, which encloses area I and lies in $\{x < I + \frac{\delta}{2}\}$ (dashed lines); the dot is the common minimum $(b_{\rm st}/2, \frac{1}{2})$. The axes x and T have different scales.

Proof. (i), (ii) Apply Lemma 11.4(iii) to the smooth positive density $\mathcal{J}_{\mathrm{st}}$ and F_{st}. The latter is given by Lemma 11.2(iv), and $\mathfrak{a}_{\mathrm{st}} \to \infty$ by Lemma 11.5(iii). The level identity follows from $R_{\mathrm{st}} \circ X_{\mathrm{st}} + V_{\mathrm{st}} \circ Y_{\mathrm{st}} = |\xi|^2$. (iii) Let $I = \pi|z|^2$ and $\mathcal{E} = \mathcal{E}_{\mathrm{st}}(I)$. By Lemma 11.5(iii), $I = \mathfrak{a}_{\mathrm{st}}(\mathcal{E}) \geq \mathcal{E} - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$. By (ii) and $V_{\mathrm{st}} \geq 0$, $R_{\mathrm{st}}(x) \leq \mathcal{E}$, so by Lemma 11.5(i)

$$x \leq x_+(\mathcal{E}) < \mathcal{E} + b_{\mathrm{st}} \leq I + b_{\mathrm{st}}\tfrac{\pi}{2} = I + \tfrac{\delta}{2}.$$

Positivity of x and $0 < T < 1$ hold because Ψ_δ takes values in the strip. (iv) Let $\mathcal{E} = \mathcal{E}_{\mathrm{st}}(\pi|z|^2)$. Since $\mathcal{E}_{\mathrm{st}}$ is increasing, $\mathcal{E} \leq \mathcal{E}_{\mathrm{st}}(I_{\max})$, and $I_{\max} = \mathfrak{a}_{\mathrm{st}}(\mathcal{E}_{\mathrm{st}}(I_{\max})) \geq \mathcal{E}_{\mathrm{st}}(I_{\max}) - b_{\mathrm{st}}(\frac{\pi}{2} - 1)$ by Lemma 11.5(iii); so $\mathcal{E} \leq \mathcal{E}_{\max}$. By (ii), $R_{\mathrm{st}}(x) \leq \mathcal{E}_{\max}$ and $V_{\mathrm{st}}(T) \leq \mathcal{E}_{\max}$. The first bound is $x \geq x_-(\mathcal{E}_{\max}) > 0$ (Lemma 11.5(i)). The second says $2\sqrt{T(1-T)} \geq (1 + \mathcal{E}_{\max}/b_{\mathrm{st}})^{-1}$, i.e. $(T - \frac{1}{2})^2 = \frac{1}{4} - T(1-T) \leq \frac{1}{4}\big[1 - (1 + \mathcal{E}_{\max}/b_{\mathrm{st}})^{-2}\big]$. □

11.3. **Enlarged capacities and the ball estimate.** Throughout the remaining layer construction we work in the situation of Theorem M2.6 with $n \geq 3$: a model input (Definition M2.3) of rank $m = n - 1$ with handle data (Definition M4.1), capacities $a_1, \dots, a_k > 0$ with $\{p \in \mathbb{R}^m_{\geq 0} : |p| \leq a_i\} \subset \tau_\Delta \Delta$ for some $\tau_\Delta < 1$, a function P and a comparison datum (K, q) with gap $\gamma > 0$, and capacities $\rho_i < a_i$. The outer faces of Δ are $\{\langle b_\nu, p\rangle = \ell_\nu\}$, $1 \leq \nu \leq N$, with $\ell_\nu > 0$. The caps h_i are those of (M17) with $\rho_i^\sharp = \hat\rho_i$ ($\hat\rho_i$ is defined in Definition 11.9 below). By Proposition M4.5(i), each h_i is smooth on $\mathbb{R}^m$ and

$$0 \leq h_i(p), \qquad (a_i + \hat\rho_i)/2 - |p| \leq h_i(p) \qquad (p \in \mathbb{R}^m). \tag{66}$$

Lemma 11.8 (Interior of the fibre polytope)**.** *Let $V_\Delta = \{w \in \mathbb{C}^m : \langle b_\nu, p(w)\rangle \leq \ell_\nu \text{ for all } \nu\}$. Then*

$$\operatorname{int} V_\Delta = \{w \in \mathbb{C}^m : \langle b_\nu, p(w)\rangle < \ell_\nu \text{ for all } \nu\}, \tag{67}$$

and $\{w : |p(w)| \leq a_i\} \subset \operatorname{int} V_\Delta$ for $1 \leq i \leq k$.

Proof. The right side of (67) is open and contained in V_Δ. If $\langle b_\nu, p(w)\rangle = \ell_\nu > 0$ for some ν, then $\langle b_\nu, p((1+t)w)\rangle = (1+t)^2\ell_\nu > \ell_\nu$ for $t > 0$, so w is not an interior point. If $|p(w)| \leq a_i$, then $p(w) \in \tau_\Delta\Delta$, so $\langle b_\nu, p(w)\rangle \leq \tau_\Delta \ell_\nu < \ell_\nu$ for all ν. □

The margin $a_i - \rho_i = 2\delta_i$ gives nested source balls and a positive gap between the strip image and the cap. We use strip parameter $\varepsilon_i = \delta_i/8$, whose excess is less than $\delta_i/16$.

Definition 11.9 (Enlarged capacities)**.** For $1 \leq i \leq k$ put

$$\delta_i = \frac{a_i - \rho_i}{2}, \qquad \bar\rho_i = a_i - \frac{3}{4}\delta_i, \qquad \hat\rho_i = a_i - \frac{\delta_i}{2}.$$

Here $\bar\rho_i$ is the closed enlarged source capacity and $\hat\rho_i$ the open domain capacity, and $\rho_i < \frac{\rho_i + a_i}{2} = a_i - \delta_i < \bar\rho_i < \hat\rho_i < a_i$. Put $\varepsilon_i = (a_i - \hat\rho_i)/4 = \delta_i/8$; the i-th ball is placed by Ψ_{ε_i}. In the four-dimensional branch (§10.9) the same letters denote different values: there $\bar\rho_i = (\rho_i + a_i)/2 = a_i - \delta_i$, $\hat\rho_i = a_i - \frac{3}{4}\delta_i$, and the i-th ball is placed by Ψ_{δ_i} (Proposition 10.29). The two branches never occur together, and each section states which values it uses.

Lemma 11.10 (Ball estimate)**.** *Let $1 \leq i \leq k$, $(z_{\mathrm{b}}, z_{\mathrm{f}}) \in \mathbb{C} \times \mathbb{C}^m$, $\varpi = \pi|z_{\mathrm{b}}|^2$, $p = p(z_{\mathrm{f}})$, and let x be the first component of $\Psi_{\varepsilon_i}(z_{\mathrm{b}})$.*

(a) If $\varpi + |p| \leq a_i - \delta_i$, then

$$x < \varpi + \frac{\delta_i}{16} \leq a_i - |p| - \frac{15\delta_i}{16} \leq h_i(p) - \frac{11\delta_i}{16} < h_i(p) - \frac{\delta_i}{2}. \tag{68}$$

(b) If $\varpi + |p| \leq \bar\rho_i$, then $x < a_i - |p| - \frac{11\delta_i}{16} \leq h_i(p) - \frac{7\delta_i}{16} < h_i(p) - \frac{\delta_i}{4}$.

(c) If $\varpi + |p| < \hat\rho_i$, then $x < a_i - |p| - \frac{7\delta_i}{16} \leq h_i(p) - \frac{3\delta_i}{16} < h_i(p)$.

In each case $|p| < a_i$, hence $z_{\mathrm{f}} \in \operatorname{int} V_\Delta$.

Proof. By (65), $x < \varpi + \varepsilon_i/2 = \varpi + \delta_i/16$. (a) $\varpi \le a_i - \delta_i - |p| = a_i - |p| - \delta_i$. (b) $\varpi \le \bar\rho_i - |p| = a_i - |p| - \frac{3}{4}\delta_i$. (c) $\varpi < \hat\rho_i - |p| = a_i - |p| - \frac{1}{2}\delta_i$. In each case the cap comparison is (66) in the form $a_i - \delta_i/4 - |p| \le h_i(p)$. Finally $|p| \le \varpi + |p| < a_i$, and the last assertion is Lemma 11.8. □

The base factor is placed by Ψ_{ε_i}, which is not toric, and the fibre level p enters only through the bound $x < h_i(p)$.

11.4. **Layer coordinates.** We recall the objects of §M4 and §10 used here. In the product coordinates of Definition M4.1, $E|_{\mathcal{A}^+} \cong \mathcal{A}^+ \times \mathbb{C}^m$ with base coordinates (s, v) and fibre coordinate w, and $\omega_V = \sum_j dp_j \wedge d\theta_j$. The one-form β equals $b(v)\,dv$ on $\mathcal{A}^+$, where $b \ge 0$ is smooth, $b > 0$ exactly on (v_-, v_+) and $\int b = 1$. The layer intervals $I_j = [l_j, r_j] \subset (0,1)$, $1 \le j \le k$, and the residual interval I_* are pairwise disjoint, with flat steps ς_j rising from 0 to 1 across I_j ($\varsigma_j' \ge 0$). By Proposition 10.10, the final coefficient G_{f} satisfies, on $I_i \times \mathbb{C}^m$,

$$G_{\mathrm{f}}(s, w) = A_i(p(w)) + \varsigma_i(s) h_i(p(w)), \qquad A_i = J + \sum_{j<i} h_j, \tag{69}$$

where $J = K - H^* = K - H_{\mathrm{b}} + \epsilon_{\mathrm{sm}}$ is the collar coefficient of (M17). The function K is smooth on an open neighbourhood of Δ in $\mathbb{R}^m$ (Definition M2.4) and the h_j are smooth on $\mathbb{R}^m$. The functions $H_{\mathrm{b}}(p) = \int_{\Sigma_\ell} \kappa(p)$ and $\epsilon_{\mathrm{sm}}(p) = \int_{\Sigma_\ell} \kappa_{\mathrm{sm}}(p)$ are smooth on an open neighbourhood of Δ as well: $\kappa(p)$ extends smoothly in the moment variables (Definition M2.3), $\kappa_{\mathrm{sm}}(p)$ is obtained from it by multiplying by the fixed cutoff χ_∂ and adding the p-independent form $(1 - \chi_\partial) e_\Lambda \Lambda$ (see equation M13), and Σ_ℓ is compact, so one may differentiate under the integral sign. Hence A_i is smooth on an open neighbourhood W of Δ in $\mathbb{R}^m$. Write $\Lambda = a_\Lambda(s, v)\, ds \wedge dv$ with $a_\Lambda > 0$ on $\mathcal{A}^+$ and put $\ell_0(s, v) = e_\Lambda a_\Lambda(s, v) > 0$. On $E|_{\mathcal{A}}$, over a neighbourhood of V_Δ, the twisted form is given by (M19) (Proposition 10.10),

$$\Omega_{\mathrm{pre}} = \omega_V + e_\Lambda \Lambda + d(G_{\mathrm{f}}\beta) = \omega_V + \ell_0(s, v)\, ds \wedge dv + d(G_{\mathrm{f}}\beta), \qquad \Omega_{\mathrm{final}} = \mathcal{D}^* \Omega_{\mathrm{pre}}. \tag{70}$$

The i-th layer is the open set

$$\Lambda_i = \{(s, v, w) : l_i < s < r_i,\ v_- < v < v_+,\ w \in \operatorname{int} V_\Delta\} \subset E|_{\mathcal{A}}.$$

On Λ_i only the i-th step is moving: the lower steps already equal 1 and contribute the toric function A_i, the upper ones vanish. The coordinate T integrates β, and the coordinate x_i integrates the background area in the s-direction and adds the height $\varsigma_i h_i$ by which the i-th step has risen.

Construction 11.11 (Layer coordinates)**.** Let $T(v) = \int_{v_-}^{v} b(t)\, dt$ for $v_- < v < v_+$, an increasing diffeomorphism onto $(0,1)$ with $dT = \beta$, and $v(T)$ its inverse. For $0 < s < 1$, $0 < T < 1$ and $p \in \mathbb{R}^m$ put

$$\ell(s, T) = \frac{\ell_0(s, v(T))}{b(v(T))}, \qquad L_i(s, T) = \int_{l_i}^{s} \ell(u, T)\, du,$$
$$x_i(s, T, p) = L_i(s, T) + \varsigma_i(s) h_i(p), \qquad x_i^+(T, p) = x_i(r_i, T, p).$$

This is the coordinate (M20). On Λ_i the layer chart is

$$\kappa_i(s, v, w) = \big(x_i(s, T(v), p(w)),\ T(v),\ w\big),$$

and

$$\mathcal{U}_i = \big\{(x, T, w) \in \mathbb{R} \times (0,1) \times \operatorname{int} V_\Delta :\ 0 < x < x_i^+(T, p(w))\big\}.$$

Lemma 11.12 (Layer root)**.** *(i) x_i is smooth on $(0,1)\times(0,1)\times\mathbb{R}^m$, and $\partial_s x_i = \ell(s,T) + \varsigma_i'(s)h_i(p) \geq \ell(s,T) > 0$. Moreover $x_i(l_i,T,p) = 0$ and $x_i^+(T,p) = L_i(r_i,T) + h_i(p) > h_i(p)$.*

(ii) For $0<T<1$, $w\in \operatorname{int} V_\Delta$ and $0 < x < x_i^+(T,p(w))$ there is a unique $s = \sigma_i(x,T,p(w)) \in (l_i,r_i)$ with $x_i(s,T,p(w)) = x$. This root is smooth as a function of (x,T,w) on that open domain, including the fibre coordinate axes.

(iii) κ_i is a diffeomorphism of Λ_i onto $\mathcal{U}_i$, with inverse $(x,T,w)\mapsto \big(\sigma_i(x,T,p(w)),\, v(T),\, w\big)$.

Proof. (i) $v(T)$ is smooth because $T'(v) = b(v) > 0$ on (v_-,v_+), so L_i is smooth; ς_i and h_i are smooth. The derivative formula is immediate, and $\varsigma_i' \geq 0$, $h_i \geq 0$ by (66). At $s = l_i$, $L_i = 0$ and $\varsigma_i = 0$; at $s = r_i$, $\varsigma_i = 1$ and $L_i(r_i,T) > 0$ because $\ell > 0$ and $r_i > l_i$. (ii) By (i), $s\mapsto x_i(s,T,p)$ is strictly increasing on $[l_i,r_i]$ from 0 to $x_i^+(T,p)$, which gives existence and uniqueness of the root in (l_i,r_i). The set of (x,T,w) in question is open because x_i^+ and $p(w)$ are continuous. Near any of its points the implicit function theorem ($\partial_s x_i > 0$) gives a smooth local solution, which coincides with the root by uniqueness. The moment map is smooth in Cartesian coordinates, so this also holds at its axes. (iii) κ_i is smooth since $p(w)$ is smooth in Cartesian coordinates. For fixed (v,w) the first component increases strictly from 0 to $x_i^+(T(v),p(w))$ as s runs over (l_i,r_i), and T is a bijection $(v_-,v_+)\to(0,1)$; hence κ_i is a bijection $\Lambda_i\to\mathcal{U}_i$, $\mathcal{U}_i$ is open, and the inverse is the stated map, which is smooth by (ii). □

Lemma 11.13 (The form on a layer)**.** *On Λ_i, $\Omega_{\mathrm{pre}} = \omega_V + dx_i\wedge dT + dA_i(p)\wedge dT$, where $x_i = x_i(s,T(v),p(w))$; equivalently $(\kappa_i^{-1})^*\Omega_{\mathrm{pre}} = \omega_V + dx\wedge dT + dA_i(p)\wedge dT$ on $\mathcal{U}_i$.*

Proof. On Λ_i we have $\beta = dT$ and $ds\wedge dv = b^{-1}ds\wedge dT$. Since β is closed and G_{f} does not depend on v, (70) and (69) give

$$\Omega_{\mathrm{pre}} = \omega_V + \ell(s,T)\,ds\wedge dT + \big(\varsigma_i' h_i\,ds + dA_i + \varsigma_i\,dh_i\big)\wedge dT,$$

with A_i, h_i evaluated at $p(w)$. On the other hand $dx_i\wedge dT = \big(\partial_s L_i + \varsigma_i' h_i\big)\,ds\wedge dT + \varsigma_i\,dh_i\wedge dT$, because the dT-component of dx_i drops out, and $\partial_s L_i = \ell(s,T)$. Subtracting gives the claim. □

11.5. **The layer maps.** The term $dA_i(p)\wedge dT$ couples the fibre to the T-direction. It is removed by the multi-angle shear $\mathrm{Sh}_{A_i}(x,T,w) = (x,T,\mathrm{Sh}_{A_i,T}(w))$ of Lemma 10.31, applied with $A = A_i$ on W:

$$\mathrm{Sh}_{A_i,T}(w)_j = \exp\big(-2\pi i\,T\,\partial_j A_i(p(w))\big)\,w_j \qquad (1\leq j\leq m).$$

It preserves x, T and p, hence maps $\mathcal{U}_i$ diffeomorphically onto itself (its inverse, the same formula with $+2\pi i$, also preserves $\mathcal{U}_i$), and $\mathrm{Sh}_{A_i}^*\big(\omega_V + dx\wedge dT + dA_i(p)\wedge dT\big) = \omega_V + dx\wedge dT$.

Construction 11.14 (Layer maps)**.** For $1\leq i\leq k$ and $(z_{\mathrm{b}},z_{\mathrm{f}})\in B^\circ(\hat\rho_i)$ put $(x,T) = \Psi_{\varepsilon_i}(z_{\mathrm{b}})$, $p = p(z_{\mathrm{f}})$ and

$$E_i(z_{\mathrm{b}},z_{\mathrm{f}}) = \Big(\sigma_i(x,T,p),\ v(T),\ \mathrm{Sh}_{A_i,T}(z_{\mathrm{f}})\Big) = \kappa_i^{-1}\circ\mathrm{Sh}_{A_i}\circ(\Psi_{\varepsilon_i}\times\mathrm{id})\,(z_{\mathrm{b}},z_{\mathrm{f}}).$$

Proposition 11.15 (Layer maps)**.** *For $1\leq i\leq k$:*

(i) E_i is well defined and smooth on $B^\circ(\hat\rho_i)$, which contains $B(\bar\rho_i)\supset B(a_i-\delta_i)$, and is a diffeomorphism onto an open subset of Λ_i, with inverse

$$E_i^{-1}(s,v,w) = \Big(\Psi_{\varepsilon_i}^{-1}\big(x_i(s,T,p(w)),\,T\big),\ \mathrm{Sh}_{A_i,T}^{-1}(w)\Big), \qquad T = T(v);$$

(ii) $E_i^*\Omega_{\mathrm{pre}} = \omega_0$;

(iii) $E_i(B^\circ(\hat\rho_i))$ lies over the base rectangle $(l_i,r_i)\times(v_-,v_+)\subset\mathcal{A}$ and in $\{|p|<a_i\}\subset\operatorname{int}V_\Delta$;

(iv) on $B(\bar\rho_i)$ the coordinate $x = x_i\circ E_i$ satisfies $x < h_i(p) - \delta_i/4$, and on $B(a_i-\delta_i)$ it satisfies $x < h_i(p) - \delta_i/2$; the set $E_i(B(\bar\rho_i))$ is a compact subset of Λ_i;

(v) the images $E_i(B^\circ(\hat\rho_i))$, $1 \le i \le k$, are pairwise disjoint.

Proof. (i) By Lemma 11.10(c), for $(z_{\mathrm b}, z_{\mathrm f}) \in B^\circ(\hat\rho_i)$ we have $z_{\mathrm f} \in \operatorname{int} V_\Delta$ and $0 < x < h_i(p) < x_i^+(T,p)$ (Lemma 11.12(i)), and $0 < T < 1$ by (65). Hence $\Psi_{\varepsilon_i} \times \mathrm{id}$, a diffeomorphism of $\mathbb{C}\times\mathbb{C}^m$ onto $(0,\infty)\times(0,1)\times\mathbb{C}^m$ (Proposition 11.7(i)), maps the open set $B^\circ(\hat\rho_i)$ diffeomorphically onto an open subset of $\mathcal{U}_i$. The shear Sh_{A_i} is a diffeomorphism of $\mathcal{U}_i$ onto itself, and κ_i^{-1} a diffeomorphism of $\mathcal{U}_i$ onto Λ_i (Lemma 11.12(iii)). So E_i is a diffeomorphism of $B^\circ(\hat\rho_i)$ onto an open subset of Λ_i, and composing the three inverses gives the formula, using that $\mathrm{Sh}_{A_i,T}^{-1}$ preserves p. The inclusions of balls hold because $a_i - \delta_i < \bar\rho_i < \hat\rho_i$.

(ii) By Lemma 11.13, $(\kappa_i^{-1})^*\Omega_{\mathrm{pre}} = \omega_V + dx\wedge dT + dA_i(p)\wedge dT$ on $\mathcal{U}_i$. By Lemma 10.31 and Proposition 11.7(i),

$$\begin{aligned} E_i^*\Omega_{\mathrm{pre}} &= (\Psi_{\varepsilon_i}\times\mathrm{id})^*\mathrm{Sh}_{A_i}^*\big(\omega_V + dx\wedge dT + dA_i(p)\wedge dT\big)\\ &= (\Psi_{\varepsilon_i}\times\mathrm{id})^*\big(\omega_V + dx\wedge dT\big)\\ &= \omega_V(z_{\mathrm f}) + \omega_{\mathbb{C}}(z_{\mathrm b}) = \omega_0. \end{aligned}$$

(iii) The s-component $\sigma_i(x,T,p)$ lies in (l_i,r_i) and $v(T) \in (v_-,v_+)$; the fibre component has the same p as $z_{\mathrm f}$, and $|p| < a_i$ by Lemma 11.10.

(iv) The two bounds are Lemma 11.10(b) and (a), since $x_i\circ E_i$ is the first component of $\Psi_{\varepsilon_i}(z_{\mathrm b})$ and the shear preserves p. The closed ball $B(\bar\rho_i)$ is compact and contained in the domain of the continuous map E_i, whose values lie in Λ_i.

(v) The s-components of points of different images lie in the disjoint open intervals (l_i,r_i). □

For $(z_{\mathrm b}, z_{\mathrm f}) \in B(\bar\rho_i)$, Proposition 11.7(iv) with $I_{\max} = \bar\rho_i$ and $b_{\mathrm{st}} = \varepsilon_i/\pi$ bounds $\Psi_{\varepsilon_i}(z_{\mathrm b})$ away from the edges $x = 0$ and $T \in \{0,1\}$ by explicit constants, and $|p| \le \bar\rho_i < a_i$. These are the margins entering the effective data of §23.

11.6. **Undoing the twist and the Moser map.** The layer maps land in $(\Lambda_i, \Omega_{\mathrm{pre}})$. To reach (E,Ω) we undo the twist $\mathcal{D}$, which turned Ω_{pre} into Ω_{final}, and then the Moser map Φ, which turned Ω_{final} into Ω.

Lemma 11.16 (The twist and the Moser map)**.** *(D) The twist $\mathcal{D}(s,v,w) = (s,v,q_{f(s)}(w))$ restricts to a diffeomorphism of $\{(s,v,w) \in E|_{\mathcal{A}} : w \in \operatorname{int} V_\Delta\}$ onto itself which fixes base points, with inverse $\mathcal{D}^{-1}(s,v,w) = (s,v,q_{f(s)}^{-1}(w))$, and $(\mathcal{D}^{-1})^*\Omega_{\mathrm{final}} = \Omega_{\mathrm{pre}}$ there.*

(Φ) $\Phi = \Phi_2\circ\Phi_1\circ\Phi_0$ is a diffeomorphism of E which maps $V_\Delta(E)$ and its interior $\operatorname{int} V_\Delta(E)$ onto themselves, and $\Phi^*\Omega_{\mathrm{final}} = \Omega$ on $V_\Delta(E)$.

Proof. (D) The map $(t,w) \mapsto q_t(w)$ is smooth and each q_t is a diffeomorphism of $\mathbb{C}^m$ with $q_t(V_\Delta) = V_\Delta$ (Definition M2.4); a homeomorphism which preserves a set preserves its interior, so $q_t(\operatorname{int} V_\Delta) = \operatorname{int} V_\Delta$. Hence $\mathcal{D}$ and the displayed inverse are smooth maps of the stated set into itself, inverse to each other, and they fix base points. Since $\Omega_{\mathrm{final}} = \mathcal{D}^*\Omega_{\mathrm{pre}}$ on $E|_{\mathcal{A}}$ by (70), $(\mathcal{D}^{-1})^*\Omega_{\mathrm{final}} = \Omega_{\mathrm{pre}}$.

(Φ) Each Φ_j is the time-one map of a smooth vector field with compact support, hence a diffeomorphism of E whose inverse is the backward flow. By Proposition M4.6 and Proposition 10.11, each Φ_j maps the model region $V_\Delta(E)$ and its interior onto themselves and satisfies $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ there, where $\Omega_{0,0} = \Omega$, $\Omega_{j,1} = \Omega_{j+1,0}$ and $\Omega_{2,1} = \Omega_{\mathrm{final}}$. Composing gives $\Phi^*\Omega_{\mathrm{final}} = \Phi_0^*\Phi_1^*\Phi_2^*\Omega_{2,1} = \Omega$ on $V_\Delta(E)$. □

Proposition 11.17 (Embeddings into the model)**.** *For $1 \le i \le k$ let*

$$\psi_i = \Phi^{-1}\circ\mathcal{D}^{-1}\circ E_i\colon B^\circ(\hat\rho_i) \to E.$$

Then:

(i) *ψ_i is a smooth symplectic embedding of $(B^\circ(\hat\rho_i),\omega_0)$ onto an open subset of $(\operatorname{int} V_\Delta(E),\Omega)$; in particular it restricts to a symplectic embedding of a neighbourhood of $B(\bar\rho_i)$, and $\psi_i(B(\bar\rho_i))$ is a compact subset of $\operatorname{int} V_\Delta(E)$;*
(ii) *the images $\psi_i(B^\circ(\hat\rho_i))$, $1\le i\le k$, are pairwise disjoint.*

Proof. (i) By Proposition 11.15(i),(iii), E_i is a diffeomorphism of $B^\circ(\hat\rho_i)$ onto an open subset of Λ_i. By Lemma 11.16(D), $\mathcal{D}^{-1}$ maps it diffeomorphically onto an open subset of $E|_{\mathcal{A}}$ lying over $(l_i,r_i)\times(v_-,v_+)$ with fibre component in $\operatorname{int} V_\Delta$. Points of this base rectangle have $b(v)>0$, so they lie in the support of β, which is contained in $\mathcal{B}\subset\operatorname{int}\Sigma_\ell$ (Definition M4.1); hence the image lies in $\operatorname{int}\Sigma_\ell\times\operatorname{int} V_\Delta=\operatorname{int} V_\Delta(E)$. By Lemma 11.16($\Phi$), Φ^{-1} maps $\operatorname{int} V_\Delta(E)$ diffeomorphically onto itself. So ψ_i is a diffeomorphism onto an open subset of $\operatorname{int} V_\Delta(E)$, and

$$\psi_i^*\Omega = E_i^*(\mathcal{D}^{-1})^*(\Phi^{-1})^*\Omega = E_i^*(\mathcal{D}^{-1})^*\Omega_{\mathrm{final}} = E_i^*\Omega_{\mathrm{pre}} = \omega_0$$

by Lemma 11.16 and Proposition 11.15(ii). Since $\bar\rho_i<\hat\rho_i$, $B^\circ(\hat\rho_i)$ is a neighbourhood of the compact set $B(\bar\rho_i)$, whose image is compact.

(ii) The sets $E_i(B^\circ(\hat\rho_i))$ are pairwise disjoint (Proposition 11.15(v)). $\mathcal{D}^{-1}$ is one injective map, the same for every i, so the sets $\mathcal{D}^{-1}E_i(B^\circ(\hat\rho_i))$ are pairwise disjoint; the same holds after applying the single injective map Φ^{-1}. □

These maps are the higher-dimensional embeddings of Theorem 10.12. For the periodic four-dimensional input, the upper caps, clocks and deck coordinates give the maps of Remark 10.30. Their images avoid the exceptional fibres by the equivariant handle and layer identities of that construction.

12. The finite-atlas toric construction

This section fixes the choices behind the higher-dimensional model maps of §M4: the finite primitive operator, the normalised primitive Γ_n, the three Moser fields and the maps ψ_i. The data are those of a higher-dimensional model input with handle data (Definition M2.3 and Definition M4.1): the physical primitive $\Gamma_{\mathrm{phys}}(p)$ with curvature $\kappa(p)=d_\Sigma\Gamma_{\mathrm{phys}}(p)$, the reference primitive Γ_{ref} with positive curvature Λ, the annular coordinate $s_{\mathrm a}$ of half-width w with working half-width $b<w$, the closed one-form β, the clock $f(s)=S(s)$, $s=(s_{\mathrm a}+b)/(2b)$, and the collar cutoff χ_∂. All forms are evaluated in the explicit coordinates of the model curve. The *correction atlas* is the finite family of rational coordinate boxes of Definition 10.3; its inner boxes cover a compact subset of $\operatorname{int}\Sigma_\ell$, the *correction core*. These data satisfy: the support of $1-\chi_\partial$ in Σ_ℓ and the handle are contained in the correction core, and the correction core is contained in the correction region of Definition M4.1(d), on which $\kappa(p)\ge\kappa_-\Lambda$ for $p\in\Delta$. Its overlap tree has root the box 0, and every other box i has a parent $j<i$. A second, larger finite atlas, the *flow atlas*, covers the whole closed model base.

12.1. **A fixed finite primitive operator.** Let ζ_i be the partition weights of the correction atlas ($\sum_i\zeta_i=1$ on the correction core, ζ_i supported in box i), η_i a unit-mass bump in box i, and η_{ij} a unit-mass bump in the overlap of a child i and its parent j. Write P_i for the rectangular primitive (54) in box i; thus $dP_i\alpha=\alpha$ for every two-form α supported in box i with zero coordinate mass, and $P_i\alpha$ is supported in box i (Lemma 10.2). Set

$$e_i=P_i(\eta_i-\eta_{ij})+P_j(\eta_{ij}-\eta_j),\qquad T_0=0,\qquad T_i=e_i+T_j.$$

For a two-form ω supported in the correction core put

$$m_i(\omega)=\int\zeta_i\omega,\quad m(\omega)=\sum_i m_i(\omega),\qquad \mathcal{P}\omega=\sum_i\big[P_i(\zeta_i\omega-m_i(\omega)\eta_i)+m_i(\omega)T_i\big],$$

where $\int \zeta_i\omega$ is the coordinate integral in box i. Each argument of P_i has zero coordinate mass, and $de_i = \eta_i - \eta_j$; by induction along the tree, $dT_i = \eta_i - \eta_0$. Hence

$$d\mathcal{P}\omega = \sum_i \big[\zeta_i\omega - m_i(\omega)\eta_i + m_i(\omega)(\eta_i - \eta_0)\big] = \omega - m(\omega)\eta_0,$$

so $d\mathcal{P}\omega = \omega$ when $m(\omega) = 0$. Every sum is finite and every integral has a fixed coordinate domain, so a smooth family $\omega(p)$ gives a smooth family $\mathcal{P}\omega(p)$ with fixed compact support in $\operatorname{int}\Sigma_\ell$.

Lemma 12.1 (Intrinsic calibration)**.** *Let $\eta = df \wedge \beta$. Then $\int_{\Sigma_\ell}\eta = 1$, $m(\eta) \neq 0$, and for every smooth two-form ω supported in the correction core,*

$$\frac{m(\omega)}{m(\eta)} = \int_{\Sigma_\ell}\omega,$$

where the right side is the intrinsic surface integral.

Proof. If α is supported in the correction core and $m(\alpha) = 0$, then $\alpha = d\mathcal{P}\alpha$ with $\mathcal{P}\alpha$ compactly supported in $\operatorname{int}\Sigma_\ell$, so $\int_{\Sigma_\ell}\alpha = 0$ by Stokes' theorem. The bump η_0 has intrinsic integral 1 (change of variables in the oriented root chart); hence $m(\eta_0) \neq 0$. For ω as in the statement the form $\alpha = m(\omega)\eta_0 - m(\eta_0)\omega$ has $m(\alpha) = 0$, so $m(\omega) = m(\eta_0)\int\omega$. The form $\eta = f'(s)b(v)\,ds\wedge dv$ is supported in the handle, and $\int\eta = (f(1) - f(0))\int_0^1 b = 1$; so $m(\eta) = m(\eta_0) \neq 0$, and dividing gives the identity. □

Construction 12.2 (The normalised primitive)**.** Let γ_* be the reserve of the comparison datum, A_{ref} the reference-area bound and $\kappa_- > 0$ the density floor of Definition M4.1, and $e_\Lambda = \min\big(\gamma_*/(32A_{\mathrm{ref}}), \kappa_-/2\big)$ as in (M11). Put

$$\omega_p = (1 - \chi_\partial)\big(\kappa(p) - e_\Lambda\Lambda\big), \qquad H(p) = \frac{m(\omega_p)}{m(\eta)}, \qquad \Gamma_r(p) = \Gamma_{\mathrm{phys}}(p) + \mathcal{P}\big(H(p)\eta - \omega_p\big),$$

and, on the annulus $|s_{\mathrm{a}}| < w$, $N_p = e_\Lambda\Gamma_{\mathrm{ref}} + H(p)f(s)\beta$. In the annular coordinates (s_{a}, v), $v \in \mathbb{R}/\mathbb{Z}$, write $\Gamma_r(p) - N_p = A_p\,ds_{\mathrm{a}} + B_p\,dv$, and set

$$c_p = \int_0^1 B_p(0,v)\,dv, \qquad u_p(s_{\mathrm{a}}, v) = \int_0^{s_{\mathrm{a}}} A_p(\sigma, 0)\,d\sigma + \int_0^v \big(B_p(s_{\mathrm{a}}, \nu) - c_p b(\nu)\big)\,d\nu.$$

With $b_1 = (b + w)/2$ and $b_2 = (b_1 + w)/2$ let $\zeta = \chi^{\mathrm{I}}_{-b_2, b_2, b_2 - b_1}(s_{\mathrm{a}})$ (Definition 2.6), which equals 1 for $|s_{\mathrm{a}}| \le b_1$ and 0 for $|s_{\mathrm{a}}| \ge b_2$, and put

$$\Gamma_n(p) = \Gamma_r(p) - c_p\beta - d_\Sigma\big(\zeta u_p\big),$$

where ζu_p is extended by zero outside the annulus and d_Σ differentiates in the base variables at fixed p. The first variation primitive is $\alpha_1 = \Gamma_n - \Gamma_{\mathrm{phys}}$.

Proposition 12.3 (Normalisation identities)**.** *$H(p)$, c_p, u_p and $\Gamma_n(p)$ are smooth in p and the base variables, and for p in a neighbourhood of Δ:*

- *(i) $H(p) = H_{\mathrm{b}}(p) - \epsilon_{\mathrm{sm}}(p) = H^*(p)$ in the notation of (M13), and $0 \le H(p)$ and $H(p) \ge H_{\mathrm{b}}(p) - \gamma_*/16$ on Δ;*
- *(ii) $d_\Sigma\Gamma_n(p) = \kappa_{\mathrm{sm}}(p) + H(p)\eta$ with $\kappa_{\mathrm{sm}} = \chi_\partial\kappa + (1 - \chi_\partial)e_\Lambda\Lambda$; in particular $d_\Sigma\Gamma_n(p) = \kappa(p)$ where $\chi_\partial = 1$ outside the handle, and $d_\Sigma\Gamma_n(p)$ is positive on Σ_ℓ for $p \in \Delta$;*
- *(iii) $\Gamma_n(p) = N_p = e_\Lambda\Gamma_{\mathrm{ref}} + H(p)f(s)\beta$ on $\{|s_{\mathrm{a}}| < b_1\} = \mathcal{A}^+$.*

Proof. Smoothness: $\kappa(p)$ is smooth in p (Definition M2.3), and m, $\mathcal{P}$, c_p and u_p are finite sums of integrals of smooth integrands over fixed compact domains.

(i) By Lemma 12.1, $H(p) = \int_{\Sigma_\ell}(1-\chi_\partial)(\kappa(p) - e_\Lambda\Lambda) = H_{\mathrm{b}}(p) - \int_{\Sigma_\ell}\kappa_{\mathrm{sm}}(p)$. The lower bound is $\epsilon_{\mathrm{sm}} \le \int \chi_\partial\kappa + e_\Lambda\int_{\Sigma_\ell}\Lambda \le \gamma_*/16$ by (M12). For $H \ge 0$: where $1-\chi_\partial \ne 0$ we are in the correction region, so for $p \in \Delta$, $\kappa(p) - e_\Lambda\Lambda \ge (\kappa_- - e_\Lambda)\Lambda \ge \frac{\kappa_-}{2}\Lambda \ge 0$ because $e_\Lambda \le \kappa_-/2$; thus $\omega_p \ge 0$ and $H(p) = \int\omega_p \ge 0$.

(ii) $H\eta - \omega_p$ is supported in the correction core and has $m(H\eta-\omega_p) = 0$ by the choice of H; so $d_\Sigma\Gamma_r = \kappa + H\eta - \omega_p = \kappa_{\mathrm{sm}} + H\eta$. The terms $c_p\beta$ and $d_\Sigma(\zeta u_p)$ are closed. On Δ, κ_{sm} is a convex combination of the positive forms $\kappa(p)$ and $e_\Lambda\Lambda$, and $H\eta = Hf'b\,ds\wedge dv \ge 0$ by (i).

(iii) The cutoff χ_∂ vanishes on the annulus (Definition M4.1(g)), so there $d_\Sigma\Gamma_r = e_\Lambda\Lambda + H\,df\wedge\beta = d_\Sigma N_p$, and $\Gamma_r - N_p$ is closed: $\partial_v A_p = \partial_{s_{\mathrm{a}}}B_p$. By Stokes' theorem its period $\int_0^1 B_p(s_{\mathrm{a}},v)\,dv$ does not depend on s_{a} and equals c_p. Hence $\partial_v u_p = B_p - c_p b$ and $\partial_{s_{\mathrm{a}}}u_p = A_p(s_{\mathrm{a}},0) + \int_0^v \partial_\nu A_p(s_{\mathrm{a}},\nu)\,d\nu = A_p$, and $u_p(s_{\mathrm{a}},v+1) - u_p(s_{\mathrm{a}},v) = c_p - c_p\int_0^1 b = 0$; so u_p is a smooth function on the annulus with $d_\Sigma u_p = \Gamma_r - N_p - c_p\beta$. Where $\zeta = 1$ this gives $\Gamma_n = N_p$. □

12.2. **The three Moser fields.** For given closed source capacities $\rho_i < a_i$ put $\rho_i^\sharp = (3a_i+\rho_i)/4$. Take the caps h_i and clocks ς_j of (M17) with this $\rho_i^\sharp$, and the strip parameter $\varepsilon_i = (a_i - \rho_i^\sharp)/4$; these are the constants $\hat\rho_i = \rho_i^\sharp$ and ε_i of Definition 11.9. The three primitive families, without the vertical Liouville form λ_V, are

$$Q_{0,t} = (1-t)\Gamma_{\mathrm{phys}} + t\Gamma_n, \qquad Q_{1,t} = \Gamma_n + \eta_t, \qquad Q_{2,t} = \Gamma_n + \eta_1 + t\xi,$$

with η_t the suspension primitive of Lemma M4.4 (built from the twist (M16)) and ξ the one-form (M18), formed with these caps and clocks and with $H^* = H$. Put $\lambda_{j,t} = \lambda_V + Q_{j,t}$ and $\Omega_{j,t} = d\lambda_{j,t}$; then $\Omega_{0,0} = \Omega$, $\Omega_{0,1} = \Omega_{1,0} = \Omega_1$, $\Omega_{1,1} = \Omega_{2,0}$ and $\Omega_{2,1} = \Omega_{\mathrm{final}}$, and $\partial_t\Omega_{j,t} = d(\partial_t Q_{j,t})$. By §M4 each $\Omega_{j,t}$, $t\in[0,1]$, is symplectic on a neighbourhood of the closed model region $K = V_\Delta(E)$.

Let w_i be the bumps of the flow atlas ($w_i = 1$ on the inner box of chart i, compactly supported in chart i; the inner boxes cover the model base) and put

$$g_j(t,z) = \sum_i w_i(z)\det\big(\Omega_{j,t}\text{ in chart }i\big)^2, \qquad \mu_j = \min\Big(1, \min_{[0,1]\times K} g_j\Big).$$

At each point of K some w_i equals 1 and the corresponding determinant is nonzero, so $g_j > 0$ on the compact set $[0,1]\times K$ and $\mu_j > 0$. Conversely, $g_j(t,z) > 0$ implies that $\Omega_{j,t}$ is nondegenerate at z. With $|w|^2$ the sum of squares of the real fibre coordinates and $R = \max_K|w|^2$, set

$$\chi_j(t,z) = S\big(4g_j(t,z)/\mu_j - 1\big)\,S\big(2(R+1-|w|^2)\big) \qquad (t\in[0,1]).$$

Then $\chi_j = 1$ on the open set $\{g_j > \mu_j/2,\ |w|^2 < R+\frac12\} \supset [0,1]\times K$; and $\chi_j(t,\cdot)$ is supported in the compact set $\{g_j(t,\cdot) \ge \mu_j/4,\ |w|^2 \le R+1\}$, on which $\Omega_{j,t}$ is nondegenerate (the base part is compact because the w_i have compact support). Define $Y_{j,t}$ by $\iota_{Y_{j,t}}\Omega_{j,t} = -\chi_j\,\partial_t Q_{j,t}$ on the nondegenerate locus and $Y_{j,t} = 0$ off the support of χ_j; the two definitions agree on the overlap, so Y_j is smooth on $[0,1]\times E$ with compact spatial support. Let Φ_j be its time-one map, a diffeomorphism of E whose inverse is the backward flow.

Construction 12.4 (The canonical toric map)**.** Let E_i be the layer map of Construction 11.14 for the constants above, and $\mathcal{D}^{-1}(s,v,w) = (s,v,q_{f(s)}^{-1}(w))$ the inverse twist. The canonical map is

$$\psi_i = \Phi_0^{-1}\circ\Phi_1^{-1}\circ\Phi_2^{-1}\circ\mathcal{D}^{-1}\circ E_i\colon B^\circ(\rho_i^\sharp) \to E.$$

This formula defines the maps of the quadric, mixed and six-dimensional models.

Lemma 12.5 (Invariance of the model region)**.** *For $j = 0,1,2$, every integral curve $\gamma\colon J\to E$ of Y_j, $J\subset[0,1]$ an interval, which meets K (resp. $\operatorname{int}K$) stays in K (resp. $\operatorname{int}K$).*

Proof. Write $K = \Sigma_\ell \times V_\Delta$ with $V_\Delta = \{\mu_\nu \le \ell_\nu \text{ for all } \nu\}$, $\mu_\nu = \langle b_\nu, p\rangle$; the conditions $p_j \ge 0$ hold everywhere.

Base boundary. The variation primitives vanish over a neighbourhood N of $\partial\Sigma_\ell$ in the base: $\partial_t Q_{0,t} = \alpha_1 = \mathcal{P}(H\eta - \omega_p) - c_p\beta - d_\Sigma(\zeta u_p)$ is supported in the union of the correction core, the support of β and the annulus; $\partial_t Q_{1,t} = \partial_t \eta_t$ is supported over the annulus (Lemma M4.4(iii)); and $\partial_t Q_{2,t} = \xi$ is supported in the support of β and over the annulus (equation M18). These are compact subsets of $\operatorname{int}\Sigma_\ell$. By Lemma 2.10(c), Y_j vanishes over N, so every point over N is stationary, and by uniqueness an integral curve meeting a point over N is constant. Hence no integral curve crosses $\partial\Sigma_\ell$ in either time direction.

Outer fibre faces. Fix ν and let $X_\nu = \sum_j b_{\nu,j}\partial_{\theta_j}$, so that $\iota_{X_\nu}\omega_V = -d\mu_\nu$; its flow ϕ_r is a torus rotation of the fibre, preserves p, and commutes with every map that depends on w only through p. By Definition M2.4(1), Q_t Poisson-commutes with μ_ν on the collar $C_\nu = \{\mu_\nu > \ell_\nu - \varsigma\}$. Hence μ_ν is constant along trajectories of X_{Q_t} while they stay in C_ν; since C_ν is a superlevel set of μ_ν, a trajectory starting in C_ν never leaves it, so q_t and q_t^{-1} map C_ν into itself and preserve μ_ν there. Also Q_t is constant along ϕ_r in C_ν, so $\phi_r^* X_{Q_t} = X_{Q_t}$ there (the form ω_V is ϕ_r-invariant). The curves $t \mapsto \phi_r(q_t(w))$ and $t \mapsto q_t(\phi_r(w))$, $w \in C_\nu$, solve the same ODE with the same initial value; hence $\phi_r \circ q_t = q_t \circ \phi_r$ on C_ν, and likewise for q_t^{-1}.

Each $Q_{j,t}$ is a horizontal one-form whose coefficients are built from the toric forms $\Gamma_{\mathrm{phys}}(p)$, $\Gamma_n(p)$, $J(p)\beta$, from the functions $Q_{\lambda f(s)} \circ q_{\lambda f(s)}$ and $H^*(p(q_\lambda^{-1} q_{\lambda f(s)} w))$ of Lemma M4.4, and from $G_{\mathrm{f}} \circ \mathcal{D}$ and $G_1 \circ \mathcal{D}$, where G_{f} involves $R^*(w) = K(p(q^{-1}w)) - J(p) - \sum_i h_i(p)$. By the commutation just proved, each of these is invariant under ϕ_r over C_ν; for instance $Q_t(q_t(\phi_r w)) = Q_t(\phi_r(q_t w)) = Q_t(q_t w)$. Hence, over C_ν: (a) $\partial_t Q_{j,t}(X_\nu) = 0$, since $Q_{j,t}$ is horizontal and X_ν is vertical; (b) $\mathcal{L}_{X_\nu} Q_{j,t} = 0$, so $\iota_{X_\nu} dQ_{j,t} = \mathcal{L}_{X_\nu} Q_{j,t} - d\big(Q_{j,t}(X_\nu)\big) = 0$ and $\iota_{X_\nu}\Omega_{j,t} = \iota_{X_\nu}\omega_V = -d\mu_\nu$. Lemma 2.10(d), with $\mathfrak{v}_t = \chi_j \partial_t Q_{j,t}$, $V = X_\nu$ and $\mu = \mu_\nu$, shows that μ_ν is constant along every integral curve of Y_j while the curve stays in C_ν. An integral curve therefore cannot pass from $\{\mu_\nu \le \ell_\nu\}$ to $\{\mu_\nu > \ell_\nu\}$, nor from $\{\mu_\nu < \ell_\nu\}$ to $\{\mu_\nu \ge \ell_\nu\}$, in either time direction: near a time at which it is on the level ℓ_ν it lies in C_ν, where μ_ν is constant along it. Applying this to every ν and combining with the base boundary gives the claim for K and for $\operatorname{int} K = \operatorname{int}\Sigma_\ell \times \{\mu_\nu < \ell_\nu \text{ for all } \nu\}$ (Lemma 11.8). □

Correctness of the construction. Since $\chi_j = 1$ near $[0,1]\times K$, Lemma 12.5 and Lemma 2.10(b),(e) with $\mathfrak{v}_t = \partial_t Q_{j,t}$ give $\Phi_j(K) = K$, $\Phi_j(\operatorname{int} K) = \operatorname{int} K$ and $\Phi_j^*\Omega_{j,1} = \Omega_{j,0}$ on K. Hence $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ satisfies $\Phi^*\Omega_{\mathrm{final}} = \Omega$ on K (Lemma 11.16), and Proposition 11.17 shows that the ψ_i are smooth symplectic embeddings of $B^\circ(\rho_i^\sharp)$ into $(\operatorname{int} K, \Omega)$ with pairwise disjoint images. Since $\rho_i < \rho_i^\sharp$, $B^\circ(\rho_i^\sharp)$ is a neighbourhood of $B(\rho_i)$. By Lemma 2.10(f), running the three fields consecutively with flat clocks gives one Moser isotopy with time-one map Φ and the same variation primitives. □

13. Return to the four-ball

We work with the torus cover of §9 and with unit representatives in $\mathbb{C}^3 \times \mathbb{C}$. Write $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$ and fix $0 < h < d_n < 1/3$. The source set is

$$\mathcal{N}_h = \{(Z,U) : |Z| = 1,\ \mathcal{F}(Z) = 0,\ \pi|U|^2 \le h\}.$$

The cover map $\mathsf{N}(u,v,w) = (Z(u,v), w)$ takes $\{p \le h\}$ into $\mathcal{N}_h$, and the three marked base points are the zeros of Z_0 on the torus; on the cover the zero set is their full $\mathbb{Z}^2$-periodic inverse image (Lemma 9.11). Two pairs are gauge equivalent if $(Z', U') = (cZ, c^3 U)$ with $|c| = 1$. We prove that the transfer $\mathcal{T}_{\mathrm{u}}$ is injective on representatives and gauge equivariant, so that $\mathcal{T}_{\mathrm{u}} \circ \mathsf{N}$ identifies exactly the deck-equivalent points.

13.1. **Conventions and the ambient form.** For $Z \neq 0$ use

$$\eta_H = \frac{\operatorname{Im}(Z^* dZ)}{2\pi |Z|^2}, \qquad \omega_{\mathrm{FS}} = d\eta_H = dd^c \log |Z|^2.$$

For a unit vector put $H_Z = \{\xi : Z^*\xi = 0\}$. On H_Z, $\omega_{\mathrm{FS}}(\xi_1, \xi_2) = \pi^{-1} \operatorname{Im}(\xi_1^* \xi_2)$: at $|Z| = 1$, $d\eta_H = \frac{1}{2\pi} \sum_j d \operatorname{Im}(\bar{Z}_j dZ_j) +$ (terms containing $d|Z|^2$), and $\frac{1}{2\pi} d \operatorname{Im}(\bar{Z}_j dZ_j)(\xi_1, \xi_2) = \frac{1}{\pi} \operatorname{Im}(\bar{\xi}_{1,j} \xi_{2,j})$, while $d|Z|^2 = 2 \operatorname{Re}(Z^* \cdot)$ vanishes on H_Z. All forms below are invariant under the gauge action and annihilate its generator, hence descend to the quotient; we evaluate them on horizontal representatives.

Definition 13.1 (Ambient potential on representatives)**.** On $(\mathbb{C}^3 \setminus 0) \times \mathbb{C}$ set

$$q(Z, W) = \frac{|W|^2}{|Z|^6}, \quad \Phi_E^{(\delta)} = \log |Z|^2 + \delta \log(1 + q), \quad \Omega_E^{(\delta)} = dd^c \Phi_E^{(\delta)}, \quad \Omega_E = \Omega_E^{(d_n)}.$$

On unit representatives put $p_W = d_n |W|^2/(1 + |W|^2)$ and $X_T = \{(Z, W) : |Z| = 1,\ \mathcal{F}(Z) = TW\}$. Use E_3 as shorthand for these representative data with gauge $(Z, W) \sim (cZ, c^3 W)$. For $W \neq 0$, write $\alpha = d \arg W/(2\pi) - 3\eta_H$.

A variable unit phase changes η_H by $d\chi/(2\pi)$ and the fibre angular form by $3d\chi/(2\pi)$. Thus α is gauge invariant and zero on the gauge generator. All calculations can be pulled back along a smooth family of unit representatives. The potential and its derivatives are explicit functions on the indicated ambient space.

Lemma 13.2 (d^c of the norm)**.** *On $\{W \neq 0\}$, $d^c \log \|W\|^2 = \alpha$. On E_3,*

$$d^c \log \left(1 + \|W\|^2\right) = \frac{\|W\|^2}{1 + \|W\|^2}\, \alpha,$$

where the right side extends smoothly across $\{W = 0\}$.

Proof. Over an affine chart of the unit representative space choose a holomorphic lift $\hat{Z}$ of $[Z]$ and the holomorphic fibre coordinate v with $[Z, W] = [\hat{Z}, v]$. Then $\|W\|^2 = |v|^2 \|\hat{Z}\|^{-6}$, so $d^c \log \|W\|^2 = d \arg v/2\pi - 3\, d^c \log \|\hat{Z}\|^2$. In the unit representative $Z = \hat{Z}/\|\hat{Z}\|$ one has $W = v/\|\hat{Z}\|^3$, so $\arg W = \arg v$; and since $\operatorname{Re}(\hat{Z}^\dagger d\hat{Z}) = \frac{1}{2} d\|\hat{Z}\|^2$,

$$\eta_H = \frac{Z^\dagger dZ}{2\pi i} = \frac{\operatorname{Im}(\hat{Z}^\dagger d\hat{Z})}{2\pi \|\hat{Z}\|^2} = d^c \log \|\hat{Z}\|^2.$$

This proves the first identity. The second follows from $d^c \phi(\mu) = \phi'(\mu)\, d^c \mu$ and $d^c \mu = \mu\, d^c \log \mu$ for $\mu = \|W\|^2$. The right side is smooth because $\|W\|^2 \alpha = \|\hat{Z}\|^{-6} \big[(\operatorname{Re} v\, d \operatorname{Im} v - \operatorname{Im} v\, d \operatorname{Re} v)/2\pi - 3|v|^2 d^c \log \|\hat{Z}\|^2\big]$. □

Lemma 13.3 (The horizontal Hermitian blocks)**.** *At $|Z| = 1$, for $(\xi, \varsigma) \in H_Z \times \mathbb{C}$,*

$$(71)\quad \Omega_E^{(\delta)}((\xi_1, \varsigma_1), (\xi_2, \varsigma_2)) = \frac{1}{\pi} \operatorname{Im}(A_\delta \xi_1^* \xi_2 + B_\delta \bar{\varsigma}_1 \varsigma_2), \quad A_\delta = 1 - \frac{3\delta |W|^2}{1 + |W|^2}, \quad B_\delta = \frac{\delta}{(1 + |W|^2)^2}.$$

For $\delta = d_n$, $A > 1 - 3d_n > 0$ and $B > 0$. The form is closed, and its restriction to every complex horizontal constraint kernel $\{(\xi, \varsigma) : d\mathcal{F}_Z \xi = T\varsigma\}$ is nondegenerate.

Proof. Write $q = |W|^2 |Z|^{-6}$, which is invariant under $(Z, W) \mapsto (\lambda Z, \lambda^3 W)$, $\lambda \in \mathbb{C}^*$; hence $\Omega_E^{(\delta)}$ is the sum of ω_{FS} and the pullback of a form from the quotient, and it suffices to evaluate on

$H_Z \times \mathbb{C}$ at $|Z| = 1$. There $\partial|Z|^2 = Z^* dZ$ vanishes, so $\partial q = \bar{W}\, dW$ and $\partial\bar{\partial} q = dW \wedge d\bar{W} - 3|W|^2 \sum_j dZ_j \wedge d\bar{Z}_j$. Hence

$$\partial\bar{\partial}\log(1+q) = \frac{\partial\bar{\partial} q}{1+q} - \frac{\partial q \wedge \bar{\partial} q}{(1+q)^2} = \frac{dW \wedge d\bar{W}}{(1+|W|^2)^2} - \frac{3|W|^2}{1+|W|^2} \sum_j dZ_j \wedge d\bar{Z}_j,$$

with no mixed terms. Since $dd^c = \frac{i}{2\pi}\partial\bar{\partial}$ and $\frac{i}{2\pi} dz \wedge d\bar{z}(\xi_1, \xi_2) = \frac{1}{\pi}\operatorname{Im}(\bar{\xi}_1 \xi_2)$, adding ω_{FS} gives (71). For $\delta = d_n$, $A \geq 1 - 3d_n \frac{|W|^2}{1+|W|^2} > 1 - 3d_n > 0$ and $B > 0$, so the Hermitian form $A\xi_1^* \xi_2 + B\bar{\varsigma}_1 \varsigma_2$ is positive definite; its restriction to any complex subspace is positive definite, so the imaginary part is nondegenerate there. Closedness is $d\, dd^c = 0$. □

13.2. **The central normal injection.** The injection matches the moment $p = \pi|U|^2 \in [0, d_n)$ of N with $p_W \in [0, d_n)$ fibrewise.

Construction 13.4 (Weight-three injection)**.** For a unit pair (Z, U) with $p < d_n$ put

$$\iota[Z, U] = [Z, W], \qquad W = \frac{\sqrt{\pi}\, U}{\sqrt{d_n - p}}, \qquad p = \pi|U|^2.$$

Lemma 13.5 (The fibre injection)**.** *At each unit Z, the radial maps*

$$U \mapsto \frac{\sqrt{\pi}\, U}{\sqrt{d_n - \pi|U|^2}}, \qquad W \mapsto W\sqrt{\frac{d_n}{\pi(1+|W|^2)}}$$

are smooth inverses from $\{\pi|U|^2 < d_n\}$ to $\mathbb{C}$ and back. They commute with the weight-three gauge action and preserve Z. Moreover $|W|^2 = p/(d_n - p)$, $p_W = p$, and on $\mathcal{N}_h$ one has $|W|^2 \leq h/(d_n - h)$.

Proof. Substitute each radial expression into the other and use positivity of its denominator. The norm and moment identities follow by squaring the positive radial factor. The last bound follows from the increasing function $p \mapsto p/(d_n - p)$. □

Proposition 13.6 (The central pullback on the cover)**.** *On $\{p < d_n\}$ in the torus cover, $(\iota \circ \mathsf{N})^* \Omega_E = \Omega_0$.*

Proof. By Lemma 13.2, on unit representatives $d^c \Phi_E = \eta_H + d_n \frac{|W|^2}{1+|W|^2}\alpha = \eta_H + p_W \alpha$. Along ι we have $p_W = p$ and $\arg W = \arg U$ (Lemma 13.5), and along N, $Z^* \eta_H = \eta$ and $U = w$. Hence $(\iota \circ \mathsf{N})^* d^c \Phi_E = \eta + p\big(\frac{d \arg w}{2\pi} - 3\eta\big) = (1 - 3p)\eta + \frac{1}{2}\operatorname{Im}(\bar{w}\, dw)$, which is smooth across $w = 0$, and its exterior derivative is Ω_0 by Lemma 9.3. □

13.3. **Transport to $T = 1$.** At a point of X_T the tangent space TX_T is the kernel of the derivative of $\mathcal{F}(Z) - TW$. The transport field is the lift of ∂_T which is Ω_E-orthogonal to TX_T and preserves the constraint.

Construction 13.7 (Transport field)**.** For $(Z, W, T) \in S^5 \times \mathbb{C} \times \mathbb{R}$ put $f = \mathcal{F}(Z)$, $\bar{Z}^2 = (\bar{Z}_0^2, \bar{Z}_1^2, \bar{Z}_2^2)$,

$$n = 3\bar{Z}^2 - 3\bar{f} Z, \qquad G = \|n\|^2, \qquad N_{\mathrm{den}} = \frac{G}{A} + \frac{T^2}{B},$$

with A, B as in Lemma 13.3. On the open set $\mathcal{O} = \{N_{\mathrm{den}} > 0\}$ define

$$\dot{Z} = \frac{W\, n}{A\, N_{\mathrm{den}}}, \qquad \dot{W} = -\frac{T\, W}{B\, N_{\mathrm{den}}}, \qquad \dot{T} = 1. \tag{72}$$

Since $A, B > 0$ everywhere, $N_{\mathrm{den}} \geq 0$, with equality only where $T = 0$ and $n = 0$; so $\mathcal{O}$ is the complement of $\{T = 0,\ n = 0\}$, and the field is smooth on $\mathcal{O}$.

Lemma 13.8 (Properties of the transport field)**.** *(i)* $Z^\dagger n = 0$, $n^\dagger \xi = d\mathcal{F}_Z(\xi)$ *for* $\xi \in H_Z$, *and* $G = 9\sum_j |Z_j|^4 - 9|f|^2$.

(ii) The field is equivariant under $(Z, W) \mapsto (e^{i\chi}Z, e^{3i\chi}W)$ *and satisfies* $Z^\dagger \dot Z = 0$*; hence it is tangent to* $S^5 \times \mathbb{C} \times \mathbb{R}$. *Its horizontal component is* $V = (\dot Z, \dot W) \in H_Z \times \mathbb{C}$.

(iii) Along the field, $\frac{d}{dT}\mathcal{F}(Z) = \frac{WG}{AN_{\mathrm{den}}} = \frac{d}{dT}(TW)$, *so* $\mathcal{F}(Z) - TW$ *is constant;* $\|Z\|$ *is constant; and* $\frac{d}{dT}|W|^2 = -\frac{2T|W|^2}{BN_{\mathrm{den}}} \le 0$ *for* $T \ge 0$.

(iv) At a point of $X_T \times \{T\}$ *in* $\mathcal{O}$, V *is* Ω_E*-orthogonal to* $\ker(d\mathcal{F}_Z, -T) \subset H_Z \times \mathbb{C}$.

Proof. (i) $Z^\dagger(3\bar Z^2) = 3\sum_j \bar Z_j^3 = 3\bar f$, so n is the orthogonal projection of $3\bar Z^2$ to H_Z; hence $Z^\dagger n = 0$ and $\|n\|^2 = \|3\bar Z^2\|^2 - |3\bar f|^2$. For $\xi \in H_Z$, $d\mathcal{F}_Z(\xi) = \sum_j 3Z_j^2\xi_j = (3\bar Z^2)^\dagger \xi = n^\dagger\xi + 3fZ^\dagger\xi = n^\dagger\xi$.

(ii) Under the gauge action $f \mapsto e^{3i\chi}f$ and $n \mapsto e^{-2i\chi}n$, while $G, A, B, N_{\mathrm{den}}$ are invariant; so $\dot Z \mapsto e^{i\chi}\dot Z$ and $\dot W \mapsto e^{3i\chi}\dot W$. By (i), $Z^\dagger \dot Z = 0$, so $\mathrm{Re}(Z^\dagger \dot Z) = 0$ and the field is tangent to $S^5 \times \mathbb{C} \times \mathbb{R}$. The kernel of $T(S^5 \times \mathbb{C}) \to TE_3$ is spanned by $(iZ, 3iW)$, and $H_Z \times \mathbb{C}$ is a complement of it, which gives the descent and the representative.

(iii) By (i), $\frac{d}{dT}\mathcal{F}(Z) = n^\dagger \dot Z = WG/(AN_{\mathrm{den}})$, and $W + T\dot W = W\big(1 - \frac{T^2}{BN_{\mathrm{den}}}\big) = W\frac{G/A}{N_{\mathrm{den}}}$. Further $\frac{d}{dT}\|Z\|^2 = 2\,\mathrm{Re}(Z^\dagger \dot Z) = 0$ and $\frac{d}{dT}|W|^2 = 2\,\mathrm{Re}(\bar W \dot W)$.

(iv) By (i), in the gauge of Lemma 13.3 the tangent space $\ker(d\mathcal{F}_Z, -T) \subset H_Z \times \mathbb{C}$ is the kernel of the complex linear functional $\ell(\xi, \varsigma) = n^\dagger\xi - T\varsigma$ on $H_Z \times \mathbb{C}$. For the Hermitian form $\mathfrak{h} = A\,\xi_1^\dagger\xi_2 + B\,\bar\varsigma_1\varsigma_2$ of (71) and $\nu_\ell = (n/A, -T/B)$ one has $\mathfrak{h}(\nu_\ell, \kappa) = \ell(\kappa)$ for all κ. Since $V = (W/N_{\mathrm{den}})\,\nu_\ell$ and $\mathfrak{h}$ is conjugate linear in the first slot, $\mathfrak{h}(V, \kappa) = 0$ for every $\kappa \in \ker \ell$, hence $\Omega_E(V, \kappa) = \frac{1}{\pi}\,\mathrm{Im}\,\mathfrak{h}(V, \kappa) = 0$. Moreover $\ell(V) = W(G/A + T^2/B)/N_{\mathrm{den}} = W$: the field is the unique Ω_E-orthogonal lift of ∂_T along which the derivative $n^\dagger \dot Z - T\dot W - \dot T W$ of the constraint vanishes. □

Lemma 13.9 (Denominator)**.** *Let* $\mathcal{C} = \{(Z, W, T) : \|Z\| = 1,\ \mathcal{F}(Z) = TW,\ 0 \le T \le 1\}$. *On* $\mathcal{C}$, $N_{\mathrm{den}} > 0$*: for* $T = 0$ *one has* $G \ge 3$ *and* $N_{\mathrm{den}} \ge 3$, *and for* $T > 0$, $N_{\mathrm{den}} \ge T^2/B > 0$. *If moreover* $|W| \le W_{\max}$ *and* $T_0 = \min\big(1, 1/(\sqrt{6}\max(1, W_{\max}))\big)$, *then* $N_{\mathrm{den}} \ge \min\big(\frac{3}{2}, T_0^2/d_n\big)$.

Proof. By Cauchy–Schwarz, $\sum_j |Z_j|^4 \ge \frac{1}{3}\big(\sum_j |Z_j|^2\big)^2 = \frac{1}{3}$, and $0 < A \le 1$, so $N_{\mathrm{den}} \ge G/A \ge G$. On $\mathcal{C}$, $|f| = T|W|$. For $T = 0$ this gives $G = 9\sum_j |Z_j|^4 \ge 3$. If $T \le T_0$, then $9|f|^2 \le 9T_0^2 W_{\max}^2 \le \frac{3}{2}$, so $G \ge \frac{3}{2}$. If $T \ge T_0$, then $N_{\mathrm{den}} \ge T^2/B \ge T_0^2/d_n$, since $B \le d_n$. □

Proposition 13.10 (Protected transport)**.** *Fix* $w_* = h/(d_n - h)$ *and let* τ_T *be the flow, from* $T = 0$, *of the cut field defined in the proof. It is a smooth injective map, commuting with the gauge action. For every unit pair* $(Z, W) \in X_0$ *with* $|W|^2 \le w_*$ *the trajectory solves* (72) *on* $[0, 1]$, *stays in* $X_T \cap \{|W|^2 \le w_*\}$, *and* τ_1 *maps* $X_0 \cap \{|W|^2 \le w_*\}$ *into* X_1 *with* $\tau_1^*(\Omega_E|_{X_1}) = \Omega_E|_{X_0}$ *there.*

Proof. Let $T_0 = T_0(\sqrt{w_*})$ and $c_* = \min(\frac{3}{2}, T_0^2/d_n) > 0$ be the bound of Lemma 13.9. For $a < b$ write $S_{a,b}(x) = S((x - a)/(b - a))$ and $P_{a,b,c,d}(x) = S_{a,b}(x)(1 - S_{c,d}(x))$, and put

$$\chi = S_{c_*/2,3c_*/4}(N_{\mathrm{den}})\, P_{1/2,3/4,5/4,3/2}(|Z|^2)\, S_{0,1}(w_* + 2 - |W|^2)\, P_{-1,-1/2,3/2,2}(T),$$

where off the unit sphere n is replaced by $3\bar Z^2 - 3\bar f Z/|Z|^2$ and N_{den} is computed with it. The field $\chi\cdot(\dot Z, \dot W, 1)$ (with $\dot T$ replaced by χ) is smooth on $\mathbb{C}^3 \times \mathbb{C} \times \mathbb{R}$, because the first factor vanishes where $N_{\mathrm{den}} \le c_*/2$, and it has compact support in $\{\frac{1}{2} \le |Z|^2 \le \frac{3}{2},\ |W|^2 \le w_* + 2,\ |T| \le 2\}$. Its flow is therefore complete, each time-T map is a diffeomorphism (hence injective), and it commutes with the gauge action because every factor is gauge invariant and the field is equivariant (Lemma 13.8(ii)).

On $\mathcal{C}_* = \{|Z| = 1,\ \mathcal{F}(Z) = TW,\ |W|^2 \le w_*,\ 0 \le T \le 1\}$ every factor of χ equals one: $N_{\mathrm{den}} \ge c_*$ by Lemma 13.9 with $W_{\max} = \sqrt{w_*}$. Along the uncut field, $|Z|$ and $\mathcal{F}(Z) - TW$ are constant and $|W|^2$ is nonincreasing for $T \ge 0$ (Lemma 13.8(iii)). Hence the uncut trajectory from a point of $X_0 \cap \{|W|^2 \le w_*\}$ stays in $\mathcal{C}_*$ while it exists, so it is also a trajectory of the cut field, which is defined for $T \in [0,1]$; by uniqueness the two coincide on $[0,1]$.

For the pullback, let $x(\sigma)$ be a smooth family of such initial points and $F(\sigma,T) = \tau_T(x(\sigma))$. Since $\mathcal{F}(Z) - TW$ vanishes identically along F, a tangent vector (κ, t) satisfies $\ell(\kappa) = tW$, where $\ell = (d\mathcal{F}_Z, -T)$. Replace κ by its horizontal representative κ_H modulo the gauge generator $G_{\mathrm{gauge}} = (iZ, 3iW)$. On X_T one has $\ell(G_{\mathrm{gauge}}) = 3i(\mathcal{F}(Z) - TW) = 0$, and Ω_E annihilates this generator, so the constraint and the pairing with Ω_E are unchanged. Now $\partial_T F = (V, 1)$, and by Lemma 13.8(iv), $\ell(V) = W$ and V is Ω_E-orthogonal to the horizontal kernel of ℓ. Writing $\kappa_H = tV + (\kappa_H - tV)$ gives $\Omega_E(V, \kappa) = \Omega_E(V, \kappa_H) = 0$. Thus $\iota_{\partial_T} F^*\Omega_E = 0$. As $F^*\Omega_E$ is closed, Cartan's formula gives $\partial_T(F_T^*\Omega_E) = d\,\iota_{\partial_T} F^*\Omega_E = 0$, so $F_1^*\Omega_E = F_0^*\Omega_E$, which is the asserted identity. □

13.4. **Return to the Fubini–Study form.** On X_1 we have $W = \mathcal{F}(Z)$, so the projection $\mathrm{pr}(Z, W) = Z$ has inverse the graph map $\mathrm{gr}[Z] = [Z, \mathcal{F}(Z)]$. Let $u = |\mathcal{F}(Z)|^2/\|Z\|^6$, a smooth function on unit representatives, and for $s \in \mathbb{R}$

$$\omega_s^{\mathrm{gr}} = \omega_{\mathrm{FS}} + s\, d_n\, dd^c \log(1+u) = \mathrm{gr}^*\Omega_E^{(sd_n)}.$$

The last identity holds because $\mathrm{gr}^*\|W\|^2 = u$; in particular $\omega_0^{\mathrm{gr}} = \omega_{\mathrm{FS}}$ and $\omega_1^{\mathrm{gr}} = \mathrm{gr}^*\Omega_E = (\mathrm{pr}^{-1})^*(\Omega_E|_{X_1})$.

The return is the Moser flow for the path ω_s^{gr}: since $\partial_s \omega_s^{\mathrm{gr}} = d_n\, dd^c \log(1+u)$, we need $\iota_{Y_s}\omega_s^{\mathrm{gr}} = -d_n\, d^c \log(1+u)$. The field Y_s below solves this in closed form because n is an eigenvector of the Hermitian matrix M_s of ω_s^{gr}; its factor $\frac{1}{2}$ is the ratio of the factor $\frac{1}{\pi}$ of the Kähler convention to the factor $\frac{1}{2\pi}$ of Lemma 13.13.

Construction 13.11 (Return field)**.** In unit representatives, with f, n, G as in Construction 13.7, put

$$A_s = 1 - 3s\, d_n \frac{u}{1+u}, \qquad B_s = \frac{s\, d_n}{(1+u)^2}, \qquad M_s = A_s I + B_s\, nn^\dagger \ \text{ on } H_Z,$$

$$Y_s = -\frac{d_n}{2}\, \frac{f\, n}{(1+u)(A_s + B_s G)} \in H_Z. \tag{73}$$

Lemma 13.12 (Graph forms)**.** *For $s \in [0,1]$ and $\xi_1, \xi_2 \in H_Z$, $\omega_s^{\mathrm{gr}}(\xi_1, \xi_2) = \frac{1}{\pi}\,\mathrm{Im}(\xi_1^\dagger M_s \xi_2)$. Moreover $u \le 1$, $A_s \ge 1 - \frac{3}{2}d_n > \frac{1}{2}$ and $B_s \ge 0$; hence $M_s \ge \frac{1}{2} I$ and every ω_s^{gr} is positive on H_Z.*

Proof. By Lemma 13.8(i), $d\,\mathrm{gr}$ maps the tangent vector represented by $\xi \in H_Z$ to the vector represented by

$$(\xi, d\mathcal{F}_Z(\xi)) = (\xi, n^\dagger \xi)$$

in the gauge of Lemma 13.3. At $W = \mathcal{F}(Z)$ one has $|W|^2 = u$, so $A_{sd_n} = A_s$ and $B_{sd_n} = B_s$, and (71) gives the Hermitian form $A_s \xi_1^\dagger \xi_2 + B_s \overline{n^\dagger \xi_1}\, n^\dagger \xi_2 = \xi_1^\dagger M_s \xi_2$. For $\|Z\| = 1$, $|f| \le \sum_j |Z_j|^3 \le \max_j |Z_j| \sum_j |Z_j|^2 \le 1$, so $u \le 1$, $u/(1+u) \le \frac{1}{2}$ and $A_s \ge 1 - \frac{3}{2}d_n > \frac{1}{2}$ because $d_n < \frac{1}{3}$. □

Thus, unlike Ω_E on E_3, whose horizontal coefficient is only bounded below by $1 - 3d_n$, the graph forms are uniformly non-degenerate as $d_n \to \frac{1}{3}$, and the denominator $A_s + B_s G$ of (73) is at least $\frac{1}{2}$ for $s \in [0,1]$.

Lemma 13.13 (d^c along the graph)**.** *For $\xi \in H_Z$ (unit representative), $d^c\log(1+u)(\xi) = \dfrac{\operatorname{Im}\left(\bar f\, n^\dagger \xi\right)}{2\pi(1+u)}$.*

Proof. Where $f \neq 0$, as in Lemma 13.2, $d^c\log(1+u) = \frac{u}{1+u}\big(d^c\log|\mathcal{F}|^2 - 3\,d^c\log\|Z\|^2\big) = \frac{u}{1+u}\big(\frac{d\arg\mathcal{F}}{2\pi} - 3\eta_H\big)$. On H_Z, $\eta_H(\xi) = 0$ and $d\arg\mathcal{F}(\xi) = \operatorname{Im}\big(d\mathcal{F}_Z(\xi)/f\big) = \operatorname{Im}(n^\dagger\xi/f)$, and $u/f = \bar f$. Both sides are smooth, so the identity extends to $\{f = 0\}$. □

Lemma 13.14 (The return field)**.** *Y_s is smooth in (s, Z) for s in a neighbourhood of $[0,1]$, gauge equivariant and tangent to S^5, hence descends to a time-dependent smooth vector field on unit representatives, and*

$$\iota_{Y_s}\omega_s^{\mathrm{gr}} = -d_n\, d^c\log(1+u) \qquad (s \in [0,1]).$$

Proof. Under the gauge action $f \mapsto e^{3i\chi}f$ and $n \mapsto e^{-2i\chi}n$, so $Y_s \mapsto e^{i\chi}Y_s$, and $Z^\dagger Y_s = 0$. The denominator is positive for $-\frac{1}{9d_n} < s < \frac{2}{3d_n}$, an interval containing $[0,1]$: for $s \geq 0$, $A_s + B_sG \geq A_s \geq 1 - \frac{3}{2}s\,d_n$ since $u/(1+u) \leq \frac{1}{2}$; for $s < 0$, $A_s \geq 1$ and, since $0 \leq G \leq 9\sum_j |Z_j|^4 \leq 9$ and $u \geq 0$, $B_sG \geq -9|s|\,d_n$, so $A_s + B_sG \geq 1 - 9|s|\,d_n$. Since $M_sn = (A_s + B_s\|n\|^2)n = (A_s + B_sG)n$, we have $M_sY_s = -\frac{d_n}{2}\frac{fn}{1+u}$, and for $\xi \in H_Z$, by Lemma 13.12 and Lemma 13.13,

$$\omega_s^{\mathrm{gr}}(Y_s,\xi) = \frac{1}{\pi}\operatorname{Im}\big((M_sY_s)^\dagger\xi\big) = -\frac{d_n}{2\pi(1+u)}\operatorname{Im}\left(\bar f\, n^\dagger\xi\right) = -d_n\, d^c\log(1+u)(\xi). \qquad \square$$

Proposition 13.15 (Unit-representative return)**.** [Lean: needed consequence verified.]

Let ψ_s be the flow from $s = 0$ of the cut field of the proof. It is defined for all s, preserves $|Z| = 1$, commutes with the gauge action, and its time-s maps are diffeomorphisms. On the unit sphere it is the flow of Y_s for $s \in [0,1]$, and $\psi_s^*\omega_s^{\mathrm{gr}} = \omega_{\mathrm{FS}}$. Consequently $(\psi_1^{-1})^*\omega_{\mathrm{FS}} = \omega_1^{\mathrm{gr}}$, and $(\psi_1^{-1}\circ\mathrm{pr})^*\omega_{\mathrm{FS}} = \Omega_E|_{X_1}$.

Proof. Put $m = 1 - \frac{3}{2}d_n > 0$ and let the cut field be

$$S_{m/2,3m/4}(D_s)\,P_{1/2,3/4,5/4,3/2}(|Z|^2)\,P_{-1,-1/2,3/2,2}(s)\,Y_s, \qquad D_s = A_s + B_sG,$$

with n extended off the sphere as in Proposition 13.10. It is smooth (the first factor vanishes where $D_s \leq m/2$) and compactly supported, so its flow is complete with diffeomorphic time maps; it is gauge equivariant because Y_s is (Lemma 13.14). On the unit sphere Y_s is horizontal, so $d|Z|^2(Y_s) = 2\operatorname{Re}(Z^\dagger Y_s) = 0$ and the sphere is invariant. For $|Z| = 1$ and $s \in [0,1]$, $D_s \geq A_s \geq 1 - \frac{3}{2}sd_n \geq m$ (Lemma 13.12), so every cutoff factor equals one and the cut field is Y_s. Since $\partial_s\omega_s^{\mathrm{gr}} = d\big(d_n\, d^c\log(1+u)\big)$ and $\iota_{Y_s}\omega_s^{\mathrm{gr}} = -d_n\, d^c\log(1+u)$ (Lemma 13.14), Cartan's formula gives $\frac{d}{ds}\psi_s^*\omega_s^{\mathrm{gr}} = \psi_s^*d\big(\iota_{Y_s}\omega_s^{\mathrm{gr}} + d_n\, d^c\log(1+u)\big) = 0$, hence $\psi_s^*\omega_s^{\mathrm{gr}} = \omega_0^{\mathrm{gr}} = \omega_{\mathrm{FS}}$. Finally $\mathrm{gr}\circ\mathrm{pr} = \mathrm{id}$ on X_1, so $\mathrm{pr}^*\omega_1^{\mathrm{gr}} = \mathrm{pr}^*\mathrm{gr}^*\Omega_E = \Omega_E|_{X_1}$. □

13.5. **Equivariant injectivity and the ball chart.** From now on, $\mathcal{T}_{\mathrm{u}}$ uses the protected cutoff $w_* = h/(d_n - h)$ and is considered on $\mathcal{N}_h$. Let $\mathsf{r}Z = (\zeta Z_0, Z_1, Z_2)$ and $\mathsf{r}(Z,W) = (\mathsf{r}Z, W)$. It preserves the norm, Fermat equation and source moment and commutes with the unit gauge action.

Lemma 13.16 (Fixed representatives modulo gauge)**.** *For a nonzero vector, $\mathsf{r}Z$ is a scalar multiple of Z exactly when $Z_0 = 0$ or $Z_1 = Z_2 = 0$. For a central unit pair (Z,U), the equation $\mathsf{r}(Z,U) = (cZ, c^3U)$ is possible exactly when $Z_0 = 0$.*

Proof. A nonzero last coordinate forces $c = 1$ and then $Z_0 = 0$; otherwise both last coordinates vanish. The latter alternative is excluded at a central unit vector by $\mathcal{F}(Z) = 0$. When $Z_0 = 0$, the pair is fixed with $c = 1$. □

Lemma 13.17 (Equivariance)**.** r *preserves* $\mathcal{F}$, ω_{FS}, $\|W\|$, Ω_E, u, *each* X_T *and* $\{(Z,U) : |Z| = 1, \mathcal{F}(Z) = 0, p < d_n\}$, *and it maps the fields* (72) *and* (73) *to themselves. Consequently* ι, τ_1, pr, ψ_1^{-1} *and* $\mathcal{T}_{\mathrm{u}}$ *commute with* r.

Proof. r is unitary on $\mathbb{C}^3$ and $\mathcal{F} \circ \mathsf{r} = \mathcal{F}$, so r preserves $\|Z\|$, ω_{FS}, $\|W\|$, Ω_E, X_T, u, p, A, B, A_s, B_s. Since $\bar{\zeta}^2 = \zeta$, $n(\mathsf{r}Z) = 3\,\overline{\mathsf{r}Z}^2 - 3\bar{f}\,\mathsf{r}Z = \mathsf{r}\,n(Z)$; so G and N_{den} are invariant, and both fields satisfy $\dot{Z}(\mathsf{r}Z, W) = \mathsf{r}\,\dot{Z}(Z,W)$, $\dot{W}(\mathsf{r}Z, W) = \dot{W}(Z,W)$. By uniqueness of solutions the flows commute with r. ι commutes with r because p is invariant, and pr commutes with r because the lift of r to E_3 covers r. □

Lemma 13.18 (Injectivity modulo gauge)**.** *For* $x, x' \in \mathcal{N}_h$ *and* $|c| = 1$, *the equality* $\mathcal{T}_{\mathrm{u}}(x) = c\mathcal{T}_{\mathrm{u}}(x')$ *implies* $x = (cZ', c^3U')$.

Proof. Each factor of $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ is injective on representatives: ι has the explicit inverse of Lemma 13.5, τ_1 and ψ_1^{-1} are time maps of complete flows, and pr is injective on X_1 because $W = \mathcal{F}(Z)$ there. Each factor commutes with $(Z,W) \mapsto (cZ, c^3W)$, $|c| = 1$, so $\mathcal{T}_{\mathrm{u}}(cZ', c^3U') = c\,\mathcal{T}_{\mathrm{u}}(Z', U')$. The hypothesis therefore reads $\mathcal{T}_{\mathrm{u}}(x) = \mathcal{T}_{\mathrm{u}}(cZ', c^3U')$, and injectivity gives $x = (cZ', c^3U')$. □

Proposition 13.19 (Avoidance and compact margin)**.** *If* $x = (Z,U) \in \mathcal{N}_h$ *and* $Z_0 \neq 0$, *then* $(\mathcal{T}_{\mathrm{u}}(x))_0 \neq 0$ *and its last two coordinates do not vanish together. On every nonempty compact family of these inputs,* $m_{\mathcal{K}} = \min_{x \in \mathcal{K}} |(\mathcal{T}_{\mathrm{u}}(x))_0|^2 > 0$. *For* $\mathcal{K} = \varnothing$ *set* $m_{\mathcal{K}} = 1$.

Proof. Let $Y = \mathcal{T}_{\mathrm{u}}(x)$, a unit vector. If $Y_0 = 0$, then $\mathsf{r}Y = Y$, so $\mathcal{T}_{\mathrm{u}}(\mathsf{r}x) = \mathsf{r}\mathcal{T}_{\mathrm{u}}(x) = Y = \mathcal{T}_{\mathrm{u}}(x)$ by Lemma 13.17, and Lemma 13.18 with $c = 1$ gives $\mathsf{r}x = x$; by Lemma 13.16, $Z_0 = 0$. If $Y_1 = Y_2 = 0$, then $\mathsf{r}Y = \zeta Y$, so $\mathcal{T}_{\mathrm{u}}(\mathsf{r}x) = \zeta\mathcal{T}_{\mathrm{u}}(x)$ and Lemma 13.18 gives $\mathsf{r}x = (\zeta Z, \zeta^3 U)$; by Lemma 13.16, again $Z_0 = 0$. Both contradict $Z_0 \neq 0$. The function $x \mapsto |(\mathcal{T}_{\mathrm{u}}(x))_0|^2$ is continuous and positive on $\mathcal{K}$, so it has a positive minimum when $\mathcal{K} \neq \varnothing$. □

Lemma 13.20 (The ball chart on representatives)**.** [Lean: needed consequence verified.]

On $S^5_* = \{Z \in \mathbb{C}^3 : |Z| = 1,\ Z_0 \neq 0\}$ *put* $b_{\mathrm{ball}}(Z) = \frac{\bar{Z}_0}{|Z_0|}\frac{(Z_1,Z_2)}{\sqrt{\pi}}$. *This map is gauge invariant,* $\pi|b_{\mathrm{ball}}(Z)|^2 = 1 - |Z_0|^2 < 1$, *and in the affine coordinates* $z = (Z_1/Z_0, Z_2/Z_0)$ *it is the chart* $\mathrm{ch}(z) = z/\sqrt{\pi(1+|z|^2)}$ *of Lemma 2.2(b). Moreover* $b_{\mathrm{ball}}^*\omega_0 = \frac{1}{\pi}\sum_{j=0}^{2} dx_j \wedge dy_j\big|_{S^5_*} = \omega_{\mathrm{FS}}|_{S^5_*}$, *where* $Z_j = x_j + iy_j$. *Hence* $(b_{\mathrm{ball}} \circ \varphi)^*\omega_0 = \varphi^*\omega_{\mathrm{FS}}$ *for every smooth map* φ *of a manifold into* S^5_*.

Proof. The unit phases cancel, and substituting $Z = e^{i\chi}(1,z)/\sqrt{1+|z|^2}$ gives the affine formula and the norm identity. Write $e = \bar{Z}_0/|Z_0| = e^{i\phi}$, $\phi = -\arg Z_0$, and $y_j = eZ_j/\sqrt{\pi}$ $(j = 1,2)$. Then $\bar{y}_j dy_j = \frac{1}{\pi}\big(|Z_j|^2\,\bar{e}\,de + \bar{Z}_j dZ_j\big)$ with $\bar{e}\,de = i\,d\phi$, so the Liouville form $\lambda_0 = \frac{1}{2}\sum_{j=1}^{2}\mathrm{Im}(\bar{y}_j dy_j)$ pulls back to $\frac{1}{2\pi}\big((1-|Z_0|^2)d\phi + \sum_{j=1}^{2}\mathrm{Im}(\bar{Z}_j dZ_j)\big)$ on S^5_*. Since $\mathrm{Im}(\bar{Z}_0 dZ_0) = |Z_0|^2 d\arg Z_0 = -|Z_0|^2 d\phi$, this equals $\frac{1}{2\pi}\big(d\phi + \mathrm{Im}(Z^\dagger dZ)\big)$, whose differential is $\frac{1}{2\pi}\sum_{j=0}^{2} d(x_j dy_j - y_j dx_j) = \frac{1}{\pi}\sum_j dx_j \wedge dy_j$. As $d\lambda_0 = \omega_0$, this proves the first identity; the second holds because $\frac{1}{2\pi}\mathrm{Im}(Z^\dagger dZ) = \eta_H$ on the sphere and $d\eta_H = \omega_{\mathrm{FS}}$. □

The vector $(\sqrt{1-\pi|y|^2}, \sqrt{\pi}\,y_1, \sqrt{\pi}\,y_2)$ is a unit vector with positive first coordinate for $y \in B^\circ(1)$, and its ball-chart image is y. Conversely, every unit vector with nonzero first coordinate has a unique unit-gauge representative with positive first coordinate, namely the vector above for its ball-chart image.

13.6. **The transfer theorem.**

Theorem 13.21 (Transfer on the cover)**.** [Lean: needed consequence verified.]
Let $w_* = h/(d_n - h)$ and $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ on $\mathcal{N}_h$. The cover map

$$\mathcal{T} = b_{\mathrm{ball}} \circ \mathcal{T}_{\mathrm{u}} \circ \mathsf{N} \quad on \quad \{p < h,\ (u,v) \notin \mathcal{Z}\}$$

is smooth, invariant under the deck maps, takes values in $B^\circ(1)$ and satisfies $\mathcal{T}^*\omega_0 = \Omega_0$. Two points have the same image if and only if they are deck equivalent. For every compact $\mathcal{K}$ in the domain, $\mathcal{T}(\mathcal{K}) \subset B(1 - m_{\mathcal{K}})$ with $m_{\mathcal{K}} > 0$ as in Proposition 13.19. In particular the restriction of $\mathcal{T}$ to an open set on which it is injective is a symplectic embedding, and $\mathcal{T}$ carries pairwise disjoint subsets of its domain, no two points of whose union are deck equivalent, to pairwise disjoint sets.

Proof. On $\{p < h\}$, N takes values in $\mathcal{N}_h$ with $p < d_n$, and $\tau_1 \circ \iota$ lands in $X_1 \cap \{|W|^2 \le w_*\}$ (Lemma 13.5, Proposition 13.10). Composing Propositions 13.6, 13.10 and 13.15 gives $(\mathcal{T}_{\mathrm{u}} \circ \mathsf{N})^*\omega_{\mathrm{FS}} = \Omega_0$. If $(u,v) \notin \mathcal{Z}$, then $Z_0(u,v) \neq 0$, so $\mathcal{T}_{\mathrm{u}}(\mathsf{N}(x)) \in S^5_*$ by Proposition 13.19; Lemma 13.20 then gives $\mathcal{T}^*\omega_0 = \Omega_0$ and $\pi|\mathcal{T}|^2 = 1 - |(\mathcal{T}_{\mathrm{u}}\mathsf{N})_0|^2 < 1$. A deck map changes $\mathsf{N}(x)$ to a gauge-equivalent pair (Construction 9.8(b)), $\mathcal{T}_{\mathrm{u}}$ is gauge equivariant and b_{ball} is gauge invariant, so $\mathcal{T}$ is deck invariant. Conversely, suppose $\mathcal{T}(x) = \mathcal{T}(x')$. A unit vector with nonzero first coordinate is determined up to gauge by its ball-chart image (see the paragraph before this subsection), so $\mathcal{T}_{\mathrm{u}}(\mathsf{N}x) = c\,\mathcal{T}_{\mathrm{u}}(\mathsf{N}x')$ with $|c| = 1$, and Lemma 13.18 gives $\mathsf{N}(x) \sim \mathsf{N}(x')$. Then $[Z(u,v)] = [Z(u',v')]$ in $\mathbb{CP}^2$, and injectivity of Φ_Γ (Lemma 9.5) gives $(u,v) - (u',v') \in \mathbb{Z}^2$. Applying the corresponding deck map T to x' we get $\mathsf{N}(x) \sim \mathsf{N}(Tx')$ with equal unit vectors $Z(u,v)$, so the gauge factor is one and the fibre coordinates agree: $x = Tx'$. The margin is $1 - \pi|\mathcal{T}|^2 = |(\mathcal{T}_{\mathrm{u}}\mathsf{N})_0|^2 \ge m_{\mathcal{K}}$. Finally, on an open set on which $\mathcal{T}$ is injective it is an injective immersion between manifolds of equal dimension, hence an open symplectic embedding, and the last assertion follows from the characterisation of equal images. □

All objects in the theorem are given by closed formulas and by the solutions of the explicit ODEs (72) and (73), the latter integrated backwards from $s = 1$ to give ψ_1^{-1}.

13.7. **The packing in dimension four.** We now compose the packing in the model with the transfer. Recall from Lemma 8.5 the data attached to capacities $0 < \rho'_1, \dots, \rho'_k < \frac{1}{3}$ with $\sum_i (\rho'_i)^2 < 1$: auxiliary capacities a_i, a height h, the core fraction $\tau_\Delta = (a_{\max} + h)/(2h)$, and $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - p)_+$ on $\Delta = [0,h]$. The transfer depends on the capacities only through h.

Construction 13.22 (Data of the four-dimensional packing)**.** Let $k \ge 1$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$. Define $(a_i, h, P, (K,q), \gamma)$ as follows.

(a) *Equal case*: $k \ge 10$, $\rho_1 = \cdots = \rho_k = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$. Put $t_0 = \frac{1}{2}(1 + \rho\sqrt{k})$, $a_i = a = t_0/\sqrt{k}$, $h = t_0/3$, $P = P_{\mathrm{b}}$, and let (K,q) be the comparison datum of Proposition M3.8, with gap $\gamma = \frac{31}{32} \cdot 3(1 - t_0)$.

(b) *General case*, otherwise. Use the auxiliary quantities of (37) for ρ. Define K by the scalar iterate and fixed convolution (13)–(17), and use the planar path (30)–(34). The cubic comparison data are (39), with $g = 3d/4$ and comparison gap g as in Proposition 6.8.

Every admissible input falls under (a) or (b); in particular equal capacities with $k \le 9$, or with $\rho\sqrt{k} \le \frac{3}{4}$, are treated by (b). (For $k = 9$, the formulas of (a) would give $a = t_0/3 = h$, violating $a < h$.)

In both cases put $d_n = \frac{1}{2}(h + \frac{1}{3})$, let $f_i = f_i^{\mathrm{cap}}$ be the maps of Remark 10.30, built by the handle construction of §10.9, put $\bar\rho_i = (\rho_i + a_i)/2$ and $\hat\rho_i = \bar\rho_i + (a_i - \bar\rho_i)/4$, and let the open domain be $B^\circ(\hat\rho_i)$. Let $\rho_i^+ < \rho_i^{\mathrm{dom}}$ be the recorded capacities of Remark 13.24. Put $\mathcal{K} = \bigcup_i f_i(B(\rho_i^+))$ and $m = \frac{1}{2}\min_{z \in \mathcal{K}}(1 - \pi|z|^2)$.

Theorem 13.23 (Packing in dimension four)**.** [Lean: needed consequence verified.]

Let $k \geq 1$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$. Then there are explicit numbers $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ and smooth maps $f_i \colon B^\circ(\rho_i^{\mathrm{dom}}) \to \mathbb{C}^2$ with symplectic derivative, injective on $B(\rho_i^+)$, such that the sets $f_i(B(\rho_i^+))$ are pairwise disjoint and contained in $B(1-\mu_{\mathrm{ball}})$ for some $\mu_{\mathrm{ball}} \in (0,1)$. More precisely, for the data of Construction 13.22:

- *(i) each f_i is a smooth symplectic embedding of $B^\circ(\hat\rho_i)$ into $(B^\circ(1), \omega_0)$, and $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}} \leq \hat\rho_i$;*
- *(ii) the sets $f_i(B^\circ(\hat\rho_i))$, $1 \leq i \leq k$, are pairwise disjoint;*
- *(iii) $m > 0$ and $f_i(B(\rho_i^+)) \subset B(1-m)$ for every i;*
- *(iv) the comparison isotopy q, the Moser map Φ, the handle map $\mathcal{D}$ and the transfer $\mathcal{T}$ are the same for all i; only the layer map E_i depends on i.*

Proof. In both cases the data $(k, \rho_i, a_i, h, d_n, g, K, q)$ are valid cubic data (Definition 10.13): in case (a) by Proposition M3.8 and Remark 13.25, in case (b) by Proposition 6.8; moreover $h < d_n = \frac{1}{2}(h + \frac{1}{3}) < \frac{1}{3}$. The maps are $f_i = \mathcal{T} \circ f_i^N$ with f_i^N of Proposition 10.29 and $\mathcal{T}$ of Theorem 13.21 (Remark 10.30).

(i) By Proposition 10.29(a), f_i^N is smooth on $B^\circ(\hat\rho_i)$ with values in $\{p < h\} \setminus \widetilde{\mathcal{F}}$, the domain of $\mathcal{T}$, and $(f_i^N)^*\Omega_0 = \omega_0$; by Theorem 13.21, f_i is smooth with values in $B^\circ(1)$ and $f_i^*\omega_0 = \omega_0$. If $f_i(X) = f_i(X')$, Theorem 13.21 gives $f_i^N(X) = T f_i^N(X')$ for a deck map T, and Proposition 10.29(c) gives $X = X'$. A symplectic map is an immersion, and an injective immersion between manifolds of equal dimension is an open embedding; so f_i is a symplectic embedding of $B^\circ(\hat\rho_i)$. The inequalities for the recorded capacities are in Remark 13.24.

(ii) If $f_i(X) = f_j(X')$, the same argument with Proposition 10.29(c) gives $i = j$ and $X = X'$.

(iii) $\mathcal{K}$ is a finite union of continuous images of compact balls, hence compact, and lies in $B^\circ(1)$; so the continuous positive function $1 - \pi|z|^2$ has a positive minimum on it, $m > 0$, and $1 - \pi|z|^2 \geq 2m > m$ on $\mathcal{K}$.

(iv) The factors q (in $\mathcal{D}$), $\mathcal{D}$, Φ, $\chi_{\mathcal{A}}$ and $\mathcal{T}$ do not depend on i.

The first statement follows from (i)–(iii) with $\mu_{\mathrm{ball}} = m < 1$, restricting f_i to $B^\circ(\rho_i^{\mathrm{dom}})$. □

Remark 13.24 (Recorded capacities). The proof gives embeddings of the open balls $B^\circ(\hat\rho_i)$ with pairwise disjoint images, so Theorem 13.23 holds for every choice of $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}} \leq \hat\rho_i$. The choice enters only the radii of the common isotopy (Definition M6.4) and hence the Hamiltonian. The construction records $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\bar\rho_i, \hat\rho_i)$ in case (b), as in (41), and $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\rho r_3^2, \rho r_4^2)$ in case (a), where $r_4 = \sqrt{\bar\rho/\rho}$ and $r_3 = (1+r_4)/2$. In case (a), $1 < r_3 < r_4$ gives $\rho < \rho r_3^2 < \rho r_4^2 = \bar\rho < \hat\rho$. For target capacity C and given capacity $c = C\rho$ the physical map $u \mapsto \sqrt{C}\, f_i(u/\sqrt{C})$ has the capacities $C\rho_i^+$ and $C\rho_i^{\mathrm{dom}}$, that is, cr_3^2 and cr_4^2 in case (a).

Remark 13.25 (The equal-capacity branch). In case (a) of Construction 13.22 let $\eta_\star = 1 - \rho\sqrt{k} \in (0, \frac{1}{4})$ be the volume defect. Then

$$t_0 = 1 - \frac{\eta_\star}{2}, \quad a = \frac{t_0}{\sqrt{k}}, \quad h = \frac{t_0}{3}, \quad d_n = \frac{1}{3} - \frac{\eta_\star}{12}, \qquad a - \rho = \frac{\eta_\star}{2\sqrt{k}}, \quad d_n - h = \frac{\eta_\star}{12}, \quad 1 - 3d_n = \frac{\eta_\star}{4},$$

the comparison function is $K(p) = (3\sqrt{k} - 9)p$, and the comparison gap of Proposition M3.8 is $\frac{31}{32} g$ with $g = 3(1 - t_0) = \frac{3\eta_\star}{2}$. With the isotopy of Theorem 5.34 and the handle construction of §10.9, the maps of this branch are

$$f_i = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i \qquad (1 \leq i \leq k).$$

For the input of Corollary F, $\rho = \sqrt{(1 - \varepsilon/2)/k}$ and $\eta_\star = 1 - \sqrt{1 - \varepsilon/2} \in (\varepsilon/4, \varepsilon/2)$; case (a) is selected exactly when $k \geq 10$ and $\rho\sqrt{k} > 3/4$, and case (b) covers the other inputs. For unit balls

(capacity π) the target capacity is $C = \pi/\rho$, and the physical rescaling of Definition M6.1 makes the target the open Euclidean ball of radius $R = \rho^{-1/2}$, so $k/R^4 = k\rho^2 = (1-\eta_\star)^2$. For the data of Theorem A, $R^4 = k/(1-\varepsilon/4)$, so $\rho\sqrt{k} = \sqrt{1-\varepsilon/4} > \frac{3}{4}$ and $\eta_\star = 1 - \sqrt{1-\varepsilon/4} \in (\varepsilon/8, \varepsilon/4)$; case (a) applies for every $\varepsilon \in (0,1)$ when $k \geq 10$. The strip constants are those of Remark 10.30 with $\delta = (a-\rho)/2$.

The defect $1-\rho\sqrt{k} = \eta_\star$ splits as $(1-t_0) + \sqrt{k}\,(a-\rho) = \frac{\eta_\star}{2} + \frac{\eta_\star}{2}$. The fibre part $1-t_0 = 1-3h$ pays for $h < \frac{1}{3}$, for the comparison gap g and for the margins $d_n - h$ and $1-3d_n$ of the transfer; the strip part $\sqrt{k}\,(a-\rho)$ pays for the enlargement from ρ to $\bar\rho$ and for the excess of the strip map. The transfer, the ball chart and the dilation are symplectic embeddings and lose no volume beyond these margins.

Remark 13.26 (Obstructions). The condition $3\rho_i < 1$ of Theorem 13.23 is not necessary (one ball of capacity $\rho < 1$ embeds by inclusion), but it cannot be replaced by the volume and two-ball inequalities: five balls of capacity $\frac{11}{25} > \frac{1}{3}$ and five of capacity $\frac{1}{100}$ satisfy $\sum_i \rho_i^2 = \frac{1937}{2000} < 1$ and $\rho_i + \rho_j \leq \frac{22}{25} < 1$, yet by the disjoint-union property of ECH capacities [Hut11] the fifth ECH capacity of their union is at least $5 \cdot \frac{11}{25} = \frac{11}{5}$, while that of $B^\circ(1)$ is 2; so they do not embed.

14. Dimension at least six: capacities below one half

Let $n \geq 3$, $m = n-1$, and $\rho_i = c_i/C$. We use a fixed affine curve and a product with m complex fibre planes. The horizontal curvature has coefficient $1-2|p|$, so the fibre simplex must stop below $|p| = 1/2$. An explicit compact exhaustion of the curve has area tending to 2^m; the integrated profiles tend to $1/n!$. The finite truncation loses a controlled amount of area, which is subtracted from the comparison profile. The transfer is a single map from the affine model into $B^\circ(1)$: a radial injection, horizontal transport, an inverse Moser return and an affine ball chart. This realises the normal-transfer mechanism of [Ope26] in affine coordinates.

14.1. **The affine curve and the finite model.**

Definition 14.1 (The diagonal affine curve)**.** Let $z = (t, y_1, \dots, y_m) \in \mathbb{C}^{m+1}$ and put

$$F_j(z) = y_j^2 - t^2 + j, \qquad \Sigma = \{z : F_1(z) = \cdots = F_m(z) = 0\}, \qquad D(z) = 1 + |z|^2.$$

On Σ set $\lambda = d^c \log D$ and $\sigma = d\lambda$, using $d^c = i(\bar\partial - \partial)/(4\pi)$. For an integer $\ell \geq 0$ put

$$R_\ell = 2 + \sum_{j=1}^{m} j + \ell, \qquad \Sigma_\ell = \{z \in \Sigma : |t| \leq R_\ell\}, \qquad A_\ell = \int_{\Sigma_\ell} \sigma.$$

Lemma 14.2 (Regularity and exhaustion)**.** *The affine locus Σ is a smooth complex curve, σ is positive, and $t : \Sigma \to \mathbb{C}$ is proper. Each Σ_ℓ is a compact connected surface with smooth boundary and connected nonempty interior. The fixed annular supports of Proposition 15.4 lie in its interior, and*

$$0 \leq 2^m - A_\ell \leq \frac{2^m(m+2)}{R_\ell^2} \leq \frac{2^m(m+2)}{\ell+2}.$$

Proof. At most one y_j vanishes. If none vanishes, the y-columns of dF are invertible. If $y_i = 0$, then $t^2 = i \neq 0$ and the columns $t, (y_j)_{j\neq i}$ are invertible. The implicit coordinate formulas are given in Lemma 15.2. Positivity follows by restricting the affine Fubini–Study form. The estimate $|y_j|^2 \leq |t|^2 + j$ proves properness. All critical values of t lie at $\pm\sqrt{j}$, strictly inside $|t| < R_\ell$, so the boundary is smooth. Loops about these branch values independently reverse the root signs and connect the sheets over the disc; thus the interior is connected. The area estimate, including its limiting constant 2^m, is proved from the explicit affine ends in Lemma 15.5. □

Definition 14.3 (The finite quadric model)**.** For $0 < h < 1/2$ take the product $E_\ell = \Sigma_\ell \times \mathbb{C}^m$, with $p_j = \pi|U_j|^2$, $|p| = \sum_j p_j$,

$$\Delta_h = \{p \geq 0 : |p| \leq h\}, \qquad \Omega_0 = \omega_V + d[(1-2|p|)\lambda], \qquad H_\ell(p) = A_\ell(1-2|p|).$$

The differential includes the fibre variables. The marked set is empty. We use $H_{\mathrm{b}}(p) = 2^m(1-2|p|)$ for the limiting profile, which is a numerical comparison budget rather than the area of a finite base. All forms and maps are defined on the surrounding affine product $\Sigma \times \mathbb{C}^m$.

Lemma 14.4 (The bordered toric model)**.** *For $0 < h < 1/2$, this is a bordered toric model of rank m. Its horizontal form at fixed moment is $(1-2|p|)\sigma > 0$, its base area is exactly H_ℓ, and its fixed handle data are those of Proposition 15.4. The finite primitive operator is applied to forms supported away from the outer collar.*

Proof. The form is closed and torus invariant, and restricts to ω_V on each fibre. In product coordinates it is

$$\Omega_0 = \omega_V + (1-2|p|)\sigma - 2\sum_j dp_j \wedge \lambda.$$

Its horizontal coefficient is positive on a neighbourhood of Δ_h. The compact base and its boundary have the properties of Lemma 14.2. The annuli, closed dual, collar cutoff and finite coordinate primitive are supplied by Proposition 15.4; the mass budget used for packing is checked below. □

Lemma 14.5 (Simplex integrals)**.** *Let $m \geq 1$ be an integer and $\Delta_h = \{p \in \mathbb{R}^m_{\geq 0} : |p| \leq h\}$.*

(a) For $0 < h < \infty$ and continuous $f\colon [0,h] \to \mathbb{R}$, $\int_{\Delta_h} f(|p|)\,dp = \frac{1}{(m-1)!}\int_0^h f(t)\,t^{m-1}\,dt$.

(b) For $0 < h \leq \frac{1}{2}$,

$$\int_{\Delta_h} 2^m(1-2|p|)\,dp = J_m(h) := \frac{2^m h^m}{m!}\Big[1 - \frac{2mh}{m+1}\Big], \qquad J_m(\tfrac{1}{2}) = \frac{1}{(m+1)!}.$$

(c) For $h = \frac{1}{2} - \delta$ with $0 < \delta < \frac{1}{2}$, $\frac{1}{(m+1)!} - J_m(h) \leq \frac{2\delta^2}{(m-1)!}$.

Proof. (a) The simplex $\{p \in \mathbb{R}^m_{\geq 0} : |p| \leq t\}$ has volume $t^m/m!$, so the distribution of $|p|$ on Δ_h has density $t^{m-1}/(m-1)!$. (b) With $f(t) = 2^m(1-2t)$, (a) gives $\frac{2^m}{(m-1)!}\big(\frac{h^m}{m} - \frac{2h^{m+1}}{m+1}\big) = J_m(h)$, and $J_m(\frac{1}{2}) = \frac{1}{m!}(1 - \frac{m}{m+1}) = \frac{1}{(m+1)!}$. (c) Using $t^{m-1} \leq 2^{-(m-1)}$ on $[h, \frac{1}{2}]$,

$$\frac{1}{(m+1)!} - J_m(h) = \frac{2^m}{(m-1)!}\int_h^{1/2}(1-2t)t^{m-1}\,dt \leq \frac{2}{(m-1)!}\int_h^{1/2}(1-2t)\,dt = \frac{2\delta^2}{(m-1)!}.$$ □

For the finite model the integrated profile is $(A_\ell/2^m)J_m(h)$. Thus the limits $\ell \to \infty$ and $h \uparrow 1/2$ recover $1/n!$. A finite packing uses one explicit ℓ and the corresponding finite area estimate.

14.2. **The affine transfer.** Put $u(z,W) = |W|^2/D(z)^2$ and use the Kähler potential

$$\Phi_b(z,W) = \log D(z) + b\log(1+u(z,W)), \qquad 0 < b < 1/2.$$

The graph family is $X_T = \{F(z) = TW\}$ in $\mathbb{C}^{m+1} \times \mathbb{C}^m$. At $T = 0$ its base is Σ; for $T > 0$ it is the graph $W = F(z)/T$. The injection is

$$\iota_b(z,U) = \left(z, \frac{D(z)\sqrt{\pi}\,U}{\sqrt{b-|p|}}\right).$$

The horizontal field is the unique solution of the linear equations

$$dF_z\dot{z} - T\dot{W} = W, \qquad dd^c\Phi_b\big((\dot{z},\dot{W}),(v,w)\big) = 0 \quad \text{if } dF_z v - Tw = 0.$$

Let τ be its flow from $T=0$ to $T=1$. On $\mathbb{C}^{m+1}$ put

$$\omega_s = dd^c\left[\log D + sb\log\left(1+\frac{|F|^2}{D^2}\right)\right], \qquad \iota_{Y_s}\omega_s = -b\,d^c\log\left(1+\frac{|F|^2}{D^2}\right),$$

and let ψ_1^{-1} be its flow from $s=1$ back to zero. Finally define $\mathrm{ch}(z) = z/\sqrt{\pi D(z)}$.

The transfer $\mathcal{T}_b = \mathrm{ch}\circ\psi_1^{-1}\circ\mathrm{pr}\circ\tau\circ\iota_b$ is a symplectic embedding of the tube $\{|p|<b\}$ into $B^\circ(1)$ (Theorem 15.12 in §15.4).

14.3. **Parameters and packing.**

Lemma 14.6 (Parameters for the quadric branch)**.** *Let $n\ge 3$ and let $a_1,\dots,a_k$ be real numbers with $0<a_i<\frac12$ and $\sum_i a_i^n<1$. Put $a_{\max}=\max_i a_i$ and*

$$\varsigma = 1-\sum_i a_i^n, \quad \delta = \min\Big(\frac{1/2-a_{\max}}{2},\ \frac{\varsigma}{4n(n-1)}\Big),$$

$$h = \tfrac12-\delta, \quad b=\frac{h+1/2}{2}, \quad \tau_\Delta = \frac{a_{\max}+h}{2h}.$$

These numbers are real (rational for rational a), and:

- *(i) $a_{\max}<h<b<\frac12$, $\tau_\Delta<1$, and $\{|p|\le a_i\}\subset\tau_\Delta\Delta_h$ for every i;*
- *(ii) $P_{\mathrm b}(p)=H_{\mathrm b}(p)-\sum_i(a_i-|p|)_+$ is a continuous, concave piecewise affine function (rational when a is rational) on Δ_h;*
- *(iii) $\int_{\Delta_h}P_{\mathrm b}\,dp > 7\varsigma/(8\,n!)$, and the mean μ of $P_{\mathrm b}$ on Δ_h satisfies $\mu>7\varsigma/(8\,n!)$;*
- *(iv) $P_{\mathrm b}\ge P_{\min}:=2^m(1-2h)=2^{m+1}\delta>0$ on $\Delta_h\setminus\tau_\Delta\Delta_h$;*
- *(v) the number $d=\frac12\min\big(\varsigma/(2\,n!),\,2^{m+1}\delta\big)$ satisfies $0<d<\min(\mu,P_{\min})$.*

The factor $\frac78$ leaves room for the profile-budget defect $\frac{1}{n!}-J_m(h)<\varsigma/(8\,n!)$ of the proof of (iii). If $0<\rho_i<\frac12$ and $\sum_i\rho_i^n<1$, rational numbers $a_i>\rho_i$ with $a_i<\frac12$ and $\sum_i a_i^n<1$ exist, and any enumeration of $\mathbb{Q}^k$ reaches one after finitely many steps.

Proof. (i) $\delta>0$, so $h<b<\frac12$; and $h\ge\frac12-\frac{1/2-a_{\max}}{2}=\frac{1/2+a_{\max}}{2}>a_{\max}$. Hence $\tau_\Delta h=\frac{a_{\max}+h}{2}\in(a_{\max},h)$, which gives $\tau_\Delta<1$ and $\{|p|\le a_i\}\subset\{|p|\le\tau_\Delta h\}$. (ii) $H_{\mathrm b}$ is affine with integer coefficients and each $(a_i-|p|)_+$ is convex and piecewise affine (rational for rational a_i). (iii) Since $a_i<h$, the support of $(a_i-|p|)_+$ in $\mathbb{R}^m_{\ge0}$ lies in Δ_h, so by Lemma M2.2 its integral over Δ_h is $a_i^n/n!$. With Lemma 14.5(b),(c) for $m=n-1$,

$$\int_{\Delta_h}P_{\mathrm b}\,dp = J_m(h)-\frac{1-\varsigma}{n!} = \frac{\varsigma}{n!}-\Big(\frac{1}{n!}-J_m(h)\Big)\ge\frac{\varsigma}{n!}-\frac{2\delta^2}{(n-2)!}.$$

As $\frac{1}{(n-2)!}=\frac{n(n-1)}{n!}$ and $\delta\le\varsigma/(4n(n-1))$, $\frac{2\delta^2}{(n-2)!}\le\frac{\varsigma^2}{8n(n-1)\,n!}<\frac{\varsigma}{8\,n!}$, because $0<\varsigma<1$ and $n(n-1)\ge6$. This gives the bound on the integral. Since $\mathrm{vol}\,\Delta_h=h^m/m!<1$ and the integral is positive, μ exceeds the integral. (iv) On $\Delta_h\setminus\tau_\Delta\Delta_h$ we have $|p|>\tau_\Delta h>a_{\max}$, so all caps vanish and $P_{\mathrm b}=H_{\mathrm b}\ge2^m(1-2h)$. (v) follows from (iii) and (iv).

For the last assertion, the set $\{a\in\mathbb{R}^k:\ a_i>\rho_i,\ a_i<\frac12,\ \sum_i a_i^n<1\}$ is open. It is nonempty, since it contains $\rho+t(1,\dots,1)$ for all sufficiently small $t>0$. Hence it contains rational points. A complete enumeration of $\mathbb{Q}^k$ therefore contains a valid tuple. □

Theorem 14.7 (Packing by small balls in dimension $2n\ge6$)**.** *Let $n\ge3$ and $0<\rho_1,\dots,\rho_k<\frac12$ with $\sum_i\rho_i^n<1$, and let $a_1,\dots,a_k$ be real numbers with $\rho_i<a_i<\frac12$ and $\sum_i a_i^n<1$. Put $\rho_i^+=(2\rho_i+a_i)/3$ and $\rho_i^{\mathrm{dom}}=(\rho_i+2a_i)/3$. The comparator construction with real coefficients, with closed source capacities ρ_i^{dom} (Construction 12.4), defines from these data smooth symplectic*

embeddings f_i of open neighbourhoods of the closed balls $B(\rho_i^{\mathrm{dom}}) \subset \mathbb{C}^n$ into $B^\circ(1)$ and a rational $\zeta > 0$, such that the sets $f_i(B(\rho_i^{\mathrm{dom}}))$ are pairwise disjoint and contained in $B(1-\zeta)$.

Proof of Theorem 14.7. Parameters and comparison. Let $\delta, h, b, \tau_\Delta, d$ be given by Lemma 14.6 for a, put $P = P_{\mathrm{b}}$. By parts (i)–(v) of that lemma, Δ_h, τ_Δ and P satisfy the hypotheses of Proposition M3.3 with $\beta = P_{\min}$ and with this d, and Theorem M3.5 provides a comparison datum (K, q) for $(\Delta_h, P_{\mathrm{b}})$ with gap $\gamma = 11d/16 > 0$.

Truncation and packing. Put $m = n - 1$ and

$$l = \left\lceil \frac{2^{m+6}(m+2)}{d} \right\rceil, \qquad R_l = 2 + \frac{m(m+1)}{2} + l, \qquad \Sigma_l = \{|t| \le R_l\} \subset \Sigma.$$

By Lemma 15.5, its exact profile is $H_l(p) = A_l(1 - 2|p|)$, with $0 \le H_{\mathrm{b}} - H_l \le d/64$. Lower the comparison profile to $P_l = P_{\mathrm{b}} - 11d/32$. The same comparison datum has gap $\gamma_l = 11d/32$, because replacing P by $P - c$ reduces its gap by c. Also

$$P_l \le H_l - \sum_i (a_i - |p|)_+.$$

The handle data are those of Construction 10.4 on Σ_l: by Lemma 10.5 the inner boxes of the correction atlas cover $\{|t| \le R_l - 1\}$, which contains the handle and the support of $1 - \zeta_\partial$, and the correction region is all of Σ_l with $\kappa_- = 1 - 2h$, as required in §12. Set $\gamma_* = 11d/64$, $\kappa_- = 1 - 2h$, and

$$c_{\mathrm{bg}} = \min\left(\frac{\gamma_*}{32\, 2^m}, \frac{\kappa_-}{2}\right), \qquad \zeta_\partial(t) = S\left(\frac{|t|^2 - (R_l - 2)^2}{(R_l - 1)^2 - (R_l - 2)^2}\right).$$

The background form is $\big[\zeta_\partial(1 - 2|p|) + (1 - \zeta_\partial)c_{\mathrm{bg}}\big]\sigma$. It agrees with the original curvature near the boundary, and it is positive, since $c_{\mathrm{bg}} \le \kappa_-/2$. The removed density $(1 - \zeta_\partial)(1 - 2|p| - c_{\mathrm{bg}})\sigma$ has compact support in $\{|t| \le R_l - 1\}$; its integral is the handle mass. The same end estimate as in Lemma 15.5, at radius $R_l - 2$, gives

$$\int_{|t| > R_l - 2} \sigma \le \frac{2^m(m+2)}{(R_l - 2)^2} \le \frac{d}{65536} < \frac{\gamma_*}{32}.$$

Here $d < 1$, $m \ge 2$ and $R_l - 2 \ge l \ge 64\, 2^m(m+2)/d$. Since $c_{\mathrm{bg}} \int_{\Sigma_l} \sigma \le \gamma_*/32$, the whole background mass is less than $\gamma_*/16$. Also $\min H_l > d$, so $\epsilon_0 = \gamma_*/16$ in (M11). This proves the collar and background budgets used in the handle construction. Put $\rho_i^+ = (2\rho_i + a_i)/3$ and $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$. Apply the layer construction with the further interior source capacity $u_i = (3a_i + \rho_i^{\mathrm{dom}})/4 < a_i$. Theorem M2.6, with the exact truncated profile and its positive remaining gap, gives smooth symplectic embeddings ψ_i on a neighbourhood of $B(\rho_i^{\mathrm{dom}})$ into $\operatorname{int} V_{\Delta_h}(E|_{\Sigma_l})$, with pairwise disjoint images. The images lie in the interior of the finite affine model.

Transfer. By Theorem 15.12, $\mathcal{T}_b$ is a symplectic embedding of a neighbourhood of $V_{\Delta_h}(E_l)$ into $B^\circ(1)$. Proposition M2.7 gives $f_i = \mathcal{T}_b \circ \psi_i$, smooth and symplectic on $B^\circ(\rho_i^{\mathrm{dom}})$, with pairwise disjoint images of $B(\rho_i^{\mathrm{dom}})$, since $\mathcal{T}_b$ is one injective map.

Margin. Let $\mathcal{K} = \bigcup_i \psi_i(B(\rho_i^{\mathrm{dom}}))$ and $m_{\mathcal{K}} = \min_{\mathcal{K}}(1 - \pi|\mathcal{T}_b|^2) > 0$. Choose the positive rational $\zeta = (2 + \lceil 1/m_{\mathcal{K}} \rceil)^{-1} < m_{\mathcal{K}}$. All images then lie in $B(1 - \zeta)$. □

If some normalised capacity is at least $\frac{1}{2}$, the cap $\{|p| \le a_i\}$ does not fit in Δ_h for any $h < \frac{1}{2}$. This case is treated in Section 16.

15. The affine quadric transfer

We use the affine locus $\Sigma \subset \mathbb{C}^{m+1}$, the forms λ, σ and the exhaustion Σ_ℓ of Definition 14.1, and write $F(z) = (y_j^2 - t^2 + j)_{j=1}^m$. In §15.1 $m = n - 1 \geq 2$; the transfer of §§15.2–15.4 is valid for every $m \geq 1$.

15.1. **Sheets and explicit coordinates.**

Lemma 15.1 (The affine sheets)**.** *The projection $t : \Sigma \to \mathbb{C}$ has 2^m points in every fibre away from $\{\pm\sqrt{j} : 1 \leq j \leq m\}$. Over $t = \pm\sqrt{i}$ it has 2^{m-1} points, with simple branching in coordinate y_i. Outside a sufficiently large disc its 2^m ends are indexed by signs $\epsilon_j \in \{\pm 1\}$ and have coordinates*

$$t = s^{-1}, \qquad y_j = \epsilon_j s^{-1}\sqrt{1 - js^2}, \qquad 0 < |s| < 1/R_0,$$

where $R_0 = 2 + \sum_j j$. Each root is the branch equal to one at zero; the inverse is $s = 1/t$.

Proof. The independent root signs count the fibres. At a branch value only one root vanishes, and the local equation is $y_i^2 = t^2 - i$, with derivative $2t \neq 0$ in t. The formula at the ends follows by factoring t^2 from every radicand. On $|s| < 1/R_0$ all perturbations have modulus less than one, so the roots above and their finite signs are unambiguous and account for every point. □

Lemma 15.2 (Explicit coordinate boxes)**.** *Let $c = (t_c, y_c) \in \Sigma$.*

(i) If every $y_{c,j} \neq 0$, a coordinate near c is $u = t$, with inverse

$$t = u, \qquad y_j = y_{c,j}\sqrt{1 + (u^2 - t_c^2)/y_{c,j}^2}.$$

(ii) If $y_{c,i} = 0$, a coordinate near c is $v = y_i$, with inverse

$$t = t_c\sqrt{1 + v^2/t_c^2}, \qquad y_i = v, \qquad y_j = y_{c,j}\sqrt{1 + v^2/y_{c,j}^2} \quad (j \neq i).$$

Here $t_c^2 = i$, and $y_{c,j}^2 = i - j \neq 0$ for $j \neq i$.

The square roots in (i)–(ii) are the branches equal to one at zero. Restricting a chart to a coordinate square on which every perturbation of its radicand has modulus less than one gives an injective holomorphic chart. The inverse coordinates are ambient coordinate projections. Transition functions are compositions of these projections with the roots above.

Proof. Squaring the coordinates verifies all equations. In (i), at the centre the ratios $y_j/y_{c,j}$ equal one; in (ii) the same holds for t/t_c and $y_j/y_{c,j}$, $j \neq i$. Requiring positive real part selects these branches uniquely. The indicated coordinate projection composed with the inverse chart is the identity, while the branch conditions give the reverse identity. Holomorphicity follows from the binomial series on the prescribed radicand domains. □

Lemma 15.3 (A finite rational family on a compact projection domain)**.** *Let $R \geq 0$ be rational and put*

$$\eta = \frac{1}{64}, \quad r = \frac{1}{262144(R+1)}, \quad h = \frac{r}{32}, \quad N = \lceil R/h \rceil + 2.$$

Consider the grid $q = h(a + ib)$ with $-N \leq a, b \leq N$. Retain the ordinary centres

$$c(q, \epsilon) = \big(q, (\epsilon_j\sqrt{q^2 - j})_{j=1}^m\big)$$

that pass the rational tests

$$|q|^2 < (R + \tfrac{1}{2})^2, \qquad |q^2 - j|^2 > (\eta^2/2)^2 \quad (1 \leq j \leq m).$$

Use all signs, ordinary coordinate squares of outer half-width r and inner half-width $r/4$, and the branch charts centred at every $t_c = \pm\sqrt{i}$, $y_{c,i} = 0$, $y_{c,j} = \pm\sqrt{i - j}$, with outer half-width $1/8$ and

inner half-width $1/32$. *The inner open squares cover* $\{|t| \le R\} \subset \Sigma$. *Every ordinary closed outer square lies in* $\{|t| < R+1\}$. *The same holds for the branch squares provided* $i + 1/16 < (R+1)^2$ *for all* i. *All index tests are comparisons of rational polynomials in* $\operatorname{Re} q$, $\operatorname{Im} q$.

Proof. The chosen constants satisfy

$$2r\big(2(R+1)+2r\big) < \eta^2/2, \qquad 4r(2R+4r) < \eta^2, \qquad 2h \le r/8, \qquad R + h < Nh.$$

If every $|y_j| \ge \eta$, choose a grid point with $|q - t| < 2h \le r/8$. Then

$$|q^2 - t^2| \le |q - t|(|q| + |t|) < \eta^2/2,$$

so $|q^2 - j| > \eta^2/2$ for each j, and the centre passes the tests. Among its finite root signs there is one for which all ratios to the coordinates of the point have positive real part. The ordinary chart formula then puts the point in the inner square. The first displayed inequality bounds every radicand perturbation on the outer square strictly below one; also $|q| + 2r < R + 1$. If some $|y_i| < \eta = (1/8)/8$, the separated parameters give $|t|^2$ close to i and $|y_j|^2$ close to $|i - j|$. The branch formula with the corresponding finite root signs applies. Its perturbations have modulus at most $4(1/8)^2 < 1$, and the point is in the inner square. On its closed outer square, $|t|^2 = |i + y_i^2| \le i + 1/16 < (R+1)^2$. □

Proposition 15.4 (Fixed handle data)**.** *On each compact truncation containing the fixed supports, the quadric model has the annular data below. Its primitive operator acts on zero-mass forms with compact support in the affine interior. The annular charts and closed dual form can be chosen independently of* h *by the following root formulas. Put* $P(z) = (z + z^{-1})/2$ *and* $M(z) = (z - z^{-1})/2$. *Two annuli, initially defined near* $|z| = 1$, *are*

$$A(z) = \big(\sqrt{1 + P(z)^2},\ P(z),\ M(z),\ (i\sqrt{j - 1 - P(z)^2})_{j=3}^{m}\big),$$
$$C(z) = \big(P(z),\ M(z),\ (i\sqrt{j - P(z)^2})_{j=2}^{m}\big).$$

Their inverses are respectively $z = y_1 + y_2$ *and* $z = t + y_1$. *The transverse cylinder is* $e(s,\theta) = C(\exp(s + 2\pi i\theta))$. *Fix* $\varepsilon = 1/8$ *and* $w = 1/4$, *and let*

$$T_\varepsilon(z) = S\left(\frac{|z|^2 - (1-\varepsilon)^2}{(1+\varepsilon)^2 - (1-\varepsilon)^2}\right), \qquad \beta_0 = A_* dT_\varepsilon,$$

with zero extension outside A. *Then* β_0 *is smooth, closed, compactly supported and has period one on the core of* C. *With* $b(\theta) = 2S'(2\theta - 1/2)$ *on* $[0,1]$, *extended periodically, the fixed smooth circle clock supported in* $[1/4, 3/4]$ *with integral one, write*

$$e^*\beta_0 = A_0(s,\theta)\, ds + B_0(s,\theta)\, d\theta.$$

The periodic function

$$F(s,\theta) = \int_0^s A_0(v,0)\, dv + \int_0^\theta (B_0(s,v) - b(v))\, dv$$

satisfies $dF = e^*\beta_0 - b(\theta)d\theta$. *With*

$$\chi(s) = S\big((7w/8 - s)4/w\big)S\big((7w/8 + s)4/w\big), \qquad \beta = \beta_0 - d\big(e_*(\chi F)\big),$$

the function $e_*(\chi F)$ *is extended by zero. The form* β *equals* $b(\theta)d\theta$ *on* $|s| < 5w/8$. *The annular handle has width* $w/2 = 1/8$. *The normalization and layer paths use the smaller working half-width* $b_H = w/4 = 1/16$ *and the radial clock* $S((s + b_H)/(2b_H))$; $w = 1/4$ *is the outer cylinder width.*

In the product fibre coordinates,

$$\Omega_0 = \omega_V + d\Gamma_{\mathcal{A}}, \qquad \Gamma_{\mathcal{A}} = (2|p| - 1)\vartheta, \qquad \vartheta = -d^c \log(1 + |t|^2 + \textstyle\sum_j |y_j|^2).$$

The auxiliary area form and the zero-mass primitive operator are obtained from the fixed finite coordinate families of Lemma 15.3, the chart bump partition and local integral primitive of Construction 10.1; their annular correction is the same two-segment formula for F.

Proof. The identity $P^2 - M^2 = 1$ verifies the two annulus formulas. On $|z| = 1$, $P(z)$ is real in $[-1, 1]$ and $M(z)$ is purely imaginary. Every remaining square-root radicand is positive, including $1 + P^2$ and $j - 1 - P^2$ for $j \geq 3$. Thus the formulas extend to annuli of positive width. The displayed inverses prove injectivity there. The two core circles meet once: y_1 is real on the first and imaginary on the second, hence zero at an intersection; the positive choice of t then gives $t = 1$, $y_2 = i$. The radial coordinate crosses the transition in T_ε with positive orientation at this point. Integrating its derivative gives one, so $\int_{C(|z|=1)} \beta_0 = 1$. The derivative of T_ε vanishes near both boundary circles; extension by zero is smooth and closed. Closedness gives $\partial_\theta A_0 = \partial_s B_0$. Differentiating the two integrals proves the formula for dF. The period-one identity makes F periodic. The cutoff equals one on $|s| \leq 5w/8$ and vanishes for $|s| \geq 7w/8$, which proves the assertions about β and smooth zero extension. All these supports are compact in the affine curve.

The fixed widths are admissible uniformly in the number of quadrics. On $3/4 \leq |z| \leq 5/4$ one has $|P(z)| \leq 4/3$ and $|\operatorname{Im} P(z)| \leq 7/24$. Hence $\operatorname{Re}(1 + P(z)^2) > 0$ and $\operatorname{Re}(j - 1 - P(z)^2) > 0$ for $j \geq 3$, so A is defined on the whole closed annulus $\frac{3}{4} \leq |z| \leq \frac{5}{4}$, which contains the support $\frac{7}{8} \leq |z| \leq \frac{9}{8}$ of dT_ε. For $|s| \leq 1/2$, the inequality $e^{1/2} < 7/4$ gives $|P(e^{s+2\pi i\vartheta})| \leq 11/8 < \sqrt{2}$; thus every $j - P^2$, $j \geq 2$, has positive real part, and C is defined on that whole closed cylinder. The chosen widths leave a neighbourhood of the supports inside both coordinate domains. The finite chart partition and zero-mass local primitives give the remaining scalar handle data by the construction of Construction 10.1. The initial boxes cover the required compact support. Additional rational subboxes in the connected open affine base connect their overlap graph: an open cover of a connected space has finite overlap chains between any two nonempty members. Each nonempty overlap contains a rational subbox with compact closure in both charts. Enumerating these subboxes and their strict inclusion certificates therefore finds finite chains. The normalized chart bump weights give the partition; the chains give the remaining finite data of that construction. Finally $\vartheta = -\lambda$, so $(2|p| - 1)\vartheta = (1 - 2|p|)\lambda$ gives the stated horizontal primitive directly. □

Lemma 15.5 (Explicit truncation error)**.** *Put $R_l = 2 + \sum_{j=1}^{m} j + l$, $\Sigma_l = \{|t| \leq R_l\} \subset \Sigma$, and $A_l = \int_{\Sigma_l} \sigma$. Then Σ_l is a compact connected surface with smooth boundary, and*

$$0 \leq 2^m - A_l \leq \frac{2^m(m+2)}{R_l^2} \leq \frac{2^m(m+2)}{l+2}.$$

In particular, for $d > 0$ and

$$l = \left\lceil \frac{2^{m+6}(m+2)}{d} \right\rceil, \qquad H_l(p) = A_l(1 - 2|p|),$$

one has $0 \leq H_{\mathrm{b}}(p) - H_l(p) \leq d/64$ on Δ_h. The handle and dual supports of Proposition 15.4 lie in the interior of Σ_l.

Proof. The projection t is proper on Σ, since $|y_j|^2 = |t^2 - j| \leq |t|^2 + j$. Its critical points have some $y_j = 0$, i.e. $t^2 = j$, so the circle $|t| = R_l$ contains no critical value and Σ_l has smooth boundary. The interior is connected: analytic continuation around a small circle about $\sqrt{j}$ changes the sign of y_j and of no other coordinate, so the monodromy acts transitively on the 2^m sheets over a regular point of the disc. The handle supports lie in $|t| \leq \frac{5}{3}$ (proof of Proposition 15.4: $|P| \leq \frac{4}{3}$ on A and $|P| \leq \frac{11}{8}$ on C), inside Σ_0 since $R_l \geq 5$.

Let $R \geq R_0 = 2 + \frac{m(m+1)}{2}$. By Lemma 15.1, $\{|t| > R\}$ is the disjoint union of 2^m punctured discs $0 < |s| < 1/R$, on each of which $[1 : t : y] = [s : 1 : v]$ with $F(s) = (s, 1, v_1(s), \dots, v_m(s))$, $v_j(s) = \epsilon_j \sqrt{1 - js^2}$. Since $\lambda = d^c \log D$ and $\log D = \log |F(s)|^2 - \log |s|^2$ with $\log |s|^2$ harmonic, $\sigma = dd^c \log |F(s)|^2$ there, and its density with respect to Lebesgue measure in s is $\pi^{-1}(|F|^2|F'|^2 - |F^\dagger F'|^2)/|F|^4 \leq |F'|^2/(\pi|F|^2)$. For $|s| \leq 1/R$ we have $|v_j|^2 = |1 - js^2| \geq 1 - j/R^2 \geq \frac{1}{2}$ and $|v_j'|^2 = j^2|s|^2/|v_j|^2 \leq 2j^2/R^2$, and $\sum_j j^2 \leq R_0^2 \leq R^2$, so $|F'|^2 \leq 3$; with $|F| \geq 1$ the density is at most $3/\pi$, and each end contributes at most $3/R^2$. Hence $\int_{|t|>R} \sigma \leq 2^m \cdot 3/R^2 \leq 2^m(m+2)/R^2$.

On the other hand, by Stokes' theorem $A_{R'} = \int_{|t|=R'} \lambda$ for $\Sigma_{R'} = \{|t| \leq R'\}$. On each end, $\lambda = d^c \log |t|^2 + d^c \log |F(s)|^2 = \frac{d \arg t}{2\pi} + d^c \log |F(s)|^2$, and $|\partial \log |F|^2| = |F^\dagger F'|/|F|^2 = O(|s|)$, so the second term contributes $O(R'^{-2})$ on the circle of length $2\pi/R'$ in s. Thus $A_{R'} \to 2^m$, and since $\sigma > 0$, $2^m - A_R = \int_{|t|>R} \sigma \in [0, 2^m(m+2)/R^2]$. For $R = R_l \geq l + 2 \geq 2$ we get $R_l^2 \geq l + 2$, which is the second bound. The ceiling choice of l gives $2^m - A_l \leq d/64$, and $0 < 1 - 2|p| \leq 1$ on Δ_h gives the profile estimate. □

15.2. **The affine potentials.** Write $D = 1 + |z|^2$ and $u = |W|^2/D^2$. On $\mathbb{C}^{m+1} \times \mathbb{C}^m$ put

$$\Phi_b = \log D + b \log(1 + u), \qquad \Omega_b = dd^c \Phi_b.$$

Lemma 15.6 (Positivity and the graph forms)**.** *For $0 < b < 1/2$, Ω_b is a smooth Kähler form. For $T > 0$, its restriction to the graph $j_T(z) = (z, F(z)/T)$ is*

$$\omega_T = dd^c \left[\log D + b \log \left(1 + \frac{|F|^2}{T^2 D^2} \right) \right].$$

The path $\omega_s^{\mathrm{ret}} = dd^c[\log D + sb \log(1 + |F|^2/D^2)]$, $0 \leq s \leq 1$, is also Kähler.

Proof. The identity

$$\Phi_b = (1 - 2b) \log D + b \log(D^2 + |W|^2)$$

exhibits its Hessian as a positive combination of Fubini–Study Hessians. Indeed $D^2 + |W|^2$ is the squared norm of the vector consisting of the ordered quadratic monomials of $(1, z)$ and the coordinates W. Its constant entry is one; its linear z entries and its W entries separate tangent vectors. The first summand is positive on nonzero base vectors, and the second is positive on nonzero pure fibre vectors, so the sum is positive definite. The graph identity is pullback by the holomorphic map j_T. The same decomposition with sb proves positivity of the return path. □

Construction 15.7 (Radial injection)**.** For $0 < b < 1/2$, let

$$\iota_b(z, U) = \left(z, \frac{D(z)\sqrt{\pi}\, U}{\sqrt{b - |p|}} \right), \qquad \iota_b^{-1}(z, W) = \left(z, \sqrt{\frac{b}{\pi}} \frac{W}{\sqrt{D(z)^2 + |W|^2}} \right).$$

The first map has domain $|p| < b$ over Σ, and the second is defined on $\Sigma \times \mathbb{C}^m$.

Lemma 15.8 (The radial pullback)**.** *These maps are mutually inverse smooth maps, and $\iota_b^* \Omega_b = \Omega_0$ on the tube $|p| < b$. Moreover $u \circ \iota_b = |p|/(b - |p|)$.*

Proof. With $|W|^2 = D^2 \pi |U|^2/(b - |p|)$ we get $u = |p|/(b - |p|)$, $1 + u = b/(b - |p|)$ and $|p| = bu/(1+u)$, which verifies the inverse formulas; both denominators are positive on their domains. On $\{W \neq 0\}$, $d^c \log(1 + u) = \frac{u}{1+u}(d^c \log |W|^2 - 2\lambda)$ and $d^c \log |W|^2 = \mathrm{Im}(W^\dagger dW)/(2\pi|W|^2)$. Along ι_b, $W = cU$ with $c > 0$ a function of $(z, |p|)$, so $\mathrm{Im}(W^\dagger dW) = c^2 \,\mathrm{Im}(U^\dagger dU)$ and $b\frac{u}{1+u} d^c \log |W|^2 = |p| \frac{\mathrm{Im}(U^\dagger dU)}{2\pi|U|^2} = \frac{1}{2} \mathrm{Im}(U^\dagger dU) = \sum_j p_j d\theta_j$. Hence

$$\iota_b^* d^c \Phi_b = \lambda + \sum_j p_j d\theta_j - 2|p|\lambda = \sum_j p_j d\theta_j + (1 - 2|p|)\lambda,$$

which is smooth on the axes. Its exterior derivative is $\omega_V + d[(1-2|p|)\lambda] = \Omega_0$. □

15.3. **Complete transport in affine coordinates.** Let $J_z = dF_z$. At $(z, W) \in X_T = \{F(z) = TW\}$ define $V_T = (\dot z, \dot W)$ by

$$J_z\dot z - T\dot W = W, \qquad \Omega_b(V_T, k) = 0 \quad (k \in \ker(J_z, -T)). \tag{74}$$

Here $\ell = (J_z, -T)$ is complex linear and onto: at $T = 0$ because J_z is onto at points of Σ (Lemma 14.2), for $T \neq 0$ because of the fibre term. The Ω_b-orthogonal complement of the complex subspace $\ker \ell$ is its orthogonal complement for the Kähler metric, on which ℓ is an isomorphism; so V_T exists, is unique, and depends smoothly on (z, W, T) on the set $\mathcal{X} = \{F(z) = TW\}$, which is a smooth submanifold of $\mathbb{C}^{m+1}\times\mathbb{C}^m\times\mathbb{R}$ near $T = 0$ and the graph of $W = F(z)/T$ for $T > 0$.

Lemma 15.9 (Extension to projective space)**.** *Let $\overline{\mathbb{C}^{m+1}} = \mathbb{CP}^{m+1}$ with homogeneous coordinates $Z = (Z_*, Z')$, $z = Z'/Z_*$, and hyperplane at infinity $H_\infty = \{Z_* = 0\}$. For $T > 0$ and $0 \le s \le 1$ the forms*

$$\omega_T = dd^c\Big[\log D + b\log\Big(1 + \frac{|F|^2}{T^2D^2}\Big)\Big], \qquad \omega_s^{\mathrm{ret}} = dd^c\Big[\log D + sb\log\Big(1 + \frac{|F|^2}{D^2}\Big)\Big]$$

extend to Kähler forms on $\mathbb{CP}^{m+1}$, and the functions $g_T = \frac{|F|^2}{T^2D^2+|F|^2}$ and $\log(1+|F|^2/D^2)$ extend smoothly. Hence the Moser fields N_T^{gr}, $\iota_{N_T^{\mathrm{gr}}}\omega_T = \frac{2b}{T}\,d^c g_T$, and Y_s, $\iota_{Y_s}\omega_s^{\mathrm{ret}} = -b\,d^c\log(1+|F|^2/D^2)$, extend to smooth vector fields on $\mathbb{CP}^{m+1}$, smooth in $T > 0$ and s, and tangent to H_∞. Their flows over any compact parameter interval are complete on $\mathbb{CP}^{m+1}$ and preserve $\mathbb{C}^{m+1}$.

Proof. Let $F_j^{\mathrm{hom}}(Z) = Z_{y_j}^2 - Z_t^2 + jZ_*^2$ and let $V_2(Z)$ be the vector of ordered quadratic monomials Z_aZ_b, so $|V_2(Z)|^2 = |Z|^4$. Put $P_T(Z) = (V_2(Z), F^{\mathrm{hom}}(Z)/T)$, a homogeneous quadratic map without nontrivial zero. At $Z = (1, z)$, $D^2 + |F|^2/T^2 = |P_T(Z)|^2$, so

$$\omega_T = dd^c\big[(1-2b)\log|Z|^2 + b\log|P_T(Z)|^2\big] = (1-2b)\,\omega_{\mathrm{FS}} + b\,P_T^*\omega_{\mathrm{FS}},$$

which is a Kähler form on $\mathbb{CP}^{m+1}$ because $1-2b > 0$ and the second term is semipositive; likewise $\omega_s^{\mathrm{ret}} = (1-2sb)\omega_{\mathrm{FS}} + sb\,P_1^*\omega_{\mathrm{FS}}$ with $1-2sb \ge 1-2b > 0$. The functions $g_T = |F^{\mathrm{hom}}/T|^2/|P_T|^2$ and $\log(1+|F|^2/D^2) = \log(|P_1|^2/|Z|^4)$ are quotients of homogeneous functions of equal degree with nonvanishing denominators, hence smooth on $\mathbb{CP}^{m+1}$. The Moser fields are therefore smooth on the compact manifold $\mathbb{CP}^{m+1}$, and their flows are complete.

The involution $\iota(Z_*, Z') = (-Z_*, Z')$ is unitary, $F^{\mathrm{hom}}\circ\iota = F^{\mathrm{hom}}$ and $|V_2\circ\iota| = |V_2|$, so it preserves all the forms and functions above, hence the fields (Lemma 2.10(g)). At a point $x \in H_\infty$, which is fixed by ι, a ι-invariant vector lies in the $+1$ eigenspace of $d\iota_x$, which is T_xH_∞, since $d\iota_x$ acts by -1 on the normal direction. So the fields are tangent to H_∞, their flows preserve H_∞, and hence its complement $\mathbb{C}^{m+1}$. □

Proposition 15.10 (Complete horizontal transport)**.** *For every point of $X_0 = \Sigma\times\mathbb{C}^m$ the field (74) has a unique trajectory to $T = 1$. Its time-one map τ is smooth and injective and satisfies $\tau^*(\Omega_b|_{X_1}) = \Omega_b|_{X_0}$. Along each trajectory the normalized fibre norm $u = |W|^2/D^2$ is nonincreasing for $T \ge 0$; more precisely,*

$$\frac{du}{dT} = -\frac{2\pi T}{b}(1+u)^2\Omega_b(V_T, iV_T) \le 0.$$

Proof. Constraint. Along a trajectory, $\frac{d}{dT}(F(z) - TW) = J_z\dot z - W - T\dot W = 0$, so the trajectory stays in X_T.

The graph equation. For $T > 0$ identify X_T with $\mathbb{C}^{m+1}$ by $j_T(z) = (z, F(z)/T)$; then $j_T^*\Omega_b = \omega_T$ (Lemma 15.6). Write $V_T = dj_T(\dot z) - (0, W/T)$, using $\dot W = (J_z\dot z - W)/T$. For $v \in \mathbb{C}^{m+1}$,

$dj_T(v) \in \ker \ell$, so $\omega_T(\dot z, v) = \Omega_b((0, W/T), dj_T(v))$. The fibre circle action is generated by $X = 2\pi i(0, W)$ and preserves Φ_b, and $d^c\Phi_b(X) = bu/(1+u)$; hence $\iota_X\Omega_b = -d(bu/(1+u))$. Using J-invariance of Ω_b and holomorphy of j_T, $\Omega_b((0,W), dj_T(v)) = \Omega_b(i(0,W), dj_T(iv)) = -\frac{1}{2\pi}d\big(b\frac{u}{1+u}\big)(dj_T(iv))$, and for real ϕ, $d\phi(iv) = -4\pi\, d^c\phi(v)$. Since $u \circ j_T = |F|^2/(T^2D^2)$, so that $\frac{u}{1+u} \circ j_T = g_T$,

$$\iota_{\dot z}\omega_T = \frac{2b}{T}\, d^c g_T,$$

that is, $\dot z = N_T^{\mathrm{gr}}(z)$ is the field of Lemma 15.9. (Equivalently $\dot z = -r_T N_T/T$ with $\iota_{N_T}\omega_T = -b\, d^c \log(1 + u \circ j_T)$ and $r_T = 2/(1 + u \circ j_T)$.)

Completeness. Let $x \in X_0$. Since V_T is smooth on $\mathcal{X}$, the trajectory exists on $[0, \delta]$ for some $\delta \in (0, 1)$. For $T \in [\delta, 1]$ it is, in the graph coordinate, an integral curve of N_T^{gr}, whose flow is complete in $\mathbb{C}^{m+1}$ by Lemma 15.9; so it extends to $[0, 1]$, and it is unique. The time-T maps are restrictions of the flow of the smooth field $V_T + \partial_T$ on $\mathcal{X}$, hence smooth and injective (backward uniqueness).

Norm. By the circle identity above, $\frac{d}{dT}\frac{bu}{2\pi(1+u)} = \Omega_b(V_T, i(0, W))$. Since $\ell(i(0, W)) = -iTW$ and $\ell(iV_T) = iW$, the vector $i(0, W) + T\, iV_T$ lies in $\ker \ell$, so $\Omega_b(V_T, i(0, W)) = -T\,\Omega_b(V_T, iV_T)$. This gives the displayed derivative, which is ≤ 0 for $T \ge 0$ because $\Omega_b(V, iV)$ is the Kähler norm of V.

Pullback. Let $\Phi(\sigma, T)$ be the transported image of a smooth family of initial points. Its tangent vectors are (κ, t) with $\ell(\kappa) = tW$, and $\partial_T\Phi = (V_T, 1)$. Writing $\kappa = tV_T + (\kappa - tV_T)$ with $\kappa - tV_T \in \ker \ell$ gives $\Omega_b(V_T, \kappa) = 0$, so $\iota_{\partial_T}\Phi^*\Omega_b = 0$. As $\Phi^*\Omega_b$ is closed, Cartan's formula gives $\partial_T(\Phi_T^*\Omega_b) = 0$, which is the pullback identity. □

15.4. **Moser return and the ball chart.**

Proposition 15.11 (Affine return)**.** *Let*

$$\omega_s^{\mathrm{ret}} = dd^c\left[\log D + sb\log\left(1 + \frac{|F|^2}{D^2}\right)\right], \qquad \iota_{Y_s}\omega_s^{\mathrm{ret}} = -b\, d^c \log\left(1 + \frac{|F|^2}{D^2}\right).$$

The smooth field Y_s has a flow through every affine point for all $s \in [0, 1]$, in both directions. If ψ_s starts at $s = 0$, then $\psi_s^*\omega_s^{\mathrm{ret}} = dd^c \log D$.

Proof. By Lemma 15.9, Y_s is a smooth field on $\mathbb{CP}^{m+1}$ tangent to H_∞, so its flow is defined for all s in both directions and preserves $\mathbb{C}^{m+1}$. Since $\partial_s\omega_s^{\mathrm{ret}} = d\big(b\, d^c \log(1 + |F|^2/D^2)\big)$, Cartan's formula gives $\frac{d}{ds}\psi_s^*\omega_s^{\mathrm{ret}} = \psi_s^* d\big(\iota_{Y_s}\omega_s^{\mathrm{ret}} + b\, d^c \log(1 + |F|^2/D^2)\big) = 0$, so $\psi_s^*\omega_s^{\mathrm{ret}} = \omega_0^{\mathrm{ret}} = dd^c \log D$. Finally $j_1^*\Omega_b = \omega_1^{\mathrm{ret}}$ by Lemma 15.6. □

Theorem 15.12 (The affine quadric transfer)**.** *For $0 < b < 1/2$, the map*

$$\mathcal{T}_b(z, U) = \mathrm{ch}\big(\psi_1^{-1}(\mathrm{pr}(\tau(\iota_b(z, U))))\big), \qquad \mathrm{ch}(z) = \frac{z}{\sqrt{\pi(1 + |z|^2)}},$$

is defined on the whole tube $U_b = \{(z, U) \in \Sigma \times \mathbb{C}^m : |p(U)| < b\}$ and is a smooth symplectic embedding of (U_b, Ω_0) into $(B^\circ(1), \omega_0)$, for every $m \ge 1$. It agrees with ch *on the zero section. For every compact $\mathcal{K} \subset U_b$ there is $m_{\mathcal{K}} > 0$ with $\mathcal{T}_b(\mathcal{K}) \subset B(1 - m_{\mathcal{K}})$. For $0 < h < b$ the map is defined on a neighbourhood of every finite model $V_{\Delta_h}(E_\ell)$; it depends on b and the fixed curve, and not on h, ℓ or $\mathcal{K}$.*

Proof. By Lemma 15.8, Proposition 15.10 and Proposition 15.11,

$$\begin{aligned}\mathcal{T}_b^*\omega_0 &= \iota_b^*\tau^*\mathrm{pr}^*(\psi_1^{-1})^*\mathrm{ch}^*\omega_0 = \iota_b^*\tau^*\mathrm{pr}^*\omega_1^{\mathrm{ret}} \\ &= \iota_b^*\tau^*(\Omega_b|_{X_1}) = \iota_b^*\Omega_b = \Omega_0,\end{aligned}$$

using $\mathrm{ch}^*\omega_0 = dd^c \log D$ (Lemma 2.2), $(\psi_1^{-1})^* dd^c \log D = \omega_1^{\mathrm{ret}}$, and $\mathrm{pr}^*\omega_1^{\mathrm{ret}} = \mathrm{pr}^* j_1^* \Omega_b = \Omega_b|_{X_1}$ since $j_1 \circ \mathrm{pr} = \mathrm{id}$ on X_1. Each factor is injective: ι_b has an inverse, τ and ψ_1^{-1} are time maps of flows, pr is injective on the graph X_1, and ch is a bijection onto $B^\circ(1)$ (Lemma 2.2(b)). A symplectic map between manifolds of equal dimension $2n$ is an immersion, so $\mathcal{T}_b$ is an injective local diffeomorphism, hence an open embedding, with values in $B^\circ(1)$ because $\pi|\mathrm{ch}(z)|^2 = 1 - D^{-1} < 1$. At $U = 0$, $W = 0$ and $V_T = 0$, so $\tau(z, 0) = (z, 0)$; and Y_s vanishes on $\Sigma = \{F = 0\}$, where $d|F|^2 = 0$; so $\mathcal{T}_b(z, 0) = \mathrm{ch}(z)$. A compact source has compact image in the open ball, at positive distance from its boundary sphere. For $0 < h < b$ the closed set $V_{\Delta_h}(E_\ell)$ lies in the open tube U_b, and no choice in the construction involves h, ℓ or $\mathcal{K}$. □

16. One large ball: the models

Throughout this section $n \geq 3$, the target is $B^\circ(1)$, and the normalised input consists of one *distinguished* capacity A and further capacities $\rho_1, \dots, \rho_k$ ($k \geq 0$) with

$$A \geq \tfrac{1}{2}, \qquad A^n + \sum_{i=1}^{k} \rho_i^n < 1, \qquad A + \rho_i < 1 \quad (1 \leq i \leq k). \tag{75}$$

We write $\rho_0 = A$ when all balls are treated alike.

The quadric model of Section 14 stacks every ball as a toric ball $\{|p| \leq a_i\}$ inside $\{|p| \leq h\}$, and its form degenerates at $|p| = \frac{1}{2}$; so it accepts only capacities below one half. The two-ball inequality allows at most one capacity $\rho_j \geq \frac{1}{2}$, and then every other capacity is smaller than $1 - \rho_j \leq \frac{1}{2}$. For $n \geq 4$ we reserve the central ball $B(a)$, $a > A$, for the distinguished ball, place it by inclusion, and pack the other balls into the shell

$$S_a = \{y \in \mathbb{C}^n : \ a < \pi|y|^2 < 1\}, \qquad \mathrm{vol}\, S_a = \frac{1 - a^n}{n!}.$$

Write $s = 1 - a$. The return to the shell (Lemma 17.7) uses coupled coordinates and the slit coordinate $x = 1 - \pi|y|^2$. A toric ball stacked in the normal direction needs capacity below s; the two-ball condition $A + \rho_i < 1$ is what makes $\rho_i < s$ possible with $a > A$. The volume condition then gives a positive integrated-profile budget for the smaller balls, which tends to $(1 - a^n)/n!$.

The two-stage mechanism follows [Ope26]. For $n \geq 4$ we use a finite bordered part of the explicit affine quadric curve, a product mixed model and a radial-integral phase lift, followed by the slit-shell map. For $n = 3$ we use the polynomial cubic model and insert the distinguished ball by a polynomial bubble before the final return to the ball.

Definition 16.1 (Shell and reserved ball)**.** For $0 < a < 1$ put $s = 1 - a$, $S_a = \{a < \pi|y|^2 < 1\} \subset \mathbb{C}^n$, and call $B(a)$ the *reserved ball*.

In this section the auxiliary capacities r_i below play the role of the capacities a_i of Theorem M2.6; the letter a is kept for the reserved capacity.

Lemma 16.2 (Auxiliary capacities)**.** *Let $n \geq 3$ and let $A, \rho_1, \dots, \rho_k > 0$. Call a tuple $(a; r_1, \dots, r_k) \in \mathbb{R}^{k+1}$ valid auxiliary data for $(A; \rho_1, \dots, \rho_k)$ if*

$$\max(A, \tfrac{1}{2}) < a < 1, \qquad \rho_i < r_i < s = 1 - a \quad (1 \leq i \leq k), \qquad a^n + \sum_{i=1}^{k} r_i^n < 1. \tag{76}$$

If the input satisfies (75)*, the valid tuples form a nonempty open subset of $\mathbb{R}^{k+1}$; in particular rational valid tuples exist. For a rational tuple, validity is a finite conjunction of strict inequalities, decided by rational arithmetic for rational input and semi-decided from names otherwise (Lemma 23.2).*

Proof. Under (75) the point $(a,r)=(A,\rho)$ satisfies the strict inequalities $a<1$, $a^n+\sum r_i^n<1$ and $a+r_i<1$, and $A\ge\frac12$. By continuity these inequalities persist on an open neighbourhood $U\subset\mathbb{R}^{k+1}$ of (A,ρ). Define the real open set

$$V=U\cap\{(a,r):a>\max(A,\tfrac12),\ r_i>\rho_i\ (1\le i\le k)\}.$$

For sufficiently small $\eta>0$, the point $(A+\eta,\rho_1+\eta,\dots,\rho_k+\eta)$ belongs to U and, since $A\ge\frac12$, also to V. Thus V is nonempty and contains a rational tuple by density of $\mathbb{Q}^{k+1}$ in $\mathbb{R}^{k+1}$. Every point of V is valid, because $a+r_i<1$ is $r_i<s$; conversely the set of all valid tuples is open, being defined by strict continuous inequalities. The last assertion is Lemma 23.2. □

The theorems below take valid auxiliary data as input. In dimension six the reserved capacity a must also be rational, so that the polynomial construction has integer weights; the remaining auxiliary capacities may be real. Construction 20.1 fixes these data by explicit formulas.

16.1. **Dimension at least eight.** Let $n\ge4$ and $m=n-2$. Use the affine curve Σ of Definition 14.1, with m equations, its positive form $\sigma=d\lambda$ and the compact domains Σ_ℓ. The second-stage transfer φ is Theorem 15.12 in dimension $2(n-1)$. It maps a tube in $\Sigma\times\mathbb{C}^m$ into $B^\circ(1)\subset\mathbb{C}^{n-1}$ and agrees with the affine ball chart on the zero section.

Definition 16.3 (The affine mixed model)**.** Take $E_0=\Sigma\times\mathbb{C}^{m+1}$, with fibre coordinates $(z_1,\dots,z_{n-1})$,

$$p_j=\pi|z_j|^2,\qquad u=p_1,\qquad w=\sum_{j=2}^{n-1}p_j,\qquad \Omega_{\mathrm{mix}}=\omega_V+d[(1-u-2w)\lambda].$$

For $0<h_1<1$ and $0<h_2<1/2$ put

$$\Delta(h_1,h_2)=\{p\ge0:u\le h_1,\ w+h_2u\le h_2\},\qquad H_{\mathrm b}(p)=2^m(1-u-2w).$$

The finite model is the restriction to $\Sigma_R\times\{p\in\Delta\}$, where Σ_R and $A_R=\int_{\Sigma_R}\sigma$ are as in Proposition 16.5; it has empty marked set and base profile $H_b^R(p)=A_R(1-u-2w)$.

The full moment-dependent form has horizontal coefficient $1-u-2w$ and mixed terms $-du\wedge\lambda-2\sum_{j\ge2}dp_j\wedge\lambda$. On Δ,

$$1-u-2w\ge(1-u)(1-2h_2)\ge(1-h_1)(1-2h_2)>0. \tag{77}$$

The finite product is therefore a bordered toric model. The annuli and primitive operator are those of Proposition 17.6, and the finite mass budget is Proposition 16.5.

The return target has coordinates $(z,\xi)\in\mathbb{C}\times B^\circ_{n-1}(1)$ and the coupled form

$$\Omega_c=d[\lambda_{\mathrm{fib}}+(1-u)\lambda_{\mathrm{std}}],\qquad \lambda_{\mathrm{std}}(\xi)=\tfrac12\operatorname{Im}(\bar\xi^{\mathsf T}d\xi),\quad u=\pi|z|^2.$$

All line coordinates are product coordinates. The phase required to preserve this form is given by a radial integral, as a smooth function of the affine base and fibre coordinates.

Lemma 16.4 (Integrated mixed profile)**.** *For $0<h_1<1$ and $0<h_2\le\frac12$,*

$$\int_{\Delta(h_1,h_2)}H_{\mathrm b}\,dp=J(h_1,h_2):=J_m(h_2)\,\frac{1-(1-h_1)^n}{n},\qquad J_m(h)=\frac{2^mh^m}{m!}\Big[1-\frac{2mh}{m+1}\Big],$$

and $J(h_1,h_2)\to J(s,\frac12)=(1-a^n)/n!$ as $(h_1,h_2)\to(s,\frac12)$, $s=1-a$. For $r>0$, $\int_{\mathbb{R}^{n-1}_{\ge0}}(r-|p|)_+\,dp=r^n/n!$.

Proof. Fix $u \in [0, h_1]$. The remaining variables $p' = (p_2, \dots, p_{n-1})$ range over $\{p' \ge 0,\ w \le h_2(1-u)\}$. With the simplex formula of Lemma 14.5(a) and the substitution $w = (1-u)t$,

$$\int 2^m(1-u-2w)\,dp' = \frac{2^m}{(m-1)!}\int_0^{h_2(1-u)} (1-u-2w)w^{m-1}\,dw$$
$$= (1-u)^{m+1}\frac{2^m}{(m-1)!}\int_0^{h_2}(1-2t)t^{m-1}\,dt,$$

and the last integral times $2^m/(m-1)!$ is $J_m(h_2)$. Integrating in u gives $\int_0^{h_1}(1-u)^{m+1}du = \frac{1-(1-h_1)^{m+2}}{m+2}$ and $m+2 = n$. By Lemma 14.5(b), $J_m(\frac{1}{2}) = 1/(m+1)! = 1/(n-1)!$, so $J(s, \frac{1}{2}) = (1-a^n)/n!$; J is continuous. The cap integral is Lemma 14.5(a) in dimension $n-1$ with $f(t) = (r-t)_+$. □

Proposition 16.5 (The affine model and its area budget)**.** *Let $n \ge 4$, $m = n-2$, and let $(a; r)$ be a valid auxiliary tuple. Take h_1, h_2, d, P from Lemma 17.1 and (K, q) from Theorem M3.5 for this profile. Put $A_m = 2^m(m+2)$, $\ell = \lceil 64A_m/d \rceil$ and $R = 2 + \sum_{j=1}^m j + \ell$. In the affine quadric coordinate t, let $\Sigma_R = \{|t| \le R\}$ and $A_R = \int_{\Sigma_R}\sigma$. The restricted mixed model has base area $H_b^R(p) = A_R(1-u-2w)$ and satisfies*

$$0 \le H_{\mathrm{b}}(p) - H_b^R(p) \le d/64 \quad (p \in \Delta).$$

The same comparison datum works for $P_R = P - 11d/32$, with gap $11d/32$ and reserve $11d/64$; moreover $P_R \le H_b^R - \sum_i (r_i - |p|)_+$. The boundary cutoff and primitive operator on Σ_R satisfy the handle hypotheses with this reserve.

Proof. The quadric tail bound gives $2^m - A_R \le A_m/R^2 \le d/64$, and $0 < 1-u-2w \le 1$ on Δ. Subtracting $11d/32$ from the comparison profile subtracts the same amount from its gap, and $d/64 < 11d/32$. The boundary and background estimates are Proposition 17.5. □

The integral of the finite base profile is $\int_\Delta H_b^R = (A_R/2^m)J(h_1, h_2)$; its difference from $J(h_1, h_2)$ is absorbed by the reserve of Proposition 16.5. The packing for $n \ge 4$ is Theorem 17.9 in §17: the comparison datum of Theorem M3.5 for this profile, the packing over Σ_R, the affine mixed return (Lemma 17.4) and the slit-shell map (Lemma 17.7) into the shell S_a, while the distinguished ball is placed by inclusion in $B^\circ(a)$.

16.2. **Dimension six.** For $n = 3$ the second stage of §16.1 would have a conic as its base. Instead we use the polynomial cubic model of [Ope26]. Its area profile still approaches the volume of the shell, but the final placement of the large ball uses a polynomial bubble. All the sections used below are given by explicit formulas.

Definition 16.6 (The cubic model)**.** Put $s = 1-a$ and $b = (2-a)/3$, where $(a; r)$ is a valid auxiliary tuple with $a \in \mathbb{Q}$, and let $A < r_0 < a$. Use the homogeneous polynomial $F(u, v, w) = vw(v-w) + u^2(w-2v)$ and its affine restriction $f(v, w) = vw(v-w) + w - 2v$. The model base is $C = \{vw \ne 0, f(v, w) = 0\}$. Its four ends have homogeneous limits $[0:1:0], [0:0:1], [0:1:1], [1:0:0]$. The same affine curve is parametrised by

$$t \in \mathbb{C}, \quad y_1^2 = t^2 - 1, \quad y_2^2 = t^2 - 2, \qquad (v, w) = (y_1 - t, y_2 - t).$$

Take the rational parameters h_1, e, d_1, h_2, d_2 of Lemma 18.2; they satisfy

$$\max_i r_i < h_1 < e < s < d_1 < h_2 < d_2 < b, \qquad \Delta = \{p \ge 0 : p_1 \le h_1,\ p_1 + p_2 \le h_2\}. \tag{78}$$

Put $Q = \lfloor 3/(a - r_0) \rfloor + 1$ and

$$L = (Q+1)\operatorname{den}(a)\operatorname{den}(e)\operatorname{den}(d_2), \qquad A_* = Le, \quad B_* = Ld_2, \quad S_* = L(1-a), \quad D = L + S_*,$$

where den is the reduced denominator; these are integers and $L > Q$. Here A_*, B_* are integer weights; the distinguished input capacity remains A. Let $\Lambda = \{(j_1, j_2) \in \mathbb{Z}^2_{\geq 0} : j_1 \leq A_*,\ j_1 + j_2 \leq B_*\}$ and put

$$d_j = D - 3(j_1 + j_2), \qquad m_j = (S_* - j_1 - j_2)_+,$$

$$q_{d,m}(v, w) = (1, v^{d-m}, w^{d-m}, v^{d-m}w^m, v^m w^{d-m}), \qquad a_j = \|q_{d_j,m_j}\|^2.$$

The central potential is

$$\phi_L(y, z) = \frac{1}{L} \log \sum_{j \in \Lambda} a_j(y)|z_1|^{2j_1}|z_2|^{2j_2}. \tag{79}$$

Its radial moment coordinates identify a neighbourhood of Δ with a toric normal model on $C \times \mathbb{C}^2$; the primitive and the identification are given in Definition 18.3.

The integer inequalities are

$$0 < A_* < S_* \leq B_*, \qquad 3B_* < D, \qquad 2S_* < L.$$

They follow from (78) and $a > 1/2$. The five entries of $q_{d,m}$ are fixed before the transport parameters.

Lemma 16.7 (Limiting profile and its integral)**.** *The limiting base area at physical moment p is*

$$H_0(p) = 9(b - |p|) - 2(s - |p|)_+, \qquad |p| = p_1 + p_2.$$

For $0 < h_1 \leq s \leq h_2 \leq b$,

$$\int_\Delta H_0 \, dp = \frac{3}{2}[b^3 - (b - h_1)^3] - \frac{9}{2}h_1(b - h_2)^2 - \frac{1}{3}[s^3 - (s - h_1)^3].$$

As $(h_1, h_2) \to (s, b)$ this tends to $(1-a^3)/6$. Moreover $H_0 \geq 9(b-h_2)$ on Δ and $\int_{p \geq 0}(r-|p|)_+ \, dp = r^3/6$.

Proof. Put $d(p) = D - 3L|p|$ and $h(p) = (S_* - L|p|)_+$. By Proposition 18.6 (the four end limits in its Step 1 and the marked-height estimate of its Step 4), the limiting pole orders at $[0:1:0], [0:0:1], [0:1:1], [1:0:0]$ are $d(p) - h(p)$, $d(p) - h(p)$, $d(p)$ and 0, and Stokes' theorem gives $H_0 = L^{-1}(3d(p) - 2h(p))$. For fixed $p_1 = t$, integration in $|p| \in [t, h_2]$ gives $\frac{9}{2}[(b-t)^2 - (b-h_2)^2] - (s-t)^2$. Integrate in $t \in [0, h_1]$ and use $3b = 1 + s$. The lower bound follows because H_0 decreases with $|p|$ and $h_2 \geq s$; the cap integral is $\int_0^r (r-t)t \, dt$. □

Proposition 16.8 (Polynomial lifts)**.** *For every $j \in \Lambda$ the five polynomials $f^{j_2} q_{d_j,m_j,i}$ have explicit homogeneous lifts to the quadric $\mathcal{Q}(X) = X_0X_1 - X_2X_3 = 0$, described by the homogeneous substitution*

$$(u, v, w) \longmapsto (vw, u^2, uv, uw).$$

Multiplication by $\mathcal{Q}^{j_1}$ gives homogeneous polynomials $P_{j,i}$ of degree L, each vanishing to order at least $L - S_*$ at $[1:0:0:0]$. For $t, \delta > 0$ set

$$P_{t,\delta}(X)_{j,i} = \delta^{-j_1} t^{-j_2} P_{j,i}(X).$$

On the tube $(v, w, z) \mapsto (v, w, vw + \delta z)$ in the chart $X_1 = 1$, the factors δ^{-j_1} cancel exactly, and the resulting polynomial family extends smoothly to $\delta = 0$. Its value there is

$$z^{j_1}(f(v, w)/t)^{j_2} q_{d_j,m_j,i}(v, w).$$

Proof. For a monomial $v^J w^K$ and $\bar n = L - 2j_1$ put $\alpha_0 = (J + K - \bar n)_+$, $\alpha_1 = \bar n + \alpha_0 - J - K$, $\alpha_2 = J - \alpha_0$, $\alpha_3 = K - \alpha_0$. Then $X_0^{\alpha_0} X_1^{\alpha_1} X_2^{\alpha_2} X_3^{\alpha_3}$ pulls back to $u^{2\bar n - J - K} v^J w^K$. The inequalities $J, K \le \bar n$ and the required order at the distinguished point follow from $j_1 < S_*$, $3(j_1 + j_2) < L + S_*$ and the degree bounds $\deg_v, \deg_w (f^{j_2} q_i) \le 2j_2 + d_j - m_j$, $\deg (f^{j_2} q_i) \le 3j_2 + d_j$. Finally $\mathcal{Q}(vw + \delta z, 1, v, w) = \delta z$. The complete inequalities are Lemma 18.5. □

Packing the smaller balls. By Lemma 18.2 the parameters (78) satisfy $\int_\Delta H_0 > \sum_i r_i^3/6$. The function $G = H_0 - \sum_i (r_i - |p|)_+$ is concave, has positive integral, and is positive off a smaller homothetic copy of Δ containing all the cap supports. The comparison datum of Theorem M3.5 for a lowered copy of G and the handle construction over a compact piece Σ_l of C give the inner packing (Proposition 19.1).

The remaining maps are applied to the compact union of these ball images. A short exact cubic transfer gives the cubic ambient form, and a second exact transfer along the tube gives the form defined by $P_{t,\delta}$. After changing to $X_0 = 1$, this compact set avoids the origin. A polynomial bubble inserts the distinguished ball near the origin while moving the smaller images within a disjoint open set. A common global Moser map and the Fubini–Study ball chart then put all the images into $B^\circ(1)$. The formulas, the two compact transfer estimates and the packing theorem (Theorem 19.7) are in §19.

In Construction 20.1(iv) these data are $r_0 = A + 2\eta$, $\rho_0^+ = A + \eta$, and $r_i = \rho_i + 3\eta$, $r_i' = \rho_i + 2\eta$, $\rho_i^+ = \rho_i + \eta$ for the smaller balls; the auxiliary ball with index j has $r_j = \eta$, $r_j' = \eta/2$, and its map is replaced by f_0.

Example. For $A = 59/100$ and eighteen smaller capacities $7/20 > 1/3$, $A^3 + 18(7/20)^3 = 0.977129$ and $A + 7/20 = 0.94$. Thus Theorem 19.7 applies beyond the range of a cubic used directly as a second stage.

17. One large ball in dimension at least eight: the shell return and the packing

Let $n \ge 4$, $m = n - 2$. The base is the affine locus $\Sigma = \{y_j^2 = t^2 - j\}$ with $\lambda = d^c \log(1 + |t|^2 + \sum_j |y_j|^2)$ and $\sigma = d\lambda$. The model is the product over the compact piece Σ_R. The second-stage quadric transfer, a radial-integral phase and the slit-shell map return the inner packing to the shell S_a. Throughout $(a; r)$ is valid auxiliary data for $(A; \rho)$, with $1/2 < a < 1$ and $s = 1 - a$.

17.1. **Parameters.** Recall from Lemma 16.4 that for $0 < h_1 < 1$ and $0 < h_2 \le \frac12$

$$J(h_1, h_2) = J_m(h_2)\,\frac{1 - (1 - h_1)^n}{n}, \qquad J_m(h) = \frac{2^m h^m}{m!}\Big[1 - \frac{2mh}{m+1}\Big],$$

is the integral of $H_{\mathrm b} = 2^m(1 - u - 2w)$ over $\Delta(h_1, h_2) = \{p \in \mathbb{R}^{n-1}_{\ge 0} : u \le h_1,\ w + h_2 u \le h_2\}$, where $u = p_1$ and $w = p_2 + \cdots + p_{n-1}$, and that $J(s, \frac12) = (1 - a^n)/n!$. The following lemma gives the parameters of the large branch by closed formulas; they are the parameters of Theorem 17.9.

Lemma 17.1 (Parameters of the large branch)**.** *Let $n \ge 4$, $k \ge 1$, and let $(a; r_1, \dots, r_k)$ be a valid auxiliary tuple. Put $s = 1 - a$, $r_* = \max_i r_i$, $\varsigma = 1 - a^n - \sum_i r_i^n$,*

$$\delta = \min\Big(\frac{s - r_*}{2},\ \frac{\varsigma}{2n!(4 + 2^n)}\Big), \qquad h_1 = s - \delta, \qquad h_2 = \tfrac12 - \delta,$$

$$b_2 = \tfrac12\big(h_2 + \tfrac12\big), \qquad h_2' = \tfrac12(h_2 + b_2), \qquad \tau_\Delta = \frac{r_* + h_1}{2h_1},$$

$$\beta_\Delta = 2^m(1 - h_1)(1 - 2h_2), \qquad d = \tfrac12 \min\Big(\beta_\Delta, \frac{J(h_1, h_2) - \sum_i r_i^n/n!}{2}\Big), \qquad \Delta = \Delta(h_1, h_2),$$

and $P = H_{\mathrm b} - \sum_i (r_i - |p|)_+$ on Δ. All these numbers are positive, and rational when the auxiliary data are rational, and:

(i) $r_* < h_1 < s$ *and* $r_* < h_2 < h_2' < b_2 < \frac{1}{2}$*;*
(ii) $J(h_1,h_2) \ge \big(1 - a^n - \frac{1}{2}\varsigma\big)/n!$*, hence* $\int_\Delta P\,dp = J(h_1,h_2) - \frac{1}{n!}\sum_i r_i^n \ge \varsigma/(2\,n!) > 0$*;*
(iii) $\tau_\Delta < 1$ *and* $\{p \in \mathbb{R}^{n-1}_{\ge 0} : |p| \le r_i\} \subset \tau_\Delta\Delta$ *for every* i*;*
(iv) P *is a concave, piecewise affine function with real coefficients on* Δ*, and* $P = H_{\mathrm{b}} \ge \beta_\Delta > 0$ *on* $\Delta \setminus \tau_\Delta\Delta$*;*
(v) $\operatorname{vol}\Delta < 1$*, and the mean* $\bar P$ *of* P *on* Δ *satisfies* $\bar P > \int_\Delta P\,dp$*;*
(vi) $0 < d < \min(\bar P, \beta_\Delta)$*.*

In particular $(\Delta, P, \tau_\Delta, d)$ *satisfies the hypotheses of Proposition M3.3, in the real-coefficient form stated after it (with rational coefficients when the auxiliary data are rational).*

Proof. The auxiliary inequalities give $r_i < s < \frac{1}{2}$ and $\varsigma > 0$, so $\delta > 0$. Rational auxiliary data give rational parameters.

(i) We have $h_1 \ge (s + r_*)/2 > r_*$ and $h_1 < s$. Since $h_2 - h_1 = 1/2 - s > 0$, this also gives $r_* < h_2$. The inequalities $h_2 < h_2' < b_2 < 1/2$ follow from their definitions.

(ii) Put $F = n!$ and $B = (1-2\delta)^m(1+2m\delta)$. Substituting the two heights in the profile formula gives

$$FJ(h_1,h_2) = B\big(1 - (a+\delta)^n\big).$$

Here $0 < \delta < 1/4$. Bernoulli's inequality and $(1-2\delta)^m \le 1$ give

$$1 - 4m^2\delta^2 \le B \le 1 + 2m\delta.$$

Also $0 \le (a+\delta)^n - a^n \le \delta n 2^n$, by expanding the difference of powers with $0 \le a, \delta \le 1$. Therefore

$$\begin{aligned} 1 - a^n - FJ(h_1,h_2) &= (1-B)(1-a^n) + B\big((a+\delta)^n - a^n\big) \\ &\le 4m^2\delta^2 + (1+2m\delta)\delta n 2^n \\ &\le F(4+2^n)\delta \le \varsigma/2. \end{aligned}$$

For the penultimate inequality use $m^2 \le n!$, $\delta \le 1/4$ and $(1+2m\delta)n \le (m+1)(m+2) \le n!$. This proves the required lower bound for J. The caps lie in Δ by (iii), and their integrals are $r_i^n/n!$, giving the formula for $\int_\Delta P$.

(iii) $\tau_\Delta < 1$ since $r_* < h_1$ and $r_* < h_2$. Let $|p| = u + w \le r_i$. Then $u \le r_* < \frac{1}{2}(h_1 + r_*) \le \tau_\Delta h_1$, and $w + h_2 u \le w + u \le r_* < \tau_\Delta h_2$: here $h_2 < 1$ and $\tau_\Delta h_2 > \tau_\Delta h_1 > r_*$. Hence $p \in \tau_\Delta\Delta = \{u \le \tau_\Delta h_1,\ w + h_2 u \le \tau_\Delta h_2\}$.

(iv) H_{b} is affine with integer coefficients and each $-(r_i - |p|)_+$ is concave and piecewise affine. If $p \in \Delta \setminus \tau_\Delta\Delta$, then $|p| > r_i$ for every i by (iii), so all caps vanish and $P = H_{\mathrm{b}}$, and $H_{\mathrm{b}} \ge \beta_\Delta$ by (77).

(v) $\Delta \subset [0, h_1] \times \{p' \in \mathbb{R}^m_{\ge 0} : |p'| \le h_2\}$, so $\operatorname{vol}\Delta \le h_1 h_2^m/m! < 1$; since $\int_\Delta P > 0$, its mean exceeds its integral.

(vi) follows because $d \le \beta_\Delta/2$ and $d \le \frac{1}{4}\int_\Delta P < \bar P$. The last assertion is (iii)–(vi) with $\beta_P = \beta_\Delta$; Δ is a polytope and P is concave and piecewise affine. □

17.2. **The second stage and radial phase.** Put $b_2 = (2h_2 + 1)/4$ and let $\varphi = \mathcal{T}_{b_2}$ be the affine quadric transfer in dimension $2(n-1)$. Write $v = (v_2, \dots, v_{n-1})$ for its fibre moments and

$$\ell = \sum_{j=2}^{n-1} v_j\,d\theta_j + (1 - 2|v|)\lambda, \qquad d\ell = \Omega_X.$$

This is a smooth primitive across the axes. Its restriction to a radial fibre tangent is zero.

The fibre contraction $\mathsf{r}_t(x,z') = (x, tz')$, $0 \le t \le 1$, preserves $\mathcal{N}_{b_2} = \{(x,z') : |v(z')| < b_2\}$.

Lemma 17.2 (The zero section)**.** *The second transfer satisfies* $\varphi(x,0)=\mathrm{ch}(x)$ *and* $i^*(\varphi^*\lambda_{\mathrm{std}}-\ell)=0$*, where* $i(x)=(x,0)$*.*

Proof. The first identity is part of Theorem 15.12. The chart ch pulls the standard Liouville form back to $\lambda=d^c\log D$ (Lemma 2.2), and the restriction of ℓ to the zero section is also λ. □

17.3. **The phase identity.**

Lemma 17.3 (The radial phase and its covector)**.** *Put* $\Xi=\varphi^*\lambda_{\mathrm{std}}-\ell$ *and*

$$\Psi(x,z')=\int_0^1\lambda_{\mathrm{std}}(\varphi(x,tz'))[\partial_t\varphi(x,tz')]\,dt.$$

Then Ψ *is smooth on* $\mathcal{N}_{b_2}$ *and*

$$d\Psi=\Xi=\varphi^*\lambda_{\mathrm{std}}-\ell. \tag{80}$$

For $\hat{\mathcal{J}}_{x,z'}(z_1)=e^{2\pi i\Psi(x,z')}z_1$ *and* $\Phi_2(x,z_1,z')=(\hat{\mathcal{J}}_{x,z'}(z_1),\varphi(x,z'))$*, one has*

$$\Phi_2^*\Omega_c=d[\lambda_{\mathrm{fib}}+(1-u)\ell]. \tag{81}$$

Proof. The transfer pullback gives $d\Xi=\varphi^*\omega_0-d\ell=0$. By Lemma 17.2, $i^*\Xi=0$. Write $\mathsf{r}^*\Xi=a_t+g\,dt$ on $[0,1]\times\mathcal{N}_{b_2}$. Closedness gives $\partial_t a_t=dg$; integrating from zero to one gives $\Xi=d\int_0^1 g(t,\cdot)dt$. Since ℓ is zero on radial fibre tangents, the last integral is exactly the displayed Ψ. Smooth parameter integration proves its smoothness, including at $z'=0$.

A phase rotation of z_1 pulls its Liouville form back to $\lambda_{\mathrm{fib}}+u\,d\Psi$. Therefore the primitive of the target coupled form pulls back to

$$\lambda_{\mathrm{fib}}+u\,d\Psi+(1-u)\varphi^*\lambda_{\mathrm{std}}=\lambda_{\mathrm{fib}}+\varphi^*\lambda_{\mathrm{std}}-u\ell.$$

Its exterior derivative is $d[\lambda_{\mathrm{fib}}+(1-u)\ell]$, because $d\varphi^*\lambda_{\mathrm{std}}=d\ell$. This proves (81). □

17.4. **The finite mixed model.** Use the affine product and form of Definition 16.3. For the moment change put

$$\chi(x,z_1,z')=(x,z_1,z'/\sqrt{1-u}),\qquad \mathcal{U}=\{u<1,\ w<b_2(1-u)\},\qquad \mathcal{R}=\Phi_2\circ\chi.$$

Lemma 17.4 (Mixed form and return)**.** *On* $u<1$ *the smooth inverse of* χ *is*

$$(x,z_1,v)\longmapsto(x,z_1,\sqrt{1-u}\,v).$$

It preserves u *and fibre angles, with tail moments* $v_j=p_j/(1-u)$*. On the interior of the finite mixed region,*

$$\mathcal{R}^*\Omega_c=\Omega_{\mathrm{mix}}=\omega_V+d[(1-u-2w)\lambda].$$

For the finite region $\mathcal{M}_R=\Sigma_R\times\{z:p(z)\in\Delta\}$*, the map* $\mathcal{R}$ *is injective on* $\mathcal{M}_R$ *and smooth on its interior. The displayed pullback identity holds on that interior. It preserves* u *and has base moment* $\mu<1$*. The compact region over* Σ_R *with moment polytope* $\Delta(h_1,h_2)$ *lies in* $\mathcal{U}$*. Its actual base profile is* $A_R(1-u-2w)$ *and its scalar integral is* $(A_R/2^m)J(h_1,h_2)$*.*

Proof. The square root is positive on $u<1$, which proves both smooth coordinate formulas across the axes. Using $v_j=p_j/(1-u)$ gives

$$\chi^*\{\lambda_{\mathrm{fib}}+(1-u)\ell\}=\sum_{j=1}^{n-1}p_j d\theta_j+(1-u-2w)\lambda.$$

Take the full exterior derivative and use (81). Injectivity follows first from injectivity of φ, then cancellation of the common nonzero phase in the first coordinate, and finally the inverse of χ. The base moment bound is the open-ball containment of φ. On Δ, $u\le h_1<1$ and

$w \le h_2(1-u) < b_2(1-u)$. The horizontal curvature at fixed moment is $(1-u-2w)\sigma$; its base integral is $A_R(1-u-2w)$. The scalar formula for J gives the last assertion. □

Proposition 17.5 (Affine mixed-model budget)**.** *Let $m = n-2 \ge 2$, and take Δ, d from Lemma 17.1. Put $A_m = 2^m(m+2)$, $\ell = \lceil 64A_m/d \rceil$, $R = 2 + \sum_{j=1}^{m} j + \ell$, $\Sigma_R = \{|t| \le R\}$ and $A_R = \int_{\Sigma_R} \sigma$. Set*

$$\gamma_* = \frac{11d}{64}, \qquad \kappa_{\min} = (1-h_1)(1-2h_2), \qquad c_{\mathrm{bg}} = \min\Big(\frac{\gamma_*}{32\,2^m}, \frac{\kappa_{\min}}{2}\Big).$$

Use the flat cutoff $\chi_R = \chi_{R-2,R-1}(t)$ of Definition 2.6, $\chi_R(t) = 1 - S\big((|t|^2-(R-2)^2)/((R-1)^2-(R-2)^2)\big)$, which equals one for $|t| \le R-2$ and zero for $|t| \ge R-1$. Set $c(p) = 1-u-2w$, $H_{\mathrm{b}}^R(p) = \int_{\Sigma_R} c(p)\sigma = A_R c(p)$ and

$$\Omega_{\mathrm{rem}}(p) = \chi_R(c(p) - c_{\mathrm{bg}})\sigma, \quad H_R(p) = \int_{\Sigma_R} \Omega_{\mathrm{rem}}(p), \quad \kappa_{\mathrm{bg}}(p) = c(p)\sigma - \Omega_{\mathrm{rem}}(p).$$

For every $p \in \Delta$, $\kappa_{\mathrm{bg}}(p) > 0$, it equals $c(p)\sigma$ near $\partial\Sigma_R$ and $c_{\mathrm{bg}}\sigma$ on the handle, and

$$0 < \int_{\Sigma_R} \kappa_{\mathrm{bg}}(p) < \gamma_*/16, \qquad H_R(p) > H_{\mathrm{b}}^R(p) - \gamma_*/16.$$

If $\eta_{\mathrm{h}} = df \wedge \beta$ is the unit-integral handle density, the form $H_R(p)\eta_{\mathrm{h}} - \Omega_{\mathrm{rem}}(p)$ has integral zero and is supported in $\Sigma_{R-1} \subset \operatorname{int}\Sigma_R$, and the finite coordinate primitive of Lemma 10.2 applied to it depends smoothly on p.

Proof. On $\Delta = \{u \le h_1,\ w \le h_2(1-u)\}$ we have $c(p) \ge (1-u)(1-2h_2) \ge (1-h_1)(1-2h_2) = \kappa_{\min} > 0$ and $c(p) \le 1$. Hence $\kappa_{\mathrm{bg}} = [c(p)(1-\chi_R) + c_{\mathrm{bg}}\chi_R]\sigma$ is positive, equals $c(p)\sigma$ for $|t| \ge R-1$, and equals $c_{\mathrm{bg}}\sigma$ for $|t| \le R-2$, which contains the handle supports ($|t| \le 3$, Lemma 15.5). Since $R-2 = R_{\ell-2}$ in the notation of Lemma 15.5 and $\int_\Sigma \sigma = 2^m$, that lemma and $c \le 1$ give

$$\int_{\Sigma_R \setminus \Sigma_{R-2}} c(p)\sigma \le 2^m - A_{\ell-2} \le \frac{2^m(m+2)}{(R-2)^2} \le \frac{d^2}{4096 \cdot 2^m(m+2)} \le \frac{d}{65536} < \frac{\gamma_*}{32},$$

because $R-2 \ge \ell \ge 64 \cdot 2^m(m+2)/d$, $2^m(m+2) \ge 16$, and $d \le 1$: indeed $d \le J(h_1,h_2)/4 \le J_m(h_2)/4 \le 1/4$, as $2^m h_2^m \le 1$. Also $A_R \le 2^m$, so $c_{\mathrm{bg}}A_R \le \gamma_*/32$. Adding, $\int_{\Sigma_R} \kappa_{\mathrm{bg}} \le \int_{\Sigma_R \setminus \Sigma_{R-2}} c\sigma + c_{\mathrm{bg}}A_R < \gamma_*/16$, and the integral is positive because $\kappa_{\mathrm{bg}} > 0$. Integrating $c(p)\sigma = \kappa_{\mathrm{bg}} + \Omega_{\mathrm{rem}}$ gives $H_R = H_{\mathrm{b}}^R - \int \kappa_{\mathrm{bg}} > H_{\mathrm{b}}^R - \gamma_*/16$. Both η_{h} and Ω_{rem} are supported in Σ_{R-1}, and $\int(H_R\eta_{\mathrm{h}} - \Omega_{\mathrm{rem}}) = H_R - H_R = 0$. The primitive is a finite sum of coordinate integrals over fixed compact boxes of smooth functions of (p,x), hence smooth in p (Lemma 10.2). □

Proposition 17.6 (Handle data for the affine mixed model)**.** *Use the base Σ_R of Proposition 17.5, the fixed annuli and dual form of Proposition 15.4, the physical primitive $\Gamma_{\mathrm{phys}}(p) = (1-u-2w)\lambda$ with curvature $\kappa(p) = (1-u-2w)\sigma$, the reference primitive λ with $\Lambda = \sigma$ and $A_{\mathrm{ref}} = 2^m$, the density floor $\kappa_- = (1-h_1)(1-2h_2)$, and the collar cutoff $\chi_\partial = 1-\chi_R$. Take as correction core Σ_{R-1}, covered by the inner boxes of the quadric presentation of Construction 10.4, and as correction region the union of these boxes, contained in $\operatorname{int}\Sigma_R$. Then:*

- *(a) the support of $1-\chi_\partial = \chi_R$ in Σ_R and the handle are contained in the correction core;*
- *(b) $\kappa(p) \ge \kappa_-\Lambda$ on all of Σ_R for $p \in \Delta$, in particular on the correction region;*
- *(c) with $P_R = P - 11d/32$, gap $11d/32$ and reserve $\gamma_* = 11d/64$, these data satisfy the bordered handle hypotheses of Theorem M2.6, and Construction 12.4 applies to them.*

Proof. (a) χ_R vanishes for $|t| \geq R-1$, and the annular supports lie in $|t| < 3 \leq R-2$ (Lemma 15.5). (b) On Δ, $1-u-2w \geq (1-h_1)(1-2h_2)$ by (77). (c) The product form is $\omega_V + d\Gamma_{\mathrm{phys}}$, including all moment derivatives. Proposition 17.5, whose background coefficient is $c_{\mathrm{bg}} = \min(\gamma_*/(32A_{\mathrm{ref}}), \kappa_-/2) = e_\Lambda$, gives the positive background, the collar bound and the compactly supported zero-mass defect, whose finite coordinate primitive depends smoothly on p. Hence the normalisation and layer construction apply with reserve γ_*. □

17.5. **The slit-shell map.** The coupled target is the explicit product $(\xi, z) \in \{\pi|\xi|^2 < 1\} \times \mathbb{C}$. Write $\mu = \pi|\xi|^2$, $u = \pi|z|^2$, $\lambda_{\mathrm{base}} = \lambda_{\mathrm{std}}(\xi)$ and $\lambda_{\mathrm{fib}} = \frac{1}{2}\operatorname{Im}(\bar{z}\,dz)$, so

$$\Omega_1 = \Omega_c = d[\lambda_{\mathrm{fib}} + (1-u)\lambda_{\mathrm{base}}]. \tag{82}$$

The strip map moves the first moment away from zero. Its compensating base dilation is exactly the square-root factor in the following shell coordinates; the factor $\sqrt{1-u}$ in (83) absorbs the change of the fibre moment from u to x, and the zero section $u = 0$ lands at depth $x > 0$ inside the shell.

Lemma 17.7 (Slit-shell map)**.** *Let $0 < a < 1$, $s = 1-a$, and let $\hat{h}, \hat{\mu}, \delta$ be real numbers with*

$$0 < \hat{h} < s, \qquad 0 < \hat{\mu} < 1, \qquad 0 < \delta < \tfrac{1}{4}\min\bigl(s - \hat{h},\ (1-\hat{h})(1-\hat{\mu})\bigr).$$

Let $\Psi = \Psi_{2\delta} \colon z \mapsto (x, T)$ be the strip map of Proposition 11.7 (Proposition 10.32) with parameter 2δ, put $b_\delta = 2\delta/\pi$ and $U = \{(\xi, z) : \mu < \hat{\mu},\ u < \hat{h}\}$, and define $\mathrm{Sl}(\xi, z) = (y_1, y_2, \dots, y_n) \in \mathbb{C}^n$ on U by

$$y_1 = e^{-2\pi i T}\sqrt{\frac{1-x-(1-u)\mu}{\pi}}, \qquad (y_2, \dots, y_n) = e^{-2\pi i T}\sqrt{1-u}\,\xi. \tag{83}$$

For $I \geq 0$ put $\mathcal{E} = I + b_\delta(\frac{\pi}{2} - 1)$ and

$$x_*(I) = \frac{b_\delta^2}{2\bigl(\mathcal{E} + b_\delta + \sqrt{\mathcal{E}(\mathcal{E} + 2b_\delta)}\bigr)}. \tag{84}$$

Then:

- *(i) On U, $0 < x < u + \delta$, and $1 - x - (1-u)\mu > (1-\hat{h})(1-\hat{\mu}) - \delta > 3\delta$; Sl is smooth on U, including along $z = 0$, and $y_1 \neq 0$.*
- *(ii) $\mathrm{Sl} = \mathrm{Hopf} \circ \mathrm{Dil}$, where $\mathrm{Dil}(\xi, z) = (\xi', x, T)$ with $\xi' = \sqrt{(1-u)/(1-x)}\,\xi$, $\pi|\xi'|^2 < 1$, and $\mathrm{Hopf}(\xi', x, T) = \sqrt{(1-x)/\pi}\,e^{-2\pi i T}\bigl(\sqrt{1-\pi|\xi'|^2}, \sqrt{\pi}\,\xi'\bigr)$.*
- *(iii) With $\lambda_0 = \frac{1}{2}\operatorname{Im}(\bar{y}^{\mathsf{T}} dy)$,*

$$\mathrm{Sl}^*\lambda_0 = (1-u)\lambda_{\mathrm{base}} - (1-x)\,dT, \qquad \mathrm{Sl}^*\omega_0 = \Omega_1.$$

- *(iv) $\pi|\mathrm{Sl}(\xi, z)|^2 = 1 - x$, and $a < 1 - \hat{h} - \delta < \pi|\mathrm{Sl}|^2 < 1$ on U.*
- *(v) Sl is injective, with inverse on its image $x = 1 - \pi|y|^2$, $e^{-2\pi i T} = y_1/|y_1|$ with $T \in (0, 1)$, $z = \Psi^{-1}(x, T)$, $\xi = e^{2\pi i T}(y_2, \dots, y_n)/\sqrt{1-u}$. Hence Sl is a symplectic embedding of (U, Ω_1) onto an open subset of S_a.*
- *(vi) If $I \geq 0$, $\mathcal{K} \subset U$ is compact and $u \leq I$ on $\mathcal{K}$, then $x \geq x_*(I) > 0$ on $\mathcal{K}$, with $x_*(I)$ given by (84), and $|T - \frac{1}{2}| \leq \frac{1}{2}\sqrt{1 - (1 + \mathcal{E}/b_\delta)^{-2}} < \frac{1}{2}$ on $\mathcal{K}$, where $\mathcal{E} = I + b_\delta(\frac{\pi}{2} - 1)$. Hence $\mathrm{Sl}(\mathcal{K})$ lies in the compact set $\{1 - \hat{h} - \delta \leq \pi|y|^2 \leq 1 - x_*(I)\} \subset S_a$.*

Proof. (i) Proposition 11.7(iii) with parameter 2δ gives $0 < x < \pi|z|^2 + \delta = u + \delta$. Hence on U

$$1 - x - (1-u)\mu > (1-u)(1-\mu) - \delta > (1-\hat{h})(1-\hat{\mu}) - \delta > 4\delta - \delta = 3\delta.$$

The functions x, T are smooth in z (Proposition 11.7(i)), and $u = \pi|z|^2$, $\mu = \pi|\xi|^2$ are polynomials; the radicands $1 - x - (1-u)\mu$ and $1 - u$ are bounded below by positive constants on U; so the formula is smooth on U, and $|y_1|^2 > 3\delta/\pi > 0$.

(ii) $1-\pi|\xi'|^2=\big(1-x-(1-u)\mu\big)/(1-x)>0$ by (i). The first component of Hopf $\circ$ Dil is

$$e^{-2\pi iT}\sqrt{(1-x)/\pi}\,\sqrt{1-\pi|\xi'|^2}=y_1,$$

and the others are

$$e^{-2\pi iT}\sqrt{(1-x)/\pi}\sqrt{\pi}\sqrt{(1-u)/(1-x)}\,\xi=e^{-2\pi iT}\sqrt{1-u}\,\xi.$$

(iii) Write $y=c\,V$ with $c=\sqrt{(1-x)/\pi}\,e^{-2\pi iT}$ and $V=(\sqrt{1-\pi|\xi'|^2},\sqrt{\pi}\,\xi')$, so $|V|=1$. Then $\bar y^{\mathsf T}dy=\bar c\,dc+|c|^2\bar V^{\mathsf T}dV$. Here $\mathrm{Im}(\bar c\,dc)=-2\pi|c|^2dT$, and $\mathrm{Im}(\bar V^{\mathsf T}dV)=\pi\,\mathrm{Im}(\bar\xi'^{\mathsf T}d\xi')=2\pi\lambda_{\mathrm{base}}(\xi')$ since the first entry of V is real. As $\pi|c|^2=1-x$,

$$\mathrm{Sl}^*\lambda_0=(1-x)\big(\lambda_{\mathrm{base}}(\xi')-dT\big).$$

For a positive real function c', $\lambda_{\mathrm{base}}(c'\xi)=\frac12\,\mathrm{Im}\big(c'\bar\xi^{\mathsf T}(\xi\,dc'+c'd\xi)\big)=c'^2\lambda_{\mathrm{base}}(\xi)$, because $\bar\xi^{\mathsf T}\xi\,c'dc'$ is real. With $c'^2=(1-u)/(1-x)$ this gives $\mathrm{Sl}^*\lambda_0=(1-u)\lambda_{\mathrm{base}}(\xi)-(1-x)\,dT$. Its differential is $d[(1-u)\lambda_{\mathrm{base}}]+dx\wedge dT$, and $dx\wedge dT=\Psi^*(dx\wedge dT)=\omega_{\mathbb C}=d\lambda_{\mathrm{fib}}$ (Proposition 11.7(i)). By (82), $\mathrm{Sl}^*\omega_0=\Omega_1$.

(iv) $\pi|y|^2=\pi|c|^2|V|^2=1-x$. By (i), $0<x<u+\delta<\hat h+\delta$, so $1-\hat h-\delta<\pi|y|^2<1$, and $1-\hat h-\delta>1-s+3\delta>a$ because $s-\hat h>4\delta$.

(v) Let $y=\mathrm{Sl}(\xi,z)$. Then $x=1-\pi|y|^2$; since $y_1\neq0$, $y_1/|y_1|=e^{-2\pi iT}$ determines T in the interval $(0,1)$; Ψ is a bijection of $\mathbb C$ onto $(0,\infty)\times(0,1)$, so $z=\Psi^{-1}(x,T)$, hence u; and ξ is read off from the last coordinates. So Sl is injective. $\Omega_1=\omega_{\mathbb C}+(1-u)\omega_{\mathrm{base}}-du\wedge\lambda_{\mathrm{base}}$ is symplectic on $\{u<1\}$: $du\wedge\omega_{\mathbb C}=0$, and $\lambda_{\mathrm{base}}\wedge\omega_{\mathrm{base}}^{n-1}=0$ for degree reasons, so $\Omega_1^n=n(1-u)^{n-1}\omega_{\mathbb C}\wedge\omega_{\mathrm{base}}^{n-1}\neq0$. Since $\mathrm{Sl}^*\omega_0=\Omega_1$, $d\,\mathrm{Sl}$ is injective; $\dim_{\mathbb R}U=2n$, so Sl is a local diffeomorphism, and an injective local diffeomorphism is a diffeomorphism onto its open image. The image lies in S_a by (iv).

(vi) If $u\le I$ on $\mathcal K$, Proposition 11.7(iv) with parameter 2δ (so $b=b_\delta$) gives $x\ge\frac12\big[\mathcal E+b_\delta-\sqrt{\mathcal E(\mathcal E+2b_\delta)}\big]$, which equals $x_*(I)$ because $(\mathcal E+b_\delta)^2-\mathcal E(\mathcal E+2b_\delta)=b_\delta^2$, and gives the bound on T. Then $\pi|y|^2=1-x\le1-x_*(I)$ on $\mathrm{Sl}(\mathcal K)$, and with (iv) this set lies in the closed annulus $\{1-\hat h-\delta\le\pi|y|^2\le1-x_*(I)\}$, which is compact and contained in S_a because $1-\hat h-\delta>a$ and $x_*(I)>0$. □

Proposition 17.8 (Natural slit domain)**.** *For every $\delta>0$, formula* (83)*, using $\Psi_{2\delta}$, defines a smooth injective symplectic map on*

$$V_\delta=\{(\xi,z):u<1,\ 1-x-(1-u)\mu>0\}.$$

It has capacity $1-x$. Let $0<h<s<1$ and $0\le M<1$. If $\delta<(1-h)(1-M)$ and $h+\delta<s$, every compact set with $u\le h$ and $\mu\le M$ lies in $V_\delta\cap\{x<s\}$, and its image is compact in S_a, $a=1-s$. The bound (84) *still holds on such a set whenever $I\ge0$ and $u\le I$.*

Proof. The smoothness, pullback and inverse calculations in the proof of Lemma 17.7 use only $u<1$ and positivity of the radicand. On the stated compact set, the strip inequality gives $1-x-(1-u)\mu>(1-h)(1-M)-\delta>0$ and $x<h+\delta<s$. The set $V_\delta\cap\{x<s\}$ is open and its image has capacity in $(a,1)$. The lower bound for x uses only the strip map and the bound for u, so is unchanged. □

17.6. **Packing.** Below, ρ_i^{dom} is the closed source capacity to which Theorem M2.6 is applied, and ρ_i^+ is the enlarged closed capacity of part (iii) of Construction 20.1.

Theorem 17.9 (One large ball, $n\ge4$)**.** *Let $n\ge4$, $k\ge0$, and let $(A;\rho_1,\dots,\rho_k)$ admit the valid auxiliary tuple $(a;r_1,\dots,r_k)$. Put $\rho_0=A$, $\rho_0^+=\frac12(A+a)$, and let f_0 be the identity map of $\mathbb C^n$. If $k\ge1$, let:*

(a) $h_1,h_2,b_2,h_2',\tau_\Delta,d,\Delta,P$ be as in Lemma 17.1;

(b) (K,q) be the comparison datum of Theorem M3.5 for (Δ,P,τ_Δ,d), with gap $\gamma=\frac{11}{16}d$;

(c) $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$, $\rho_i^+ = \frac{1}{2}(\rho_i + \rho_i^{\mathrm{dom}}) = (11\rho_i + 5r_i)/16$, *and* ψ_i *the maps of Construction 12.4 for the mixed model over* Σ_R *with the handle data of Proposition 17.6, the profile* $P_R = P - 11d/32$ *with the datum* (K, q)*, auxiliary capacities* r_i*, closed source capacities* ρ_i^{dom} *and internal capacities* $(3r_i + \rho_i^{\mathrm{dom}})/4$*;*

(d) $\mathcal{K} = \bigcup_{i\geq 1} \psi_i(B(\rho_i^{\mathrm{dom}}))$, $\mathcal{K}' = \Phi_2\chi(\mathcal{K})$, $\mu_{\mathcal{K}} = \max_{\mathcal{K}'} \mu$, *and*

$$\hat{\mu} = \tfrac{1}{2}(1 + \mu_{\mathcal{K}}), \qquad D = \min\big((1-h_1)(1-\hat{\mu}), s - h_1\big), \qquad \delta_{\mathrm{sl}} = D/2;$$

(e) $f_i = \mathrm{Sl} \circ \Phi_2 \circ \chi \circ \psi_i$ *for* $1 \leq i \leq k$*, with* Sl *the map of Proposition 17.8 for* $\delta = \delta_{\mathrm{sl}}$*, that is, built from* Ψ_D*.*

Put $\zeta = x_*(h_1)$ *of* (84) *with* $\delta = \delta_{\mathrm{sl}}$ *if* $k \geq 1$*, and* $\zeta = 1 - \rho_0^+$ *if* $k = 0$*. Then* $\mu_{\mathcal{K}} < 1$*,* $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ *for* $1 \leq i \leq k$*,* $A < \rho_0^+ < a$*,* $0 < \zeta \leq 1 - \rho_0^+$*, and:*

(i) f_0 *is a symplectic embedding of* $\mathbb{C}^n$*, and* f_i *(*$i \geq 1$*) is a smooth symplectic embedding of an open neighbourhood of* $B(\rho_i^{\mathrm{dom}})$ *into* $B^\circ(1)$*;*

(ii) $f_0(B(\rho_0^+)) = B(\rho_0^+) \subset B^\circ(a)$*, and* $f_i(B(\rho_i^{\mathrm{dom}})) \subset \mathcal{C} = \{1 - h_1 - \delta_{\mathrm{sl}} \leq \pi|y|^2 \leq 1 - \zeta\} \subset S_a$ *for* $i \geq 1$*;*

(iii) the sets $f_0(B(\rho_0^+))$ *and* $f_i(B(\rho_i^{\mathrm{dom}}))$*,* $1 \leq i \leq k$*, are pairwise disjoint and contained in* $B(1-\zeta)$*.*

The same holds with ρ_i^+ *in place of* ρ_i^{dom} *(*$i \geq 1$*), since* $B(\rho_i^+) \subset B(\rho_i^{\mathrm{dom}})$*. Moreover* f_0 *restricts to an embedding of every ball* $B^\circ(\rho)$*,* $\rho \leq 1$*, into* $B^\circ(1)$*; in particular it is defined on* $B^\circ(a)$*, the open domain* $B^\circ(\rho_j^{\mathrm{dom}})$ *of Construction 20.1(iii).*

Proof. The distinguished ball. The auxiliary inequalities give $A < a$, so $A < \rho_0^+ < a$. The identity is symplectic and maps $B(\rho_0^+)$ into $B^\circ(a) \subset B^\circ(1)$. If $k = 0$ this proves everything, since $\zeta = 1 - \rho_0^+$.

Comparison and packing in the model. Let $k \geq 1$. By Lemma 17.1(iii)–(vi), $(\Delta, P, \tau_\Delta, d)$ satisfies the hypotheses of Proposition M3.3, so Theorem M3.5 provides the comparison datum (K, q) for (Δ, P) with gap $\gamma = \frac{11}{16}d > 0$. By Lemma 17.4 and Proposition 17.6 the restricted mixed model on Σ_R is a bordered toric model with handle data, including the correction-core and density-floor conditions of Section 12. By Proposition 16.5, the same datum compares P_R with gap $11d/32$, reserve $11d/64$, and $P_R \leq H_b^R - \sum_i (r_i - |p|)_+$; and $\{|p| \leq r_i\} \subset \tau_\Delta\Delta$ by Lemma 17.1(iii). Since $\rho_i < \rho_i^{\mathrm{dom}} < r_i$, Theorem M2.6 gives smooth symplectic embeddings ψ_i of neighbourhoods of $B(\rho_i^{\mathrm{dom}})$ into $(\operatorname{int} V_\Delta(E_0), \Omega_{\mathrm{mix}})$ with pairwise disjoint images in the interior of the finite model. So $\mathcal{K}$ is a compact subset of that interior.

Affine return. By Lemma 17.4, $\Phi_2 \circ \chi$ is smooth and injective on $\mathcal{U} = \operatorname{int} \mathcal{M}_R$, with symplectic derivative; by the inverse function theorem it is an embedding into the coupled product $(\mathbb{C} \times B^\circ_{n-1}(1), \Omega_1)$, preserving u. Hence $\mathcal{K}'$ is compact, $u \leq h_1$ and $\mu < 1$ on $\mathcal{K}'$, and the maximum $\mu_{\mathcal{K}}$ of the continuous function μ is attained, so $\mu_{\mathcal{K}} < 1$.

Slit-shell map. Here $0 < h_1 < s < 1$, $\mu_{\mathcal{K}} < \hat{\mu} < 1$, $D \leq (1-h_1)(1-\hat{\mu})$ and $D \leq s - h_1$, so $\delta_{\mathrm{sl}} < (1-h_1)(1-\mu_{\mathcal{K}})$ and $h_1 + \delta_{\mathrm{sl}} < s$. By Proposition 17.8 with $h = h_1$, $M = \mu_{\mathcal{K}}$, the open set $W = V_{\delta_{\mathrm{sl}}} \cap \{x < s\}$ contains $\mathcal{K}'$, and $\mathcal{T}_{\mathcal{K}} = \mathrm{Sl} \circ \Phi_2 \circ \chi$ is a smooth symplectic embedding of the open set $\mathcal{U} \cap (\Phi_2\chi)^{-1}(W)$ into S_a. On $\mathcal{K}'$, $\pi|y|^2 = 1 - x > 1 - u - \delta_{\mathrm{sl}} \geq 1 - h_1 - \delta_{\mathrm{sl}}$, and the strip estimate of Proposition 17.8 with $I = h_1$ gives $x \geq \zeta = x_*(h_1) > 0$. Hence $\mathcal{T}_{\mathcal{K}}(\mathcal{K}) \subset \mathcal{C}$. Also $\zeta \leq x < h_1 + \delta_{\mathrm{sl}} < s$, so $1 - \zeta > a > \rho_0^+$ and $1 - h_1 - \delta_{\mathrm{sl}} > a$; thus $\mathcal{C} \subset S_a$.

Assembly. For $i \geq 1$, $\psi_i^{-1}\big(\mathcal{U} \cap (\Phi_2\chi)^{-1}(W)\big)$ is an open neighbourhood of $B(\rho_i^{\mathrm{dom}})$ in the domain of ψ_i, and $f_i = \mathcal{T}_{\mathcal{K}} \circ \psi_i$ is a smooth symplectic embedding of it (Proposition M2.7). The sets $f_i(B(\rho_i^{\mathrm{dom}}))$, $i \geq 1$, are pairwise disjoint because the $\psi_i(B(\rho_i^{\mathrm{dom}}))$ are and $\mathcal{T}_{\mathcal{K}}$ is injective. They lie in $\mathcal{C} \subset \{\pi|y|^2 > a\}$, so they are disjoint from $f_0(B(\rho_0^+)) = B(\rho_0^+)$. Finally $\mathcal{C} \subset B(1-\zeta)$ and $B(\rho_0^+) \subset B(1-\zeta)$. □

18. Dimension six: the polynomial model

We use the cubic and quadric polynomials of [Ope26]. The finite list of coefficients is fixed explicitly; all later choices concern positive margins, inverse roots and flows on compact sets. We retain the convention $d^c = i(\bar\partial - \partial)/(4\pi)$. Thus the Hopf primitive at a nonzero vector Z is

$$\lambda_c(Z) = \frac{c}{2\|Z\|^2}\,\mathrm{Im}\langle Z, dZ\rangle = \pi c\, d^c \log \|Z\|^2.$$

Only this primitive and its exterior derivative enter the affine construction.

18.1. **The curve and the parameters.**

Lemma 18.1 (Affine cubic coordinates)**.** *Let $f(v,w) = vw(v-w) + w - 2v$ and $C = \{(v,w) \in \mathbb{C}^2 : vw \neq 0,\ f(v,w) = 0\}$. The affine curve C is identified with*

$$\Sigma = \{(t, y_1, y_2) : y_1^2 = t^2 - 1,\ y_2^2 = t^2 - 2\}$$

by $(v,w) = (y_1 - t, y_2 - t)$ in the chart $u = 1$. The inverse is

$$t = -\frac{v^2+1}{2v}, \qquad y_1 = t + v, \qquad y_2 = t + w.$$

Its four ends, in homogeneous coordinate notation, have limits $[0:1:0], [0:0:1], [0:1:1], [1:0:0]$.

Proof. The two equations on Σ imply $v, w \neq 0$ and $f(v,w) = 0$. Conversely $f = 0$ and $vw \neq 0$ give the displayed inverse by substitution. The four sign choices in the end coordinates of Lemma 15.1, with $m = 2$, give the stated limits. The formulas are holomorphic on their affine domains; smoothness follows from the explicit quadric charts. □

Write $s = 1 - a$, $b = (2-a)/3$ and $V = 1 - a^3 - \sum_i r_i^3 > 0$, with $a \in \mathbb{Q}$, $1/2 < a < 1$ and $r_i < s$. For a nonempty smaller family let $R = \max_i r_i$; the empty family needs no cubic model.

Lemma 18.2 (Explicit truncation parameters)**.** *Set*

$$\mu = \min(s - R, (2a-1)/6, V/100), \qquad M = \lceil 1/\mu \rceil + 1, \qquad \delta_0 = 1/M,$$

$$h_1 = s - \delta_0, \quad e = s - \delta_0/2, \quad d_1 = s + \delta_0/2, \quad h_2 = b - \delta_0, \quad d_2 = b - \delta_0/2.$$

Then these parameters satisfy (78) *and $\int_\Delta H_0 > \sum_i r_i^3/6$. For any integer $Q \geq 0$ one may take*

$$L = (Q+1)\,\mathrm{den}(a)\,\mathrm{den}(e)\,\mathrm{den}(d_2), \quad A_* = Le,\ B_* = Ld_2,\ D = L(2-a),\ S_* = L(1-a).$$

These are integers, $L > Q$, and $0 < A_* < S_* \leq B_*$, $A_* \leq B_*$, $3B_* < D$, $3S_* < D$, $2S_* < L$.

Proof. All three entries defining μ are positive and $0 < \delta_0 < \mu$. Also $b - s = (2a-1)/3 > 2\delta_0$, which gives every ordering assertion. Expanding the integrated-profile formula of Lemma 16.7 at $(h_1, h_2) = (s - \delta_0, b - \delta_0)$ gives

$$\int_\Delta H_0 \geq (1 - a^3)/6 - 3\delta_0.$$

Indeed, let $I(h_1, h_2)$ be that formula. Then $\partial_{h_1} I = \frac{9}{2}(b - h_1)^2 - \frac{9}{2}(b - h_2)^2 - (s - h_1)^2$ and $\partial_{h_2} I = 9h_1(b - h_2)$. On the segment from (s, b) to $(s - \delta_0, b - \delta_0)$, $0 \leq b - h_2 \leq \delta_0 < \frac{1}{12}$, $0 < h_1 \leq s < \frac{1}{2}$ and $b < \frac{1}{2}$, so $|\partial_{h_1} I| < 2$ and $|\partial_{h_2} I| < 1$. Hence $I(h_1, h_2) \geq I(s, b) - 3\delta_0$, and $I(s, b) = (1 - a^3)/6$ (Lemma 16.7). Since $3\delta_0 < 3V/100 < V/6$, the required integral is positive. The denominator product clears the three rational denominators. The integer inequalities follow by multiplying $0 < e < s < d_2 < b$ by L and using $a > 1/2$. □

18.2. **The fixed coefficient list.**

Definition 18.3 (Central potential and physical primitive)**.** Use the integers of Lemma 18.2 and set

$$\Lambda = \{k \in \mathbb{Z}^2_{\geq 0} : k_1 \leq A_*,\ k_1 + k_2 \leq B_*\}, \quad d_k = D - 3|k|, \quad m_k = (S_* - |k|)_+.$$

For $d > m \geq 0$ let

$$q_{d,m} = (1, v^{d-m}, w^{d-m}, v^{d-m}w^m, v^m w^{d-m}), \quad a_k = \|q_{d_k,m_k}\|^2.$$

For $r = (r_1, r_2) \geq 0$ define

$$Z_a(r) = \sum_{k\in\Lambda} a_k r_1^{k_1} r_2^{k_2}, \qquad \mathsf{M}_a(r)_j = r_j \partial_{r_j} \log Z_a(r).$$

For $P \geq 0$ with $P_1 < A_*$ and $P_1 + P_2 < B_*$, let $r_a(P)$ be the unique nonnegative solution of $\mathsf{M}_a(r) = P$, and set

$$\vartheta_k(y, P) = \frac{a_k(y) r_a(P)_1^{k_1} r_a(P)_2^{k_2}}{Z_a(r_a(P))}, \qquad \beta_{d,m} = \frac{1}{L} d^c \log \|q_{d,m}\|^2,$$

$$\Gamma(p) = \sum_{k\in\Lambda} \vartheta_k(y, Lp)\, \beta_{d_k,m_k}, \qquad \Omega_{\mathrm{phys}} = \omega_{\mathbb{C}^2} + d[\Gamma(p(z))], \quad p_j(z) = \pi|z_j|^2.$$

In the last differential p varies with z; the primitive is horizontal in the base variable. The toric coordinates of the central potential are related to these physical coordinates by the radial disk map of Lemma 18.4.

Lemma 18.4 (Radial coordinates)**.** *The inverse $r_a(P)$ exists and is unique on the stated lower trapezoid. The functions defining the radial disk map and its inverse are smooth across both axes. The central potential* (79)*, expressed in physical coordinates, has the form Ω_{phys} of Definition 18.3. At fixed p its base curvature $d_\Sigma \Gamma(p)$ is positive.*

Proof. For positive radii put $r_j = e^{x_j}$. Then M_a is the gradient of $\log \sum_k a_k e^{\langle k,x\rangle}$. Its Hessian is the covariance matrix of k, which is positive definite because $(0,0), (1,0), (0,1)$ occur. Its gradient maps onto the interior of the trapezoid: strict convexity gives uniqueness, and subtraction of $\langle P, x\rangle$ is coercive when P is interior. On an axis the same argument in one variable applies. Write $\mathsf{M}_a(r)_j = r_j s_j(a, r)$. The polynomials giving the numerators of s_j have positive coefficients and include a positive constant. Thus $s_j > 0$ on the closed quadrant. The disk map is

$$(z_1, z_2) \longmapsto (\sqrt{s_1(a, |z|^2)}z_1, \sqrt{s_2(a, |z|^2)}z_2).$$

Its inverse is obtained by the unique quadrant root, and its derivative is invertible also on the axes; the parameter implicit-function theorem gives joint smoothness. The bridge from physical to cone coordinates uses the inverse disk map:

$$\Xi(y, z) = \big(y, \mathsf{R}^{-1}_{a(y)}(\sqrt{\pi L}\, z)\big),$$

where R_a is the displayed radial disk map. For a prescribed compact source this formula is multiplied away from a neighbourhood of that source by a plateau inside the two outer moment faces. Differentiating the logarithmic potential separates the angular primitive $\sum_j p_j d\theta_j$ from the horizontal primitive $\Gamma(p)$. Reduction of the positive central Kähler form gives positivity of the base form. The radial disk pullback identity includes the vanishing angular terms on the axes. □

18.3. **Explicit homogeneous lifts.** Put

$$Q(X) = X_0X_1 - X_2X_3, \qquad \mathsf{q}(u,v,w) = (vw, u^2, uv, uw).$$

For a plane monomial v^Jw^K with $J, K \le n$, define

$$\alpha = (J+K-n)_+, \quad \beta = n+\alpha-J-K, \quad \gamma = J-\alpha, \quad \nu = K-\alpha, \qquad \mathcal{L}_n(v^Jw^K) = X_0^\alpha X_1^\beta X_2^\gamma X_3^\nu,$$

and extend $\mathcal{L}_n$ linearly to polynomials satisfying those separate degree bounds. All four exponents are nonnegative integers.

Lemma 18.5 (Polynomial lifts and vanishing orders)**.** *For $k \in \Lambda$ and each entry q_i put*

$$P_{k,i}(X) = Q(X)^{k_1}\mathcal{L}_{L-2k_1}(f^{k_2}q_{d_k,m_k,i})(X).$$

This is a homogeneous polynomial of degree L, vanishing to order at least $L - S_*$ at $o = [1:0:0:0]$. If

$$P_{t,\delta}(X)_{k,i} = \delta^{-k_1}t^{-k_2}P_{k,i}(X), \qquad T_\delta(v,w,z) = (v,w,vw+\delta z),$$

then $P_{t,\delta}(vw+\delta z, 1, v, w)$ has a polynomial extension in δ at zero, with value

$$z^{k_1}(f(v,w)/t)^{k_2}q_{d_k,m_k,i}(v,w).$$

Proof. The exponent identities are $\alpha+\gamma = J$, $\alpha+\nu = K$, $2\beta+\gamma+\nu = 2n-J-K$ and $\alpha+\beta+\gamma+\nu = n$. Hence $\mathcal{L}_n(v^Jw^K)(\mathsf{q}(u,v,w)) = u^{2n-J-K}v^Jw^K$. Since $2m_k \le d_k$, each coefficient monomial has separate degrees at most $d_k - m_k$ and total degree at most d_k. The separate degrees of $f^{k_2}q_i$ are at most $2k_2 + d_k - m_k$, while its total degree is at most $3k_2 + d_k$. Writing $n = L - 2k_1$, the integer inequalities give

$$2k_2 + d_k - m_k \le n, \quad L - S_* - k_1 \le n, \quad 3k_2 + d_k + (L - S_* - k_1) \le 2n.$$

The last two imply $\beta+\gamma+\nu \ge L - S_* - k_1$ for every monomial in its lift. The factor Q^{k_1} contributes another k_1 to the order at o. The homogeneity follows from $2k_1 + n = L$. Finally $Q(vw+\delta z, 1, v, w) = \delta z$, so the negative powers of δ cancel term by term. At zero use $\mathsf{q}(1,v,w) = (vw, 1, v, w)$. □

18.4. **Area and exhaustion.** In the diagonal-quadric coordinates set $\Sigma_l = \{|t| \le 5 + l\}$, $l \in \mathbb{Z}_{\ge 0}$. These are compact, nested bordered surfaces exhausting C; the explicit handle and its dual form lie in $|t| < 3$, hence in $\operatorname{int}\Sigma_0$. Let $H_l(p) = \int_{\Sigma_l} d_\Sigma\Gamma(p)$.

Proposition 18.6 (Area profile with a quantitative tail)**.** *On Δ one has*

$$0 \le H_0(p) - H_l(p) \le E_l \quad (l \ge 32), \qquad H_0(p) = 9(b - |p|) - 2(s - |p|)_+.$$

Put $R_* = 1/16$ and fix the end cutoff

$$\chi_{\mathrm{end}}(q) = S\left(\frac{q + R_*}{R_*/2}\right)\left[1 - S\left(\frac{q - R_*/2}{R_*/2}\right)\right].$$

Let $K = 1 + \max(0, \sup_{q\in\mathbb{R}} |\chi'_{\mathrm{end}}(q)|)$, let M_* bound the finitely many regular end primitives, and let $B_{\log}$ bound the absolute logarithms of the regular coefficient norms, on the four end discs $|\zeta|^2 \le R_*$. These bounds are the maxima plus one. With $q_* = |c|(2B_{\log} + \#\Lambda)$ and $c = 1/(L\pi)$, one may take

$$E_l = 64KR_*M_*\sqrt{\frac{R_*}{l+1}} + \frac{16q_*KR_*}{-\log(R_*/(l+1))}.$$

In particular, if $C_* = 1 + 64KR_*M_* + 16q_*KR_*$ and $\eta > 0$, then

$$\mathfrak{m}(\eta) = \max\left(32, \left\lceil (2C_*/\eta)^2 \right\rceil, \left\lceil e^{2C_*/\eta} \right\rceil\right)$$

satisfies $|H_l - H_0| < \eta$ for $l \ge \mathfrak{m}(\eta)$, uniformly on Δ.

Proof. We write Ω_p for the base form $d_\Sigma\Gamma(p)$ and Γ_p for its primitive, a convex combination of the weighted Fubini–Study primitives of the five homogeneous coefficients, with weights the selection probabilities below. The proof has five steps.

Step 1: end orders. At the ends [0:1:0], [0:0:1], [0:1:1], [1:0:0] one has $(v,w)\sim(-2t,-1/t)$, $(-1/(2t),-2t)$, $(-2t,-2t)$, $(-1/(2t),-1/t)$ as $t\to\infty$ (Lemma 18.1). The pole orders of the five entries of $q_{d,m}$ are read from

$$(u^d, u^m v^{d-m}, u^m w^{d-m}, v^{d-m}w^m, v^m w^{d-m}).$$

so the maximal order is $o=d-m$ at the marked ends [0:1:0], [0:0:1], $o=d$ at [0:1:1] and $o=0$ at [1:0:0] (recall $2m\le d$). Steps 2–5 show that the total curvature is $H_0(p)$, with the stated rate.

Step 2: end charts. The curve has four ends, indexed by the sign choices $\varepsilon\in\{\pm\}^2$ of the two square roots $y_j=\pm t\sqrt{1-j\,t^{-2}}$. On each end use $\zeta=t^{-1}$. If $|\zeta|^2\le R_*=\frac{1}{16}$, then $|j\zeta^2|\le\frac18$ for $j=1,2$, so $\mathrm{Re}(1-j\zeta^2)\ge\frac78>0$; the principal roots $\sqrt{1-j\zeta^2}$ are smooth and nonvanishing, and the end map $\zeta\mapsto(t,y)$ is a diffeomorphism of the punctured disc $\{0<|\zeta|^2\le R_*\}$ onto a neighbourhood of the end. On this disc every coefficient is ζ^{-o} times a smooth nonvanishing function of ζ, where o is its end order from Step 1.

Step 3: decomposition on an end. On the punctured disc of end ε,

$$\Gamma_p=\beta_\varepsilon-\tfrac12\,d_\varepsilon\,d\theta,\qquad d\theta=\text{the angular form of }\zeta,$$

where β_ε is the selection-weighted sum of the regular parts of the coefficient primitives, a smooth one-form with $\|\beta_\varepsilon\|\le M_*$, and $d_\varepsilon(\zeta)=c\sum_k\pi_k(\zeta)\,o_k(\varepsilon)$ is the residue: the average of the weighted end orders $o_k(\varepsilon)$ under the selection probabilities $\pi_k(\zeta)$. The probabilities are the weights $\pi_k=\vartheta_k(y,Lp)$ of Definition 18.3; they are positive, sum to one and have mean $\sum_k\pi_k k=Lp$. The decomposition holds because $d^c\log a_k=d^c\log c_k-o_k\,d^c\log|\zeta|^2$ with $c_k=|\zeta|^{2o_k}a_k$ smooth and positive and $d^c\log|\zeta|^2=d\theta/(2\pi)$; thus $\beta_\varepsilon=L^{-1}\sum_k\pi_k d^c\log c_k$ and $\|\beta_\varepsilon\|\le M_*$.

Step 4: the residue. Put $\ell=-\log|\zeta|^2>0$ and $P=Lp$. On end ε write $a_k=c_k\,|\zeta|^{-2o_k}=c_k e^{\ell o_k}$, with $c_k>0$ the regular coefficient norm of Step 2, so $|\log c_k|\le B_{\log}$. The selection probabilities $\pi_k\propto a_k r^k$, $r=r_a(P)$, have mean $\sum_k\pi_k k=P$. They are the unique maximiser of

$$\mathcal{J}(\mu)=\sum_k\mu_k\log a_k-\sum_k\mu_k\log\mu_k$$

over probability vectors μ on Λ with $\sum_k\mu_k k=P$: $\mathcal{J}$ is strictly concave, and the Lagrange condition $\log\mu_k=\log a_k+\langle x,k\rangle+\text{const}$ at an interior critical point is the defining form $\pi_k\propto a_k r^k$ with $r_j=e^{x_j}$ (if some $P_j=0$, the same holds on the face $k_j=0$).

At the ends [1:0:0] and [0:1:1] the order o_k is 0, resp. $d_k=D-3|k|$, an affine function of k, so $d_\varepsilon=c\sum_k\pi_k o_k$ is c times this affine function at P; this is c_ε, with no error. At the two marked ends $o_k=d_k-m_k$ with $m_k=(S_*-|k|)_+$. Let $A(k)=S_*-|k|$ if $|P|<S_*$ and $A=0$ otherwise; then A is affine, $A(k)\le m_k$ on Λ and $A(P)=(S_*-|P|)_+$. Put $e_k=m_k-A(k)\ge0$. The set $F=\{k\in\Lambda:e_k=0\}$ consists of the lattice points of the polygon $\{k\ge0,\ k_1\le A_*,\ |k|\le S_*\}$, resp. $\{k\ge0,\ k_1\le A_*,\ S_*\le|k|\le B_*\}$, whose vertices are integral; since $P_1<A_*$ and $|P|<B_*$, P lies in this polygon, so some probability vector ν supported on F has mean P. Since $\log a_k=\log c_k+\ell(d_k-A(k))-\ell e_k$ and $d_k-A(k)$ is affine, $\sum_k\mu_k\ell(d_k-A(k))$ is the same for all admissible μ, and $\mathcal{J}(\pi)\ge\mathcal{J}(\nu)$ gives

$$\ell\sum_k\pi_k e_k\le\sum_k(\pi_k-\nu_k)\log c_k-\sum_k\pi_k\log\pi_k+\sum_k\nu_k\log\nu_k\le2B_{\log}+\log\#\Lambda.$$

As $\sum_k \pi_k e_k = \sum_k \pi_k m_k - (S_* - |P|)_+$, and Jensen's inequality for the convex function $k \mapsto m_k$ gives the lower bound,

$$0 \le \sum_k \pi_k m_k - (S_* - L|p|)_+ \le \frac{2B_{\log} + \#\Lambda}{-\log|\zeta|^2}.$$

With $c_\varepsilon = c\,(d(p) - h(p))$ at the marked ends, $d(p) = D - 3L|p|$, $h(p) = (S_* - L|p|)_+$, this is

$$|d_\varepsilon(\zeta) - c_\varepsilon| \le \frac{q_*}{-\log|\zeta|^2}, \qquad q_* = |c|\,(2B_{\log} + \#\Lambda),$$

uniformly in $p \in \Delta$. The limiting orders are $d-h$, $d-h$, d and 0, so $\pi \sum_\varepsilon c_\varepsilon = L^{-1}(3d(p) - 2h(p)) = H_0(p)$.

Step 5: the cutoff and Stokes. Let $\chi_l(\zeta) = \chi_{\text{end}}((l+1)|\zeta|^2)$ on each end disc and $\psi_l = 1 - \chi_l$ on the curve, extended by 1 off the ends. Since $\chi_{\text{end}} = 1$ on $[-R_*/2, R_*/2]$, ψ_l vanishes where $(l+1)|t|^{-2} < R_*/2 = \frac{1}{32}$, that is for $|t|^2 > 32(l+1)$. For $l \ge 32$ the radius $R_l = 5 + l$ of Σ_l satisfies $R_l^2 \ge (l+3)^2 \ge 32(l+1)$, because $(l+3)^2 - 32(l+1) = l^2 - 26l - 23 \ge 0$ for $l \ge 27$. Hence $\operatorname{supp}\psi_l \subset \Sigma_l$. On each end, $d\psi_l$ is supported in the annulus $A_l = \{R_*/(2(l+1)) \le |\zeta|^2 \le R_*/(l+1)\}$, and $|\chi'_{\text{end}}| \le K$. Stokes' theorem gives

$$\int_\Sigma \psi_l\,\Omega_p = -\int_\Sigma d\psi_l \wedge \Gamma_p = \sum_\varepsilon \int_{A_l} \Big(d\chi_l \wedge \beta_\varepsilon - \tfrac{1}{2} d_\varepsilon\, d\chi_l \wedge d\theta \Big).$$

For the first term, $|d\chi_l| \le 2K(l+1)|\zeta|$ and the area of A_l is at most $\pi R_*/(l+1)$, so it contributes at most $16KR_*M_*\sqrt{R_*/(l+1)}$ on each end. For the second, $-\frac{1}{2}\int_{A_l} d\chi_l \wedge d\theta = \pi$, since χ_l is radial and decreasing on A_l in the orientation induced from Σ, because χ_l runs once between its boundary values 1 and 0 across the annulus; so replacing d_ε by c_ε produces πc_ε with an error at most $4KR_*\,q_*/(-\log(R_*/(l+1)))$ by Step 4, since $|\zeta|^2 \le R_*/(l+1)$ on A_l. Summing over the four ends,

$$\Big| \int_\Sigma \psi_l\,\Omega_p - H_0(p) \Big| \le 64KR_*M_*\sqrt{\frac{R_*}{l+1}} + \frac{16q_*KR_*}{-\log(R_*/(l+1))} = E_l.$$

Since $E_l \to 0$ and $\psi_l \uparrow 1$, monotone convergence gives $\int_\Sigma \Omega_p = H_0(p)$. As Ω_p is positive, $0 \le \psi_l \le 1$ and $\operatorname{supp}\psi_l \subset \Sigma_l$, we get $\int_\Sigma \psi_l \Omega_p \le H_l(p) \le H_0(p)$, and therefore $0 \le H_0(p) - H_l(p) \le E_l$.

For $l \ge \mathfrak{m}(\eta)$ the first term of E_l is at most $C_*/(4\sqrt{l+1}) < \eta/2$ and the second at most $C_*/\log(16(l+1)) < \eta/2$, which proves the uniform modulus. □

19. Dimension six: compact transfers and the large ball

The smaller balls are first packed into the polynomial model of Section 18. We transfer their compact images through the cubic and quadric polynomial families, insert the large ball by a bubble at the distinguished point, and apply one common map into the unit ball. Each transfer preserves the form exactly.

19.1. **Packing over a bordered base.** Let $r_i' \in (\rho_i, r_i)$ and use the parameters of Lemma 18.2. Put $G = H_0 - \sum_i (r_i - |p|)_+$, $\tau = (1 + \max_i r_i/h_1)/2$ and $d_0 = 9(b - h_2)$. The cap supports lie in $\tau\Delta$, and $G \ge d_0$ off $\tau\Delta$. The function G is concave: it is the sum of the concave profile H_0 and the negatives of convex caps. Write $\overline{G} = |\Delta|^{-1}\int_\Delta G$ and set

$$\eta = \frac{1}{32}\min(d_0, \overline{G}) > 0.$$

Set

$$d = \min\big((\overline{G} - \eta)/2, d_0/4\big), \quad \gamma = 11d/16, \quad r_* = 11d/64, \quad l = \max\big(\mathfrak{m}(\eta), \mathfrak{m}(r_*/32) + 2\big).$$

These are positive; the level l reserves two outer collar levels.

Proposition 19.1 (Inner packing)**.** *The physical form Ω_{phys} admits pairwise disjoint symplectic embeddings ϕ_i of neighbourhoods of $B(r_i')$, with compact images in $\Sigma_l \times \{z : p(z) \in \Delta\}$ and with moments strictly inside the two outer faces. All maps are obtained from the finite comparator for $G - \eta$ and the common handle construction.*

Proof. Lemma 18.2 gives $\overline{G} > 0$; the positivity bound for H_0 gives the other positive entries of η. Proposition 18.6 gives $H_l > H_0 - \eta$. For the finite comparator use $P = G - \eta$, outer lower bound $d_0/2$, parameter d and gap γ. The factor $1/32$ ensures $P \geq d_0/2$ off the core and positive mean. Also $P < H_l - \sum_i (r_i - |p|)_+$. To represent $-P$ as a maximum of affine functions, take every sum consisting of one negative affine piece of H_0, one of the two pieces of each cap, and the constant η. The maximum of these sums is $-H_0 + \sum_i (r_i - |p|)_+ + \eta$. The handle lies in $\operatorname{int} \Sigma_0$. Put $R = 5 + l$. The second entry in the maximum defining l ensures that the curvature mass between Σ_{l-2} and Σ_l is less than $r_*/32$, uniformly on Δ: positivity and the tail bound give $0 \leq H_l - H_{l-2} \leq H_0 - H_{l-2} < r_*/32$. Use the fixed boundary cutoff and background coefficient

$$\chi_\partial(t) = S\Big(\frac{|t|^2 - (R-2)^2}{(R-1)^2 - (R-2)^2}\Big), \qquad c_{\mathrm{bg}} = \min\Big(\frac{r_*}{128}, \frac{\kappa_{\min}}{2}\Big) > 0.$$

The correction atlas is the family of Lemma 15.3 with $m = 2$ and $R = R - 1$; its inner boxes cover $\{|t| \leq R - 1\}$, which contains the support of $1 - \chi_\partial$ and the handle, and its closed outer boxes lie in $\operatorname{int} \Sigma_l$ (Lemma 10.5). Here $\kappa_{\min} > 0$ is the minimum of the ratio $d_\Sigma \Gamma(p)/\sigma$, $\sigma = dd^c \log(1 + |t|^2 + |y_1|^2 + |y_2|^2)$, over the union of these closed boxes times Δ, a compact set on which $d_\Sigma \Gamma(p) > 0$ (Lemma 18.4). Thus the correction region of Definition M4.1(d) is this union, with $\Lambda = \sigma$ and $\kappa_- = \kappa_{\min}$, and it contains the correction core, as required in §12. The reference mass is $\int_{\Sigma_l} \sigma \leq 2^m = 4$ (Lemma 15.5), so the removed form

$$(1 - \chi_\partial)\big(d_\Sigma \Gamma(p) - c_{\mathrm{bg}} \sigma\big)$$

has handle mass $H(p)$ satisfying

$$H(p) \geq H_l(p) - r_*/16.$$

Indeed the physical collar costs less than $r_*/32$ and the background costs at most $4c_{\mathrm{bg}} \leq r_*/32$. The compact-coordinate primitive and annular correction are then the fixed formulas of Construction 12.2. In the cylinder use the working half-width $b_H = 1/16$ and circle-clock parameter $1/4$. For $N = k + 1$ the layer step is $D_H = 2b_H/(3N + 1)$, with layer intervals $[-b_H + (3i + 1)D_H, -b_H + (3i + 2)D_H]$, $0 \leq i < N$. For the enlarged inner capacities $\widetilde{r}_i = (3r_i + r_i')/4$, use the cap smoothing scale $(r_i - \widetilde{r}_i)/2$ and thin-strip parameter $(r_i - \widetilde{r}_i)/4$. All these layer parameters are positive. The comparison inequality and the cap bound give a residual of at least $\gamma - r_*/16 > 0$ after redistribution. Thus the common layer and Moser formulas apply to H without assuming its concavity and give the stated neighbourhood embeddings. Positivity and the strict outer face margins persist on neighbourhoods of the compact ball images. □

Let $K_1 = \bigcup_i \phi_i(B(r_i'))$. In the subsequent steps only this compact set, and its successive compact images, need to be controlled.

19.2. **Compact exact transfer.** We use a complete auxiliary flow before choosing its stopping time. Its cutoff and the stopping rule are described here; all extrema below are of continuous scalar functions on the displayed compact sets.

For a product of the base curve with a vector space, take the finite coordinate boxes of the quadric atlas covering the compact base projection. Write ρ_i for their coordinate plateaux: $\rho_i = 1$ on the inner box and its closed support lies inside the chart. Let $G_i(t, x)$ be the squared

determinant of $d\Gamma_t$ in that product chart, evaluated at x, and put

$$g(t,x)=\sum_i \rho_i(x_{\mathrm{base}})G_i(t,x), \qquad \mu=\min\bigl(1,\min_K g(0,\cdot)\bigr)>0,$$

$$R=\max\bigl(0,\max_K |x_{\mathrm{fibre}}|^2_{\mathrm{coord}}\bigr).$$

The summands are extended by zero outside their charts. This extension is smooth because the closed support of ρ_i lies strictly inside the chart. Use

$$\chi(t,x)=S(4g(t,x)/\mu-1)\,S(2-t^2)\,S\bigl(2(R+1-|x_{\mathrm{fibre}}|^2_{\mathrm{coord}})\bigr). \tag{85}$$

Its closed support lies where $g\geq \mu/4>0$, hence where the form is nondegenerate and at least one of the finite charts applies. It is spatially compact: the base lies in the union of the closed boxes, and $|x_{\mathrm{fibre}}|^2_{\mathrm{coord}}\leq R+1$.

In each chart, pull back the full ambient covector representing Γ_t, including its vertical components, differentiate it, and solve the resulting finite linear system

$$\iota_{X_t}d\Gamma_t=-\partial_t\Gamma_t.$$

Take the first applicable chart in the fixed finite order and multiply its field by χ. The chart solutions agree on overlaps. Consequently this rule defines a globally smooth, spatially compact field W_t, extended by zero away from the cutoff support. Spatial compactness gives its flow Ψ_s on $-2<s<2$, determined by the ambient-coordinate ODE and continuity of the lifted curve. The displayed chart inverses prove smoothness of this lift.

Lemma 19.2 (Exact transfer with a finite stopping rule)**.** *Let Γ_t be a smooth one-form family, with $d\Gamma_0$ nondegenerate on a compact set K. Let $A_t:U\to N$ be a smooth family, an embedding for $0<t\leq\delta$, with $A_t^*\Omega_t=d\Gamma_t$ on the open neighbourhood U of K. Let $T\subset N$ be open. Suppose the complete field W above comes with finitely many smooth scalar control functions $b_j(t,x)$ such that*

$$b_j(0,x)>0 \quad (x\in K), \qquad \bigl[\,b_j(t,x)>0 \text{ for all } j\,\bigr]\Longrightarrow \chi(t,x)=1,\ x\in U,\ A_t(x)\in T.$$

Define

$$c_*=\min\bigl(1,\min_{j,x\in K} b_j(0,x)\bigr), \tag{86}$$

$$V_*=1+\max\left(0,\max_{j,(s,x)\in[-1,1]\times K}\bigl|\partial_s\bigl[b_j(s,\Psi_s x)\bigr]\bigr|\right), \tag{87}$$

$$\tau_*=\min\left(\delta/2,\,1/4,\,c_*/(4V_*)\right). \tag{88}$$

For $0<t\leq\tau_*$, the map $A_t\circ\Psi_t$ is a symplectic embedding of a neighbourhood of K into T. Empty minima and maxima are interpreted by the inserted constants 1 and 0.

Proof. The field is defined before τ_*, so the derivative in (87) is continuous on the compact set $[-1,1]\times K$. We have $c_*>0$ and $V_*>0$. For $|s|\leq 2\tau_*$, the mean-value estimate gives

$$b_j(s,\Psi_s x)\geq b_j(0,x)-V_*|s|\geq c_*/2>0 \qquad (x\in K).$$

Thus the image of $[-\tau_*,2\tau_*]\times K$ under $(s,x)\mapsto(s,\Psi_s x)$ is a compact set on which every control function is positive. Positivity persists on a neighbourhood of this compact set. On that neighbourhood $W=X$, so Cartan's formula gives

$$\frac{d}{ds}\Psi_s^*d\Gamma_s=\Psi_s^*d(\partial_s\Gamma_s+\iota_{X_s}d\Gamma_s)=0.$$

Consequently $(A_t\Psi_t)^*\Omega_t=d\Gamma_0$. The flow map Ψ_t is a diffeomorphism; composing its restriction with the embedding A_t gives the required neighbourhood embedding. The target control functions give its image in T. □

On a vector space the same construction uses a fixed basis, with $\ell = \min_K \det(d\Gamma_0)^2$ and $R = \max_K |x|^2_{\mathrm{coord}}$. Its cutoff is

$$S(4\det(d\Gamma_t)^2/\ell - 2)\, S\big(2(R + 1 - |x|^2_{\mathrm{coord}})\big)\, S(2 - t^2).$$

The three plateau control functions are $4\det(d\Gamma_t)^2/\ell - 3$, $2(R + 1 - |x|^2_{\mathrm{coord}}) - 1$, and $1 - t^2$. Together with the displayed source and target control functions, they enter (86)–(88). The same rule, applied uniformly, gives a common bubble parameter in Proposition 19.5.

19.3. **The cubic transfer.** Here the smaller family is nonempty. The empty-family case in Theorem 19.7 is the direct inclusion and uses no cubic transfer.

Define the cone polynomial on $\mathbb{C}^2 \times \mathbb{C}^2$ by

$$Z(v, w, z_1, z_2)_{k,i} = z_1^{k_1} z_2^{k_2} q_{d_k, m_k, i}(v, w), \qquad \lambda_{\mathrm{cone}} = Z^* \lambda_{1/(L\pi)}.$$

The $(k, i) = ((0, 0), 0)$ coordinate equals one, so this primitive is everywhere smooth. For $t > 0$ define

$$\mathcal{N}_t(v, w, z) = (v, w, z, f(v, w)/t), \qquad \lambda_t^{\mathrm{cub}} = \mathcal{N}_t^* \lambda_{\mathrm{cone}}, \quad \Omega_t^{\mathrm{cub}} = d\lambda_t^{\mathrm{cub}}.$$

For a nonempty compact physical source K_1, write $\zeta = \sqrt{\pi L}\, z$ and put

$$\delta_b = \frac{1}{2} \min_{(y,z)\in K_1} \min\big(A_* - |\zeta_1|^2,\ B_* - |\zeta_1|^2 - |\zeta_2|^2\big) > 0,$$

$$\chi_b(\zeta) = S_{\delta_b/2, \delta_b}(A_* - |\zeta_1|^2) S_{\delta_b/2, \delta_b}(B_* - |\zeta_1|^2 - |\zeta_2|^2).$$

The bridge is $\Xi(y, z) = (y, \chi_b(\zeta) \mathsf{R}^{-1}_{a(y)}(\zeta))$ on the disk domain and zero in its fibre coordinate outside that domain. Its base coordinate remains y. The plateau is zero near the boundary of the disk domain and one on a neighbourhood of K_1. Thus this is an explicit globally smooth extension of the inverse radial map, with its exact pullback on that neighbourhood.

Proposition 19.3 (Cubic transfer)**.** *There are an explicit $t > 0$ and a smooth symplectic embedding g_1 of a neighbourhood of K_1 into $\{(v, w, z) : vw \neq 0\}$, with $g_1^* \Omega_t^{\mathrm{cub}} = \Omega_{\mathrm{phys}}$.*

Proof. Use a normal coordinate ξ to the cubic and write $g(z) = |f_v(z)|^2 + |f_w(z)|^2$ for $z = (v, w)$. Let K_C and H_C be the base and disk projections of $\Xi(K_1)$, and set

$$\mathcal{D} = \{(\iota y, \xi) : (y, z_1, \xi) \in K_C \times H_C\}, \qquad m = \frac{1}{2} \min_{y\in K_C} g(\iota y) > 0, \qquad R_1 = 1 + \max_{Y\in\mathcal{D}} \mathrm{nsq}(Y),$$

where $\mathrm{nsq}(v, w, \xi) = |v|^2 + |w|^2 + |\xi|^2$. For the displacement and derivative bounds below use the product maximum norm. Define

$$r_N = \sqrt{R_1} + 1, \qquad R_N = 3r_N^2, \qquad G_N = \max_{\|z\|\le r_N} \|Dg(z)\|,$$

$$h_m(s) = sS_{m/4, m/2}(s) + \frac{m}{2}\big(1 - S_{m/4, m/2}(s)\big), \qquad c_R(s) = 1 - S_{R_N, R_N+1}(s),$$

$$X(v, w, \xi) = c_R(\mathrm{nsq}(v, w, \xi)) \left(h_m(g(v, w))^{-1}(\overline{f_v}\xi, \overline{f_w}\xi), 0\right).$$

The denominator is at least $m/4$, and X is spatially compact. Let

$$C_N = \sup_Y \|X(Y)\|, \qquad d_N = \min\left(1, \frac{m}{2(G_N + 1)}\right), \qquad \epsilon_N = \min\left(1, \frac{d_N}{C_N + 1}\right) > 0.$$

Take ψ_t to be the complete flow of $S^{\mathrm{plateau}}_{-2,-1,1,2}(t)X$. For initial $Y = (v, w, \xi)$ in the open tube

$$\mathcal{W} = \{Y : \mathrm{nsq}(Y) < R_1,\ g(v, w) > m\},$$

the displacement on $|t| \le \epsilon_N$ is less than $d_N \le 1$. It keeps the product maximum norm below r_N, the gradient square above $m/2$ and $\mathrm{nsq} < 3r_N^2 = R_N$. The regularized field therefore agrees

with the normal lift, and $f(\psi_t(v,w,\xi)_{\mathbb{C}^2}) = f(v,w) + t\xi$ on this tube. The initial data $\mathcal{D}$ lie strictly inside $\mathcal{W}$, since $g \geq 2m$ and nsq $\leq R_1 - 1$. In particular the constraint is $f = t\xi$ when $f(v,w) = 0$. Set $A_t(y, z_1, \xi) = (\psi_t(\iota y, \xi)_{\mathbb{C}^2}, z_1)$. For $t \neq 0$, $\xi = f/t$ is recovered from the image, and reversing ψ_t recovers y, so A_t is an embedding on a common neighbourhood. The map $\mathcal{N}_t \circ A_t$ has the smooth extension $(\psi_t(\iota y, \xi)_{\mathbb{C}^2}, z_1, \xi)$ at $t = 0$. Pull back λ_{cone} along this extension, then along Ξ, to obtain Γ_t. Its initial curvature is Ω_{phys} by the radial bridge identity. The target $vw \neq 0$ contains $A_0\Xi(K_1)$. For the stopping rule, use the three plateau control functions $4g(t,x)/\mu - 2$, $1 - t^2$ and $2(R + 1 - |x_{\mathrm{fibre}}|^2_{\mathrm{coord}}) - 1$ of (85), whose positivity implies $\chi = 1$, together with

$$A_* - |\zeta_1|^2 - \delta_b, \quad B_* - |\zeta_1|^2 - |\zeta_2|^2 - \delta_b, \quad R_1 - |y|^2,$$
$$|\nabla f(y_{\mathrm{base}})|^2 - m, \quad |(A_s\Xi(x))_v|^2, \quad |(A_s\Xi(x))_w|^2.$$

Here ζ is the scaled physical disk coordinate, δ_b is the bridge margin, and

$$y = (\iota(\Xi(x)_{\mathrm{base}}), \Xi(x)_{\mathrm{normal}}).$$

The constants R_1, m, δ_b are the explicit margins above. These nine smooth control functions are positive at $s = 0$ on K_1. Use $\delta = \epsilon_N$ in Lemma 19.2. Extend Ξ to the ambient affine coordinates using the same positive coefficient formulas. If $\widetilde{F}_s$ is the resulting smooth cone-coordinate map, the ambient covector used to compute the field is

$$\alpha_s(p) = \lambda_{\mathrm{cone}}(\widetilde{F}_s(p)) \circ D\widetilde{F}_s(p).$$

Its full Jacobian includes the dependence on both fibre coordinates. Use its complete cutoff flow in (87), and set $t = \tau_*$ from (88). Lemma 19.2 gives the assertion. □

19.4. **The quadric tube.** Let $K_2 = g_1(K_1)$ and use $P_{t,\delta}$ of Lemma 18.5. Define

$$\lambda^{\mathrm{out}}_{t,\delta} = [P_{t,\delta}(z, 1, v, w)]^* \lambda_{1/(L\pi)}, \qquad \Omega^{\mathrm{out}}_{t,\delta} = d\lambda^{\mathrm{out}}_{t,\delta}.$$

Again the zero-weight constant coordinate is one. Thus the primitive is smooth everywhere in this chart, including points where some other coefficients vanish.

Proposition 19.4 (The quadric tube map)**.** *There are an explicit $\delta > 0$ and a symplectic embedding g_2 of a neighbourhood of K_2 into $\{z \neq 0\}$, with $g_2^*\Omega^{\mathrm{out}}_{t,\delta} = \Omega^{\mathrm{cub}}_t$.*

Proof. Put $A_\delta = T_\delta$. Lemma 18.5 supplies a smooth extension $\Gamma_\delta = T_\delta^* \lambda^{\mathrm{out}}_{t,\delta}$ at zero, and its value at zero is λ^{cub}_t. For $\delta > 0$, T_δ is an affine shear with invertible last coordinate. Moreover $T_0(v,w,z)$ has last coordinate vw, which is nonzero on K_2. The initial form is nondegenerate there because g_1 is a symplectic neighbourhood embedding. Lemma 19.2, with target $\{z \neq 0\}$, gives the assertion. □

Change from the chart $X_1 = 1$ to $X_0 = 1$ by

$$J(v,w,z) = (z^{-1}, v/z, w/z).$$

It is a diffeomorphism on $z \neq 0$. The compact set $K' = Jg_2(K_2)$ avoids the origin. Set $\rho = \min_{x \in K'} \max_j |x_j| > 0$ and $e(x) = \sum_{j=0}^{2} |x_j|^2$. Take the disjoint open sets $T_{\mathrm{out}} = \{e(x) > \rho^2/4\}$ and $T_{\mathrm{in}} = \{e(x) < \rho^2/4\}$. If the smaller family is empty, omit these steps.

19.5. **The polynomial bubble.** Let $A < r_0 < a$, and take $L = (Q+1)\operatorname{den}(a)\operatorname{den}(e)\operatorname{den}(d_2)$ in Lemma 18.2 with $Q = \lfloor 3/(a-r_0)\rfloor + 1$, so that $L > Q$. Set

$$m_0 = \lfloor r_0 L \rfloor, \quad M_1 = m_0 + 1, \quad M_2 = m_0 + 2, \quad M_3 = m_0 + 3, \qquad d(\epsilon) = (1-\epsilon)/L.$$

Then $r_0 < M_1/L$ and $M_1 < M_2 < L - S_*$, $M_2 < M_3$. Let V_j denote the ordered-monomial Veronese map, whose coordinates are indexed by words of length j and satisfy $\|V_j(X)\|^2 = \|X\|^{2j}$. For $X = (X_0, X')$ put $B_\epsilon X = (\epsilon X_0, X')$ and define the degree-L polynomial vector

$$\mathcal{P}_\epsilon(X) = \Big(P_{t,\delta}(X),\ \epsilon^{M_2 - M_1} X_0^{L-M_1} V_{M_1}(B_\epsilon X),\ \epsilon^{M_3} V_L(X) \Big). \tag{89}$$

For $\epsilon \neq 0$ the last block has no nontrivial zero; hence neither does $\mathcal{P}_\epsilon$. In the distinguished affine chart define

$$\omega_\epsilon = \epsilon\, dd^c \log(1+|x|^2) + d(\epsilon)\, dd^c \log \|\mathcal{P}_\epsilon(1,x)\|^2. \tag{90}$$

It is Kähler for $0 < \epsilon < 1$.

Proposition 19.5 (Separated insertion of the large ball)**.** *There is an explicit $\epsilon \in (0, 1/2)$ such that the smaller ball maps deform to symplectic neighbourhood embeddings for ω_ϵ with compact images in T_{out}, and $B(r_0)$ has a symplectic neighbourhood embedding for the same form with image in T_{in}.*

Proof. On the outer compact set, (90) has a smooth primitive family at $\epsilon = 0$ with initial curvature the outer polynomial form. Indeed $P_{t,\delta}$ is nonzero in the $X_1 = 1$ chart, and both added blocks vanish at zero. The uniform compact Moser transfer, with A_ϵ the identity, uses the two target control functions

$$|z|^2, \qquad 1 + |v|^2 + |w|^2 - (\rho^2/4)|z|^2.$$

They express $z \neq 0$ and $J(v,w,z) \in T_{\mathrm{out}}$ without division. Equations (86)–(88) give a threshold $\tau > 0$ and the smaller maps in T_{out} for every $0 < \epsilon \leq \tau$.

For the central chart put $x = \epsilon y$ and divide $\mathcal{P}_\epsilon(1, \epsilon y)$ by ϵ^{M_2}. Every entry of $P_{t,\delta}$ vanishes to order at least $L - S_* > M_2$ at o, so its divided block extends polynomially and vanishes at zero. The middle block becomes exactly $V_{M_1}(1,y)$, while the final block vanishes because $M_3 > M_2$. The resulting vector is therefore nonzero at zero for every y, and its Hopf primitive extends smoothly in ϵ. The first term of (90), pulled back by $y \mapsto \epsilon y$, also extends smoothly and vanishes at zero. The limiting central form is

$$\frac{M_1}{L}\, dd^c \log(1+|y|^2).$$

Its affine chart contains a symplectic image of $B(r_0)$, since $r_0 < M_1/L$. Apply the uniform compact transfer to this compact image, with $A_\epsilon(y) = \epsilon y$ and the target control function $\rho^2/4 - e(\epsilon y) > 0$; it gives a threshold $\sigma > 0$ by the same finite stopping rule. Set $\epsilon = \min(\tau, \sigma)/2$, taking both initial time bounds at most $1/2$. The two target sets are disjoint, so the images are disjoint. □

19.6. **A common map to the ball.**

Lemma 19.6 (Polynomial target form)**.** *Let $P : \mathbb{C}^{n+1} \to \mathbb{C}^q$ be a homogeneous polynomial map of degree L, with no nontrivial zero. If $0 \leq d$ and $Ld < 1$, there is an explicit smooth symplectic embedding of*

$$\big(\mathbb{C}^n, (1-Ld)dd^c \log(1+|x|^2) + d\, dd^c \log \|P(1,x)\|^2\big) \quad \textit{into } (B^\circ(1), \omega_0).$$

Proof. Put $c = 1/\pi$ and, on $\mathbb{C}^{n+1} \setminus 0$, $h_c(X) = c\,\mathrm{Im}\langle X, dX\rangle/(2\|X\|^2) = d^c \log \|X\|^2$; thus $\|X\|^2 h_c = c\lambda_0$, with λ_0 the standard Liouville form. Fix the plateaux $\chi = S^{\mathrm{plateau}}_{1/4,\,3/8,\,3,\,4}$ and $\psi = S^{\mathrm{plateau}}_{-2,\,-1,\,1,\,2}$, and put

$$\Gamma_s(X) = \|X\|^2\big[(1 - Lsd\,\chi(\|X\|^2))\,h_c(X) + sd\,\chi(\|X\|^2)\,P^*h_c(X)\big], \qquad \Omega_s = d\Gamma_s,$$

extended by $c\lambda_0$ near 0, where $\chi = 0$. Outside the shell $\frac{1}{4} \le \|X\|^2 \le 4$, $\Omega_s = c\,\omega_0$.

Nondegeneracy. All data are invariant under $X \mapsto e^{i\theta}X$, with generator iX. Since $h_c(iX) = c/2$ and $P^*h_c(iX) = h_c(iLP) = cL/2$, we get $\Gamma_s(iX) = c\|X\|^2/2$ for all s, so $\iota_{iX}\Omega_s = -d(c\|X\|^2/2)$, i.e. $\Omega_s(iX, V) = -c\,\mathrm{Re}\langle X, V\rangle$. On the complex orthogonal complement E_X of $\mathbb{C}X$, $d\|X\|^2 = 0$, so $\Omega_s|_{E_X} = \|X\|^2\big[(1 - Lsd\chi)\,dh_c + sd\chi\,P^*dh_c\big]|_{E_X}$, which is positive for $s \in [0,1]$ because dh_c restricts to a positive form on E_X, $P^*dh_c \ge 0$ and $1 - Lsd\chi \ge 1 - Ld > 0$. If $\iota_V\Omega_s = 0$ with $V = aX + b\,iX + \xi$, $\xi \in E_X$, then pairing with iX gives $a = 0$, pairing with E_X gives $\xi = 0$, and $\Omega_s(iX, X) = -c\|X\|^2 \ne 0$ gives $b = 0$. So Ω_s is symplectic on $\mathbb{C}^{n+1}$ for $s \in [0,1]$; the same argument shows that the complex hyperplane $E = \{X_0 = 0\}$, which contains $\mathbb{C}X$ for $X \in E$, is a symplectic subspace.

Relative Moser field. Put $\alpha = \Gamma_1 - \Gamma_0$, supported in the shell, with $\alpha(iX) = 0$. For $b \in \mathbb{C}^n$ let $\mathsf{e}(b) = (0, b)$, and let $\mathsf{N}_t(b, u) = (u, \mathsf{Y}_t(b, u))$ be the unique vector with X_0-component $u \in \mathbb{C}$ that is $\Omega_{S(t)}$-orthogonal to E at $\mathsf{e}(b)$; it is real linear in u and smooth in (t, b), because E^Ω is a real two-dimensional complement of E projecting isomorphically to the X_0-coordinate. Put $\mathrm{pr}(X) = X'$ and

$$h_t(X) = -\psi(|X_0|^2)\,\alpha_{\mathsf{e}(\mathrm{pr}\,X)}\big(\mathsf{N}_t(\mathrm{pr}\,X, X_0)\big), \qquad \beta_t = \alpha + dh_t,$$

$\iota_{W_t}\Omega_{S(t)} = -\beta_t$ and $V_t = S'(t)W_t$. Then h_t vanishes on E and is real linear in X_0 near E, so at $\mathsf{e}(b)$, $dh_t(\mathsf{N}_t(b, u)) = -\alpha(\mathsf{N}_t(b, u))$: β_t annihilates $E^{\Omega_{S(t)}}$, hence $W_t \in (E^\Omega)^\Omega = E$ along E. The function h_t is circle invariant (all data are) and $\beta_t(iX) = \alpha(iX) + dh_t(iX) = 0$, so $\Omega_{S(t)}(iX, W_t) = -\beta_t(iX) = 0$ gives $\mathrm{Re}\langle X, V_t\rangle = 0$. The field V_t is smooth, vanishes for $t \notin [0,1]$, and has compact support in $\{X : \|X\|^2 \le 6\}$. Indeed, α is supported in the shell $\frac{1}{4} \le \|X\|^2 \le 4$, whereas the support of h_t is contained in $\{|X_0|^2 \le 2,\ \|X'\|^2 \le 4\}$, and differentiation does not enlarge support. Thus its flow ϕ_t is complete. It preserves every sphere $\|X\| = r$, the hyperplane E and therefore $\{X_0 \ne 0\}$, and commutes with the circle action. Since $\frac{d}{dt}\Omega_{S(t)} = S'(t)\,d\alpha = S'(t)\,d\beta_t$, Cartan's formula gives $\frac{d}{dt}\phi_t^*\Omega_{S(t)} = \phi_t^* d(\iota_{V_t}\Omega_{S(t)} + S'(t)\beta_t) = 0$, so $\phi_1^*\Omega_1 = \Omega_0$.

Descent. On the unit sphere $\chi = 1$, and Ω_s restricted to the sphere is the pullback under the Hopf map of $(1 - Lsd)\,dd^c \log \|X\|^2 + sd\,dd^c \log \|P(X)\|^2$; in the chart $X_0 = 1$ this is $\omega_s := (1 - Lsd)dd^c \log(1 + |x|^2) + sd\,dd^c \log \|P(1, x)\|^2$ (the difference of the potentials is $\log |X_0|^{2(1-Lsd)+2Lsd} = \log |X_0|^2$, which is pluriharmonic). Since ϕ_1 preserves the unit sphere and $\{X_0 \ne 0\}$ and commutes with the circle action, it descends to a diffeomorphism $\bar\phi$ of $\mathbb{C}^n$ with $\bar\phi^*\omega_1 = \omega_0 = dd^c \log(1 + |x|^2)$. Put $G = \bar\phi^{-1}$, so $G^*dd^c \log(1 + |x|^2) = \omega_1$. The map $x \mapsto x/\sqrt{\pi(1 + |x|^2)}$ is a diffeomorphism of $\mathbb{C}^n$ onto $B^\circ(1)$ pulling ω_0 back to $dd^c \log(1 + |x|^2)$ (Lemma 13.20); its composite with G is the required embedding. □

19.7. **Assembly.**

Theorem 19.7 (One large ball, $n = 3$)**.** *Let $A, \rho_1, \dots, \rho_k > 0$ have valid auxiliary data $(a; r)$ with $a \in \mathbb{Q}$, and let $\rho_i < r_i' < r_i$ and $A < r_0 < a$. Then there are smooth symplectic embeddings f_0 of an open neighbourhood of $B(r_0)$ and f_i of open neighbourhoods of $B(r_i')$, $1 \le i \le k$, into $B^\circ(1)$, given by the formulas below, such that the sets $f_0(B(r_0))$ and $f_i(B(r_i'))$ are pairwise*

disjoint and contained in $B(1-\zeta)$, where

$$\zeta = \tfrac{1}{2}\min\Big\{1-\pi|y|^2 :\ y \in f_0(B(r_0)) \cup \textstyle\bigcup_{i=1}^{k} f_i(B(r_i'))\Big\} > 0.$$

Proof. Take L and S_* from Definition 16.6, so that $L > Q = \lfloor 3/(a-r_0)\rfloor + 1$, and put

$$m_0 = \lfloor r_0 L\rfloor, \qquad M_1 = m_0+1, \quad M_2 = m_0+2, \quad M_3 = m_0+3.$$

Then $r_0 < M_1/L$ and $M_1 < M_2 < L - S_*$, $M_2 < M_3$. The vanishing order in Proposition 16.8 permits the bubble construction of Proposition 19.5 with the orders M_1, M_2, M_3, in which $\epsilon = \min(\tau,\sigma)/2$ for the two explicit thresholds $\tau,\sigma > 0$ of that proposition. Its central chart contains a symplectic image of $B(r_0)$ and its outer chart contains the images of Proposition 19.1: starting with the maps ϕ_i of that proposition, apply in order g_1 (Proposition 19.3), g_2 (Proposition 19.4), the outer bubble deformation and J. The resulting maps and the large-ball map of Proposition 19.5 have pairwise disjoint compact images for the common degree-L polynomial form (90), with coefficient $(1-\epsilon)/L$. Lemma 19.6 applies because $1 - Ld(\epsilon) = \epsilon > 0$, and its common embedding into $B^\circ(1)$ preserves all disjointness. The union of the images of $B(r_0)$ and the $B(r_i')$ is compact and lies in $B^\circ(1)$, so the minimum defining ζ is attained and positive, and $\pi|y|^2 \le 1 - 2\zeta < 1 - \zeta$ on it. If there are no smaller balls, f_0 is the inclusion. □

20. The normalised embeddings

This section gives the case distinction of §M6.1 in full.

Construction 20.1 (The normalised embeddings)**.** Let $(n,k,1,\rho)$ be admissible. The *normalised embeddings* $f_i^\natural$, enlarged capacities ρ_i^+ and domain capacities ρ_i^{dom} are defined as follows.

(i) If $n = 2$, they are those of Theorem 13.23: for $k \ge 10$, equal $\rho_i = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$ it uses the equal-capacity maps of Remark 13.25; otherwise it uses the explicit scalar data and maps of Proposition 6.8 and (41).

(ii) If $n \ge 3$ and $\rho_i < \frac{1}{2}$ for all i, put

$$\varepsilon_i = \min\left(\frac{1/2-\rho_i}{2}, \frac{1-\sum_j \rho_j^n}{2nk}\right), \qquad a_i = \rho_i + \varepsilon_i.$$

Use the quadric construction of Theorem 14.7 with these auxiliary capacities. Its enlarged closed capacity and open domain capacity are, respectively, $\rho_i^+ = \rho_i + \varepsilon_i/3$ and $\rho_i^{\mathrm{dom}} = \rho_i + 2\varepsilon_i/3$.

(iii) If $n \ge 4$ and $\rho_j \ge \frac{1}{2}$, the index j is unique, since $\rho_i + \rho_j < 1$ for $i \ne j$. Write $A = \rho_j$, $r_* = \max(\{0\} \cup \{\rho_i : i \ne j\})$, $V = 1 - \sum_i \rho_i^n$, and put

$$t = \min\left(\frac{1-A-r_*}{4},\ 1,\ \frac{V}{2nk2^n}\right), \qquad a = A + t, \qquad r_i = \rho_i + t \quad (i \ne j).$$

Use the construction of Theorem 17.9. For the distinguished ball set $(\rho_j^+, \rho_j^{\mathrm{dom}}) = (A + t/2, A + t)$. For $i \ne j$, set $(\rho_i^+, \rho_i^{\mathrm{dom}}) = ((11\rho_i + 5r_i)/16, (3\rho_i + 5r_i)/8)$.

(iv) If $n = 3$ and $\rho_j \ge \frac{1}{2}$, put $A = \rho_j$ and

$$\gamma_0 = \min\left(1 - \sum_i \rho_i^3,\ 1 - A,\ \min_{i\ne j}(1 - A - \rho_i)\right), \qquad \eta = \frac{\gamma_0}{100(k+1)}.$$

The minimum over an empty set is omitted. Set

$$M = \lceil 1/\eta\rceil + 1, \qquad a = \frac{\lfloor (A+4\eta)M\rfloor + 1}{M}.$$

Use the polynomial construction of Theorem 19.7, with distinguished capacity $r_0 = A + 2\eta$. The auxiliary small family retains all k indices: for $i \neq j$ its outer and inner capacities are $r_i = \rho_i + 3\eta$ and $r_i' = \rho_i + 2\eta$, while the j-th auxiliary ball has input capacity $\eta/4$ and capacities $r_j = \eta$, $r_j' = \eta/2$, so that the small family is nonempty and indexed like the input. The comparison, the weights and the transfer are those of this auxiliary family of k balls, including the j-th ball in every parameter and compact union. The final j-th map is the bubble map of the distinguished ball; the map of the auxiliary j-th ball is not used. Set $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\rho_i + \eta, \rho_i + 2\eta)$ for every final ball.

For the toric maps in (ii)–(iv), use the finite-atlas normalization and three inverse Moser endpoints of Construction 12.4, applied to the closed source balls $B(\rho_i^{\mathrm{dom}})$. In particular the cap smoothing and layer strip use its internal capacity $(3a_i + \rho_i^{\mathrm{dom}})/4$, with a_i replaced by r_i in cases (iii) and (iv); the distinguished ball, placed by inclusion or by the bubble, is exempt. In case (i) the pairs $(\rho_i^+, \rho_i^{\mathrm{dom}})$ are those of Remark 13.24.

Verification of the hypotheses. All increments ε_i, t, η are positive: admissibility gives $\sum_i \rho_i^n < 1$, $\rho_i < \frac{1}{2}$ in case (ii), and $A + \rho_i < 1$ for $i \neq j$ together with $A < 1$ in cases (iii) and (iv). We use $(x+u)^n - x^n = n\xi^{n-1}u$ for some $\xi \in [x, x+u]$.

Case (ii). $a_i \leq \rho_i + (\frac{1}{2} - \rho_i)/2 < \frac{1}{2}$, and, since $0 \leq \rho_i < a_i < 1$, $\sum_i a_i^n \leq \sum_i \rho_i^n + n\sum_i \varepsilon_i \leq \sum_i \rho_i^n + (1 - \sum_j \rho_j^n)/2 < 1$; these are the hypotheses of Theorem 14.7, and $\rho_i^+ = (2\rho_i + a_i)/3$, $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$ are the capacities named there.

Case (iii). $(a; r)$ is valid auxiliary data (Lemma 16.2): $a = A + t > A \geq \frac{1}{2}$; $a < 1$ because $t < 1 - A$; $r_i + a = \rho_i + A + 2t \leq r_* + A + \frac{1}{2}(1 - A - r_*) < 1$, i.e. $r_i < 1 - a$; and with $0 \leq x, t \leq 1$, $(x+t)^n \leq x^n + n2^{n-1}t$, so $a^n + \sum_{i \neq j} r_i^n \leq \sum_i \rho_i^n + kn2^n t \leq \sum_i \rho_i^n + V/2 < 1$. Theorem 17.9 gives $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$ and $\rho_i < (11\rho_i + 5r_i)/16 < \rho_i^{\mathrm{dom}} < r_i$; for the distinguished ball, the inclusion is injective on all of $\mathbb{C}^n$ and $B(A + t/2) \subset B^\circ(a)$, which is disjoint from the shell S_a containing the other images.

Case (iv). First $1/M < \eta$, so $A + 4\eta < a \leq A + 4\eta + 1/M < A + 5\eta$, and $a \in \mathbb{Q}$. With $\gamma_0 \leq 1 - A$, $\gamma_0 \leq 1 - A - \rho_i$ and $\eta \leq \gamma_0/200$: $a < 1$; $r_0 = A + 2\eta \in (A, a)$; $\rho_i < r_i' < r_i$ and $r_i + a < \rho_i + A + 8\eta < 1$ for $i \neq j$; and for the auxiliary ball $\eta/4 < \eta/2 < \eta$ and $\eta + a < A + 6\eta < 1$. By the mean value bound with $n = 3$ and $x, u \leq 1$, $a^3 \leq A^3 + 60\eta$, $r_i^3 \leq \rho_i^3 + 36\eta$ and $\eta^3 \leq \eta$, so $a^3 + \sum_{i \neq j} r_i^3 + \eta^3 \leq \sum_i \rho_i^3 + (36k + 25)\eta < \sum_i \rho_i^3 + \gamma_0 \leq 1$. Thus $(a; r)$, with the auxiliary j-th entry, is valid auxiliary data with a rational, and Theorem 19.7 applies with r_0 and the r_i'. Its domains are $B(r_0) = B(\rho_j^{\mathrm{dom}})$ and $B(r_i') = B(\rho_i^{\mathrm{dom}})$.

The data may have real coefficients, which is allowed at level L1. The polyhedral construction applies over real coefficients by the observation following Proposition M3.3; the finite-output algorithm certifies its polynomial directly.

21. The common isotopy and the pushforward of the source primitive

We construct, for k disjoint balls of arbitrary radii and centres in $\mathbb{R}^{2n}$, a single smooth family $F_{i,t}$ of symplectic embeddings from the translations $s_i + u$ to the given embeddings f_i, keeping the images disjoint at all times (Proposition 21.14). The balls are first translated to far centres along a route on which they never meet; there each is deformed linearly to $A_i = Df_i(0)$, then to the rescaled embedding $u \mapsto \tau^{-1} f_i(\tau u)$, and finally one dilation parameter common to all balls runs from τ to 1. Every stage is symplectic: translations and the maps $M_{i,\sigma} \in \mathrm{Sp}(2n)$ trivially, and the shape and tail stages because their derivatives are values of Df_i. Disjointness comes from separation during the route, linear and shape stages and from the common dilation during the tail. Every ingredient of $F_{i,t}$ is an explicit function of the input: the maxima defining L, the minimum defining δ_c, the positive square root and the least admissible angle in the linear path,

and the ranks in the route. We then show that the pushforward of a cut-off source primitive along F is smooth with compact support (Lemma 21.16) and give the primitive in closed form on each stage (Lemma 21.15).

Conventions. We use the conventions of §2.1: $\omega_0(\xi,\xi') = \langle \xi, J_0\xi'\rangle$ with $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ in the coordinates (x,y), and $\mathrm{Sp}(2n)$ is the group of real $2n\times 2n$ matrices with $A^{\mathsf T}J_0A = J_0$. Multiplication by i on $\mathbb{C}^n$ is $J_{\mathbb C} = -J_0 = \left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)$. Matrices are written in $n\times n$ blocks with respect to (x,y). Vector norms are Euclidean, $\|\cdot\|$ is the operator norm, and a linear map is identified with its matrix. $\mathbb{B}(y,r)$ and $\mathbb{B}^\circ(y,r)$ are the closed and open Euclidean balls of radius r about y, and $\mathbb{B}(r) = \mathbb{B}(0,r)$. S is the flat step of Definition 2.3. For a matrix-valued function of $\sigma\in\mathbb{R}$, *smooth entries* means that every entry is a C^∞ function of σ. A symplectic matrix is invertible ($A^{-1} = J_0^{-1}A^{\mathsf T}J_0$), so a map with symplectic derivative is a local diffeomorphism.

21.1. **The input.** The isotopy uses the following properties of the embeddings and nothing else.

Definition 21.1 (Isotopy input)**.** Let $n\ge 1$ and $k\ge 1$. An *isotopy input* consists of centres $s_1,\dots,s_k\in\mathbb{R}^{2n}$, radii $0<r_{1,i}<r_{2,i}<r_{3,i}<r_{4,i}<\bar r_i$ for $1\le i\le k$, a number $R_0>0$, and maps $f_i\colon \mathbb{B}^\circ(\bar r_i)\to\mathbb{R}^{2n}$ such that

(i) f_i is C^∞ and $Df_i(u)\in\mathrm{Sp}(2n)$ for every $u\in\mathbb{B}^\circ(\bar r_i)$;
(ii) f_i is injective on $\mathbb{B}(r_{4,i})$;
(iii) the sets $f_i(\mathbb{B}(r_{4,i}))$, $1\le i\le k$, are pairwise disjoint and contained in $\mathbb{B}^\circ(R_0)$;
(iv) $|s_i-s_j|>r_{3,i}+r_{3,j}$ for $i\ne j$.

We write $y_i = f_i(0)$ and $A_i = Df_i(0)\in\mathrm{Sp}(2n)$.

By Lemma M6.5, an embedding datum together with the radii $r_{\ell,i}$ of Definition M6.4 and $R_0=\sqrt{C/\pi}$ is an isotopy input. From now on we fix an isotopy input. The open ball $\mathbb{B}^\circ(r_{4,i})$ is the domain of every stage, the closed ball $\mathbb{B}(r_{3,i})$ is the set whose images are kept disjoint, and the cutoff of the Hamiltonian interpolates between $r_{1,i}$ and $r_{2,i}$.

21.2. **An explicit path in** $\mathrm{Sp}(2n)$**.** The linear stage needs, for every $A\in\mathrm{Sp}(2n)$, an explicit smooth path in $\mathrm{Sp}(2n)$ from I to A. We factor A as a unitary matrix times a block upper triangular symplectic matrix and deform each factor separately; the unitary factor is deformed by a Cayley transform after a rotation by one of $n+1$ fixed angles. Apart from one positive square root and the choice of one of $n+1$ angles, every entry of the path is a rational function of σ, $\sin^2(\pi\sigma/2)$, $\cos(\sigma\theta_A)$, $\sin(\sigma\theta_A)$, $\cos\theta_A$, $\sin\theta_A$ and the entries of A and of the square root; no branch of a matrix logarithm has to be chosen.

Definition 21.2 (The linear path)**.** For $n\times n$ matrices T,a,e and $\theta,\sigma\in\mathbb{R}$ let

$$\mathrm{Up}(T)=\begin{pmatrix} I & T\\ 0 & I\end{pmatrix},\qquad \mathrm{Dg}(a,e)=\begin{pmatrix} a & 0\\ 0 & e\end{pmatrix},\qquad \mathrm{rot}(\theta)=\cos\theta\, I+\sin\theta\, J_{\mathbb C},$$

and, for a $2n\times 2n$ matrix X with $X^{\mathsf T}=-X$, $\mathrm{Cay}(X,\sigma)=(I+\sigma X)(I-\sigma X)^{-1}$. Let $A=\left(\begin{smallmatrix} a & b\\ c & d\end{smallmatrix}\right)\in\mathrm{Sp}(2n)$. Put

$$G_A = a^{\mathsf T}a+c^{\mathsf T}c,\qquad P_A=\mathrm{the}\{P:\ P \text{ symmetric positive semidefinite},\ P^2=G_A\},$$

$$U_A=\begin{pmatrix} aP_A^{-1} & -cP_A^{-1}\\ cP_A^{-1} & aP_A^{-1}\end{pmatrix},\qquad T_A = P_A^{-1}\left(U_A^{\mathsf T}A\right)_{12},$$

where $(\cdot)_{12}$ is the upper right block. Let $\theta_m=\pi m/(n+1)$ for $m=0,\dots,n$, let m_A be the least $m\in\{0,\dots,n\}$ with $\det\left(\mathrm{rot}(\theta_m)U_A+I\right)\ne 0$, $\theta_A=\theta_{m_A}$, $V_A=\mathrm{rot}(\theta_A)U_A$, and $X_A=(V_A-I)(V_A+I)^{-1}$. Finally, for $\sigma\in\mathbb{R}$,

$$P_{A,\sigma}=I+\sin^2\!\left(\tfrac{\pi\sigma}{2}\right)(P_A-I),\qquad M_{A,\sigma}=\mathrm{rot}(-\sigma\theta_A)\ \mathrm{Cay}(X_A,\sigma)\ \mathrm{Dg}\left(P_{A,\sigma},P_{A,\sigma}^{-1}\right)\ \mathrm{Up}(\sigma T_A),$$

and $M_{i,\sigma} = M_{A_i,\sigma}$. By Lemma 21.3(a),(b), P_A is invertible, m_A exists and $V_A + I$ is invertible, so these formulas make sense.

The index m_A is obtained by $n+1$ case distinctions on propositions about the input, an operation of level L1 (Definition M1.3).

We say that a $2n \times 2n$ matrix Q is of *unitary type* if $Q^{\mathsf{T}}Q = I$ and $QJ_{\mathbb{C}} = J_{\mathbb{C}}Q$. A matrix commutes with $J_{\mathbb{C}}$ exactly when it has the block form $\left(\begin{smallmatrix} \mathfrak{w}_1 & -\mathfrak{w}_2 \\ \mathfrak{w}_2 & \mathfrak{w}_1 \end{smallmatrix}\right)$, and then it is the realification of the complex $n \times n$ matrix $\mathfrak{w} = \mathfrak{w}_1 + i\mathfrak{w}_2$; realification is an injective ring homomorphism which takes $\mathfrak{w}^*$ to the transpose. So the matrices of unitary type are the realifications of the unitary matrices, and $\mathrm{rot}(\theta)$ is the realification of $e^{i\theta}I$. Moreover

$$\det \begin{pmatrix} \mathfrak{w}_1 & -\mathfrak{w}_2 \\ \mathfrak{w}_2 & \mathfrak{w}_1 \end{pmatrix} = \left|\det(\mathfrak{w}_1 + i\mathfrak{w}_2)\right|^2, \tag{91}$$

because, over $\mathbb{C}$, $\left(\begin{smallmatrix} I & 0 \\ iI & I \end{smallmatrix}\right)\left(\begin{smallmatrix} \mathfrak{w}_1 & -\mathfrak{w}_2 \\ \mathfrak{w}_2 & \mathfrak{w}_1 \end{smallmatrix}\right)\left(\begin{smallmatrix} I & 0 \\ -iI & I \end{smallmatrix}\right) = \left(\begin{smallmatrix} \mathfrak{w}_1+i\mathfrak{w}_2 & -\mathfrak{w}_2 \\ 0 & \mathfrak{w}_1-i\mathfrak{w}_2 \end{smallmatrix}\right)$.

Lemma 21.3 (The linear path)**.** *Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{Sp}(2n)$.*

- *(a) G_A is positive definite, P_A is well defined, symmetric and positive definite, U_A is of unitary type, T_A is symmetric, and $A = U_A \,\mathrm{Dg}(P_A, P_A^{-1})\,\mathrm{Up}(T_A)$.*
- *(b) m_A exists. X_A is skew-symmetric and commutes with $J_{\mathbb{C}}$; for every $\sigma \in \mathbb{R}$ the matrix $I - \sigma X_A$ is invertible and $\mathrm{Cay}(X_A, \sigma)$ is of unitary type; and $\mathrm{Cay}(X_A, 1) = V_A$.*
- *(c) $M_{A,\sigma} \in \mathrm{Sp}(2n)$ for every $\sigma \in \mathbb{R}$, $M_{A,0} = I$, $M_{A,1} = A$, and $M_{A,\sigma}$ has smooth entries; hence $(\sigma, u) \mapsto M_{A,\sigma}u$ is C^∞ on $\mathbb{R} \times \mathbb{R}^{2n}$.*

Proof. Block criterion. Writing out $A^{\mathsf{T}}J_0A$ in blocks shows that $A \in \mathrm{Sp}(2n)$ if and only if

$$a^{\mathsf{T}}c = c^{\mathsf{T}}a, \qquad b^{\mathsf{T}}d = d^{\mathsf{T}}b, \qquad a^{\mathsf{T}}d - c^{\mathsf{T}}b = I. \tag{92}$$

In particular $\mathrm{Up}(T) \in \mathrm{Sp}(2n)$ if T is symmetric, and $\mathrm{Dg}(P, P^{-1}) \in \mathrm{Sp}(2n)$ if P is symmetric and invertible. A matrix Q of unitary type is symplectic: $Q^{\mathsf{T}}J_0Q = -Q^{\mathsf{T}}J_{\mathbb{C}}Q = -Q^{\mathsf{T}}QJ_{\mathbb{C}} = -J_{\mathbb{C}} = J_0$. Products of matrices of unitary type are of unitary type.

(a) $G_A = \iota^{\mathsf{T}}A^{\mathsf{T}}A\iota$ with $\iota = \left(\begin{smallmatrix} I \\ 0 \end{smallmatrix}\right)$, and $A\iota$ has trivial kernel because A is invertible; so $\xi^{\mathsf{T}}G_A\xi = |A\iota\xi|^2 > 0$ for $0 \neq \xi \in \mathbb{R}^n$. A positive definite symmetric matrix has a unique positive semidefinite symmetric square root, and it is positive definite (diagonalise orthogonally and take square roots of the eigenvalues; uniqueness because a positive semidefinite square root commutes with G_A and hence preserves its eigenspaces, on each of which it is the positive multiple of the identity determined by the eigenvalue). So P_A is well defined, symmetric and positive definite. Put $\hat{a} = aP_A^{-1}$ and $\hat{c} = cP_A^{-1}$. Then $\hat{a}^{\mathsf{T}}\hat{a} + \hat{c}^{\mathsf{T}}\hat{c} = P_A^{-1}G_AP_A^{-1} = I$, and $\hat{a}^{\mathsf{T}}\hat{c} = P_A^{-1}a^{\mathsf{T}}cP_A^{-1}$ is symmetric by (92), so $\hat{c}^{\mathsf{T}}\hat{a} = \hat{a}^{\mathsf{T}}\hat{c}$. Hence

$$U_A^{\mathsf{T}}U_A = \begin{pmatrix} \hat{a}^{\mathsf{T}}\hat{a} + \hat{c}^{\mathsf{T}}\hat{c} & \hat{c}^{\mathsf{T}}\hat{a} - \hat{a}^{\mathsf{T}}\hat{c} \\ \hat{a}^{\mathsf{T}}\hat{c} - \hat{c}^{\mathsf{T}}\hat{a} & \hat{a}^{\mathsf{T}}\hat{a} + \hat{c}^{\mathsf{T}}\hat{c} \end{pmatrix} = I,$$

and $U_A = \left(\begin{smallmatrix} \hat{a} & -\hat{c} \\ \hat{c} & \hat{a} \end{smallmatrix}\right)$ commutes with $J_{\mathbb{C}}$; so U_A is of unitary type. The first block column of $U_A^{\mathsf{T}}A$ is $\left(\begin{smallmatrix} \hat{a}^{\mathsf{T}}a + \hat{c}^{\mathsf{T}}c \\ -\hat{c}^{\mathsf{T}}a + \hat{a}^{\mathsf{T}}c \end{smallmatrix}\right) = \left(\begin{smallmatrix} P_A^{-1}G_A \\ P_A^{-1}(a^{\mathsf{T}}c - c^{\mathsf{T}}a) \end{smallmatrix}\right) = \left(\begin{smallmatrix} P_A \\ 0 \end{smallmatrix}\right)$. So $U_A^{\mathsf{T}}A = \left(\begin{smallmatrix} P_A & B' \\ 0 & D' \end{smallmatrix}\right)$, and this matrix is symplectic. By (92), $P_AD' = I$, so $D' = P_A^{-1}$, and $B'^{\mathsf{T}}P_A^{-1} = P_A^{-1}B'$, which says that $T_A = P_A^{-1}B'$ satisfies $T_A^{\mathsf{T}} = B'^{\mathsf{T}}P_A^{-1} = T_A$. Since $\mathrm{Dg}(P_A, P_A^{-1})\,\mathrm{Up}(T_A) = \left(\begin{smallmatrix} P_A & P_AT_A \\ 0 & P_A^{-1} \end{smallmatrix}\right) = U_A^{\mathsf{T}}A$ and U_A is orthogonal, $A = U_A\,\mathrm{Dg}(P_A, P_A^{-1})\,\mathrm{Up}(T_A)$.

(b) Let $\mathfrak{u}$ be the unitary $n \times n$ matrix whose realification is U_A. Then $\mathrm{rot}(\theta)U_A + I$ is the realification of $e^{i\theta}\mathfrak{u} + I$, and by (91) its determinant is $|\det(e^{i\theta}\mathfrak{u} + I)|^2$, which vanishes exactly when $-e^{-i\theta}$ is an eigenvalue of $\mathfrak{u}$. The numbers $-e^{-i\theta_m}$, $m = 0, \dots, n$, are pairwise distinct because $0 \le \theta_m < \pi$, and $\mathfrak{u}$ has at most n eigenvalues; so some $m \le n$ is admissible and m_A

exists. $V = V_A$ is of unitary type and $V + I$ is invertible. The matrices V, $V^{\mathsf{T}} = V^{-1}$ and $(V+I)^{-1}$ commute with each other and with $J_{\mathbb{C}}$. Hence X_A commutes with $J_{\mathbb{C}}$, and

$$X_A^{\mathsf{T}} = (V^{-1}+I)^{-1}(V^{-1}-I) = \big(V^{-1}(I+V)\big)^{-1}V^{-1}(I-V) = (I+V)^{-1}(I-V) = -X_A.$$

For skew-symmetric X and $\xi \in \mathbb{R}^{2n}$, $\langle \xi, (I-\sigma X)\xi\rangle = |\xi|^2$, so $I - \sigma X$ is invertible. The matrices $I \pm \sigma X_A$ commute, so $Q_\sigma = \operatorname{Cay}(X_A, \sigma)$ satisfies $Q_\sigma^{\mathsf{T}} Q_\sigma = (I+\sigma X_A)^{-1}(I-\sigma X_A)(I+\sigma X_A)(I-\sigma X_A)^{-1} = I$, and Q_σ commutes with $J_{\mathbb{C}}$; so it is of unitary type. Finally $I + X_A = 2V(V+I)^{-1}$ and $I - X_A = 2(V+I)^{-1}$, so $\operatorname{Cay}(X_A, 1) = 2V(V+I)^{-1} \cdot \frac{1}{2}(V+I) = V$.

(c) $\operatorname{rot}(\theta)$ is of unitary type for every θ, and $\operatorname{rot}(\alpha)\operatorname{rot}(\beta) = \operatorname{rot}(\alpha+\beta)$ because $J_{\mathbb{C}}^2 = -I$. The matrix $P_{A,\sigma} = (1-\beta_\sigma)I + \beta_\sigma P_A$, $\beta_\sigma = \sin^2(\pi\sigma/2) \in [0,1]$, is symmetric with eigenvalues $(1-\beta_\sigma) + \beta_\sigma\mu$, μ an eigenvalue of P_A, which lie between 1 and μ; so it is positive definite. By the block criterion, (a) and (b), each of the four factors of $M_{A,\sigma}$ is symplectic, hence so is $M_{A,\sigma}$. At $\sigma = 0$ all four factors are I. At $\sigma = 1$, $M_{A,1} = \operatorname{rot}(-\theta_A)V_A \operatorname{Dg}(P_A, P_A^{-1}) \operatorname{Up}(T_A) = U_A \operatorname{Dg}(P_A, P_A^{-1}) \operatorname{Up}(T_A) = A$ by (a). The entries of $\operatorname{rot}(\pm\sigma\theta_A)$, $I \pm \sigma X_A$, $P_{A,\sigma}$ and σT_A are C^∞ in σ, and the inverses of $I - \sigma X_A$ and $P_{A,\sigma}$ have smooth entries by the adjugate formula, their determinants being nonzero for every σ. Products of matrices with smooth entries have smooth entries, and $(\sigma, u) \mapsto M_{A,\sigma}u$ is bilinear in the entries and u.

□

21.3. **Parameters and the stage maps.**

Definition 21.4 (Parameters)**.** Put

$$L = 1 + \sum_{i=1}^{k} \Big(\max_{u \in \mathbb{B}(r_{3,i})} \|Df_i(u)\| + \max_{(\sigma,u)\in[0,1]\times\mathbb{B}(r_{3,i})} |M_{i,\sigma}u| \Big), \qquad r_{\mathrm{tot}} = 1 + \sum_{i=1}^{k} r_{3,i},$$

$\delta_c = \frac{1}{2}\min_{i\neq j}|y_i - y_j|$ if $k \geq 2$ and $\delta_c = 1$ if $k = 1$, and

$$\tau = \min\Big(\frac{1}{2},\ \frac{\delta_c}{2Lr_{\mathrm{tot}}}\Big), \qquad Y_i = \frac{y_i}{\tau}.$$

The maxima exist: Df_i is continuous on $\mathbb{B}^\circ(\bar{r}_i) \supset \mathbb{B}(r_{3,i})$, and $(\sigma, u) \mapsto M_{i,\sigma}u$ is continuous by Lemma 21.3(c); both are maximised over compact sets.

Lemma 21.5 (Parameters)**.** *(a) $L \geq 1$ and $r_{\mathrm{tot}} \geq 1$.*
(b) $\|Df_i(u)\| \leq L$ and $|M_{i,\sigma}u| \leq L$ for $\sigma \in [0,1]$ and $u \in \mathbb{B}(r_{3,i})$.
(c) $r_{3,i} + r_{3,j} < r_{\mathrm{tot}}$ for $i \neq j$.
(d) The y_i are pairwise distinct, $\delta_c > 0$, and $\delta_c \leq \frac{1}{2}|y_i - y_j|$ for $i \neq j$.
(e) $0 < \tau \leq \frac{1}{2}$, and $|Y_i - Y_j| \geq 4Lr_{\mathrm{tot}} > r_{3,i} + r_{3,j}$ for $i \neq j$.

Proof. All summands of L and r_{tot} are nonnegative; this gives (a), (b) and $r_{\mathrm{tot}} \geq 1 + r_{3,i} + r_{3,j}$, hence (c). Since $0 \in \mathbb{B}(r_{4,i})$ and the sets $f_i(\mathbb{B}(r_{4,i}))$ are pairwise disjoint, $y_i \neq y_j$ for $i \neq j$; so δ_c is half the minimum of finitely many positive numbers (or 1), which gives (d). Therefore $0 < \tau \leq \frac{1}{2}$, and $\tau \leq \delta_c/(2Lr_{\mathrm{tot}})$ gives $|Y_i - Y_j| = |y_i - y_j|/\tau \geq 2\delta_c \cdot 2Lr_{\mathrm{tot}}/\delta_c = 4Lr_{\mathrm{tot}}$. Finally $4Lr_{\mathrm{tot}} > r_{\mathrm{tot}} > r_{3,i} + r_{3,j}$ by (a) and (c). □

Definition 21.6 (The stage maps)**.** For $1 \leq i \leq k$ define:

(i) the *linear stage* $G_{i,\sigma}^{\mathrm{lin}}(u) = Y_i + M_{i,\sigma}u$ for $\sigma \in \mathbb{R}$, $u \in \mathbb{R}^{2n}$;
(ii) the *shape family* $\kappa_{i,\rho}(u) = \int_0^1 Df_i(\phi\rho u)\, u\, d\phi$ for $(\rho, u) \in \Omega_i = (-1,1) \times \mathbb{B}^\circ(r_{4,i})$, and the *shape stage* $G_{i,\sigma}^{\mathrm{sh}}(u) = Y_i + \kappa_{i,\tau\sigma}(u)$ for $\sigma \in (-2,2)$, $u \in \mathbb{B}^\circ(r_{4,i})$;
(iii) the *tail* $T_{i,v}(u) = v^{-1}f_i(vu)$ for $v > 0$ and $u \in \mathbb{R}^{2n}$ with $vu \in \mathbb{B}^\circ(\bar{r}_i)$.

The shape stage is defined because $|\tau\sigma| < 1$ for $|\sigma| < 2$, and the integrand of $\kappa_{i,\rho}$ is defined because $|\phi\rho u| < r_{4,i} < \bar{r}_i$.

Lemma 21.7 (The linear stage)**.** *$(\sigma, u) \mapsto G^{\mathrm{lin}}_{i,\sigma}(u)$ is C^∞, $G^{\mathrm{lin}}_{i,0}(u) = Y_i + u$ and $G^{\mathrm{lin}}_{i,1}(u) = Y_i + A_i u$. For $\sigma \in [0,1]$, $G^{\mathrm{lin}}_{i,\sigma}$ is an affine symplectic isomorphism of $\mathbb{R}^{2n}$ which maps $\mathbb{B}(r_{3,i})$ into $\mathbb{B}(Y_i, L) \subset \mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, and the balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, $1 \le i \le k$, are pairwise disjoint.*

Proof. Smoothness and the endpoints are Lemma 21.3(c); the linear part $M_{i,\sigma}$ is symplectic, hence invertible. The inclusion is $|M_{i,\sigma}u| \le L \le Lr_{\mathrm{tot}}$ (Lemma 21.5(a),(b)). Two closed balls of radius Lr_{tot} whose centres are at least $4Lr_{\mathrm{tot}} > 2Lr_{\mathrm{tot}}$ apart are disjoint (Lemma 21.5(e)). □

21.4. **The shape stage and the tail.** The shape stage deforms $Y_i + A_i u$ into $\tau^{-1} f_i(\tau u)$ with the far centre fixed. The difference quotient $\rho^{-1}(f_i(\rho u) - y_i)$ is written as an average of derivatives, so that it is visibly smooth through $\rho = 0$.

Lemma 21.8 (The shape stage)**.** *(a) $(\rho, u) \mapsto \kappa_{i,\rho}(u)$ is C^∞ on Ω_i.*

(b) $\kappa_{i,0}(u) = A_i u$, and $\rho\,\kappa_{i,\rho}(u) = f_i(\rho u) - y_i$; in particular $\kappa_{i,\rho}(u) = \rho^{-1}(f_i(\rho u) - y_i)$ for $\rho \neq 0$.

(c) $D_u \kappa_{i,\rho}(u) = Df_i(\rho u)$; in particular it is symplectic.

(d) $|\kappa_{i,\rho}(u)| \le L|u|$ for $|\rho| < 1$ and $u \in \mathbb{B}(r_{3,i})$.

(e) $\partial_\rho \kappa_{i,\rho}(u) = \int_0^1 \phi\, D^2 f_i(\phi\rho u)[u,u]\, d\phi$.

(f) For $\sigma \in [0,1]$, $G^{\mathrm{sh}}_{i,\sigma}$ is injective on $\mathbb{B}(r_{3,i})$ and maps it into $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$; $G^{\mathrm{sh}}_{i,0} = G^{\mathrm{lin}}_{i,1}$ and $G^{\mathrm{sh}}_{i,1} = T_{i,\tau}$ on $\mathbb{B}^\circ(r_{4,i})$.

Proof. For $(\phi, \rho, u) \in [0,1] \times \Omega_i$ we have $|\phi\rho u| < r_{4,i}$, so the integrand $g(\phi, \rho, u) = Df_i(\phi\rho u)u$ is defined and C^∞ on the open set $\{(\phi, \rho, u) : |\phi\rho u| < \bar{r}_i\}$, which contains $[0,1] \times \Omega_i$. For a compact neighbourhood $N \subset \Omega_i$ of a point of Ω_i, every partial derivative of g in (ρ, u) is bounded on the compact set $[0,1] \times N$; hence the integral over $[0,1]$ may be differentiated under the integral sign any number of times, which proves (a), and (e) is the derivative of the integrand, $\partial_\rho\big(Df_i(\phi\rho u)u\big) = D^2 f_i(\phi\rho u)[\phi u, u]$. (b) At $\rho = 0$ the integrand is $A_i u$. For general ρ, $\phi \mapsto f_i(\phi\rho u)$ is C^1 on $[0,1]$ with derivative $Df_i(\phi\rho u)(\rho u) = \rho\, Df_i(\phi\rho u)u$, and the fundamental theorem of calculus gives $f_i(\rho u) - f_i(0) = \rho\kappa_{i,\rho}(u)$. (c) For $\rho = 0$, $\kappa_{i,0} = A_i$ is linear. For $\rho \neq 0$, (b) holds on the open set $\mathbb{B}^\circ(r_{4,i})$, so $D_u\kappa_{i,\rho}(u) = \rho^{-1} Df_i(\rho u)\,\rho = Df_i(\rho u)$, which is symplectic by Definition 21.1(i). (d) $|\phi\rho u| \le |u| \le r_{3,i}$, and $\|Df_i\| \le L$ on $\mathbb{B}(r_{3,i})$ (Lemma 21.5(b)); so the integrand has norm at most $L|u|$. (f) If $\sigma = 0$, $G^{\mathrm{sh}}_{i,0}(u) = Y_i + A_i u = G^{\mathrm{lin}}_{i,1}(u)$, which is injective. If $\sigma \in (0,1]$, put $\rho = \tau\sigma \in (0, \frac{1}{2}]$; by (b), $G^{\mathrm{sh}}_{i,\sigma}(u) = Y_i + \rho^{-1}(f_i(\rho u) - y_i)$, which is injective on $\mathbb{B}(r_{3,i})$ because $\rho\mathbb{B}(r_{3,i}) \subset \mathbb{B}(r_{4,i})$ and f_i is injective there. For every $\sigma \in [0,1]$ and $u \in \mathbb{B}(r_{3,i})$, (d) with $\rho = \tau\sigma$ gives $|G^{\mathrm{sh}}_{i,\sigma}(u) - Y_i| \le Lr_{3,i} \le Lr_{\mathrm{tot}}$. At $\sigma = 1$, (b) gives $G^{\mathrm{sh}}_{i,1}(u) = Y_i + \tau^{-1}(f_i(\tau u) - y_i) = \tau^{-1} f_i(\tau u) = T_{i,\tau}(u)$, as $Y_i = \tau^{-1} y_i$. □

Lemma 21.9 (The tail stage)**.** *The map $(v, u) \mapsto T_{i,v}(u)$ is C^∞ on the open set $\{(v, u) : v > 0,\ |vu| < \bar{r}_i\}$, which contains $(0,1] \times \mathbb{B}^\circ(r_{4,i})$. For $v \in (0,1]$:*

(a) $D_u T_{i,v}(u) = Df_i(vu)$ is symplectic, and $\partial_v T_{i,v}(u) = v^{-2}\big(v\, Df_i(vu)u - f_i(vu)\big)$;

(b) $T_{i,v}$ is injective on $\mathbb{B}(r_{3,i})$, and the sets $T_{i,v}(\mathbb{B}(r_{3,i}))$, $1 \le i \le k$, are pairwise disjoint;

(c) $|T_{i,v}(u)| < R_0/v$ for $u \in \mathbb{B}(r_{3,i})$;

(d) $T_{i,1} = f_i$, and $T_{i,\tau}(u) = G^{\mathrm{sh}}_{i,1}(u) = Y_i + \kappa_{i,\tau}(u)$ for $u \in \mathbb{B}^\circ(r_{4,i})$.

Proof. Smoothness: $(v, u) \mapsto v^{-1}$ and $(v, u) \mapsto vu$ are smooth, and f_i is smooth on $\mathbb{B}^\circ(\bar{r}_i)$; if $0 < v \le 1$ and $|u| < r_{4,i}$ then $|vu| < r_{4,i} < \bar{r}_i$. (a) is the chain and product rules. (b) Since $v\mathbb{B}(r_{3,i}) \subset \mathbb{B}(r_{4,i})$, injectivity follows from Definition 21.1(ii), and the sets $T_{i,v}(\mathbb{B}(r_{3,i})) =$

$v^{-1}f_i(v\mathbb{B}(r_{3,i}))$ are the images of the pairwise disjoint sets $f_i(v\mathbb{B}(r_{3,i})) \subset f_i(\mathbb{B}(r_{4,i}))$ (Definition 21.1(iii)) under the one injective map $y \mapsto v^{-1}y$. (c) holds because $vu \in \mathbb{B}(r_{4,i})$, so $|f_i(vu)| < R_0$ by Definition 21.1(iii). (d) is Lemma 21.8(b),(f). □

This is why all balls share one dilation parameter v: disjointness of the sets $f_i(\mathbb{B}(r_{4,i}))$ passes through a single dilation $y \mapsto v^{-1}y$, whereas a ball-by-ball comparison would need the sets $f_i(\mathbb{B}(r_{3,i})) - y_i$ to be star-shaped, which the isotopy input does not provide.

21.5. **The translation route.** The route moves the round balls from the centres s_i to the far centres Y_i. Translations are symplectic, so only collisions matter. Every ball is first lifted in the x_1-direction to a level given by its rank, then in the y_1-direction; at these levels any two balls are separated by one of the two offsets, so all balls can move in a straight line to their targets at once; then the offsets are removed in the reverse order. Ranks are chosen so that lifting and lowering never decrease distances.

Let e_1 and e_{n+1} be the unit vectors in the x_1- and y_1-directions; they are orthonormal. For real numbers $\gamma_1, \dots, \gamma_k$ let

$$\mathrm{rk}_\gamma(i) = \#\{j : \ \gamma_j < \gamma_i \text{ or } (\gamma_j = \gamma_i \text{ and } j < i)\} \in \{0, \dots, k-1\},$$

the rank of i in the lexicographic order of (γ_j, j). It is defined by finitely many case distinctions on propositions about the input (Definition M1.3).

Definition 21.10 (The route)**.** Let $\nu_i = \mathrm{rk}_\gamma(i)$ for $\gamma_j = \langle s_j, e_1\rangle$, and $\nu'_i = \mathrm{rk}_{\gamma'}(i)$ for $\gamma'_j = \langle Y_j, e_{n+1}\rangle$. Put

$$\Lambda_{\mathrm{rt}} = 1 + \sum_{j=1}^{k} 2\big(r_{3,j} + |s_j| + |Y_j|\big),$$

and, for $\boldsymbol{\sigma} = (\sigma_0, \dots, \sigma_4) \in \mathbb{R}^5$,

$$z_i(\boldsymbol{\sigma}) = s_i + \sigma_2(Y_i - s_i) + (\sigma_0 - \sigma_3)\Lambda_{\mathrm{rt}}\nu_i\, e_1 + (\sigma_1 - \sigma_4)\Lambda_{\mathrm{rt}}\nu'_i\, e_{n+1}.$$

A vector $\boldsymbol{\sigma}$ is *on the route* if there is $m \in \{0, \dots, 4\}$ with $\sigma_l = 1$ for $l < m$, $\sigma_l = 0$ for $l > m$, and $\sigma_m \in [0, 1]$ (*phase* m).

Lemma 21.11 (The route)**.** *(a) For $\gamma \in \mathbb{R}^k$ and $i \neq j$: $\mathrm{rk}_\gamma(i) \neq \mathrm{rk}_\gamma(j)$, so $|\,\mathrm{rk}_\gamma(i) - \mathrm{rk}_\gamma(j)| \geq 1$, and $(\gamma_i - \gamma_j)\big(\mathrm{rk}_\gamma(i) - \mathrm{rk}_\gamma(j)\big) \geq 0$.*

(b) $\boldsymbol{\sigma} \mapsto z_i(\boldsymbol{\sigma})$ is affine, $z_i(0, \dots, 0) = s_i$ and $z_i(1, \dots, 1) = Y_i$.

(c) If $\boldsymbol{\sigma}$ is on the route, then $|z_i(\boldsymbol{\sigma}) - z_j(\boldsymbol{\sigma})| > r_{3,i} + r_{3,j}$ for all $i \neq j$.

(d) If $\boldsymbol{\sigma} \in [0, 1]^5$, then $|z_i(\boldsymbol{\sigma})| \leq \max(|s_i|, |Y_i|) + 2(k-1)\Lambda_{\mathrm{rt}}$.

Proof. (a) Write $j \prec i$ if $\gamma_j < \gamma_i$, or $\gamma_j = \gamma_i$ and $j < i$. This is a strict total order on $\{1, \dots, k\}$, and $\mathrm{rk}_\gamma(i)$ is the number of j with $j \prec i$. If $j \prec i$, then every $l \prec j$ satisfies $l \prec i$, and $j \prec i$ but not $j \prec j$; so $\mathrm{rk}_\gamma(j) < \mathrm{rk}_\gamma(i)$, and $\gamma_j \leq \gamma_i$. Exchanging the roles gives the other case.

(b) z_i is affine in $\boldsymbol{\sigma}$ by its formula; at $\boldsymbol{\sigma} = 0$ all terms but s_i vanish, and at $\boldsymbol{\sigma} = (1, \dots, 1)$ both offsets cancel and $s_i + (Y_i - s_i) = Y_i$.

(c) We use two elementary facts. For $\xi \in \mathbb{R}^{2n}$, a unit vector e and $x \in \mathbb{R}$ with $x\langle \xi, e\rangle \geq 0$,

$$|\xi + xe|^2 = |\xi|^2 + 2x\langle \xi, e\rangle + x^2 \geq |\xi|^2; \tag{93}$$

and $|\xi| \geq |\langle \xi, e\rangle|$. Fix $i \neq j$, put $d = r_{3,i} + r_{3,j}$, $\Delta s = s_i - s_j$, $\Delta Y = Y_i - Y_j$, $\Delta\nu = \nu_i - \nu_j$, $\Delta\nu' = \nu'_i - \nu'_j$ and $\Sigma = |s_i| + |s_j| + |Y_i| + |Y_j|$. By Definition 21.1(iv) and Lemma 21.5(e), $|\Delta s| > d$ and $|\Delta Y| > d$. By (a), $|\Delta\nu|, |\Delta\nu'| \geq 1$, $\langle \Delta s, e_1\rangle\Delta\nu \geq 0$ and $\langle \Delta Y, e_{n+1}\rangle\Delta\nu' \geq 0$. Also $\Lambda_{\mathrm{rt}} \geq 1 + 2d + 2\Sigma$, so $\Lambda_{\mathrm{rt}} > d$ and $\Lambda_{\mathrm{rt}} - \Sigma > d$. Let $\boldsymbol{\sigma}$ be in phase m, and $w = z_i(\boldsymbol{\sigma}) - z_j(\boldsymbol{\sigma})$.

Phase 0. $w = \Delta s + \sigma_0\Lambda_{\mathrm{rt}}\Delta\nu\, e_1$ with $\sigma_0 \geq 0$; by (93), $|w| \geq |\Delta s| > d$.

Phase 1. $w = \Delta s + \Lambda_{\mathrm{rt}}\Delta\nu\, e_1 + \sigma_1\Lambda_{\mathrm{rt}}\Delta\nu'\, e_{n+1}$, and $e_{n+1} \perp e_1$, so $\langle w, e_1\rangle = \langle \Delta s, e_1\rangle + \Lambda_{\mathrm{rt}}\Delta\nu$. The two terms do not have opposite signs, so $|w| \geq |\langle w, e_1\rangle| \geq \Lambda_{\mathrm{rt}}|\Delta\nu| \geq \Lambda_{\mathrm{rt}} > d$.

Phase 2. $\langle w, e_1\rangle = (1-\sigma_2)\langle \Delta s, e_1\rangle + \sigma_2\langle \Delta Y, e_1\rangle + \Lambda_{\mathrm{rt}}\Delta\nu$ with $\sigma_2 \in [0,1]$. The sum of the first two terms is a convex combination of $\langle \Delta s, e_1\rangle$ and $\langle \Delta Y, e_1\rangle$, so its absolute value is at most Σ. Since $|\Delta\nu| \geq 1$, $|w| \geq |\langle w, e_1\rangle| \geq \Lambda_{\mathrm{rt}} - \Sigma > d$.

Phase 3. $w = \Delta Y + (1-\sigma_3)\Lambda_{\mathrm{rt}}\Delta\nu\, e_1 + \Lambda_{\mathrm{rt}}\Delta\nu'\, e_{n+1}$, so $\langle w, e_{n+1}\rangle = \langle \Delta Y, e_{n+1}\rangle + \Lambda_{\mathrm{rt}}\Delta\nu'$, and as in phase 1, $|w| \geq \Lambda_{\mathrm{rt}} > d$.

Phase 4. $w = \Delta Y + (1-\sigma_4)\Lambda_{\mathrm{rt}}\Delta\nu'\, e_{n+1}$ with $1-\sigma_4 \geq 0$; by (93), $|w| \geq |\Delta Y| > d$.

(d) $s_i + \sigma_2(Y_i - s_i)$ is a convex combination of s_i and Y_i; $|\sigma_0 - \sigma_3|, |\sigma_1 - \sigma_4| \leq 1$ and $\nu_i, \nu_i' \leq k-1$. □

21.6. **Concatenation and the common isotopy.** The stages run in eight time slots of length $\frac{1}{8}$: the five route phases, the linear stage, the shape stage and the tail. Each is driven by its own flat clock $\sigma_m(t) = S(8t-m)$, $m = 0, \dots, 7$, which is 0 for $t \leq m/8$ and 1 for $t \geq (m+1)/8$. We write $\boldsymbol{\sigma}(t) = (\sigma_0(t), \dots, \sigma_4(t))$ and $v(t) = \tau + (1-\tau)\sigma_7(t)$.

Construction 21.12 (The common isotopy)**.** Each bracket in (94) below vanishes while its clock is 0, and once its clock is 1 its first term cancels the subtracted term of the next bracket. So on slot m only the m-th stage moves. For $t \in \mathbb{R}$ and $u \in \mathbb{B}^\circ(r_{4,i})$ let

$$F_{i,t}(u) = z_i(\boldsymbol{\sigma}(t)) + u + \big(M_{i,\sigma_5(t)}u - u\big) + \big(\kappa_{i,\tau\sigma_6(t)}(u) - A_i u\big) + \big(T_{i,v(t)}(u) - T_{i,\tau}(u)\big). \tag{94}$$

The stage maps are $G_{i,m}(\sigma, u) = z_i(1, \dots, 1, \sigma, 0, \dots, 0) + u$ (σ in position m) for $m \leq 4$, $G_{i,5}(\sigma, u) = G^{\mathrm{lin}}_{i,\sigma}(u)$, $G_{i,6}(\sigma, u) = G^{\mathrm{sh}}_{i,\sigma}(u)$ and $G_{i,7}(\sigma, u) = T_{i,\tau+(1-\tau)\sigma}(u)$.

Each term of (94) is defined: $\tau\sigma_6(t) \in [0, \frac{1}{2}]$ and $v(t) \in [\tau, 1]$.

Lemma 21.13 (Flat concatenation)**.** *(a) Each σ_m is C^∞ with values in $[0,1]$. If $t \in [m/8, (m+1)/8]$, then $\sigma_l(t) = 1$ and $\dot\sigma_l(t) = 0$ for $l < m$, $\sigma_l(t) = 0$ and $\dot\sigma_l(t) = 0$ for $l > m$, and $\boldsymbol{\sigma}(t)$ is on the route if $m \leq 4$. All clocks and their derivatives vanish for $t \leq 0$; all clocks equal 1 and their derivatives vanish for $t \geq 1$.*

(b) For $t \in [m/8, (m+1)/8]$ and $u \in \mathbb{B}^\circ(r_{4,i})$, $F_{i,t}(u) = G_{i,m}(\sigma_m(t), u)$. Explicitly: $F_{i,t}(u) = z_i(\boldsymbol{\sigma}(t)) + u$ if $m \leq 4$; $Y_i + M_{i,\sigma_5(t)}u$ if $m = 5$; $Y_i + \kappa_{i,\tau\sigma_6(t)}(u)$ if $m = 6$; $T_{i,v(t)}(u)$ if $m = 7$. Moreover $G_{i,m}(1, \cdot) = G_{i,m+1}(0, \cdot)$ for $m < 7$.

(c) $(t, u) \mapsto F_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$.

(d) $F_{i,t} = F_{i,0}$ for $t \leq 0$, $F_{i,t} = F_{i,1}$ for $t \geq 1$, $F_{i,0}(u) = s_i + u$ and $F_{i,1} = f_i$ on $\mathbb{B}^\circ(r_{4,i})$.

(e) For $t \in [m/8, (m+1)/8]$,

$$\partial_t F_{i,t}(u) = \dot\sigma_m(t)\, \partial_\sigma G_{i,m}(\sigma_m(t), u),$$
$$D_u F_{i,t}(u) = D_u G_{i,m}(\sigma_m(t), u).$$

In particular $\partial_t F_{i,t} = 0$ for $t \notin (0,1)$.

Proof. (a) By Lemma 2.4, S is C^∞ with values in $[0,1]$, $S = 0$ on $(-\infty, 0]$, $S = 1$ on $[1, \infty)$, and $S' = 0$ on both half-lines. For $t \in [m/8, (m+1)/8]$ and $l < m$, $8t - l \geq 1$; for $l > m$, $8t - l \leq 0$. For $t \leq 0$ and $l \geq 0$, $8t - l \leq 0$; for $t \geq 1$ and $l \leq 7$, $8t - l \geq 1$.

(b) Insert the clock values of (a) into (94), and use $M_{i,0} = I$, $M_{i,1} = A_i$ (Lemma 21.3(c)), $\kappa_{i,0} = A_i$ (Lemma 21.8(b)) and $z_i(1, \dots, 1) = Y_i$ (Lemma 21.11(b)). If $m \leq 4$, then $\sigma_5 = \sigma_6 = \sigma_7 = 0$ and the last three brackets vanish. If $m \geq 5$, then $z_i(\boldsymbol{\sigma}(t)) = Y_i$. If $m = 5$, the last two brackets vanish. If $m = 6$, $\sigma_5 = 1$ and $u + (A_i u - u) + (\kappa_{i,\tau\sigma_6}(u) - A_i u) = \kappa_{i,\tau\sigma_6}(u)$. If $m = 7$, $\sigma_5 = \sigma_6 = 1$ and $F_{i,t}(u) = Y_i + \kappa_{i,\tau}(u) + T_{i,v(t)}(u) - T_{i,\tau}(u) = T_{i,v(t)}(u)$ by Lemma 21.9(d). In each case the result is $G_{i,m}(\sigma_m(t), u)$. The matching conditions are $z_i(1, \dots, 1) = Y_i = G^{\mathrm{lin}}_{i,0}(u) - u$

for $m = 4$, Lemma 21.8(f) for $m = 5$ and Lemma 21.9(d) for $m = 6$; for $m < 4$ both sides are z_i at the same point.

(c) The clocks are smooth, and each term of (94) is a smooth map composed with $(t, u) \mapsto$ (clock values, u): z_i is affine; $(\sigma, u) \mapsto M_{i,\sigma}u$ is smooth on $\mathbb{R} \times \mathbb{R}^{2n}$ (Lemma 21.3(c)); $(\rho, u) \mapsto \kappa_{i,\rho}(u)$ is smooth on Ω_i, which contains $(\tau\sigma_6(t), u)$ (Lemma 21.8(a)); and $(v, u) \mapsto T_{i,v}(u)$ is smooth on an open set containing $(v(t), u)$ and (τ, u) (Lemma 21.9).

(d) For $t \le 0$ all clocks are 0 and for $t \ge 1$ all are 1, so $F_{i,t}$ depends on t only through constant clock values. At $t = 0$, $F_{i,0}(u) = z_i(0) + u = s_i + u$ by (b). At $t = 1$, $F_{i,1} = T_{i,1} = f_i$ by (b) with $m = 7$ and Lemma 21.9(d).

(e) By the chain rule, $\partial_t F_{i,t}(u)$ is the sum over l of $\dot\sigma_l(t)$ times the partial derivative of the l-th term of (94) with respect to its clock; by (a) only $l = m$ contributes on the closed slot $[m/8, (m+1)/8]$, and that partial derivative is $\partial_\sigma G_{i,m}(\sigma_m(t), u)$. The identity of (b) holds for all u in the open ball, so D_u may be computed from it. For $t \notin (0, 1)$ all $\dot\sigma_l(t)$ vanish. □

Proposition 21.14 (The common symplectic isotopy)**.** *For every isotopy input, the maps $F_{i,t}$ of Construction 21.12 satisfy:*

- *(a) $(t, u) \mapsto F_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$; $F_{i,0}(u) = s_i + u$, $F_{i,1} = f_i$; $F_{i,t} = F_{i,0}$ for $t \le 0$ and $F_{i,t} = F_{i,1}$ for $t \ge 1$;*
- *(b) $D_u F_{i,t}(u) \in \mathrm{Sp}(2n)$ for all $t \in \mathbb{R}$ and $u \in \mathbb{B}^\circ(r_{4,i})$;*
- *(c) each $F_{i,t}$ is injective on $\mathbb{B}(r_{3,i})$, and for every t the sets $F_{i,t}(\mathbb{B}(r_{3,i}))$, $1 \le i \le k$, are pairwise disjoint;*
- *(d) $F_{i,t}(\mathbb{B}(r_{3,i})) \subset \mathbb{B}(R_F)$ for all i and t, where $R_F = 2k\Lambda_{\mathrm{rt}} + Lr_{\mathrm{tot}} + R_0/\tau$.*

Proof. (a) is Lemma 21.13(c),(d). Every $t \in \mathbb{R}$ lies in a closed slot $[m/8, (m+1)/8]$, $0 \le m \le 7$, or in $(-\infty, 0]$ or $[1, \infty)$, where $F_{i,t}$ equals $F_{i,0}$ (slot 0 at $\sigma_0 = 0$) or $F_{i,1}$ (slot 7 at $\sigma_7 = 1$). By Lemma 21.13(b),(e) it suffices to check (b)–(d) for $F_{i,t} = G_{i,m}(\sigma, \cdot)$ with $\sigma = \sigma_m(t) \in [0, 1]$, and for $u \in \mathbb{B}(r_{3,i})$ in (c), (d).

Route slots, $m \le 4$. $F_{i,t}(u) = z_i + u$ with $z_i = z_i(\boldsymbol{\sigma}(t))$ and $\boldsymbol{\sigma}(t)$ on the route. The derivative is I; a translation is injective; the images are the balls $\mathbb{B}(z_i, r_{3,i})$, which are pairwise disjoint because $|z_i - z_j| > r_{3,i} + r_{3,j}$ (Lemma 21.11(c)). By Lemma 21.11(d) and $\max(|s_i|, |Y_i|) + r_{3,i} \le \Lambda_{\mathrm{rt}}$, $|F_{i,t}(u)| \le \Lambda_{\mathrm{rt}} + 2(k-1)\Lambda_{\mathrm{rt}} \le 2k\Lambda_{\mathrm{rt}}$.

Linear slot, $m = 5$. Lemma 21.7: the derivative $M_{i,\sigma}$ is symplectic, the map is injective, the images lie in the pairwise disjoint balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, and $|F_{i,t}(u)| \le |Y_i| + L \le \Lambda_{\mathrm{rt}} + Lr_{\mathrm{tot}}$.

Shape slot, $m = 6$. Lemma 21.8(c),(f): the derivative $Df_i(\tau\sigma u)$ is symplectic, the map is injective on $\mathbb{B}(r_{3,i})$, the images lie in the pairwise disjoint balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$, and $|F_{i,t}(u)| \le \Lambda_{\mathrm{rt}} + Lr_{\mathrm{tot}}$.

Tail slot, $m = 7$. Lemma 21.9 with $v = v(t) \in [\tau, 1]$: symplectic derivative, injectivity, disjointness, and $|F_{i,t}(u)| < R_0/v \le R_0/\tau$.

Since $k \ge 1$ and all terms of R_F are positive, each bound is at most R_F. □

21.7. **The source primitive on each stage and the pushforward.** Let $\mathfrak{a}_{i,t}$ and $\mathfrak{h}_{i,t}$ be the source one-form and primitive of Definition M6.7 for the isotopy of Construction 21.12: $\mathfrak{a}_{i,t}(u)[\xi] = \omega_0(\partial_t F_{i,t}(u), D_u F_{i,t}(u)\xi)$ and $\mathfrak{h}_{i,t}(u) = \int_0^1 \mathfrak{a}_{i,t}(\vartheta u)[u]\, d\vartheta$. On each slot they have a closed form.

Lemma 21.15 (The primitive on each stage)**.** *Let $t \in [m/8, (m+1)/8]$, $\sigma = \sigma_m(t)$ and $u \in \mathbb{B}^\circ(r_{4,i})$. Then $\mathfrak{h}_{i,t}(u) = \dot\sigma_m(t)\, \hat h_{i,m,\sigma}(u)$, where:*

- *(a) for $m \le 4$, $\hat h_{i,m,\sigma}(u) = \omega_0(w_{i,m}, u)$ with the constant vectors $w_{i,0} = -w_{i,3} = \Lambda_{\mathrm{rt}}\nu_i e_1$, $w_{i,1} = -w_{i,4} = \Lambda_{\mathrm{rt}}\nu_i' e_{n+1}$ and $w_{i,2} = Y_i - s_i$;*
- *(b) for $m = 5$, $\hat h_{i,5,\sigma}(u) = \frac{1}{2}\,\omega_0\big(\dot M_{i,\sigma}u, M_{i,\sigma}u\big)$, $\dot M_{i,\sigma} = \frac{d}{d\sigma}M_{i,\sigma}$;*

(c) for $m=6$, $\hat h_{i,6,\sigma}=\tau\,h^{\mathrm{sh}}_{i,\tau\sigma}$, where for $|\rho|<1$

$$h^{\mathrm{sh}}_{i,\rho}(u)=\int_0^1\!\!\int_0^1 \vartheta^2\phi\;\omega_0\big(D^2f_i(\phi\rho\vartheta u)[u,u],\;Df_i(\rho\vartheta u)u\big)\,d\phi\,d\vartheta;$$

(d) for $m=7$, $\hat h_{i,7,\sigma}=(1-\tau)\,h^{\mathrm{tl}}_{i,v}$ with $v=\tau+(1-\tau)\sigma$, where for $0<v\le 1$

$$h^{\mathrm{tl}}_{i,v}(u)=\frac{1}{v^2}\int_0^1\omega_0\big(Df_i(v\vartheta u)u,\;f_i(v\vartheta u)\big)\,d\vartheta.$$

Proof. By Lemma 21.13(e), on the slot $\partial_tF_{i,t}=\dot\sigma_m(t)\,\partial_\sigma G_{i,m}(\sigma,\cdot)$ and $D_uF_{i,t}=D_uG_{i,m}(\sigma,\cdot)$; $\mathfrak a$ and $\mathfrak h$ are linear in ∂_tF, so $\mathfrak h_{i,t}=\dot\sigma_m(t)\hat h$ with $\hat h(u)=\int_0^1\omega_0\big(\partial_\sigma G_{i,m}(\sigma,\vartheta u),D_uG_{i,m}(\sigma,\vartheta u)u\big)d\vartheta$. (a) $\partial_\sigma G_{i,m}=\partial_{\sigma_m}z_i=w_{i,m}$ by Definition 21.10, and $D_uG=I$. (b) $\partial_\sigma G(\vartheta u)=\vartheta\dot M_{i,\sigma}u$ and $D_uG=M_{i,\sigma}$, so $\hat h(u)=\int_0^1\vartheta\,d\vartheta\cdot\omega_0(\dot M_{i,\sigma}u,M_{i,\sigma}u)$. (c) With $\rho=\tau\sigma$, $\partial_\sigma G(x)=\tau\partial_\rho\kappa_{i,\rho}(x)=\tau\int_0^1\phi D^2f_i(\phi\rho x)[x,x]\,d\phi$ (Lemma 21.8(e)); at $x=\vartheta u$ the bilinear term contributes the factor ϑ^2, and $D_uG(\vartheta u)u=Df_i(\rho\vartheta u)u$ (Lemma 21.8(c)). (d) $\partial_\sigma G=(1-\tau)\partial_vT_{i,v}$, and by Lemma 21.9(a), at $x=\vartheta u$ with $\xi=Df_i(v\vartheta u)u$, $\partial_vT_{i,v}(\vartheta u)=v^{-2}\big(v\vartheta\xi-f_i(v\vartheta u)\big)$ and $D_uT_{i,v}(\vartheta u)u=\xi$. Since $\omega_0(\xi,\xi)=0$, $\omega_0\big(v^{-2}(v\vartheta\xi-f_i(v\vartheta u)),\xi\big)=v^{-2}\omega_0\big(\xi,f_i(v\vartheta u)\big)$. □

In (c) the difference quotient of Lemma 21.8(b) has disappeared, so $h^{\mathrm{sh}}_{i,\rho}$ is smooth through $\rho=0$.

The Hamiltonian of Definition M6.10 is a sum of pushforwards of cut-off source functions. The following lemma, stated for one ball, gives their smoothness and support.

Lemma 21.16 (The pushforward of a cut-off source function)**.** [Lean: needed consequence verified.]

Let $0<r_2<r_3<r_4$, and let $F\colon\mathbb R\times\mathbb B^\circ(r_4)\to\mathbb R^{2n}$, $(t,u)\mapsto F_t(u)$, be C^∞ with $D_uF_t(u)\in\mathrm{Sp}(2n)$, F_t injective on $\mathbb B(r_3)$ for every t, $F_t=F_0$ for $t\le 0$ and $F_t=F_1$ for $t\ge 1$. Let $g\colon\mathbb R\times\mathbb B^\circ(r_4)\to\mathbb R$ be C^∞ with $g(t,u)=0$ whenever $|u|\ge r_2$, and define

$$\mathrm{Push}(g)(t,y)=\begin{cases} g(t,u), & \text{if } u \text{ is the unique point of } \mathbb B(r_3) \text{ with } F_t(u)=y,\\ 0, & \text{if there is no such unique point.}\end{cases}$$

Let $\Gamma(t,u)=(t,F_t(u))$, $W=\mathbb R\times\mathbb B^\circ(r_3)$ and $Z=\Gamma(\mathbb R\times\mathbb B(r_2))$.

(a) $O=\Gamma(W)$ is open in $\mathbb R\times\mathbb R^{2n}$, and Γ restricts to a C^∞ diffeomorphism $W\to O$.

(b) Z is closed, $Z\subset O$, $\mathrm{Push}(g)=g\circ(\Gamma|_W)^{-1}$ on O, and $\mathrm{Push}(g)=0$ outside Z.

(c) $\mathrm{Push}(g)$ is C^∞ on $\mathbb R\times\mathbb R^{2n}$.

(d) For $t\in\mathbb R$, $|u|<r_3$ and $\xi\in\mathbb R^{2n}$: $d_y\,\mathrm{Push}(g)(t,F_t(u))\big[D_uF_t(u)\xi\big]=d_ug(t,u)[\xi]$.

(e) $\mathrm{Push}(g)(t,\cdot)$ vanishes outside $F_t(\mathbb B(r_2))$, a subset of the compact set

$$K=\bigcup_{t\in[0,1]}F_t(\mathbb B(r_2)).$$

If $g(t,\cdot)=0$ for $t\notin(0,1)$, then $\mathrm{Push}(g)(t,\cdot)=0$ for $t\notin(0,1)$.

Proof. (a) Γ is C^∞ on $\mathbb R\times\mathbb B^\circ(r_4)$ with derivative $\left(\begin{smallmatrix}1 & 0\\ \partial_tF & D_uF\end{smallmatrix}\right)$, which is invertible because D_uF is symplectic. By the inverse function theorem Γ is a local C^∞ diffeomorphism at every point. It is injective on $\mathbb R\times\mathbb B(r_3)$: $\Gamma(t,u)=\Gamma(t',u')$ gives $t=t'$ and $F_t(u)=F_t(u')$, so $u=u'$. An injective local diffeomorphism of the open set W is an open map and a diffeomorphism onto its open image, its inverse being given near each point by a local inverse.

(b) Z is the union of $(-\infty,0]\times F_0(\mathbb B(r_2))$, $\Gamma([0,1]\times\mathbb B(r_2))$ and $[1,\infty)\times F_1(\mathbb B(r_2))$, because F_t is constant in t outside $[0,1]$. The middle set and the sets $F_0(\mathbb B(r_2))$, $F_1(\mathbb B(r_2))$ are compact as continuous images of compact sets, so all three sets are closed and Z is closed. $Z\subset O$ because $r_2<r_3$. If $(t,y)\in O$, then $y=F_t(u)$ with $(t,u)=(\Gamma|_W)^{-1}(t,y)$, and $u\in\mathbb B(r_3)$ is the unique

such point by injectivity; so $\mathrm{Push}(g)(t,y) = g((\Gamma|_W)^{-1}(t,y))$. If $(t,y) \notin Z$ and $u \in \mathbb{B}(r_3)$ satisfies $F_t(u) = y$, then $|u| > r_2$ and $g(t,u) = 0$; otherwise $\mathrm{Push}(g)(t,y) = 0$ by definition.

(c) On the open set O, $\mathrm{Push}(g)$ is the composition of the smooth maps $(\Gamma|_W)^{-1}$ and g; on the open set $(\mathbb{R} \times \mathbb{R}^{2n}) \setminus Z$ it vanishes. The two open sets cover $\mathbb{R} \times \mathbb{R}^{2n}$ because $Z \subset O$.

(d) On W, $\mathrm{Push}(g) \circ \Gamma = g$ by (b). Differentiate at (t,u) in the direction $(0,\xi)$, using $D\Gamma(t,u)(0,\xi) = (0, D_u F_t(u)\xi)$.

(e) The first statement is (b) at fixed t; for $t \notin [0,1]$, $F_t(\mathbb{B}(r_2))$ equals $F_0(\mathbb{B}(r_2))$ or $F_1(\mathbb{B}(r_2))$, both contained in K. K is the image of the compact set $[0,1] \times \mathbb{B}(r_2)$ under the continuous map $(t,u) \mapsto F_t(u)$. The last statement is immediate from the definition. □

For ball i take $r_\ell = r_{\ell,i}$, $F = F_i$ (Proposition 21.14(a)–(c)) and $g = \chi_i\, \mathfrak{h}_i$ with the cutoff $\chi_i = \chi_{r_{1,i}, r_{2,i}}$ of Definition M6.10. By Lemma M6.9, $\mathfrak{h}_i$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$ and vanishes for $t \notin (0,1)$; by Lemma 2.7, χ_i is C^∞ and vanishes for $|u| \geq r_{2,i}$. Then $\mathrm{Push}(g)$ is the function H_i of Definition M6.10, and Lemma 21.16 together with Proposition 21.14(d) gives Lemma M6.11.

22. A worked instance: ten balls in dimension four

This section substitutes $k = 10$ and $\varepsilon = \frac{1}{10}$ into Theorem A and follows the data through every stage of the construction. It is a substitution of parameters into the closed formulas of the paper and of this supplement, together with a check of the inequalities that the proofs require. No numerical evaluation of the Hamiltonian, of the flows or of the maps f_i is reported, and the quantities that the construction defines as maxima of continuous functions over compact sets are not evaluated; they are listed at the end.

Conventions. The tables give exact expressions in radicals and π, followed by decimal approximations rounded to seven significant digits. The displayed decimals are for orientation; the inequalities use the exact expressions. The smallest gap in the displayed chains exceeds $4 \cdot 10^{-5}$.

22.1. **Input, normalisation and branch.** The input is $n = 2$, $k = 10$, $c_i = \pi$, $s_i = (3i - 3, 0, 0, 0)$ $(1 \leq i \leq 10)$ and

$$R^4 = \frac{k}{1 - \varepsilon/4} = \frac{400}{39}, \qquad R \approx 1.789571, \qquad C = \pi R^2 = \frac{20\pi}{\sqrt{39}} \approx 10.06115.$$

The closed balls $B(s_i, \pi)$ are the Euclidean unit balls about the s_i; distinct centres are at distance at least $3 > 2$, so they are pairwise disjoint. The normalised capacity (Definition M6.1) is

$$\rho = \frac{\pi}{C} = \frac{1}{R^2} = \frac{\sqrt{39}}{20} \approx 0.3122499, \qquad k\rho^2 = \frac{39}{40} = 1 - \frac{\varepsilon}{4}, \qquad \rho\sqrt{k} = \sqrt{\frac{39}{40}} \approx 0.9874209.$$

Admissibility: $k\rho^2 < 1$ and $3\rho \approx 0.9367497 < 1$. Since $n = 2$, $k \geq 10$, the capacities are equal and $\frac{3}{4} < \rho\sqrt{k} < 1$, Construction 20.1(i) takes the equal-capacity maps of Remark 13.25, that is, case (a) of Construction 13.22 with the linear comparison datum of Proposition M3.8. The same holds for every $k \geq 10$ and $\varepsilon \in (0,1)$, because then $\rho\sqrt{k} = \sqrt{1 - \varepsilon/4} \in (\frac{\sqrt{3}}{2}, 1)$.

The defect is $\eta_\star = 1 - \rho\sqrt{k} = 1 - \sqrt{39/40} \approx 0.01257912$ (Remark 13.25).

22.2. **Model parameters.** Table 5 lists the parameters of the model, the comparison datum, the handle and the transfer. The sources are Remarks 13.24 and 13.25, §M3.4, Definition 3.10, Definitions 10.14 and 10.16, Construction 10.27, Lemma 13.9, Lemma 13.12 and Remark 10.30.

Checks. The chain required by Definition 10.13(a) and Proposition 10.29 reads

$$\rho \approx 0.3122499 < \rho^+ \approx 0.3127469 < \bar{\rho} \approx 0.3132444 < \hat{\rho} \approx 0.3134930$$
$$< a \approx 0.3142388 < h \approx 0.3312368 < d_n \approx 0.3322851 < \tfrac{1}{3},$$

Quantity	Closed form	Value
$t_0 = \frac{1}{2}(1+\rho\sqrt{k})$	$1 - \frac{\eta_\star}{2}$	0.9937104
auxiliary capacity a	$t_0/\sqrt{10}$	0.3142388
height h	$t_0/3$	0.3312368
$d_n = \frac{1}{2}(h+\frac{1}{3})$	$(1+t_0)/6 = \frac{1}{3} - \frac{\eta_\star}{12}$	0.3322851
$a - \rho$	$\eta_\star/(2\sqrt{10})$	$1.988933 \cdot 10^{-3}$
$1 - 3h$	$\eta_\star/2$	$6.289559 \cdot 10^{-3}$
$d_n - h$, $1 - 3d_n$	$\eta_\star/12$, $\eta_\star/4$	$1.048260 \cdot 10^{-3}$, $3.144779 \cdot 10^{-3}$
profile $P_{\mathrm{b}}(p)$	$3(1-3p) - 10(a-p)_+$ on $[0,h]$	$P_{\mathrm{b}}(0) \approx -0.1423883$, $P_{\mathrm{b}}(h) = g$
$\int_0^h P_{\mathrm{b}}$	$t_0(1-t_0) = \frac{1}{4}(1-k\rho^2) = \frac{\varepsilon}{16}$	0.00625
comparison function $K(p) = c_s p$	$c_s = 3\sqrt{10} - 9$	$c_s \approx 0.4868330$
slopes of the tent Λ	$A = \sqrt{10}/3$, $B = 10 + 3\sqrt{10}$	1.054093, 19.48683
$g = 3(1-t_0) = H_{\mathrm{b}}(h)$	$3\eta_\star/2$	0.01886868
$e_q = \min(\frac{h}{100}, \frac{g}{128 c_s})$	$g/(128 c_s)$ (as $\frac{h}{100} \approx 3.312368 \cdot 10^{-3}$)	$3.027969 \cdot 10^{-4}$
planar cost $4 c_s e_q$	$g/32$	$5.896461 \cdot 10^{-4}$
comparison gap γ	$\frac{31}{32} g$	0.01827903
planar support P_{supp}	$h - \frac{31}{8} e_q < P_{\mathrm{supp}} < h$ (Lemmas 5.31, 4.12)	$\in$ (0.3300635, 0.3312368)
background e_h, cap scale μ_h	$g/16$, $g/160$	$1.179292 \cdot 10^{-3}$, $1.179292 \cdot 10^{-4}$
cap excess $k\mu_h/2$; residual	$g/32$; $\frac{7}{8} g$	$5.896461 \cdot 10^{-4}$; 0.01651009
$\delta_m = \min(\frac{d_n - h}{4}, \frac{g}{100})$	$g/100 = 3\eta_\star/200$ (as $\frac{d_n-h}{4} = \frac{\eta_\star}{48}$)	$1.886868 \cdot 10^{-4}$
moment cutoff $\mathfrak{c}$	1 for $p \le h + \frac{3}{2}\delta_m$, 0 for $p \ge h + \frac{7}{4}\delta_m$	0.3315198, 0.3315670
safe region bound $h + 2\delta_m$	$h + 3\eta_\star/100$	0.3316142
$\frac{15}{16} g - 18\delta_m$	lower bound for H^* on the safe region	0.01429302
nondegeneracy of Ω_λ	$p < \frac{1}{3} - \frac{e_h}{9}$	0.3332023
clock, layers	$u_0 = \frac{7}{48}$, $u_1 = \frac{3}{16}$, $N_\ell = 11$, $\ell_s = \frac{1}{264}$	$\ell_s \approx 3.787879 \cdot 10^{-3}$
layer i	$l_i = \frac{7}{48} + \frac{i-1}{264} + \frac{1}{1056}$, $r_i = l_i + \frac{1}{528}$	$l_1 \approx 0.1467803$, $r_{10} \approx 0.1827652$
strip slack $\delta = \frac{1}{2}(a - \rho)$	$\eta_\star/(4\sqrt{10})$	$9.944665 \cdot 10^{-4}$
Morse parameter of Ψ_δ	δ/π	$3.165485 \cdot 10^{-4}$
$\bar\rho = \frac{1}{2}(\rho + a)$	$\rho + \delta$	0.3132444
$\hat\rho = a - \frac{3}{4}\delta$	$\rho + \frac{5}{4}\delta$	0.3134930
$r_4^2 = \bar\rho/\rho$, $r_3 = \frac{1}{2}(1 + r_4)$	$r_4^2 = \frac{4 - 3\eta_\star}{4(1-\eta_\star)}$	$r_4 \approx 1.001591$, $r_3 \approx 1.000796$
enlarged capacity $\rho^+ = \rho r_3^2$	$\frac{\rho}{4}(1 + r_4)^2$	0.3127469
transport: $w_* = \frac{h}{d_n - h}$	$4t_0/\eta_\star$	315.9873
T_0, $c_* = \min(\frac{3}{2}, \frac{T_0^2}{d_n})$	$T_0 = (6w_*)^{-1/2}$, $c_* = \frac{\eta_\star}{4t_0(1+t_0)}$	0.02296622, $1.587334 \cdot 10^{-3}$
return: $1 - \frac{3}{2} d_n$	$(3 - t_0)/4$	0.5015724

TABLE 5. Model parameters for $k = 10$, $\varepsilon = \frac{1}{10}$; $\eta_\star = 1 - \sqrt{39/40}$.

and $0 < g \le H_{\mathrm{b}}(h)$ holds with equality. The comparison identity $F = K - P_{\mathrm{b}} = -g + c_s\Lambda$ of Proposition M3.8 uses $c_s A = k - 3\sqrt{k}$, $c_s B = 3\sqrt{k}$ and $3\sqrt{k}\, a = 3 - g$; numerically $c_s A = 10 - 3\sqrt{10} \approx 0.5131670$ and $c_s B = 3\sqrt{10} \approx 9.486833$. The safe region lies below d_n and below the nondegeneracy bound: $h + 2\delta_m \approx 0.3316142 < d_n < \frac{1}{3} - \frac{e_h}{9}$. The volume of $B^\circ(1)$ not filled by the ten balls is $\frac{1}{2}(1 - k\rho^2) = \frac{\varepsilon}{8}$. Half of it, $\int_0^h P_{\mathrm{b}} = \frac{\varepsilon}{16}$, remains as the integral of the profile,

from which the gap is paid; the other half, $\frac{1}{2}k(a^2-\rho^2)+\frac{1}{2}(1-3h)^2$, is taken by the capacity slack and the height defect (Remark 13.25).

22.3. **The common isotopy and the Hamiltonian.** The physical maps are $f_i(u) = \sqrt{C}\, f_i^{\natural}(u/\sqrt{C})$ with $\sqrt{C} \approx 3.171931$ (Proposition M6.3). The embedding datum has $c_i' = \pi r_3^2$ and $\bar{c}_i = \pi r_4^2$. Table 6 lists the radii of Definition M6.4; here the source radius is $r_i = 1$ and the centre gap is $\min_{i\neq j}(|s_i - s_j| - 2) = 1$.

Quantity	Closed form	Value
c_i', $\bar{c}_i$	πr_3^2, πr_4^2	3.146593, 3.151598
r_i', $\bar{r}_i$	r_3, r_4	1.000796, 1.001591
step $\min(\frac{1}{4},\, r_i' - r_i)$	$r_3 - 1$	$7.955775 \cdot 10^{-4}$
$r_{1,i}, r_{2,i}, r_{3,i}, r_{4,i}$	$1 + \frac{\ell}{5}(r_3 - 1)$	1.000159, 1.000318, 1.000477, 1.000636
target radius $R_C = \sqrt{C/\pi}$	R	1.789571

TABLE 6. Radii of the common isotopy for $k = 10$, $\varepsilon = \frac{1}{10}$.

The Hamiltonian $H = H[2, 10, C, (\pi, \dots, \pi), s]$ of Definition M6.16 is assembled as follows.

(1) *Model embeddings.* For $1 \le i \le 10$,
$$f_i^{\natural} = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i$$
on $B^\circ(\hat{\rho})$, with the parameters of Table 5: the layer map E_i uses Ψ_δ, the shear with $A_i'(p)$ and the root in (l_i, r_i); the twist $\mathcal{D}$ uses the planar isotopy q_t of Proposition M3.8 with e_q above; Φ is the time-one map of the cut field of Construction 10.27 with δ_m above; the transfer uses d_n and w_* above. Only E_i depends on i (Theorem 13.23(iv)).

(2) *Route translations* (slots $[\frac{m}{8}, \frac{m+1}{8}]$, $0 \le m \le 4$): the balls $s_i + u$ are moved, by translations with ranks fixed by their starting and target heights, to the centres $Y_i = y_i/\tau$, $y_i = f_i(0)$ (Definition 21.10).

(3) *Linear symplectic stage* (slot 5): $Y_i + M_{i,\sigma}u$, with $M_{i,\sigma}$ the path of Definition 21.2 from I to $A_i = Df_i(0)$.

(4) *Shape stage* (slot 6): $Y_i + \kappa_{i,\tau\sigma}(u)$, from $Y_i + A_i u$ to $\tau^{-1} f_i(\tau u)$.

(5) *Tail* (slot 7): $v^{-1} f_i(vu)$ for v from τ to 1, ending at f_i; the slots are joined by flat clocks (Construction 21.12).

(6) *Radial primitive and cutoff*: $\mathfrak{h}_{i,t}(u) = \int_0^1 \omega_0(\dot{F}_{i,t}(\vartheta u), D_u F_{i,t}(\vartheta u)u)\, d\vartheta$, the radial cutoff χ_i equal to one on $\mathbb{B}(r_{1,i})$ and zero outside $\mathbb{B}^\circ(r_{2,i})$, and $H(t,y) = \sum_i \chi_i(u)\mathfrak{h}_{i,t}(u)$ with u the unique preimage of y under $F_{i,t}$ in $\mathbb{B}(r_{3,i})$, and 0 where there is none (Definition M6.10).

By Theorem M6.14, $\phi_H^1(B(s_i, \pi)) = f_i(B(\pi))$; these ten sets are pairwise disjoint and lie in the compact set $\bigcup_i f_i(B(c_i')) \subset B^\circ(\pi R^2)$, so in the Euclidean ball of radius $R \approx 1.789571$, and $k/R^4 = \frac{39}{40} > 1 - \varepsilon$.

Not evaluated. The following numbers enter the definition but are maxima or minima of continuous functions over compact sets, or depend on such numbers, and the proofs show only that they are positive or finite: the base speed C_b of the handle flow (Proposition 10.28); the target margin $m = \frac{1}{2}\min_{\mathcal{K}}(1 - \pi|z|^2)$ of Construction 13.22; the points $y_i = f_i(0)$ and matrices $A_i = Df_i(0)$; the bound L, the scale τ, the separation δ_c and the support radius R_F of Definition 21.4 and Proposition 21.14; and, for the finite form, C_0, R_*, b_R, L_X and the Bernstein degree of §M7.1. The planar support area P_{supp} is a closed expression in the constants of Definitions 5.27 and 5.32; only the bracket of Table 5 is recorded.

23. Input names and elementary programs

The finite verifier reads the capacity and centre names, the radius names derived from them, and performs integer arithmetic. This section gives the programs for the derived names. None of them evaluates the geometric construction of level L1.

Definition 23.1 (Names)**.** Put $\varepsilon_p = 2^{-p}$. A rational stream q *names* $x \in \mathbb{R}$ if $|q(p) - x| \le \varepsilon_p$ for every $p \in \mathbb{N}$. Rationals are coded by the natural triples of Definition 25.1, so a name is a function $\mathbb{N} \to \mathbb{N}$. A family $(x_c)_{c\in\mathbb{N}}$ is named by one stream $q(\langle c, p\rangle)$, where $\langle\cdot,\cdot\rangle$ is the pairing bijection $\mathbb{N}^2 \to \mathbb{N}$. A program *computes relative to* given names if it is a partial recursive function relative to those streams as oracles (`Nat.RecursiveIn`); an oracle-free program results when the streams are themselves computable.

Lemma 23.2 (Strict certification)**.** *If q, r name x, y, the rational test $q(p) + \varepsilon_p < r(p) - \varepsilon_p$ implies $x < y$, and if $x < y$ it holds for every p with $4\varepsilon_p < y - x$. Consequently, for a family of finite conjunctions of strict inequalities between named reals, given uniformly in the candidate index, a search over pairs (candidate, precision) halts if some conjunction is true, and every pair it accepts has a true conjunction.*

Proof. The first assertion follows from the two name bounds. If $4\varepsilon_p < y - x$, then $q(p) + \varepsilon_p \le x + 2\varepsilon_p < y - 2\varepsilon_p \le r(p) - \varepsilon_p$. For a true conjunction one precision suffices for all its finitely many inequalities, so the search reaches an accepted pair. □

Proposition 23.3 (Arithmetic on names)**.** *Sums, negatives, products, absolute values and maxima have uniform total name programs. If an integer $a \ge 0$ with $2^{-a} \le |x|$ is supplied, $1/x$ has one as well: at precision p it inverts the approximation of x at precision $p + 2a + 2$. Each program makes finitely many oracle queries, at indices computed from p and earlier answers, and is total.*

Proof. Addition reads both inputs at precision $p+1$; negation changes sign. Absolute value and maximum are 1-Lipschitz, so they read at precision p. For multiplication let q, r name x, y and put $M_q = \lfloor |q(0)| \rfloor + 2$, $M_r = \lfloor |r(0)| \rfloor + 2$, so $|x| \le M_q$ and $|y| \le M_r$, and $K = M_q + M_r + 1$. The program returns $u'v'$ with $u' = q(p + K)$, $v' = r(p + K)$; since $|v'| \le M_r + 1$,

$$|xy - u'v'| \le |x|\,|y - v'| + |v'|\,|x - u'| \le K 2^{-(p+K)} \le 2^{-p}.$$

For the reciprocal, the queried rational u satisfies $|u| \ge 2^{-a} - 2^{-a-2} \ge 2^{-a-1}$, so

$$|u^{-1} - x^{-1}| = \frac{|u - x|}{|ux|} \le 2^{2a+1} 2^{-(p+2a+2)} < 2^{-p}.$$

All operations on rationals are integer operations on their codes. □

Lemma 23.4 (Quadrature)**.** *Let f be L-Lipschitz on $[0, 1]$ and let $N \ge 1$. Then the left Riemann sum $N^{-1} \sum_{j<N} f(j/N)$ differs from $\int_0^1 f$ by at most L/N. If every sample is replaced by a rational within θ, the error is at most $L/N + \theta$.*

Proof. On $[j/N, (j + 1)/N]$ the integrand differs from $f(j/N)$ by at most L/N, so the integral differs from $f(j/N)/N$ by at most L/N^2. Sum over j. The average of the sample errors is at most θ. □

Lemma 23.5 (A name of π)**.** *The identity $\pi = \int_0^1 4/(1 + t^2)\,dt$ gives a computable rational name of π, and the lower bound $1 \le \pi$ gives a computable name of $1/\pi$.*

Proof. The identity holds because $4/(1+t^2)$ is the derivative of $4\arctan t$. On $[0,1]$ the integrand is 8-Lipschitz:

$$\left|\frac{4}{1+x^2}-\frac{4}{1+y^2}\right| = \frac{4|x-y|(x+y)}{(1+x^2)(1+y^2)} \le 8|x-y|.$$

Its values at rational points are rational and are computed exactly, so Lemma 23.4 with $\theta = 0$ and any $N \ge 2^{p+3}$ gives error at most 2^{-p}. Proposition 23.3 with $a = 0$ gives the reciprocal. □

Square roots of names are computed by the following direct program, which is total.

Lemma 23.6 (Square root)**.** *Let q name $x \ge 0$. For $p \in \mathbb{N}$ put $S = 2^{p+2}$, $u = q(2p+4)$, $v = \max(u,0)$, $c = \lceil vS^2 \rceil \in \mathbb{N}$, and let the output be $\lfloor\sqrt{c}\rfloor/S$, where $\lfloor\sqrt{c}\rfloor$ is the integer square root. This is a name of $\sqrt{x}$.*

Proof. Since $x \ge 0$, $|v-x| \le |u-x| \le 2^{-2p-4} = S^{-2}$, hence $|\sqrt{v}-\sqrt{x}| \le \sqrt{|v-x|} \le S^{-1}$. Put $w = \lfloor\sqrt{c}\rfloor/S$. From $\sqrt{c}-1 < \lfloor\sqrt{c}\rfloor \le \sqrt{c}$ and $vS^2 \le c < vS^2+1$,

$$\sqrt{v}-S^{-1} < w \le \sqrt{v+S^{-2}} \le \sqrt{v}+S^{-1}.$$

Hence $|w-\sqrt{x}| \le 2S^{-1} = 2^{-p-1}$. □

Corollary 23.7 (Radius names)**.** *Names of positive capacities c_i and C yield, by total programs with no further oracle, names of $r_i = \sqrt{c_i/\pi}$ and $R = \sqrt{C/\pi}$.*

Proof. Multiply each capacity name by the name of $1/\pi$ of Lemma 23.5 (Proposition 23.3) and apply Lemma 23.6. □

24. Finite scalar compilation

This section compiles a Hamiltonian H satisfying $\mathrm{Pack}(n,k,C,c,s)$ into the finite form $H_{\mathrm{fin}} = \beta_B \cdot P$, with $B \in \mathbb{N}$, β_B an explicit product of flat cutoffs and $P \in \mathbb{Q}[t,z]$, and proves that the time-one map of H_{fin} still carries every closed source ball into the open target ball. It contains the proofs of Lemma M7.1 and Theorem M7.2.

The argument has three steps. A tensor Bernstein polynomial in the $2n+1$ variables (t,z), with rounded rational coefficients, approximates H in C^1 on a box (Proposition 24.7, Lemma 24.9). Multiplication by the box cutoff makes the error estimate global, at the price of the term $(P-H)\nabla\beta_B$, which is controlled by the value error (Lemma 24.12). Because H is smooth with compact spatial support, the Lipschitz bound of X_H also holds on all of $\mathbb{R}^{2n}$, so a single Grönwall estimate on $[0,1]\times\mathbb{R}^{2n}$ bounds the displacement of every starting point, with no prior information on where the approximate trajectory goes (Lemma 24.14). The displacement is paid from the positive gap between the exact images and the boundary of the target (Theorem 24.15).

Throughout, $|v|$ is the Euclidean norm of $v \in \mathbb{R}^d$, also for values of vector fields, $|v|_\infty = \max_l |v_l|$, so that $|v| \le \sqrt{d}\,|v|_\infty$, and $\|A\|$ is the operator norm of a linear or bilinear map. The real coordinates of $z \in \mathbb{R}^{2n}$ are $\zeta_1,\dots,\zeta_{2n}$. For $G\colon \mathbb{R}\times\mathbb{R}^{2n} \to \mathbb{R}$, $X_G = J_0\nabla_z G$ is the Hamiltonian field and φ_G^t its flow from time 0 to time t (§2.1); since J_0 is orthogonal, $|X_G - X_{G'}| = |\nabla_z G - \nabla_z G'|$. Fix $n \ge 2$, $k \ge 1$, $C > 0$, $c \in (0,\infty)^k$, $s \in (\mathbb{R}^{2n})^k$ and a function H satisfying $\mathrm{Pack}(n,k,C,c,s)$ (Definition M1.1); thus H is C^∞, $H(t,z) = 0$ for all t and all z outside a compact set $K \subset \mathbb{R}^{2n}$, and φ_H^1 maps every $B(s_i,c_i)$ into $B^\circ(C)$. A *compilation datum* for H is a tuple (C1)–(C7) as in §M7.1; we use the numbers C_0, R_*, R_C, a_{box}, B, L_X, C_β, b_R, η, M_0, δ, N_{B} and $\tilde{H}_m$ named there.

24.1. **The box cutoff and global Lipschitz bounds.**

Definition 24.1 (Box cutoff)**.** Let S be the flat step (Definition 2.3). For a real $B > 0$, $x \in \mathbb{R}$ and $z \in \mathbb{R}^{2n}$ put

$$\beta_{B,1}(x) = 1 - S\Big(\frac{x^2 - (B/2)^2}{(3B/4)^2 - (B/2)^2}\Big), \qquad \beta_B(z) = \prod_{l=1}^{2n} \beta_{B,1}(\zeta_l).$$

The compilation uses the integer B of (C2).

Lemma 24.2 (Support in the box)**.** *For every real $B > 0$, the function β_B is C^∞ on $\mathbb{R}^{2n}$, $0 \le \beta_B \le 1$, $\beta_B = 1$ on $[-B/2, B/2]^{2n}$, $\beta_B = 0$ on the closed set $\{|z|_\infty \ge 3B/4\} = \mathbb{R}^{2n} \setminus (-3B/4, 3B/4)^{2n}$, and $\nabla\beta_B = 0$ on the open set $\{|z|_\infty > 3B/4\}$. For the integer B of (C2), $\beta_B(z)H(t,z) = H(t,z)$ for all $(t,z) \in \mathbb{R} \times \mathbb{R}^{2n}$.*

Proof. By Lemma 2.4, S is C^∞ with values in $[0,1]$, $S = 0$ on $(-\infty, 0]$ and $S = 1$ on $[1, \infty)$. The argument of S in $\beta_{B,1}(x)$ is a polynomial in x; it is ≤ 0 for $|x| \le B/2$ and ≥ 1 for $|x| \ge 3B/4$. Hence $\beta_{B,1}$ is C^∞, $0 \le \beta_{B,1} \le 1$, $\beta_{B,1}(x) = 1$ for $|x| \le B/2$ and $\beta_{B,1}(x) = 0$ for $|x| \ge 3B/4$. The statements on β_B follow factor by factor; on the open set $\{|z|_\infty > 3B/4\}$ the function β_B vanishes identically, so its gradient does. If $\beta_B(z) \ne 1$, then $|z|_\infty > B/2 > a_{\mathrm{box}}$ by (C2), so $H(t,z) = 0$ and both sides of $\beta_B H = H$ vanish. □

Lemma 24.3 (Global Lipschitz bounds)**.** *The suprema $\sup_{[0,1]\times\mathbb{R}^{2n}} \|D_z^2 H\|$ and $\sup_{\mathbb{R}^{2n}} |\nabla\beta_B|$ are finite, so (C3) can be met. For L_X, C_β as in (C3) and every $t \in [0,1]$, the field $X_H(t,\cdot)$ is L_X-Lipschitz on $\mathbb{R}^{2n}$ and $|\nabla\beta_B| \le C_\beta$.*

Proof. The set $\mathbb{R} \times (\mathbb{R}^{2n} \setminus K)$ is open and H vanishes on it, so $D_z^2 H$ vanishes on it; and $D_z^2 H$ is continuous on the compact set $[0,1] \times K$. Hence the first supremum is a maximum over $[0,1] \times K$. Likewise $\nabla\beta_B$ is continuous and vanishes outside $[-3B/4, 3B/4]^{2n}$ (Lemma 24.2). Since $D_z X_H = J_0 D_z^2 H$ and J_0 is orthogonal, $\|D_z X_H(t,z)\| = \|D_z^2 H(t,z)\| < L_X$ for $t \in [0,1]$ and every z, and the mean value inequality on the convex set $\mathbb{R}^{2n}$ gives $|X_H(t,z) - X_H(t,z')| \le L_X|z - z'|$. □

24.2. **Tensor Bernstein approximation.**

Definition 24.4 (Tensor Bernstein operator)**.** For $N \ge 1$ and $0 \le j \le N$ let $b_{N,j}(x) = \binom{N}{j} x^j (1-x)^{N-j}$, and $b_{N,j} = 0$ for $j \notin \{0, \dots, N\}$. For $d \ge 1$, $m \in \{0, \dots, N\}^d$ and $x \in [0,1]^d$ put $W_m(x) = \prod_{\ell=1}^d b_{N,m_\ell}(x_\ell)$, and for $g\colon [0,1]^d \to \mathbb{R}$

$$\mathcal{B}_N g(x) = \sum_{m \in \{0,\dots,N\}^d} g(m/N)\, W_m(x).$$

The weights $W_m(x)$ are nonnegative and $\sum_m W_m(x) = \prod_\ell (x_\ell + 1 - x_\ell)^N = 1$.

Lemma 24.5 (Mean absolute deviation)**.** *For $N \ge 1$ and $x \in [0,1]$,*

$$\sum_{j=0}^{N} b_{N,j}(x) \Big|x - \frac{j}{N}\Big| \le \sqrt{\frac{x(1-x)}{N}} \le \frac{1}{2\sqrt{N}}, \qquad \sum_{j=0}^{N-1} b_{N-1,j}(x)\, \big|j - (N-1)x\big| \le \frac{\sqrt{N}}{2}.$$

Proof. The numbers $b_{M,j}(x)$, $0 \le j \le M$, are the probabilities of the binomial distribution with parameters M and x, whose mean is Mx and whose variance is $Mx(1-x)$ [Lor86, §1.1]. By the Cauchy–Schwarz inequality against these probabilities, $\sum_j b_{M,j}(x)|j - Mx| \le \sqrt{Mx(1-x)}$. With $M = N$ and division by N this is the first bound, and $x(1-x) \le 1/4$ gives the second; with $M = N - 1$ it gives $\sqrt{(N-1)x(1-x)} \le \sqrt{N}/2$. □

Lemma 24.6 (Value error)**.** *Let $g\colon [0,1]^d \to \mathbb{R}$ and $M \ge 0$ satisfy $|g(y)-g(y')| \le M\sum_\ell |y_\ell - y'_\ell|$ for all $y, y' \in [0,1]^d$. Then $|\mathcal{B}_N g(x) - g(x)| \le dM/(2\sqrt{N})$ for all $N \ge 1$ and $x \in [0,1]^d$.*

Proof. Since the weights sum to 1,

$$|\mathcal{B}_N g(x) - g(x)| \le \sum_m W_m(x)\,|g(m/N) - g(x)| \le M \sum_{\ell=1}^{d} \sum_m W_m(x) \Big|\frac{m_\ell}{N} - x_\ell\Big|.$$

Summing out the coordinates other than ℓ, whose one-variable weights sum to 1, the inner sum is the first sum of Lemma 24.5 at x_ℓ, hence at most $1/(2\sqrt{N})$. □

Proposition 24.7 (Tensor Bernstein estimates)**.** *Let $d \ge 1$, $N \ge 1$, and let g be the restriction to $[0,1]^d$ of a C^2 function on a neighbourhood of $[0,1]^d$ whose first and second partial derivatives are bounded in absolute value by M_0 on $[0,1]^d$. Then for every $x \in [0,1]^d$ and every $j \in \{1,\dots,d\}$:*

(a) $|\mathcal{B}_N g(x) - g(x)| \le dM_0/(2\sqrt{N})$;
(b) $|\partial_j \mathcal{B}_N g(x) - \partial_j g(x)| \le M_0\big[d/(2\sqrt{N}) + 2/N\big]$;
(c) if $(e_m)_{m\in\{0,\dots,N\}^d}$ are reals with $|e_m| \le a$ for all m, then $\big|\sum_m e_m W_m(x)\big| \le a$ and $\big|\partial_j \sum_m e_m W_m(x)\big| \le 2Na$.

Proof. (a) For $y, y' \in [0,1]^d$ the segment between them lies in the cube, and the mean value theorem along it gives $|g(y) - g(y')| \le M_0 \sum_\ell |y_\ell - y'_\ell|$. Apply Lemma 24.6.

(b) From $b'_{N,i} = N(b_{N-1,i-1} - b_{N-1,i})$ and summation by parts in the j-th index [Lor86, §1.4],

(95)
$$\partial_j \sum_m h_m W_m(x) = N \sum_{m:\, m_j \le N-1} \big(h_{m+e_j} - h_m\big)\, W_m^{(j)}(x), \qquad W_m^{(j)}(x) = b_{N-1,m_j}(x_j) \prod_{\ell \ne j} b_{N,m_\ell}(x_\ell),$$

for arbitrary coefficients h_m, where e_j is the j-th unit vector. The weights $W_m^{(j)}(x)$ are nonnegative and sum to 1. For $h_m = g(m/N)$ and $m_j \le N-1$ the points $(m + se_j)/N$, $s \in [0,1]$, lie in $[0,1]^d$, and

$$N\Big[g\Big(\frac{m+e_j}{N}\Big) - g\Big(\frac{m}{N}\Big)\Big] = \int_0^1 \partial_j g\Big(\frac{m+se_j}{N}\Big) ds.$$

The bound on second derivatives gives, as in (a), $|\partial_j g(y) - \partial_j g(x)| \le M_0 \sum_\ell |y_\ell - x_\ell|$ on the cube. Hence

$$|\partial_j \mathcal{B}_N g(x) - \partial_j g(x)| \le M_0 \sum_{m_j \le N-1} W_m^{(j)}(x) \int_0^1 \Big(\sum_{\ell \ne j} \Big|\frac{m_\ell}{N} - x_\ell\Big| + \Big|\frac{m_j + s}{N} - x_j\Big|\Big) ds.$$

Each term with $\ell \ne j$ contributes at most $1/(2\sqrt{N})$ by Lemma 24.5. For the differentiated coordinate write $\frac{m_j+s}{N} - x_j = \frac{m_j - (N-1)x_j}{N} + \frac{s - x_j}{N}$; the second bound of Lemma 24.5 and $|s - x_j| \le 1$ bound its contribution by $1/(2\sqrt{N}) + 1/N$. Summing gives $|\partial_j \mathcal{B}_N g(x) - \partial_j g(x)| \le M_0[d/(2\sqrt{N}) + 1/N]$; we state the weaker constant $2/N$, which is the one used in (C6).

(c) The first bound holds because the weights $W_m(x)$ are nonnegative with sum 1. By (95), $\partial_j \sum_m e_m W_m(x)$ is N times an average, with nonnegative weights of sum 1, of numbers $e_{m+e_j} - e_m$ of absolute value at most $2a$. □

Definition 24.8 (The finite Hamiltonian)**.** Let a compilation datum (C1)–(C7) be given, put $d = 2n+1$ and $N = N_{\mathrm{B}}$, and let $\Phi_B\colon \mathbb{R}^d \to \mathbb{R}^d$,

$$\Phi_B(t, z) = \Big(t, \frac{\zeta_1 + B}{2B}, \dots, \frac{\zeta_{2n} + B}{2B}\Big),$$

which maps $[0,1]\times[-B,B]^{2n}$ onto $[0,1]^d$. We index the coordinates of $[0,1]^d$ as $\xi=(\xi_0,\dots,\xi_{2n})$; then $\tilde H = H\circ\Phi_B^{-1}$ is the function of (C5). With the rationals $\tilde H_m$ of (C7) put

$$\hat P(\xi)=\sum_{m\in\{0,\dots,N\}^d}\tilde H_m\,W_m(\xi),\qquad P=\hat P\circ\Phi_B,\qquad H_{\mathrm{fin}}(t,z)=\beta_B(z)\,P(t,z).$$

This P is the polynomial displayed after (C7) in §M7.1. Since $\tilde H_m\in\mathbb{Q}$ and $B\in\mathbb{N}$, $P\in\mathbb{Q}[t,z]$, of degree at most N in each variable. We write $a_N=\max_m|\tilde H_m-\tilde H(m/N)|$, so that $a_N<\delta/(2N+1)$ by (C7).

Lemma 24.9 (Polynomial error in physical coordinates)**.** *For every compilation datum, every $t\in[0,1]$, every $z\in[-B,B]^{2n}$ and every $l\in\{1,\dots,2n\}$,*

$$|P(t,z)-H(t,z)|<\delta,\qquad |\partial_{\zeta_l}(P-H)(t,z)|\le\frac{\delta}{B},$$

and hence $|\nabla_z P(t,z)-\nabla_z H(t,z)|\le\sqrt{2n}\,\delta/B<\delta$.

Proof. On $[0,1]^d$ we have $\hat P-\tilde H=(\mathcal{B}_N\tilde H-\tilde H)+\sum_m e_m W_m$ with $e_m=\tilde H_m-\tilde H(m/N)$, $|e_m|\le a_N$. The function $\tilde H$ is the restriction of the C^∞ function $H\circ\Phi_B^{-1}$ on $\mathbb{R}^d$, and M_0 bounds its first and second partial derivatives on $[0,1]^d$ by (C5). By (C6), $\sqrt{N}>dM_0/\delta$ and $N>4M_0/\delta$, so $dM_0/(2\sqrt{N})<\delta/2$ and $2M_0/N<\delta/2$. Proposition 24.7 gives, on $[0,1]^d$,

$$|\hat P-\tilde H|<\tfrac{\delta}{2}+a_N<\delta,\qquad |\partial_{\xi_j}(\hat P-\tilde H)|<\delta+2Na_N<\delta+\tfrac{2N}{2N+1}\delta<2\delta\quad(0\le j\le 2n).$$

Composing with Φ_B, which maps $[0,1]\times[-B,B]^{2n}$ onto $[0,1]^d$, gives the value bound. The chain rule gives $\partial_{\zeta_l}(P-H)=(2B)^{-1}\big(\partial_{\xi_l}(\hat P-\tilde H)\big)\circ\Phi_B$ for $1\le l\le 2n$, so each spatial partial derivative of $P-H$ is at most δ/B in absolute value, and $|\nabla_z(P-H)|\le\sqrt{2n}\,|\nabla_z(P-H)|_\infty\le\sqrt{2n}\,\delta/B$. Finally $B>2n\ge\sqrt{2n}$ by (C2). □

The condition $B>2n$ in (C2) absorbs the factor $\sqrt{2n}$ from the passage to the Euclidean norm, so the bound on δ in (C6) has no dimensional factor.

Remark 24.10 (Grid values). Condition (C7) asks only for rationals near the finitely many grid values, and these exist by density of $\mathbb{Q}$. Compilation data are used only to prove that the search of Section 26 terminates.

24.3. **The global field error.**

Lemma 24.11 (Product rule estimate)**.** *Let E be a real normed space, $b,p,h,C',\varepsilon\in\mathbb{R}$ and $g_b,g_p,g_h\in E$ with $g_h=bg_h+hg_b$, $0\le b\le 1$, $\|g_b\|\le C'$, $|p-h|<\varepsilon$ and $\|g_p-g_h\|<\varepsilon$. Then $\|(bg_p+pg_b)-g_h\|<\varepsilon(1+C')$.*

Proof. By the first hypothesis $(bg_p+pg_b)-g_h=b(g_p-g_h)+(p-h)g_b$, whose norm is at most $b\,\|g_p-g_h\|+|p-h|\,\|g_b\|<\varepsilon+\varepsilon C'$. □

Lemma 24.12 (Global field error)**.** *For every compilation datum, H_{fin} is C^∞ on $\mathbb{R}\times\mathbb{R}^{2n}$, $H_{\mathrm{fin}}(t,z)=0$ for all t whenever $z\notin(-3B/4,3B/4)^{2n}$, and for all $t\in[0,1]$ and $z\in\mathbb{R}^{2n}$*

$$|X_{H_{\mathrm{fin}}}(t,z)-X_H(t,z)|=|\nabla_z H_{\mathrm{fin}}(t,z)-\nabla_z H(t,z)|<\delta(1+C_\beta)<\eta/16.$$

Proof. H_{fin} is the product of the C^∞ function β_B and a polynomial, and vanishes where β_B does (Lemma 24.2). Fix $t\in[0,1]$.

Let $z\in[-B,B]^{2n}$. Differentiating the identity $\beta_B H=H$ of Lemma 24.2 gives $\nabla_z H=\beta_B\nabla_z H+H\nabla\beta_B$, and $\nabla_z H_{\mathrm{fin}}=\beta_B\nabla_z P+P\nabla\beta_B$. Hence

$$\nabla_z H_{\mathrm{fin}}-\nabla_z H=\beta_B\,\nabla_z(P-H)+(P-H)\,\nabla\beta_B.$$

Where β_B is not constant, $H = 0$, so the second term is $P\nabla\beta_B$ and is controlled by the value error of P. By Lemma 24.9, $|P - H| < \delta$ and $|\nabla_z(P - H)| < \delta$ at (t, z); with $0 \le \beta_B \le 1$ and $|\nabla\beta_B| \le C_\beta$ (Lemmas 24.2 and 24.3), Lemma 24.11 with $E = \mathbb{R}^{2n}$, $b = \beta_B(z)$, $p = P(t, z)$, $h = H(t, z)$, $g_b = \nabla\beta_B(z)$, $g_p = \nabla_z P(t, z)$, $g_h = \nabla_z H(t, z)$, $C' = C_\beta$ and $\varepsilon = \delta$ gives $|\nabla_z H_{\mathrm{fin}} - \nabla_z H| < \delta(1 + C_\beta)$.

Let $z \notin (-B, B)^{2n}$. Then z lies in the open set $\{|z|_\infty > 3B/4\}$, on which β_B, hence $H_{\mathrm{fin}}(t, \cdot)$, vanishes, and z lies in the open set $\{|z|_\infty > a_{\mathrm{box}}\}$, on which $H(t, \cdot)$ vanishes by (C2). Both gradients are 0 at z.

Finally $\delta(1 + C_\beta) < \eta/16$ by (C6). □

24.4. **The Grönwall estimate.**

Lemma 24.13 (Grönwall)**.** *Let E be a finite-dimensional real normed space, $L > 0$, $\eta \ge 0$, and let $v, w\colon \mathbb{R}\times E \to E$ be continuous, with $v(t, \cdot)$ L-Lipschitz for every $t \in \mathbb{R}$ and $\|w(t, e)-v(t, e)\| \le \eta$ for all (t, e). If $x, y\colon [0, 1] \to E$ are differentiable with $\dot x(t) = w(t, x(t))$, $\dot y(t) = v(t, y(t))$ for $t \in [0, 1]$ (one-sided derivatives at the end points) and $x(0) = y(0)$, then*

$$\|x(1) - y(1)\| \le \eta\,\frac{e^L - 1}{L}.$$

If moreover $b > 0$ and $\eta = be^{-L}/2$, then $\eta(e^L - 1)/L < b/2$.

Proof. The derivatives of x and y are continuous, so $u = x - y$ is C^1 on $[0, 1]$ with $u(0) = 0$ and

$$\|\dot u(s)\| \le \|w(s, x(s)) - v(s, x(s))\| + \|v(s, x(s)) - v(s, y(s))\| \le \eta + L\|u(s)\|.$$

Put $\Omega(t) = \int_0^t (\eta + L\|u(s)\|)\,ds$. Then $\|u(t)\| \le \int_0^t \|\dot u(s)\|ds \le \Omega(t)$, and Ω is C^1 with $\Omega(0) = 0$ and $\Omega' = \eta+L\|u\| \le \eta+L\Omega$. Hence $(e^{-Lt}\Omega)' \le \eta e^{-Lt}$, and integration gives $\Omega(t) \le \eta(e^{Lt}-1)/L$. At $t = 1$ this is the first claim. If $\eta = be^{-L}/2$, then $\eta(e^L - 1)/L = b(1 - e^{-L})/(2L) < b/2$, since $1 - e^{-L} < L$ for $L > 0$. □

Proof of Lemma M7.1. Let $\pi_{[0,1]}(t) = \min(1, \max(0, t))$ and put

$$v(t, e) = X_H(\pi_{[0,1]}(t), e), \qquad w(t, e) = X_G(\pi_{[0,1]}(t), e).$$

These fields are continuous, and hypotheses (i), (ii) of the lemma, which concern only $t \in [0, 1]$, give the hypotheses of Lemma 24.13 for all $t \in \mathbb{R}$ with $E = \mathbb{R}^{2n}$ and $L = L_X$. On $[0, 1]$ the fields v, w agree with X_H, X_G, so γ_H, γ_G are integral curves of v, w there. Lemma 24.13 with $x = \gamma_G$, $y = \gamma_H$ gives both conclusions. □

Lemma 24.14 (Flow error of the compiled Hamiltonian)**.** *Let a compilation datum be given and let G be H or H_{fin}.*

- *(a) For every $(t_0, z_0) \in \mathbb{R}\times\mathbb{R}^{2n}$ there is an integral curve $\gamma\colon \mathbb{R} \to \mathbb{R}^{2n}$ of X_G with $\gamma(t_0) = z_0$, and every integral curve of X_G defined on an interval $I \ni t_0$ with value z_0 at t_0 is the restriction of γ to I. In particular φ_G^t is defined on all of $\mathbb{R}^{2n}$ for every t, and φ_G^1 is injective.*
- *(b) For every $z \in \mathbb{R}^{2n}$,*

$$|\varphi^1_{H_{\mathrm{fin}}}(z) - \varphi^1_H(z)| \le \eta\,\frac{e^{L_X} - 1}{L_X} < b_R.$$

Proof. (a) The field $X_G = J_0\nabla_z G$ is C^∞ and vanishes on $\mathbb{R} \times (\mathbb{R}^{2n} \setminus K_G)$ for a compact set K_G independent of t: $K_H = K$, and $K_{H_{\mathrm{fin}}} = [-3B/4, 3B/4]^{2n}$ (Lemma 24.12). For every compact interval $[T_1, T_2] \subset \mathbb{R}$, the field X_G and its spatial derivative are continuous and vanish outside the compact set $[T_1, T_2] \times K_G$, so they are bounded on $[T_1, T_2] \times \mathbb{R}^{2n}$. Hence X_G is Lipschitz in z uniformly in $t \in [T_1, T_2]$ and bounded there. By the Picard–Lindelöf theorem, through

every (t_0, z_0) there is a unique maximal integral curve, and every integral curve through (t_0, z_0) is a restriction of it. A maximal curve with a finite end point T would have bounded velocity on a bounded interval ending at T, hence a limit at T, and could be continued past T; so the maximal curve is defined on $\mathbb{R}$. If $\varphi^1_G(z) = \varphi^1_G(z')$, the curves $t \mapsto \varphi^t_G(z)$ and $t \mapsto \varphi^t_G(z')$ pass through the same point at $t = 1$, so they coincide, and $z = z'$.

(b) Let $\pi_{[0,1]}$ be as in the proof of Lemma M7.1 and put $v(t, e) = X_H(\pi_{[0,1]}(t), e)$, $w(t, e) = X_{H_{\mathrm{fin}}}(\pi_{[0,1]}(t), e)$. These fields are continuous; $v(t, \cdot)$ is L_X-Lipschitz for every t by Lemma 24.3, and $|w - v| \le \eta/16 \le \eta$ everywhere by Lemma 24.12. On $[0, 1]$ the curves $x(t) = \varphi^t_{H_{\mathrm{fin}}}(z)$ and $y(t) = \varphi^t_H(z)$ are integral curves of w and v, with $x(0) = y(0) = z$. Lemma 24.13 with $L = L_X$ and $\eta = b_R e^{-L_X}/2$ (C4) gives the claim. □

24.5. **The finite output.**

Theorem 24.15 (Finite output)**.** *Let H satisfy* Pack(n, k, C, c, s)*. For every compilation datum (C1)–(C7) for H, the function $H_{\mathrm{fin}} = \beta_B \cdot P$ of Definition 24.8 is C^∞, $H_{\mathrm{fin}}(t, z) = 0$ for all t whenever $z \notin (-3B/4, 3B/4)^{2n}$, and $|X_{H_{\mathrm{fin}}} - X_H| < \eta$ on $[0, 1] \times \mathbb{R}^{2n}$. For every $z \in \bigcup_i B(s_i, c_i)$,*

$$|\varphi^1_{H_{\mathrm{fin}}}(z) - \varphi^1_H(z)| < b_R, \qquad |\varphi^1_{H_{\mathrm{fin}}}(z)| < R_* + b_R < R_C.$$

In particular H_{fin} satisfies Pack(n, k, C, c, s)*, and if the balls $B(s_i, c_i)$ are pairwise disjoint, so are their images under $\varphi^1_{H_{\mathrm{fin}}}$. This is Theorem M7.2.*

Proof. Smoothness, the support statement and the field error are Lemma 24.12; the support statement is condition (a) of Definition M1.1.

Let $z \in B(s_i, c_i)$. By (C1), $\varphi^1_H(z) \in B(C_0)$, that is $|\varphi^1_H(z)| \le R_*$. By Lemma 24.14(b), $|\varphi^1_{H_{\mathrm{fin}}}(z) - \varphi^1_H(z)| < b_R$, hence, by (C4),

$$|\varphi^1_{H_{\mathrm{fin}}}(z)| < R_* + b_R < R_C,$$

that is $\pi|\varphi^1_{H_{\mathrm{fin}}}(z)|^2 < C$. By Lemma 24.14(a) the integral curve of $X_{H_{\mathrm{fin}}}$ through $(0, z)$ exists on $\mathbb{R}$, and every integral curve $\gamma\colon [0, 1] \to \mathbb{R}^{2n}$ of $X_{H_{\mathrm{fin}}}$ with $\gamma(0) = z$ is its restriction, so $\gamma(1) = \varphi^1_{H_{\mathrm{fin}}}(z) \in B^\circ(C)$. This is condition (b) of Definition M1.1. Since $\varphi^1_{H_{\mathrm{fin}}}$ is injective (Lemma 24.14(a)), it maps pairwise disjoint balls to pairwise disjoint sets. □

25. Finite trajectory certificates

The algorithm searches finite rational data. Its verifier uses the input names, symbolic differentiation of a rational polynomial and exact rational Euler iteration; the geometric construction enters only the proof that an accepted certificate exists (Theorem 25.7). Put $d = 2n$. In this section $\|\cdot\|_\infty$ is the maximum norm and $|\cdot|$ the Euclidean norm, so that $\|x\|_\infty \le |x| \le \sqrt{d}\,\|x\|_\infty \le d\|x\|_\infty$.

Definition 25.1 (Polynomial programs)**.** A polynomial program is a finite expression formed from rational constants, the variables $x_0 = t, x_1 = z_1, \dots, x_d = z_d$, addition and multiplication; a variable x_j with $j > d$ denotes zero. A rational constant is coded by a triple $(a, b, c) \in \mathbb{N}^3$, with value $(a - b)/(c + 1)$. Every rational polynomial in (t, z) is P_e for some program e. For an integer $B \ge 1$ the *budget* of a program is the integer

$$h_B(a, b, c) = a + b + 2, \qquad h_B(x_j) = B + 2,$$
$$h_B(e + f) = h_B(e) + h_B(f), \qquad h_B(ef) = 4h_B(e)h_B(f).$$

Write P_e for the polynomial of e and $\Theta = h_B(e)$. Symbolic differentiation $\partial_j e$ is defined by $\partial_j x_i = \delta_{ij}$, $\partial_j(\text{constant}) = 0$ and the sum and product rules.

Lemma 25.2 (Field bounds)**.** *Let* $\mathrm{clip}(t,z) \in [0,1] \times [-B/2, B/2]^d$ *be the point obtained by clipping* t *to* $[0,1]$ *and each* z_l *to* $[-B/2, B/2]$*, and put*

$$F_e(t,z) = J_0 \nabla_z P_e\big(\mathrm{clip}(t,z)\big).$$

Then, in the maximum norm, $\|F_e\|_\infty \le \Theta$*,* $\|F_e(t,z) - F_e(t,z')\|_\infty \le \Theta \|z - z'\|_\infty$ *and* $\|F_e(t,z) - F_e(t',z)\|_\infty \le \Theta |t - t'|$*. Through every point* F_e *has a unique integral curve on* $[0,1]$*. At rational arguments* F_e *is evaluated exactly by rational arithmetic on the programs* $\partial_j e$*.*

Proof. Let $X = [-B, B]^{d+1}$ with the maximum norm. Define integers $\mathrm{bd}(e)$ and $\mathrm{lip}(e)$ recursively: $\mathrm{bd} = a + b + 1$ and $\mathrm{lip} = 0$ for a constant (a, b, c); $\mathrm{bd} = B$, $\mathrm{lip} = 1$ for a variable; both are additive under sums; and for products $\mathrm{bd}(ef) = \mathrm{bd}(e)\,\mathrm{bd}(f)$, $\mathrm{lip}(ef) = \mathrm{bd}(e)\,\mathrm{lip}(f) + \mathrm{bd}(f)\,\mathrm{lip}(e)$. By induction, $|P_e| \le \mathrm{bd}(e)$ on X and P_e is $\mathrm{lip}(e)$-Lipschitz on X; for products use $P_e P_f(x) - P_e P_f(y) = P_e(x)(P_f(x) - P_f(y)) + P_f(y)(P_e(x) - P_e(y))$. A second induction shows that $h_B(e)$ bounds the four numbers $\mathrm{bd}(e)$, $\mathrm{lip}(e)$, $\mathrm{bd}(\partial_j e)$ and $\mathrm{lip}(\partial_j e)$ for every j. Constants and variables are immediate (note $\mathrm{bd}(\partial_j x_i) \le 2$). Sums are additive. For a product, $\partial_j(ef) = (\partial_j e) f + e\,\partial_j f$, so each of the four numbers for ef is a sum of at most four products of a number for e and a number for f, each at most $h_B(e) h_B(f)$.

Symbolic differentiation agrees with the partial derivative by the sum and product rules, so the components of F_e are $\pm P_{\partial_j e} \circ \mathrm{clip}$. Clipping maps into X (as $B \ge 1$) and is 1-Lipschitz in each coordinate. Hence the speed bound follows from $\mathrm{bd}(\partial_j e) \le \Theta$, and both Lipschitz bounds from $\mathrm{lip}(\partial_j e) \le \Theta$. The field F_e is continuous and globally Lipschitz in z, uniformly in t, so the Picard–Lindelöf theorem gives a unique integral curve on $[0,1]$ through every point. □

Lemma 25.3 (Euler tubes)**.** *Let* $N \ge 1$*,* $a \in \mathbb{Q}^d$*, and form the exact rational table*

$$x_0 = a, \qquad x_{j+1} = x_j + N^{-1} F_e(j/N, x_j) \quad (0 \le j < N).$$

Let γ *be an integral curve of* F_e *on* $[0,1]$ *with* $\|\gamma(0) - a\|_\infty \le \rho$*, and put*

$$E = \left(\rho + \frac{\Theta + \Theta^2}{N}\right) 4^\Theta + \frac{\Theta}{N}. \tag{96}$$

Then $\|\gamma(j/N) - x_j\|_\infty \le E$ *for* $0 \le j \le N$*, and for every* $t \in [0,1]$ *some* $j \le N$ *satisfies* $\|\gamma(t) - x_j\|_\infty \le E$*.*

Proof. Put $t_j = j/N$ and $e_j = \|\gamma(t_j) - x_j\|_\infty$. For $s \in [t_j, t_{j+1}]$ the speed bound gives $\|\gamma(s) - \gamma(t_j)\|_\infty \le \Theta(s - t_j)$, so by the two Lipschitz bounds

$$\|F_e(s, \gamma(s)) - F_e(t_j, x_j)\|_\infty \le (\Theta + \Theta^2)(s - t_j) + \Theta e_j.$$

Integrating $\gamma(t_{j+1}) - x_{j+1} = \gamma(t_j) - x_j + \int_{t_j}^{t_{j+1}} \big(F_e(s,\gamma(s)) - F_e(t_j, x_j)\big) ds$ gives $e_{j+1} \le (1 + \Theta/N) e_j + (\Theta + \Theta^2)/N^2$. Since $e_0 \le \rho$ and $(1 + \Theta/N)^j \le e^\Theta$ for $j \le N$, induction gives $e_j \le (\rho + (\Theta + \Theta^2)/N) e^\Theta$. For $t \in [t_j, t_{j+1}]$ the curve moves by at most Θ/N from $\gamma(t_j)$. Finally $e^\Theta \le 4^\Theta$. □

Let $q_{i,l}(p)$ name the coordinates of s_i, $q_i^r(p)$ name $r_i = \sqrt{c_i/\pi}$ and $q^R(p)$ name $R = \sqrt{C/\pi}$, all with error at most 2^{-p}. The radius names are computed from the capacity names (Corollary 23.7); they are not additional input. Put

$$s_{i,l}^p = q_{i,l}(p), \qquad r_{i,p}^+ = q_i^r(p) + (d+2)2^{-p}, \qquad R_p^- = \max(0, q^R(p) - 2^{-p}). \tag{97}$$

Since $|s_i^p - s_i| \le \sqrt{d}\, 2^{-p}$ and $r_i \le q_i^r(p) + 2^{-p}$, the closed ball $\overline{B}(s_i^p, r_{i,p}^+)$ contains the closed source ball $\overline{B}(s_i, r_i) = B(s_i, c_i)$; and $0 \le R_p^- \le R$.

Definition 25.4 (Finite checks)**.** A candidate is (B, e, Q, p, m, N) with integers $B, Q, N \geq 1$, $p, m \geq 0$ and a polynomial program e. It passes if the following finitely many strict rational inequalities hold. First the source-box tests

$$|s_{i,l}^p| + r_{i,p}^+ < Q \qquad (1 \leq i \leq k,\ 1 \leq l \leq d).$$

Next, let the grid nodes be the $(2^m+1)^d$ points $-Q + 2Q\nu/2^m$, $\nu \in \{0, \dots, 2^m\}^d$; every point of $[-Q, Q]^d$ lies within maximum-norm distance $\rho = 2Q2^{-m}$ of a node. For every node a and every source i, either the discard test

$$\sum_l (a_l - s_{i,l}^p)^2 > (r_{i,p}^+ + d\rho)^2 \tag{98}$$

holds, or the Euler table of a (Lemma 25.3, with E computed from this ρ) satisfies

$$|x_{j,l}| + E < B/2 \qquad (0 \leq j \leq N,\ 1 \leq l \leq d), \tag{99}$$

$$R_p^- - dE > 0, \qquad \sum_l x_{N,l}^2 < (R_p^- - dE)^2. \tag{100}$$

Theorem 25.5 (Soundness)**.** *If a candidate passes, $\beta_B P_e$ satisfies* Pack(n, k, C, c, s).

Proof. The function $\beta_B P_e$ is C^∞ and vanishes for $z \notin (-3B/4, 3B/4)^d$ (Lemma 24.2), which is condition (a) of Definition M1.1. Its field is smooth and vanishes outside $\mathbb{R} \times [-3B/4, 3B/4]^d$, so its integral curves exist on $\mathbb{R}$ and are unique, by the argument of Lemma 24.14(a).

Fix $y \in B(s_i, c_i)$. Then $|y - s_i^p| \leq r_{i,p}^+$, so the source-box tests give $|y_l| < Q$, and some node a has $\|a - y\|_\infty \leq \rho$. Then $|a - s_i^p| \leq r_{i,p}^+ + \sqrt{d}\rho \leq r_{i,p}^+ + d\rho$, so (98) fails and (99), (100) hold for a. Let γ be the integral curve of F_e on $[0, 1]$ with $\gamma(0) = y$. By Lemma 25.3 and (99), $|\gamma_l(t)| < B/2$ for all $t \in [0, 1]$ and all l. On the open cube $(-B/2, B/2)^d$ clipping is the identity and $\beta_B \equiv 1$ on a neighbourhood, so $F_e = X_{\beta_B P_e}$ along γ; thus γ is the integral curve of the output Hamiltonian from y. By Lemma 25.3 and (100),

$$|\gamma(1)| \leq |x_N| + \sqrt{d}\, E \leq |x_N| + dE < R_p^- \leq \sqrt{C/\pi},$$

that is $\gamma(1) \in B^\circ(C)$. By uniqueness every integral curve on $[0, 1]$ from y is γ, which is condition (b). □

Lemma 25.6 (Retained nodes)**.** *If a node a is not discarded for source i, its maximum-norm distance from $B(s_i, c_i)$ is at most*

$$\delta_p = (2d + 3)2^{-p} + d\rho.$$

Proof. Failure of (98) gives $|a - s_i^p| \leq r_{i,p}^+ + d\rho$. Since $|s_i^p - s_i| \leq d2^{-p}$ and $r_{i,p}^+ \leq r_i + (d+3)2^{-p}$, we get $|a - s_i| \leq r_i + \delta_p$. Radial projection to $\overline{B}(s_i, r_i)$ moves a by at most δ_p in the Euclidean norm, hence also in the maximum norm. □

Theorem 25.7 (Nonemptiness)**.** *Let (n, k, C, c, s) satisfy the hypotheses of Theorem C, and let names of the coordinates of the s_i and of the radii r_i, R be given. Then some candidate passes the checks of Definition 25.4.*

Proof. A compiled packing with margins. The Hamiltonian $H = H[n, k, C, c, s]$ of Theorem C satisfies Pack(n, k, C, c, s). The set $\bigcup_i \phi_H^1(B(s_i, c_i))$ is compact and contained in $B^\circ(C)$, hence in $B^\circ(C_0)$ for some $C_0 \in (0, C)$; put $C_1 = (C_0 + C)/2$, so H satisfies Pack(n, k, C_1, c, s) with (C1) for C_0. The exact tube $\mathcal{T} = \{\phi_H^t(y) : t \in [0, 1],\ y \in \bigcup_i B(s_i, c_i)\}$ is compact. Choose a compilation datum (C1)–(C7) for H with target C_1, with $b_R \leq 1$, and with $B > 4 + 2\max_{\mathcal{T}} \|\cdot\|_\infty$; this is possible because each condition is a strict lower or upper bound. Let $H_{\mathrm{fin}} = \beta_B P$ be the compiled function and e a program with $P_e = P$. By the proof of Lemma 24.13, applied

on $[0,t]$, the compiled and exact curves from the same point stay within Euclidean distance $\eta(e^{L_X}-1)/L_X < b_R \le 1$ on $[0,1]$. Hence for every source point y the compiled curve γ_y satisfies $\|\gamma_y(t)\|_\infty < B/2-1$ on $[0,1]$. On this set $F_e = X_{H_{\mathrm{fin}}}$, so γ_y is also the integral curve of F_e from y. By Theorem M7.2 with target C_1, $|\gamma_y(1)| < \sqrt{C_1/\pi} < R$.

A source neighbourhood. Let $\Theta = h_B(e)$. For integral curves of F_e, the spatial Lipschitz bound of Lemma 25.2 and Grönwall's inequality give $\|\gamma_a(t)-\gamma_y(t)\|_\infty \le e^{\Theta}\|a-y\|_\infty$. In particular $y \mapsto \gamma_y(1)$ is continuous, so $\mu = \frac{1}{2}\big(R - \max_y |\gamma_y(1)|\big) > 0$, the maximum being over the compact union of the sources. Choose $\nu > 0$ with $e^{\Theta}\nu \le \frac{1}{2}$ and $de^{\Theta}\nu \le \mu$. If $\|a-y\|_\infty \le \nu$ for a source point y, then

$$|\gamma_{a,l}(t)| < B/2 - \tfrac{1}{2} \quad (t \in [0,1],\ 1 \le l \le d), \qquad |\gamma_a(1)| \le R - \mu. \tag{101}$$

The finite parameters. Since $|s^p_{i,l}| \le |s_{i,l}| + 1$ and $r^+_{i,p} \le r_i + d + 3$ for all p, an integer $Q > \max_{i,l}(|s_{i,l}| + r_i) + d + 4$ passes the source-box tests at every precision. Now $\delta_p \to 0$ as $p, m \to \infty$, and $E \to 0$ as $m, N \to \infty$ with Θ fixed. Choose p, m, N with

$$\delta_p \le \nu, \qquad E < \tfrac{1}{8}, \qquad dE < \tfrac{\mu}{8}, \qquad 2^{1-p} < \tfrac{\mu}{4}.$$

Let a be a node that is not discarded for source i. By Lemma 25.6 it lies within ν of a source point, so (101) holds for γ_a. Lemma 25.3 with initial error 0 (whose bound is at most E) gives $|x_{j,l}| \le |\gamma_{a,l}(j/N)| + E$, hence $|x_{j,l}| + E < B/2 - \frac{1}{2} + \frac{1}{4} < B/2$; and

$$|x_N| + dE \le |\gamma_a(1)| + 2dE < R - \mu + \tfrac{\mu}{4} < R - 2^{1-p} \le R^-_p,$$

using $R^-_p \ge q^R(p) - 2^{-p} \ge R - 2^{1-p}$. Hence $R^-_p - dE > |x_N| \ge 0$ and $|x_N|^2 < (R^-_p - dE)^2$. Discarded pairs pass trivially, so the candidate (B, e, Q, p, m, N) passes. □

26. The algorithm and its termination

The verifier of Section 25 reads finitely many approximations of the input and performs rational arithmetic. This section fixes the encoding, describes the machine and proves that it halts. The dimension and the number of balls are runtime inputs of one machine.

26.1. **Input and output.**

Definition 26.1 (Input)**.** An input consists of integers $n \ge 2$, $k \ge 1$ and names (Definition 23.1) of $C, c_1, \dots, c_k$ and of the $2nk$ coordinates of $s_1, \dots, s_k$. They are supplied as two oracle streams: the capacity stream, whose value at $\langle i, p\rangle$ is the precision-p approximation of c_i for $1 \le i \le k$ and of C for $i = 0$ and for $i > k$; and the centre stream, whose value at $\langle\langle i, p\rangle, l\rangle$ is the precision-p approximation of the l-th coordinate of the $(i+1)$-st centre. A rational input is given by its constant name. The promises are that (n, k, C, c) is admissible and the closed balls $B(s_i, c_i)$ are pairwise disjoint (Theorem C); they are hypotheses of the termination theorem and are not tested.

The output is a natural number $\langle B, \ulcorner e \urcorner\rangle$, where $B \ge 1$ and e is a polynomial program (Definition 25.1). It denotes $H_{\mathrm{fin}} = \beta_B P_e$; expanding the sums and products of e gives the rational coefficients of P_e.

An Epoch input is an integer $k \ge 9$ and a name of $\varepsilon \in (0,1)$. It stands for

$$n = 2, \quad R = \left(\frac{k}{1-\varepsilon/2}\right)^{1/4}, \quad C = \pi R^2, \quad c_i = \pi, \quad s_i = (3i-3, 0, 0, 0). \tag{102}$$

26.2. **Finite encoding.** Tuples and finite lists of natural numbers are coded by iterated pairing. Rationals have the triple code of Definition 25.1. Addition, multiplication, negation, absolute value, maximum, the inverse of a nonzero rational and strict comparison are integer operations on codes; comparison is cross multiplication by the positive denominators, and no reduction to lowest terms is needed.

A polynomial program is coded as a list of instructions $\langle \tau, v \rangle$. The list is read from right to left with a stack of expressions: tag $\tau = 0$ pushes the constant with code v, tag 1 pushes the variable x_v, tag 2 replaces the top two entries by their sum, and any other tag replaces them by their product; a missing entry counts as 0. The program is the top of the final stack, or 0 if the stack is empty. This reading is total, and every expression has a code. The budget $h_B(e)$ and the derivative programs $\partial_j e$ are computed by the same stack recursion.

A vector of runtime dimension d is coded as a list of d rational codes; a code that is not such a list is read as the zero vector. Evaluation of F_e at a rational point clips the arguments and interprets $\partial_j e$; an Euler step is one such evaluation and d rational operations, and the table is computed by primitive recursion on N. The grid node with index $g < (2^m+1)^d$ has coordinates $-Q + 2Q\nu_l/2^m$, where $\nu_1, \dots, \nu_d$ are the base-(2^m+1) digits of g. Table 7 lists the data of the finite verifier and where each is defined.

The radius names $r_i = \sqrt{c_i/\pi}$ and $R = \sqrt{C/\pi}$ are computed from the capacity stream by the total programs of Corollary 23.7. For an Epoch input the source radius name is the constant 1, the centres are rational constants, and the target radius name is obtained from the name of ε: the name of $1-\varepsilon/2$ is formed by Proposition 23.3, its reciprocal uses the lower bound $1-\varepsilon/2 > 2^{-1}$, the product with k gives a name of $k/(1-\varepsilon/2)$, and two applications of Lemma 23.6 give a name of R. This machine never forms names of C or c_i.

Object	Defining rule	Effective data	Reference
Input enclosures	(97)	error 2^{-p}; outer source radius, inner target radius	Def. 25.4
Polynomial field	$F_e = J_0 \nabla P_e$ at clipped arguments	exact rational interpreter; speed and both Lipschitz bounds $\Theta = h_B(e)$	Lem. 25.2
Spatial grid	$-Q + 2Q\nu/2^m$	$(2^m+1)^d$ nodes; covering radius $\rho = 2Q2^{-m}$	Def. 25.4
Euler table	$x_{j+1} = x_j + F_e(j/N, x_j)/N$	tube radius E of (96) on $[0,1]$	Lem. 25.3
Tests	plateau and terminal tests	margins $B/2 - \lvert x_{j,l}\rvert - E$ and $R_p^- - \lvert x_N\rvert - dE$	Thm. 25.5
Existence	smaller-target packing and Bernstein compilation	neighbourhood radius ν, plateau margin $\frac{1}{2}$, terminal margin μ	Thm. 25.7

TABLE 7. Data of the finite verifier. Every quantity used by a test is rational.

26.3. **The machine.**

Construction 26.2 (Algorithm)**.** On input (n,k) and the oracle streams, put $d = 2n$. For $\mathrm{st} = 0, 1, 2, \dots$ decode st as a candidate (p, B, Q, m, N, e) and compute its failure count $\Phi(n,k,\mathrm{st}) \in \mathbb{N}$, a sum of failure indicators: one for the condition $B, Q, N \geq 1$, one for each source-box test, and one for each node-source pair that passes neither the discard test nor both trajectory tests of Definition 25.4, all evaluated with the names at precision p. Thus $\Phi = 0$ exactly when the candidate passes. Stop at the first st with $\Phi(n,k,\mathrm{st}) = 0$ and output $\langle B, \ulcorner e \urcorner \rangle$. The Epoch machine is the same search with $n = 2$ and the names described above; its runtime input is k and its only oracle is the name of ε.

Lemma 26.3 (Totality of the checker)**.** *For fixed oracle streams,* $(n, k, \mathrm{st}) \mapsto \Phi(n, k, \mathrm{st})$ *is a total function, recursive in the oracles. Each value uses finitely many oracle queries.*

Proof. Decoding is primitive recursive, and every natural number decodes to a candidate. The radius names are total programs (Corollary 23.7), each value of which makes finitely many queries; the centre approximations are single queries. The checker then performs: kd source-box tests; for each of the $k(2^m+1)^d$ node-source pairs, one discard test and, if it fails, an Euler table of N steps with $N+1$ plateau rows and one terminal test. The budget Θ, the radius ρ and the tube radius E are computed from the candidate by rational arithmetic and powers. All loop bounds are computed from (n, k, st), every elementary step is an integer operation on codes, and the interpretation of programs is total. Hence Φ is obtained from the oracles by composition and bounded recursion. □

Theorem 26.4 (Termination and correctness)**.** *For every input satisfying the promises of Definition 26.1, Construction 26.2 halts, and its output* $\langle B, \ulcorner e \urcorner \rangle$ *satisfies:* $\beta_B P_e$ *is* C^∞*, vanishes for* $z \notin (-3B/4, 3B/4)^{2n}$*, and satisfies* $\mathrm{Pack}(n, k, C, c, s)$*. The machine is partial recursive relative to the oracle streams, uniformly in* (n, k)*. For an Epoch input it halts with* $\beta_B P_e$ *satisfying* $\mathrm{Pack}(2, k, \pi R^2, \pi, s)$ *for the data* (102)*, and* $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$*.*

Proof. The machine is an unbounded minimisation of the total oracle-recursive function Φ of Lemma 26.3, followed by a primitive recursive output map; hence it is partial recursive in the oracles, with (n, k) as inputs. By Theorem 25.7 there is a passing candidate; its code st_0 satisfies $\Phi(n, k, \mathrm{st}_0) = 0$, since decoding inverts encoding. The minimisation therefore stops at some $\mathrm{st} \le \mathrm{st}_0$, after finitely many evaluations of Φ, each of which halts. The candidate at which it stops passes all tests, so its output is correct by Theorem 25.5.

For an Epoch input, the data (102) are admissible with disjoint closed source balls by Corollary F. The names formed by the machine name the centres, the source radius $\sqrt{\pi/\pi} = 1$ and the target radius $\sqrt{C/\pi} = R$, so the argument above applies with $n = 2$. Finally $R^4 = k/(1 - \varepsilon/2)$ and $\varepsilon > 0$. □

Remark 26.5 (Epoch form). The same verifier serves the Epoch input and all four cases of the construction (Table M8): the cases enter only the proof that a passing certificate exists.

Appendix A. The explicitness check

This appendix describes the Lean command `#check_explicit`, which checks the syntactic form of level L1 (Definition M1.3) on the actual construction terms. It complements the kernel's axiom report: the kernel checks that a theorem proves its stated proposition, while the command audits the data dependencies of a definition. It is not used in any proof of this supplement. Its implementation is in `SymplecticPacking/Explicit.lean`.

A.1. **Data dependencies.** Starting at the named definition, the command follows constants appearing in data positions, including the bodies of lambda expressions and local values. It skips theorem bodies and arguments whose types are propositions, so a classical existence proof that justifies a formula does not make the formula an unspecified choice; a choice that supplies a value is inspected.

Project and vendored definitions are traversed. The audit rejects direct witness selection by `Classical.choose`, `Function.invFun` and the other recorded choice operators unless it belongs to an accepted case below. A missing proof or value represented by `sorryAx` is rejected. The source scan separately excludes new axioms and unchecked implementation mechanisms. The kernel audit of the main results permits only `propext`, `Classical.choice` and `Quot.sound`.

A.2. **Definite operations and searches.** `Explicit.the` supplies definite descriptions with proofs of uniqueness. Its uses in the construction specify integral curves, inverse images, scalar roots and linear solves. Each defining predicate and its domain are part of the mathematical construction. Classical decisions on propositions are allowed as case distinctions at level L1; they do not establish computability from real names.

For a search over naturals, rationals or finite encoded data, the command checks both the search type and genuine decidability of its predicate. A decidability argument reaching classical propositional decision is rejected. Requirements on a wrapper's arguments are propagated to its call sites, so wrapping a search in another definition does not remove the requirement. Search predicates on function spaces are rejected. The implementation also checks natural infima, filtered finite sets and definite descriptions over search types.

Mathlib uses classical choice to define limits, integrals and other definite mathematical operations. The audit accepts only the exact reviewed declarations listed in `Explicit/Whitelist.lean`; it records their dependency paths in its output. These entries are library foundations. They do not permit witness selection in a project or vendored construction. The finite charts, primitives, cutoffs and flows of the construction are therefore audited through their actual data dependencies.

A.3. **Definite descriptions in the construction.** The descriptions over continuous types that the command accepts are reviewed by hand (§M8). The report of `#check_explicit` on `General.packingHamiltonian` lists 26 declarations containing `Explicit.the`. Each describes one of the following objects; in every case the defining property is an explicit equation or inequality in the data of the construction, uniqueness is proved, and none refers to a packing specification.

- *Inverses of injective maps* (the unique preimage, with a fixed fallback value off the image): `Inner.invR`, `Inner.TrotInv`, `Qhat.ordInv`, `Qhat.qhatInv`, `HighDim.ToricPackingStages.circleClockInv`, `layerChartInv` and `nativeInverse`, `BranchLarge3.CubicBridge.expInv` (inverse of a radial moment map), and `General.Ham.pushforward` (the unique source point $u \in \mathbb{B}(r)$ with $F_t(u) = y$, Definition M6.10).
- *Unique roots and critical points*: `Strip.layerRoot` and `Strip.Cubic.layerRoot` (the layer roots of Lemma 11.12 and of the four-dimensional layer maps of §10.9), `Strip.vOfT`, `BranchLarge3.PrimExt.rootExt` (the root closest to an explicit centre), `Branch2.Planar.PlanarInput.pstar` and `HighDim.PlanarGen.Input.vstar` (the unique critical point of a strictly convex potential).
- *Solutions of linear equations*: `BranchLarge3.TargetMoser.solveL` (an invertible bilinear form), `BranchSmall.VT.transportField` and `returnField` (the Moser fields defined by nondegenerate linear systems, §15), and `General.Ham.Linear.posSqrt` (the positive square root of a positive semidefinite matrix).
- *Solutions of ordinary differential equations* (the unique integral curve of a Lipschitz field): `Qhat.flowOf`, `Return.flowFrom`, `BranchSmall.VT.flowI`, `BranchLarge3.UMoser.curve`, `HighDim.ToricPackingStages.nativeAmbientFlow`.
- *Extrema and measures*: `HighDim.ToricPackingStages.finiteMinimumOrZero` (the least value of finitely many continuous functions on compact sets, 0 if there is none) and `Explicit.planeVolume` (Lebesgue measure on $\mathbb{R}^2$, characterised by its values on rectangles).

A.4. **Vendored declarations used.** The 3810 declarations of the vendored library in the dependency closure of the statement and algorithm theorems (§M8) are listed by module in

`verification/Audit/VendoredUsage.lean`. They are explicit charts and coordinates on the diagonal quadrics and the Fermat cubic, Fubini–Study and Liouville primitives, toric and Hopf coordinates, Moser fields and their naturality under pullback, planar area and rearrangement bounds, and the moment polynomials of the models. The closure contains nothing from the curve-analysis directories `Compactness`, `Continuation`, `Degree`, `Cauchy` and `Campanato`.

A.5. **Formal statements that differ in form from the text.** Table 8 records the statements of the paper and of this supplement whose Lean counterparts have a different form. The complete printed statements are proved in the text; where Lean records only the consequence needed by the construction, the statement carries the visible label *needed consequence verified*. The label *full statement verified* is reserved for full mathematical counterparts, up to the stated presentation differences. Unqualified source tags give references without a full-coverage claim. The scope labels record a human comparison, not a kernel check of the prose. Lean names are relative to `SymplecticPacking`.

Table 8: Formal statements that differ in form from the text.

statement	formal form
flat step S (Def. 2.3)	Mathlib's `Real.smoothTransition`, built from `expNegInvGlue` $= a_\flat$; the definite description the(h) is `Explicit.the`
real coordinates	the coordinate ζ_l of $\mathbb{R}^{2n}$ (Table 3) is the coordinate of index $l-1$ of `Problem.E` n; the coordinates are called (q,p) there, with p a coordinate and not a moment
Thm. M3.4 (Thm. 7.13)	the formal statement records (c) with real data, that every $q_t^{(j)}$ maps V_Δ into itself, and, in place of (b), one collar width on which $Q_t^{(j)}$ Poisson-commutes with every face moment; the remaining assertions of (a) and (b) are proved in Theorem 7.13 only
Thm. 4.21	stated for a single ordinary path, without the parameter y, as used in dimension four; the smooth dependence of the planar flow on a finite-dimensional parameter is a separate statement, `HighDim.PlanarGen.compact_param_flow_smooth`
Cor. 4.22	the case of Theorem 5.34(e): $g(I) = I + 4e$, $\varepsilon = 0$, $\Upsilon \le h$ on $[0,h]$, and the core hypothesis $\Upsilon(\pi\lvert\mathfrak{e}_1(z)\rvert^2) < \pi\lvert z\rvert^2 + \frac{13}{24}e$ on $\bar{D}(h-4e)$
Def. 10.13	valid cubic data admit for (q_t) only the whole-disc isotopy $\hat{q}$ of an ordinary path of disc area h, which both four-dimensional branches use
Prop. 10.28	items (a)–(d) and the smoothness of Φ and Φ^{-1}; the decomposition $\Phi = \Phi_2 \circ \Phi_1 \circ \Phi_0$ is not formalised
Prop. 10.29(c)	in the form used in Theorem 13.23: if $\mathcal{T} f_i^N(X) = \mathcal{T} f_j^N(X')$ for the transfer $\mathcal{T}$ of Theorem 13.21, then $f_i^N(X) = f_j^N(X')$
Prop. 13.15, Lem. 13.20, Thm. 13.21	there is no single statement about $\mathcal{T}$ on its whole domain; the theorem is formalised along each embedding $\mathcal{T} \circ f_i^N$. The pullback identities are stated along smooth families of unit vectors (return and ball chart) and for $\mathcal{T}_{\mathrm{u}}$ composed with smooth families with values in $\mathcal{N}_h$, together with Proposition 13.6; injectivity is stated as in the previous row
Thm. 13.23	for the physical maps $u \mapsto \sqrt{C}\, f_i(u/\sqrt{C})$: smoothness with symplectic derivative on the open domain $B^\circ(C\rho_i^{\mathrm{dom}})$, injectivity on $B(C\rho_i^+)$, pairwise disjoint images of these closed balls, and images in $B^\circ(C)$; the margin m is not stated, and (iv) is structural. Injectivity, disjointness and $\pi\lvert f_i\rvert^2 < 1$ on all of $B^\circ(\hat{\rho}_i)$ are proved for the capacity-scale maps in both cases

Continued on the next page

Table 8 (continued)

statement	formal form
Rem. 13.25	the equal-capacity data form the structure `Params`, whose field `Params.`ε is $4\eta_\star$ (not the ε of Theorem A) and whose `Params.rho` $= (1-\eta_\star)/\sqrt{k}$ is ρ; `Branch2.CubicData.ofParams` turns them into cubic data, so both cases pass through `Branch2.cubicEmbeddings_spec`, the map $u \mapsto \sqrt{C}\, f_i(u/\sqrt{C})$. The data of Theorem A are those of `General.epochHamiltonianQ`, and `General.epochQ_equalCase` proves `Branch2.EqualCase` for them
support of H (§M6)	`General.Ham.isotopySupport` is the union of $F_{i,t}(B(r_{2,i}))$ over i and $t \in [0,1]$; `isotopySupport_compact` proves compactness and `ofIsotopy_support_precise` proves that H vanishes outside it, under the common-isotopy hypotheses. For $H[n,k,C,c,s]$, `General.packingSupportSet_compact`, `packingHamiltonian_support_precise` and `packingSupportSet_subset_radius` give the compact swept set K_H and its containment in the closed ball of radius `General.packingSupportRadius`. The resulting vanishing outside that ball is also stated by `main_support_radius`

A.6. **Reproduction.** The statement file runs the check on the Hamiltonian of Theorem C and on the Hamiltonian of the Epoch problem (Remark M7.7); further checks inspect each embedding constructor and, for the finite output, the polynomial interpreter, the rational verifier and the output construction. A nonempty violation list fails the build. Besides the rejected searches listed in §M8, the regression file `test/ExplicitTest.lean` contains genuine finite rational searches that must be accepted, a natural infimum, filtering and a definite description. These tests exercise the search rule; the soundness and termination of the finite packing search are proved in §§25 and 26.

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Email address: `sl12160010@gmail.com`